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Several Complex Variables and Complex Manifolds

Parts I and II

Chapters 1–7

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Preface to Part I

These notes, in two parts, are intended to provide a self-contained and relatively elementary introduction to functions of several complex variables and complex manifolds. They are based on courses on complex analysis that I have given at symposia at the International Centre for Theoretical Physics, Trieste, in 1972 and 1974 and various postgraduate and seminar courses held at Warwick and Sydney. Prerequisites for the reading of Part I are minimal and, in particular, I have made no significant use of differential forms, algebraic topology, differential geometry or sheaf theory. As these notes are primarily directed towards graduate and advanced undergraduate students I have included some exercises. There are also a number of references for further reading which may serve as a suitable starting point for graduate assignments or projects. I have endeavoured to give at least one reference for any result stated but not proved in the text. For the more experienced reader, who is not a specialist in complex analysis, I have included references to related topics not directly within the scope of these notes.

My aim in these notes was to give a broad introduction to several complex variables and complex manifolds and, in particular, achieve a synthesis of the theories of compact and non-compact complex manifolds. This approach is perhaps best exemplified by the conclusion of Part II where we present Grauert's pseudoconvexity proof of the Kodaira embedding theorem. I would hope that parts I and II together comprise a useful introduction to more advanced works on complex analysis. Notably, the books by Grauert and Remmert on Stein spaces [1] and coherent analytic sheaves (forthcoming) and that of Griffiths and Harris on the Principles of Algebraic Geometry [1].

Chapter 1 of the text is devoted to functions of one complex variable and Riemann surfaces with particular emphasis on the $\bar{\partial}$ -operator and the construction of meromorphic functions with specified pole and zero sets, themes that run throughout parts I and II. The presentation is geared towards generalisations to several variables and complex manifolds and most of the results, though perhaps not the proofs, should be familiar to all readers. Section 5 of the Chapter (on vector bundles) can safely be omitted on first reading. In Chapter 2, we develop the basic theory of analytic functions of several complex variables. Amongst the results and concepts discussed are Hartog's theorem on extension of analytic functions, domains of holomorphy, holomorphic convexity, pseudoconvexity, Levi pseudoconvexity and the Levi problem, the Bergman kernel function, the Cousin problems. In section 5, I have given a fairly complete treatment of boundary invariants of domains in $\mathbb{R}^n$ with C^2 boundary. In part this was because of the incomplete treatment of the topic in other texts on several complex variables. In Chapter 3 we prove the Weierstrass Division and Preparation theorems and give applications to the algebraic structure of power series rings and the local structure theory of analytic sets. Here, as elsewhere in the notes, I have concentrated on the structure theory of hypersurfaces leaving the much harder general structure theory of analytic sets to the references (for example, Gunning-Rossi [1], Narasimhan [3], Whitney [1] and the forthcoming text by Grauert and Remmert on coherent analytic sheaves). The chapter concludes with a section on modules over power series rings, the reading of which may be deferred until Chapter 6 of Part II. In Chapter 4 we describe a number of basic examples of complex manifolds, both compact

and non-compact, and conclude with sections on the structure theory of analytic hypersurfaces and blowing up.

Part II of these notes consists of three chapters which we now briefly describe. Chapter 5 covers calculus on complex manifolds including the construction of the $\bar{\partial}$ -operator and the Dolbeault–Grothendieck lemma. Chapter 6 is a self-contained introduction to the theory of sheaves in complex analysis. Chapter 7 is devoted to coherence and the cohomology vanishing theorems of Cartan, Grauert and Serre. Applications include Grauert’s proof of the Kodaira embedding theorem.

When I originally started these notes I had intended to include chapters on complex differential geometry and the elliptic theory of the complex Laplace–Beltrami operator applied to compact and non-compact complex manifolds. For reasons of length I eventually decided to omit these topics from Parts I and II. However, the reader will find references to chapters 8 through 12 scattered throughout the text. At some future time I hope it may be possible to complete the project with these additional chapters.

A few words of guidance to the reader of Part I: There is more than enough material in these notes for a one semester course. As we make the most substantial use of Chapter 3 in Part II, the reader may prefer to omit Chapter 3 at first reading, together with those parts of Chapter 4 on meromorphic functions and analytic sets (in particular, section 6). An alternative approach would be to read Chapter 3 (omitting section 6) and conclude with selected sections of Chapter 4 including section 6 on the structure theory of analytic hypersurfaces (this last section plays an important role in Part II).

I would like to acknowledge the great debt I owe in the preparation of these notes to many authors. I especially would like to mention the books by Grauert and Remmert on Stein Spaces, Gunning and Rossi on Analytic functions of Several Complex Variables and Hörmander on Complex Analysis in Several Variables. This last work has perhaps had the most decisive influence on the final form of my lecture notes.

On a more personal level, it is a great pleasure for me to express thanks to Jim Eells for interesting me in the field of complex analysis back in 1970 and for his continued help and encouragement since then. Thanks also to Tzee-Char Kuo for his advice and encouragement and to the postgraduate students here at Sydney who have been so helpful with their stimulating comments, assignments and critical questioning. Last, but by no means least, may I thank Cathy Kicinski for her beautiful job of typing the bulk of my manuscript.

Mike Field
Sydney,
September, 1981.

Preface to Part II

Since we have already given a general outline of the aim and scope of these notes in the preface to *Several Complex Variables and Complex Manifolds I*, we shall do no more here than provide a brief description of the contents of this volume together with a few notes of guidance to the reader.

Chapter 5 of the present notes is devoted to calculus on complex manifolds. The first four sections cover basic linear algebra and calculus on a differential manifold. Most of the material in these sections should be familiar (though perhaps not the notation) and I would suggest reading through them quickly, referring back to them later, if necessary, for specific results and notation. The next three sections lead up to the construction of the $\bar{\partial}$ -operator on an arbitrary complex manifold and also describe the $\bar{\partial}$ -operator for holomorphic bundle valued forms. In section 8 we prove the Dolbeault–Grothendieck lemma and solve the Cousin problems for polydiscs in $\mathbb{C}^n$. In section 9 we show how holomorphic vector bundles naturally enter into the study of compact complex manifolds. We discuss, for example, the Euler sequence for projective space; the geometric genus; theta functions for complex tori. Finally in section 10, we discuss various pseudoconvexity conditions for non-compact complex manifolds.

Chapter 6 is a self-contained introduction to the theory of sheaves in complex analysis. Section 1 is devoted to sheaves and presheaves with many examples: In section 2 we show how sheaf theory can be used to construct the envelope of holomorphy of a domain spread in $\mathbb{C}^n$. This section is not used elsewhere in the text and may be omitted at first reading. In section 3 we define sheaf cohomology using fine resolutions. After proving Leray’s theorem, we go on to define Čech cohomology and prove that it is naturally isomorphic to cohomology computed using fine resolutions. There are many important illustrations of sheaf cohomology arguments in this section. For example: The de Rham isomorphism between singular and de Rham cohomology; The Dolbeault isomorphism theorem; the first Chern class and classification of complex line bundles.

In Chapter 7 we prove a number of foundational results in the theory of complex manifolds. In section 1 we define coherence and prove Oka’s theorem on the coherence of the sheaf of relations. In section 2 we prove Cartan’s theorems A and B granted the exactness of the $\bar{\partial}$ -sequence for locally free sheaves (A proof of this result will be included in the projected part III of these notes; proofs may also be found in Hörmander [1] or Vesentini [1]). In section 3 we prove the finiteness theorem of Cartan and Serre and in section 4 the finiteness theorem of Grauert. In section 5 we prove Serre’s theorems A and B and give a number of applications. In section 6 we prove Grauert’s vanishing theorem and, following Grauert, show how it may be used to prove the Kodaira embedding theorem.

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Notations and Conventions

Throughout these notes $\mathbb{R}^n$ will always denote real n -space and $\mathbb{C}^n$ complex n -space. We shall often identify $\mathbb{C}^n$ and $\mathbb{R}^{2n}$ by letting $(z_1, \dots, z_n) \in \mathbb{C}^n$ correspond to $(x_1, y_1, \dots, x_n, y_n) \in \mathbb{R}^{2n}$, where $z_j = x_j + iy_j$, $1 \leq j \leq n$. We let $\mathbb{C}^\bullet$ denote the multiplicative group of non-zero complex numbers. We let $\mathbb{Z}, \mathbb{N}$ denote the integers and positive integers respectively.

If E, F are finite dimensional vector spaces over the field $\mathbb{K}$, we let $L_{\mathbb{K}}(E, F)$ denote the $\mathbb{K}$ -vector space of $\mathbb{K}$ -linear maps from E to F . We often drop the subscript $\mathbb{K}$ when it is implicit from the context. If $\mathbb{K} = \mathbb{R}$, we set $E' = L_{\mathbb{R}}(E, \mathbb{R})$ and if $\mathbb{K} = \mathbb{C}$, we set $E^* = L_{\mathbb{C}}(E, \mathbb{C})$. If $A \in L_{\mathbb{R}}(E, F)$, we let $A' \in L_{\mathbb{R}}(F', E')$ denote the transpose of A . We let A^* denote the transpose of A in case A is $\mathbb{C}$ -linear. We let $\text{GL}(E)$ denote the group of linear isomorphisms of E . In case $E = \mathbb{R}^n$, we often use the notation $\text{GL}(n, \mathbb{R})$ for $\text{GL}(\mathbb{R}^n)$. Similarly, we often write $\text{GL}(n, \mathbb{C})$ instead of $\text{GL}(\mathbb{C}^n)$.

A *domain* will always refer to a connected open subset.

If Ω is a domain in $\mathbb{C}^n$, E is a finite dimensional vector space (over $\mathbb{R}$ or $\mathbb{C}$) and $f : \Omega \rightarrow E$ we say that f is C^r if it is r -times continuously differentiable. That is, we identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$ and require that all partial derivatives of f of order less than or equal to r exist and are continuous on Ω . If f is C^r for all positive integers r , we say that f is C^∞ (or smooth). We let $C^r(\Omega, E)$ denote the space of all E -valued C^r maps on Ω . In case $E = \mathbb{C}$ we abbreviate to $C^r(\Omega)$ and if $E = \mathbb{R}$, we abbreviate to $C_{\mathbb{R}}^r(\Omega)$.

If f is a vector valued map we define the (closed) support of f , $\text{supp}(f)$, to be the closure of the set of points where f is non-zero. If $\text{supp}(f)$ is compact we shall say that f has compact support. We denote the set of C^r E -valued maps on a domain Ω which have compact support by $C_c^r(\Omega, E)$.

Suppose that Ω is a domain in $\mathbb{C}^n$ or $\mathbb{R}^n$ and $f \in C^r(\Omega, E)$, $r > 0$. We use either of the notations $D^s f_x$, $D^s f(x)$ to denote the s th. derivative of f at x . Thus, $D^s f_x$ will be an s -linear E -valued map (see also Dieudonné [1] and Field [1]).

Given $r > 0$, $z \in \mathbb{C}$, we let $D_r(z)$, $\overline{D}_r(z)$ denote the open and closed discs, centre z , radius r in $\mathbb{C}$ respectively. We let $D_r(z)^*$ denote the punctured disc $D_r(z) \setminus \{z\}$.

For $z = (z_1, \dots, z_n) \in \mathbb{C}^n$ we define

$$|z| = \max_i |z_i|,$$

$$\|z\| = \left\{ \sum_{i=1}^n |z_i|^2 \right\}^{1/2}, \quad \text{“Euclidean norm”}.$$

For $r > 0$, we let $D(z; r)$, $E(z; r)$ respectively denote the open discs, centre z , radius r in $\mathbb{C}^n$ relative to norms $|\cdot|$, $\|\cdot\|$. Given $r_1, \dots, r_n > 0$, we let $D(z; r_1, \dots, r_n)$ denote the open polydisc $\prod_{j=1}^n D_{r_j}(z_j) \subset \mathbb{C}^n$.

If f is a continuous $\mathbb{C}$ - or $\mathbb{R}$ -valued function defined on a neighbourhood of a compact set K , we define

$$\|f\|_K = \sup_{x \in K} |f(x)|.$$

Other notations will be defined in the text. We remark here only that from Chapter 5, $C^p(M)$ will refer to the space of smooth differential p -forms on the differential manifold M and that from Chapter 6, $\mathbb{C}$ may also refer to the constant sheaf of complex numbers.

Finally, we remark that Hermitian forms are always complex linear in the *first* variable.

Part I

Functions of Complex Variables and Complex Manifolds

Several Complex Variables and Complex Manifolds I

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Chapter 1

Functions of One Complex Variable

Introduction

Our aim in this chapter is to develop the familiar theories of analytic functions of one complex variable and Riemann surfaces in a way that generalises well to the several variable theory. In §3 we see how the existence theory of the Cauchy–Riemann equations can be used to prove the Mittag-Leffler theorem and also how the topology of domains in $\mathbb{C}$ naturally enters into the proof of the Weierstrass theorem. In §§4,5 we show how the theory of holomorphic line bundles may be used to reformulate some of the classical problems in Riemann surface theory. We also define the Cauchy–Riemann equations on an arbitrary Riemann surface and indicate how they are related to the problem of constructing meromorphic functions with specified divisors. In an appendix we prove a number of classical results, including the Runge approximation theorem. We use the Runge theorem to construct solutions of the Cauchy–Riemann equations.

1 Analytic functions and power series

Let Ω be a domain in $\mathbb{C}$. We recall that a function $f : \Omega \rightarrow \mathbb{C}$ is said to be *analytic* or *holomorphic* if it is complex differentiable on Ω . Writing f in real and imaginary parts, $f = u + iv$, analyticity implies that u and v satisfy the Cauchy–Riemann equations on Ω :

$$\partial u / \partial x = \partial v / \partial y; \quad \partial u / \partial y = -\partial v / \partial x.$$

Recalling that a real 2×2 -matrix $[a_{ij}]$ induces a complex linear endomorphism of $\mathbb{C}$ if and only if $a_{11} = a_{22}$ and $a_{12} = -a_{21}$, we may interpret the Cauchy–Riemann equations as saying that if f is analytic then f is differentiable in the real sense and the (real) derivative of f is everywhere a complex linear map (see, for example, Field [1, Example 3, page 133]).

Let $A(\Omega)$ denote the set of all analytic functions on Ω .

We now introduce a pair of partial differential operators which, together with their generalisations to several variables, will be of the utmost importance in the sequel. We set

$$\partial / \partial z = \frac{1}{2}(\partial / \partial x - i \partial / \partial y); \quad \partial / \partial \bar{z} = \frac{1}{2}(\partial / \partial x + i \partial / \partial y).$$

For $r \geq 1$,

$$\partial / \partial z, \partial / \partial \bar{z} : C^r(\Omega) \longrightarrow C^{r-1}(\Omega).$$

The significance of these operators may be gauged from

Lemma 1.1.1. A function $f \in C^1(\Omega)$ is analytic if and only if $\partial f / \partial \bar{z} = 0$.

Proof. The reader may verify that $\partial f / \partial \bar{z} = 0$ iff the Cauchy–Riemann equations hold. Since f is assumed to be C^1 , the Cauchy–Riemann equations hold iff f is analytic. $\square$

Next we recall the basic theorem on the local representation of analytic functions by power series.

Theorem 1.1.2. Let $f \in A(\Omega)$. Given $\zeta \in \Omega$ and $r > 0$ such that $D_r(\zeta) \subset \Omega$, we have

$$f(z) = \sum_{j=0}^{\infty} a_j (z - \zeta)^j, \quad z \in D_r(\zeta),$$

where $a_j = \partial^j f / \partial z^j(\zeta) / j!$, and convergence is uniform on compact subsets of $D_r(\zeta)$.

Remark. Simple examples, such as $f(z) = (1 - z)^{-1}$, $\Omega = \mathbb{C} \setminus \{1\}$, show that the power series at ζ need not converge on the whole of Ω .

Corollary 1.1.3. An analytic function is C^∞ .

Remark. Notice that $A(\Omega) = \text{Kernel}(\partial / \partial \bar{z})$. Now $\partial / \partial \bar{z}$ is an example of an *elliptic* differential operator and it can be shown that the kernel of any elliptic operator consists of C^∞ functions. We shall return to this type of question in later chapters.

Corollary 1.1.4. If $f \in A(\Omega)$, then $\partial^j f / \partial z^j \in A(\Omega)$, $j \geq 0$.

Corollary 1.1.5. Let Ω be a domain in $\mathbb{C}$. Suppose $f \in A(\Omega)$ and that at some point $\zeta \in \Omega$, $\partial^j f / \partial z^j(\zeta) = 0$, $j \geq 0$. Then f vanishes identically on Ω .

Proof. Let $X = \{z \in \Omega : \partial^j f / \partial z^j = 0, \text{ all } j \geq 0\}$. X is open by the power series representation of analytic functions given by Theorem 1.1.2 and X is certainly non-empty since $\zeta \in X$. Since X is the intersection of the closed sets $\{z \in \Omega : \partial^j f / \partial z^j(z) = 0\}$, X is also closed. Since Ω is connected, $X = \Omega$. $\square$

Remark. Another way of stating this corollary is that the value of an analytic function and all its derivatives at a single point of a domain determine the function uniquely. This type of behaviour does not, of course, hold for C^∞ or continuous functions.

Proposition 1.1.6. (Uniqueness of analytic continuation). Let U, V be connected open subsets of $\mathbb{C}$ and suppose $U \cap V \neq \emptyset$. If $f \in A(U)$ and h is an analytic extension of f to $U \cup V$ (that is, $h \in A(U \cup V)$ and $h|_U = f$), then h is unique.

Proof. If h_1, h_2 are analytic extensions of f to $U \cup V$ then $h_1 - h_2$ is an analytic extension of the zero function on U to $U \cup V$. Therefore, $h_1 - h_2$ is identically zero by Corollary 1.1.5. $\square$

Remark. Once we have uniqueness of analytic continuation it is natural to try to construct the largest domain to which any given analytic function may be extended. It turns out of course that we have to enlarge our class of domains to include Riemann surfaces spread over $\mathbb{C}$. We return to this question in §4 of this chapter and again in §2 of Chapter 6.

Exercises. These exercises are revision of basic theory of functions of one complex variable. Proofs may be found in any of the many introductory texts on complex analysis.

- (1) (Laurent series). Let f be analytic on the annulus $r < |z - z_0| < R$. Derive the Laurent series of f at z_0 ,

$$f(z) = \sum_{n=-\infty}^{n=+\infty} a_n(z - z_0)^n, \quad r < |z - z_0| < R$$

where $a_n = (2\pi i)^{-1} \int_{|\zeta - z_0|=s} f(\zeta)/(\zeta - z_0)^{n+1} d\zeta$, $r < s < R$, and convergence is uniform on compact subsets of the annulus.

- (2) (Cauchy's inequalities). Continuing with the notation and assumptions of question 1, show that if $M(t) = \sup\{|f(z)| : |z - z_0| = t\}$, $r < t < R$, then

$$|a_n| \leq M(t)/t^n, \quad n \in \mathbb{Z}.$$

In particular, show that if f is holomorphic on the disc $D_R(z_0)$ then

$$|\partial^n f / \partial z^n(z_0)| \leq M(t)n!/t^n, \quad n \geq 0.$$

- (3) (Riemann removable singularities theorem). Suppose that f is an analytic function in the punctured disc $D_r(z_0)^* = \{z : 0 < |z - z_0| < r\}$. Show that a necessary and sufficient condition for f to extend analytically to $D_r(z_0)$ is that f is bounded on some neighbourhood of z_0 . (Use the result of question 2).
- (4) (Open mapping theorem). Let f be analytic and not identically zero on the domain U in $\mathbb{C}$. Prove that $f(U)$ is open in $\mathbb{C}$.
- (5) (Monodromy theorem). Let Ω be a domain in $\mathbb{C}$, $z_0, y_0 \in \Omega$ and suppose f is analytic on some neighbourhood U of z_0 . Let C be a continuous path in Ω parametrized by $\phi : [0, 1] \rightarrow \Omega$ with $\phi(0) = z_0$, $\phi(1) = y_0$. We say f can be analytically continued along C if we can find discs $D_i = D_{r_i}(\phi(t_i))$, $0 = t_0 < \dots < t_k = 1$, covering C , $h_i \in A(D_i)$, $i = 0, \dots, k$, such that $h_0 = f$ on $U \cap D_0$ and $h_i = h_{i+1}$ on $D_i \cap D_{i+1}$, $i \geq 0$. Define $f_C(y_0)$ to be $h_k(y_0)$. Show that

- (a) $f_C(y_0)$ depends only on C and not on any of the choices we have made.
- (b) If C, C' are homotopic curves in Ω joining z_0 to y_0 then $f_C(y_0) = f_{C'}(y_0)$.

Give examples to show that if Ω is not simply connected and C, C' are non-homotopic curves joining z_0 to y_0 then $f_C(y_0)$ may not equal $f_{C'}(y_0)$.

- (6) (Maximum principle). Let f be analytic on the domain Ω in $\mathbb{C}$. If $|f|$ has a maximum in Ω then f is constant.
- (7) (Schwarz' lemma). Suppose $f \in A(D_1(0))$ and satisfies $f(0) = 0$ and $|f(z)| \leq 1$, $z \in D_1(0)$. Then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$. Equality holds if and only if $f(z) = cz$, where $|c| = 1$ (Hint: Apply the maximum principle to the function $f(z)/z$).

2 Meromorphic functions

Roughly speaking meromorphic functions are the analytic analogue of rational functions and in this section we briefly review their definition and elementary properties. Throughout the section Ω will denote a domain in $\mathbb{C}$.

Let $\zeta \in \Omega$ and $\Omega' = \Omega \setminus \{\zeta\}$. Suppose $f \in A(\Omega')$. By Laurent's expansion we have

$$f(z) = \sum_{j=-\infty}^{j=+\infty} a_j(z - \zeta)^j, \quad z \in D_r(\zeta)^* \subset \Omega.$$

There are three possibilities:

- (a) f is bounded on some neighbourhood of ζ . In this case $a_j = 0$, $j < 0$, and f extends uniquely to an analytic function on Ω (Riemann removable singularities theorem).
- (b) $f(z) \rightarrow \infty$ as $z \rightarrow \zeta$. Here one can show that there exists a strictly positive integer N such that $a_j = 0$, $j < -N$ and $a_{-N} \neq 0$.
- (c) Neither a) or b) occurs (ζ is an *essential singularity* of f).

In case b), f is an example of a *meromorphic function* on Ω with a single pole of order N at ζ . We may write $f(z) = u(z)/(z - \zeta)^N$, $z \in D_r(\zeta)^*$, where $u \in A(D_r(\zeta)^*)$ and the power series of u at ζ is given explicitly as the Laurent series of f at ζ multiplied by $(z - \zeta)^N$. We may extend u analytically to Ω by taking $u(z) = (z - \zeta)^N f(z)$, $z \neq \zeta$. In this way we may represent f as the quotient $u(z)/(z - \zeta)^N$ of analytic functions defined on all of Ω .

There are some problems in giving a satisfactory general definition of a meromorphic function. If we attempt to define a meromorphic function on Ω as a quotient f/g , $f, g \in A(\Omega)$, $g \not\equiv 0$, we are faced with the difficulty that f and g may have infinitely many common zeros. If this happens we cannot cancel the zeros using the elementary power series techniques of the previous paragraph to obtain the maximal subdomain of Ω on which the meromorphic function is defined as an analytic function. That is, the representation f/g may be rather non-canonical. We start by giving a definition that is rather special to functions of one complex variable and then show how to reformulate the definition in a way that generalises well to functions of several complex variables.

Definition 1.2.1. We say that m is a *meromorphic function* on Ω if there exists a discrete subset X of Ω such that

- (i) m is an analytic function on $\Omega \setminus X$.
- (ii) Every point of X is a pole of m .

We notice that the definition excludes essential singularities and so, for example, $e^{-1/z}$ would not define a meromorphic function on $\mathbb{C}$.

We denote the set of meromorphic functions on Ω by $M(\Omega)$.

Locally a meromorphic function can be expressed as a quotient of analytic functions. That is, given $m \in M(\Omega)$ and $z \in \Omega$, we can find an open neighbourhood U of z and $f, g \in A(U)$ such that g is not identically zero and $m|_U = f/g$ outside of any poles of m in U . Of course, if U does not contain any poles of m , we can take $g = 1$.

We now work towards an alternative definition of a meromorphic function which is framed in terms of local information and requires no explicit information about the pole set.

Suppose that $\{U_i : i \in I\}$ is an open cover of Ω and that for each $i \in I$ we are given $f_i, g_i \in A(U_i)$ with g_i not vanishing identically on any connected component of U_i . Then $\{(f_i, g_i) : i \in I\}$ defines a meromorphic function m on Ω provided that for all $i, j \in I$ we have

$$f_i/g_i = f_j/g_j$$

at all points of $U_i \cap U_j$ where both g_i and g_j are non-zero. (Equivalently, $f_i g_j = f_j g_i$ on $U_i \cap U_j$). We omit the routine construction of m . If $\{V_j : j \in J\}$ is another open cover of Ω and $\{(a_j, b_j) : j \in J\}$ a corresponding set of analytic functions satisfying the above conditions, it is easily verified that $\{(f_i, g_i) : i \in I\}$ and $\{(a_j, b_j) : j \in J\}$ define the same meromorphic function if and only if

$$f_i/g_i = a_j/b_j$$

at all points of $U_i \cap V_j$ where both g_i and b_j are non-zero.

We can now use this condition to define an equivalence relation on all sets of pairs of analytic functions $\{(f_i, g_i) : i \in I\}$ satisfying the requisite compatibility conditions. The equivalence classes of this relation are then defined to be meromorphic functions. This is essentially the approach that we adopt in later chapters.

One immediate consequence of our local description of meromorphic functions is that $M(\Omega)$ is a field (the connectedness of Ω is essential here to avoid zero divisors).

Suppose $m \in M(\Omega)$ and $\zeta \in \Omega$. We define the *order* of m at ζ , $\text{ord}(m, \zeta)$, to be the smallest index with non-zero coefficient in the Laurent expansion of m at ζ . That is, if

$$m(z) = \sum_{j=N}^{\infty} a_j(z - \zeta)^j$$

on some neighbourhood of ζ and $a_N \neq 0$, then $\text{ord}(m, \zeta) = N$. Clearly, if $m = f/g$ in some neighbourhood of ζ , $\text{ord}(m, \zeta) = \text{ord}(f, \zeta) - \text{ord}(g, \zeta)$ though of course the terms on the right hand side depend on the choice of local representation for m !

If $\text{ord}(m, \zeta) > 0$, we say that m has a *zero of order* $\text{ord}(m, \zeta)$ at ζ and if $\text{ord}(m, \zeta) < 0$, we say that m has a *pole of order* $-\text{ord}(m, \zeta)$ at ζ . We set

$$\begin{aligned} Z(m) &= \{z \in \Omega : \text{ord}(m, z) > 0\} = \{z \in \Omega : m(z) = 0\}, \\ P(m) &= \{z \in \Omega : \text{ord}(m, z) < 0\} = \{z \in \Omega : m \text{ has a pole at } z\}. \end{aligned}$$

$Z(m)$ and $P(m)$ are called the zero and pole set of m respectively. Clearly $Z(m)$ and $P(m)$ are disjoint discrete subsets of Ω (assuming $m \neq 0$).

We now introduce some useful definitions and notation. Suppose $p : \Omega \rightarrow \mathbb{Z}$ and $\{z \in \Omega : p(z) \neq 0\}$ is a discrete subset of Ω . We call the formal sum $\sum_{z \in \Omega} p(z) \cdot z$ a *divisor* on Ω . We denote the set of divisors on Ω by $\mathcal{D}(\Omega)$. $\mathcal{D}(\Omega)$ has the structure of an ordered abelian group if we define

$$\begin{aligned} \left(\sum_{z \in \Omega} p(z) \cdot z \pm \sum_{z \in \Omega} q(z) \cdot z \right) &= \sum_{z \in \Omega} (p \pm q)(z) \cdot z \\ \sum_{z \in \Omega} p(z) \cdot z &> \sum_{z \in \Omega} q(z) \cdot z \quad \text{if and only if} \\ p(z) &\geq q(z) \text{ for all } z \in \Omega \text{ with strict inequality for at least one point of } \Omega. \end{aligned}$$

$p(z) \geq q(z)$ for all $z \in \Omega$ with strict inequality for at least one point of Ω .

Let $M^*(\Omega)$ denote the group of invertible elements of $M(\Omega)$. Since Ω is assumed connected, $M^*(\Omega)$ is all of $M(\Omega)$ except the zero function. Given $m \in M^*(\Omega)$, the divisor of m , $\text{div}(m)$, is defined to be

$$\sum_{z \in \Omega} \text{ord}(m, z) \cdot z.$$

$\text{div} : M^*(\Omega) \rightarrow \mathcal{D}(\Omega)$ is a homomorphism (relative to the multiplicative structure on $M^*(\Omega)$). We note that $\text{div}(m) \geq 0$ if and only if $m \in A(\Omega)$.

Suppose $m \in M(\Omega)$ and $\zeta \in \Omega$. Let m have Laurent series

$$\sum_{j=N}^{\infty} a_j(z - \zeta)^j$$

at ζ , where we suppose $a_N \neq 0$. The principal part of m at ζ is defined to be

$$\sum_{j=N}^{-1} a_j (z - \zeta)^j$$

if $N \leq -1$ and zero otherwise. We note that $m, m' \in M(\Omega)$ have the same principal part at ζ if and only if $m - m'$ is analytic on some neighbourhood of ζ .

To conclude this section we remark that Proposition 1.1.6 generalises to meromorphic functions and therefore we can discuss meromorphic continuation. We leave details to the reader.

Exercises

1. Verify that

$$\begin{aligned} \operatorname{div}(mm') &= \operatorname{div}(m) + \operatorname{div}(m'), \quad m, m' \in M^*(\Omega), \\ \operatorname{div}(m^{-1}) &= -\operatorname{div}(m), \quad m \in M^*(\Omega), \end{aligned}$$

$\operatorname{div}(m) = 0$ if and only if m is a nowhere vanishing analytic function on Ω .

2. Let $m \in M^*(\mathbb{C})$. Show that if $\operatorname{div}(m) = 0$ then either m is constant or m has an essential singularity at infinity.
3. Under what conditions is the composition of two meromorphic functions meromorphic?

3 Theorems of Weierstrass and Mittag-Leffler

In the preceding sections we have reviewed some of the basic elementary properties of analytic and meromorphic functions. However, we have as yet given no means of constructing such functions so as to satisfy specified properties. For example, if X is any closed subset of $\mathbb{C}$ it is not difficult to construct a C^∞ function on $\mathbb{C}$ with zero set X . Can we find an analytic function whose zero set is equal to X ? Clearly we cannot unless X is a discrete subset of $\mathbb{C}$. It turns out though that if X is discrete we can always find an analytic function on $\mathbb{C}$ with zero set X . This is exactly the type of result we need if our study of analytic functions is to amount to much more than a study of polynomials, rational functions and the standard analytic functions such as $\log$ and $\exp$.

Our aim in this section will be to show the importance of the theory of the partial differential operator $\partial/\partial\bar{z}$ and the topology of domains in $\mathbb{C}$ in questions involving the construction of analytic and meromorphic functions with specified behaviour at prescribed poles and zeros. We adopt this approach because it generalises well to functions of several complex variables and complex manifolds. We must emphasise, however, that the Mittag-Leffler and Weierstrass theorems can be given rather more elementary proofs than those presented here which do not depend on the theory of $\partial/\partial\bar{z}$ (see, for example, Heins [1] or Hille [1]).

Throughout this section Ω will denote an open subset of $\mathbb{C}$.

We give the proof of the following basic existence theorem in the appendix to this chapter.

Theorem 1.3.1. Let $f \in C^\infty(\Omega)$. Then there exists $u \in C^\infty(\Omega)$ such that

$$\partial u / \partial \bar{z} = f.$$

Remark. An equivalent formulation of Theorem 1.3.1 is that the sequence

$$0 \longrightarrow A(\Omega) \xrightarrow{i} C^\infty(\Omega) \xrightarrow{\partial/\partial\bar{z}} C^\infty(\Omega) \longrightarrow 0$$

is exact for every open subset Ω of $\mathbb{C}$ (i denotes inclusion).

The Mittag-Leffler theorem gives conditions under which there exists a meromorphic function on Ω with specified principal parts.

Before stating the Mittag-Leffler theorem we introduce some useful notation. Suppose U_i, U_j are sets, then we let U_{ij} denote the intersection $U_i \cap U_j$. We use the obvious generalisation of this notation for intersections of more than two indexed sets.

Theorem 1.3.2. (Mittag-Leffler theorem) Let $\{U_i : i \in I\}$ be an open cover of Ω and suppose we are given $m_i \in M(U_i)$ for each $i \in I$. Then, provided that $m_i - m_j \in A(U_{ij})$ for all $i, j \in I$, there exists $m \in M(\Omega)$ such that

$$m - m_i \in A(U_i), \quad i \in I.$$

Remark. An alternative formulation of this theorem would be: Suppose X is a discrete subset of Ω and that for each $z \in X$ we are given a meromorphic function m^z which is defined on some neighbourhood of z and has a single pole at z . Then there exists $m \in M(\Omega)$ with pole set X and such that the principal part of m at z equals the principal part of m^z at z for all $z \in X$.

Proof of theorem. (Following Hörmander [1]). Observe that it suffices to construct $f_i \in A(U_i)$ such that for all i, j

$$m_i + f_i = m_j + f_j \quad \text{on } U_{ij}.$$

Indeed, if these compatibility conditions hold we can define $m \in M(\Omega)$ by setting $m|_{U_i} = m_i + f_i$ and m will then obviously satisfy the requirements of the theorem.

What we shall do is to start by constructing $h_i \in C^\infty(U_i)$ such that $m_i + h_i = m_j + h_j$ on U_{ij} . This will only use the theory of C^∞ functions and no complex analysis. Using Theorem 1.3.1, we then construct a “correction” term $u \in C^\infty(\Omega)$ such that for all $i \in I$, $h_i + u \in A(U_i)$. It is then enough to take $f_i = h_i + u$.

Step 1. Choose a partition of unity $\{\theta_\alpha : \alpha \in \Lambda\}$ subordinate to the cover $\{U_i : i \in I\}$. That is, each θ_α is a positive C^∞ function with compact support, $\text{supp}(\theta_\alpha) \subset U_{\tau(\alpha)}$ for some $\tau(\alpha) \in I$, only finitely many θ_α are non-zero on any given compact subset of Ω and $\sum_{\alpha \in \Lambda} \theta_\alpha = 1$ on Ω (for the construction of partitions of unity we refer to Lang [1], Spivak [1] or de Rham [1]).

Set $f_{ij} = m_i - m_j \in A(U_{ij})$ and observe that for all $i, j, k \in I$

$$f_{ij} + f_{jk} + f_{ki} = 0 \quad \text{on } U_{ijk} \quad \text{and} \quad f_{ij} = -f_{ji} \quad \text{on } U_{ij}. \quad (\text{A})$$

We define

$$h_i = \sum_{\alpha \in \Lambda} \theta_\alpha f_{\tau(\alpha)i}.$$

Setting $f_{\tau(\alpha)i} = 0$ outside $U_{\tau(\alpha)i}$ it is clear that $h_i \in C^\infty(U_i)$. Moreover

$$\begin{aligned} h_j - h_i &= \sum_{\alpha \in \Lambda} \theta_\alpha (f_{\tau(\alpha)j} - f_{\tau(\alpha)i}) \\ &= \sum_{\alpha \in \Lambda} \theta_\alpha f_{ij}, \quad \text{by (A)} \\ &= f_{ij} = m_i - m_j. \end{aligned}$$

Hence we have found $h_i \in C^\infty(U_i)$ satisfying $m_i + h_i = m_j + h_j$ on U_{ij} .

Step 2. Since $h_i - h_j \in A(U_{ij})$,

$$\partial h_i / \partial \bar{z} = \partial h_j / \partial \bar{z} \quad \text{on } U_{ij}.$$

Hence we may define $F \in C^\infty(\Omega)$ by requiring that $F|_{U_i} = \partial h_i / \partial \bar{z}$.

To construct $u \in C^\infty(\Omega)$ satisfying $u + h_i \in A(U_i)$ for all $i \in I$, we require $\partial u / \partial \bar{z} + \partial h_i / \partial \bar{z} = 0$ on U_i . That is,

$$\partial u / \partial \bar{z} = -F \quad \text{on } \Omega.$$

By Theorem 1.3.1, there exists $u \in C^\infty(\Omega)$ satisfying this equation. Finally we take $f_i = u + h_i \in A(U_i)$ and define m as indicated in the introduction to the proof. $\square$

The relation (A) occurring in the proof of Theorem 1.3.2 is usually referred to as a *cocycle* condition and $\{f_{ij} : i, j \in I\}$ as a cocycle. We shall meet this type of relation frequently in the sequel. The proof of Theorem 1.3.2 clearly proves the slightly stronger

Theorem 1.3.2'. Let $f_{ij} \in A(U_{ij})$ satisfy the cocycle conditions

$$f_{ij} + f_{jk} + f_{ki} = 0 \quad \text{on } U_{ijk} \quad \text{and} \quad f_{ij} = -f_{ji} \quad \text{on } U_{ij}, \quad \text{all } i, j, k.$$

Then there exist $g_i \in A(U_i)$ such that for all i, j ,

$$f_{ij} = g_j - g_i \quad \text{on } U_{ij}.$$

Notation. If U is an open subset of $\mathbb{C}$, we shall let $A^*(U)$ denote the set of nowhere vanishing analytic functions on U . That is, the units in $A(U)$.

Theorem 1.3.3. (Weierstrass theorem) Let $\mathcal{U} = \{U_i : i \in I\}$ be an open cover of Ω and suppose that we are given $m_i \in M^*(U_i)$ for each $i \in I$. Then, provided that $m_i/m_j \in A^*(U_{ij})$ for all $i, j \in I$, there exists $m \in M^*(\Omega)$ such that

$$m/m_i \in A^*(U_i), \quad \text{for all } i \in I.$$

Equivalently: The map $\text{div} : M^*(\Omega) \rightarrow \mathcal{D}(\Omega)$ is surjective. That is, subject to the requirement that pole and zero sets are discrete, we may construct a meromorphic function on Ω with prescribed zeros and poles of given orders.

Proof. By taking a refinement of the cover $\mathcal{U}$, it is no loss of generality to assume that each $U_i \in \mathcal{U}$ is convex (for example, an open disc).

Given $i, j \in I$, set $h_{ij} = m_i/m_j$. Fix a branch of $\log h_{ij}$ on U_{ij} and define

$$c_{ijk} = \frac{1}{2\pi i} (\log h_{ij} + \log h_{jk} + \log h_{ki}).$$

Since $h_{ij}h_{jk}h_{ki} = 1$, $c_{ijk} \in \mathbb{Z}$ for all i, j, k (note that U_{ijk} is convex and therefore connected and so c_{ijk} is constant on U_{ijk}).

Suppose that we can choose the branches of $\log h_{ij}$ so that $c_{ijk} = 0$ for all $i, j, k \in I$. Then if we set $f_{ij} = \log h_{ij} \in A(U_{ij})$, we see that the conditions of Theorem 1.3.2' are satisfied. Hence there exist $g_i \in A(U_i)$ such that for all $i, j \in I$

$$\log h_{ij} = g_j - g_i \quad \text{on } U_{ij}.$$

Now define $a_i = \exp(g_i) \in A^*(U_i)$. Clearly $a_j/a_i = h_{ij} = m_i/m_j$ and so

$$m_i a_i = m_j a_j \quad \text{on } U_{ij}.$$

Hence we can define $m \in M^*(\Omega)$ by taking $m|_{U_i} = m_i a_i$. Clearly m satisfies the required conditions.

To complete the proof it is sufficient to show that we can choose the branches of $\log h_{ij}$ so that $c_{ijk} = 0$ for all i, j, k . This is essentially a topological problem. First we observe that $\{c_{ijk}\}$ defines a class in $H^2(\mathcal{U}, \mathbb{Z})$ (For those unfamiliar with Čech theory we refer to Chapter 6). Now any finite intersection U of open sets of the cover $\mathcal{U}$ is convex and so $H^p(U, \mathbb{Z}) = 0$, $p \neq 0$. Hence by Leray's theorem (Theorem 6.3.16)

$$H^2(\mathcal{U}, \mathbb{Z}) \cong H^2(\Omega, \mathbb{Z}).$$

But $H^2(\Omega, \mathbb{Z}) = 0$ for all open domains in $\mathbb{C}$ (in fact for all non-compact oriented 2-manifolds—see the remarks following this proof). Hence $\{c_{ijk}\}$ is a coboundary and there exist integers n_{ij} , $i, j \in I$, such that

$$c_{ijk} = n_{ij} + n_{jk} + n_{ki}, \quad \text{for all } i, j, k \in I.$$

Now define a new branch of $\log h_{ij}$ by subtracting $2\pi i n_{ij}$ from the original choice. For the new choice of branches we have $c_{ijk} = 0$, for all i, j, k . $\square$

Notice that the proof of Theorem 1.3.3 actually proves the slightly stronger

Theorem 1.3.3'. Let $h_{ij} \in A^*(U_{ij})$ satisfy the cocycle conditions

$$h_{ij} h_{jk} h_{ki} = 1 \text{ on } U_{ijk} \quad \text{and} \quad h_{ij} = h_{ji}^{-1} \text{ on } U_{ij}, \quad i, j, k \in I.$$

Then there exist $h_i \in A^*(U_i)$ such that for all i, j

$$h_{ij} = h_j h_i^{-1} \quad \text{on } U_{ij}.$$

Remarks

1. The reader should observe the similarity between the proofs of Theorems 1.3.2 and 1.3.3. Indeed the last part of the above proof can be written as a problem in C^∞ functions if we note that $c_{ijk} = 0$ for all i, j, k iff there exists $b_i \in C^\infty(U_i)$ such that $\log h_{ij} = b_j/b_i$. Later, in Chapter 6, we develop machinery that handles all arguments of this type in a particularly simple and elegant way.
2. The explicit use of cohomology can of course be avoided in the proof of Weierstrass' theorem. See, for example, the proof given in Hörmander [1]. The reader might wish to attempt a direct construction of the integers n_{ij} without using any general facts about the cohomology of 2-manifolds.
3. For the vanishing of $H^2(\Omega, \mathbb{Z})$, we note that $H^2(\Omega, \mathbb{Z}) \cong H_2(\Omega, \mathbb{Z})$ (computation using universal coefficient theorems and the fact that Ω is a 2-manifold). Since Ω is assumed oriented, Poincaré duality implies that $H_2(\Omega, \mathbb{Z}) \cong H_c^0(\Omega, \mathbb{Z})$, where c denotes compact supports. Since Ω is non-compact, $H_c^0(\Omega, \mathbb{Z}) = 0$. For further details and references we refer the reader to the appendix in Milnor and Stasheff [1].

Corollary 1.3.4. Let $m \in M(\Omega)$. Then there exist $f, g \in A(\Omega)$ such that

- (i) Off the pole set of m , $m = f/g$.
- (ii) $Z(m) = Z(f)$, $P(m) = Z(g)$ and $Z(f) \cap Z(g) = \emptyset$.

Proof. Suppose m has poles z_j with orders p_j . Then, by Weierstrass' theorem there exists $g \in A(\Omega)$ such that $\text{div}(g) = \sum_j p_j \cdot z_j$. Since $\text{div}(mg) \geq 0$, $mg \in A(\Omega)$. Taking $f = mg$, it is clear that f and g satisfy the conditions of the corollary. $\square$

Remark. f and g are unique up to multiplication by elements of $A^*(\Omega)$.

Corollary 1.3.5. There exists $f \in A(\Omega)$ which cannot be extended as an analytic or meromorphic function to any open subset strictly containing Ω .

Proof. Let $d(z, \partial\Omega)$ denote the distance between $z \in \mathbb{C}$ and the boundary of Ω , $\partial\Omega$. Define

$$X = \left\{ \left(\frac{p}{2^n}, \frac{q}{2^n} \right) \in \Omega : \begin{array}{l} p, q, n \in \mathbb{Z}, n \geq 0 \text{ and} \\ d((p/2^n, q/2^n), \partial\Omega) < 2^{-n+2} \end{array} \right\}.$$

X is a discrete subset of Ω and every point of $\partial\Omega$ is the limit of some subsequence of X . By Weierstrass' theorem there exists $f \in A(\Omega)$ such that

$$\operatorname{div}(f) = \sum_{x \in X} 1 \cdot x.$$

Since the zeros of an analytic or meromorphic function are isolated, we see that f cannot be extended to any domain strictly containing Ω . Indeed, such a domain would contain at least one point of $\partial\Omega$ and this point would then be a non-isolated zero of the extension. $\square$

Remark. The proof of Corollary 1.3.5 actually shows that we can find an analytic function on Ω which cannot be continued (locally) across any point of the boundary of Ω .

Exercise. Let $\{z_i : i \geq 0\}$ be a discrete subset of $\Omega \subset \mathbb{C}$ and suppose that for each i we are given a polynomial $P_i(z) = \sum_{j=0}^{k_i} a_{ij}(z-z_i)^j$. Show that there exists an analytic function f on Ω such that the Taylor series of f at each z_i agrees with P_i to order k_i . (Hint: Find a meromorphic function m on Ω such that at each z_i the principal part of m equals $(z-z_i)^{-k_i-1}P_i(z)$. Then use Weierstrass' theorem).

4 Riemann surfaces

In sections 4 and 5 we make a preliminary investigation of possible generalisations of the results of §3 to Riemann surfaces. As we shall see both the topology and complex structure of a Riemann surface play a crucial role as is also the case in the theory of higher dimensional complex manifolds.

Let U and V be open subsets of $\mathbb{C}$ and $f : U \rightarrow V$. We shall say that f is *biholomorphic* if f is a homeomorphism onto V and both f and f^{-1} are analytic. We remark that, by the inverse function theorem, for f to be biholomorphic it is sufficient for f to be bijective and everywhere analytic with non-vanishing derivative.

Before recalling the definition of a Riemann surface, we wish to stress that all topological spaces considered in this book will be Hausdorff and paracompact. Unless the contrary is clearly indicated they will also be connected. The Hausdorff and paracompactness assumptions imply metrizable and, together with connectedness, imply separability (see, for example, Matsushima [1]).

A *chart* on a topological space M consists of a pair (U, ϕ) where U is an open subset of M and ϕ is a homeomorphism of U onto an open subset of Euclidean space.

Definition 1.4.1. Let M be a topological space. Suppose that we are given a set $\mathcal{A} = \{(U_i, \phi_i) : i \in I\}$ of charts on M satisfying

1. $\{U_i\}$ is an open cover of M .
2. For each $i \in I$, ϕ_i is a homeomorphism of U_i onto an open subset of $\mathbb{C}$.

3. For all $i, j \in I$, $\phi_i \phi_j^{-1}$ is a biholomorphism of $\phi_j(U_{ij})$ with $\phi_i(U_{ij})$.

Then $\mathcal{A}$ is called a (*complex analytic*) *atlas* on M and M , together with the atlas $\mathcal{A}$, is called a *Riemann surface*.

Remarks.

1. We refer to charts $(U, \phi) \in \mathcal{A}$ as complex analytic charts on M or just charts on M if the analyticity is clear from the context.
2. If $\mathcal{A}$ is an atlas on M , we shall generally assume that $\mathcal{A}$ is maximal in the sense that if (U, ϕ) is a chart on M such that $\phi \phi_i^{-1}$ is biholomorphic for all $(U_i, \phi_i) \in \mathcal{A}$ then $(U, \phi) \in \mathcal{A}$. Of course, every atlas can be completed to a maximal atlas.

Before giving examples and motivation for the study of Riemann surfaces we shall generalise some of the definitions of §§1,2.

Suppose that M is a Riemann surface with atlas $\mathcal{A}$. Let W be an open subset of M . If $f : W \rightarrow \mathbb{C}$, we say f is analytic if $f \phi^{-1} \in A(\phi(U \cap W))$ for all $(U, \phi) \in \mathcal{A}$. We let $A(W)$ denote the ring of analytic functions on W . We say that m is a meromorphic function on W if $m \phi^{-1} \in M(\phi(U \cap W))$ for all $(U, \phi) \in \mathcal{A}$ (a more careful definition can be made along the lines of Definition 1.2.1). We let $M(W)$ denote the ring of meromorphic functions on W and $M^*(W)$, $A^*(W)$ denote the groups of units in $M(W)$, $A(W)$ respectively. Note that if W is connected $M(W)$ is a field.

Let $m \in M^*(M)$. We define the zero set $Z(m)$ of m by

$$Z(m) = \bigcup \phi^{-1} Z(m \phi^{-1}),$$

where the union is over all charts $(U, \phi) \in \mathcal{A}$. We similarly define the pole set $P(m)$ of m . We define the *order of m at z* , $\text{ord}(m, z)$ to be $\text{ord}(m \phi^{-1}, \phi(z))$, where $(U, \phi) \in \mathcal{A}$ contains z , and the *divisor* of m , $\text{div}(m)$, by

$$\text{div}(m) = \sum_{z \in M} \text{ord}(m, z) \cdot z.$$

We leave it to the reader to verify that $Z(m)$ and $P(m)$ are well-defined, discrete, disjoint subsets of M ; that $M \setminus P(m)$ is the largest subset of M on which m is analytic; that $\text{ord}(m, z)$ does not depend on the choice of chart containing z and that if M is compact, $Z(m)$ and $P(m)$ are finite.

Suppose N is another Riemann surface with atlas $\mathcal{B}$ and $f : M \rightarrow N$. We say that f is analytic or holomorphic if $\psi f \phi^{-1} : \phi(U \cap f^{-1}(V)) \rightarrow \mathbb{C}$ is analytic for all $(U, \phi) \in \mathcal{A}$, $(V, \psi) \in \mathcal{B}$ and that f is biholomorphic if f is bijective and both f and f^{-1} are analytic. If there exists a biholomorphic map between M and N , we say that M and N are biholomorphically equivalent or analytically isomorphic.

Examples.

1. The *Riemann sphere*. The Riemann sphere is defined to be the 1-point compactification of $\mathbb{C}$, $\mathbb{C} \cup \{\infty\}$, and we shall denote it by either S^2 or $P^1(\mathbb{C})$ (see Chapter 4 for the second notation). We define an atlas $\{(U, \phi), (V, \psi)\}$ on S^2 by taking $U = \mathbb{C}$, $V = \mathbb{C}^\bullet \cup \{\infty\}$ ($\mathbb{C}^\bullet$ here denotes non-zero complex numbers) and $\phi(z) = z$; $\psi(z) = 1/z$, $\psi(\infty) = 0$.

Suppose Ω is an open subset of $\mathbb{C}$ and $m \in M(\Omega)$. Then m induces an analytic map $\tilde{m} : \Omega \rightarrow S^2$ if we take $\tilde{m}|_{\Omega \setminus P(m)} = m$ and set $\tilde{m}(p) = \infty$, for all $p \in P(m)$. The verification that $\tilde{m}$ is analytic is easy using our explicit atlas for S^2 .

2. If M is a *simply connected* Riemann surface then M is biholomorphic to precisely one of: The Riemann sphere, the open unit disc, the complex plane. This is the fundamental (and difficult!) theorem of Riemann surfaces known as the *Uniformization Theorem* (for a proof and more details we refer to Ahlfors and Sario [1], Siegel [1], Springer [1]).
3. Let $f : M \rightarrow N$ be a local homeomorphism between topological spaces M and N . That is, for each $x \in M$ there exists an open neighbourhood U of x such that $f(U)$ is an open subset of N and $f|_U$ maps U homeomorphically onto $f(U)$. Suppose that N is a Riemann surface with atlas $\mathcal{A}$. Then f induces the structure of a Riemann surface on M in such a way that f becomes a holomorphic map which is locally biholomorphic. Indeed, suppose U is an open subset of M which is mapped homeomorphically by f onto an open subset of N . Suppose $(W, \phi) \in \mathcal{A}$ and $W \supset f(U)$. Then $(U, \phi f)$ is a chart on M . The set of all charts on M constructed in this way is easily seen to define a complex analytic atlas on M with respect to which f is locally biholomorphic.
4. By examples 2 and 3, the simply connected covering surface of a Riemann surface has the natural structure of a Riemann surface and is biholomorphic to one of: the Riemann sphere, the open unit disc, the complex plane (see also Chapter 4).
5. If M is a compact Riemann surface, M is diffeomorphic to a compact orientable surface. Such surfaces are classified, up to diffeomorphism by their *genus* or number of handles (see Hirsch [1]). However, genus does not classify M up to biholomorphic equivalence (unless M is simply connected). For a specific example, we refer to Chapter 4, §4.
6. Suppose $\pi : M \rightarrow \mathbb{C}$ is a local homeomorphism. Then M has the structure of a non-compact Riemann surface with respect to which π is locally biholomorphic. The pair (M, π) is called a *Riemann domain* or *spreading* of M over $\mathbb{C}$. Riemann domains occur as the domains of maximal analytic continuation of analytic functions defined on open subsets of $\mathbb{C}$. Indeed, if Ω is a domain in $\mathbb{C}$ and $f \in A(\Omega)$, it is not hard to construct a Riemann domain (M_f, π) such that f extends analytically to M_f and (M_f, π) is the maximal Riemann domain to which f can be continued analytically (The converse is also true but much harder: Any Riemann domain is the domain of maximal analytic continuation of an analytic function. This result, due to Behnke and Stein [1], generalises Corollary 1.3.5 of the Weierstrass theorem. See also R. Narasimhan [1] and Gunning and Rossi [1]). We give one explicit example here and leave the general construction to Chapter 6.

Let Log denote the principal branch of the logarithm defined off the negative real axis. We let $M \subset \mathbb{C}^2$ denote the set of all points $(z, \log |z| + i \arg z)$, $z \in \mathbb{C}^\bullet$, where $\arg z$ is a value of the argument of z . If we let π denote the restriction to M of the projection of $\mathbb{C}^2$ on the first factor, we may easily verify that (M, π) is a Riemann domain. Furthermore, M is the maximal Riemann domain to which Log extends analytically. Moreover, if we let $\log : M \rightarrow \mathbb{C}$ denote the restriction to M of the projection of $\mathbb{C}^2$ on the second factor, we see that $\log$ is the analytic extension of Log to M and that if $y \in \pi^{-1}(z)$, $z \in \mathbb{C}^\bullet$, then $\log(y) = \log |z| + i\phi$, where ϕ is a value of $\arg z$.

7. There is no natural way of compactifying the Riemann domain associated to the logarithm that we constructed in the previous example. This is a reflection of the transcendental (non-algebraic) character of $\log$ (see Chapter 7). In the case of algebraic functions we can always compactify the Riemann domain associated to the function in a particularly nice way. Here we shall consider a simple example and refer the reader to Ahlfors [1] or Siegel [1] for the general theory.

Let us regard $P(y, z) = y^3 - z^2$ as a polynomial in y with coefficients depending on z . The zero set of P defines a subset X of $\mathbb{C}^2$. Now X will not be a complex submanifold of

$\mathbb{C}^2$ as there is a *singularity* or *branch point* at $z = 0$ (see Chapter 4 for terminology). Let $\pi : X \rightarrow \mathbb{C}$ denote the restriction to X of the projection of $\mathbb{C}^2$ on the z -axis. If X_P denotes the 1-point compactification of X then, if we define $\pi(\infty) = \infty$, π extends continuously to X_P and $\pi : X_P \rightarrow S^2$ is a 3-fold branched cover of S^2 with branch points at 0 and ∞ . In particular, if we set $X^* = X \setminus \{0, \infty\}$, $\pi : X^* \rightarrow \mathbb{C}^\bullet$ is a local homeomorphism and $\pi^{-1}(z)$ contains precisely three points for all $z \in \mathbb{C}^\bullet$. Now, by example 3, (X^*, π) is a Riemann domain and we denote its atlas by $\mathcal{A}^*$. We now construct a complex analytic atlas for the whole of X_P with respect to which π is analytic. Let D denote the open unit disc centre zero in the ζ -plane. We define

$$\begin{aligned} \phi : D \rightarrow X_P \quad & \text{by } \phi(\zeta) = (\zeta^2, \zeta^3), \\ \psi : D \rightarrow X_P \quad & \text{by } \phi(\zeta) = \begin{cases} (\zeta^{-2}, \zeta^{-3}), & \zeta \neq 0, \\ \infty, & \zeta = 0. \end{cases} \end{aligned}$$

Notice that ϕ and ψ are homeomorphisms onto open neighbourhoods of $\pi^{-1}(0)$ and ∞ respectively. As atlas on X_P we take

$$\{(\phi(D), \phi^{-1}), (\psi(D), \psi^{-1})\} \cup \mathcal{A}^*.$$

We leave it to the reader to verify that this atlas defines on X_P the structure of a compact Riemann surface, biholomorphic to the Riemann sphere, with respect to which π is analytic.

Let $p : X_P \rightarrow S^2$ denote the restriction to X_P of the projection of $\mathbb{C}^2$ onto the second factor where we have defined $p(\infty) = \infty$. Then p is a meromorphic function on X_P with a pole of order 3 at ∞ and a zero of order 3 at 0. p gives all the roots of $y^3 = z^2$ in the sense that $\{p(x) : x \in \pi^{-1}(z)\}$ is the set of all roots of $y^3 = z^2$ for all $z \in \mathbb{C}$.

For the remainder of this section we wish to investigate the possible generalisation of Weierstrass' theorem to compact Riemann surfaces. So from now on assume M is a compact Riemann surface.

Lemma 1.4.2. Every analytic function on M is constant.

Proof. If $f \in A(M)$, $|f|$ has a maximum on M , say at z_0 . A simple application of the maximum modulus theorem shows that f is constant in some neighbourhood of z_0 . Uniqueness of analytic continuation then implies that f is constant on M . $\square$

In view of this Lemma it becomes a matter of some importance to prove that there exist plenty of non-constant meromorphic functions on M . Specifically: Let $d \in \mathcal{D}(M)$. Under what conditions on d , if any, do there exist $m \in M^*(M)$ such that $\text{div}(m) = d$?

For the Riemann sphere this question is easily solved: If $d = \sum_{i=1}^n p_i \cdot z_i \in \mathcal{D}(S^2)$, then there exists $m \in M^*(S^2)$ with $\text{div}(m) = d$ if and only if $\sum_{i=1}^n p_i = 0$.

Indeed, if $z_1, \dots, z_{n-1} \in \mathbb{C}$, $z_n = \infty$, we may define

$$m(z) = \prod_{i=1}^{n-1} (z - z_i)^{p_i}, \quad z \in \mathbb{C}.$$

m extends to a meromorphic function on S^2 with singularities of order p_i at z_i , $1 \leq i \leq n-1$, and of order $-\sum_{i=1}^{n-1} p_i$ at ∞ .

The converse is included in Lemma 1.4.3 below.

In future if $d = \sum_{z \in M} p(z) \cdot z$ is a divisor on M , we set

$$\deg(d) = \sum_{z \in M} p(z) \in \mathbb{Z}$$

and call $\deg(d)$ the *degree* of d .

We have the following topological restriction on divisors of meromorphic functions.

Lemma 1.4.3. A necessary condition for $d \in \mathcal{D}(M)$ to be the divisor of a meromorphic function is that $\deg(d) = 0$.

Proof. Let $m \in M^*(M)$ and set $\text{div}(m) = \sum_{i=1}^n n_i \cdot z_i$. For $i = 1, \dots, n$ choose charts (U_i, ϕ_i) containing z_i and such that $\phi_i(U_i)$ contains $\overline{D}_1(0)$. Set $D_i = \phi_i^{-1}(\overline{D}_1(0))$ and $\gamma_i = \partial D_i = \phi_i^{-1}(\partial \overline{D}_1(0))$. We shall assume that the charts are chosen so that the sets D_i are mutually disjoint.

$\log m$ is defined up to integer multiples of $2\pi i$ on $M' = M \setminus \bigcup_i D_i$. Hence the 1-form $\phi = d(\log m)$ is well defined on M' . By Stokes' theorem

$$\int_{M'} d\phi = \sum_{i=1}^n \int_{\gamma_i} \phi.$$

But $d\phi = d^2(\log m) = 0$, and so the left hand integral must vanish. Now

$$\begin{aligned} \sum_{i=1}^n \int_{\gamma_i} \phi &= \sum_{i=1}^n \int_{|z|=1} \partial(\log m_i)/\partial z \, dz, \quad \text{where } m_i = m\phi_i^{-1} \\ &= \sum_{i=1}^n \int_{|z|=1} m'_i/m_i \, dz \\ &= \sum_{i=1}^n n_i, \quad \text{by the residue theorem.} \end{aligned}$$

Hence $\deg(d) = 0$. □

The reader is warned that the vanishing of the degree of d is *not* generally sufficient for d to be the divisor of a meromorphic function (see the end of §5 and also Chapter 4, §4).

In view of the importance of Theorem 1.3.1 in the proofs of the Mittag-Leffler and Weierstrass theorems it is natural to seek a generalisation to Riemann surfaces. The first difficulty we encounter is that of finding a suitable definition of $\partial/\partial\bar{z}$ for a Riemann surface. Indeed, suppose $f \in C^\infty(M)$ and (U, ϕ) and (V, ψ) are complex analytic charts on M . In general, $[\partial(f\phi^{-1})/\partial\bar{z}]\phi \neq [\partial(f\psi^{-1})/\partial\bar{z}]\psi$ and so we cannot expect to define $\partial/\partial\bar{z}$ as an endomorphism of $C^\infty(M)$. The correct generalisation of $\partial/\partial\bar{z}$ involves the introduction of “twisted” functions and tensor fields on M . These concepts are most easily discussed in the general framework of vector bundles and we shall defer further consideration of the construction of meromorphic functions until we have developed sufficient of the elementary theory of vector bundles in §5. The reader who is largely unfamiliar with the theory of vector bundles may prefer to go straight to Chapter 2 and return to section 5 after the end of Chapter 4.

Exercises.

1. Construct the (compact) Riemann surface of $z^2 = (y-a)(y-b)$, $a \neq b$, and show that it is biholomorphic to the Riemann sphere. (Hint: prove that the Riemann surface is simply connected).
2. Construct the Riemann surface of $y^2 = (z-a)(z-b)(z-c)$, a, b, c distinct, and show that it is diffeomorphic to the two dimensional real torus.
3. Formalise and prove a version of the Mittag-Leffler theorem valid for the Riemann sphere.
4. Let $f : M \rightarrow N$ be a holomorphic map between Riemann surfaces. Show
 - (a) Either f is constant or $f(M)$ is an open subset of N .
 - (b) In case M is compact, either f is constant or $f(M) = N$.

- (c) Given $z_0 \in N$, $f^{-1}(z_0)$ is a discrete subset of M (possibly empty).
5. Let $f : M \rightarrow N$ be a non-constant map between Riemann surfaces M and N and suppose $d \in \mathcal{D}(N)$. Show how to define $f^*(d) \in \mathcal{D}(M)$ and verify that if $m \in M^*(N)$ then $\text{div}(mf) = f^*(\text{div}(m))$.

5 Vector bundles

The first part of this section constitutes a summary of the theory of real and complex vector bundles over a topological space. For more comprehensive treatments the reader may refer to Abraham and Marsden [1], Husemoller [1] or Lang [1].

Definition 1.5.1. Let X be a topological space. An n -dimensional real vector bundle E over X consists of a topological space E and continuous map $\pi : E \rightarrow X$ which satisfy

- (a) The fibre $\pi^{-1}(x) = E_x$ has the structure of a real n -dimensional vector space for every $x \in X$.
- (b) There exists an open cover $\{U_i : i \in I\}$ of X and homeomorphisms $\theta_i : \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^n$ such that for all $i \in I$, $x \in U_i$, θ_i maps E_x linearly and isomorphically onto $\{x\} \times \mathbb{R}^n$.

We denote the vector bundle by $\pi : E \rightarrow X$ or just E .

The topological space E is called the *total space* of the vector bundle, X is called the *base space* and π the *projection map*. The maps θ_i are called *trivialisations* of E .

Example 1. Let $\pi : X \times \mathbb{R}^n \rightarrow X$ denote projection on the first factor. Then $\pi : X \times \mathbb{R}^n \rightarrow X$ is an n -dimensional vector bundle over X called the *trivial n -dimensional vector bundle* over X . We often denote the trivial bundle over X with fibre $\mathbb{R}^n$ by $\underline{\mathbb{R}}^n$.

Condition b) on a vector bundle implies that a vector bundle is locally a trivial bundle. Globally it may be twisted.

Definition 1.5.2. A (continuous) *section* of the vector bundle $\pi : E \rightarrow X$ is a (continuous) map $s : X \rightarrow E$ such that $\pi s = \text{identity}$. That is, $s(x) \in E_x$ for all $x \in X$.

Example. If $f : X \rightarrow \mathbb{R}^n$, then the graph map $x \mapsto (x, f(x))$ is a section of the trivial bundle $X \times \mathbb{R}^n$. Conversely, every section of the trivial bundle $X \times \mathbb{R}^n$ defines an $\mathbb{R}^n$ -valued function on X .

We shall let $C^0(E)$ denote the set of all continuous sections of E and remark that $C^0(E)$ has the structure of a real vector space with addition and scalar multiplication induced from the vector space structure on the fibres of E .

If s is a section of the vector bundle E , then $\theta_i s : U_i \rightarrow U_i \times \mathbb{R}^n$ is the graph map of a function $S_i : U_i \rightarrow \mathbb{R}^n$. Thus we may think of sections of E as *locally* being $\mathbb{R}^n$ -valued functions. Globally, they may be “twisted” $\mathbb{R}^n$ -valued functions.

Example. Let $S^1 \dot{\times} \mathbb{R}$ denote the twisted cylinder. $S^1 \dot{\times} \mathbb{R}$ is homeomorphic to the Möbius band and may be explicitly parametrized as the subset $\{(\cos \theta, \sin \theta, t \cos \theta/2, t \sin \theta/2) : \theta \in [0, 2\pi), t \in \mathbb{R}\}$ of $\mathbb{R}^4$. The space $S^1 \dot{\times} \mathbb{R}$ has the structure of a 1-dimensional real vector bundle over S^1 if we define projection onto S^1 in the obvious way.

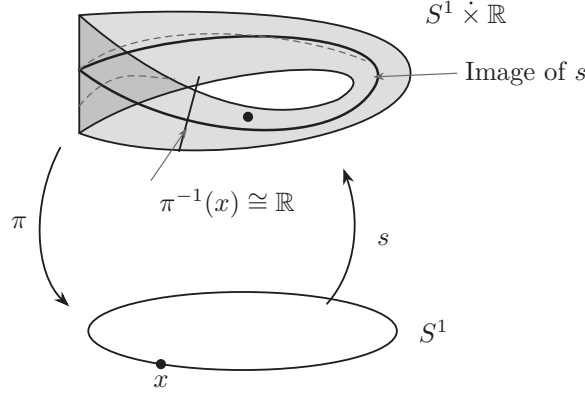


Figure I.1

We observe that every continuous section of $S^1 \times \mathbb{R}$ is zero at at least one point in sharp contrast to what happens for sections of $S^1 \times \mathbb{R}$. Thus although *locally*, twisted and untwisted functions on S^1 possess the same properties, their global behaviour may be rather different.

We shall now give an alternative description of vector bundles in terms of *transition functions*. We follow the notation of Definition 1.5.1. For each $i \in I$, $x \in U_i$, we let

$$\theta_{ix} : E_x \longrightarrow \mathbb{R}^n$$

denote the restriction of θ_i to E_x composed with the projection on $\mathbb{R}^n$.

Define

$$\theta_{ij} : U_{ij} \longrightarrow \text{GL}(\mathbb{R}^n)$$

by $\theta_{ij}(x) = \theta_{ix}\theta_{jx}^{-1}$, $x \in U_{ij}$, $i, j \in I$ ($\text{GL}(\mathbb{R}^n)$ denotes the group of linear isomorphisms of $\mathbb{R}^n$ which is commonly given the alternative notation $\text{GL}(n, \mathbb{R})$).

It may easily be verified that the θ_{ij} are continuous maps which satisfy the cocycle conditions

$$\left. \begin{aligned} \theta_{ij} &= \theta_{ji}^{-1} \\ \theta_{ij} \cdot \theta_{jk} &= \theta_{ik} \end{aligned} \right\} \quad (\text{A})$$

(Inversion and multiplication are in $\text{GL}(\mathbb{R}^n)$). The θ_{ij} are called *transition functions* of the vector bundle E . Suppose s is a section of E and $S_i : U_i \rightarrow \mathbb{R}^n$ is the local representation of s on U_i as described above. Then for all i, j

$$\theta_{ij}S_j = S_i \quad \text{on } U_{ij}. \quad (\text{B})$$

Conversely, any set of functions $S_i : U_i \rightarrow \mathbb{R}^n$ satisfying B defines a unique section of E .

If E and F are vector bundles over X we shall say that a continuous map $A : E \rightarrow F$ is a *vector bundle map* if for all $x \in X$, $A(E_x) \subset F_x$ and $A|_{E_x}$ is linear. If A is bijective and A^{-1} is a vector bundle map we shall say that A is a *vector bundle isomorphism* and that E and F are *isomorphic vector bundles*.

Suppose that E, F have trivialising maps θ_i, η_i respectively, relative to some (common) cover $\{U_i : i \in I\}$ of X . Then for each $i \in I$, A induces a continuous map $A_i : U_i \times \mathbb{R}^n \rightarrow U_i \times \mathbb{R}^n$ given by $A_i = \eta_i A \theta_i^{-1}$. In turn, A_i induces a continuous map

$$a_i : U_i \longrightarrow L(\mathbb{R}^n, \mathbb{R}^n)$$

such that for all i, j

$$\eta_{ij}a_j = a_i\theta_{ij} \quad \text{on } U_{ij}.$$

Conversely, a family of continuous maps $a_i : U_i \rightarrow L(\mathbb{R}^n, \mathbb{R}^m)$ determines a vector bundle map from E to F provided that for all i, j

$$\eta_{ij}a_j = a_i\theta_{ij} \quad \text{on } U_{ij}. \quad (\text{C})$$

Notice that E will be a trivial vector bundle (that is, isomorphic to a trivial vector bundle) if and only if there exist $a_i : U_i \rightarrow \text{GL}(\mathbb{R}^n)$ such that for all i, j

$$a_j = a_i \theta_{ij} \quad \text{on } U_{ij}.$$

Suppose that $\{U_i : i \in I\}$ is an open cover of the topological space X and that we are given continuous maps $\theta_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{R}^n)$ satisfying the cocycle condition A. Then the θ_{ij} determine a unique—up to isomorphism— n -dimensional real vector bundle over X whose transition functions are the θ_{ij} . To see this we let Z denote the disjoint union over I of the products $U_i \times \mathbb{R}^n$. We define an equivalence relation on Z by $(i, x, U) \sim (j, y, V)$ iff $x = y$ and $U = \theta_{ij}(x)V$. It is an easy matter to show that $Z/\sim$ has the structure of an n -dimensional real vector bundle over X with transition functions θ_{ij} and we leave details to the reader. In the sequel we usually construct vector bundles by specifying a set of transition functions. From C above, it follows that two sets $\theta_{ij}, \eta_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{R}^n)$ of transition functions define isomorphic vector bundles if and only if there exist $a_i : U_i \rightarrow \text{GL}(\mathbb{R}^n)$ such that for all i, j , $\eta_{ij} a_j = a_i \theta_{ij}$ on U_{ij} .

Notation. Let $\mathbb{R}^{m'}$ denote the real dual of $\mathbb{R}^m$. If $A \in L(\mathbb{R}^m, \mathbb{R}^n)$, then the *transpose* of A , A' , is the linear map from $\mathbb{R}^{n'} \rightarrow \mathbb{R}^{m'}$ characterised by $A'(\phi)(e) = \phi(A(e))$, $\phi \in \mathbb{R}^{n'}$, $e \in \mathbb{R}^m$.

Example 4. Let E be an n -dimensional real vector bundle with transition functions θ_{ij} . We define $\theta'_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{R}^{n'})$ by

$$\theta'_{ij}(x) = [\theta_{ij}(x)']^{-1}, \quad x \in U_{ij}.$$

The θ'_{ij} satisfy the cocycle condition A and so are the transition functions for a vector bundle which we call the *dual* vector bundle of E and denote by E' . We remark that the trivialisations θ'_i of E' map naturally to $U_i \times \mathbb{R}^{n'}$ (rather than $U_i \times \mathbb{R}^n$).

Motivated by the previous example, we shall now extend the possible “fibre models” of real vector bundles to include any finite combination of direct sum, tensor product, exterior and symmetric power of $\mathbb{R}^n$ and $\mathbb{R}^{n'}$. All these vector space operations extend immediately to vector bundles over a fixed base space X . We give one example to illustrate the method and refer the reader to the references for more details. Suppose that E and F are vector bundles over X with transition functions $\theta_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{R}^m)$ and $\eta_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{R}^n)$ respectively. $\bigwedge^r F \otimes (E \oplus (\odot^s F'))$ will then denote the vector bundle over X with fibre model $\bigwedge^r \mathbb{R}^n \otimes (\mathbb{R}^m \oplus (\odot^s \mathbb{R}^{n'}))$ and transition functions $\psi_{ij} : U_{ij} \rightarrow \text{GL}(\bigwedge^r \mathbb{R}^n \otimes (\mathbb{R}^m \oplus (\odot^s \mathbb{R}^{n'})))$ defined by

$$\psi_{ij}(x) = \left(\bigwedge^r \eta_{ij}(x) \right) \otimes (\theta_{ij}(x) \oplus (\odot^s \eta'_{ij}(x))), \quad x \in U_{ij}.$$

($\bigwedge^r$ and $\odot^s$ denote the operations of r th. exterior and s th. symmetric power respectively).

Suppose that X is a differential manifold (with C^∞ structure). We can define smooth (that is C^∞) real vector bundles over X in the obvious way and everything we have said above generalises immediately to the smooth case. If E is a smooth vector bundle over X , we let $C^r(E)$ denote the vector space of C^r sections of E , $1 \leq r \leq \infty$.

Example 5. Let M be an m -dimensional differential manifold with smooth atlas $\{(U_i, \phi_i) : i \in I\}$. We define C^∞ maps $\phi_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{R}^m)$ by $\phi_{ij}(x) = D(\phi_i \phi_j^{-1})(\phi_j(x))$. The ϕ_{ij} are the transition functions for the (real) *tangent bundle* of M which we denote in the sequel by $\mathcal{T}M$. The (real) *cotangent bundle* of M , $\mathcal{T}'M$, is the dual bundle of $\mathcal{T}M$ with transition functions ϕ'_{ij} (See Abraham and Marsden [1], Chillingworth [1] and Lang [1] for more details).

We now turn our attention to vector bundles with fibre a complex vector space. An m -dimensional complex vector bundle E over X is defined exactly as in Definition 1.5.1 except that “real” and “ $\mathbb{R}^n$ ” are everywhere replaced by “complex” and “ $\mathbb{C}^m$ ” respectively. We leave the

writing out of the formal definitions of complex vector bundles, sections, transition functions, etc. to the reader. Sections of an m -dimensional complex vector bundle will locally be $\mathbb{C}^m$ -valued functions: globally they may be twisted.

One important feature of complex vector bundles is the greater complexity of the fibre models which involve not only duals but also conjugates. But first we need to review some complex linear algebra.

If $J : \mathbb{C}^m \rightarrow \mathbb{C}^m$ denotes scalar multiplication by i , then J is a $\mathbb{C}$ -linear isomorphism of $\mathbb{C}^m$ satisfying $J^2 = -I$. The usual complex structure on $\mathbb{C}^m$ may be defined in terms of J and the underlying real structure on $\mathbb{C}^m$ by

$$(a + ib)Z = aZ + bJ(Z), \quad a, b \in \mathbb{R}, \quad Z \in \mathbb{C}^m.$$

The *conjugate complex structure* on $\mathbb{C}^m$ is defined by

$$(a + ib)Z = aZ - bJ(Z), \quad a, b \in \mathbb{R}, \quad Z \in \mathbb{C}^m.$$

We denote the resulting complex vector space by $\overline{\mathbb{C}^m}$. Suppose $A \in L(\mathbb{C}^m, \mathbb{C}^n)$, we define $\overline{A} \in L(\overline{\mathbb{C}^m}, \overline{\mathbb{C}^n})$ by simply requiring that $A = \overline{A}$ on the common underlying set of $\mathbb{C}^m$ and $\overline{\mathbb{C}^m}$ (We should emphasise that all linear maps here are assumed *complex* linear unless the contrary is explicitly stated).

We let $\mathbb{C}^{m*} = L(\mathbb{C}^m, \mathbb{C})$ denote the *conjugate dual space* of $\mathbb{C}^m$. If $A \in L(\mathbb{C}^m, \mathbb{C}^n)$, we define the *transpose* $A^* \in L(\mathbb{C}^{n*}, \mathbb{C}^{m*})$ exactly as in the real case.

We let $\overline{\mathbb{C}^{m*}}$ denote the complex vector space of conjugate complex linear maps on $\mathbb{C}^m$. We call $\overline{\mathbb{C}^{m*}}$ the *conjugate dual space* of $\mathbb{C}^m$. Note that $f \in \overline{\mathbb{C}^{m*}}$ if $f : \mathbb{C}^m \rightarrow \mathbb{C}$ is $\mathbb{R}$ -linear and $f(\lambda Z) = \overline{\lambda} f(Z)$, $\lambda \in \mathbb{C}$, $Z \in \mathbb{C}^m$. Clearly, $\overline{\mathbb{C}^{m*}} \cong L(\overline{\mathbb{C}^m}, \mathbb{C})$. Moreover, $\overline{\mathbb{C}^{m*}} \cong \mathbb{C}^{m*}$ where we map $\phi \in \mathbb{C}^{m*}$ to $\overline{\phi} \in \overline{\mathbb{C}^{m*}}$ and $\overline{\phi}(e) = \overline{\phi(e)}$, $e \in \mathbb{C}^m$. If $A \in L(\mathbb{C}^m, \mathbb{C}^n)$, we define the *conjugate transpose* $\overline{A}^* \in L(\overline{\mathbb{C}^{n*}}, \overline{\mathbb{C}^{m*}})$ by

$$\overline{A}^*(\phi)(e) = \phi(A(e)), \quad e \in \mathbb{C}^m, \quad \phi \in \overline{\mathbb{C}^{n*}}.$$

Finally, if $A \in L(\mathbb{C}^m, \mathbb{C}^n)$ has matrix $[a_{ij}]$ relative to the standard bases of $\mathbb{C}^m$ and $\mathbb{C}^n$, the reader may easily verify that the matrices of $\overline{A}$, A^* and $\overline{A}^*$ are respectively $[\overline{a_{ij}}]$, $[a_{ji}]$ and $[\overline{a_{ji}}]$ relative to the naturally induced bases on $\overline{\mathbb{C}^m}, \dots, \overline{\mathbb{C}^{n*}}$.

Example 6. Let E be a complex vector bundle with transition functions $\theta_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{C}^m)$. We define maps

$$\begin{aligned} \overline{\theta}_{ij} : U_{ij} &\longrightarrow \text{GL}(\overline{\mathbb{C}^m}), \\ \theta_{ij}^* : U_{ij} &\longrightarrow \text{GL}(\mathbb{C}^{m*}), \\ \overline{\theta}_{ij}^* : U_{ij} &\longrightarrow \text{GL}(\overline{\mathbb{C}^{m*}}) \end{aligned}$$

by $\overline{\theta}_{ij}(x) = \overline{\theta_{ij}(x)}$, $\theta_{ij}^*(x) = [\theta_{ij}(x)^*]^{-1}$, $\overline{\theta}_{ij}^*(x) = [\overline{\theta_{ij}(x)^*}]^{-1}$, $x \in U_{ij}$.

Now $\overline{\theta}_{ij}$, θ_{ij}^* , $\overline{\theta}_{ij}^*$ satisfy the cocycle condition A and so are the transition functions for complex vector bundles which we call the *conjugate bundle*, $\overline{E}$, *dual bundle*, E^* , and *conjugate dual bundle*, $\overline{E}^*$, of E respectively.

Exactly as for the real case we may now extend the possible fibre models of complex vector bundles to include finite combinations of direct sum, tensor product, exterior and symmetric powers of $\mathbb{C}^m$ and its conjugate and dual spaces. All these operations extend immediately to complex vector bundles over a fixed base space and we leave formal details to the reader.

Remarks.

1. We shall show in Chapter 9 that, as complex vector bundles, E is isomorphic to $\overline{E}^*$ and E^* is isomorphic to $\overline{E}$. However, these isomorphisms are not natural and in the sequel it will often be convenient to regard these bundles as all being distinct.
2. If E is a 1-dimensional complex vector bundle, we shall usually refer to E as a *complex line bundle*. Since $\text{GL}(\mathbb{C}) \cong \mathbb{C}^\bullet$, the transition functions for a complex line bundle may be equivalently given by non-vanishing complex valued functions: $\theta_{ij} : U_{ij} \rightarrow \mathbb{C}^\bullet$.

Examples.

7. Let $\text{CLB}(X)$ denote the set of isomorphism classes of complex line bundles over X . We claim that $\text{CLB}(X)$ has the natural structure of an abelian group with identity the trivial bundle $\underline{\mathbb{C}}$ and product and inversion respectively defined by

$$\begin{aligned} E \cdot F &= E \otimes F, \quad E, F \in \text{CLB}(X), \\ E^{-1} &= E^*, \quad E \in \text{CLB}(X). \end{aligned}$$

Certainly $E \cdot \underline{\mathbb{C}} = \underline{\mathbb{C}} \cdot E = E$ for all $E \in \text{CLB}(X)$. We verify that E^* is the inverse of E with respect to $\otimes$ and leave the remaining details to the reader. To verify that E^* is the inverse of E it is enough to note that there is a natural isomorphism $E \otimes E^* \cong \underline{\mathbb{C}}$ induced from the dual pairing $E_x \otimes E_x^* \cong \mathbb{C}$, $x \in X$ (Perhaps we should remark that for the group $\text{RLB}(X)$ of *real* line bundles on X , L is isomorphic to L' , $L \in \text{RLB}(X)$, and so $L^2 \cong \underline{\mathbb{R}}$, for all $L \in \text{RLB}(X)$).

2. Let M be a Riemann surface with complex analytic atlas $\{(U_i, \phi_i) : i \in I\}$. We define $\phi_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{C})$ by $\phi_{ij}(z) = \partial(\phi_i \phi_j^{-1}) / \partial z(\phi_j(z))$, $z \in U_{ij}$. The ϕ_{ij} are complex analytic maps and are the transition functions for the *holomorphic tangent bundle* of M which we denote in the sequel by TM .

$\phi_{ij}^* : U_{ij} \rightarrow \text{GL}(\mathbb{C}^*)$ are also analytic maps and define the transition functions for the *holomorphic cotangent bundle*, TM^* , of M .

$\overline{\phi}_{ij} : U_{ij} \rightarrow \text{GL}(\overline{\mathbb{C}})$ are C^∞ maps (not analytic!) and are the transition functions for the *anti-holomorphic tangent bundle*, $\overline{TM}$, of M .

$\overline{\phi}_{ij}^* : U_{ij} \rightarrow \text{GL}(\overline{\mathbb{C}}^*)$ are C^∞ maps and are the transition functions for the *anti-holomorphic cotangent bundle*, $\overline{TM}^*$, of M .

We shall now show how the operator $\partial/\partial\bar{z}$ extends to a Riemann surface M as a map $\bar{\partial} : C^\infty(M) \rightarrow C^\infty(\overline{TM}^*)$. First we need to note the composite mapping formula for $\partial/\partial z$, $\partial/\partial\bar{z}$:

Suppose h and g are C^1 complex valued functions defined on open subsets of $\mathbb{C}$. Then

$$\partial(gh)/\partial z = (\partial g/\partial z)(\partial h/\partial z) + (\partial g/\partial\bar{z})\overline{\partial h/\partial\bar{z}}, \quad (*)$$

$$\partial(gh)/\partial\bar{z} = (\partial g/\partial z)(\partial h/\partial\bar{z}) + (\partial g/\partial\bar{z})\overline{\partial h/\partial z}. \quad (**)$$

The proof is most easily done by taking real and imaginary parts and applying the chain rule for $\partial/\partial x$ and $\partial/\partial y$.

Let (U_i, ϕ_i) , (U_j, ϕ_j) be charts for M and $f \in C^\infty(M)$. Then

$$\begin{aligned} \partial(f\phi_i^{-1})/\partial\bar{z} &= \partial(f\phi_j^{-1}\phi_j\phi_i^{-1})/\partial\bar{z} \\ &= \partial(f\phi_j^{-1})/\partial\bar{z} \overline{\partial(\phi_j\phi_i^{-1})/\partial z}, \quad \text{using } ** \text{ above} \\ &= \overline{\phi}_{ij}^* \partial(f\phi_j^{-1})/\partial\bar{z}, \end{aligned}$$

where $\bar{\phi}_{ij}^*$ are the transition functions for $\overline{TM}^*$. Hence $\partial(f\phi_i^{-1})/\partial\bar{z}$ are the local representatives of a C^∞ section of $\overline{TM}^*$ which in the sequel we shall denote by $\bar{\partial}f$. The operator $\bar{\partial} : C^\infty(M) \rightarrow C^\infty(\overline{TM}^*)$ clearly has kernel equal to $A(M)$. We shall return to the question of how far $\bar{\partial}$ fails to be surjective at the end of this section.

The fibre model for $\overline{TM}^*$ is $\mathbb{C}^*$ and we denote the standard basis of $\mathbb{C}^*$ by $d\bar{z}$. Suppose that $g \in C^\infty(\overline{TM}^*)$, then the local representative of g with respect to a chart (U_i, ϕ_i) is given by $g_i d\bar{z}$, where $g_i = \bar{\phi}_i^* g \in C^\infty(U_i)$. If $f \in C^\infty(M)$, the local representative of $\bar{\partial}f$ is $\partial f_i / \partial \bar{z} d\bar{z}$, where $f_i = f\phi_i^{-1} \in C^\infty(U_i)$.

We may similarly define $\partial : C^\infty(M) \rightarrow C^\infty(TM)$ and the local representative of ∂f with respect to a chart (U_i, ϕ_i) will be $\partial f_i / \partial z dz$, where dz denotes the standard basis of $\mathbb{C}^*$.

We remark that $df = \partial f + \bar{\partial}f$ as is easily verified by the local computation $\partial f / \partial z dz + \partial f / \partial \bar{z} d\bar{z} = \partial f / \partial x dx + \partial f / \partial y dy$.

Definition 1.5.3. Let $\theta_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{C})$ be the transition functions for a complex line bundle E over the Riemann surface M . If the θ_{ij} are all analytic, we shall say that E is a *holomorphic line bundle* over M .

Examples.

9. If M is a Riemann surface, TM and TM^* are holomorphic line bundles over M .

10. Let $\text{HLB}(M)$ denote the set of isomorphism classes of holomorphic line bundles over the Riemann surface M (isomorphism here is, of course, analytic line bundle isomorphism, and may be given explicitly in terms of analytic functions using condition C above).

As in the previous example on $\text{CLB}(M)$, $\text{HLB}(M)$ has the natural structure of an abelian group with composition defined as tensor product and inverse as dual (see also below).

Holomorphic line bundles and their generalisation, holomorphic vector bundles, will be of the greatest importance in the sequel. As we shall see they provide a particularly potent geometric framework for many problems in complex analysis.

Suppose that E is a holomorphic line bundle on the Riemann surface M with transition functions θ_{ij} relative to the cover $\{U_i\}$. A family $S_i \in A(U_i)$ will define an *analytic section* of E if, for all i, j ,

$$S_i = \theta_{ij} S_j \quad \text{on } U_{ij}.$$

A family $M_i \in M(U_i)$ will define a *meromorphic section* m of E if, for all i, j ,

$$M_i = \theta_{ij} M_j \quad \text{on } U_{ij}.$$

We denote the sets of analytic and meromorphic sections of E by $\Omega(E)$ and $M(E)$ respectively. We let $M^*(E)$ denote the set of non-zero meromorphic sections of E . As in §§3,4, we define $\text{ord}(m, z)$, $m \in M(E)$, $z \in M$; $\text{div}(m)$, $m \in M^*(E)$ and, if M is compact, $\deg(\text{div}(m))$, $m \in M^*(E)$.

With the above formalism out of the way we can now rapidly achieve one of the primary goals of this section: The geometric formulation of the problem of generalising Weierstrass' theorem to compact Riemann surfaces.

For the remainder of this section we shall assume that M is a compact Riemann surface.

Proposition 1.5.4. There is a natural homomorphism $[\] : \mathcal{D}(M) \rightarrow \text{HLB}(M)$ satisfying the property that for any $d \in \mathcal{D}(M)$ there exists $s(d) \in M^*([d])$ such that $\text{div}(s(d)) = d$. Furthermore, $s(d)$ is determined uniquely up to multiplication by elements of $\mathbb{C}^\bullet$.

Proof. As we showed in §3, any divisor may be uniquely specified by an open cover $\{U_i\}$ of M and $m_i \in M^*(U_i)$ such that, for all i, j , $m_i/m_j \in A^*(U_{ij})$. Define $\theta_{ij} = m_i/m_j \in A^*(U_{ij})$.

Clearly the θ_{ij} are the transition functions for a holomorphic line bundle on M which we denote by $[d]$. We leave it to the reader to verify that $d \mapsto [d]$ is a group homomorphism.

We define $s(d)_i = m_i \in M^*(U_i)$ and observe that $\theta_{ij}s(d)_j = s(d)_i$ on U_{ij} . Hence the $s(d)_i$ are the local representatives of a meromorphic section $s(d)$ of $[d]$. Obviously $\text{div}(s(d)) = d$. We leave the remaining assertions as an exercise—see also Lemma 1.5.6. $\square$

Remark. Proposition 1.5.4 gives a “formal” solution to the problem of which divisors are divisors of meromorphic functions: Every divisor is the divisor of a “twisted” meromorphic function.

Proposition 1.5.5. A divisor d on M is the divisor of a meromorphic function if and only if $[d]$ is trivial as a holomorphic line bundle.

Proof. Suppose $d = \text{div}(m)$, $m \in M^*(M)$. Let $\{U_i\}$ be any open cover of M . Set $m_i = m|_{U_i}$ and $\theta_{ij} = m_i/m_j = 1$. The m_i then define the divisor d and, since $\theta_{ij} = 1$, $[d]$ is holomorphically trivial. Conversely, if $[d]$ is holomorphically trivial, there exists a nowhere zero analytic section s of $[d]$. In terms of local representatives $s_i \in A^*(U_i)$ we must have $\theta_{ij}s_i = s_j$, where θ_{ij} are the transition functions of $[d]$. But $\theta_{ij} = m_i/m_j$, where the m_i define the divisor d . Therefore, we have $m_i/s_i = m_j/s_j$ on U_{ij} and we may define $m \in M^*(M)$ by $m|_{U_i} = m_i/s_i$. Obviously, $\text{div}(m) = d$. $\square$

Lemma 1.5.6. Suppose that $L \in \text{HLB}(M)$ and $s, t \in M^*(L)$. Then s/t naturally defines a non-zero meromorphic function on M .

Proof. Suppose that L has transition functions θ_{ij} relative to some open cover $\{U_i\}$ of M . Then, s, t are locally given by $s_i, t_i \in M^*(U_i)$ subject to the relations $\theta_{ij}s_{ij} = s_i$, $\theta_{ij}t_j = t_i$ for all i, j . Therefore, $s_j/t_j = s_i/t_i$ on U_{ij} and so we may define $m \in M^*(M)$ by $m|_{U_i} = s_i/t_i$. $\square$

Remark. Notice that t^{-1} naturally defines a meromorphic section of $L^{-1} = L^*$ and so s/t may equivalently be defined by using the pairing $M^*(L) \times M^*(L^{-1}) \rightarrow M^*(M)$ induced from the dual pairing $L \otimes L^* \cong \mathbb{C}$.

Proposition 1.5.7. Let $L \in \text{HLB}(M)$ and suppose $M^*(L) \neq \emptyset$. Given $s \in M^*(L)$, set $\deg(s) = \deg(\text{div}(s))$. Then $\deg(s)$ is independent of s and depends only on L .

Proof. Let $s, t \in M^*(L)$. Clearly, $\text{div}(s/t) = \text{div}(s) - \text{div}(t)$. But $s/t \in M^*(M)$ and so by Lemma 1.4.3, $\deg(s/t) = 0$. Hence $\deg(s) = \deg(t)$. $\square$

For every holomorphic line bundle L on M that admits a non-trivial meromorphic section we may define the *degree of L* , $\deg(L)$, to be $\deg(s)$, where s is any non-trivial meromorphic section of L . Notice that if $\deg(L) \neq 0$, then L cannot be holomorphically trivial. It is also clear that if $d \in \mathcal{D}(M)$ then $\deg(d) = \deg([d])$.

The degree of L is an important topological invariant of L . In fact we shall show later (Example 14, §3, Chapter 6) that $\deg(L) = 0$ if and only if L is trivial as a *complex* line bundle.

What we have achieved above is to shift the problem of determining which divisors are divisors of meromorphic functions to the apparently more general problem of determining which holomorphic line bundles admit non-trivial meromorphic sections. We already know that a divisor d can be the divisor of a meromorphic function on M only if $\deg([d]) = \deg(d) = 0$. Let $\text{HLB}_0(M)$ denote the subgroup of $\text{HLB}(M)$ consisting of line bundles of degree zero. Two problems naturally arise at this stage.

1. Show that $[\] : \mathcal{D}(M) \rightarrow \text{HLB}(M)$ is onto.

2. Describe $\mathrm{HLB}_0(M)$.

To show that $[\]$ is onto amounts to proving that every holomorphic line bundle on M has a non-trivial meromorphic section. The existence of such meromorphic sections follows from a rather general finiteness theorem that we prove in Chapter 7 (see the exercises at the end of §5, Chapter 7). Once we know that $[\]$ is onto it is an easy matter to prove Riemann's part of the *Riemann–Roch* theorem:

For any divisor d on M we have

$$\dim_{\mathbb{C}} \Omega([d]) \geq \deg(d) + 1 - g,$$

where g denotes the genus of M .

This result is already sufficient to prove useful existence theorems for meromorphic functions on compact Riemann surfaces. The full Riemann–Roch theorem depends on Serre-duality which we establish in Chapter 10 (see also §5, Chapter 7).

As far as the second problem goes, $\mathrm{HLB}_0(M)$ may be shown to have the structure of a compact complex Lie group of complex dimension g . The group $\mathrm{HLB}_0(M)$ is usually referred to as the *Picard variety* of M and denoted by $\mathrm{Pic}(M)$. Whilst the dimension of $\mathrm{Pic}(M)$ depends only on the genus on M its complex structure depends on the complex structure of M . Abel's theorem gives explicit conditions on a divisor in order that it determine the identity of $\mathrm{Pic}(M)$, that is the trivial holomorphic line bundle (We refer to Gunning [1] for details).

We conclude this section by briefly indicating the role that $\bar{\partial}$ plays in the study of meromorphic functions on a compact Riemann surface.

Suppose that E is a holomorphic line bundle on M with transition functions θ_{ij} . We shall investigate conditions under which E is holomorphically trivial. First observe that, as in the proof of Theorem 1.3.3, $\{(2\pi i)^{-1} \log \theta_{ij}\}$ defines a class γ_E in $H^2(M, \mathbb{Z})$. The class γ_E will be zero if and only if E is trivial as a complex line bundle. From now on we assume E is complex trivial, that is $E \in \mathrm{Pic}(M)$. We may choose the branches of $\log \theta_{ij}$ so that $\log \theta_{ij} = h_{ij} \in A(U_{ij})$ determine an additive cocycle (see the proof of Theorem 1.3.3). Now E is trivial as a complex line bundle if there exist $g_i \in C^\infty(U_i)$ such that $h_{ij} = g_j - g_i$ (exponentiate!). Define $F \in C^\infty(\overline{TM}^*)$ by $F|U_i = \bar{\partial} g_i$. We can solve $\bar{\partial} u = -F$ if the class of F in $C^\infty(\overline{TM}^*)/\bar{\partial} C^\infty(M)$ is zero. Given such a solution, the functions $a_i = \exp(g_i + u) \in A^*(U_i)$ will then give a holomorphic trivialisation of E . The basic result now is that the complex vector space $C^\infty(\overline{TM}^*)/\bar{\partial} C^\infty(M)$ is finite dimensional (the dimension is actually g , the genus of M). If we are more careful in our arguments we see that to prove E is holomorphically trivial it is enough to find $u \in C^\infty(M)$ such that $\bar{\partial} u + F$ exponentiates to zero. Further analysis then shows that $H^1(M, \mathbb{Z}) (\cong \mathbb{Z}^{2g})$ determines an integer lattice in $C^\infty(\overline{TM}^*)/\bar{\partial} C^\infty(M)$ and that a necessary and sufficient condition for E to be holomorphically trivial is that F determines a lattice point. The Picard variety is then isomorphic to the torus $(C^\infty(\overline{TM}^*)/\bar{\partial} C^\infty(M))/H^1(M, \mathbb{Z})$. We refer to Gunning [1] for details. In Chapter 7 we shall show that $C^\infty(\overline{TM}^*)/\bar{\partial} C^\infty(M)$ is finite dimensional using rather general arguments about sheaves. In Chapter 10 we give a more direct proof of finiteness and much more explicit description of the quotient $C^\infty(\overline{TM}^*)/\bar{\partial} C^\infty(M)$ using Hodge theory.

Exercises.

1. Let $f : X \rightarrow Y$ be a continuous map between topological spaces and E be a vector bundle on Y with transition functions θ_{ij} relative to some cover $\{U_i\}$ of Y . Show that $\theta_{ij}f$ are transition functions for a vector bundle f^*E over X (the “pull-back bundle of E by f ”). Verify that, up to vector bundle isomorphism, f^*E depends only on E and f and not on any choices we have made. Show also that in case f is a holomorphic map between Riemann surfaces our construction determines a homomorphism $f^* : \mathrm{HLB}(Y) \rightarrow \mathrm{HLB}(X)$.

2. Let $f : X \rightarrow Y$ be a holomorphic map between Riemann surfaces and let $d \in \mathcal{D}(Y)$. Prove that $f^*[d] \cong [f^*(d)]$ (see Exercise 5, §4).
3. Let M be a Riemann surface and $d, d' \in \mathcal{D}(M)$. We say d and d' are *linearly equivalent* if $d - d' = \text{div}(m)$ for some $m \in M^*(M)$. Show that d and d' are linearly equivalent if and only if $[d] \cong [d']$ as holomorphic line bundles.
4. Let Ω be a domain in $\mathbb{C}$. Prove that every holomorphic line bundle on Ω is holomorphically trivial.
5. Let E, F be vector bundles over the differential manifold M . Prove that $C^\infty(E' \otimes F)$ is naturally isomorphic to the space of smooth vector bundle maps from E to F , $\text{Hom}(E, F)$.
6. Suppose that E, F are vector bundles over the differential manifold M and $\phi : C^\infty(E) \rightarrow C^\infty(F)$ is a $C^\infty(M)$ -linear map. Prove that ϕ is induced from a vector bundle map $\Phi : E \rightarrow F$. That is, show that $\phi(s)(x) = \Phi(s(x))$, $x \in M$, $s \in C^\infty(E)$. Deduce that ϕ determines a unique section γ of $C^\infty(E' \otimes F)$ characterised by $\langle \gamma, X \rangle = \phi(X)$, $X \in C^\infty(E)$, where $\langle \cdot, \cdot \rangle$ is induced from the dual pairing between E and E' .

Appendix to Chapter 1

In the first part of this appendix we prove a number of results related to Cauchy's integral formula that will be used later in the text as well as in the proof of Theorem 1.3.1. The main element in the proof of Theorem 1.3.1 is the Runge approximation theorem and this, together with the proof of Theorem 1.3.1, constitute the remainder of the appendix.

In the statement of the next three results we shall assume that Ω is a bounded open domain in $\mathbb{C}$ with $\partial\Omega$ a finite union of piecewise C^1 curves. We use the notation $C^1(\overline{\Omega})$ to denote the set of functions on $\overline{\Omega}$ which are restrictions of C^1 functions defined on some neighbourhood of $\overline{\Omega}$.

Lemma A1.1. Let $f \in C^1(\overline{\Omega})$. Then

$$\int_{\partial\Omega} f dz = \int_{\Omega} \partial f / \partial \bar{z} d\bar{z} dz = 2i \int_{\Omega} \partial f / \partial \bar{z} dx dy.$$

Proof. Stoke's or Green's theorem in the plane (take real and imaginary parts). □

Theorem A1.2. (Cauchy's Integral Formula). Let $u \in C^1(\overline{\Omega})$. Then

$$u(\zeta) = (2\pi i)^{-1} \int_{\partial\Omega} \frac{u(z)}{z - \zeta} dz + \frac{1}{\pi} \int_{\Omega} \partial u / \partial \bar{z} / (z - \zeta) dx dy, \quad \zeta \in \Omega.$$

Proof. Let $\Omega_\varepsilon = \{z \in \Omega : |z - \zeta| > \varepsilon\}$, where $\varepsilon < d(\zeta, \partial\Omega)$. Apply Lemma A1.1 with $f(z) = u(z)/(z - \zeta)$ and let $\varepsilon \rightarrow 0$. □

Corollary A1.3. Let $u \in C^0(\overline{\Omega})$ and suppose u is analytic in Ω . Then

$$u(\zeta) = (2\pi i)^{-1} \int_{\partial\Omega} \frac{u(z)}{z - \zeta} dz, \quad \zeta \in \Omega.$$

Corollary A1.4. Let Ω be a domain in $\mathbb{C}$ and suppose that u_n , $n \geq 1$, is a sequence in $A(\Omega)$ which converges uniformly on compact subsets of Ω to a function u on Ω . Then $u \in A(\Omega)$.

Proof. Apply Corollary A1.3 to $u_n - u_m$ restricted to closed discs contained in Ω to deduce that $\partial u_n / \partial \bar{z}$ converges uniformly on compact subsets of Ω . Since $\partial u_n / \partial \bar{z} = 0$, it follows that u is C^1 on Ω with $\partial u / \partial \bar{z} = 0$. □

Corollary A1.5. (Montel's theorem). Let Ω be a domain in $\mathbb{C}$ and suppose that u_n , $n \geq 1$, is a sequence in $A(\Omega)$ which is uniformly bounded on every compact subset of Ω . Then there is a subsequence of the u_n which converges uniformly on compact subsets of Ω to a limit $u \in A(\Omega)$.

Proof. By Cauchy's inequalities (Exercise 2, §1) the sequence $\partial u_n / \partial z$ is uniformly bounded on any closed disc $D_r(z) \subset \Omega$. Hence $\partial u_n / \partial z$ is uniformly bounded on any compact subset of Ω . Therefore the sequence u_n is an equicontinuous family and, by Ascoli's theorem, has a subsequence converging uniformly on compact subsets of Ω to a function u . By Corollary A1.4, $u \in A(\Omega)$. $\square$

Let μ be a (finite, complex regular, Borel) measure on $\mathbb{C}$. The *support* of μ , $\text{supp}(\mu)$, is defined to be the smallest closed subset of $\mathbb{C}$ outside of which μ is zero.

Theorem A1.6. Let μ be a measure on $\mathbb{C}$ with compact support. Then

$$u(\zeta) = \int_{\mathbb{C}} (z - \zeta)^{-1} d\mu(z)$$

defines an analytic function outside $\text{supp}(\mu)$. If $d\mu = \frac{1}{\pi} \phi dx dy$, $\phi \in C_c^\infty(\mathbb{C})$, then $u \in C^\infty(\mathbb{C})$ and $\partial u / \partial \bar{z} = \phi$.

Proof. Certainly u is C^1 outside $\text{supp}(\mu)$ —in fact C^∞ —and differentiating under the integral sign we see that $\partial u / \partial \bar{z} = 0$. Hence u is analytic outside $\text{supp}(\mu)$. For the second statement we see by changing variables that

$$u(\zeta) = -\frac{1}{\pi} \int_{\mathbb{C}} \phi(\zeta - z) z^{-1} dx dy.$$

Now z^{-1} is integrable on compact sets and so $u \in C^\infty(\mathbb{C})$. Moreover,

$$\begin{aligned} \partial u / \partial \bar{\zeta} &= \frac{1}{\pi} \int_{\mathbb{C}} \partial \phi / \partial \bar{\zeta} (\zeta - z) / z dx dy \\ &= \frac{1}{\pi} \int_{\mathbb{C}} \partial \phi / \partial \bar{\zeta} (z) / (\zeta - z) dx dy. \end{aligned}$$

Apply Theorem A1.2 with Ω any disc containing $\text{supp}(\mu)$ to deduce that $\partial u / \partial \bar{\zeta} = \phi$. $\square$

Remark. Theorem A1.6 proves Theorem 1.3.1 in the special case that the right hand side is a C^∞ function with compact support. The solution u given by Theorem A1.6 need not have compact support as is easily seen by choosing ϕ so that $\int_{\mathbb{C}} \phi \neq 0$. This is in sharp distinction to what happens for more than one variable as we shall see in Chapter 2.

We shall now prove a special case of the Runge Approximation Theorem. The proof we give is modelled on that given in Gamelin [1] and Hörmander [1] and uses the Riesz representation theorem in combination with the Hahn–Banach theorem. An alternative proof may be found in Hille [1].

Theorem A1.7. Let Ω be an open subset of $\mathbb{C}$ and suppose that K is a compact subset of Ω such that $\Omega \setminus K$ has no component which is relatively compact in Ω . Then every function which is analytic on a neighbourhood of K can be uniformly approximated on K by functions in $A(\Omega)$.

Proof. By the Hahn–Banach and Riesz representation theorems, it is sufficient to show that every measure μ on K which is orthogonal to $A(\Omega)$ is also orthogonal to every function analytic on a neighbourhood of K .

Suppose f is analytic on a neighbourhood of K . Since we are interested in uniform approximation on K we can assume that $f \in C_c^\infty(\mathbb{C})$ (multiply f by $\phi \in C_c^\infty(\mathbb{C})$ where $\phi \equiv 1$ on K). By Theorem A1.2,

$$\begin{aligned} f(\zeta) &= \frac{1}{\pi} \int_{\mathbb{C}} \partial f / \partial \bar{z} (z - \zeta)^{-1} dx dy \\ &= \frac{1}{\pi} \int_{\mathbb{C} \setminus K} \partial f / \partial \bar{z} (z - \zeta)^{-1} dx dy, \end{aligned}$$

since f is analytic on a neighbourhood of K .

By Fubini's theorem

$$\int_{\mathbb{C}} f(\zeta) d\mu(\zeta) = \frac{1}{\pi} \int_{\mathbb{C} \setminus K} \partial f / \partial \bar{z} \left[\int_{\mathbb{C}} (z - \zeta)^{-1} d\mu(\zeta) \right] dx dy.$$

Set $u(z) = \int_{\mathbb{C}} (z - \zeta)^{-1} d\mu(\zeta)$. Then u is analytic on $\mathbb{C} \setminus K$, Theorem A1.6. We claim that $u \equiv 0$ on $\mathbb{C} \setminus K$. If so, it then follows that $\int_{\mathbb{C}} f d\mu = 0$ and the proof is complete.

Let $R = \sup\{|z| : z \in K\}$. Then $(z - \zeta)^{-1}$ can be expanded as a power series in ζ^n which converges uniformly on K provided that $|z| > R$. By our orthogonality assumption on μ , $\int_{\mathbb{C}} \zeta^n d\mu(\zeta) = 0$, $n \geq 0$. Hence $u(z) = 0$, $|z| > R$ and so, by uniqueness of analytic continuation, u vanishes identically on the unbounded component of $\mathbb{C} \setminus K$. On the other hand if $z \in \mathbb{C} \setminus \Omega$, $\partial^k u / \partial z^k = (-1)^k k! \int_{\mathbb{C}} (z - \zeta)^{-k-1} d\mu(\zeta) = 0$ since $(z - \zeta)^{-k-1} \in A(\Omega)$ for $z \in \mathbb{C} \setminus \Omega$. Consequently, $u \equiv 0$ on every component of $\mathbb{C} \setminus K$ which intersects $\Omega \setminus K$. But, by assumption, $\Omega \setminus K$ has no components which are relatively compact in Ω and so $u \equiv 0$ on $\mathbb{C} \setminus K$. $\square$

Theorem A1.8. Let $f \in C^\infty(\Omega)$. Then there exists $u \in C^\infty(\Omega)$ such that

$$\partial u / \partial \bar{z} = f.$$

Proof. (Following Hörmander [1]). We already know from Corollary A1.5 that we can solve $\partial u / \partial \bar{z} = f$ if f has compact support. What we shall do is to write f as an infinite sum of C^∞ functions with compact support and then use Runge's theorem to modify the resulting infinite set of solutions so that they converge to a solution of $\partial u / \partial \bar{z} = f$.

For $n = 1, 2, \dots$, define $K_n = \{z \in \Omega : |z| \leq n \text{ and } d(z, \partial\Omega) \geq 1/n\}$. Then, the K_n are compact; $\Omega \setminus K_n$ has no components which are relatively compact in Ω ; every compact subset of Ω is contained in some K_n . For $n = 1, 2, \dots$, choose $\theta_n \in C_c^\infty(\Omega, \mathbb{R})$ so that $\theta_n \equiv 1$ on a neighbourhood of K_n . Set $\phi_n = \theta_n - \theta_{n-1}$, $n > 1$. Clearly $\theta_n \equiv 0$ on a neighbourhood of K_{n-1} and $\sum \phi_n \equiv 1$ on Ω . Let $u_n \in C^\infty(\Omega)$ be a solution of $\partial u / \partial \bar{z} = \phi_n f$ given by Theorem A1.6. By the Runge Theorem, there exists $a_n \in A(\Omega)$ such that $|a_n - u_n| < 2^{-n}$ on K_{n-1} . Set $u = \sum_n (u_n - a_n)$ and note that the sum converges uniformly on any compact subset of Ω . For fixed N , the sum from N to ∞ consists of terms which are analytic on a neighbourhood of K_N and so defines an analytic function on the interior of K_N . Hence $u \in C^\infty(\Omega)$ and, differentiating term-by-term, $\partial u / \partial \bar{z} = f$. $\square$

Remark. As we shall see later, Theorems A1.7 and A1.8 do not generalise to arbitrary domains in $\mathbb{C}^n$, $n > 1$.

Exercises.

1. Let Ω be a domain in $\mathbb{C}$. Suppose that $\omega \subset \Omega$ is an open neighbourhood of the compact subset K of Ω . For integers $p \geq 1$ and $u \in A(\Omega)$ let

$$|u|_p^\omega = \left\{ \int_\omega |u|^p dx dy \right\}^{1/p} \quad (|u|_p^\omega \text{ may be infinite}).$$

Show that, given s , $0 < s < d(K, \partial\Omega)$, there exists $C > 0$ such that for all $u \in A(\Omega)$ we have

$$\|\partial^j u / \partial z^j\|_K \leq \frac{C j!}{s^j} |u|_p^\omega, \quad j \geq 0.$$

(Hint: Choose $\phi \in C_c^\infty(\Omega)$ such that $\phi(z) = 1$ whenever $d(z, K) \leq s$ and apply Theorem A1.2 to ϕu^p).

2. Let Ω be a domain in $\mathbb{C}$ and for integers $p \geq 1$ let $L^p(\Omega) = \{f \in A(\Omega), |f|_p^\Omega < \infty\}$. Show that $L^p(\Omega)$ is a Banach space and, in case $p = 2$, a Hilbert space (Use the result of Q1. in combination with Corollary A1.4).
3. Show that if Ω is a simply connected domain in $\mathbb{C}$ and $f \in A(\Omega)$ then there is a sequence of polynomials converging to f uniformly on compact subsets of Ω . Show that if Ω is not simply connected, then there is a sequence of rational functions converging uniformly to f on compact subsets of Ω and that we may assume that the rational functions have all their poles outside Ω .

Chapter 2

Functions of Several Complex Variables

Introduction

In this Chapter we develop the basic theory of analytic functions of several complex variables with particular reference to the problem of extension of analytic functions. Section 1 is a straightforward generalisation of the one variable theory and contains such theorems as the power series representation of an analytic function. In section 2 we prove a Riemann removable singularities theorem for analytic functions of more than one variable. The material in section 3 is in sharp contrast to the one variable theory and includes the theorem of Hartog's that implies, for example, that every analytic function on the punctured Euclidean disc in $\mathbb{C}^n$, $n > 1$, extends analytically to the whole disc. In section 4 we define domains of holomorphy and prove their equivalence with holomorphically convex domains. Section 5 is devoted to various pseudoconvexity properties that a domain of holomorphy possesses. In section 6 we discuss the Bergman kernel function of a domain and in section 7 we make a preliminary study of the Cousin problems.

1 Elementary theory of analytic functions of several complex variables

In this section we shall make a preliminary study of analytic functions of more than one complex variable. Our development will follow that of §1 of Chapter 1 rather closely.

For $j = 1, \dots, n$, we define the differential operators

$$\partial/\partial z_j = \frac{1}{2}(\partial/\partial x_j - i\partial/\partial y_j), \quad \partial/\partial \bar{z}_j = \frac{1}{2}(\partial/\partial x_j + i\partial/\partial y_j).$$

Just as in §4 of Chapter 1 we have a composite mapping formula for these operators.

Proposition 2.1.1. (Composite mapping formula) Let $U \subset \mathbb{C}^m$, $V \subset \mathbb{C}^n$ be open and $g \in C^1(U, \mathbb{C}^n)$, $f \in C^1(V)$. Suppose $g(U) \subset V$. Then

$$\begin{aligned} \partial(fg)/\partial z_j &= \sum_{i=1}^n \left(\frac{\partial f}{\partial z_i} \frac{\partial g_i}{\partial w_j} + \frac{\partial f}{\partial \bar{z}_i} \overline{\frac{\partial g_i}{\partial \bar{w}_j}} \right), \\ \partial(fg)/\partial \bar{z}_j &= \sum_{i=1}^n \left(\frac{\partial f}{\partial z_i} \frac{\partial g_i}{\partial \bar{w}_j} + \frac{\partial f}{\partial \bar{z}_i} \overline{\frac{\partial g_i}{\partial w_j}} \right), \end{aligned}$$

where we denote coordinates on $\mathbb{C}^m$, $\mathbb{C}^n$ by $(z_1, \dots, z_m)$, $(w_1, \dots, w_n)$ respectively.

Proof. As in §5 of Chapter 1, we take real and imaginary parts and apply the usual composite mapping formula. We omit the tedious but elementary computations. $\square$

For the remainder of this section Ω will always denote a domain in $\mathbb{C}^n$.

Definition 2.1.2. Let $f \in C^1(\Omega)$. We say that f is *analytic* or *holomorphic* on Ω if f is analytic in each variable separately. That is, if f satisfies the following system of first order partial differential equations:

$$\partial f / \partial \bar{z}_j = 0, \quad 1 \leq j \leq n.$$

Remarks.

1. Let $f \in C^1(\Omega)$. The derivative of f at $z \in \Omega$, $Df(z) : \mathbb{C}^n \rightarrow \mathbb{C}$, is an $\mathbb{R}$ -linear map. It is easy to verify that f is analytic on Ω iff $Df(z)$ is $\mathbb{C}$ -linear for all $z \in \Omega$. If f is analytic then the matrix of $Df(z)$, relative to the standard basis of $\mathbb{C}^n$, is $[\partial f / \partial z_1, \dots, \partial f / \partial z_n]$.
2. If $f = (f_1, \dots, f_m) \in C^1(\Omega, \mathbb{C}^m)$, we say that f is analytic if each f_i is analytic. It follows from the previous remark that $Df(z) : \mathbb{C}^n \rightarrow \mathbb{C}^m$ is $\mathbb{C}$ -linear with matrix $[\partial f_i / \partial z_j]$.
3. A consequence of Proposition 2.1.1 is that the composite of analytic maps is analytic.
4. Since the inverse of a $\mathbb{C}$ -linear map is $\mathbb{C}$ -linear, a holomorphic diffeomorphism is necessarily biholomorphic. For the same reason, the inverse function theorem, implicit function theorem and rank theorem all hold for analytic mappings (see Dieudonné [1] or Field [1]).

Notation. We denote the set of analytic functions on Ω by $A(\Omega)$.

As in the 1-variable theory, the main technical tool used in developing the elementary theory of analytic functions of several complex variables is an integral representation formula. First, however, some notational conventions.

Given $a = (a_1, \dots, a_n) \in \mathbb{C}^n$ and $r_1, \dots, r_n > 0$, we set

$$\begin{aligned} D(a; r_1, \dots, r_n) &= \prod_{i=1}^n D_{r_i}(a_i) \\ &= \{z \in \mathbb{C}^n : |z_j - a_j| < r_j\}. \end{aligned}$$

We call $D(a; r_1, \dots, r_n)$ an open *polydisc* with centre a . We may similarly define closed polydiscs. In case $r_1 = \dots = r_n = r$, we set $D(a; r_1, \dots, r_n) = D(a; r)$ and call $D(a; r)$ the open polydisc of radius r , centre a . Note that $D(a; r)$ is just the open disc, radius r , centre a relative to the norm $|z| = \max_i |z_i|$ on $\mathbb{C}^n$.

Suppose $D = \prod_{i=1}^n D_i$ is a polydisc in $\mathbb{C}^n$. Thus each D_i will be a disc in $\mathbb{C}$. We let ∂D_j denote the boundary of D_j and define

$$\partial_0 D = \prod_{i=1}^n \partial D_i.$$

We call $\partial_0 D$ the *distinguished boundary* of D . We remark that $\partial_0 D$ is a proper subset of ∂D and is homeomorphic to a real n -dimensional torus.

Given $a = (a_1, \dots, a_n) \in \mathbb{C}^n$ and $r > 0$, we set

$$E(z; r) = \{z \in \mathbb{C}^n : \sum_{i=1}^n |z_i - a_i|^2 < r^2\}.$$

We call $E(z; r)$ the open *Euclidean disc*, centre a , radius r . In case $z = 0$, we often abbreviate $E(z; r)$ to $E(r)$.

Both polydiscs and Euclidean discs will play an important role in the sequel.

Next we briefly review some multi-index notation. If $\mathbb{N}$ denotes the positive integers, including zero, and n is a strictly positive integer, $m \in \mathbb{N}^n$ is to be thought of as an n -tuple $(m_1, \dots, m_n)$ of positive integers. Then $m!$ is shorthand for $m_1! \cdots m_n!$; $|m|$ for $|m_1 + \cdots + m_n|$; ∂^m for $\partial^{m_1}/\partial z_1^{m_1} \cdots \partial^{m_n}/\partial z_n^{m_n}$; r^m for $r_1^{m_1} \cdots r_n^{m_n}$; z^m for $z_1^{m_1} \cdots z_n^{m_n}$; $f_m(z)$ for $f_{m_1 \cdots m_n}(z)$.

Theorem 2.1.3. (Cauchy's integral formula for polydiscs). Let D be an open polydisc in $\mathbb{C}^n$ with centre a and suppose $f \in C^0(\bar{D})$ is analytic on D . Then for all $z = (z_1, \dots, z_n) \in D$, we have

$$f(z) = (2\pi i)^{-n} \int_{\partial_0 D} f(\zeta_1, \dots, \zeta_n) \prod_{i=1}^n (\zeta_i - z_i)^{-1} d\zeta_1 \cdots d\zeta_n.$$

In particular, f is C^∞ and

$$\partial^m f(a) = (2\pi i)^{-n} m! \int_{\partial_0 D} f(\zeta_1, \dots, \zeta_n) \prod_{i=1}^n (\zeta_i - a_i)^{-m_i-1} d\zeta_1 \cdots d\zeta_n.$$

Proof. Repeated application of the Cauchy formula in 1-variable (Corollary A1.3) gives

$$f(z) = (2\pi i)^{-n} \int_{\partial D_1} \frac{d\zeta_1}{\zeta_1 - z_1} \cdots \int_{\partial D_n} \frac{d\zeta_n}{\zeta_n - z_n} f(\zeta_1, \dots, \zeta_n), \quad z \in D.$$

For fixed z , the integrand is continuous on a compact domain of integration and so the integral formula follows from Fubini's theorem. The remaining statements are immediate by differentiation under the integral sign. $\square$

Corollary 2.1.4. Every analytic function on an open subset of $\mathbb{C}^n$ is C^∞ .

Proof. Immediate from Theorem 2.1.3 since infinite differentiability is a local property. $\square$

Remarks.

1. The proof of Theorem 2.1.3 only uses the analyticity of f in each variable *separately* together with the continuity of f on $\bar{D}$. This observation, together with Corollary 2.1.4, proves Osgood's Lemma: If $f \in C^0(\Omega)$ and is separately analytic on Ω then f is analytic on Ω . Much less trivial is the amazing theorem of Hartogs which states that if f is separately analytic, with *no* assumption of continuity on f , then f is analytic. This result is definitely false for real analytic functions as the simple example $f(x, y) = xy/(x^2 + y^2)$, $(x, y) \neq (0, 0)$; $f(0, 0) = 0$ shows. For proofs of Hartogs theorem we refer the reader to Hörmander [1] or R. Narasimhan [2].
2. The reader should note that we only integrate over the proper subset $\partial_0 D$ of ∂D in Cauchy's integral formula. Another, related, way of seeing the significance of $\partial_0 D$ is to observe that f must always take its maximum on $\partial_0 D$. Indeed, this is a simple consequence of the maximum principle for analytic functions of one variable ($\partial_0 D$ is actually the *Šilov boundary* of D . For general facts about Šilov boundaries see Fuks [1] or Gamelin [1]).
3. Although we shall not need them in the sequel, we wish to point out that there are several other integral representation formulae for analytic functions of more than one variable that are of considerable importance in some applications. We mention in particular the Bochner–Martinelli, Bergmann–Cauchy–Weil and Cauchy–Fantappiè integral formulae. For the first two formulae we refer to Fuks [2] or Vladimirov [1] and for the third to Leray [1] or Aizenberg [1]. For the relations between these formulae and Cauchy's integral formula we refer to Harvey [1].

Theorem 2.1.5. Let $f \in A(\Omega)$ and $D \subset \Omega$ be a polydisc, centre a . Then

$$f(z) = \sum_m a_m (z - a)^m, \quad z \in D,$$

where convergence is uniform on compact subsets of D and the coefficients a_m are given by $m!a_m = \partial^m f(a)$, $m \in \mathbb{N}^n$.

Proof. Let $D' \subset D$ be any open polydisc, centre a , such that $\bar{D}' \subset \Omega$ (that is, D' is relatively compact in Ω). By Theorem 2.1.3, we have

$$f(z) = (2\pi i)^{-n} \int_{\partial_0 D'} f(\zeta_1, \dots, \zeta_n) \prod_{i=1}^n (\zeta_i - z_i)^{-1} d\zeta_1 \cdots d\zeta_n, \quad z \in D'.$$

Now if $z \in D'$, $\zeta \in \partial_0 D'$ we have

$$\prod_{i=1}^n (\zeta_i - z_i)^{-1} = \sum_m \frac{(z - a)^m}{(\zeta - a)^m} (\zeta - a)^{-1},$$

where $(\zeta - a)^{-1}$ is shorthand for $(\zeta_1 - a_1)^{-1} \cdots (\zeta_n - a_n)^{-1}$. The convergence of the sum on the right is uniform on compact subsets of D' . Multiply by $f(\zeta_1, \dots, \zeta_n)$ and integrate term by term to obtain

$$\begin{aligned} f(z) &= (2\pi i)^{-n} \sum_m (z - a)^m \int_{\partial_0 D} \frac{f(\zeta_1, \dots, \zeta_n)}{(\zeta - a)^m} (\zeta - a)^{-1} d\zeta_1 \cdots d\zeta_n \\ &= \sum_m a_m (z - a)^m. \end{aligned}$$

Differentiation of the power series and evaluation at a give the formulae for a_m . □

The proofs of the next four results parallel those of their one variable counterparts and we omit them.

Corollary 2.1.6. If $f \in A(\Omega)$, then $\partial^m f \in A(\Omega)$ for all $m \in \mathbb{N}^n$.

Corollary 2.1.7. (Cauchy's inequalities). If f is analytic on the polydisc $D = D(a; r_1, \dots, r_n)$ and $|f| \leq M$ on D , then

$$|\partial^m f(a)| \leq M m! r^{-m}, \quad m \in \mathbb{N}^n.$$

Corollary 2.1.8. Let u_n , $n \geq 1$, be a sequence of analytic functions on Ω which converges uniformly on compact subsets of Ω to a function u . Then $u \in A(\Omega)$.

Corollary 2.1.9. (Montel's theorem). Let u_n , $n \geq 1$, be a sequence in $A(\Omega)$ which is uniformly bounded on every compact subset of Ω . Then there is a subsequence of the u_n which converges uniformly on compact subsets of Ω to a limit $u \in A(\Omega)$.

Examples.

1. Let $P(z) = \sum a_m z^m$, where the $a_m \in \mathbb{C}$ and only finitely many are non-zero. Then P is analytic on $\mathbb{C}^n$ and is called a *polynomial* (in n -variables).
2. Let $f, g \in A(\Omega)$ and suppose g is not identically zero and has zero set $Z(g)$. Then f/g defines an analytic function on $\Omega \setminus Z(g)$.
3. If $f \in A(\mathbb{C}^n)$, we call f an *entire* analytic function. Every polynomial is entire. So also are the Laplace transforms of functions (or distributions) with compact support (see, for example, Hörmander [2] or Vladimirov [2]).

Theorem 2.1.10. (Laurent series). Suppose f is analytic in the annular region $A = \{z \in \mathbb{C}^n : r_j < |z_j| < R_j, j = 1, \dots, n\}$. Then

$$f(z) = \sum_{|m|=-\infty}^{+\infty} a_m z^m, \quad z \in A,$$

where the a_m are determined uniquely by f and A and convergence is uniform on compact subsets of A .

Proof. Exactly as for the 1-variable case using Cauchy's integral formula. Uniqueness of the coefficients a_m follows by induction on n . $\square$

Proposition 2.1.11. (Uniqueness of analytic continuation). Let U, V be connected open subsets of $\mathbb{C}^n$ and suppose $U \cap V \neq \emptyset$. If $f \in A(U)$ and h is an analytic extension of f to $U \cup V$ then h is unique.

Proof. Exactly as for Proposition 1.1.6. $\square$

Theorem 2.1.12. (Maximum Principle). Let $f \in A(\Omega)$ and suppose there exists $\zeta \in \Omega$ such that $|f(z)| \leq |f(\zeta)|$ for all $z \in \Omega$. Then f is constant.

Proof. Choose $r > 0$ so that the Euclidean disc $E = E(\zeta; r)$ is contained in Ω . If L denotes any (affine) complex line through ζ , the Maximum Principle for analytic functions of one complex variable applied to $f|_{(L \cap E)}$ implies that f is constant on $L \cap E$. This holds for all complex lines through ζ and so f is constant on E . Uniqueness of analytic continuation now implies that f is constant on Ω . $\square$

Finally, another simple application of one-variable theory.

Theorem 2.1.13. (Open Mapping Theorem). Let $f \in A(\Omega)$ and suppose that f is not constant. Then f is an open mapping. That is, f maps open subsets of Ω to open subsets of $\mathbb{C}$.

Proof. It is enough to prove that if $E \subset \Omega$ is any (Euclidean) open disc neighbourhood of $\zeta \in \Omega$, then $f(\zeta)$ is an interior point of $f(E)$. Since f is not constant, it follows by uniqueness of analytic continuation that $f|_{E \cap L}$ is not constant on every complex line L through ζ (compare the proof of Theorem 2.1.12). But if $f|_{E \cap L}$ is not constant, $f(E \cap L)$ is a neighbourhood of $f(\zeta)$ by the well-known one-variable result (Exercise 4, §1, Chapter 1). $\square$

Exercises.

1. Let Ω be a domain in $\mathbb{C}^n$. We say Ω is a *Reinhardt domain* if $(z_1, \dots, z_n) \in \Omega$ implies $(e^{i\theta_1} z_1, \dots, e^{i\theta_n} z_n) \in \Omega$ for all $\theta_1, \dots, \theta_n \in \mathbb{R}$. Show that if Ω is a Reinhardt domain containing the origin and $f \in A(\Omega)$ then

$$f(z) = \sum a_m z^m, \quad z \in \Omega,$$

where convergence is uniform on compact subsets of Ω . (Hint: Given $\epsilon > 0$, let $\Omega_\epsilon = \{z \in \Omega : d(z, \partial\Omega) > \epsilon|z|\}$ and Ω'_ϵ denote the connected component of Ω_ϵ containing the origin. For $z \in \Omega'_\epsilon$, define $g(z) = (2\pi i)^{-n} \int_{\partial_0 D} f(t_1 z_1, \dots, t_n z_n) (t-1)^{-1} dt_1 \cdots dt_n$, where $(t-1)^{-1}$ is shorthand for $(t_1-1)^{-1} \cdots (t_n-1)^{-1}$ and $D = D(0; 1+\epsilon)$. Note that if $z \in \Omega'_\epsilon$, then $(1+\epsilon)z \in \Omega$. Certainly g is analytic in Ω'_ϵ . Show that $f(z) = g(z)$ for $|z|$ sufficiently small. Expand $(t-1)^{-1}$ as a Laurent series in t and integrate term by term to obtain a power series for f . Then let $\epsilon \rightarrow 0$ and note that $\Omega = \bigcup_{\epsilon>0} \Omega'_\epsilon$).

Deduce that if Ω is a Reinhardt domain containing the origin then the polynomials are dense in $A(\Omega)$ (topology of uniform convergence on compact subsets). In particular, the polynomials are dense in $A(E(0; r))$, $r > 0$.

2. Let Ω be a domain in $\mathbb{C}^n$ and $\omega \subset \Omega$ be an open neighbourhood of the compact subset K of Ω . For integers $p \geq 1$ and $u \in A(\Omega)$ let

$$|u|_p^\omega = \left(\int_\omega |u|^p d\lambda \right)^{1/p},$$

where $d\lambda$ denotes Lebesgue measure on $\mathbb{C}^n$ ($|u|_p^\omega$ may be infinite). Show that for all $m \in \mathbb{N}^n$ there exist constants C_m such that

$$\sup_{z \in K} |\partial^m u(z)| = \|\partial^m u\|_K \leq C_m |u|_p^\omega, \quad u \in A(\Omega).$$

Deduce that the spaces $L^p(\Omega) = \{f \in A(\Omega) : |f|_p^\Omega < \infty\}$ are Banach spaces and that $L^2(\Omega)$ is a Hilbert space. (Hint for the first part: Repeated application of Corollary A1.4, see also Exercises 1,2 in the Appendix to Chapter 1).

2 Removable Singularities

The Riemann removable singularities theorem for functions of one complex variable (§2, Chapter 1) implies that if X is a discrete subset of $\Omega \subset \mathbb{C}$ then every analytic function on $\Omega \setminus X$ which is locally bounded on Ω has a unique analytic extension to Ω . In this section we generalise this result to analytic functions of more than one variable.

Definition 2.2.1. Let X be a subset of the domain Ω in $\mathbb{C}^n$. We say that X is an *analytic subset* of Ω if for every $z \in \Omega$ there exists an open neighbourhood U of z in Ω and analytic map $f : U \rightarrow \mathbb{C}^p$ such that $X \cap U = f^{-1}(0)$ (p may depend on z).

We shall undertake a more systematic study of analytic sets in Chapters 3 and 4 and for the present we remark only that our definition implies that if X is an analytic subset of Ω then X is a closed subset of Ω .

Theorem 2.2.2. (Riemann removable singularities theorem). Let X be a proper analytic subset of the domain Ω in $\mathbb{C}^n$ and $f \in A(\Omega \setminus X)$. Suppose that for every $x \in \Omega$, there exists an open neighbourhood U of x in Ω such that $f|_{U \setminus X}$ is bounded. Then f has a unique analytic extension to Ω .

Proof. It is clearly sufficient to show that we can find an open neighbourhood D of every $z_0 \in \Omega$ such that $f|_{D \setminus X}$ extends uniquely to D . The case $z_0 \notin X$ being trivial we shall suppose $z_0 \in X$. Choose an open connected neighbourhood U of z_0 so that $X \cap U = g^{-1}(0)$ for some $g \in A(U, \mathbb{C}^p)$. Writing $g = (g_1, \dots, g_p)$, we may suppose $h = g_1 \not\equiv 0$. Since h cannot be identically zero on every affine complex line through z_0 , we may make an affine linear change of coordinates and suppose that $z_0 = 0$ and $h(0, \dots, 0, z_n) \not\equiv 0$ on a neighbourhood of $z_n = 0$. Since $z_n = 0$ is an isolated zero of $h(0, \dots, 0, z_n)$, there exists $\delta > 0$ such that $h(0, \dots, 0, z_n) \neq 0$ for $0 < |z_n| \leq \delta$. Denote the variable $(z_1, \dots, z_{n-1}) \in \mathbb{C}^{n-1}$ by z' and set $|z'| = \max_i |z_i|$. By the continuity of h we may choose $r > 0$ so that $h(z', z_n) \neq 0$ for $|z'| \leq r$ and $|z_n| = \delta$. Let D denote the polydisc $\{(z', z_n) : |z'| < r, |z_n| < \delta\}$. We define

$$g(z', z_n) = (2\pi i)^{-1} \int_{|\zeta|=\delta} \frac{f(z', \zeta)}{\zeta - z_n} d\zeta, \quad (z', z_n) \in D.$$

Since $f(z', \zeta)$ is holomorphic in z' for $|z'| \leq r$ and $|\zeta| = \delta$, $g \in A(D)$.

For fixed z' , $|z'| < r$, the function $f(z', \zeta)$ extends by the one variable Riemann removable singularities theorem to the disc $|\zeta| < \delta$. Hence, by the Cauchy integral formula, $g(z) = f(z)$ if $z \in D \setminus X$. $\square$

Corollary 2.2.3. Let X be a proper analytic subset of the domain Ω in $\mathbb{C}^n$. Then $\Omega \setminus X$ is an open, connected and dense subset of Ω .

Proof. If $\Omega \setminus X$ were not connected we could define $f \in A(\Omega \setminus X)$ by taking $f \equiv 0$ on one connected component of $\Omega \setminus X$ and $f \equiv 1$ on all the other connected components. The Riemann removable singularities theorem then implies that f extends uniquely to Ω and we obtain a contradiction by the uniqueness of analytic continuation. We leave the remaining assertion as an exercise. $\square$

Next we state an important removable singularities theorem due to Rado.

Theorem 2.2.4. (Rado's Theorem). Let Ω be a domain in $\mathbb{C}^n$ and $f \in C^0(\Omega)$. Suppose that f is analytic on $\Omega \setminus f^{-1}(0)$. Then f is analytic on Ω .

Proof. One possible proof of this theorem depends on a reduction to the one-variable case using Hartog's theorem (Remark 1, §1). The one-variable proof uses the theory of harmonic and subharmonic functions. We refer the reader to R. Narasimhan [2] for more details. An alternative elementary proof which avoids the use of Hartogs theorem and makes minimal use of subharmonic functions is given in Whitney [1]. $\square$

Remark. We shall not make any use of Rado's theorem in these notes. Rado's theorem does, however, have important applications, notably to the theory of biholomorphic maps and automorphisms. By way of example, we mention Osgood's theorem: If $f : \Omega \subset \mathbb{C}^n \rightarrow \mathbb{C}^n$ is an injective holomorphic map then $f(\Omega)$ is open and f is biholomorphic onto its image (The example $f(x) = x^3$, shows that this result is not true for real analytic maps). For a proof of Osgood's theorem and further examples and references, we refer to R. Narasimhan [2].

We conclude this section by stating an important result on the singularities of analytic functions due to Hartog's. Suppose that Z is a subset of the domain Ω in $\mathbb{C}^n$ and f is holomorphic on $\Omega \setminus Z$. We say that f is *singular* at $z \in Z$ if there is no holomorphic function defined on a neighbourhood U of z in Ω whose restriction to $U \setminus Z$ is $f|_{U \setminus Z}$.

Theorem 2.2.5. (Hartogs). Let Ω be a domain in $\mathbb{C}^{n-1}$ and suppose $\phi : \Omega \rightarrow D_R(0)$ is a map. Let $\Sigma = \{(z, t) \in \Omega \times D_R(0) : \phi(z) = t\}$ and suppose $f \in A(\Omega \times D_R(0) \setminus \Sigma)$. If every point of Σ is singular for f then ϕ is holomorphic. In particular, Σ is an analytic subset of $\Omega \times D_R(0)$.

Proof. A proof, depending on the theory of subharmonic functions, may be found in R. Narasimhan [2]. $\square$

As a straightforward corollary of Theorem 2.2.5 we have

Corollary 2.2.6. Let Σ be a (real) k -dimensional submanifold of the domain Ω in $\mathbb{C}^n$. Then

1. If $k < 2n - 2$, every analytic function on $\Omega \setminus \Sigma$ extends to Ω .
2. If $k = 2n - 2$ and there do not exist any points $x \in \Sigma$ which have a neighbourhood U such that $U \cap \Sigma$ is an analytic subset of U , then every analytic function on $\Omega \setminus \Sigma$ extends to Ω .

Exercises.

1. Let A be a subset of the domain Ω in $\mathbb{C}^n$. We say that A is *thin* if for every $z \in \Omega$, we may find a polydisc $D(z; r)$ and $f \in A(D(z; r))$, not identically zero, such that $A \cap U \subset f^{-1}(0)$. Show that if A is a thin subset of Ω and f is analytic on $\Omega \setminus A$ and locally bounded on Ω , then f extends uniquely to an analytic function on Ω .
2. Prove part 1 of Corollary 2.2.6 directly without using Theorem 2.2.5 (Use Cauchy's integral formula).

3 Extension of analytic functions

Thus far our development of the theory of analytic functions of more than one complex variable has paralleled the one-variable theory very closely. In this section, we start investigating phenomena peculiar to analytic functions of two or more complex variables.

Examples.

1. Let f be an analytic function on $\mathbb{C}^n \setminus \{0\}$, $n > 1$. Then f extends analytically to $\mathbb{C}^n$. Indeed, for $(z_1, \dots, z_n) \in \mathbb{C}^{n-1} \times D_1(0)$ we may define

$$F(z_1, \dots, z_n) = (2\pi i)^{-1} \int_{|\zeta|=1} f(z_1, \dots, z_{n-1}, \zeta) / (\zeta - z_n) d\zeta.$$

Certainly $F \in A(\mathbb{C}^{n-1} \times D_1(0))$ and by Cauchy's integral formula $F(z_1, \dots, z_n) = f(z_1, \dots, z_n)$ provided $(z_1, \dots, z_{n-1}) \neq 0$. Hence, by uniqueness of analytic continuation, $F = f$ on $(\mathbb{C}^{n-1} \times D_1(0)) \setminus \{0\}$. Setting $F = f$ outside $\mathbb{C}^{n-1} \times D_1(0)$ we see that F is the required analytic extension of f to $\mathbb{C}^n$. This result is definitely false if $n = 1$ (take $f(z) = z^{-1}$!) and shows that Corollary 1.3.5 fails for arbitrary domains in $\mathbb{C}^n$, $n > 1$. That is, an arbitrary open subset of $\mathbb{C}^n$ will not generally be the domain of existence of an analytic function if $n > 1$.

2. Let D be an open polydisc in $\mathbb{C}^n$ and suppose that K is a compact subset of D such that $D \setminus K$ is connected. Then every analytic function on $D \setminus K$ extends uniquely to an analytic function on D . The proof uses the same idea as in example 1 and we leave details to the reader. Notice that in this example K may have interior points. Later in this section we prove the far more general theorem of Hartogs: If K is a compact subset of $\Omega \subset \mathbb{C}^n$, $n > 1$, and $\Omega \setminus K$ is connected, then every analytic function on $\Omega \setminus K$ extends uniquely to Ω .

3. Let $D = \{(z_1, \dots, z_n) : |z_i| < 1, 1 \leq i \leq n\}$ and

$$P = \{(z_1, \dots, z_n) \in D : |z_i| < \frac{1}{2}, 1 \leq i \leq n-1 \text{ or } |z_n| > \frac{1}{2}\}.$$

The real parts of D and P are pictured in Figure I.2.

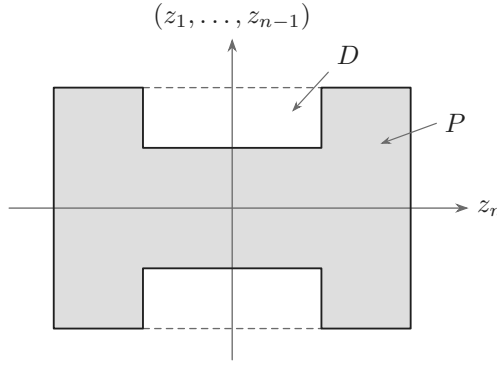


Figure I.2

We claim that every analytic function on P extends analytically to D . Our proof is similar to that of example 1. Suppose $f \in A(P)$, we define

$$F(z_1, \dots, z_n) = (2\pi i)^{-1} \int_{|\zeta|=3/4} f(z_1, \dots, z_{n-1}, \zeta) / (\zeta - z_n) d\zeta,$$

where $|z_i| < 1$, $1 \leq i \leq n-1$ and $|z_n| < 3/4$. Certainly F is analytic and if $|z_1|, \dots, |z_{n-1}| < \frac{1}{2}$ we see that $F(z_1, \dots, z_n) = f(z_1, \dots, z_n)$ by Cauchy's integral formula. It now follows by uniqueness of analytic continuation that $F = f$ wherever both are defined and setting $F = f$ for $1 > |z_n| \geq 3/4$ we see that F is the required analytic extension of f to D . The pair (D, P) is called a *Euclidean Hartogs figure*.

4. Suppose (D, P) is as in example 3 and $g : D \rightarrow \mathbb{C}^n$ is biholomorphic onto the image of g . Set $\tilde{D} = g(D)$, $\tilde{P} = g(P)$. We claim that every analytic function on $\tilde{P}$ extends uniquely to an analytic function on $\tilde{D}$. Indeed, if $f \in A(\tilde{P})$, $gf \in A(P)$. Now fg extends by example 3 to $F \in A(D)$. Clearly $Fg^{-1} \in A(\tilde{D})$ is the required analytic extension of f to $\tilde{D}$. The pair $(\tilde{D}, \tilde{P})$ is called a *generalised Hartogs figure*. As we shall see later in this chapter we can use generalised Hartogs figures as a test to determine whether or not every analytic function on a given domain can be extended to a larger domain.

The remainder of this section is devoted to a proving the Theorem of Hartogs referred to in example 2 above. The proof follows Ehrenpreis [1] and Hörmander [1] and uses an existence theorem for the $\partial/\partial\bar{z}_j$ operators.

Suppose $f_j \in C^\infty(\mathbb{C}^n)$, $j = 1, \dots, n$ and that we wish to solve the system of partial differential equations

$$\partial u / \partial \bar{z}_j = f_j, \quad 1 \leq j \leq n.$$

We first remark that this system is overdetermined in that for solvability the additional conditions

$$\partial f_i / \partial \bar{z}_j = \partial f_j / \partial \bar{z}_i, \quad 1 \leq i, j \leq n$$

obviously have to be satisfied.

Theorem 2.3.1. Let $f_j \in C_c^\infty(\mathbb{C}^n)$, $1 \leq j \leq n$ and suppose that $\partial f_i / \partial \bar{z}_j = \partial f_j / \partial \bar{z}_i$, $1 \leq i, j \leq n$. Then the system of equations

$$\partial u / \partial \bar{z}_j = f_j$$

has a solution $u \in C_c^\infty(\mathbb{C}^n)$ provided that $n > 1$.

Proof. Define

$$u(z) = (2\pi i)^{-1} \int_{\mathbb{C}} f_1(\zeta, z_2, \dots, z_n) / (\zeta - z_1) d\zeta d\bar{\zeta}.$$

Changing variables we see that

$$u(z) = -(2\pi i)^{-1} \int_{\mathbb{C}} f_1(z_1 - \zeta, z_2, \dots, z_n) \zeta^{-1} d\zeta d\bar{\zeta}$$

and so $u \in C^\infty(\mathbb{C}^n)$. Furthermore, since f_1 has compact support, $u(z) = 0$ provided that $|z_2| + \dots + |z_n|$ is sufficiently large.

Theorem A1.6 implies that $\partial u / \partial \bar{z}_1 = f_1$. Differentiating under the integral sign with respect to $\bar{z}_j$ and using the relation $\partial f_j / \partial \bar{z}_1 = \partial f_1 / \partial \bar{z}_j$, we obtain

$$\begin{aligned} \partial u / \partial \bar{z}_j &= (2\pi i)^{-1} \int_{\mathbb{C}} (\zeta - z_1)^{-1} \partial f_j / \partial \bar{\zeta}(\zeta, z_2, \dots, z_n) d\zeta d\bar{\zeta} \\ &= f_j, \text{ by Theorem A1.2.} \end{aligned}$$

Hence u is a solution of the given system of equations. Now let K denote the union of the supports of the f_j . Then $u \in A(\mathbb{C}^n \setminus K)$ and u is zero for $|z_2| + \dots + |z_n|$ sufficiently large. Hence, by uniqueness of analytic continuation, u is zero on the unbounded component of $\mathbb{C}^n \setminus K$ and so u has compact support. $\square$

Remarks.

1. As we remarked in the appendix to Chapter 1, Theorem 2.3.1 is false for $n = 1$.
2. Later, in Chapter 5, we shall rewrite the system of partial differential equations occurring in Theorem 2.3.1 in the framework of differential forms.

Theorem 2.3.2. (Hartog's theorem). Let Ω be a domain in $\mathbb{C}^n$, $n > 1$, and K be a compact subset of Ω such that $\Omega \setminus K$ is connected. Then every analytic function on $\Omega \setminus K$ extends uniquely to an analytic function on Ω .

Proof. Choose $\theta \in C_c^\infty(\Omega)$ so that $\theta \equiv 1$ on K . Define $f_0 \in C^\infty(\Omega)$ by setting

$$f_0|_K = 0; \quad f_0|_{\Omega \setminus K} = (1 - \theta)f.$$

We shall construct $v \in C^\infty(\mathbb{C}^n)$ so that $f_0 + v$ is the required continuation of f . First notice that $f_0 + v$ will be analytic iff v satisfies

$$\begin{aligned} \partial v / \partial \bar{z}_j &= -\partial f_0 / \partial \bar{z}_j, \quad j = 1, \dots, n \\ &= -\partial \theta / \partial \bar{z}_j f. \end{aligned}$$

Now $-\partial \theta / \partial \bar{z}_j f \in C_c^\infty(\mathbb{C}^n)$ and so we may apply Theorem 2.3.1 to find $v \in C_c^\infty(\mathbb{C}^n)$ such that $f_0 + v \in A(\Omega)$. Now observe that since v has compact support, v vanishes on the unbounded component of the complement of the support of θ —uniqueness of analytic continuation. Since $\text{supp}(\theta) \subset \Omega$, there exists an open set in $\Omega \setminus K$ where $v = 0$ and so $f = f_0$. Since $\Omega \setminus K$ is connected, $f = f_0 + v$ on $\Omega \setminus K$ and so $f_0 + v$ is the required analytic extension of f . $\square$

Remark. An alternative proof of Hartog's theorem, based on the Bochner–Martinelli integral formula, may be found in Bochner [1]. See also Harvey [2, page 355].

Corollary 2.3.3. Let Ω be a domain in $\mathbb{C}^n$, $n > 1$, and $f \in A(\Omega)$. Then the zero set $Z(f)$ of f is never a compact subset of Ω .

Proof. If $Z(f)$ were compact, $1/f \in A(\Omega \setminus Z(f))$ would extend analytically to Ω by Theorem 2.3.2. $\square$

The corollary emphasises an important difference between the one- and several-variable theory of analytic functions. The zero set of an analytic function of more than one variable always propagates to the boundary of the domain on which the function is defined. It is this essential non-compactness that makes the study of the zero and pole sets so much harder than for functions of one complex variable.

Exercise. Let Ω, ω be domains in $\mathbb{C}^n, \mathbb{C}$ respectively. A map $f : \Omega \rightarrow \omega$ is said to be *proper* if $f^{-1}(K)$ is compact whenever $K \subset \omega$ is compact. Prove that if $n > 1$, there are no proper holomorphic maps of Ω into ω (Hint: Let $z_0 \in f(\Omega)$ and consider $g(z) = (f(z) - z_0)^{-1} \in A(\Omega \setminus f^{-1}(z_0))$).

4 Domains of Holomorphy

In §3 we gave a large class of examples of domains in $\mathbb{C}^n, n > 1$, that possessed the property that every analytic function on them extended to a larger domain. In this section we wish to look at domains where this phenomenon does not happen and the generalisation of Corollary 1.3.5 is true.

Throughout this section Ω will denote a domain in $\mathbb{C}^n$.

Definition 2.4.1. A domain in $\mathbb{C}^n$ is called a *domain of holomorphy* if whenever we are given open connected subsets $U \subset V \subset \mathbb{C}^n$, with $U \subset \Omega$, such that $f|_U$ extends analytically to V for all $f \in A(\Omega)$, then $V \subset \Omega$.

For Ω to be a domain of holomorphy we require that we cannot extend every analytic function on Ω locally across any point of the boundary of Ω . We frame the definition in this rather complicated fashion to exclude the following type of domain:

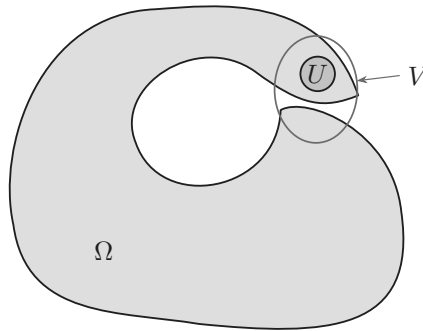


Figure I.3

Referring to the figure, suppose that $f|_U$ extends to $F \in A(V)$ for all $f \in A(\Omega)$. Then Ω will not be a domain of holomorphy. Notice though that $F|_{\Omega \cap V}$ need not equal $f|_{\Omega \cap V}$ —we give an explicit example below.

Examples.

1. $\mathbb{C}^n$ is a domain of holomorphy, $n \geq 1$.
2. If K is a compact subset of $\Omega \subset \mathbb{C}^n, n > 1$, such that $\Omega \setminus K$ is connected then $\Omega \setminus K$ is not a domain of holomorphy (Theorem 2.3.2).
3. Every domain Ω in $\mathbb{C}$ is a domain of holomorphy. This is trivial: Given $a \in \partial\Omega$, define $f(z) = (z - a)^{-1} \in A(\Omega)$ and observe that f cannot extend to any open neighbourhood of a .

4. A necessary condition for Ω to be a domain of holomorphy is that if $(\tilde{D}, \tilde{P})$ is any generalised Hartog's figure (Example 4, §3) with $\tilde{P} \subset \Omega$, then $\tilde{D} \subset \Omega$. Actually this condition turns out to be sufficient (see the discussion in §5).
5. Given open domains $U_i \subset \mathbb{C}$, $1 \leq i \leq n$, $U = \prod_{i=1}^n U_i$ is a domain of holomorphy. Indeed, $\partial U = \bigcup_i U_1 \times \cdots \times \partial U_i \times \cdots \times U_n$ and so if $a = (a_1, \dots, a_n) \in \partial U$, we must have at least one $a_i \in \partial U_i$. Now define $f(z_1, \dots, z_n) = (z_i - a_i)^{-1}$. Clearly f does not extend to any open neighbourhood of a . Alternatively, choose $f_i \in A(U_i)$ satisfying the conditions of Corollary 1.3.5. Define $f(z_1, \dots, z_n) = \sum_{i=1}^n f_i(z_i)$. In this case we see that U is the “domain of existence” for f .
6. Let $V = \{z \in D(0;1) : z_1 = \cdots = z_{n-1} = 0, \operatorname{Im}(z_n) = 0, \operatorname{Re}(z_n) \geq 0\}$ and set $\Omega = D(0;1) \setminus V$. Every analytic function on Ω extends analytically to $D(0;1)$ (Exercise 2, §2 or use Laurent series at 0). On the other hand Ω is homeomorphic to $D(0;1)$ and so we see that the property of being a domain of holomorphy is not a topological invariant.
7. We now present an example to show that the phenomenon alluded to after definition 2.4.1 can occur. Our example is in $\mathbb{C}^2$, though it is easily generalised to $\mathbb{C}^n$, $n > 2$. Let $A \subset \mathbb{C}^2$ be the product of the “cut” annulus $\{z \in \mathbb{C} : \frac{1}{2} < |z - 2| < 3/2\} \setminus \{z \in \mathbb{C} : |z| = \frac{1}{2} \text{ and } \arg(z) \in (0, \pi)\}$ with the disc $|w| < \frac{1}{4}$. We define Ω to be the union of A with $P = \{(z, w) \in D(0;1) : |z| < \frac{1}{2} \text{ or } 1 > |w| > \frac{1}{2}\}$. Let $D = D(0;1)$.

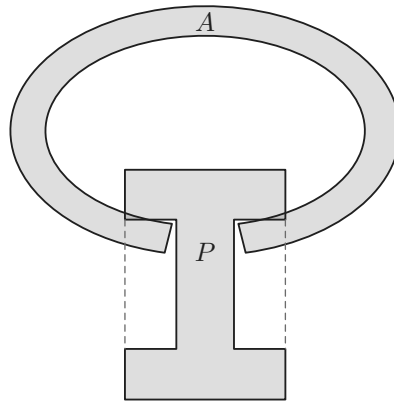


Figure I.4

Since (D, P) is a Euclidean Hartog's figure, $f|P$ extends analytically to $F \in A(D)$ for all $f \in A(\Omega)$. But in general $F|D \cap \Omega \neq f|D \cap \Omega$ as is seen by taking $f(z, w) = \sqrt{(z - 2)}$.

Example 5 suggests that a domain of holomorphy might be the domain of existence of an analytic function (note that the converse is trivially true). Much of the remainder of this section will be devoted to proving that every domain of holomorphy is the domain of existence of some analytic function. In the course of our proof we shall derive other important characterizations of domains of holomorphy.

Example 3 and 5 also suggest that if Ω is a domain of holomorphy, $z \in \partial\Omega$ and $(z_n) \subset \Omega$ converges to z , then there exists $f \in A(\Omega)$ which is unbounded on the sequence (z_n) . We make a formal definition.

Definition 2.4.2. We say that the domain Ω possesses property (S) if given any sequence $(z_n) \subset \Omega$ which converges to a point $z \in \partial\Omega$, there exists $f \in A(\Omega)$ which is unbounded on the sequence (z_n) .

Lemma 2.4.3. If the domain Ω possesses property (S), Ω is a domain of holomorphy.

Proof. Suppose Ω is not a domain of holomorphy. Then there exists $z \in \partial\Omega$ and open connected sets $U \subset V \subset \mathbb{C}^n$ such that $U \subset \Omega$, $z \in V$ and $f|_U$ extends to $F \in A(V)$ for every $f \in A(\Omega)$. Now let $(z_n) \subset \Omega \cap V$ converge to z . We see immediately that F is bounded on (z_n) . Therefore f is bounded on (z_n) for all $f \in A(\Omega)$ and Ω does not possess property (S). $\square$

Examples.

7. The Euclidean disc $E = \{z \in \mathbb{C}^n : \sum_{i=1}^n |z_i|^2 < r^2\}$ is a domain of holomorphy. We show E possesses property (S). Suppose $a = (a_1, \dots, a_n) \in \partial E$. Define $f(z) = (r^2 - \langle z, a \rangle)^{-1}$, where $\langle \cdot, \cdot \rangle$ denotes the standard Hermitian inner product on $\mathbb{C}^n$. Clearly $f \in A(E)$ and f is not bounded on any sequence of points of E converging to a .
8. Let $f_j \in A(\mathbb{C}^n)$, $1 \leq j \leq q$. The *analytic polyhedron* $P = \{z \in \mathbb{C}^n : |f_j(z)| < 1, 1 \leq j \leq q\}$ possesses property (S) and is a domain of holomorphy. Indeed, if $a \in \partial P$, $|f_i(a)| = 1$ for some i . Now define $F(z) = (f_i(z) - f_i(a))^{-1} \in A(P)$. Clearly F is not bounded on any sequence of points of P converging to a .

It turns out that in some ways domains of holomorphy are the complex analogue of convex sets. The next definition reflects these convexity properties particularly well. Before giving the definition we recall that if K is a compact subset of $\Omega \subset \mathbb{C}^n$ and $f : \Omega \rightarrow \mathbb{C}$, then $\|f\|_K$ is, by definition, $\sup_{z \in K} |f(z)|$.

Definition 2.4.4. A domain Ω is said to be *holomorphically convex* if given any compact subset $K \subset \Omega$ the set

$$\widehat{K} = \{z \in \Omega : |f(z)| \leq \|f\|_K \text{ for all } f \in A(\Omega)\}$$

is compact.

Remark. We call $\widehat{K}$ the $A(\Omega)$ -hull of K . If $K = \widehat{K}$, we say K is $A(\Omega)$ -convex.

Examples.

9. Every domain Ω in $\mathbb{C}$ is holomorphically convex. In fact it is easy to see that $d(K, \partial\Omega) = d(\widehat{K}, \partial\Omega)$ for every compact subset K of Ω : Just consider functions $f(z) = (z - \zeta)^{-1} \in A(\Omega)$, where $\zeta \in \partial\Omega$.
10. The domain $\mathbb{C}^n \setminus \{0\}$, $n > 1$, is not holomorphically convex. Indeed suppose $K = \{z \in \mathbb{C}^n : |z| = 1\}$ (the boundary of the unit polydisc). Then $\widehat{K} = \{z : 0 < |z| \leq 1\}$ (the punctured unit polydisc). To see this we notice that by Theorem 2.3.2 any $f \in A(\mathbb{C}^n \setminus \{0\})$ extends to $F \in A(\mathbb{C}^n)$. By the Maximum Principle (Theorem 2.1.12), the maximum value of F on $\{z : |z| \leq 1\}$ is taken on its boundary, K . Since F is an extension of f the same is true for f . This implies $\widehat{K} \supseteq \{z : 0 < |z| \leq 1\}$. The reverse inclusion is trivial. Notice that if $n = 1$, $\widehat{K} = K$ as is seen by taking $f(z) = z^{-1}$.

Before stating the next lemma we wish to review a few elementary facts about convex subsets of $\mathbb{R}^n$. Suppose $X \subset \mathbb{R}^n$. We say that X is *convex* if, given $x, y \in X$, $tx + (1 - t)y \in X$ for all $t \in [0, 1]$. Now suppose X is bounded. The (closed) *convex hull*, X_c , of X is defined to be

$$X_c = \{u \in \mathbb{R}^n : \phi(u) \leq \sup_{x \in X} \phi(x) \text{ for all } \phi \in \mathbb{R}^{n'}\}.$$

The following properties of the convex hull are easily verified:

$$\begin{aligned} X_c &\text{ is closed, convex and bounded} \\ X_{cc} &= X_c \\ X_c \supset X &\text{ and } X_c \text{ is the smallest closed convex set containing } X \\ X_c = X &\text{ if and only if } X \text{ is closed and convex.} \end{aligned}$$

Lemma 2.4.5. Let K be a compact subset of the domain Ω in $\mathbb{C}^n$. Then

1. $\widehat{K} \supset K$.
2. $\widehat{K}$ is a closed subset of Ω .
3. $\widehat{\widehat{K}} = \widehat{K}$.
4. $\widehat{K} \subset K_c$. In particular, $\widehat{K}$ is bounded.

Proof. 1, 2 and 3 are trivial. Let us prove 4.

$$K_c = \{u \in \mathbb{C}^n : \phi(u) \leq \sup_{z \in K} \phi(z) \text{ for all } \phi \in L_{\mathbb{R}}(\mathbb{C}^n, \mathbb{R})\}.$$

Given $\phi \in L_{\mathbb{R}}(\mathbb{C}^n, \mathbb{R})$, we may write

$$\phi(z_1, \dots, z_n) = \sum_{i=1}^n a_i z_i + \sum_{i=1}^n b_i \bar{z}_i$$

where $a_i, b_i \in \mathbb{C}$. Since ϕ is real valued we must have $b_i = \bar{a}_i$, $i = 1, \dots, n$, and so

$$\phi(z_1, \dots, z_n) = 2 \operatorname{Re} \sum_{i=1}^n a_i z_i.$$

Define $f(z) = \exp(2 \sum_{i=1}^n a_i z_i) \in A(\Omega)$. If $z \in \widehat{K}$,

$$|f(z)| \leq \|f\|_K.$$

That is,

$$|\exp \phi(z)| \leq \|\exp \phi\|_K, \text{ since } |f(z)| = |\exp \phi(z)|.$$

This implies $\phi(z) \leq \sup_K \phi$, since ϕ is real. Hence $z \in K_c$. □

Remark. The reader should note the close analogy between properties of the $A(\Omega)$ -hull and the convex hull. It should also be noted that an open subset Ω of $\mathbb{R}^n$ is convex if and only if K_c is a compact subset of Ω for every compact subset K of Ω . Furthermore, convexity is a local property of the boundary of Ω (Every disc centered on the boundary of Ω meets Ω in a convex set if Ω is convex and this property characterises the convexity of Ω). We shall see in §5 that we can also formulate holomorphic convexity in terms of local properties of the boundary.

Theorem 2.4.6. A domain Ω in $\mathbb{C}^n$ is holomorphically convex if and only if it possesses property (S). In particular, if Ω is holomorphically convex it is a domain of holomorphy.

Proof. We start by showing that (S) implies holomorphic convexity. Suppose Ω possesses property (S). If Ω is not holomorphically convex there exists a compact subset K of Ω such that $\widehat{K}$ is not compact. Since $\widehat{K}$ is bounded we can therefore find a sequence $(z_n) \subset \widehat{K}$ which converges to some point of $\partial\Omega$. Now $|f(z_n)| \leq \|f\|_K$ for all $f \in A(\Omega)$ by definition of $\widehat{K}$. Hence every $f \in A(\Omega)$ must be bounded on (z_n) . This is contrary to our assumption that Ω possesses property (S) and so Ω is holomorphically convex.

Suppose Ω is holomorphically convex. Let $(z_n) \subset \Omega$ be a sequence converging to some point of $\partial\Omega$. It is sufficient to construct $f \in A(\Omega)$ which is unbounded on (z_n) .

First we choose a sequence (K_n) of compact subsets of Ω satisfying

1. $K_n \subset K_{n+1}$, $n \geq 1$.
2. $\bigcup_n K_n = \Omega$.
3. $K_n = \widehat{K}_n$, $n \geq 1$.
4. Every point of Ω is an interior point of some K_n .

To see that such sequences exist, we first construct a sequence K'_n of compact subsets of Ω satisfying 1, 2 and 4 just as in the proof of Theorem A1.8. We then define $K_n = \widehat{K}'_n$ and note that 1 continues to hold since the operation of forming the $A(\Omega)$ -hull clearly preserves inclusions. Notice that condition 4 implies that every compact subset of Ω is contained in K_n for n sufficiently large.

Choose an infinite subsequence (x_k) of (z_n) which possesses the property that $x_k \in K_{n(k)+1} \setminus K_{n(k)}$, $k \geq 1$, where the sequence $(n(k))$ is strictly increasing. We leave the construction of such a sequence as an easy exercise for the reader (note that $(K_{n+1} - K_n) \cap (z_n)$ is always finite). It is clearly enough to construct $f \in A(\Omega)$ which is unbounded on (x_k) . For $k \geq 1$, let $L_k = K_{n(k)}$, $\|g\|_k = \|g\|_{L_k}$ and observe that the sequence (L_k) satisfies conditions 1 to 4 above.

We construct inductively a sequence $f_k \in A(\Omega)$ satisfying

$$f_k(x_k) = k + 1 + \sum_{j=1}^{k-1} |f_j(x_k)| \quad \text{and} \quad \|f_k\|_k \leq 2^{-k+2}. \quad (*)$$

Take $f_1 \equiv 2$. Suppose $f_1, \dots, f_{k-1}$ are constructed. Since $x_k \notin L_k$, there exists $g \in A(\Omega)$ such that $|g(x_k)| > \|g\|_k$. Dividing by $g(x_k)$ we may assume

$$1 = g(x_k) > \|g\|_k.$$

Choose $p(k)$ so large that

$$\|g^{p(k)}\|_k \leq 2^{-k+2} / \left(k + 1 + \sum_{j=1}^{k-1} |f_j(x_k)| \right).$$

Set $f_k = (k + 1 + \sum_{j=1}^{k-1} |f_j(x_k)|)g^{p(k)}$. The inductive step is completed. We now define

$$f = \sum_{k=1}^{\infty} f_k.$$

Conditions (*) imply that the sum converges uniformly on L_k for $k \geq 1$ (compare with the proof of Theorem A1.8) and also that $|f(x_k)| \geq k - 1$, $k \geq 1$. Since every compact subset of Ω is contained in L_k for k large enough it follows by Corollary 2.1.8 that $f \in A(\Omega)$. Since f is unbounded on (x_k) the proof is complete. $\square$

Remarks.

1. The technique used to construct the function in the second part of the proof is very effective and we use it again shortly. Holomorphic convexity is well adapted to the construction of analytic functions.
2. The sequence (K_n) of compact subsets of Ω constructed in the proof of Theorem 2.4.6 is called a *normal exhaustion* of Ω .

Example 11. Any open convex subset of $\mathbb{C}^n$ is a domain of holomorphy. Indeed, since $\widehat{K} \subset K_c$, any open convex subset of $\mathbb{C}^n$ must be holomorphically convex. Of course, using the technique of the proof of Lemma 2.4.5, it can easily be proved directly that an open convex subset of $\mathbb{C}^n$ is a domain of holomorphy.

Remark. Holomorphic convexity is much weaker than convexity: Any open subset of $\mathbb{C}$ is holomorphically convex as are products of open subsets of $\mathbb{C}$.

Definition 2.4.7. Let $f \in A(\Omega)$. We say that Ω is the *domain of existence* of f , if f cannot be analytically continued across any point of the boundary of Ω .

Remarks.

1. Every domain of existence is trivially a domain of holomorphy.
2. If Ω is a domain in $\mathbb{C}$ then Ω is a domain of existence by Corollary 1.3.5.

Theorem 2.4.8. Every holomorphically convex domain Ω in $\mathbb{C}^n$ is the domain of existence of some analytic function on Ω .

Proof. Our proof is rather similar to that of Theorem 2.4.6.

Let (K_n) be a normal exhaustion of Ω and M be a discrete subset of Ω such that $\bar{M} \supset \partial\Omega$ (see the proof of Corollary 1.3.5 for the construction of M). For each $n \geq 1$, there exist only finitely many points of M in $K_{n+1} \setminus K_n$, say $x_1, \dots, x_p$. As in the proof of Theorem 2.4.6 there exist $g_1, \dots, g_p \in A(\Omega)$ such that

$$1 = g_1(x_1) = \dots = g_p(x_p) > \|g_1\|_{K_n}, \dots, \|g_p\|_{K_n}.$$

By perturbing each g_j by a term $\theta(z - x_j)$, where $\theta \in \mathbb{C}^{n*}$ is small, it is clear that we can assume that $|g_j(x_k)| \neq 1$, $j \neq k$.

Set $a_{jk}(z) = (g_j(z) - g_j(x_k))/(1 - g_j(x_k))$, $j \neq k$. Taking sufficiently high powers of the g_j 's, we may further require that

$$\|g_j\|_{K_n} \leq 2^{-n-p}/p \quad \text{and} \quad \|a_{jk}\|_{K_n} \leq 2.$$

We now define

$$f_n(z) = \sum_{j=1}^p \left(\prod_{k \neq j} a_{jk}(z) \right) g_j(z), \quad z \in \Omega.$$

Our estimates imply that

$$f_n(x_i) = 1, \quad 1 \leq i \leq p$$

$$\|f_n\|_{K_n} \leq 2^{-n}.$$

Define

$$f = \prod_{n=1}^{\infty} (1 - f_n)^n.$$

Since $\sum_{n=1}^{\infty} n2^{-n}$ is convergent, the infinite product converges uniformly on compact subsets of Ω and so $f \in A(\Omega)$. The convergence of the infinite product together with the fact that no f_n is identically equal to 1 implies that f is not identically zero.

Let $(x_k) \subset M$ be a sequence converging to $x \in \partial\Omega$. If $x_k \in K_{n(k)+1} \setminus K_{n(k)}$, f and its first $n(k)$ derivatives vanish at x_k . Since $n(k) \rightarrow \infty$ as $k \rightarrow \infty$, we see that any analytic extension of f to a neighbourhood of x would have all its derivatives vanishing at x . But this implies, by uniqueness of analytic continuation, that $f \equiv 0$. Hence f does not extend across any point of $\partial\Omega$ and Ω is the domain of existence of f . $\square$

Remark. Using the method of proof of Theorems 2.4.6, 2.4.8 it is not hard to show that if M is an infinite discrete subset of a holomorphically convex domain Ω , then there exists $f \in A(\Omega)$ which is unbounded on *every* infinite subset of M . Clearly this very strong form of Property (S) implies Theorem 2.4.8 as well as the non-trivial part of Theorem 2.4.6.

Let us summarise what we have proved so far. We have shown that for a domain Ω in $\mathbb{C}^n$, Property (S) is equivalent to holomorphic convexity. We have also proved that if Ω is holomorphically convex then Ω is a domain of existence which in turn implies that Ω is a domain of holomorphy. We next show that these four properties are equivalent by proving that every domain of holomorphy is holomorphically convex.

First some notation. If $K \subset \Omega$ is compact, we let

$$\begin{aligned} d_{\Omega}(K) &= d(\partial\Omega, K) \\ &= \inf\{|z - \zeta| : z \in \partial\Omega, \zeta \in K\}. \end{aligned}$$

Provided that $\Omega \neq \mathbb{C}^n$, $d_{\Omega}(K) < \infty$. We remark that if $d(z, \partial\Omega) = r$ and $z \in \Omega$, then $D(z; r) \subset \Omega$ and indeed is the largest open polydisc $D(z; s)$ contained in Ω and centered at z .

Proposition 2.4.9. Let K be a compact subset of $\Omega \subset \mathbb{C}^n$ and $z_0 \in \widehat{K}$. If $f \in A(\Omega)$, the power series for f at z_0 ,

$$f(z) = \sum_m \frac{1}{m!} \partial^m f(z_0) (z - z_0)^m,$$

converges on $D(z_0; d_{\Omega}(K))$.

Proof. Choose r , $0 < r < d_{\Omega}(K)$ and define $K_r = \bigcup_{z \in K} \bar{D}(z; r)$. K_r is certainly compact and contained in Ω . Let $M = \|f\|_{K_r}$. By Corollary 2.1.8 applied to $\bar{D}(z; r)$ we have

$$|\partial^m f(z)| \leq M m! r^{-|m|}$$

for $z \in K$ ($|m| = m_1 + \dots + m_n$). Hence

$$\|\partial^m f\|_K \leq M m! r^{-|m|}.$$

Since $\partial^m f \in A(\Omega)$ we have, by definition of $\widehat{K}$,

$$\|\partial^m f\|_{\widehat{K}} \leq M m! r^{-|m|}.$$

Since $z_0 \in \widehat{K}$, this estimate implies

$$|\partial^m f(z_0)| \leq M m! r^{-|m|}$$

and so the series $\sum \frac{1}{m!} \partial^m f(z_0) (z - z_0)^m$ converges for $|z - z_0| < r$. Since this is true for all $r < d_{\Omega}(K)$, the Proposition follows. $\square$

Theorem 2.4.10. (Cartan–Thullen). A domain of holomorphy is holomorphically convex.

Proof. It is clearly enough to prove that for any compact subset K of a domain of holomorphy Ω , $d_\Omega(K) = d_\Omega(\widehat{K})$. Obviously $d_\Omega(K) \geq d_\Omega(\widehat{K})$. Suppose that for some compact subset K of Ω , $d_\Omega(K) > d_\Omega(\widehat{K})$. Pick $z_0 \in \widehat{K}$ such that $d(z_0, \partial\Omega) < d_\Omega(K)$. By Proposition 2.4.9, $\sum \frac{1}{m!} \partial^m f(z_0)(z - z_0)^m$ converges on $D(z_0; d_\Omega(K))$. But $D(z_0; d_\Omega(K)) \not\subset \Omega$ (see remarks immediately preceding Proposition 2.4.9). Hence we have constructed an analytic extension of every $f \in A(\Omega)$ to some open set not contained in Ω . Contradiction. Hence Ω is holomorphically convex. $\square$

Examples.

12. Suppose $\Omega, \Omega' \subset \mathbb{C}^n$ and $f : \Omega \rightarrow \Omega'$ is biholomorphic. If one of Ω, Ω' is a domain of holomorphy so is the other. This follows since holomorphic convexity is obviously preserved by biholomorphic maps. Notice that it is *not* obvious from the definitions that domains of holomorphy or existence are preserved by biholomorphic maps.
13. Suppose that $\Omega \subset \mathbb{C}^n, \Omega' \subset \mathbb{C}^m$ are domains of holomorphy and $f : \Omega \rightarrow \Omega'$ is analytic. Then $f^{-1}(\Omega')$ is a domain of holomorphy. We show that $f^{-1}(\Omega')$ possesses property (S). Let (z_n) be a sequence of points of $f^{-1}(\Omega')$ converging to some point $z \in \partial f^{-1}(\Omega')$. If $z \in \partial\Omega$, then there exists $g \in A(\Omega)$ which is unbounded on (z_n) . Restricting to $f^{-1}(\Omega')$, we have found an analytic function on $f^{-1}(\Omega')$ unbounded on (z_n) . If $z \notin \partial\Omega$, then $(f(z_n))$ converges to $f(z) \in \mathbb{C}^m$. Clearly, $f(z) \in \partial\Omega'$, and so there exists $g \in A(\Omega')$ which is unbounded on $(f(z_n))$. But then $gf \in A(f^{-1}(\Omega'))$ is unbounded on (z_n) . We remark that we can weaken the conditions on Ω to require only that Ω possesses property (S) at all points of $\partial\Omega \cap \partial f^{-1}(\Omega')$.
14. Let $\Omega \subset \mathbb{C}^n$ be a domain of holomorphy and $f_1, \dots, f_q \in A(\Omega)$. Then $P = \{z \in \Omega : |f_j(z)| < 1, j = 1, \dots, q\}$ is a domain of holomorphy. Indeed, set $f = (f_1, \dots, f_q) : \Omega \rightarrow \mathbb{C}^q$. Then $P = f^{-1}(D(0; 1))$ and example 13 applies. We recall that P is an analytic polyhedron.
15. Let $\Omega_i \subset \mathbb{C}^n, i \in I$, be domains of holomorphy. Then $\Omega = \text{Interior}(\bigcap_{i \in I} \Omega_i)$ is a domain of holomorphy. We prove that Ω is holomorphically convex. Let K be a compact subset of Ω and $\widehat{K}$ and $\widehat{K}_i$ respectively denote the $A(\Omega)$ - and $A(\Omega_i)$ -hulls of K . Certainly $\widehat{K} \subset \widehat{K}_i, i \in I$. Hence, with the notation of Theorem 2.4.10, $d_\Omega(K) \leq d_{\Omega_i}(\widehat{K}), i \in I$. This being true for all $i \in I$, we must have $d_\Omega(\widehat{K}) \geq d_\Omega(K)$ and so $\widehat{K}$ is a compact subset of Ω .

Exercises.

1. Let Ω be a domain of holomorphy and M be an infinite discrete subset of Ω . Show that there exists an analytic function on Ω which is unbounded on every infinite subset of M (see the remarks following Theorem 2.4.8).
2. Let Ω be holomorphically convex. Show that there exist normal exhaustions (K_n) of Ω that satisfy the additional property $K_n \subset \overset{\circ}{K}_{n+1}, n \geq 1$.

5 Pseudoconvexity

The results of §4 show that there is a close parallel between the convexity of domains in $\mathbb{R}^n$ and holomorphic convexity of domains in $\mathbb{C}^n$. In this section we shall explore this analogy further. First we shall review some facts about convex domains in $\mathbb{R}^n$. A standard reference is Bonnesen and Fenchel [1].

Let Ω be a proper subdomain of $\mathbb{R}^n$ and suppose $\partial\Omega$ is a C^r submanifold of $\mathbb{R}^n$, $r \geq 1$. Using partitions of unity it is not hard to construct $\phi \in C^r(\mathbb{R}^n, \mathbb{R})$ satisfying

- A) $\Omega = \{x \in \mathbb{R}^n : \phi(x) < 0\}$
- B) $\partial\Omega = \{x \in \mathbb{R}^n : \phi(x) = 0\}$
- C) $d\phi \neq 0$ on $\partial\Omega$.

We call a map ϕ satisfying these conditions a (C^r -) defining function for Ω . The following technical lemma will be important in defining boundary invariants of Ω .

Lemma 2.5.1. Let Ω be a proper subdomain of $\mathbb{R}^n$ with C^2 boundary and suppose that ϕ and θ are defining functions for Ω . Then there exists $h \in C^1(\mathbb{R}^n, \mathbb{R})$ satisfying

- a) $\phi = h\theta$
- b) h is strictly positive
- c) h is C^2 off $\partial\Omega$
- d) Given $1 \leq i, j \leq n$, let

$$a_{ij}(x) = \begin{cases} d(x, \partial\Omega) \frac{\partial^2 h}{\partial x_i \partial x_j}(x), & x \notin \partial\Omega, \\ 0, & x \in \partial\Omega. \end{cases}$$

Then a_{ij} is continuous.

($d(x, \partial\Omega)$ is the distance from x to $\partial\Omega$, relative to some norm on $\mathbb{R}^n$).

Proof. Clearly we may define $h = \phi/\theta$ on $\mathbb{R}^n \setminus \partial\Omega$ and h is C^2 on $\mathbb{R}^n \setminus \partial\Omega$. The problem is to show that h extends as a positive C^1 function across $\partial\Omega$ satisfying property d). Fix $z \in \partial\Omega$. It is enough to find an open neighbourhood U of z in $\mathbb{R}^n$ and $h \in C^1(U, \mathbb{R})$ such that h satisfies properties a)–d) in U . Choose local coordinates on some open neighbourhood U of z so that z corresponds to zero, $\partial\Omega \cap U = \{x \in U : x_1 = 0\}$ and, in the new coordinates, U is convex. In the new coordinate system we have $\phi(0, x_2, \dots, x_n) = 0$ and $\frac{\partial\phi}{\partial x_1}(0, x_2, \dots, x_n) \neq 0$ for $(0, x_2, \dots, x_n) \in U$. Similarly for θ . Now

$$\begin{aligned} \phi(x_1, \dots, x_n) &= \int_0^1 \frac{d}{dt}(\phi(tx_1, \dots, x_n)) dt \\ &= \int_0^1 x_1 \frac{\partial\phi}{\partial x_1}(tx_1, \dots, x_n) dt \\ &= x_1 \tilde{\phi}(x_1, \dots, x_n). \end{aligned}$$

Since $\frac{\partial\phi}{\partial x_1}(0, x_2, \dots, x_n) \neq 0$, $\tilde{\phi}(0, x_2, \dots, x_n) \neq 0$ on $\partial\Omega \cap U$. Furthermore $\tilde{\phi}$ is C^1 on U . Similarly, $\theta(x_1, \dots, x_n) = x_1 \tilde{\theta}(x_1, \dots, x_n)$ and so we may extend h to a C^1 function on U by setting $h = \tilde{\phi}/\tilde{\theta}$.

For property d), it is enough to show that the function

$$A_{ij}(x) = \begin{cases} x_1 \frac{\partial^2 \tilde{\phi}}{\partial x_i \partial x_j}(x), & x \notin \partial\Omega \cap U, \\ 0, & x \in \partial\Omega \cap U \end{cases}$$

is continuous on U for $1 \leq i, j \leq n$. We shall prove this is so in case $i = j = 1$, leaving the remaining (easier) cases $i = 1, j \neq 1$ and $i, j \neq 1$ to the reader.

Since $\tilde{\phi}$ is C^2 provided $x_1 \neq 0$, we may differentiate the identity $\phi = x_1 \tilde{\phi}$ twice to obtain

$$\begin{aligned} \frac{\partial \phi}{\partial x_1} &= \tilde{\phi} + x_1 \frac{\partial \tilde{\phi}}{\partial x_1} \\ \frac{\partial^2 \phi}{\partial x_1^2} &= 2 \frac{\partial \tilde{\phi}}{\partial x_1} + x_1 \frac{\partial^2 \tilde{\phi}}{\partial x_1^2}, \quad x_1 \neq 0. \end{aligned}$$

Eliminating $\partial \tilde{\phi} / \partial x_1$ from the second relation we obtain

$$x_1 \tilde{\phi}_{11} = \frac{\frac{1}{2} x_1^2 \phi_{11} - x_1 \phi_1 + \phi}{\frac{1}{2} x_1^2}, \quad x_1 \neq 0, \quad (*)$$

where we have used the abbreviated notation ϕ_{11} and ϕ_1 for $\partial^2 \phi / \partial x_1^2$ and $\partial \phi / \partial x_1$ respectively.

By Taylor's theorem, there exists ξ , $0 < \xi < 1$, such that

$$\begin{aligned} \phi(x_1, x_2, \dots, x_n) &= \phi(0, x_2, \dots, x_n) + x_1 \phi_1(0, x_2, \dots, x_n) \\ &\quad + \frac{1}{2} x_1^2 \phi_{11}(\xi x_1, x_2, \dots, x_n) \\ &= x_1 \phi_1(0, x_2, \dots, x_n) + \frac{1}{2} x_1^2 \phi_{11}(\xi x_1, x_2, \dots, x_n). \end{aligned}$$

Substituting, we see that the numerator of the right hand side of *) is equal to

$$\frac{1}{2} x_1^2 \phi_{11}(x_1, \dots, x_n) - x_1 \phi_1(x_1, \dots, x_n) + x_1 \phi_1(0, \dots, x_n) + \frac{1}{2} x_1^2 \phi_{11}(\xi x_1, \dots, x_n).$$

By the Mean value theorem there exists ρ , $0 < \rho < 1$, such that

$$x_1 \phi_1(x_1, \dots, x_n) - x_1 \phi_1(0, \dots, x_n) = x_1^2 \phi_{11}(\rho x_1, \dots, x_n).$$

Substituting in the above expression for the numerator of the right hand side of *), we arrive at the following formula for $x_1 \tilde{\phi}_{11}$:

$$x_1 \tilde{\phi}_{11}(x_1, \dots, x_n) = \frac{x_1^2}{2} (\phi_{11}(x_1, \dots, x_n) + \phi_{11}(\xi x_1, \dots, x_n) - 2 \phi_{11}(\rho x_1, \dots, x_n)),$$

provided $x_1 \neq 0$. The continuity of A_{11} now follows from the continuity of ϕ_{11} . $\square$

Remark. It must be emphasised that even though ϕ and θ are C^2 , h may not be C^2 . For example, if $n = 1$, $\phi(x) = x + x^2|x|$, $\theta(x) = x$, then ϕ/θ will only be of class C^1 .

Let $Q_n(\mathbb{R})$ denote the space of real quadratic forms on $\mathbb{R}^n$. Taking the standard basis on $\mathbb{R}^n$, we may identify $Q_n(\mathbb{R})$ with the space of $n \times n$ real symmetric matrices and we shall give $Q_n(\mathbb{R})$ the topology induced from that on the space of $n \times n$ real matrices. If $q \in Q_n(\mathbb{R})$ corresponds to $[a_{ij}]$, we set

$$q(v) = \sum_{i,j=1}^n a_{ij} v_i v_j, \quad v = (v_1, \dots, v_n) \in \mathbb{R}^n.$$

Let $\phi \in C^2(\mathbb{R}^n, \mathbb{R})$. The *Hessian* of ϕ is the continuous map $H(\phi) : \mathbb{R}^n \rightarrow Q_n(\mathbb{R})$ defined by

$$H(\phi)(x) = \left[\frac{\partial^2 \phi}{\partial x_i \partial x_j}(x) \right].$$

Avoiding coordinates, we may equivalently define $H(\phi)(x) = D^2\phi_x$ —the second derivative of ϕ at x .

Suppose $\Omega \subset \mathbb{R}^n$ has C^2 boundary and that ϕ is a C^2 defining function for Ω . We say that $H(\phi)$ is positive semi-definite on tangent vectors to $\partial\Omega$ if for all $x \in \partial\Omega$

$$H(\phi)(x)(v) \geq 0, \text{ for all } v \in \mathbb{R}^n \text{ such that } d\phi(x)(v) = 0.$$

We say $H(\phi)$ is positive definite on tangent vectors to $\partial\Omega$ if for all $x \in \partial\Omega$

$$H(\phi)(x)(v) > 0, \text{ for all non-zero } v \in \mathbb{R}^n \text{ such that } d\phi(x)(v) = 0.$$

Lemma 2.5.2. Suppose Ω is a proper subdomain of $\mathbb{R}^n$ with C^2 boundary. If there exists a C^2 defining function ϕ for Ω such that $H(\phi)$ is positive (semi-) definite on tangent vectors to $\partial\Omega$ then the same is true for the Hessian of every C^2 defining function for Ω .

Proof. Suppose θ is another C^2 defining function for Ω . By Lemma 2.5.1 there exists a strictly positive C^1 function h on $\mathbb{R}^n$ such that $\phi = h\theta$. Since h is C^2 off $\partial\Omega$, we certainly have

$$H(\phi)(x) = \theta(x)H(h)(x) + 2(d\theta(x), dh(x)) + h(x)H(\theta)(x), \quad x \in \mathbb{R}^n \setminus \partial\Omega.$$

By part d) of Lemma 2.5.1, the function $x \mapsto \theta(x)H(h)(x)$ extends to a continuous function on $\mathbb{R}^n$ which vanishes on $\partial\Omega$. Hence

$$H(\phi)(x)(v) = 2(dh(x)(v)d\theta(x)(v)) + h(x)H(\theta)(x)(v),$$

$x \in \partial\Omega$, $v \in \mathbb{R}^n$. If v is a tangent vector to $\partial\Omega$, $d\theta(x)(v) = 0$ and so we see that

$$H(\phi)(x)(v) = h(x)H(\theta)(x)(v), \quad x \in \partial\Omega, \quad v \text{ a tangent vector to } \partial\Omega \text{ at } x.$$

Since $h(x) > 0$, the result follows. $\square$

Theorem 2.5.3. Let Ω be a proper subdomain of $\mathbb{R}^n$ with C^2 boundary. Then Ω is convex if and only if for some defining function ϕ for Ω , $H(\phi)$ is positive semi-definite on tangent vectors to $\partial\Omega$.

Proof. We shall prove that the positive semi-definiteness of $H(\phi)$ on tangent vectors to $\partial\Omega$ implies the convexity of Ω and leave the converse as an easy exercise for the reader. We first remark that it is sufficient to prove that $\bar{\Omega}$ is locally convex. That is, for each $z \in \partial\Omega$ there exists a (convex) neighbourhood N of z in $\mathbb{R}^n$ such that $N \cap \bar{\Omega}$ is convex. A proof of this well known characterisation of convex domains may be found in Valentine [1]. Take the Euclidean norm on $\mathbb{R}^n$ and let $\bar{E}_r(x)$ denote the closed disc, radius r and centre x in $\mathbb{R}^n$. For local convexity it is sufficient to show that for each $z \in \partial\Omega$, there exists $r > 0$ such that $T_y\partial\Omega \cap \bar{E}_r(z) \cap \Omega = \emptyset$, for all $y \in \partial\Omega \cap \bar{E}_r(z)$. Indeed, this condition implies that $\bar{E}_r(z) \cap \bar{\Omega}$ is convex (see the characterisation of closed convex hull given in §4).

Fix $z \in \partial\Omega$. By an affine linear change of coordinates we may assume that $z = 0$, the tangent plane to $\partial\Omega$ at 0 is the hyperplane $x_n = 0$ and $\frac{\partial\phi}{\partial x_n}(0) < 0$.

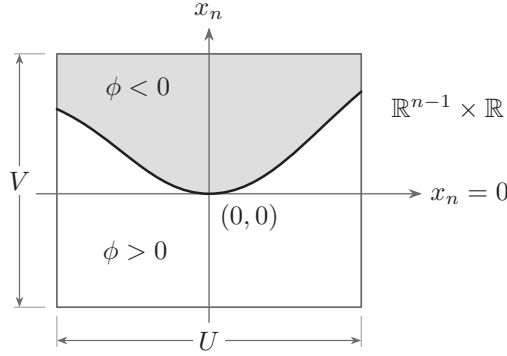


Figure I.5

By the implicit function theorem we can choose an open neighbourhood $U \times V$ of $(0, 0) \in \mathbb{R}^{n-1} \times \mathbb{R}$ such that $(U \times V) \cap \partial\Omega$ can be represented as the graph of a C^2 function $\psi : U \rightarrow V$. That is, $\phi(x, \psi(x)) = 0$, $x \in U$, and $(U \times V) \cap \partial\Omega = \{(x, \psi(x)) : x \in U\}$.

Differentiating the identity $\phi(x, \psi(x)) = 0$ twice we obtain

$$0 = D_1^2 \phi_X + 2D_{12} \phi_X \cdot D\psi_x + D_2^2 \phi_X \cdot (D\psi_x, D\psi_x) + D_2 \phi_X \cdot D^2 \psi_x, \quad x \in U.$$

Here we have set $X = (x, \psi(x))$ and we follow the derivative notation of Field [1].

Now any tangent vector to $\text{graph}(\psi)$ at x is of the form $(v, D\psi_x(v))$, $v \in \mathbb{R}^{n-1}$. Set $V = (v, D\psi_x(v))$, $v \in \mathbb{R}^{n-1}$. Evaluating the above quadratic form on V , we obtain

$$0 = H(\phi)(X)(V) + \frac{\partial \phi}{\partial x_n}(X) D^2 \psi_x(v, v), \quad v \in \mathbb{R}^{n-1}, \quad x \in U.$$

Now $H(\phi)(X)(V) \geq 0$ by assumption. Furthermore, since $\frac{\partial \phi}{\partial x_n}(0) < 0$, we can find $s > 0$ such that $\frac{\partial \phi}{\partial x_n}(X) < 0$, $\|(x_1, \dots, x_{n-1})\| < s$ ($\|\cdot\|$ denotes the Euclidean norm). Therefore, $D^2 \psi_x(v, v) \geq 0$, $\|x\| < s$, $v \in \mathbb{R}^{n-1}$.

Pick a unit vector $u \in \mathbb{R}^{n-1}$ and let L_u denote the line through the origin of $\mathbb{R}^{n-1}$ defined by u . Let $\psi^u = \psi|_{L_u}$. The positivity of $D^2 \psi_x$ implies that

$$\frac{d^2 \psi^u}{du^2}(x) \geq 0, \quad x \in L^u, \quad \|x\| < s.$$

As is well known this implies that the graph of ψ^u is convex. In particular, $L^u \cap \text{graph}(\psi^u) \neq \emptyset$. This is true for all unit vectors u in $\mathbb{R}^{n-1}$ and so the open disc in $\mathbb{R}^{n-1}$ of radius s , centre 0, does not intersect $\Omega \cap \bar{E}_s(0)$. A straightforward argument based on the continuity of $\text{grad}(\phi)$ shows that we can extend this result to find r , $0 < r \leq s$, such that $T_y \partial\Omega \cap \bar{E}_r(0) \cap \Omega = \emptyset$, for all $y \in \partial\Omega \cap \bar{E}_r(0)$. As we pointed out above, this is sufficient to prove the convexity of Ω . $\square$

Definition 2.5.4. Let Ω be a proper subdomain of $\mathbb{R}^n$ with C^2 boundary. We say that Ω is *strictly convex* if for some defining function ϕ for Ω , $H(\phi)$ is positive definite on tangent vectors to $\partial\Omega$.

Proposition 2.5.5. Let Ω be a strictly convex bounded domain in $\mathbb{R}^n$ with C^2 boundary. Then there exists a defining function ϕ for Ω such that $H(\phi)(x)$ is a positive definite quadratic form for all $x \in \partial\Omega$.

Proof. Let θ be any C^2 defining function for Ω . For $\lambda \in \mathbb{R}$, set $\phi_\lambda = \theta e^{\lambda\theta}$. Clearly for all values of λ , ϕ_λ is a C^2 defining function on Ω which is positive definite on tangent vectors to $\partial\Omega$. Computing the Hessian of ϕ_λ and restricting to $\partial\Omega$ we find

$$H(\phi_\lambda)(x)(v) = H(\theta)(x)(v) + 2\lambda(d\theta(x)(v))^2, \quad v \in \mathbb{R}^n, \quad x \in \partial\Omega.$$

Restrict $H(\phi_\lambda)$ to the compact set $\Gamma = \partial\Omega \times S^{n-1}$, where S^{n-1} denotes the unit sphere in $\mathbb{R}^n$. Let $K = \{(x, u) \in \Gamma, d\theta(x)(v) = 0\}$ (unit tangent bundle of $\partial\Omega$). $H(\theta)$ is strictly positive on K and so, by continuity, is strictly positive on some open neighbourhood U of K in Γ . Let m be the infimum of $2(d\theta(x)(v))^2$ over $\Gamma \setminus U$ and M the supremum of $H(\theta)(x)(v)$ over $\Gamma \setminus U$. Since the compact set $\Gamma \setminus U$ does not meet the zero set of $d\theta$, $m > 0$. Taking $\lambda_0 = 2M/m$, we see immediately that $H(\phi_\lambda)$ is strictly positive on Γ , $\lambda \geq \lambda_0$. $\square$

This completes our review of convex domains in $\mathbb{R}^n$. Our aim is now to obtain a characterisation of domains of holomorphy with C^2 boundary in terms of local properties of the boundary. For the remainder of this section Ω will always denote an open, connected subset of $\mathbb{C}^n$.

We let $H_n(\mathbb{C})$ denote the space of Hermitian quadratic forms on $\mathbb{C}^n$. Taking the standard complex basis on $\mathbb{C}^n$, we may identify $H_n(\mathbb{C})$ with the space of $n \times n$ Hermitian matrices and we give $H_n(\mathbb{C})$ the corresponding topology.

Definition 2.5.6. Let D be an open subset of $\mathbb{C}^n$, and $\phi \in C^2(D, \mathbb{R})$. The *Levi form* of ϕ is the map

$$L(\phi) : D \rightarrow H_n(\mathbb{C})$$

defined by

$$L(\phi)(z) = \left[\frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j} \right], \quad z \in D.$$

Remarks.

1. Given $Z = (Z_1, \dots, Z_n) \in \mathbb{C}^n$, $z \in D$,

$$L(\phi)(z)(Z) = \sum_{i,j=1}^n \frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j}(z) Z_i \bar{Z}_j.$$

2. In Chapter 5 we shall give an invariant definition of the Levi form.

As we shall soon see the Levi form is the complex analogue of the Hessian.

Lemma 2.5.7. Let Ω, Ω' be domains in $\mathbb{C}^n$ and $h : \Omega \rightarrow \Omega'$ be biholomorphic. If $\phi \in C^2(\Omega', \mathbb{R})$ then

$$L(\phi h) = L(\phi)(Dh, \overline{Dh}).$$

(Here $[Dh] = [\partial h_i / \partial z_j]$; $[\overline{Dh}] = [\overline{\partial h_i / \partial z_j}]$).

Proof. We have to show that for $1 \leq i, j \leq n$,

$$\frac{\partial^2 \phi h}{\partial z_i \partial \bar{z}_j} = \sum_{\alpha, \beta=1}^n \frac{\partial^2 \phi}{\partial z_\alpha \partial \bar{z}_\beta} \frac{\partial h_\alpha}{\partial z_i} \overline{\frac{\partial h_\beta}{\partial z_j}}.$$

This is a straightforward computation using Proposition 2.1.1 and we omit details. $\square$

Lemma 2.5.7 shows that the Levi form is invariant under holomorphic changes of variables. Of course, the Hessian is only invariant under affine linear changes of variables.

Lemma 2.5.8. Let $\phi \in C^2(\Omega, \mathbb{R})$. Then

$$L(\phi) = \frac{1}{4}(H(\phi) + H(\phi)(J)).$$

That is, for $z \in \Omega$, $Z \in \mathbb{C}^n$,

$$L(\phi)(z)(Z) = \frac{1}{4}(H(\phi)(z)(Z) + H(\phi)(z)(JZ)),$$

where on the right hand side we use the standard identification of $\mathbb{C}^n$ with $\mathbb{R}^{2n}$ and regard $Z \in \mathbb{R}^{2n}$ (J denotes the standard complex structure on $\mathbb{C}^n$ (see §5, Chapter 1)).

Proof. A straightforward, if tedious, calculation and we omit details. $\square$

Example 1. Let $\Omega \subset \mathbb{C}^n$ be a bounded convex domain. We know from §4 that every biholomorphic image of Ω is a domain of holomorphy. Suppose that Ω has C^2 boundary and that $g \in C^2(\bar{\Omega}, \mathbb{C}^n)$ is a diffeomorphism of some neighbourhood U of $\bar{\Omega}$ into $\mathbb{C}^n$ which maps Ω biholomorphically onto a domain $\Omega' \subset \mathbb{C}^n$. Certainly Ω' has C^2 boundary and $\partial\Omega' = g(\partial\Omega)$. Set $h = (g|U)^{-1}$. If ϕ is a defining function for Ω , then ϕh is a defining function for Ω' (ϕh is only defined on a neighbourhood of Ω' , but we can always extend to a C^2 function on the whole of $\mathbb{C}^n$ which is strictly positive outside Ω'). Since Ω is convex, $H(\phi)$ is positive semi-definite on tangent vectors to $\partial\Omega$. Suppose Z is a holomorphic tangent vector to Ω' at x . That is,

$$\sum_{i=1}^n \frac{\partial(\phi h)}{\partial z_i}(x) Z_i = 0, \quad Z = (Z_1, \dots, Z_n).$$

Taking real and imaginary parts, we find that Z and JZ are *real* tangent vectors to $\partial\Omega'$ at x . Therefore, by Lemmas 2.5.7, 2.5.8,

$$L(\phi h)(x)(Z) = \frac{1}{4}(H(\phi)(x)(Dh_x(Z)) + H(\phi)(x)(JDh_x(Z))) \geq 0.$$

In other words, the Levi form is positive semi-definite on holomorphic tangent vectors to $\partial\Omega'$. Furthermore, if Ω is strictly convex, the Levi form will be positive definite on holomorphic tangent vectors to $\partial\Omega'$.

Remark. Of course a domain of holomorphy need not be the biholomorphic image of a convex set. However, if Ω is a domain in $\mathbb{C}^n$ with C^2 boundary and defining function ϕ such that $L(\phi)$ is positive definite at a point $z \in \partial\Omega$, then we may clearly make a local ($\mathbb{C}$ -linear) holomorphic change of coordinates on some neighbourhood U of z such that, in the new coordinates, $U \cap \Omega$ is convex. In particular, if $L(\phi)$ is positive definite at all points of $\partial\Omega$, we see that Ω is locally the biholomorphic image of a convex set. Consequently, by example 1, Ω will locally be a domain of holomorphy. That is, for every $z \in \bar{\Omega}$, there will exist an open neighbourhood U of z in $\mathbb{C}^n$ such that $U \cap \Omega$ is a domain of holomorphy. However, if $L(\phi)$ is not positive definite and only positive semi-definite, even at just one point of $\partial\Omega$, it is not generally the case that Ω is locally the biholomorphic image of a convex set (it will however, be locally a domain of holomorphy). For examples and further references we refer the reader to the survey article by Y-T Siu [1], especially §7. We should mention that the question of boundary invariants of domains with real analytic boundary is currently a topic of much interest. See, for example, the paper of Chern and Moser [1].

Definition 2.5.9. Let Ω be a domain in $\mathbb{C}^n$ with C^2 boundary. Suppose that there exists a defining function ϕ for Ω such that $L(\phi)$ is positive semi-definite on holomorphic tangent vectors to $\partial\Omega$. Then we say that Ω is *Levi pseudoconvex* (abbreviated L.p.). In case $L(\phi)$ is positive definite on holomorphic tangent vectors to $\partial\Omega$, we say Ω is *strictly Levi pseudoconvex* (abbreviated s.L.p.).

Exactly as we did for convex subsets of $\mathbb{R}^n$ we may prove

Lemma 2.5.10. Suppose Ω is a domain in $\mathbb{C}^n$ with C^2 boundary. If Ω is L.p. then $L(\phi)$ is positive semi-definite on holomorphic tangent vectors to $\partial\Omega$ for *all* C^2 defining functions ϕ for Ω . Similarly for s.L.p. domains.

Lemma 2.5.11. If $\Omega \subset \mathbb{C}^n$ is a bounded, strictly Levi pseudoconvex domain with C^2 boundary then there exists a defining function ϕ for such that $L(\phi)$ is positive definite on $\partial\Omega$.

Just as for convexity, Levi pseudoconvexity is a local property of the boundary. More precisely we have

Proposition 2.5.12. Let Ω be a bounded domain in $\mathbb{C}^n$ with C^2 boundary. Suppose that for each $x \in \partial\Omega$, there exists an open neighbourhood U of x and $\phi \in C^2(U, \mathbb{R})$ such that

- a) $\Omega \cap U = \{x \in U : \phi(x) < 0\}$.
- b) $d\phi \neq 0$ on $\partial\Omega \cap U$.
- c) $L(\phi)(y)(v) \geq 0$ for all $y \in \partial\Omega \cap U$ and holomorphic tangent vectors v to $\partial\Omega$ at y .

Then Ω is Levi pseudoconvex. If in c), $L(\phi)$ is positive definite on holomorphic tangent vectors then Ω is strictly Levi pseudoconvex.

Proof. Since $\partial\Omega$ is compact, we can find finitely many open subsets U_j of $\mathbb{R}^n$ and $\phi_j \in C^2(U_j, \mathbb{R})$ such that $\partial\Omega \subset \bigcup U_j$ and the ϕ_j satisfy the conditions of the proposition. Set $U = \bigcup U_j$ and choose $\theta_i \in C_c^\infty(U_i)$ such that $\theta_i \geq 0$ and $\bigcup_i \text{Interior}(\text{supp}(\theta_i)) \supset \partial\Omega$. Define $\tilde{\phi} = \sum \theta_i \phi_i \in C_c^\infty(U, \mathbb{R})$. Since $\tilde{\phi}(x) < 0$ if $x \in \Omega \cap U$ and $\tilde{\phi}(x) = 0$ if $x \in \partial\Omega$, we may choose $\psi \in C^\infty(\mathbb{C}^n, \mathbb{R})$ such that $\psi \equiv 0$ on a neighbourhood of $\partial\Omega$ and $\Omega = \{x \in \mathbb{C}^n : (\tilde{\phi} + \psi)(x) < 0\}$, $\partial\Omega = \{x \in \mathbb{C}^n : (\tilde{\phi} + \psi)(x) = 0\}$. Set $\phi = \tilde{\phi} + \psi$. On $\partial\Omega$,

$$\begin{aligned} d\phi &= d\tilde{\phi} = \sum \theta_i d\phi_i \\ &\neq 0. \end{aligned}$$

Computing $L(\phi)$, we find that on $\partial\Omega$

$$\begin{aligned} L(\phi)(x)(v) &= \sum \theta_i(x) L(\phi_i)(x)(v), \quad \text{if } v \text{ is a holomorphic tangent vector to } \partial\Omega \text{ at } x, \\ &\geq 0. \end{aligned}$$

□

Theorem 2.5.13. (E.E. Levi). Suppose Ω is a holomorphically convex domain in $\mathbb{C}^n$ with C^2 boundary. Then Ω is Levi pseudoconvex.

Proof. Let ϕ be a defining function for Ω and suppose that at some point $x \in \partial\Omega$, $L(\phi)$ is not positive semi-definite on holomorphic tangent vectors to $\partial\Omega$ at x . By a complex affine linear change of coordinates we may suppose that $x = 0$, $d\phi(0) = dx_1(0)$ and $L(\phi)(u) < 0$, where u is the unit basis vector in the z_2 -direction. Thus, if for $1 \leq i, j \leq n$ we set $\phi_i(0) = \frac{\partial \phi}{\partial z_i}(0)$, $\phi_{\bar{j}}(0) = \frac{\partial \phi}{\partial \bar{z}_j}(0)$, $\phi_{i\bar{j}}(0) = \frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j}(0)$, etc., we will have

$$\phi_1(0) = \phi_{\bar{1}}(0) = \frac{1}{2}; \quad \phi_i(0) = \phi_{\bar{i}}(0) = 0, \quad i > 1; \quad \phi_{2\bar{2}}(0) < 0.$$

For $t \in \mathbb{R}$, we define the analytic embedding $\phi_t : \mathbb{C} \rightarrow \mathbb{C}^n$ by

$$\phi_t(s) = (-2\phi_{2\bar{2}}(0)s^2 - t, s, 0, 0, \dots, 0).$$

For $r > 0$, set $V_t(r) = \phi_t(D_r(0))$. $V_t(r)$ is an embedded disc in $\mathbb{C}^n$. We observe that $0 \in V_0(r)$. We claim that there exists $r > 0$ such that

$$\overline{V_t(r)} \subset \Omega, \quad t \in (0, r]$$

$$\overline{V_0(r)} \setminus \{0\} \subset \Omega.$$

On the assumption that such an r exists we now show that Ω cannot be holomorphically convex. Define

$$K = \bigcup_{0 \leq t \leq r} \phi_t(\partial D_r(0)) = \bigcup_{0 \leq t \leq r} \partial V_t(r).$$

K is a compact subset of Ω . If $f \in A(\Omega)$, $0 < t \leq r$, then $f\phi_t \in A(D_r(0))$ and is continuous on $\bar{D}_r(0)$. Hence, by the maximum principle, $\|f\phi_t\|_{\bar{D}_r(0)} = \|f\phi_t\|_{\partial D_r(0)}$. This implies that $\overline{V_t(r)} \subset \widehat{K}$, $0 < t \leq r$. By continuity, $\overline{V_0(r)} \setminus \{0\} \subset \widehat{K}$. But therefore $\widehat{K}$ cannot be compact and so Ω is not holomorphically convex.

To construct r we proceed as follows. By Taylor's theorem we may write

$$\phi(z) = x_1 + \sum_{i,j} (\phi_{ij}(0)z_i z_j + \phi_{i\bar{j}}(0)z_i \bar{z}_j + \phi_{\bar{i}j}(0)\bar{z}_i z_j) + o(|z|^2).$$

Restricting ϕ to the sets V_t , we may regard ϕ as a function of s and t . Substituting we find

$$\phi(s, t) = \phi_{2\bar{2}}(0)|s|^2 + o(|s|^2) + tg(t, s),$$

where g is a continuous function of s and t satisfying $g(0, 0) = -1$. Since $\phi_{2\bar{2}}(0) < 0$, we see that we can choose $r > 0$ so that $\phi(s, t) < 0$ for $t \in (0, r]$, $|s| \leq r$ or $t = 0$, $0 < |s| \leq r$. $\square$

We may now state one of the most famous problems in several complex variables.

Levi's problem (E.E. Levi [1]). Show that every Levi pseudoconvex domain in $\mathbb{C}^n$ is a domain of holomorphy.

Levi's problem was answered affirmatively by Oka [1] in case $n = 2$ and later, independently, by Bremermann [1], Norguet [1] and Oka [2] for general n . We shall give a proof of a generalisation of Levi's problem, due to Grauert, in Chapter 7.

For the remainder of this section we shall investigate some alternative pseudoconvexity definitions. First we need a technical lemma, the proof of which was shown to us by D.B.A. Epstein.

Lemma 2.5.14. Let Ω be a bounded domain in $\mathbb{R}^n$ with C^r boundary, $r \geq 2$. Define $d : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$d(x) = \begin{cases} d(x, \partial\Omega), & x \in \Omega \\ -d(x, \partial\Omega), & x \notin \Omega, \end{cases}$$

where distances are computed relative to the Euclidean metric on $\mathbb{R}^n$. Then d is C^r on some neighbourhood of $\partial\Omega$ in $\mathbb{R}^n$.

Proof. Let $n(z)$ denote the unit inward normal to $\partial\Omega$ at $z \in \partial\Omega$ and $\phi : \partial\Omega \times \mathbb{R} \rightarrow \mathbb{R}^n$ be the map $\phi(z, \lambda) = z + \lambda n(z)$. Thus ϕ is the exponential map of the normal bundle of $\partial\Omega$, relative to the Euclidean norm on $\mathbb{R}^n$. In particular, ϕ defines a C^{r-1} diffeomorphism of some open neighbourhood of $\partial\Omega \times \{0\} \subset \partial\Omega \times \mathbb{R}$ onto an open neighbourhood U of $\partial\Omega$ in $\mathbb{R}^n$ (note that

$n(\cdot)$ is only of class C^{r-1} on $\partial\Omega$). If $X \in U$ we may write $\phi^{-1}(X) = (\pi(X), \lambda(X))$, where $\pi : U \rightarrow \partial\Omega$, $\lambda : U \rightarrow \mathbb{R}$ are C^{r-1} functions. Thus

$$X - \pi(X) = \lambda(X)n(\pi(X)), \quad X \in U.$$

Differentiating this expression with respect to X and evaluating at $y \in \mathbb{R}^n$ we see that

$$y - D\pi_X(y) = D\lambda_X(y)n(\pi(X)) + \lambda(X)Dn_{\pi(X)}(D\pi_X(y)).$$

Take the inner product with $n(\pi(X))$ to obtain

$$\langle y, n(\pi(X)) \rangle = D\lambda_X(y) + \lambda(X)\langle Dn_{\pi(X)}(D\pi_X(y)), n(\pi(X)) \rangle.$$

Now

$$\langle Dn_{\pi(X)}(D\pi_X(y)), n(\pi(X)) \rangle = 0$$

as is seen by differentiating the identity $\langle n(\pi(X)), n(\pi(X)) \rangle = 1$. Hence

$$D\lambda_X(y) = \langle y, n(\pi(X)) \rangle.$$

But $\langle y, n(\pi(X)) \rangle$ is a C^{r-1} function of X . Therefore, $D\lambda$ is C^{r-1} and so λ is C^r . Since $d(x) = \lambda(x)$, the result follows. $\square$

Continuing with the notation of the proof of Lemma 2.5.14, we see that $Dd_x = \langle \cdot, n(\pi(x)) \rangle$, for $x \in \partial\Omega$ and so Dd is non-vanishing on $\partial\Omega$. Hence $-d$ is a continuous defining function for Ω which is C^r on some neighbourhood of $\partial\Omega$ in $\mathbb{R}^n$.

Now suppose Ω is a bounded domain in $\mathbb{C}^n$ with C^2 boundary. As in Lemma 2.5.14, we let $d : \mathbb{C}^n \rightarrow \mathbb{R}$ denote the signed Euclidean distance function to the boundary of Ω .

Proposition 2.5.15. Let Ω be a bounded domain in $\mathbb{C}^n$ with C^2 boundary and suppose that $d : \mathbb{C}^n \rightarrow \mathbb{R}$ is C^2 on the open neighbourhood U of $\partial\Omega$ in $\mathbb{C}^n$. Then the following conditions are equivalent:

- i) Ω is Levi pseudoconvex.
- ii) The Levi form of $-\log(d)$ is positive semi-definite on $U \cap \Omega$.

Proof. First suppose that $L(-\log d)$ is positive semi-definite on $U \cap \Omega$. Set $\delta = -\log d = \log d^{-1}$. Computing we find that

$$\frac{\partial^2 \delta}{\partial z_i \partial \bar{z}_j} = -d^{-1} \frac{\partial^2 d}{\partial z_i \partial \bar{z}_j} + d^{-2} \frac{\partial d}{\partial z_i} \frac{\partial \bar{d}}{\partial \bar{z}_j} \quad \text{on } U.$$

Hence for $z \in U \cap \Omega$, we have

$$\sum_{i,j} \frac{\partial^2 d}{\partial z_i \partial \bar{z}_j}(z) Z_i \bar{Z}_j \geq 0, \quad \text{provided that } \sum_i \frac{\partial d}{\partial z_i}(z) Z_i = 0.$$

By continuity, this result holds on $\partial\Omega$ and so Ω is Levi pseudoconvex.

For the converse, we first note that since $-d$ is a C^2 defining function for Ω , at least on U , Lemma 2.5.10 implies that $L(-d)$ is positive semi-definite on holomorphic tangent vectors to $\partial\Omega$. Suppose that there exists $z \in U \cap \Omega$ such that the Levi form of $-\log d$ is not positive semi-definite at z . This implies that there exists a unit vector $u \in \mathbb{C}^n$ such that

$$\left. \frac{\partial^2}{\partial t \partial \bar{t}} (\log d(z + tu)) \right|_{t=0} > 0.$$

As in the proof of Theorem 2.5.13, Taylor's theorem implies that

$$\log d(z + tu) = \log d(z) + \operatorname{Re}(at + bt^2) + c|t|^2 + O(|t|^3),$$

for constants $a, b \in \mathbb{C}$. Choose $\zeta \in \mathbb{C}^n$ such that $z + \zeta \in \partial\Omega$ and $d(z) = \|\zeta\|$. Consider the holomorphic curve

$$\phi(t) = z + tu + \zeta \exp(at + bt^2)$$

and note that $\phi(0) = z + \zeta \in \partial\Omega$. By the triangle inequality we have

$$d(\phi(t)) \geq d(z + tu) - \|\zeta\| \exp(at + bt^2), \quad t \in \mathbb{C}.$$

Using the expression above for $\log d(z + tu)$ it follows that for sufficiently small values of $|t|$ we have

$$d(\phi(t)) \geq \|\zeta\|(\exp(c|t|^2/2) - 1) \exp(at + bt^2).$$

But this estimate implies that at $t = 0$ we have

$$\frac{\partial}{\partial t} d(\phi(t)) = 0 \quad \text{and} \quad \frac{\partial^2}{\partial t \partial \bar{t}} d(\phi(t)) > 0$$

and so

$$\sum_i \frac{\partial d}{\partial z_i}(z + \zeta) \phi'_i(0) = 0, \quad \sum_{i,j} \frac{\partial^2 d}{\partial z_i \partial \bar{z}_j}(z + \zeta) \phi'_i(0) \overline{\phi'_j(0)} < 0,$$

contradicting the positive semi-definiteness of $L(d)$ on holomorphic tangent vectors to $\partial\Omega$. $\square$

Definition 2.5.16. Let Ω be a domain in $\mathbb{C}^n$ and $\phi : \Omega \rightarrow \mathbb{R}$ be C^2 . We say that ϕ is *plurisubharmonic* (psh for short) if $L(\phi)$ is positive semi-definite on Ω . We say ϕ is *strictly plurisubharmonic* if $L(\phi)$ is positive definite on ∂ .

Remark. Notice that if $\phi : \Omega \rightarrow \mathbb{R}$ is C^2 and psh (respectively, strictly psh) and $h : \Omega' \rightarrow \Omega$ is biholomorphic then h is psh (respectively, strictly psh).

In practice, it is useful to remove the differentiability requirement on psh functions. Thus suppose $\phi : \Omega \rightarrow [-\infty, +\infty)$ is semi-continuous from above. We say ϕ is psh if for arbitrary, $z, w \in \mathbb{C}^n$, the function $t \mapsto \phi(z + tw)$ is *subharmonic* wherever defined. We recall that a function u on a domain $U \subset \mathbb{C}$ is said to be subharmonic if given any compact subset K of U and continuous function h on K which is harmonic on $\mathring{K}$ and $\geq u$ on ∂K then $u \leq h$ on K . Equivalently, ϕ is psh if $L(\phi)$ is positive semi-definite in the distributional sense. That is, if $\int_{\Omega} L(u)(x)(X)\phi(x) d\lambda(x) \geq 0$ for all $u \in C_c^\infty(\Omega)$, $X \in \mathbb{C}^n$. We refer the reader to Hörmander [1] or Vladimirov [1] for the basic properties of subharmonic and psh functions that are used in several complex variables. We should stress that the only reference to subharmonic functions in these notes will be in this section.

The next result is a generalisation of Proposition 2.5.15 to domains which do not have smooth boundary.

Proposition 2.5.17. If Ω is a domain of holomorphy in $\mathbb{C}^n$ then $-\log d(x)$ is psh.

Proof. See Bremermann [2], Hörmander [1] or Vladimirov [1]. $\square$

Definition 2.5.18. A domain Ω in $\mathbb{C}^n$ is said to be *pseudoconvex* if $-\log d(x)$ is psh.

We now define a very strong form of pseudoconvexity.

Definition 2.5.19. A domain Ω in $\mathbb{C}^n$ is said to be *0-complete* or *holomorphically complete* if there exists $\phi \in C_{\mathbb{R}}^{\infty}(\Omega)$ such that

1. ϕ is strictly psh.
2. For every $a \in \mathbb{R}$, $\Omega_a = \{z \in \Omega : \phi(z) < a\}$ is relatively compact in Ω .

It is not hard to show that a domain Ω in $\mathbb{C}^n$ is pseudoconvex if and only if it is 0-complete. See, for example, Hörmander [1] or Vladimirov [1]. We shall prove directly that every domain of holomorphy is 0-complete.

Theorem 2.5.20. Let Ω be a domain of holomorphy. Then Ω is 0-complete.

Proof. Since Ω is holomorphically convex we may find a normal exhaustion $(K_n)_{n \geq 1}$ of Ω . Choose an increasing sequence (U_n) of relatively compact open subsets of Ω such that for all n , $K_n \subset U_n$ and U_n is relatively compact in U_{n+1} . Since $\widehat{K_n} = K_n$ and $\bar{U}_{n+1} \setminus U_n$ is compact there exist $f_{nk} \in A(\Omega)$, $1 = 1, \dots, k(n)$, such that

$$\|f_{nk}\|_{K_n} < 1 \quad \text{and} \quad \max_k |f_{nk}(z)| > 1, \quad z \in \bar{U}_{n+1} \setminus U_n.$$

Raising the f_{nk} to sufficiently high powers we may further assume that

$$\sum_{n=1}^{k(n)} |f_{nk}(z)|^2 < 2^{-n}, \quad z \in K_n. \quad (\text{A})$$

$$\sum_{k=1}^{k(n)} |f_{nk}(z)|^2 > n, \quad z \in \bar{U}_{n+1} \setminus U_n. \quad (\text{B})$$

Since $\Omega \subset \mathbb{C}^n$, we may further require that for every $z \in K_n$ there exist n of the f_{nk} which gives local holomorphic coordinates at z . Define

$$\phi(z) = \sum_{n,k} |f_{nk}(z)|^2, \quad z \in \Omega.$$

The sum converges by estimate A for all $z \in \Omega$. We claim that $\phi \in C_{\mathbb{R}}^{\infty}(\Omega)$. This follows since $\sum f_{nk}(z) \overline{f_{nk}(\zeta)}$ is uniformly convergent on compact subsets of $\Omega \times \Omega$ and so, by Corollary 2.1.8, the sum is analytic in z and $\bar{\zeta}$. Next notice that estimate B implies that $\phi(z) > n$, for $z \in \Omega \setminus U_n$. Hence, for all $a \in \mathbb{R}$, $\Omega_a = \{z : \phi(z) < a\}$ is relatively compact. Finally, we have

$$L(\phi) = \sum \frac{\partial f_{nk}}{\partial z_i} \overline{\frac{\partial f_{nk}}{\partial z_j}}$$

and so $L(\phi)$ is certainly positive semi-definite. To prove positive definiteness suppose that $L(\phi)(z)(Z) = 0$, $z \in K_n$. This implies that $\sum_{i,k} \frac{\partial f_{nk}}{\partial z_i}(z) Z_i = 0$. But it is possible to find n of the f_{nk} which give a local coordinate system at z . Hence $Z = 0$. $\square$

Remarks.

1. We call functions ϕ that satisfy the conditions of Definition 2.5.19 *strictly psh exhaustion functions* for Ω . Notice that our definition of 0-completeness is invariant under biholomorphic maps and makes no explicit mention of the boundary of Ω . Later, in Chapter 5, we shall generalise our 0-completeness definition to complex manifolds.
2. It follows easily from Sard's theorem that if Ω is 0-complete then Ω is the limit of an increasing sequence of s.l.p. domains (see also the discussion in §4, Chapter 7).

Next we wish to give a brief discussion of some “continuity” principles that hold for pseudoconvex domains. Motivation for the next definition is given by the proof of Theorem 2.5.13.

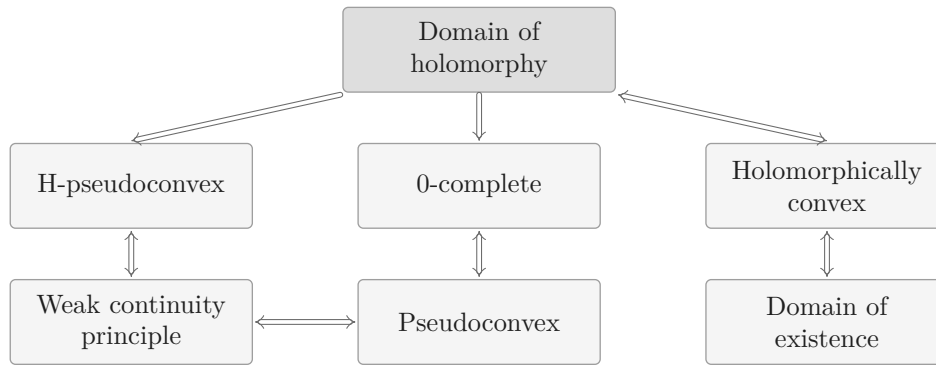
Definition 2.5.21. A domain Ω in $\mathbb{C}^n$ is said to satisfy the *weak continuity principle* if, given any sequence $\{V_n\}$ of holomorphically embedded discs in Ω which satisfy

1. For all n , $V_n \cup \partial V_n$ is a relatively compact subset of Ω .
2. $\lim V_n = V_0$ exists and ∂V_0 is relatively compact subset of Ω .

Then, provided that V_0 is bounded, V_0 is a relatively compact subset of Ω .

Using subharmonic functions it is not too hard to show that a domain is pseudoconvex if and only if it satisfies the weak continuity principle (see Bremermann [2], Vladimirov [1]). If we say that a domain Ω in $\mathbb{C}^n$ is H-pseudoconvex if it contains all its generalised Hartogs figures, then it is straightforward to show that Ω is H-pseudoconvex if and only if it satisfies the weak continuity principle (see Grauert and Fritzsche [1] and also Andreotti and Grauert [1, page 217ff]).

Let us summarise the relations between the various definitions we have been considering.



In Chapter 12 we shall prove the fundamental result that every 0-complete domain in $\mathbb{C}^n$ is a domain of holomorphy. As a consequence, the definitions above will have all been shown to be equivalent.

Finally the reader interested in pseudoconvexity and its relationship with domains of holomorphy should certainly consult the original works of Oka [3] and the H. Cartan seminar [2].

Exercises.

1. Show that Lemma 2.5.10 is false if $\partial\Omega$ is only C^1 . Can you find an example of a domain in $\mathbb{C}^n$ with C^1 boundary such that $d(x)$ is not differentiable at any point of $\mathbb{C}^n$?
2. Show
 - a) The interior of an arbitrary intersection of pseudoconvex domains is pseudoconvex (cf. Example 15, §4).
 - b) If Ω is a domain in $\mathbb{C}^n$ such that every point $z \in \partial\Omega$ has an open neighbourhood U such that $U \cap \Omega$ is pseudoconvex then Ω is pseudoconvex (the same result is true for domains of holomorphy but depends on the equivalence between domains of holomorphy and pseudoconvex domains).

(Both a) and b) use elementary properties of subharmonic functions).

3. (Kohn and Nirenberg [1]). Show that the domain Ω in $\mathbb{C}^2$ defined by $\operatorname{Re}(w) + |zw|^2 + |z|^8 + \frac{15}{7}|z|^2 \operatorname{Re}(z^6) < 0$ is L.p. and s.L.p. at every point of $\partial\Omega$ except 0.

(Ω is not convex at 0 in any local system of holomorphic coordinates, See Kohn and Nirenberg [1] and also Y.T. Siu [1]).

6 The Bergman kernel function

In this section we describe how some domains of holomorphy in $\mathbb{C}^n$ have an intrinsic strictly psh exhaustion function (granted the Euclidean metric structure on $\mathbb{C}^n$). For the most part we either omit or give very brief indications of proofs. Full details may be found in Bergman [1, 2] or Fuks [1]. We shall also assume some elementary Hilbert space theory.

Throughout this section we shall suppose that Ω is a bounded domain in $\mathbb{C}^n$. We have already shown in Exercise 2, §1, that $L^2(\Omega)$ is a Hilbert space with inner product defined by $(f, g) = \int_{\Omega} \bar{f}g \, d\lambda$, where $d\lambda$ is Lebesgue measure on $\mathbb{C}^n$. We set $|f| = (f, f)^{\frac{1}{2}}$, $f \in L^2(\Omega)$. Since $L^2(\Omega)$ is separable, we may find a countable orthonormal (Hilbert space) basis $\{\phi_j : j \geq 1\}$ for $L^2(\Omega)$. We remark that if $\{\psi_j\}$ is an arbitrary Hilbert space basis for $L^2(\Omega)$ then we can construct an orthonormal basis for $L^2(\Omega)$ from $\{\psi_j\}$ using the Gram–Schmidt orthogonalisation process.

Proposition 2.6.1. Let $\{\phi_j\}$ be an orthonormal basis for $L^2(\Omega)$. Given $f \in L^2(\Omega)$, set $a_j = (\phi_j, f)$, $j \geq 1$. Then

1. $f = \sum_{j=1}^{\infty} a_j \phi_j$, where convergence is uniform on compact subsets of $\partial\Omega$.
2. $|f|^2 = \sum_{j=1}^{\infty} |a_j|^2$ (Parseval's equality).

Proof. Straightforward and based on the estimates of Exercise 2, §1. See also the proof of Proposition 2.6.2 below. $\square$

Proposition 2.6.2. Let $\{\phi_j\}$ be an orthonormal basis for $L^2(\Omega)$. Then the series

$$\sum_{j=1}^{\infty} \phi_j(z) \overline{\phi_j(\zeta)}$$

converges uniformly on compact subsets of $\Omega \times \Omega$ to a function $K_{\Omega}(z, \bar{\zeta})$ which is analytic in $(z, \bar{\zeta})$. In particular, K_{Ω} is C^{∞} .

Proof. Let $D_0 = D(z_0; r)$, $D_1 = D(\zeta_0; s)$ be relatively compact polydiscs in Ω . By Exercise 2, §1, there exists $C_0 \geq 0$ such that for all $f \in L^2(\Omega)$, $\|f\|_{\bar{D}_0} \leq C_0 |f|$. Therefore for $z \in D_0$ we have

$$\begin{aligned} \sum_{j=1}^m |\phi_j(z)|^2 &= \int_{\Omega} \left| \sum_{j=1}^m \phi_j(\tau) \overline{\phi_j(z)} \right|^2 d\lambda(\tau) \\ &\geq C_0 \left\| \sum_{j=1}^m \phi_j(\tau) \overline{\phi_j(z)} \right\|_{D_0}^2 \\ &\geq C_0 \left\| \sum_{j=1}^m |\phi_j(z)|^2 \right\|_{D_0}^2. \end{aligned}$$

Hence

$$\sum_{j=1}^m |\phi_j(z)|^2 \leq C_0^{-1}, \quad z \in D_0.$$

Similarly,

$$\sum_{j=1}^m |\phi_j(\zeta)|^2 \leq C_1^{-1}, \quad \zeta \in D_1.$$

By the Cauchy–Schwarz inequality

$$\left\{ \sum_{j=1}^m |\phi_j(z) \overline{\phi_j(\zeta)}| \right\}^2 \leq \left\{ \sum_{j=1}^m |\phi_j(z)|^2 \right\}^2 \left\{ \sum_{j=1}^m |\phi_j(\zeta)|^2 \right\}^2$$

and so

$$\sum_{j=1}^m |\phi_j(z) \overline{\phi_j(\zeta)}| \leq (C_0 C_1)^{-1}, \quad z \in D_0, \zeta \in D_1.$$

Therefore $\sum_{j=1}^{\infty} \phi_j(z) \overline{\phi_j(\zeta)}$ converges uniformly on compact subsets of $\Omega \times \Omega$ and so the result follows by Corollary 2.1.8. $\square$

Definition 2.6.3. Let $\{\phi_j\}$ be an orthonormal basis for $L^2(\Omega)$. Then the function $K_{\Omega}(z, \bar{\zeta}) = \sum_{j=1}^{\infty} \phi_j(z) \overline{\phi_j(\zeta)}$ is called the *(Bergman) kernel function* of the domain Ω , relative to the orthonormal basis $\{\phi_j\}$.

Theorem 2.6.4. The kernel function $K_{\Omega}(z, \bar{\zeta})$ is independent of the choice of orthonormal basis for $L^2(\Omega)$. It satisfies the following characteristic variational property: Given $z_0 \in \Omega$, the function

$$f(z) = K_{\Omega}(z, \bar{z}_0) / K_{\Omega}(z_0, \bar{z}_0)$$

is the unique function in $L^2(\Omega)$ which minimises the integral $|f|$, subject to the normalising condition $f(z_0) = 1$.

Proof. Straightforward and we refer to the references. $\square$

In view of Theorem 2.6.4 we may now talk about *the* kernel function of the domain Ω .

Theorem 2.6.5. (Reproducing property of the kernel function). For all $f \in L^2(\Omega)$, we have

$$f(z) = \int_{\Omega} f(\zeta) K_{\Omega}(z, \bar{\zeta}) d\lambda(\zeta).$$

Proof. Take the Fourier series expansion of f relative to an orthonormal basis of $L^2(\Omega)$ and integrate term by term. $\square$

Example. Set $D = D(0; r_1, \dots, r_n)$, $E = E(0; r)$. Then $\{z^m/m! : m \in \mathbb{N}^n\}$ is easily seen to be an orthogonal basis for both $L^2(D)$ and $L^2(E)$. Normalising and computing the kernel functions we find

$$K_D(z, \bar{\zeta}) = \pi^{-n} r_1^2 \cdots r_n^2 \prod_{j=1}^n (r_j - z_j \bar{\zeta}_j)^{-2}$$

$$K_E(z, \bar{\zeta}) = \pi^{-n} r^2 n! \left(r^2 - \sum_{j=1}^n z_j \bar{\zeta}_j \right)^{-n-1}.$$

Observe that $K_D(z, \bar{z})$, $K_E(z, \bar{z})$ are strictly psh exhaustion functions for their respective domains. The reader may find computations of the kernel functions of the “classical domains” (see Chapter 4, §2) in Hua [1].

Next we examine how the kernel function transforms under biholomorphic maps.

Proposition 2.6.6. Let $h : \Omega \rightarrow \Omega'$ be a biholomorphic map between bounded domains in $\mathbb{C}^n$. Then

$$K_{\Omega}(z, \bar{z}) = K_{\Omega'}(h(z), \overline{h(z)}) |\det_{\mathbb{C}} Dh_z|^2.$$

Proof. A straightforward computation using the change of variables formula for multiple integrals and the fact that $\det_{\mathbb{R}} Dh_z = |\det_{\mathbb{C}} Dh_z|^2$ (see §4, Chapter 5). $\square$

Remarks.

1. Proposition 2.6.6 implies that the kernel function is invariantly defined as a section of an appropriate tensor bundle on Ω . We return to this point in Chapter 5.
3. Note that Proposition 2.6.6 in combination with the computations of the Example, shows that the open polydisc and Euclidean disc in $\mathbb{C}^n$ are biholomorphically inequivalent, $n > 1$. See also §2 of Chapter 4.

Corollary 2.6.7. The Levi form of $\log K_\Omega(z, \bar{z})$ is invariant under biholomorphic transformations.

Because of Corollary 2.6.7 and the analogy with pseudoconvexity we prefer to work with $\log K_\Omega(z, \bar{z})$ rather than $K_\Omega(z, \bar{z})$.

Proposition 2.6.8. Let Ω be a bounded domain in $\mathbb{C}^n$. Then $\log K_\Omega(z, \bar{z})$ is a C^∞ strictly psh function.

Proof. A straightforward computation that makes use of the fact that since Ω is bounded we can define local coordinate systems at any point of Ω using elements of a fixed orthonormal basis for $L^2(\Omega)$. $\square$

The question now arises as to the conditions under which $\log K_\Omega(z, \bar{z})$ gives a strictly psh exhaustion function for a domain of holomorphy Ω . In general it does not (see Bremermann [3]). However, it is certainly true that a sufficient condition for Ω to be a domain of holomorphy is that $\log K_\Omega(z, \bar{z})$ is a strictly psh exhaustion function for Ω . Moreover, it can be shown that every domain of holomorphy is the limit of an increasing sequence of domains of holomorphy for which $\log K(z, \bar{z})$ is a strictly psh exhaustion function (see Bremermann [2] and compare with the corresponding approximation of domains of holomorphy by s.l.p. domains). Finally we should mention that if we define a weighted inner product $(f, g)_\phi = \int_\Omega \bar{f} g e^{-\phi} d\lambda$ and let $L^2(\Omega; \phi)$ denote the corresponding Hilbert space of analytic functions on Ω , then the kernel theory continues to hold. Sharp estimates on the growth of the associated kernel function at the boundary of Ω are given in Hörmander [3]. See also Bremermann [2]. Notice that the introduction of weighting functions amounts to a change of metric on Ω .

For a survey of recent results on the kernel function and its applications to complex analysis see Diederich [1].

Exercises.

1. Verify that the kernel function of a domain Ω is Hermitian: $K_\Omega(z, \zeta) = \overline{K_\Omega(\zeta, z)}$, $z, \zeta \in \Omega$.
2. Verify the expressions for the kernel functions of the polydisc and Euclidean disc given in the example.

7 The Cousin problems

In this section we wish to consider the problem of constructing meromorphic functions with specified principle parts or zero and pole sets on a given domain in $\mathbb{C}^n$, $n > 1$. These questions were first raised by P. Cousin around the turn of the century and, following H. Cartan [1], now bear his name. As yet, of course, we have not defined meromorphic functions of more than one variable (see Chapter 3) but, as we showed in Chapter 1, it is possible to formulate both the Mittag-Leffler and Weierstrass theorems in a way that avoids explicit mention of meromorphic functions. We adopt this approach here but we must caution the reader that our consequent definition of the Cousin problems is *not* the classical one (see the remarks below).

Cousin's problem A. Let Ω be a domain in $\mathbb{C}^n$ and $\{U_i : i \in I\}$ be an open cover of Ω . Suppose we are given $f_{ij} \in A(U_{ij})$ such that for all i, j, k

$$f_{ij} = -f_{ji} \text{ on } U_{ij}; \quad f_{ij} + f_{jk} + f_{ki} = 0 \text{ on } U_{ijk}.$$

When can we find $f_i \in A(U_i)$ such that $f_{ij} = f_j - f_i$ on U_{ij} for all $i, j \in I$?

We say Ω is a *Cousin A domain* if we can always solve the Cousin A problem on Ω .

Cousin's problem B. Let Ω be a domain in $\mathbb{C}^n$ and $\{U_i : i \in I\}$ be an open cover of Ω . Suppose we are given $f_{ij} \in A^*(U_{ij})$ such that for all i, j, k

$$f_{ij} = f_{ji}^{-1} \text{ on } U_{ij}; \quad f_{ij}f_{jk}f_{ki} = 1 \text{ on } U_{ijk}.$$

When can we find $f_i \in A(U_i)$ such that $f_{ij} = f_j/f_i$ on U_{ij} for all $i, j \in I$?

We say Ω is a *Cousin B domain* if we can always solve the Cousin B problem on Ω .

Proposition 2.7.1. Let Ω be a domain in $\mathbb{C}^n$. Then

1. Ω is a Cousin A domain if and only if we can solve the Cauchy–Riemann equations on Ω .
2. Ω is a Cousin B domain if and only if every holomorphic line bundle on Ω is holomorphically trivial.

Proof. To say that we can solve the Cauchy–Riemann equations on Ω means that if we are given $f_j \in C^\infty(\Omega)$, $1 \leq j \leq n$, such that $\partial f_j / \partial \bar{z}_i = \partial f_i / \partial \bar{z}_j$, $1 \leq i, j \leq n$, then there exists $u \in C^\infty(\Omega)$ such that $\partial u / \partial \bar{z}_j = f_j$, $1 \leq j \leq n$ (see §3, especially Theorem 2.3.1). Suppose that we can solve the Cauchy–Riemann equations on Ω . Then, exactly as in the proof of Theorem 1.3.2, we can solve the Cousin A problem on Ω . We defer the proof of the converse to Chapter 6. The second statement is an immediate consequence of the definition of a holomorphically trivial line bundle (see §5, Chapter 1). $\square$

We know from Chapter 1 that every domain in $\mathbb{C}$ is a Cousin A and Cousin B domain. Notice that if Ω is a domain in $\mathbb{C}$ and $f_{ij} \in A(U_{ij})$ is the data for a Cousin A problem on Ω then we can find $m_j \in M(U_j)$ such that $f_{ij} = m_j - m_i$. Indeed, we can choose $m_j \in A(U_j)$ satisfying these conditions since Ω is a Cousin A domain! Hence, for domains in $\mathbb{C}$, there is really no difference between the Cousin A problem and the problem of constructing meromorphic functions with specified principal parts. Similarly for the multiplicative problem. It turns out that this equivalence between the Cousin A and B problems and the problem of finding meromorphic functions with specified principal parts or pole and zero sets no longer holds for domains in $\mathbb{C}^n$, $n > 1$. As we shall see in Chapter 12, it is possible, for example, to have domains in $\mathbb{C}^2$ for which we can always construct meromorphic functions with specified pole and zero sets but which are nevertheless not Cousin B (or A) domains. To clarify this point we now give the original definition of a Cousin II domain.

Definition 2.7.2. We say that a domain Ω in $\mathbb{C}^n$ is a *Cousin II domain* if given any open cover $\{U_i\}$ of Ω and analytic functions $f_i \in A^*(U_i)$ such that $f_i/f_j \in A^*(U_{ij})$ then there exists $F \in A(\Omega)$ such that $Ff_i^{-1} \in A^*(U_i)$ for all i .

Remark. We shall see in §4 of Chapter 3, if Ω is a Cousin II domain then we can construct meromorphic functions on Ω with specified pole and zero sets. However, a Cousin II domain need not be a Cousin B domain if $n > 1$.

Examples.

1. Every open polydisc $D \subset \mathbb{C}^n$ is a Cousin A domain. To prove this we shall show that the Cauchy–Riemann equations are solvable in D . Our proof will be very close to that of Theorem 1.3.1. Let $D = D(0; r_1, \dots, r_n)$ and for $s \in (0, 1]$, set $D_s = sD$. Suppose $f \in A(D)$. By Theorem 2.1.5

$$f(z) = \sum \frac{1}{m!} \partial^m f(0) z^m, \quad z \in D,$$

with uniform convergence on compact subsets of D . Hence, given $s \in (0, 1)$ and $\epsilon > 0$, there exists $N > 0$ such that $\|f - f_N\|_{\bar{D}_s} \leq \epsilon$, where $f_N(z) = \sum_{|m| \leq N} \frac{\partial^m f(0)}{m!} z^m$. Hence polynomials are dense in $A(D)$. Using this approximation result together with Theorem 2.3.1 we can now use the method of proof of Theorem A1.8 (Theorem 1.3.1) to show that the Cauchy–Riemann equations are solvable on D . We leave details to the reader (see also Theorem 5.8.2).

This example can be generalised in the following way: We say that $\Omega \subset \mathbb{C}^n$ is a *Runge domain* if polynomials are dense in $A(\Omega)$. It may be proved, using an ingenious inductive argument due to Oka, that the Cauchy–Riemann equations may always be solved on a Runge domain. For further details and references we refer the reader to Hörmander [1], Gunning and Rossi [1].

2. The domain $\Omega = \mathbb{C}^2 \setminus \{0\}$ is not a Cousin A domain. We shall show that the Cauchy–Riemann equations are not always solvable on Ω . Let $z = (z_1, z_2) \in \Omega$ and set $\|z\|^2 = |z_1|^2 + |z_2|^2$. We define functions f_1, f_2 by

$$\begin{aligned} f_1(z) &= \frac{\partial}{\partial \bar{z}_1} (\bar{z}_2 / z_1 \|z\|^2), \quad z_1 \neq 0 \\ &= \frac{\partial}{\partial \bar{z}_1} (\bar{z}_1 / z_2 \|z\|^2), \quad z_2 \neq 0 \\ f_2(z) &= \frac{\partial}{\partial \bar{z}_2} (\bar{z}_2 / z_1 \|z\|^2), \quad z_1 \neq 0 \\ &= \frac{\partial}{\partial \bar{z}_2} (\bar{z}_1 / z_2 \|z\|^2), \quad z_2 \neq 0. \end{aligned}$$

The identity $\frac{1}{z_1 z_2} = \bar{z}_2 / z_1 \|z\|^2 + \bar{z}_1 / z_2 \|z\|^2$, $z_1 z_2 \neq 0$, implies that $f_1, f_2 \in C^\infty(\Omega)$. Clearly we also have $\partial f_1 / \partial \bar{z}_2 = \partial f_2 / \partial \bar{z}_1$ on Ω . Suppose that there exists $u \in C^\infty(\Omega)$ such that $\partial u / \partial \bar{z}_j = f_j$, $j = 1, 2$. The function $G = z_1 u - \bar{z}_2 / \|z\|^2$ is holomorphic for $z_1 \neq 0$ since $z_1^{-1} \partial G / \partial \bar{z}_j = \partial u / \partial \bar{z}_j - f_j = 0$, $j = 1, 2$. Now G is locally bounded on Ω and so, by the Riemann removable singularities theorem, G extends to an analytic function on Ω . Hartogs theorem implies that G extends to an analytic function on $\mathbb{C}^2$. But $G(0, z_2) = 1/z_2$, $z_2 \neq 0$, and so G cannot be holomorphic at 0. Contradiction and so we cannot solve this set of Cauchy–Riemann equations on Ω . We shall show in Chapter 12 that $\mathbb{C}^n \setminus \{0\}$ is a Cousin A domain for $n \geq 3$.

Proposition 2.7.3. Let Ω be a domain in $\mathbb{C}^n$. Then Ω is a Cousin B domain if and only if we can solve the Cauchy–Riemann equations on Ω and $H^2(\Omega, \mathbb{Z}) = 0$.

Proof. If we can solve the Cauchy–Riemann equations on Ω and $H^2(\Omega, \mathbb{Z}) = 0$ then, exactly as in the Proof of Theorem 1.3.2 we can solve the Cousin B problem on Ω . We defer the proof of the converse to Chapter 6. $\square$

Example 3. Every open polydisc in $\mathbb{C}^n$ is a Cousin B domain.

The next theorem is very deep and we defer the proof until Chapter 12.

Theorem. The Cauchy–Riemann equations are solvable on every domain of holomorphy.

Corollary. If Ω is a domain of holomorphy, Ω is a Cousin A domain. If, in addition, $H^2(\Omega, \mathbb{Z}) = 0$, Ω is a Cousin B domain.

As we have already seen there are Cousin A domains which are not domains of holomorphy. However, in case $\Omega \subset \mathbb{C}^2$ it is true that a Cousin A(B) domain is a domain of Holomorphy (see H. Cartan [2] and Chapter 12). To actually characterise domains of holomorphy in terms of Cauchy–Riemann equations we have to take account of higher order versions of the Cauchy–Riemann equations. See Chapter 12.

Exercise. Show that Cauchy–Riemann equations are solvable on the Euclidean disc $E(r)$ (Use Exercise 1, §1).

Chapter 3

Local Rings of Analytic Functions

Introduction

In this chapter we shall make a study of algebraic and analytic properties of the ring $\mathbb{C}\{z_1 - a_1, \dots, z_n - a_n\}$ of convergent power series at a point $(a_1, \dots, a_n) \in \mathbb{C}^n$. It turns out that there is a rich and deep relationship between algebraic properties of power series rings and local properties of analytic functions and analytic sets. This relationship is expressed in its most elegant and powerful form using the concept of “coherence” which we introduce later in Chapter 7. In this chapter we show how to define meromorphic functions of more than one complex variable. We also prove some elementary facts about local properties of analytic sets and some not so elementary results about modules over $\mathbb{C}\{z_1 - a_1, \dots, z_n - a_n\}$. The main technical result we need is the most important Weierstrass Preparation theorem which we prove in section 2.

1 Elementary properties of power series rings

Throughout this Chapter $\mathbb{C}\{z_1 - a_1, \dots, z_n - a_n\}$ or just $\mathbb{C}\{z - a\}$ will denote the ring of convergent power series at a point $(a_1, \dots, a_n) = a \in \mathbb{C}^n$. We let $\mathcal{U}_a$ denote the set of all open neighbourhoods of a point $a \in \mathbb{C}^n$.

We first remark that $\mathbb{C}\{z - a\}$ has the structure of a commutative ring with identity under the obvious definitions of multiplication and addition. Given $a, b \in \mathbb{C}^n$, there is a natural isomorphism between $\mathbb{C}\{z - a\}$ and $\mathbb{C}\{z - b\}$ defined by mapping $\sum a_m(z - a)^m$ to $\sum a_m(z - b)^m$. Because of this isomorphism we may safely take $a = 0$ and this will result in no loss of generality as far as algebraic properties of these rings are concerned.

We now show another way to construct the ring $\mathbb{C}\{z - a\}$ in a “coordinate-free” manner. We define an equivalence relation on the set of all analytic functions which are defined on some neighbourhood of a . Suppose $f \in A(U)$, $g \in A(V)$, $U, V \in \mathcal{U}_a$. We write $f \sim_a g$ if there exists $W \in \mathcal{U}_a$ such that $f|_W = g|_W$. That is, $f \sim_a g$ if and only if $f = g$ on some neighbourhood of a . Clearly $\sim_a$ is an equivalence relation and we let $\mathcal{O}_a$ denote the set of equivalence classes of $\mathcal{O}_a$ (“ $\mathcal{O}$ ” in honour of the Japanese mathematician K. Oka who proved many of the fundamental results in the theory of several complex variables). If $f \in A(U)$, $U \in \mathcal{U}_a$, we let f_a denote the $\sim_a$ equivalence class of f in $\mathcal{O}_a$. We call f_a the *germ* of f at a . Obviously, $\mathcal{O}_a$ inherits the structure of a commutative ring with 1 from the corresponding structures on the $A(U)$. We claim that $\mathbb{C}\{z - a\}$ and $\mathcal{O}_a$ are naturally isomorphic. Indeed, if $P \in \mathbb{C}\{z - a\}$, we may define $\theta(P) \in \mathcal{O}_a$ to be the germ at a of the analytic function defined on some neighbourhood of a by the power series P . Conversely, if $f_a \in \mathcal{O}_a$ is the germ of an analytic function $f \in A(U)$, $U \in \mathcal{U}_a$, we define $\theta^{-1}(f_a)$ to be the convergent power series of f at a . Clearly θ is a ring isomorphism. In the sequel, we identify the rings $\mathbb{C}\{z - a\}$, $\mathcal{O}_a$ via θ and generally use the

more abbreviated notation $\mathcal{O}_a$ in preference to $\mathbb{C}\{z - a\}$. We generally omit reference to n . If essential we write ${}_n\mathcal{O}_a$ to denote the ring of germs of analytic functions at a point $a \in \mathbb{C}^n$.

Remark. Our construction of $\mathcal{O}_a$ as the ring of germs of analytic functions at a works equally well for other classes of functions. For example, we may construct the ring C_a of *germs of continuous functions* at a or the ring $\mathcal{D}_a$ of *germs of C^∞ functions* at a . Clearly $\mathcal{O}_a \subset \mathcal{D}_a \subset C_a$. However, the rings C_a and $\mathcal{D}_a$ do not have a simple representation in terms of power series rings nor do they have the rich algebraic structure of $\mathcal{O}_a$ and most of the results we prove for $\mathcal{O}_a$ fail for C_a and $\mathcal{D}_a$.

Lemma 3.1.1. The ring $\mathcal{O}_a$ is an integral domain. That is, if $f, g \in \mathcal{O}_a$ and $fg = 0$, then either $f = 0$ or $g = 0$.

Proof. We may choose a connected open neighbourhood U of a and $F, G \in A(U)$ such that $F_a = f$ and $G_a = g$. Now $fg = 0$ implies that $FG = 0$ on U by uniqueness of analytic continuation. Suppose $F \neq 0$. Then F is non-zero on a non-empty subset V of U . Hence G is zero on V and, by uniqueness of analytic continuation, identically zero on U . Hence $g = 0$. $\square$

Before giving the next lemma we recall

Definition 3.1.2. Let R be an integral domain with identity. An element $u \in R$ is said to be a *unit* if there exists $u^{-1} \in R$ such that $uu^{-1} = 1$.

Lemma 3.1.3.

1. An element $u \in \mathcal{O}_a$ is a unit if and only if $u(a) \neq 0$.
2. $\mathcal{O}_a$ has a unique maximal ideal $\mathfrak{m}_a$ which is characterised as being the set of non-units of $\mathcal{O}_a$. That is, $f \in \mathfrak{m}_a$ if and only if $f(a) = 0$.

Proof. We first remark that if $f \in \mathcal{O}_a$ then $f(a)$ is, of course, just the value of any representative function for f at a ; equivalently, the constant term in the power series for f . The lemma follows from the observations that if f is an analytic function which is non-zero at a , then $1/f$ is defined and analytic on some neighbourhood of a and any ideal of $\mathcal{O}_a$ which contains a unit is equal to $\mathcal{O}_a$. $\square$

Exercises

1. Let U be an open subset of $\mathbb{C}^n$. Prove that $A(U)$ is an integral domain if and only if U is connected.
2. Show that $\mathcal{D}_a$, the ring of germs of C^∞ complex valued functions at a point $a \in \mathbb{C}^n$, has a unique maximal ideal. Show also that $\mathcal{D}_a$ is not an integral domain.
3. Prove that $\mathcal{O}_a/\mathfrak{m}_a^{k+1}$ is naturally isomorphic to the complex vector space of complex valued complex polynomials on $\mathbb{C}^n$ of degree $\leq k$. (Hint: Look at the Taylor polynomial of degree k of an analytic function).
4. If $\mathfrak{m}_a^\infty$ denotes the maximal ideal of $\mathcal{D}_a$, prove that $\mathcal{D}_a/(\mathfrak{m}_a^\infty)^{k+1}$ is naturally isomorphic to the vector space of complex valued real polynomials on $\mathbb{C}^n$ of degree $\leq k$.
5. Show that there is a natural isomorphism of $\mathfrak{m}_a^k/\mathfrak{m}_a^{k+1}$ onto the complex vector space of complex valued homogeneous polynomials on $\mathbb{C}^n$ of degree k .

2 Weierstrass Division and Preparation Theorems

In this section we prove the basic technical result which allows us to prove in §3 that $\mathcal{O}_a$ is a Noetherian unique factorization domain. Before giving the main result, we remark that we take coordinates $z = (z', z_n) = (z_1, \dots, z_{n-1}, z_n)$ on $\mathbb{C}^n$, always using the prime to denote the $\mathbb{C}^{n-1}$ variable.

Theorem 3.2.1. (The Weierstrass Division Theorem) Let f be analytic on a neighbourhood Ω of $0 \in \mathbb{C}^n$ and suppose that $f(0, \dots, 0, z_n)$ has a zero of multiplicity $p \geq 0$ at $z_n = 0$. Then we may find an open polydisc neighbourhood $D \subset \Omega$ of 0 such that every bounded analytic function g on D may be written uniquely in the form

$$g = qf + h,$$

where q and h are bounded analytic functions on D and h is a polynomial in z_n of degree $\leq p - 1$ with coefficients depending analytically on z' . q and h depend continuously on g in the sense that there exists a constant M , independent of g , such that

$$\|q\|_D, \|h\|_D \leq M\|g\|_D.$$

Proof. Choose a polydisc neighbourhood ω of 0, contained in Ω , on which f is bounded. On ω we may write

$$f(z) = \sum_{j=0}^{\infty} f_j(z') z_n^j$$

and, as is easily seen using Cauchy's inequalities (Corollary 2.1.7), each f_j is bounded on ω . Our hypotheses on f imply that $f_p(0) \neq 0$ and $f_j(0) = 0$, $j < p$. From now on we shall always work inside ω . Choose $d, s > 0$ such that if

$$\begin{aligned} D' &= \{z' \in \mathbb{C}^{n-1} : |z_j| < d, 1 \leq j \leq n-1\}, \\ D &= \{z \in \mathbb{C}^n : z' \in D' \text{ and } |z_n| < s\}, \end{aligned}$$

then f_p is nonvanishing on D' and $1/f_p$ is bounded on D' . Multiplying f by $1/f_p$, it is no loss of generality to suppose that $f_p = 1$ on D' .

Let $A^B(D)$ denote the algebra of bounded analytic functions on D . Define $\|u\| = \|u\|_D = \sup_D |u(z)|$, $u \in A^B(D)$. Then $A^B(D)$ is complete with respect to $\|\cdot\|$ and so has the structure of a Banach algebra.

Since $f_j(0) = 0$, $j < p$, there exists $C_1 \geq 0$ such that $\|f_j(z') z_n^j\| \leq C_1 d s^j / p$, $j < p$. Here C_1 is independent of d and s and depends only on, say, $\|f_j\|_\omega$. Similarly there exists $C_2 \geq 0$, independent of d and s , such that

$$\left\| \sum_{j>p} f_j(z') z_n^j \right\| \leq C_2 s^{p+1}.$$

Hence

$$\|f - z_n^p\| \leq C_1 d(1 + s^p) + C_2 s^{p+1}.$$

Now choose d and s sufficiently small so that

$$\|f - z_n^p\| \leq s^p / (2(p+1)) \tag{1}$$

From now on d, s and the corresponding polydiscs D', D will remain fixed. We now prove the division theorem in the special, trivial, case $f = z_n^p$. We derive the general case by regarding f as a perturbation of z_n^p , using estimate 1 above.

Given $g \in A^B(D)$, write

$$g = q(g)z_n^p + h(g),$$

where $q(g), h(g) \in A^B(D)$ and $h(g)$ is a polynomial in z_n of degree $\leq p-1$. $q(g)$ and $h(g)$ are unique. Moreover

$$h(g)(z) = \sum_{j=0}^{p-1} \frac{\partial^j g}{\partial z_n^j}(z', 0) \frac{z_n^j}{j!}$$

and so it follows from Cauchy's inequalities (Corollary 2.1.7) that

$$\|h(g)\| \leq p\|g\| \quad (2)$$

Hence, by the triangle inequality, $\|q(g)z_n^p\| \leq (p+1)\|g\|$ and so by Schwarz' Lemma (exercise 7, §1, Chapter 1)

$$\|q(g)\| \leq (p+1)s^{-p}\|g\| \quad (3)$$

(Equivalently, argue using the maximum modulus theorem on polydiscs εD , $0 < \varepsilon < 1$, and let $\varepsilon \rightarrow 1$).

Estimates 2 and 3 imply that the linear endomorphisms $g \mapsto h(g), q(g)$ of $A^B(D)$ are continuous.

We define a continuous linear map $A : A^B(D) \rightarrow A^B(D)$ by

$$A(u) = q(u)f + h(u).$$

Now

$$\begin{aligned} \|(A - I)u\| &= \|q(u)f + h(u) - u\| \\ &= \|q(u)(f - z_n^p)\| \\ &\leq \|q(u)\| \|f - z_n^p\| \\ &\leq \frac{1}{2}\|u\|, \end{aligned}$$

using estimates 1 and 3. Hence $\|A - I\| \leq \frac{1}{2}$ and so A is a linear isomorphism (for a proof of this elementary result from Banach space theory see Dieudonné [1, page 148] or Field [1, page 189]). It follows that for any $g \in A^B(D)$, there exists $u \in A^B(D)$ such that

$$g = A(u) = q(u)f + h(u).$$

This proves the first part of the theorem. Now $q(u) = q(A^{-1}(g))$ and so by estimate 3 we have

$$\begin{aligned} \|q(u)\| &\leq (p+1)s^{-p}\|A^{-1}(g)\| \\ &\leq \|A^{-1}\|s^{-p}/(p+1)\|g\|. \end{aligned}$$

Similarly $\|h(u)\| \leq p\|A^{-1}\|\|g\|$ and so we have obtained the required estimates on $\|q\|$ and $\|h\|$. Finally the uniqueness of q and h is a trivial consequence of the injectivity of A . $\square$

Remarks.

1. The proof we give here is based on one due to Grauert and Remmert (See R. Narasimhan [3] and also Hörmander [1]). For alternative proofs based on Cauchy's integral formula, we refer the reader to Gunning and Rossi [1], R. Narasimhan [3], Whitney [1].
2. Notice that if f is any non-identically zero analytic function defined on a neighbourhood of $0 \in \mathbb{C}^n$, we may always find a $\mathbb{C}$ -linear change of coordinates on $\mathbb{C}^n$ such that the conditions of the Theorem hold for f . We then say f is *normalised in direction z_n* . Of course, we can always simultaneously normalise any finite set of analytic functions.

3. An immediate consequence of Theorem 3.2.1 is the Weierstrass Division Theorem for germs of analytic functions: Suppose $f \in \mathcal{O}_0$ and that for some representative F of f , $F(0, z_n)$ has a zero of multiplicity p at $z_n = 0$. Then any $g \in \mathcal{O}_0$ may be written uniquely in the form $g = qf + h$, where $q \in \mathcal{O}_0$ and h is the germ of a polynomial in z_n of degree less than p and with coefficients analytic in z' .

Examining the proof of Theorem 3.2.1, we see that the result remains true if we replace $D = D' \times D_s(0)$ by $D'' \times D_s(0)$ where D'' is any open neighbourhood of 0 in $\mathbb{C}^{n-1}$ which is contained in D' . Indeed estimates 1 and 3 obviously continue to hold on the smaller neighbourhood $D'' \times D_s(0)$. Moreover, given a finite set $f_1, \dots, f_k$ of analytic functions defined on a neighbourhood of 0 in $\mathbb{C}^n$, we may, by Remark 2, above, assume that the functions are simultaneously normalised in the z_n -direction. We can then choose a polydisc $D' \times D_s(0)$ in $\mathbb{C}^n$ for which estimates 1, 2 and 3 are valid for all the functions f_j . A straightforward consequence of these observations is the following strengthened form of the Division Theorem that we shall need in §6.

Theorem 3.2.1'. Let $f_1, \dots, f_k$ be analytic functions defined on a neighbourhood of 0 in $\mathbb{C}^n$ and suppose that each f_j is normalised in direction z_n . Fix a complementary subspace $\mathbb{C}^{n-1}$ of $\mathbb{C}z_n$. Then we may find an open neighbourhood W of 0 in $\mathbb{C}^{n-1}$ and $s > 0$ such that if U is any open neighbourhood of 0 in $\mathbb{C}^{n-1}$ contained in W then the conclusions of the Weierstrass Division theorem hold for all bounded analytic functions on $U \times D_s(0)$ with respect to each divisor f_j , $1 \leq j \leq k$.

Remark. As an exercise the reader may verify that the Division Theorem does not hold for arbitrary neighbourhoods of 0 contained in $W \times D_s(0)$.

Our first application of the Weierstrass Division theorem will be to put the germ of an analytic function in a normalised form. First, we give a definition.

Definition 3.2.2. A germ $P \in \mathcal{O}_0$ is said to be a *Weierstrass polynomial (of degree p)* if (on some neighbourhood of 0)

$$P(z_1, \dots, z_n) = z_n^p + \sum_{j=0}^{p-1} a_j(z') z_n^j$$

where the a_j are analytic functions defined on some neighbourhood of $0 \in \mathbb{C}^{n-1}$ and vanishing at 0 for every j .

We similarly define a Weierstrass polynomial $P \in \mathcal{O}_a$, $a \in \mathbb{C}^n$.

In what follows we let $\mathcal{O}'_0$ denote the ring of germs of analytic functions at $0 \in \mathbb{C}^{n-1}$.

Theorem 3.2.3. (The Weierstrass Preparation Theorem). Let $f \in \mathcal{O}_0$ and suppose that $f(0, z_n)$ has a zero of order $p \geq 0$ at $z_n = 0$. Then there exists a unique unit $h \in \mathcal{O}_0$ and Weierstrass polynomial W of degree p such that $f = hW$. That is, on some neighbourhood of $0 \in \mathbb{C}^n$ we have

$$f(z) = h(z) \left(z_n^p + \sum_{j=0}^{p-1} a_j(z') z_n^j \right)$$

where $h(0) \neq 0$ and $a_j(0) = 0$, $j < p$.

Proof. Take $g = z_n^p$ in Theorem 3.2.1. □

Remarks.

1. There are analogues of the Division and Preparation theorems for differentiable functions which were first proved by Malgrange [1]. These results are important in the theory of singularities of differentiable mappings and we refer to Malgrange [1] and Tougeron [1] for further details.

2. In Whitney [2, §10] there is a discussion about the set of directions z_n which lead to a Weierstrass polynomial of minimal degree. In Levinson [1, 2] the reader may find results showing that we can make a local holomorphic change of coordinates to put an analytic function in the standard Weierstrass polynomial form.

3 Factorization and finiteness properties of $\mathcal{O}_0$

Let us start by recalling some definitions from algebra.

Definition 3.3.1. Let R be a commutative ring with identity

1. $f \in R$ is said to be *irreducible* or *prime* if any relation $f = gh$, $g, h \in R$, implies that one or other of f and g is a unit.
2. R is said to be a *unique factorization domain* if every $f \in R$ can be written as a finite product of irreducible factors and this decomposition of f is unique up to the order of the factors and multiplication by units.
3. R is said to be *Noetherian* if every ideal $I \triangleleft R$ is finitely generated. That is, if there exist $g_1, \dots, g_k \in I$ such that

$$I = \{\sum a_j g_j : \text{all } a_j \in R\}.$$

4. If $M \subset R^m$ (R^m denotes the module of m -tuples of elements in R), M is said to be a *submodule* of R^m if M is closed under (coordinate-wise) addition and scalar multiplication by elements of R .

Notation. Given a commutative ring R , $R[X]$ will always denote the ring of polynomials in the indeterminate X and with coefficients in R .

Lemma 3.3.2. Let $W \in \mathcal{O}'_0[z_n]$ be a Weierstrass polynomial and suppose that

$$W = P_1 \cdots P_q,$$

where $P_j \in \mathcal{O}'_0[z_n]$. Then each P_j is a Weierstrass polynomial up to multiplication by units in $\mathcal{O}'_0$.

Proof. Set $p = \text{degree}(W)$, $p_j = \text{degree}(P_j)$, $1 \leq j \leq q$. Taking $z' = 0$ we have

$$z_n^p = \prod_{i=1}^q P_j(0, z_n).$$

Since $p = \sum p_j$, we must have the coefficient of $z_n^{p_j}$ in P_j non-vanishing at $z' = 0$. Since $\mathbb{C}[z_n]$ is a unique factorization domain, the coefficients of all the lower order terms in P_j vanish at $z' = 0$. Hence $P_j = u_j W_j$, where W_j is a Weierstrass polynomial and u_j , the coefficient of $z_n^{p_j}$ in $P_j(0, z_n)$, is a unit in $\mathcal{O}'_0$. $\square$

Lemma 3.3.3. Let $W \in \mathcal{O}'_0[z_n]$ be a Weierstrass polynomial. Then W is irreducible in the ring $\mathcal{O}'_0[z_n]$ if and only if it is irreducible in $\mathcal{O}_0$.

Proof. The other implication being trivial, it is enough to show that if W is not irreducible in $\mathcal{O}_0$ then it is not irreducible in $\mathcal{O}'_0[z_n]$. Suppose then that $W = fg$, where $f, g \in \mathcal{O}_0$ are not units. By the Weierstrass Preparation theorem we may write $f = UW_1$, $g = VW_2$, where $W_1, W_2 \in \mathcal{O}'_0[z_n]$ are Weierstrass polynomials and U and V are units. Thus $W = uW_1W_2$ for some unit $u \in \mathcal{O}_0$. Since W_1W_2 is a Weierstrass polynomial and $W = uW_1W_2 = 1W$, the uniqueness part of the preparation theorem implies that $u = 1$ and $W = W_1W_2$. Hence we have shown that W is not irreducible in $\mathcal{O}'_0[z_n]$. $\square$

Theorem 3.3.4. $\mathcal{O}_0$ is a unique factorization domain.

Proof. Our proof goes by induction on n . For $n = 0$, $\mathcal{O}_0 = \mathbb{C}$ and the result is trivial. Suppose true for $n - 1$. By Gauss' theorem (see Van der Waerden [1, page 70]) $\mathcal{O}'_0[z_n]$ is a unique factorization domain. Let $f \in \mathcal{O}_0$. Then $f = uW$, where u is a unit in $\mathcal{O}_0$ and W is a Weierstrass polynomial. Since $\mathcal{O}'_0[z_n]$ is a unique factorization domain, W may be written as a unique, up to order, product of irreducible polynomials $W_j \in \mathcal{O}'_0[z_n]$: $W = W_1 \cdots W_q$. By Lemma 3.3.3, each W_j is irreducible in $\mathcal{O}_0$ and so we have expressed f as a finite product of irreducible elements of $\mathcal{O}_0$. We leave the proof of uniqueness, up to order and multiplication by units, as an exercise for the reader. $\square$

Before proving that $\mathcal{O}_0$ is Noetherian, we recall

Lemma 3.3.5. Let R be a Noetherian ring and M be a submodule of R^p , then M is a finitely generated R -module.

Proof. Let $\pi : R^p \rightarrow R$ denote the projection on the first factor and set $M_1 = \pi M \subset R$ and $M_2 = \text{Kernel } \pi|_M \subset R^{p-1}$. M_1 is an ideal and so we may find $f_1, \dots, f_m \in M$ such that the set $\pi(f_1), \dots, \pi(f_m)$ generates M_1 . Let $\widetilde{M}_1$ denote the submodule of M generated by $f_1, \dots, f_m$. The inductive hypothesis implies that M_2 is finitely generated. Since $M = \widetilde{M}_1 + M_2$, it therefore follows that M is finitely generated. $\square$

Theorem 3.3.6. The ring $\mathcal{O}_0$ is Noetherian.

Proof. Our proof goes by induction on n . For $n = 1$, the result is trivially true since every ideal is generated by a power of z_1 . Suppose the result is true for $n - 1$ and let I be a non-zero ideal of $\mathcal{O}_0$. Changing coordinates if necessary and applying the Preparation theorem we may find a Weierstrass polynomial $W \in I$. For any $f \in I$ we then have by the Division theorem

$$f = q(f)W + r(f),$$

where $r(f)$ is a polynomial in z_n of degree less than p , the degree of W . Let $M = \{r(f) : f \in I\}$. If we write $r(f) = a_1(z')z_n^{p-1} + \cdots + a_p(z')$, where $a_j \in \mathcal{O}'_0$, we see that the set of coefficients $(a_1, \dots, a_p)$ corresponding to polynomials $r(f)$, $f \in I$, defines a submodule of $\mathcal{O}'_0{}^p$. By Lemma 3.3.5 and our inductive hypothesis, we may find a finite set of generators $(h_{11}, \dots, h_{1p}), \dots, (h_{s1}, \dots, h_{sp})$ for this submodule. In other words, the set of polynomials

$$p_j(z', z_n) = h_{j1}(z')z_n^{p-1} + \cdots + h_{jp}(z'), \quad 1 \leq j \leq s,$$

generates M over $\mathcal{O}'_0$. Therefore $\{W, p_1, \dots, p_s\}$ is a finite set of generators for I . $\square$

Exercises

1. Let U be a connected subset of $\mathbb{C}^n$. Show that if $n = 1$ or U is a domain of holomorphy then $A(U)$ is not a unique factorization domain. (Hint: For the case $n = 1$, use the Weierstrass theorem. For domains of holomorphy see the proof of Theorem 2.4.8). In fact $A(U)$ is never a unique factorization domain. For further details on uniqueness of factorization in $A(U)$ we refer the reader to Whitney [1, pages 37–40, 332].
2. Let f be a representative of $f_0 \in \mathcal{O}_0$. Show that if $Df(0) \neq 0$, then f_0 is irreducible (Hint: Apply the implicit function theorem).
3. Let $f(y, z) = y^2 - z^2(1 - z)$. Show that f_0 is not irreducible even though f is irreducible as a polynomial in $\mathbb{C}[y, z]$.

4 Meromorphic functions

In this section we apply the theory of §§2,3 to show how we may give a satisfactory definition of a meromorphic function of more than one complex variable.

Lemma 3.4.1. Let $W, P \in \mathcal{O}'_0[z_n]$, $U \in \mathcal{O}_0$ and suppose that $P = WU$. If W is a Weierstrass polynomial then $U \in \mathcal{O}'_0[z_n]$.

Proof. The leading coefficient of W is a unit in $\mathcal{O}'_0$ and so we may apply the polynomial division algorithm to obtain $P = WU' + R$, where $U', R \in \mathcal{O}'_0[z_n]$ and R is of degree less than W . But the uniqueness part of the Weierstrass Division theorem implies that $R = 0$, $U' = U$. $\square$

Proposition 3.4.2. Let f and g be analytic functions defined on some neighbourhood of $0 \in \mathbb{C}^n$. Suppose that f_0 and g_0 are relatively prime in $\mathcal{O}_0$ (that is, they have no common irreducible factor). Then we may find an open neighbourhood D of $0 \in \mathbb{C}^n$ such that $f_a, g_a \in \mathcal{O}_a$ are relatively prime for all $a \in D$.

Proof. Without loss of generality we may suppose that f and g are Weierstrass polynomials. By Lemma 3.3.3, f_0 and g_0 are relatively prime in $\mathcal{O}'_0[z_n]$. Since $\mathcal{O}'_0$ is an integral domain, we may form its quotient field $\mathcal{M}'_0$. It follows from Gauss' lemma that f_0 and g_0 are relatively prime in $\mathcal{M}'_0[z_n]$. Hence there exist $a_0, b_0 \in \mathcal{O}'_0[z_n]$ and $h_0 \in \mathcal{O}'_0$, $h_0 \neq 0$, such that $h_0 = a_0 f_0 + b_0 g_0$. On some neighbourhood of $0 \in \mathbb{C}^n$ we therefore have

$$h(z') = a(z)f(z) + b(z)g(z).$$

(As usual, a, b and h denote representatives for the germs a_0, b_0 and h_0 respectively). Since f and g are Weierstrass polynomials we may choose a polydisc neighbourhood D' of $0 \in \mathbb{C}^{n-1}$ such that for all $z' \in D'$ the polynomials $f(z', z_n)$ and $g(z', z_n)$ do not vanish identically. Choose $r > 0$ so that $f, g, a, b, h \in A(D)$, where $D = \{z \in \mathbb{C}^n : z' \in D' \text{ and } |z_n| < r\}$. Let $\zeta = (\zeta', \zeta_n) \in D$. Making the substitution $z_n = (z_n - \zeta_n) + \zeta_n$ in a, b, f and g we may consider the relation $h_\zeta = a_\zeta f_\zeta + b_\zeta g_\zeta$ as defining an equation in $\mathcal{O}'_0[z_n - \zeta_n]$. Since f_ζ and g_ζ are not identically zero, we may assume that any common factor of f_ζ and g_ζ is the germ of a Weierstrass polynomial W in $\mathcal{O}'_0[z_n - \zeta_n]$. But then by Lemma 3.4.1, this gives a factorization of h_ζ into a product WP , where $P \in \mathcal{O}'_0[z_n - \zeta_n]$. Hence W must be of degree zero and so equal to 1. Therefore W is a unit. $\square$

Remark. Proposition 3.4.2 gives a beautiful example of a characteristic “coherence” result in complex analysis: An algebraic result that holds at a point tends to hold in a neighbourhood of the point. In other words we have a transition from algebra (results holding in the ring of germs at a point) to a local result (statements holding on an open set).

From now on we shall let $\mathcal{M}_a$ denote the quotient field of $\mathcal{O}_a$. Thus $\mathcal{M}_a = \{f_a/g_a : f_a, g_a \in \mathcal{O}_a \text{ with } g_a \neq 0\}$. Since $\mathcal{O}_a$ is a unique factorization domain each $m \in \mathcal{M}_a$ may be written uniquely (up to multiplication by units) in the form f_a/g_a where f_a and g_a are relatively prime.

Suppose that $m \in \mathcal{M}_a$. Write $m = f_a/g_a$, where f_a and g_a are relatively prime. If $g_a(a) \neq 0$, we may define the value of m at a , $m(a)$, to be $f_a(a)/g_a(a)$. Clearly $m(a)$ does not depend on our choices for f_a and g_a provided only that they are relatively prime. In case $n = 1$, if $g_a(a) = 0$ then $f_a(a) \neq 0$ and we can define $m(a) = \infty$ (see Chapter 1, §4, Example 1). For $n > 1$, it is certainly possible for us to have $g_a(a) = f_a(a) = 0$ and then there is no sensible way to define $m(a)$ as the next lemma shows.

Lemma 3.4.3. Let f and g be analytic on some neighbourhood of $0 \in \mathbb{C}^n$, $n > 1$, and suppose $f(0) = g(0) = 0$ and f_0 and g_0 are relatively prime. Then given any $\zeta \in \mathbb{C}$, there exist z arbitrarily close to 0 such that $g(z) \neq 0$ and $f(z)/g(z) = \zeta$.

Proof. Replacing f by $f - \zeta g$, it is no loss of generality to suppose that $\zeta = 0$. We may then assume that f and g are Weierstrass polynomials and, exactly as in the proof of Proposition 3.4.2, we may find $a, b \in \mathcal{O}'_0[z_n]$ and $h \in \mathcal{O}'_0$, $h \neq 0$, such that on some neighbourhood of 0 we have

$$h(z') = a(z)f(z', z_n) + b(z)g(z', z_n) \quad (1)$$

If the lemma is false, there exists an open neighbourhood D of 0 such that $f(z) = 0$ implies $g(z) = 0$, $z \in D$. But for small z' , the polynomial $f(z', z_n)$ certainly has a small zero z_n and so it follows from 1 that $h(z') = 0$ for all z' in some neighbourhood of 0. Contradiction. $\square$

Let us set

$$\mathcal{M} = \bigcup_{a \in \mathbb{C}^n} \mathcal{M}_a.$$

(Disjoint union over $\mathbb{C}^n$). $\mathcal{M}$ is called the *sheaf of germs of meromorphic functions* on $\mathbb{C}^n$.

Definition 3.4.4. Let Ω be an open subset of $\mathbb{C}^n$. A map $m : \Omega \rightarrow \mathcal{M}$ is said to define a *meromorphic function* on Ω if

1. $m(z) \in \mathcal{M}_z$ for all $z \in \Omega$ (m is a “section” of $\mathcal{M}$)
2. For every $z \in \Omega$, there exists $V \in \mathcal{U}_z$ and $f, g \in A(V)$ such that $m(a) = f_a/g_a$, for all $a \in V$.

We denote the set of meromorphic functions on Ω by $M(\Omega)$.

Example 1. Let $\Omega \subset \mathbb{C}^n$ be open and $f, g \in A(\Omega)$, where g is not identically zero on any connected component of Ω . We may define $m \in M(\Omega)$ by

$$m(z) = f_z/g_z, \quad z \in \Omega.$$

We often write $m = f/g$.

Proposition 3.4.5. Let $\Omega \subset \mathbb{C}^n$ be open and $m \in M(\Omega)$. We may find an open cover $\{U_j\}$ of Ω and $m_j \in M(U_j)$ such that

1. $m_j = m|_{U_j}$
2. For every j , there exist $f_j, g_j \in A(U_j)$ such that $m_j = f_j/g_j$ and $f_{j,a}$ and $g_{j,a}$ are relatively prime for all $a \in U_j$.

Proof. Immediate from Proposition 3.4.2. $\square$

Remark. Proposition 3.4.5 is, of course, relatively trivial if $n = 1$: Every meromorphic function m on $\Omega \in \mathbb{C}$ may be written in the form $u(z)(z - a)^p$ on some neighbourhood of each point $a \in \Omega$ with $u(a) \neq 0$ and $p \in \mathbb{Z}$. In fact by Corollary 1.3.4 we can write $m = f/g$, with f_a and g_a relatively prime at every point $a \in \Omega$. It is true, though hard to prove, that a meromorphic function on an arbitrary domain in $\mathbb{C}^n$ can be written as a quotient f/g (problem of Poincaré). However, as we shall see later in Chapter 12, it is not true for arbitrary domains in $\mathbb{C}^n$ that we can require f_a and g_a to be everywhere relatively prime.

Notation. If $f_a, g_a \in \mathcal{O}_a$ are relatively prime we shall write $(f_a, g_a) = 1$. More generally, if $f, g \in A(U)$ and $(f_a, g_a) = 1$ for all $a \in U$ we shall write $(f, g) = 1$.

The local description of meromorphic functions given by Proposition 3.4.5 is essentially unique as the next lemma shows.

Lemma 3.4.6. Let $\Omega \subset \mathbb{C}^n$ be open and $m \in M(\Omega)$. Suppose that for some open subset $V \subset \Omega$ we may find $f, f', g, g' \in A(V)$ such that

1. $m|_V = f/g = f'/g'$.
2. $(f, g) = 1, (f', g') = 1$.

Then there exists $u \in A(V)$ such that

- A. $f = uf', g = ug'$.
- B. u is non-vanishing on V . That is, u_a is a unit for all $a \in V$.

Proof. At each point $a \in V$ we have $f_a/g_a = f'_a/g'_a$. Hence $f_ag'_a = g_af'_a$. If f_a is not a unit, f_a must divide f'_a , since $(f_a, g_a) = 1$. Hence there exists a unique $u_a \in \mathcal{O}_a$ such that $u_af_a = f'_a$. Since f'_a is therefore not a unit and $(f'_a, g'_a) = 1$, there exists $v_a \in \mathcal{O}_a$ such that $v_af'_a = f_a$. Substituting, we find that $u_av_a = 1$ and so u_a is a unit. Hence f'_a/f_a is a unit in $\mathcal{O}_a$. Similarly, if g_a is not a unit, g'_a/g_a is a unit in $\mathcal{O}_a$. It follows that for all $a \in V$, there exists a unique unit $u_a \in \mathcal{O}_a$ such that $f'_a/f_a = g'_a/g_a = u_a$. But the equation $f_a u_a = f'_a$ uniquely defines an analytic function u on V satisfying conditions A and B of the lemma. $\square$

Using the above results we are now able to define the pole and zero sets of a meromorphic function and prove that they are analytic subsets (see Definition 2.2.1).

Suppose that $\Omega \subset \mathbb{C}^n$ is open and $m \in M(\Omega)$. By Proposition 3.4.5, we may find an open cover $\{U_j\}$ of Ω and $m_j \in M(U_j)$ such that $m_j = m|_{U_j}$ and $m_j = f_j/g_j$ with $(f_j, g_j) = 1$. We define the subsets $Z(m), P(m), T(m)$ by

$$\begin{aligned} Z(m) \cap U_j &= \{z \in U_j : f_j(z) = 0\}, \\ P(m) \cap U_j &= \{z \in U_j : g_j(z) = 0\}, \\ T(m) \cap U_j &= \{z \in U_j : f_j(z) = g_j(z) = 0\}. \end{aligned}$$

We claim that $Z(m), P(m)$ and $T(m)$ are well defined analytic subsets of Ω which depend only on m (not on the local representation of m on U_j). To see this, choose another open cover $\{U'_k\}$ of Ω together with relatively prime $f'_k, g'_k \in A(U'_k)$ satisfying $m|_{U'_k} = f'_k/g'_k$. On the overlap $U_j \cap U'_k$ we have $f_j/g_j = f'_k/g'_k$ and so, by Lemma 3.4.6, there exists a nowhere zero $u \in A(U_j \cap U'_k)$ such that on $U_j \cap U'_k$

$$f_j = uf'_k \quad \text{and} \quad g_j = ug'_k.$$

But this implies that $f_j(z) = 0$ if and only if $f'_k(z) = 0$ and $g_j(z) = 0$ if and only if $g'_k(z) = 0$. Hence $Z(m), P(m)$ and $T(m)$ are well defined analytic sets.

Definition 3.4.7. Let m be a meromorphic function defined on a domain in $\mathbb{C}^n$. The analytic sets $Z(m), P(m)$ and $T(m)$ constructed above are called the *zero set* of m , *pole set* of m and *indeterminacy set* of m respectively.

Example 2. For $n = 1$, the indeterminacy set is always empty (see the comments immediately preceding Lemma 3.4.3). For $n > 1$, the indeterminacy set may be non-empty: Let $f(z_1, z_2) = z_1^2 - z_2^3, g(z_1, z_2) = z_1 z_2$. Define $m \in M(\mathbb{C}^2)$ by $m = f/g$. Then $Z(m) = \{(t^3, t^2) : t \in \mathbb{C}\}, P(m) = \{(z_1, z_2) : z_1 = 0 \text{ or } z_2 = 0\}$ and $T(m) = \{(0, 0)\}$.

As follows from Lemma 3.4.3 there is no way to define the value of a meromorphic function at points of its indeterminacy set. However, we do have

Proposition 3.4.8. Let m be a meromorphic function on the domain Ω in $\mathbb{C}^n$. Then m defines an analytic function on $\Omega \setminus P(m)$.

Proof. Let $\{U_j\}$ be an open cover of Ω and $f_j, g_j \in A(U_j)$ satisfy the conditions of Proposition 3.4.5. We may define $m : \Omega \setminus P(m) \rightarrow \mathbb{C}$ by

$$m(z) = f_j(z)/g_j(z), \quad z \in U_j \setminus P(m).$$

We leave it to the reader to check that this construction gives a well defined analytic function on $\Omega \setminus P(m)$. $\square$

To conclude this section, we briefly return to the problem of constructing meromorphic functions with specified principal parts or pole and zero sets (see §7 of Chapter 2).

Definition 3.4.9. Let Ω be a domain in $\mathbb{C}^n$ and $\{U_i : i \in I\}$ be an open cover of Ω . Suppose we are given $m_i \in M(U_i)$ such that $m_i - m_j \in A(U_{ij})$ for all $i, j \in I$. The *Cousin I problem* is to construct $m \in M(\Omega)$ such that $m - m_i \in A(U_i)$ for all $i \in I$.

If we can always solve the Cousin I problem on Ω , we call Ω a *Cousin I domain*.

Remark. If Ω is a Cousin A domain, it is certainly a Cousin I domain.

Notation. Let $A^*(U)$ and $M^*(U)$ denote the groups of units in $A(U)$ and $M(U)$ respectively.

Definition 3.4.10. Let Ω be a domain in $\mathbb{C}^n$ and $\{U_i : i \in I\}$ be an open cover of Ω . Suppose we are given $m_i \in M^*(U_i)$ such that $m_i m_j^{-1} \in A^*(U_{ij})$ for all $i, j \in I$. The *Cousin II problem* is to construct $m \in M(\Omega)$ such that $m m_i^{-1} \in A^*(U_i)$ for all $i \in I$.

If we can always solve the Cousin II problem on Ω , we call Ω a *Cousin II domain*.

Remarks.

1. If Ω is a Cousin B domain, it is certainly a Cousin II domain.
2. Suppose that we can construct $m \in M(\Omega)$ satisfying the conditions of the Cousin II problem. Then, with the notation of Definition 3.4.10, we see that

$$Z(m) = \bigcup_{i \in I} Z(m_i)$$

and similarly for the pole and indeterminacy sets. We return in Chapter 5 to the question of multiplicities.

3. The definition of Cousin II domain we have given above is equivalent to that given in Definition 2.7.2, as the reader may easily verify using Proposition 3.4.5.

Exercises

1. Show that Lemma 3.4.1 need not be true if W is a polynomial in z_n but not a Weierstrass polynomial.
2. Let U be an open subset of $\mathbb{C}^n$ and $f, g \in A(U)$. Suppose that for some $a \in U$, $(f_a, g_a) = 1$. Show that in general $(f, g) \neq 1$.

5 Local properties of analytic sets

In this section we show how the unique factorization and Noetherian properties of the ring $\mathcal{O}_0$ are reflected in the local structure theory of analytic sets. We start by examining the case of the zero set of an analytic function f defined on some neighbourhood of 0 in $\mathbb{C}^n$. By uniqueness of factorization, we may write $f_0 = p_1^{r_1} \cdots p_k^{r_k}$, where $p_1, \dots, p_k \in \mathcal{O}_0$ are distinct primes. Suppose that $f, p_1, \dots, p_k$ are defined as analytic functions on some neighbourhood U of 0. Since $f = p_1^{r_1} \cdots p_k^{r_k}$ on U we have

$$Z(f) = \bigcup_{j=1}^k Z(p_j).$$

Therefore as a first step in the local study of the zero sets of analytic functions it is natural to examine the zero sets of primes in $\mathcal{O}_0$. Suppose then that $p \in \mathcal{O}_0$ is prime. Changing coordinates if necessary and multiplying by a unit, we may find a polydisc neighbourhood D' of 0 in $\mathbb{C}^{n-1}$ and $s > 0$ such that if $D = \{z \in \mathbb{C}^n : z' \in D' \text{ and } |z_n| < s\}$, then p is a Weierstrass polynomial on D :

$$p(z, z_n) = z_n^p + \sum_{j=0}^{p-1} a_j(z') z_n^j.$$

Let $\delta \in A(D')$ denote the discriminant of p (we recall that δ is the resultant of p and $p' = \partial p / \partial z_n$, δ is a polynomial in the coefficients of p and $\delta(z') = 0$ if and only if $p(z', z_n)$ has a multiple root. For further details we refer the reader to Van der Waerden [1]). Since p is prime, $\delta \neq 0$ (for otherwise p and p' would have a non-constant common factor see Van der Waerden [1]). Let $\Sigma = \{z' \in D' : \delta(z') = 0\}$ (Σ is the *discriminant locus* of p). We let $\pi : D \rightarrow D'$ denote the projection on the first $n - 1$ coordinates. Figure I.6 describes $Z(p) \subset D$.

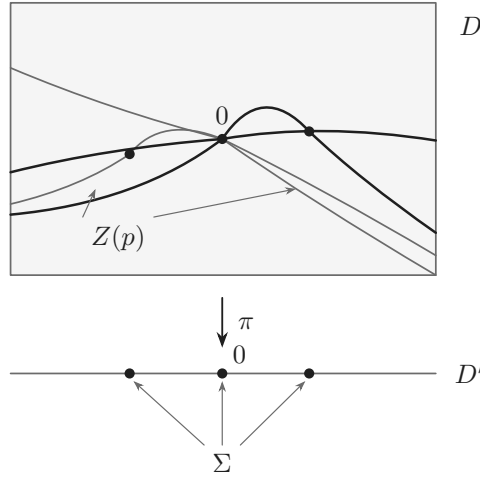


Figure I.6

At each point $z'_0 \in D' \setminus \Sigma$, $p(z'_0, z_n)$ has precisely p distinct roots. Denote these roots by $\lambda_1(z'_0), \dots, \lambda_p(z'_0)$. Since $\delta(z'_0) \neq 0$, $\frac{\partial p}{\partial z_n}(z'_0, \lambda_j(z'_0)) \neq 0$, $1 \leq j \leq p$, and we may apply the implicit function theorem to deduce that the roots λ_j depend analytically on z' , $z' \notin \Sigma$ (see Remark 4, §1 of Chapter 2). It follows that $Z(p) \setminus \pi^{-1}(\Sigma)$ is a (complex) submanifold of D of complex codimension 1 (see Chapter 4 for terminology). Moreover, π clearly restricts to a p -fold covering map of $Z(p) \setminus \pi^{-1}(\Sigma)$ over $D' \setminus \Sigma$. The zero set $Z(p)$ may have “singularities” (points where $Z(p)$ fails to be a submanifold of D) at points lying over Σ .

Examples.

1. Let $f(z, w) = z^2 - w^3$. Then f_0 is prime and is a Weierstrass polynomial in z . The discriminant locus of f is the origin of $\mathbb{C}$ (the w -axis). The reader may verify that at the origin of $\mathbb{C}^2$ it is not possible to find a submanifold chart (analytic or C^∞) for $Z(f)$. However, $Z(f)$ is a topological manifold as the map $t \mapsto (t^3, t^2)$ defines a homeomorphism of $\mathbb{C}$ onto $Z(f)$.
2. Let $f(u, v, w) = u^{2p} - vw$, for some positive integer p . Then f_0 is prime and is a Weierstrass polynomial in u . The discriminant locus of f is the origin of the (v, w) -plane. We claim that $Z(f)$ is not a topological manifold. If it were, it would have to be modelled on $\mathbb{C}^2$ since $Z(f) \setminus \{0\}$ is of complex dimension 2. Moreover, since $\mathbb{C}^2 \setminus \{0\}$ is simply connected, it would be possible to find simply connected neighbourhoods of 0 in $Z(f) \setminus \{0\}$. But the map $\phi : \mathbb{C}^2 \setminus \{0\} \rightarrow Z(f) \setminus \{0\}$ defined by $\phi(t, s) = (ts, t^p, s^p)$ is a p -fold covering map. Hence there is no neighbourhood of zero in $Z(f) \setminus \{0\}$ which is simply connected and so $Z(f)$ cannot be a topological manifold.

We shall now make a more general study of the local structure of analytic sets. Our approach is close to that given in Gunning and Rossi [1]. We start by defining the germ at a point of a subset of $\mathbb{C}^n$. Suppose X, Y are subsets of $\mathbb{C}^n$. We say that X and Y are equivalent at $a \in \mathbb{C}^n$ if there exists $U \in \mathcal{U}_a$ such that $X \cap U = Y \cap U$. This relation is certainly an equivalence relation on subsets of $\mathbb{C}^n$. We denote the equivalence class of a subset X by X_a and refer to X_a as the *germ* of X at a .

If α and β are germs of sets at a , we define

$$\alpha \cup \beta = (X \cup Y)_a; \quad \alpha \cap \beta = (X \cap Y)_a; \quad \alpha - \beta = (X \setminus Y)_a,$$

where X and Y are representatives for α and β respectively. Just as for germs of functions, it is easy to verify that the finite union, intersection and difference of germs of sets are well defined. As an exercise, the reader may show that arbitrary unions or intersections of germs of sets need not be well defined.

We write $\alpha \subset \beta$, if there exist representatives X and Y of α and β respectively such that $X \subset Y$.

Suppose $f_a \in \mathcal{O}_a^p$. Choose a representative $f = (f_1, \dots, f_p)$ for f_a . We let $Z(f_a)$ denote the germ of $Z(f)$ at a . That is, $Z(f_a) = f^{-1}(0)_a$. $Z(f_a)$ clearly depends only on f_a and not on the choice of representative function for f_a . We let $\mathcal{A}_a$ denote the set of all germs at a of analytic sets. Thus $X_a \in \mathcal{A}_a$ if and only if there exists $f_a \in \mathcal{O}_a^p$ such that $X_a = Z(f_a)$.

Lemma 3.5.1. Suppose $f = (f_1, \dots, f_p)$ is a representative function for $f_a \in \mathcal{O}_a^p$. Then

$$\begin{aligned} Z(f_a) &= \bigcap_{i=1}^p Z(f_i)_a \\ &= \bigcap_{i=1}^p Z(f_{i,a}). \end{aligned}$$

(Here $f_{i,a}$ denotes the germ of f_i at a).

Proof. Trivial. □

Notation. If $f = (f_1, \dots, f_p) \in A(U)^p$, we shall adopt the conventions that $Z(f_1, \dots, f_p) = Z(f)$ and $Z(f_{1,a}, \dots, f_{p,a}) = Z(f_1, \dots, f_p)_a = Z(f)_a$.

Proposition 3.5.2. Let $X_a, Y_a \in \mathcal{A}_a$. Then $X_a \cup Y_a, X_a \cap Y_a \in \mathcal{A}_a$.

Proof. We may suppose that $X_a = Z(f_1, \dots, f_p)$, $Y_a = Z(g_1, \dots, g_q)$ for suitable $f_i, g_j \in \mathcal{O}_a$. Clearly

$$\begin{aligned} X_a \cup Y_a &= Z(f_i g_j : 1 \leq i \leq p, 1 \leq j \leq q), \\ X_a \cap Y_a &= Z(f_1, \dots, f_p, g_1, \dots, g_q). \end{aligned}$$

□

Definition 3.5.3. Suppose $X_a \in \mathcal{A}_a$. We say $f_a \in \mathcal{O}_a$ *vanishes on* X_a if for some representatives X of X_a and f of f_a we have $f|X \equiv 0$. We write $f_a = 0$ on X_a .

Definition 3.5.4. Let $X_a \in \mathcal{A}_a$. We define

$$I(X_a) = \{f_a \in \mathcal{O}_a : f_a = 0 \text{ on } X_a\}.$$

Proposition 3.5.5. Let $X_a \in \mathcal{A}_a$. Then $I(X_a)$ is an ideal of $\mathcal{O}_a$.

Proof. Trivial. □

Notation. We write $I \triangleleft \mathcal{O}_a$ to signify that I is an ideal of $\mathcal{O}_a$.

Example. Let X be an analytic subset of the domain Ω in $\mathbb{C}^n$. For each $a \in \Omega$, we set $I_a(X) = I(X_a)$. Then $I_a(X) \triangleleft \mathcal{O}_a$ for every $a \in \Omega$. If $a \notin X$, $I_a(X) = \mathcal{O}_a$.

Before stating the next proposition recall that if R is a commutative ring and $I \triangleleft R$, then the *radical* of I , $\text{Rad}(I)$, is defined by

$$\text{Rad}(I) = \{r \in R : \text{For some positive integer } p, r^p \in I\}.$$

Proposition 3.5.6. For every germ $X_a \in \mathcal{A}_a$

$$I(X_a) = \text{Rad}(I(X_a)).$$

Proof. If f_a^p vanishes on X_a then certainly f_a vanishes on X_a . □

Proposition 3.5.7. Let I be an ideal in $\mathcal{O}_a$ and let $\{a_1, \dots, a_p\}$ be a finite set of generators for I . Then

1. If $f \in I$, f vanishes on $Z(a_1, \dots, a_p)$.
2. $Z(a_1, \dots, a_p)$ depends only on I and not on the particular choice of generators for I .

Proof. Let $f \in I$. Then $f = \sum_{j=1}^p f_j a_j$, $f_j \in \mathcal{O}_a$, $1 \leq j \leq p$. Hence $f \in Z(a_1, \dots, a_p)$, proving 1. Statement 2 follows straightforwardly from 1. □

Definition 3.5.8. Suppose $I \triangleleft \mathcal{O}_a$. We define

$$Z(I) = Z(a_1, \dots, a_p),$$

where $\{a_1, \dots, a_p\}$ is any finite set of generators for I .

It follows from Propositions 3.5.6 and 3.5.7 that we have a correspondence between radical ideals in $\mathcal{O}_a$ and germs of analytic sets. In the next theorem we list further properties of this correspondence.

Theorem 3.5.9.

1. If $X_a, Y_a \in \mathcal{A}_a$ and $X_a \subset Y_a$ then $I(X_a) \supset I(Y_a)$.
- 1'. If $I, J \triangleleft \mathcal{O}_a$ and $I \subset J$ then $Z(I) \supset Z(J)$.
2. If $X_a, Y_a \in \mathcal{A}_a$ and $X_a \neq Y_a$ then $I(X_a) \neq I(Y_a)$.
3. If $X_a \in \mathcal{A}_a$, $I(X_a) = \text{Rad}(I(X_a))$.
- 3'. If $I \triangleleft \mathcal{O}_a$, $Z(I) = Z(\text{Rad}(I))$.
4. If $X_a \in \mathcal{A}_a$, then $Z(I(X_a)) = X_a$.
- 4'. If $I \triangleleft \mathcal{O}_a$, $I(Z(I)) \supseteq \text{Rad}(I)$.

Proof. We shall prove statements 2, 4 and 4'. Statement 3 is Proposition 3.5.6. We leave the remaining cases as exercises for the reader (the full proof may be found in Gunning and Rossi [1]).

Proof of 2. For some $U \in \mathcal{U}_a$ and $f_j \in A(U)$, $1 \leq j \leq p+q$, we may write $X = Z(f_1, \dots, f_p)$, $Y = Z(f_{p+1}, \dots, f_{p+q})$, where X, Y are representatives for X_a, Y_a respectively. Since $X_a \neq Y_a$, we may find $z_W \in ((X \setminus Y) \cup (Y \setminus X)) \cap W$ for every $W \in \mathcal{U}_a$. Hence there exists $j(W) \in \{1, \dots, p+q\}$ such that $f_{j(W)}(z_W) \neq 0$. This is true for all sufficiently small neighbourhoods W of a and so there must exist a j_0 such that f_{j_0} does not vanish identically on $((X \setminus Y) \cup (Y \setminus X)) \cap W$ for any $W \in \mathcal{U}_a$. If $1 \leq j_0 \leq p$, this implies $f_{j_0} \neq 0$ on Y_a and if $p+1 \leq j_0 \leq p+q$, $f_{j_0} \neq 0$ on X_a . Hence $I(X_a) \neq I(Y_a)$.

Proof of 4. If $f_a \in I(X_a)$ then certainly $f_a = 0$ on X_a and so $Z(I(X_a)) \supseteq X_a$. On the other hand we may write $X_a = Z(f_1, \dots, f_k)$ for some $f_j \in \mathcal{O}_a$. Since $f_j = 0$ on X_a , $f_j \in I(X_a)$ and so $Z(f_j) \supseteq Z(I(X_a))$. This holds for all j and so $X_a = Z(f_1, \dots, f_k) \supseteq Z(I(X_a))$.

Proof of 4'. Obviously, $I(Z(I)) \supseteq I$ for any $I \triangleleft \mathcal{O}_a$. By 3', $Z(I) = Z(\text{Rad}(I))$ and so $I(Z(I)) = I(Z(\text{Rad}(I))) \supseteq \text{Rad}(I)$. $\square$

Remark. A consequence of Theorem 3.5.9 is that the map $X_a \rightarrow I(X_a)$ is injective into the set of radical ideals in $\mathcal{O}_a$ with left inverse defined by mapping I to $Z(I)$, for radical $I \triangleleft \mathcal{O}_a$. To show that this map is onto we need to know that $I(Z(I)) = I$ for all radical ideals I of $\mathcal{O}_a$ and not just that $I(Z(I)) \supseteq I$ (condition 4'). In fact we do have equality and this fundamental result is the *Nullstellensatz* for germs of analytic sets. The proof of the Nullstellensatz lies much deeper than the other results of this section and may be found in Gunning and Rossi [1], Gunning [1], Hervé [1] or R. Narasimhan [3]. We shall prove a special case, the Nullstellensatz for principal ideals, later in this section and this will suffice for our intended applications.

Definition 3.5.10. We say that a germ $X_a \in \mathcal{A}_a$ is *irreducible* if whenever $X_a = Y_a \cup Z_a$, $Y_a, Z_a \in \mathcal{A}_a$, we have either $X_a = Y_a$ or $X_a = Z_a$.

Theorem 3.5.11. A germ X_a is irreducible if and only if $I(X_a)$ is prime.

Proof. Suppose $I(X_a)$ is not prime. Then there exist $f, g \in \mathcal{O}_a$ such that $fg \in I(X_a)$ but neither f nor g lie in $I(X_a)$. Now $X_a = X_a \cap Z(fg) = X_a \cap (Z(f) \cup Z(g)) = (X_a \cap Z(f)) \cup (X_a \cap Z(g))$. Since $f, g \notin I(X_a)$, $X_a \cap Z(f)$ and $X_a \cap Z(g)$ are strictly contained in X_a and so X_a is not irreducible. Conversely, suppose that X_a is not irreducible. Then $X_a = Y_a \cup Z_a$, where Y_a and Z_a are strictly contained in X_a . By 2 of Theorem 3.5.9, $I(Y_a), I(Z_a) \neq I(X_a)$ and so, by 1 of Theorem 3.5.9, $I(Y_a), I(Z_a)$ strictly contain $I(X_a)$. Pick $f \in I(Y_a) \setminus I(X_a)$, $g \in I(Z_a) \setminus I(X_a)$. Since $fg \in I(Y_a) \cap I(Z_a) = I(X_a)$, we see that $I(X_a)$ is not prime. $\square$

Theorem 3.5.12. Let $X_a \in \mathcal{A}_a$. We may decompose X_a into a finite union $X_a^1 \cup \cdots \cup X_a^k$ of irreducible germs satisfying

1. $X_a^j \not\subset \bigcup_{i \neq j} X_a^i$ for $j = 1, \dots, k$.
2. The X_a^i are uniquely determined (up to order) by X_a .

Proof. Set $I = I(X_a)$. It follows from the theory of Noetherian rings that we may write $I = \bigcap_{j=1}^s I_j$, where the I_j are primary ideals (that is, $\text{rad}(I_j)$ is prime). See, for example, Van der Waerden [2] or Zariski and Samuel [1]. Set $P_j = \text{Rad}(I_j)$. By 4 of Theorem 3.5.9, $X_a = Z(I) = \bigcup_{j=1}^s Z(P_j)$. We may throw away any germs $Z(P_j)$ which are contained in the union of the remaining germs to obtain the required decomposition of X_a as a union of irreducible germs. Suppose that $X_a = X_a^1 \cup \cdots \cup X_a^k = Y_a^1 \cup \cdots \cup Y_a^p$ are two decompositions of X_a as unions of irreducible germs which satisfy condition 1 of the theorem. For $j \in \{1, \dots, k\}$, $X_a^j = \bigcup_{i=1}^p (Y_a^i \cap X_a^j)$ and since X_a^j is irreducible, we must have $X_a^j = Y_a^{\tau(j)} \cap X_a^j$ for some $\tau(j) \in \{1, \dots, p\}$. Hence $X_a^j \subset Y_a^{\tau(j)}$, $1 \leq j \leq k$. Similarly, $Y_a^i \subset X_a^{\rho(i)}$ for some $\rho(i) \in \{1, \dots, k\}$, $1 \leq i \leq p$. It follows that for all i, j we have

$$X_a^j \subset Y_a^{\tau(j)} \subset X_a^{\rho\tau(j)} \quad \text{and} \quad Y_a^i \subset X_a^{\rho(i)} \subset Y_a^{\tau\rho(i)}.$$

Since the decompositions satisfy condition 1, $\rho\tau$ and $\tau\rho$ are the identity maps. Therefore $k = p$ and $X_a^j = Y_a^{\tau(j)}$, $1 \leq j \leq k$. $\square$

We conclude this section by proving three results about germs of analytic sets whose ideals are principal.

Theorem 3.5.13. (Nullstellensatz for principal prime ideals). Let $f_a \in \mathcal{O}_a$ be irreducible. Then $I(Z(f_a)) = (f_a)$.

Proof. From 4' of Theorem 3.5.9 we already know that $I(Z(f_a)) \supset (f_a)$. If $g_a \notin (f_a)$, then $(f_a, g_a) = 1$ since f_a is irreducible. Let f and g be representatives for f_a and g_a respectively. By Lemma 3.4.3, we may find z arbitrarily close to a such that $f(z) = 0$ and $g(z) \neq 0$. Hence $g_a \notin I(Z(f_a))$. Therefore $I(Z(f_a)) \subset (f_a)$. $\square$

Corollary 3.5.14. (Nullstellensatz for principal ideals). Let $f_a \in \mathcal{O}_a$. Then $I(Z(f_a)) = \text{Rad}((f_a))$.

Proof. Write $f_a = p_1^{r_1} \cdots p_s^{r_s}$, where the $p_j \in \mathcal{O}_a$ are irreducible and distinct. We certainly have $Z(f_a) = \bigcup_{i=1}^s Z(p_i)$ and so, by Theorem 3.5.13, $I(Z(f_a)) = \bigcap_{i=1}^s (p_i) = \text{Rad}(f_a)$. $\square$

Corollary 3.5.15. Let $p_1^{r_1} \cdots p_s^{r_s}$ be the prime factorization of $f_a \in \mathcal{O}_a$. Then $Z(f_a) = \bigcup_{i=1}^s Z(p_i)$ is the unique decomposition of $Z(f_a)$ into irreducible germs given by Theorem 3.5.12.

Proof. All we need to check is that for every i , $Z(p_i) \not\subset \bigcup_{j \neq i} Z(p_j)$. By the Nullstellensatz for principal prime ideals, this condition is equivalent to $(p_i) \not\supset (p_1 \cdots p_{i-1} p_{i+1} \cdots p_s)$. This is certainly true since p_i is not divisible by p_j , $i \neq j$. $\square$

Theorem 3.5.16. Let U be a domain in $\mathbb{C}^n$ and suppose that $f \in A(U)$ is not identically zero. Set $X = f^{-1}(0)$. Given $x \in X$, suppose that p_x is a generator of the principal ideal $I_x(X)$. Then we can choose a representative p for p_x , defined on some neighbourhood V of x in U , such that $I_y(X) = (p_y)$ for all $y \in V$.

Proof. It is clearly sufficient to prove the result for the special case when p_x is irreducible. Changing coordinates and multiplying by a unit we may assume that $x = 0$ and that we may find a polydisc neighbourhood D' of 0 in $\mathbb{C}^{n-1}$ and $s > 0$ such that if $D = \{z \in \mathbb{C}^n : z' \in D', |z_n| < s\}$, then p_0 is the germ of a Weierstrass polynomial p on D :

$$p(z', z_n) = z_n^p + \sum_{j=0}^{p-1} a_j(z') z_n^j.$$

Assuming that $D \subset U$, we have $p^{-1}(0) = X \cap D$. The discriminant locus Σ of p is a proper analytic subset of D and so, in particular, $D' \setminus \Sigma$ is an open dense subset of D' . If $y = (y', y_n) \in p^{-1}(0)$ and $y' \notin \Sigma$, then $\partial p / \partial z_n(y', y_n) \neq 0$. Consequently p_y is irreducible (See Exercise 2, §3). On the other hand suppose that $y = (y', y_n) \in p^{-1}(0)$ and $y' \in \Sigma$. Let ϕ_y be a generator of $I_y(X)$. Then for some positive integer k and unit $u_y \in \mathcal{O}_y$, $p_y = u_y \phi_y^k$. This equation holds on a neighbourhood of y in D and so holds for points $z = (z', z_n) \in p^{-1}(0)$ with $z' \notin \Sigma$. Consequently, $k = 1$. Our arguments prove that $I_y(X) = (p_y)$ for all $y \in D$. $\square$

Remark. Theorem 3.5.16 is a special case of the fundamental coherence theorem of H. Cartan which asserts that if $g_x^1, \dots, g_x^k$ generate $I_x(X)$ then $g_y^1, \dots, g_y^k$ generate $I_y(X)$ for y in some neighbourhood of x . Here X is an arbitrary analytic subset. The proof of this result is considerably harder than the special case given above and the reader may find proofs in H. Cartan [2], Gunning and Rossi [1], R. Narasimhan [3] and Whitney [1].

Exercises

1. Suppose $p(z', z_n)$ is an irreducible Weierstrass polynomial of degree p defined on the polydisc $D = D' \times D_s(0)$, with discriminant $\delta \in A(D')$ and discriminant locus Σ . Let $\pi : D \rightarrow D'$ denote the projection on D' and set $X = p^{-1}(0)$.
 - i) Suppose Y is a connected component of $X \setminus \pi^{-1}(\Sigma)$. Show that $\pi|_Y$ is an s -fold cover of $D' \setminus \Sigma$ for some s , $1 \leq s \leq p$. (Hint: Fix $z'_0 \in D' \setminus \Sigma$ and a root $\lambda_j(z'_0) \in Y$. Consider the unique lifts through $\lambda_j(z'_0)$ of all closed paths in $D' \setminus \Sigma$ starting at z'_0 . Note that $D' \setminus \Sigma$ is connected (Corollary 2.2.3)).
 - ii) Using the analyticity of the roots of $p(z', z_n)$ together with the Riemann removable singularities theorem, prove that $\bar{Y}$ is the zero locus of a polynomial in z_n with coefficients analytic in $z' \in D'$.
 - iii) Deduce from ii) that $\bar{Y} = X$ and that $X \setminus \pi^{-1}(\Sigma)$ is connected.
2. Suppose that $f(y, z)$ is a Weierstrass polynomial in z of degree p defined on an open polydisc $D_a(0) \times D_b(0) \subset \mathbb{C}^2$ and that the discriminant locus is equal to $\{0\} \subset D_a(0)$. Set $c = a^{1/p}$. Show that
 - i) There exists an analytic function $\phi : D_c(0) \rightarrow \mathbb{C}$ such that $f(t^p, \phi(t)) = 0$, $t \in D_c(0)$.
 - ii) If $\omega = \exp(2\pi i/p)$, then given $y = t^p \in D_a(0)$, $f(y, z) = 0$ if and only if $z = \phi(\omega^j t)$ for some integer j , $0 \leq j \leq p-1$. (Hint for i): Define $\gamma : D_c(0) \rightarrow D_a(0)$ by $\gamma(t) = t^p$. Now show using Q1 and the Riemann removable singularities theorem that lifts to an analytic map $\tilde{\gamma} : D_c(0) \rightarrow D_a(0) \times D_b(0)$ with image X . (Observe that if $y = t^p$ and we define $y^{r/p} = t^r$ then we obtain the *fractional power series* or *Puiseux series* solution $z = \phi(y^{1/p})$ to $f(y, z) = 0$. Further details may be found in Walker [1]).
3. Let $f = (f_1, \dots, f_p) \in \mathcal{O}_0^p$. Show that if $Df(0)$ is of maximal rank then $Z(f)$ is irreducible.

6 Modules over $\mathcal{O}_0$

Suppose that M is a finitely generated $\mathcal{O}_0$ -module. Let $U_1^0, \dots, U_{p(0)}^0$ be a set of generators for M and define $U^0 : \mathcal{O}_0^{p(0)} \rightarrow M$ by $U^0(f_1, \dots, f_{p(0)}) = \sum_{j=1}^{p(0)} f_j U_j^0$. The sequence $\mathcal{O}_0^{p(0)} \xrightarrow{U^0} M \rightarrow 0$ is exact. The kernel of U^0 , $\text{Ker}(U^0)$, is a submodule of $\mathcal{O}_0^{p(0)}$ and so, by Lemma 3.3.5, is finitely generated. If $U_1^1, \dots, U_{p(1)}^1$ is a set of generators for $\text{Ker}(U^0)$ we may repeat the above construction to obtain an exact sequence $\mathcal{O}_0^{p(1)} \xrightarrow{U^1} \mathcal{O}_0^{p(0)} \xrightarrow{U^0} M \rightarrow 0$. Iterating this construction we obtain a long exact sequence

$$\dots \xrightarrow{U^{n+1}} \mathcal{O}_0^{p(n)} \xrightarrow{U^n} \dots \xrightarrow{U^1} \mathcal{O}_0^{p(0)} \xrightarrow{U^0} M \rightarrow 0.$$

Such a sequence is called a *free resolution* of M (it is also referred to as a *chain of syzygies* for M —see Gunning and Rossi [1] or Nagata [1]). Suppose that for some s , $\text{Ker}(U^{s-1})$ is a *free* $\mathcal{O}_0$ -module. That is, $\text{Ker}(U^{s-1}) \cong \mathcal{O}_0^q$ for some integer q . Then we may replace the original free resolution by the finite free resolution

$$0 \rightarrow \mathcal{O}_0^q \xrightarrow{\mu} \mathcal{O}_0^{p(s-1)} \rightarrow \dots \rightarrow \mathcal{O}_0^{p(0)} \rightarrow M \rightarrow 0,$$

where μ denotes an isomorphism of $\mathcal{O}_0^q$ onto $\text{Ker}(U^{s-1})$. Such a finite free resolution of M is said to have *length* s .

Theorem 3.6.1. (Hilbert Syzygy theorem). Every $\mathcal{O}_0$ -module has a free resolution of length n .

Proof. It is sufficient to show that for any free resolution of M ,

$$\xrightarrow{U^s} \mathcal{O}_0^{p(s-1)} \xrightarrow{U^{s-1}} \dots \xrightarrow{U^1} \mathcal{O}_0^{p(0)} \xrightarrow{U^0} M \rightarrow 0,$$

$\text{Ker}(U^{n-1})$ is free.

To simplify notation, we set $K_s = \text{Ker}(U^s)$ and let M_s denote the ideal of $\mathcal{O}_0$ generated by $z_1, \dots, z_s$. In particular, $M_n = \mathfrak{m}_0$, the maximal ideal of $\mathcal{O}_0$.

Step 1. For $s \geq j$, we have $K_s \cap M_j \mathcal{O}_0^{p(s)} = M_j K_s$. Obviously, we have $K_s \cap M_j \mathcal{O}_0^{p(s)} \supset M_j K_s$ and so it remains to verify the reverse inclusion. Our proof goes by induction on j . Suppose $j = 1$, and that $f \in K_s \cap M_1 \mathcal{O}_0^{p(s)}$. Then $f = z_1 g$, $g \in \mathcal{O}_0^{p(s)}$. Since $U^s(f) = 0$, $z_1 U^s(g) = 0$ and so $U^s(g) = 0$. Therefore $g \in K_s$ and $f \in M_1 K_s$. Now suppose that we have proved the result for $j - 1$ and all $s \geq j - 1$. Let $f \in K_s \cap M_j \mathcal{O}_0^{p(s)}$. Then $f = z_1 g_1 + \dots + z_j g_j$, $g_1, \dots, g_j \in \mathcal{O}_0^{p(s)}$. Since $f \in K_s$, $U^s(f) = z_1 U^s(g_1) + \dots + z_j U^s(g_j) = 0$. Therefore, $z_j U^s(g_j) = -(z_1 U^s(g_1) + \dots + z_{j-1} U^s(g_{j-1}))$ and so, dividing through by z_j we see that $U^s(g_j) \in M_{j-1} \mathcal{O}_0^{p(s-1)}$. Now $U^{s-1} U^s = 0$ and so $U^s(g_j) \in K_{s-1} \cap M_{j-1} \mathcal{O}_0^{p(s-1)}$. By the inductive hypothesis, $U^s(g_j) \in M_{j-1} K_{s-1}$. Since $K_{s-1} = \text{Image}(U^s)$, there exist $h_1, \dots, h_{j-1} \in \mathcal{O}_0^{p(s)}$ such that

$$U^s(g_j) = z_1 U^s(h_1) + \dots + z_{j-1} U^s(h_{j-1}).$$

Now set $h_j = g_j - z_1 h_1 - \dots - z_{j-1} h_{j-1}$. Certainly $h_j \in K_s$. We claim that $f - z_j h_j \in M_{j-1} K_s$. This follows from the inductive hypothesis since $f - z_j h_j = \sum_{i=1}^{j-1} z_i (g_i + z_j h_i)$ and so $f - z_j h_j \in K_s \cap M_{j-1} \mathcal{O}_0^{p(s)} = K_s M_{j-1}$. Since $h_j \in K_s$, $f \in M_j K_s$.

Step 2. Choose a set of generators $U_1, \dots, U_q$ for K_{n-1} which is minimal—that is, no proper subset of $U_1, \dots, U_q$ generates K_{n-1} . Define $U : \mathcal{O}_0^q \rightarrow \mathcal{O}_0^{p(n-1)}$ by $U(f_1, \dots, f_q) = \sum_{j=1}^q f_j U_j$. We claim that U is an isomorphism onto K_{n-1} . For this it is sufficient to verify that $\text{Ker}(U)$ is zero.

Modify the original free resolution of M at the n th step to obtain the new free resolution

$$\mathcal{O}_0^q \xrightarrow{U} \mathcal{O}_0^{p(n-1)} \xrightarrow{U^{n-1}} \dots \xrightarrow{U^0} M \rightarrow 0.$$

Applying Step 1 to this resolution with $j = n$ we see that

$$\text{Ker}(U) \cap \mathfrak{m}_0 \mathcal{O}_0^q = \mathfrak{m}_0 \text{Ker}(u).$$

In fact, $\text{Ker}(U) \subset \mathfrak{m}_0 \mathcal{O}_0^q$. To see this, suppose $(f_1, \dots, f_q) \in \text{Ker}(U) \subset \mathcal{O}_0^q$. Then

$$0 = U(f_1, \dots, f_q) = \sum_{j=1}^q f_j U_j.$$

Since $U_1, \dots, U_q$ is a minimal set of generators, no f_j can be a unit. That is, $f_j \in \mathfrak{m}_0$, $1 \leq j \leq q$.

We have now shown that $\text{Ker}(U) = \mathfrak{m}_0 \text{Ker}(U)$. This implies that $\text{Ker}(U) = 0$, for otherwise we may find $a_{ij} \in \mathfrak{m}_0$ such that

$$U_i = \sum_{j=1}^q a_{ij} U_j.$$

Therefore $\det[\delta_{ij} - a_{ij}] = 0$ and so, expanding the determinant, we see that $1 \in \mathfrak{m}_0$ which is a contradiction. $\square$

Remark. The last paragraph of the proof of Theorem 3.6.1 is a special case of

Nakayama's Lemma: Let R be a local ring with maximal ideal $\mathfrak{m}$ and M be a finite R -module. Then

- a) If $M = \mathfrak{m}M$, $M = 0$.
- b) If N is a submodule of M such that $M = N + \mathfrak{m}M$, $N = M$.

The proof of a) is the same as that used in the last paragraph of the proof of Theorem 3.6.1. Statement b) follows by applying a) to M/N .

Suppose that M is a submodule of $\mathcal{O}_0^p$. Since $\mathcal{O}_0$ is Noetherian, M has a finite set of generators $U_1, \dots, U_q$ and so we have an exact sequence $\mathcal{O}_0^q \xrightarrow{U} M \rightarrow 0$ where U is defined by mapping $(f_1, \dots, f_q)$ to $\sum_{j=1}^q f_j U_j$. The question naturally arises as to whether this sequence is *split* exact. Certainly it is not generally split exact as a sequence of $\mathcal{O}_0$ -modules! However, it makes sense to ask whether it splits as a sequence of vector spaces. That is, can we find a $\mathbb{C}$ -linear map $\xi : M \rightarrow \mathcal{O}_0^q$ such that $U\xi$ is the identity map of M ? Equivalently, can we write every $m \in M$ in the form $\sum \xi_i(m) U_i$, where ξ_i is linear, $1 \leq i \leq q$? In view of the continuity conditions in the Weierstrass Division Theorem it is reasonable to ask whether we can topologise the spaces M and $\mathcal{O}_0^q$ and further require that ξ is a *continuous* linear map. In fact we shall prove a result of this type but instead of working with spaces of germs we shall consider modules of bounded analytic functions defined on some neighbourhood of zero in $\mathbb{C}^n$ (for the splitting of sequences of germs and the topologization of $\mathcal{O}_0^p$ and M we refer to Hörmander [1], H. Cartan [2, Seminar 11] and the exercises at the end of this section).

Suppose U is an open neighbourhood of zero in $\mathbb{C}^n$. We let $A_B(U)^p$ denote the $A_B(U)$ -module of p -tuples of bounded analytic functions on U . We define a norm on $A^B(U)^p$ by

$$\|f\| = \max_{1 \leq i \leq p} \|f_i\|_U,$$

where $f = (f_1, \dots, f_p) \in A_B(U)^p$. Clearly $A_B(U)^p$ is a Banach space in this norm. If M is a submodule of $\mathcal{O}_0^p$, we define

$$M_B(U) = \{f \in A_B(U)^p : f_0 \in M\}.$$

$M_B(U)$ is a normed vector subspace of $A_B(U)^p$. However, it is by no means evident that $M_B(U)$ need be closed in $A_B(U)^p$.

Theorem 3.6.2. Let M be a submodule of $\mathcal{O}_0^p$ and W be an open neighbourhood of zero in $\mathbb{C}^n$. Suppose that we are given $U_1, \dots, U_q \in A(W)^p$ such that the germs of these functions at zero generate M . Then we may find an open neighbourhood $D \subset W$ of zero in $\mathbb{C}^n$ such that the U_j are bounded analytic functions on D such that if we define

$$U : A_B(D)^q \rightarrow M_B(D)$$

by $U(f_1, \dots, f_q) = \sum_{j=1}^q f_j U_j$, then the sequence

$$A_B(D)^q \xrightarrow{U} M_B(D) \rightarrow 0$$

is split exact as a sequence of normed vector spaces.

Moreover, given any finite set of submodules and generators we can choose an open neighbourhood D of zero that works for all the submodules and generators simultaneously.

Before starting the proof, we remark that the splitting assumption implies that there exists a continuous linear map $\xi = (\xi_1, \dots, \xi_q) : M_B(D) \rightarrow A_B(D)^q$ such that every $m \in M_B(D)$ may be written as a sum $\sum_{i=1}^q \xi_i(m) U_i$, where $\xi_i : M_B(D) \rightarrow A_B(D)$ is continuous linear, $1 \leq i \leq q$. In particular, there exists $M_i \geq 0$ such that $\|\xi_i(m)\| \leq M_i \|m\|$, for all $m \in M_B(D)$, $1 \leq i \leq q$.

Proof. Our proof goes by induction on n and p . Certainly the theorem is trivially true if $n = 0$. So assume the result for $n - 1$ and all positive integers p .

Step 1. The theorem is true for n and $p = 1$. We suppose U_1 is normalised in the z_n -direction and we fix a complimentary subspace $\mathbb{C}^{n-1}$ to $\mathbb{C}z_n$ in $\mathbb{C}^n$. By the Preparation theorem it is no loss of generality to suppose that U_1 is a Weierstrass polynomial in z_n of degree r , say. By the Division theorem for germs (Remark 3, following Theorem 3.2.1), every $f \in M$ may be written uniquely in the form $f = qU_1 + h$, where $q \in \mathcal{O}_0$ and $h \in \mathcal{O}'_0[z_n] \cap M$ is a polynomial of degree $< r$. The set of such germs h defines a $\mathcal{O}'_0$ -module M^* which we may regard as a submodule of $\mathcal{O}'_0{}^r$ (see the proof of Theorem 3.3.6). Let $P_1, \dots, P_s$ be a set of generators for this submodule. We may write

$$P_i = \sum_{j=1}^q a_{ij} U_j,$$

for suitable $a_{ij} \in \mathcal{O}_0$, since $M^* \subset M$. Choose an open neighbourhood $W^* \subset W$ of zero such that all the germs P_i, a_{ij} are represented by bounded analytic functions on W^* . By Theorem 3.2.1', we can find $s > 0$ and an open neighbourhood V of 0 in $\mathbb{C}^{n-1}$ such that the conditions for the Division theorem hold for the divisor U_1 on any open neighbourhood $D' \times D_s(0)$ of 0 in $\mathbb{C}^n$, provided that $D' \subset V$. By the induction hypothesis applied to M^* , we may find an open neighbourhood $D^* \subset V \cap W^*$ of 0 in $\mathbb{C}^{n-1}$ such that if we define $P(f_1, \dots, f_s) = \sum_{j=1}^s f_j P_j$, then the sequence

$$A_B(D^*)^s \xrightarrow{P} M_B^*(D^*) \rightarrow 0$$

is split exact. That is, there exists a continuous linear $\xi^* : M_B^*(D^*) \rightarrow A_B(D^*)^s$ such that $P\xi^*$ is the identity. We now set $D = D^* \times D_s(0)$. The conditions of the Weierstrass Division theorem hold on D for the divisor U_1 . Hence there exist continuous linear maps

$$q, h : M_B(D) \rightarrow A_B(D)$$

such that $F = q(F)U_1 + h(F)$, $F \in A_B(D)$. We now define $\xi = (\xi_1, \dots, \xi_q) : M_B(D) \rightarrow A_B(D)^q$ by

$$\begin{aligned}\xi_1(F) &= q(F) + \sum_{i=1}^s \xi_i^*(h(F))a_{i1}, \\ \xi_j(F) &= \sum_{i=1}^s a_{ij}\xi_i^*(h(F))z_n^i.\end{aligned}$$

Clearly ξ is a continuous linear map and $U\xi = 1$. In particular, since $U\xi = 1$, U is onto and so we see that the sequence

$$A_B(D)^q \xrightleftharpoons[\xi]{U} M_B(D) \rightarrow 0$$

is split exact. Since Theorem 3.2.1' holds for a finite set of divisors, it is clear that the proof we have given above goes through for any finite set of $\mathcal{O}_0$ -submodules and generators. Hence the first inductive step is proved.

Step 2. The theorem is true for n and $p > 1$. Let $\pi : \mathcal{O}_0^p \rightarrow \mathcal{O}_0^{p-1}$ denote the projection on the first $p-1$ coordinates. Set $M' = \pi M$ and $U'_j = \pi U_j$, $1 \leq j \leq q$. We define $M'' = \{g \in \mathcal{O}_0 : (0, \dots, 0, g) \in M\} \cong \ker(\pi|M)$. M' is a submodule of $\mathcal{O}_0^{p-1}$ generated by $U'_1, \dots, U'_q$ and M'' is an ideal of $\mathcal{O}_0$. Let $P_1, \dots, P_s$ be a set of generators for M'' . Since $\text{Ker}(\pi|M) \subset M$, there exist $a_{ij} \in \mathcal{O}_0$ such that

$$(0, \dots, 0, P_i) = \sum_{j=1}^q a_{ij}U_j, \quad 1 \leq i \leq s.$$

Choose an open neighbourhood $V \subset W$ of zero such that a_{ij}, P_i are represented by bounded analytic functions in V . By the inductive hypothesis we may find an open neighbourhood $D \subset V$ of zero such that the conclusions of the theorem hold in D for M' and M'' . That is, the sequences of normed vector spaces

$$A_B(D)^{p-1} \xrightarrow{U'} M'_B(D) \rightarrow 0 \quad \text{and} \quad A_B(D)^s \xrightarrow{P} M''_B(D) \rightarrow 0$$

are split exact. We denote the splitting maps for these sequences by

$$\begin{aligned}\xi' &= (\xi'_1, \dots, \xi'_{p-1}) : M'_B(D) \rightarrow A_B(D)^{p-1} \quad \text{and} \\ \xi'' &= (\xi''_1, \dots, \xi''_s) : M''_B(D) \rightarrow A_B(D)^s.\end{aligned}$$

We now define $\xi = (\xi_1, \dots, \xi_p) : M_B(D) \rightarrow A_B(D)^p$ by

$$\xi_i(m) = \xi'_i(\pi(m)) + \sum_{j=1}^s \xi''_j(m - \pi(m))a_{ji}, \quad 1 \leq i \leq p.$$

ξ is obviously a continuous linear map and $U\xi = \text{identity}$. In particular, U is onto and so the sequence

$$A_B(D)^p \xrightleftharpoons[\xi]{U} M_B(D) \rightarrow 0$$

is split exact.

It is clear from the proof that we may choose D so that the result holds in D for any finite set of $\mathcal{O}_0$ -submodules and generators. This proves the second inductive step. $\square$

The following remarkable result is a straightforward consequence of Theorem 3.6.2.

Theorem 3.6.3. Let Ω be a domain in $\mathbb{C}^n$ and F be any subset of $A(\Omega)$. Then $Z(F) = \bigcap_{f \in F} Z(f)$ is an analytic subset of Ω .

Proof. Fix $z \in \Omega$. Let I_z denote the ideal generated by $\{f_z : f \in F\}$. $\mathcal{O}_z$ is Noetherian and so we may find a finite set $g_1, \dots, g_p$ of generators for I_z . Since I_z is generated by the germs f_z , $f \in F$, there exists $f_1, \dots, f_q$ and $h_{ij} \in \mathcal{O}_z$ such that

$$g_i = \sum_{j=1}^q h_{ij} f_j, \quad i = 1, \dots, p, \quad (1)$$

Choosing representatives for g_i and h_{ij} we may suppose that (1) holds on some neighbourhood U of z contained in Ω . By Theorem 3.6.2, we may find an open relatively compact neighbourhood D of z , with $\overline{D} \subset U$, such that for all $f \in F$, there exist bounded analytic functions a_j on D such that $f = \sum_{i=1}^p a_i g_i$. But therefore by (1), $f = \sum_{j=1}^q a'_j f_j$ on D , where $a'_j = \sum a_i h_{ij}$. It now follows that $Z(f) \cap D \supset Z(f_1, \dots, f_q) \cap D$. This holds for all $f \in F$ and so $Z(F) \cap D \supset Z(f_1, \dots, f_q) \cap D$. The reverse inclusion is trivial, since $f_1, \dots, f_q \in F$. $\square$

Remarks.

1. An alternative proof of Theorem 3.6.3 may be found in Whitney [1, page 100].
2. Theorem 3.6.2 is a special case of a general splitting theorem for coherent sheaves over “privileged” polycylinders due to Douady. See Douady [1, 2].
3. As an example of an important splitting theorem in differential analysis we might mention Mather’s splitting theorem for smooth invariants [1]. For more examples and references see Bierstone and Schwarz [1].

Exercises

1. Show that a finitely generated $\mathcal{O}_0$ -module need not be isomorphic to any submodule of $\mathcal{O}_0^p$, $p \geq 0$ (Hint: $\mathbb{C}$ is a finitely generated $\mathcal{O}_0$ -module if we define $f.z = f(0)z$, $f \in \mathcal{O}_0$).
2. Let M be a submodule of $\mathcal{O}_0^p$ with generators $U_1, \dots, U_q$. Show that, with the notation of Theorem 3.6.2, if $A_B(D)^q \xrightarrow{U} M_B(D) \rightarrow 0$ is split exact for some open neighbourhood D of zero then $M_B(D)$ is a closed normed subspace of $A_B(D)^p$.
3. (Closure of modules theorem). Suppose M is a submodule of $\mathcal{O}_0^p$ and $F \in A(U)^p$ for some open neighbourhood U of zero in $\mathbb{C}^n$. Show that if F can be uniformly approximated on compact subsets of U by functions whose germs at zero belong to M then $F_0 \in M$ (Hint: Use Q2 and Theorem 3.6.2).
4. Let $f = \sum a_m z^m \in \mathcal{O}_0$. Define $q_m(f) = |a_m|$. Show that
 - i) q_m defines a semi-norm on $\mathcal{O}_0$, $m \in \mathbb{N}^n$.
 - ii) $d(f, g) = \sum 2^{-|m|} q_m(f - g) / (1 + q_m(f - g))$ defines a metric on $\mathcal{O}_0$.
 - iii) The completion of $\mathcal{O}_0$ with respect to the metric defined in ii) is the ring $\mathbb{C}[[z_1, \dots, z_n]]$ of formal power series in $z_1, \dots, z_n$.
- 5*. Suppose M is a submodule of $\mathcal{O}_0^p$ and that $U_1, \dots, U_q$ is a set of generators for M . Prove that the sequence $\mathcal{O}_0^q \xrightarrow{U} M \rightarrow 0$ is split exact as a sequence of topological vector spaces. Deduce that M is a closed subspace of $\mathcal{O}_0^p$.

Chapter 4

Complex Manifolds

Introduction

Much of this chapter is taken up with the discussion of specific examples of complex manifolds. After general definitions given in section 1, we consider complex submanifolds of $\mathbb{C}^n$ in section 2. We prove that the open Euclidean disc and polydisc are biholomorphically inequivalent. Included in this section are also definitions of the classical domains and Stein manifolds. In section 3 we consider projective algebraic manifolds and in section 5 complex manifolds defined as the quotient of a properly discontinuous group action. In section 4 we define complex tori and prove that every one dimensional complex torus is biholomorphic to a cubic curve. In section 6 we develop the structure theory of complex hypersurfaces and in section 7 we define blowing up and give some indication of its use in desingularising analytic sets and in the classification theory of complex manifolds.

1 Generalities on complex manifolds and analytic sets

In this section we shall review the definitions of complex manifold and submanifold, analytic map and analytic set. For the most part our definitions are straightforward adaptations of those from differential manifolds and our presentation will therefore be brief.

Throughout this section all topological spaces will be assumed Hausdorff, paracompact and, unless otherwise indicated, connected. If M is a topological space, we let $\mathcal{U}$ denote the topology of M and $\mathcal{U}_a$ denote the subset of $\mathcal{U}$ consisting of all open neighbourhoods of a given point $a \in M$.

Definition 4.1.1. Let M be a topological space. M is said to have the structure of an n -dimensional complex manifold if there exists an atlas $\mathcal{A} = \{(U_i, \phi_i) : i \in I\}$ of charts on M such that

1. ϕ_i is a homeomorphism of U_i onto the open subset $\phi_i(U_i)$ of $\mathbb{C}^n$ for all $i \in I$.
2. For all $i, j \in I$, $\phi_i \phi_j^{-1}$ is a biholomorphic map of $\phi_j(U_{ij})$ onto $\phi_i(U_{ij})$.

Remarks.

1. We often refer to an n -dimensional complex manifold as being a complex manifold modelled on $\mathbb{C}^n$. We also say that the atlas defines a complex analytic structure on M . It is always assumed that a complex manifold M comes with a specific complex analytic structure. As we shall see later, a topological space may admit many distinct complex analytic structures.

2. We generally assume that the atlas $\mathcal{A}$ is maximal. That is, if (U, ϕ) is any chart on M such that $\phi\phi_j^{-1}$ is biholomorphic for all $j \in I$, then $(U, \phi) \in \mathcal{A}$.
3. Every n -dimensional complex manifold has the structure of a $2n$ -dimensional real analytic (therefore, differential) manifold: Forget the complex structure!

Definition 4.1.2. Let M be an m -dimensional complex manifold with atlas $\mathcal{A}$ and N be a connected subset of M . We say that N is an n -dimensional complex submanifold of M if for every $x \in N$, there exists $(U, \phi) \in \mathcal{A}$ such that ϕ maps $U \cap N$ homeomorphically onto an open subset of $\mathbb{C}^n \times \{0\} \subset \mathbb{C}^n \times \mathbb{C}^{m-n} \cong \mathbb{C}^m$.

Remark. If N is an n -dimensional complex submanifold of M then N has the structure of an n -dimensional complex manifold. Indeed, an atlas for N is implicit in the definition.

Example. An open (connected) subset of an n -dimensional complex manifold M is an n -dimensional complex submanifold of M .

Definition 4.1.3. Let M, N be complex manifolds with atlases $\mathcal{A}, \mathcal{B}$ respectively. A map $f : M \rightarrow N$ is said to be *holomorphic* or *analytic* if $\zeta f \phi^{-1} : \phi(U \cap f^{-1}(V)) \rightarrow \zeta(V)$ is analytic for all $(U, \phi) \in \mathcal{A}$, $(V, \zeta) \in \mathcal{B}$. If f is a homeomorphism and both f and f^{-1} are analytic we say that f is *biholomorphic* and that M and N are *biholomorphic* or *analytically equivalent*.

Notation. Let M be an m -dimensional complex manifold with topology $\mathcal{U}$. If $U \in \mathcal{U}$, we let $A(U)$ denote the ring of $\mathbb{C}$ -valued analytic functions on U . Just as in §1, Chapter 3, we let $\mathcal{O}_x$ denote the ring of germs of analytic functions at a point $x \in M$. Using a chart containing x , it is easy to see that $\mathcal{O}_x \cong \mathbb{C}\{z_1, \dots, z_m\}$ for all $x \in M$ (of course, the isomorphism is no longer canonical). We let $\mathcal{M}_x$ denote the quotient field of $\mathcal{O}_x$ and $\mathcal{M}$ denote the disjoint union of the fields $\mathcal{M}_x$ over M . We call $\mathcal{M}$ the sheaf of germs of meromorphic functions on M .

Definition 4.1.4. Let M be a complex manifold. A map $m : M \rightarrow \mathcal{M}$ is said to define a *meromorphic function* on M if

1. For all $x \in M$, $m(x) \in \mathcal{M}_x$.
2. We may find an open neighbourhood U of every point $x \in M$ and $f, g \in A(U)$ such that $m(y) = f_y/g_y$, $y \in U$.

Notation. We let $\mathcal{M}(M)$ denote the field of meromorphic functions on M . Given $U \in \mathcal{U}$, we let $A^*(U)$, $\mathcal{M}^*(U)$ denote the multiplicative groups of units in $A(U)$, $\mathcal{M}(U)$ respectively.

Let us now indicate some of the main problems in the theory of complex manifolds. The fundamental problem is, of course, the classification of complex manifolds up to biholomorphism. As we shall soon see this is unrealistically hard, even for domains in $\mathbb{C}^n$. Next we might try to find all complex structures, if any, on a given differential manifold. Again, this appears an intractable problem at present (see §§4, 5). Finally, we might ask for necessary conditions on a differential manifold for it to admit a complex structure. Although some important results have been proved, especially about compact 4-manifolds, relatively little is known. For example, it is still unknown whether the six dimensional sphere admits a complex structure though it is known that the only other sphere that can admit a complex structure is the Riemann sphere. We do, however, have one elementary result.

Proposition 4.1.5. If the differential manifold M admits a complex structure then M is even dimensional and orientable.

Proof. M must obviously be even dimensional as $\mathbb{C}^n$ is always of even (real) dimension. As for the orientability, notice that $\mathbb{C}^n$ has a canonical orientation defined by the complex structure J on $\mathbb{C}^n$ (§5, Chapter 1). If M has complex analytic atlas $\{(U_i, \phi_i)\}$, then $D(\phi_i \phi_j^{-1})(\phi_j(x))$ is $\mathbb{C}$ -linear for all i, j and so commutes with J . Consequently, M is orientable. Alternatively, the reader may verify that the real determinant of a complex linear map is always positive (see §4, Chapter 5). $\square$

Remark. If a differential manifold is two dimensional then it admits a complex structure if and only if it is orientable. For a proof of this result, depending on the theorem of Korn and Lichtenstein on the existence of isothermal parameters, see Chern [1].

We conclude this section by giving the straightforward generalisation of Definition 2.2.1 to complex manifolds.

Definition 4.1.6. Let Y be a subset of the complex manifold M . We say that Y is an *analytic subset* of M if for each $x \in M$ there exist $U \in \mathcal{U}_x$ and an analytic function $f : U \rightarrow \mathbb{C}^p$ such that $Y \cap U = f^{-1}(0)$ (p may depend on x).

Remarks.

1. Notice that an analytic subset of M is necessarily closed.
2. In Chapter 6 we define what is meant by an analytic function on an analytic set as well as giving an “intrinsic” definition of an analytic set.

Exercises

1. Show that if M and N are complex manifolds then $M \times N$ has the structure of a complex manifold.
2. Let M be a complex manifold and $m \in \mathcal{M}^*(M)$. Show that the zero, pole and indeterminacy sets of m are well defined analytic subsets of M .
3. Show that the zero set of $y^2 = z^3$ is not a complex submanifold of $\mathbb{C}^2$.

2 Complex submanifolds of $\mathbb{C}^n$

Every domain in $\mathbb{C}^n$ has the structure of an *open complex submanifold* of $\mathbb{C}^n$. As we discussed in §4 of Chapter 1, every proper simply connected domain of $\mathbb{C}$ is biholomorphic to the open unit disc. We have no such simple classification of simply connected domains in $\mathbb{C}^n$, $n > 1$.

Example 1. Let D denote the open unit polydisc, centre zero, in $\mathbb{C}^n$, $n > 1$, and set $D^* = D \setminus \{(0, \dots, 0, t) : 0 \leq t < 1\}$. D and D^* are certainly homeomorphic but cannot be biholomorphic since D is a domain of holomorphy whilst D^* is not (see example 6, §4, Chapter 2).

Even if Ω and Ω' are homeomorphic domains of holomorphy they need not be biholomorphic.

Example 2. (Theorem of Poincaré) Let D and E respectively denote the open unit polydisc, centre zero, and the open unit Euclidean disc, centre zero, in $\mathbb{C}^n$. Then for $n > 1$, D and E are not biholomorphic. Our proof of this inequivalence uses an elementary argument due to Simha [1]. First we remark that if there exists a biholomorphic map of D onto E then there certainly exists a biholomorphic map of D onto E which preserves the origin. This is just a consequence of the existence of biholomorphisms of D taking any prescribed point of D to the origin (That such maps exist is an immediate consequence of their existence for $n = 1$). We shall show that if $f : D \rightarrow E$ and $g : E \rightarrow D$ are holomorphic origin preserving maps then $Df(0)(D) \subset E$ and $Dg(0)(E) \subset D$. Assuming this, we see that if f is a biholomorphism with inverse g then we must have $Df(0)(D) = E$ —since $Df(0) \cdot Dg(0) = I$. Hence $Df(0)(\partial D) = \partial E$. But this is absurd since ∂D contains linear pieces of positive dimension whilst ∂E does not.

1. $Df(0)(D) \subset E$.

Let $f = (f_1, \dots, f_n)$, $v \in D$, $u = (u_1, \dots, u_n) \in E$ and $\langle \cdot, \cdot \rangle$ denote the standard Hermitian inner product on $\mathbb{C}^n$. Applying Schwarz's lemma to the function $u \mapsto \sum u_i f_i(tv)$, we see that $|\langle \bar{u}, Df(0)(v) \rangle| \leq 1$. This holds for all $u \in E$, therefore $Df(0)(v) \in E$.

2. $Dg(0)(E) \subset D$.

Let $g = (g_1, \dots, g_n)$ and $u = (u_1, \dots, u_n) \in E$. Applying Schwarz's lemma to the function $u \mapsto f_j(tu_1, \dots, tu_n)$ we see that $|\sum u_i \partial f_j / \partial z_i(0)| \leq 1$ for $1 \leq j \leq n$. Hence $Df(0)(u) \in D$.

Remarks.

1. It is not hard to show that there exist no proper holomorphic maps of D into E . See Alexander [1] (By *proper* we mean a map such that inverse images of compact sets are compact).
2. For more inequivalence results of the above type and further references see Alexander [1] or R. Narasimhan [2, Chapter 5].
3. In spite of the negative character of the above examples there certainly are some topological restrictions on domains of holomorphy in $\mathbb{C}^n$. For example, if Ω is a domain of holomorphy then $H_r(\Omega, \mathbb{Z}) = 0$, $r > n$, and Ω has the homotopy type of an n -dimensional CW complex (see Andreotti and Frankel [1] and R. Narasimhan [4, 5] for further details). If Ω is a Runge domain (§7, Chapter 2) then $H^r(\Omega, \mathbb{C}) = 0$, $r \geq n$ (Theorem of Serre [1], see also Hörmander [1, Theorem 2.7.11]).

Next we turn to the “classical” domains in $\mathbb{C}^n$ which generalise the open unit disc or upper half plane in $\mathbb{C}$. As we shall not need any results from this theory in the remainder of these notes we shall be rather brief and sketchy in our presentation referring the reader to the references for further details and proofs.

Let Ω be a bounded domain in $\mathbb{C}^n$. We let $\text{Aut}(\Omega)$ denote the group of biholomorphisms of Ω . A fundamental result of H. Cartan [4] is that $\text{Aut}(\Omega)$ is a locally compact Lie group (see also R. Narasimhan [2]).

Definition 4.2.1. A domain Ω in $\mathbb{C}^n$ is said to be *homogeneous* if $\text{Aut}(\Omega)$ acts transitively on Ω . That is, given $y, z \in \Omega$, there exists $f \in \text{Aut}(\Omega)$ such that $f(y) = z$. We say Ω is *symmetric* if, in addition, given $z \in \Omega$ there exists $f \in \text{Aut}(\Omega)$ such that $f(z) = z$, $f(y) \neq y$, $y \neq z$, and $f^2 = I$.

E. Cartan [1] classified all the bounded homogeneous domains in $\mathbb{C}^2$ and $\mathbb{C}^3$ proving that any such domain in $\mathbb{C}^2$ is biholomorphic to either the unit polydisc or the unit Euclidean disc and that in $\mathbb{C}^3$ any such domain is biholomorphic to one of the unit polydisc, the unit

Euclidean disc, the product of the 2-dimensional unit Euclidean disc with the unit disc, the domain $\text{Im}(z_3) > \sqrt{\text{Im}(z_1)^2 + \text{Im}(z_2)^2}$. These domains are all symmetric. However, it was later shown by Pyatetskii-Shapiro [1] that there exist bounded homogeneous domains in $\mathbb{C}^n$, $n \geq 4$, which are not symmetric (non-countably many for $n \geq 7$). Proofs of these facts may be found in Pyatetskii-Shapiro [1], Siegel [2] and Hua [1]. We say that a bounded symmetric domain is *irreducible* if it cannot be written as a product of symmetric domains of lower dimension. E. Cartan [1] proved that there exist six classes of irreducible symmetric domain. Four of these are referred to as classical since their automorphism groups are classical semi-simple Lie groups. The remaining two classes are exceptional in that one is encountered only in $\mathbb{C}^{16}$, the other only in $\mathbb{C}^{27}$.

We now briefly describe the classical irreducible symmetric domains.

A. Classical domains of the first type. Suppose $n = pq$. Write an element $z \in \mathbb{C}^n$ as a $p \times q$ matrix $Z = [z_{ij}]$. Let I_q denote the identity in $\mathbb{C}^q$ and $\bar{Z}^*$ denote the conjugate transpose of Z . The domain $D_I = \{Z : I_q - \bar{Z}^* Z \text{ is positive definite}\}$ is a representative classical domain of the first type. Notice that if $p = n$, $q = 1$, we recover the Euclidean unit disc.

B. Classical domains of the second type. Suppose $n = p(p+1)/2$. In this case we may identify $\mathbb{C}^n$ with the space of $p \times p$ symmetric matrices. The domain $D_{II} = \{Z : I_p - Z\bar{Z} \text{ is positive definite}\}$ is a representative classical domain of the second type.

C. Classical domains of the third type. Suppose $n = p(p-1)/2$. In this case we may identify $\mathbb{C}^n$ with the space of $p \times p$ skew-symmetric matrices. The domain $D_{III} = \{Z : I_p + Z\bar{Z} \text{ is positive definite}\}$ is a representative classical domain of the third type.

D. Classical domains of the fourth type. These exist in all dimensions and a representative domain is given by

$$D_{IV} = \left\{ z \in \mathbb{C}^n : \left| \sum z_i^2 \right| > \sqrt{2 \sum |z_i|^2 - 1} \right\}.$$

Finally we describe Siegel domains of the second kind and their relationship with the classical domains (for the general theory we refer to Pyatetskii-Shapiro [1] or Siegel [2]).

Definition 4.2.2. Let $V \subset \mathbb{R}^n$ be an open convex cone which does not contain any real line. We say that a function $F : \mathbb{C}^m \times \mathbb{C}^m \rightarrow \mathbb{C}^n$ is a *V-Hermitian form* if

1. $F(u, v) = \overline{F(v, u)}$, $u, v \in \mathbb{C}^m$.
2. F is $\mathbb{C}$ -linear in the first variable.
3. $F(u, u) = 0$ if and only if $u = 0$.
4. $F(u, u) \in \bar{V}$ for all $u \in \mathbb{C}^m$.

Definition 4.2.3. A *Siegel domain of the second kind* is a domain in $\mathbb{C}^{n+m}$ consisting of points (y, z) such that $\text{Im}(z) - F(y, y) \in V$ for some fixed V -Hermitian form F on $\mathbb{C}^m \times \mathbb{C}^m$.

Example 3. Let V denote the set of strictly positive real numbers and F the standard Hermitian inner product on $\mathbb{C}^m$. The corresponding Siegel domain of the second kind is the (unbounded) domain $\{(z, t) \in \mathbb{C}^{m+1} : \text{Im}(t) - \langle z, z \rangle > 0\}$. In fact this domain is biholomorphic to a classical domain of the first type, in this case the unit Euclidean disc $\{u \in \mathbb{C}^{m+1} : \sum |u_i|^2 < 1\}$.

1}. The reader may easily verify that a biholomorphic map between the two domains is given by

$$u_1 = (t - i)/(t + i), \quad u_{j+1} = z_j \sqrt{2}/(t + i), \quad 1 \leq j \leq m.$$

More generally it may be shown that any Siegel domain of the 2nd. kind is biholomorphic to a bounded domain. A fundamental result due to Pyatetskii-Shapiro [1] states that every bounded homogeneous domain in $\mathbb{C}^n$ is biholomorphic to an (affinely homogeneous) Siegel domain of the second kind.

Siegel domains are of considerable importance in Harmonic analysis, automorphic function theory and the representation theory of semi-simple Lie groups. For further details and references see Pyatetskii-Shapiro [1] and Hua [1]. For an indication of more recent developments, heavily based on several complex variables, see Wells and Wolf [1]. Finally we remark that the kernel functions and automorphism groups of the classical domains are computed in Hua [1].

Next we shall consider closed submanifolds of $\mathbb{C}^n$. First a general result about compact complex manifolds.

Proposition 4.2.4. Let M be a compact complex manifold. Then $A(M) = \mathbb{C}$. That is, every holomorphic function on M is constant.

Proof. Exactly as for the proof of Lemma 1.4.2. □

Corollary 4.2.5. The only compact complex submanifolds of $\mathbb{C}^n$ are isolated points.

Proof. Let M be a compact complex submanifold of $\mathbb{C}^n$. The coordinate functionals $z \mapsto z_j$, $1 \leq j \leq n$, restrict to analytic functions on M . Now apply Proposition 4.2.4 to these analytic functions on M . □

A consequence of Corollary 4.2.5 is that the only interesting closed submanifolds of $\mathbb{C}^n$ are non-compact.

Proposition 4.2.6. Let M be a non-compact closed submanifold of $\mathbb{C}^n$. Then

1. Given $x, y \in M$, $x \neq y$, there exists $f \in A(M)$ such that $f(x) \neq f(y)$.
2. M is holomorphically convex: Given a compact subset K of M , $\widehat{K} = \{z \in M : |f(z)| \leq \|f\|_K \text{ for all } f \in A(M)\}$ is a compact subset of M .
3. We can define local analytic coordinates on M by means of globally defined analytic functions. That is, given $z \in M$, there exists a neighbourhood U of z in M and $f_1, \dots, f_m \in A(M)$ such that, when restricted to U , $(f_1, \dots, f_m) : U \rightarrow \mathbb{C}^m$ is a complex analytic chart for M .

Proof. 1 and 2 are obvious since $A(M)$ contains the affine $\mathbb{C}$ -linear maps on $\mathbb{C}^n$ restricted to M . For 3, just project M onto the tangent space to M at z . □

Definition 4.2.7. A complex manifold which satisfies properties 1, 2 and 3 of Proposition 4.2.6 is called a *Stein manifold*.

Examples.

4. Every domain of holomorphy is a Stein manifold. We may think of Stein manifolds as being a natural generalisation of domains of holomorphy.

5. Every non-compact Riemann surface is Stein by a theorem of Behnke and Stein [1].

Remark. As we shall see later, Stein manifolds have a rich function theory. Roughly speaking, what we can do continuously or differentiably on a Stein manifold we can do complex analytically (“Oka’s principle”—see also Chapters 7 and 12).

We have the following basic result analogous to Whitney’s embedding theorem.

Theorem 4.2.8. (Bishop [1], R. Narasimhan [6]). Every n -dimensional Stein manifold has a proper analytic embedding in $\mathbb{C}^{2n+1}$. In particular, every Stein manifold is biholomorphic to a closed submanifold of some $\mathbb{C}^n$.

Proof. The proof of this result is considerably harder than that of Whitney’s embedding theorem. The main difficulty lies in constructing proper maps from a Stein manifold into $\mathbb{C}^m$. For example, if the manifold is 2 complex dimensional there do not exist any proper analytic $\mathbb{C}$ -valued maps. We shall not give a proof of the embedding theorem but instead refer the reader to the original references or to Gunning and Rossi [1] and Hörmander [1]. $\square$

Example 6. The open unit disc D in $\mathbb{C}$ is a Stein manifold. We shall construct a proper embedding of D in $\mathbb{C}^3$. The main part of our construction will be to find a proper analytic map $F : D \rightarrow \mathbb{C}^2$. For $n > 1$, we let D_n denote the open unit disc, centre zero, and radius $1 - 1/n$. We first construct an analytic map $f : D \rightarrow \mathbb{C}$ such that

$$|f(z)| > n + 1, \quad z \in \partial D_n, \quad n > 1.$$

To do this we shall construct inductively a sequence $f_n \in A(D)$ satisfying

$$\|f_n\|_{D_{n-1}} \leq 2^{-n}; \quad \|f_n\|_{\partial D_n} \geq n + 2 + \sum_{j=2}^{n-1} \|f_j\|_{D_n}. \quad (*)$$

Suppose the f_j are constructed for $j < n$. Set $f_n(z) = \left(\frac{n-1}{n}b_n z\right)^{p(n)}$. Choose $b_n \in \left(\frac{n}{n-1}, \frac{n}{n-2}\right)$ and $p(n)$ very large so that $(*)$ holds. This completes the inductive step. Now define $f = \sum_{j=1}^{\infty} f_j$. $(*)$ guarantees that f is analytic and that $|f(z)| > n + 1$, $z \in \partial D_n$.

For $n > 1$, set $X_n = \{z \in D_{n+1} \setminus D_n : |f(z)| \leq n + 1\}$ and $Y_n = \{z \in D_n : |f(z)| \leq n + 1\}$. X_n and Y_n are disjoint compact subsets of D and we clearly have $\hat{X}_n = X_n$, $\hat{Y}_n = Y_n$. We now inductively construct a sequence $h_n \in A(D)$ satisfying

$$\|h_n\|_{Y_n} < 2^{-n}; \quad \|h_n\|_{X_n} > 2 + n + \sum_{j=2}^{n-1} \|h_j\|_{X_n}. \quad (\#)$$

Suppose the h_j are constructed for $j < n$. Define an analytic function q_n on a neighbourhood of $X_n \cup Y_n$ by requiring it to be zero on Y_n and equal to $3 + n + \sum_{j=2}^{n-1} \|h_j\|_{X_n}$ on X_n . The conditions of the Runge Approximation theorem (Theorem A1.8) hold in D for the compact set $X_n \cup Y_n$ and so we can approximate q_n by an analytic function h_n on D such that $\|h_n - q_n\|_{X_n \cup Y_n} \leq 2^{-n}$. Since h_n satisfies the required conditions the inductive step is completed. Define $h = \sum_{j=2}^{\infty} h_j$. Then $(\#)$ guarantees that h is an analytic function on D and that

$$|h(z)| > n + 1 \quad \text{on } X_n, \quad n > 1.$$

Define $F = (f, h) : D \rightarrow \mathbb{C}^2$. We see that

$$|F(z)| = \max\{|f(z)|, |h(z)|\} > n + 1, \quad z \in D_{n+1} \setminus D_n, \quad n > 1.$$

Therefore, F is a proper map. To construct the required proper embedding of D in $\mathbb{C}^3$ we define $\phi : D \rightarrow \mathbb{C}^3$ by $\phi(z) = (z, F(z))$.

Remarks.

1. It can be shown that there exists a proper embedding of D in $\mathbb{C}^2$. Since every proper map is closed, there can be no proper embedding of D in $\mathbb{C}$.
2. The technique used in the example to construct a proper map is actually a special case of the method used to construct proper maps on an arbitrary Stein manifold. The reader should note that the difficulty in generalising the example lies in filling up the space with domains to which we can apply an appropriate version of the Runge Approximation theorem.

Exercises

1. Construct a proper embedding of the polydisc $D(z; r_1, \dots, r_n)$ in $\mathbb{C}^{2n+1}$.
2. Show that if we work with real analytic maps then the real Euclidean disc $E(r)$ in $\mathbb{R}^n$ is real analytically diffeomorphic to $\mathbb{R}^n$.
- 3*. Construct a proper embedding of the Euclidean disc $E(r)$ in $\mathbb{C}^n$ in $\mathbb{C}^{2n+1}$.
- 4*. Show that the unit disc in $\mathbb{C}$ admits a proper embedding in $\mathbb{C}^2$.
5. (2nd. Riemann removable singularities theorem). Let X be a complex submanifold of the domain Ω in $\mathbb{C}^n$. Show that if $\text{codim}(X) \geq 2$, then every analytic function on $\Omega \setminus X$ extends to Ω (Show that every analytic function on $\Omega \setminus X$ is locally bounded on Ω).
- 6*. Generalise the result of question 5 to the case when X is an analytic subset of Ω of codimension at least 2 at every point.

3 Projective algebraic manifolds

The most important examples of compact complex manifolds are given by the complex projective spaces and their closed submanifolds. We start this section by constructing complex n -dimensional projective space.

Let $P^n(\mathbb{C})$ denote the set of complex lines through the origin of $\mathbb{C}^{n+1}$. We have a natural projection map $q : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow P^n(\mathbb{C})$ defined by: $q(z) = \text{Complex line through } z \text{ and the origin of } \mathbb{C}^{n+1}$. We give $P^n(\mathbb{C})$ the quotient topology. We call $P^n(\mathbb{C})$, with this topology, *complex n -dimensional projective space*.

Proposition 4.3.1. Complex n -dimensional projective space is a compact, connected Hausdorff space, $n \geq 1$.

Proof. Let S^{2n+1} denote the unit sphere in $\mathbb{C}^{n+1}$ relative to the standard Hermitian inner product on $\mathbb{C}^{n+1}$. Every complex line L in $\mathbb{C}^{n+1}$ meets S^{2n+1} in a set homeomorphic to the unit circle S^1 . Consequently, $q(S^{2n+1}) = P^n(\mathbb{C})$. Since S^{2n+1} is compact and connected so therefore is $P^n(\mathbb{C})$. We leave the verification that $P^n(\mathbb{C})$ is Hausdorff as an easy exercise. $\square$

Remark. The map $q : S^{2n+1} \rightarrow P^n(\mathbb{C})$ is called the *Hopf fibration* of S^{2n+1} . In case $n = 1$, it is a well known fact that the map $q : S^3 \rightarrow S^2$ defines a non-trivial element of $\pi_3(S^2)$ (See, for example, Hirsch [1, page 131]).

Observe that $(z_0, \dots, z_n), (z'_0, \dots, z'_n) \in \mathbb{C}^{n+1} \setminus \{0\}$ define the same point of $P^n(\mathbb{C})$ if and only if there exists $\lambda \in \mathbb{C}^*$ such that $z_i = \lambda z'_i$, $0 \leq i \leq n$. We may think of non-zero $(n+1)$ -tuples $(z_0, \dots, z_n)$ as defining *homogeneous coordinates* on $P^n(\mathbb{C})$. Here it is understood that two non-zero $(n+1)$ -tuples define the same point of $P^n(\mathbb{C})$ if and only if they are non-zero complex multiples of one another.

Proposition 4.3.2. Complex n -dimensional projective space has the natural structure of a complex manifold.

Proof. For $0 \leq i \leq n$, define open subsets U_i of $P^n(\mathbb{C})$ by $U_i = \{(z_0, \dots, z_n) \in P^n(\mathbb{C}) : z_i \neq 0\}$. We define homeomorphisms $\phi_i : U_i \rightarrow \mathbb{C}^n$ by

$$\phi_i(z_0, \dots, z_n) = (z_0/z_i, \dots, \widehat{z_i/z_i}, \dots, z_n/z_i)$$

($\widehat{}$ denotes omission). Clearly $\{U_i : 0 \leq i \leq n\}$ covers $P^n(\mathbb{C})$ and the set $\{(U_i, \phi_i) : 0 \leq i \leq n\}$ is an atlas on $P^n(\mathbb{C})$. A simple computation shows that $\phi_i \phi_j^{-1}$ is biholomorphic (even rational) for all i, j . $\square$

Remark. It is an outstanding problem to determine whether the complex structure on $P^n(\mathbb{C})$ given above is the only one that $P^n(\mathbb{C})$ admits compatible with its standard differential structure. For $n = 1$ this is well known by the Riemann mapping theorem. In higher dimensions only partial results are known. See Hirzebruch and Kodaira [1], Frankel [1], Hartshorne [1], Mori [1], Siu and Yau [1].

Suppose P is a homogeneous polynomial on $\mathbb{C}^{n+1}$. Then $M = \{z \in \mathbb{C}^{n+1} : P(z) = 0\}$ defines a complex cone in $\mathbb{C}^{n+1}$. That is, if $z \in M$ so does λz , for all $\lambda \in \mathbb{C}$. Thus, in a natural way, M determines a subset M^* of $P^n(\mathbb{C})$. Any point in M^* corresponds to a line contained in M . Put another way, M^* is just the zero locus of P in homogeneous coordinates.

Definition 4.3.3. Any subset of $P^n(\mathbb{C})$ that may be represented as the common zero locus of a set (not necessarily finite) of homogeneous polynomials defined on $\mathbb{C}^{n+1}$ is called (projective) *algebraic*. We say that a complex submanifold of $P^n(\mathbb{C})$ is algebraic if it is an algebraic subset of $P^n(\mathbb{C})$. We say that a complex manifold is algebraic if it is biholomorphic to an algebraic submanifold of some complex projective space.

Remark. It follows from Hilbert's basis theorem that every projective algebraic set is the common zero locus of a finite set of homogeneous polynomials.

Examples.

1. Let $(a_0, \dots, a_n) \in P^n(\mathbb{C})$. The zero set of the polynomial $a_0 z_0 + \dots + a_n z_n$ is called a *complex hyperplane*. Every complex hyperplane is biholomorphic to $P^{n-1}(\mathbb{C})$ and the set of complex hyperplanes is in bijective correspondence with $P^n(\mathbb{C})$. Taking the hyperplane $z_0 = 0$, we see that $P^n(\mathbb{C}) = P^{n-1}(\mathbb{C}) \cup U_0$ and iterating this construction we obtain a cell decomposition of $P^n(\mathbb{C})$ from which we can compute the cohomology ring of $P^n(\mathbb{C})$.
2. If $F : \mathbb{C}^{n+1} \rightarrow \mathbb{C}$ is a homogeneous polynomial then $F(z) = 0$ defines an algebraic submanifold of $P^n(\mathbb{C})$ if (and only if) $DF_z \neq 0$ for all $z \in F^{-1}(0) \setminus \{0\}$. This is a straightforward consequence of the implicit function theorem and the homogeneity of F (Observe that by Euler's theorem, $DF_z(z) = 0$, $z \in F^{-1}(0)$). In particular, the hyperquadric $z_0^2 + \dots + z_n^2 = 0$ is an algebraic submanifold of $P^n(\mathbb{C})$.

3. The cubic curve $y^2z - 4x^3 + g_2xz^2 + g_3z^3 = 0$ defines an algebraic submanifold of $P^2(\mathbb{C})$ provided that $g_2^3 - 27g_3^2 \neq 0$ (that is, the discriminant of the cubic is non-zero). We shall return to this important class of algebraic curves in §4.
4. Every compact Riemann surface is algebraic. We prove this for surfaces of genus 1 in §4 and in general in Chapter 10 (see also the exercises at the end of §6, Chapter 7). Proofs may also be found in Siegel [1, Chapter 2] and Griffiths and Harris [1, page 213].

Some indication of the significance of projective space may be gauged from the famous theorem of Chow that we shall prove in Chapter 7:

Theorem. Every complex analytic subset of $P^n(\mathbb{C})$ is algebraic.

Remark. We say that a meromorphic function on $P^n(\mathbb{C})$ is *rational* if it can be expressed as the quotient of two homogeneous polynomials of the same degree. One consequence of Chow's theorem is that every meromorphic function on $P^n(\mathbb{C})$ is rational (we have already proved this in §4 of Chapter 1 for the Riemann sphere, $P^1(\mathbb{C})$). We give a proof of the rationality of meromorphic functions on $P^n(\mathbb{C})$, assuming Chow's theorem, in Example 3 of §6.

For the remainder of this section we shall construct a number of important examples of algebraic manifolds. In each case we shall construct a biholomorphic map of the space onto a submanifold of projective space. We shall not verify that the submanifold of projective space is algebraic. This can either be done directly—usually a rather tedious and long computation—or indirectly by citing Chow's theorem.

Examples.

5. For $m, n \geq 1$, $P^m(\mathbb{C}) \times P^n(\mathbb{C})$ is algebraic. We take homogeneous coordinates $(z_0, \dots, z_m)$ and $(w_0, \dots, w_n)$ on $P^m(\mathbb{C})$ and $P^n(\mathbb{C})$ respectively. We define an embedding ϕ of the product in $P^{nm+n+m}(\mathbb{C})$ by

$$\phi(z_0, \dots, z_m; w_0, \dots, w_n) = (z_0w_0, z_0w_1, \dots, z_0w_n, z_1w_0, \dots, z_mw_n).$$

We leave it to the reader to verify that ϕ is an embedding (the *Segre embedding*).

6. Grassmann manifolds. For $0 \leq k \leq n$, we let $G_{k,n}(\mathbb{C})$ denote the set of k -dimensional complex linear subspaces of $\mathbb{C}^n$. We call $G_{k,n}(\mathbb{C})$ the *Grassmann manifold* of k -planes in $\mathbb{C}^n$.

We shall first show that $G_{k,n}(\mathbb{C})$ is in bijective correspondence with a closed subset of $P(\wedge^k \mathbb{C}^n)$ (For any complex vector space V , $P(V)$ will denote the projective space of complex lines through the origin of V . Thus $P(\mathbb{C}^n) = P^{n-1}(\mathbb{C})$). Suppose $E \in G_{k,n}(\mathbb{C})$ and let $v_1, \dots, v_k$ and $w_1, \dots, w_k$ be bases for E . Then $v_1 \wedge \dots \wedge v_k = aw_1 \wedge \dots \wedge w_k$ for some $a \in \mathbb{C}^*$. Conversely, if such a relation holds, $v_1, \dots, v_k$ and $w_1, \dots, w_k$ span the same k -dimensional linear subspace of $\mathbb{C}^n$. Consequently we may define an injection $\theta : G_{k,n}(\mathbb{C}) \rightarrow P(\wedge^k \mathbb{C}^n)$ by setting $\theta(E) = v_1 \wedge \dots \wedge v_k$, where $v_1, \dots, v_k$ is any basis for E . $\theta(G_{k,n}(\mathbb{C}))$ is obviously a closed subset of $P(\wedge^k \mathbb{C}^n)$ and we give $G_{k,n}(\mathbb{C})$ the induced topology.

If we let $e_1, \dots, e_n$ denote the standard basis for $\mathbb{C}^n$ and choose a basis $v_1, \dots, v_k$ for E we can write

$$\theta(E) = \sum a_{i_1 \dots i_k} e_{i_1} \wedge \dots \wedge e_{i_k}$$

where the $a_{i_1 \dots i_k}$ are skew-symmetric in the indices and are defined uniquely by E up to multiplication by elements of $\mathbb{C}^*$. The scalars $a_{i_1 \dots i_k}$ are called the *Cayley–Plücker–Grassmann coordinates* on $G_{k,n}(\mathbb{C})$.

Next we shall construct a complex analytic atlas for $G_{k,n}(\mathbb{C})$. We let $\underline{k}$ denote the set of (ordered) subsets of $\{1, \dots, n\}$ containing precisely k elements. If $\alpha = \{\alpha_1, \dots, \alpha_k\} \in \underline{k}$, we

set $c\alpha = \{1, \dots, n\} \setminus \alpha$ and let E_α denote the k -dimensional subspace of $\mathbb{C}^n$ spanned by the coordinate vectors $e_{\alpha_1}, \dots, e_{\alpha_k}$. Let V_α denote the vector space of $\mathbb{C}$ -linear maps from E_α to $E_{c\alpha}$. Clearly $V_\alpha \cong \mathbb{C}^{k(n-k)}$. We have a natural homeomorphism ψ_α of V_α onto an open subset U_α of $G_{k,n}(\mathbb{C})$ defined by mapping a linear function to its graph. Set $\phi_\alpha = \psi_\alpha^{-1}$. Then it is not difficult to show that $\{(U_\alpha, \phi_\alpha) : \alpha \in \underline{k}\}$ is a complex analytic atlas on $G_{k,n}(\mathbb{C})$. Relative to the complex structure defined by this atlas it is now a straightforward matter to verify that the bijection θ defined above is a complex analytic embedding of $G_{k,n}(\mathbb{C})$ onto a compact complex submanifold of $P(\wedge^k \mathbb{C}^n)$.

We shall now give two other descriptions of $G_{k,n}(\mathbb{C})$ in terms of Lie groups. Observe that $GL(n, \mathbb{C})$ acts transitively on $G_{k,n}(\mathbb{C})$. Given $E \in G_{k,n}(\mathbb{C})$ we let $I(E)$ denote the isotropy subgroup of $GL(n, \mathbb{C})$ at E (that is, $I(E) = \{A \in GL(n, \mathbb{C}) : A(E) = E\}$). It follows from the elementary theory of homogeneous spaces (or directly, in this case) that $G_{k,n}(\mathbb{C}) = GL(n, \mathbb{C})/I(E)$ (Kobayashi and Nomizu [2]). Let $GL(k, n-k, \mathbb{C})$ denote the subgroup of $GL(n, \mathbb{C})$ consisting of all matrices of the form $\begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$ where $A \in GL(k, \mathbb{C})$, $C \in GL(n-k, \mathbb{C})$ and B is any $k \times (n-k)$ -matrix. If we take E to be the k -plane defined by the first k coordinate vectors, it is easy to see that $I(E) = GL(k, n-k, \mathbb{C})$. Hence,

$$G_{k,n}(\mathbb{C}) \cong GL(n, \mathbb{C})/GL(k, n-k, \mathbb{C}).$$

Consequently, $G_{k,n}(\mathbb{C})$ is a complex manifold of dimension $n^2 - k^2 - (n-k)^2 - k(n-k) = k(n-k)$.

Similarly, taking the standard Hermitian inner product on $\mathbb{C}^n$, we see that

$$G_{k,n}(\mathbb{C}) \cong U(n)/(U(k) \times U(n-k))$$

($U(p)$ denotes the unitary group of $\mathbb{C}^p$). Since the unitary groups are compact, it follows from this representation that $G_{k,n}(\mathbb{C})$ is compact.

Grassmann manifolds are particularly important as they are classifying spaces for vector bundles. See the exercises at the end of the section.

The homology of a Grassmann manifold is described using the ‘‘Schubert Calculus’’. See Griffiths and Harris [1] and also Chern [1].

7. Flag manifolds. Let $0 < p_1 < p_2 < \dots < p_k \leq n$ be a sequence of integers. The *flag manifold* $F(p_1, \dots, p_k, n)$ is defined to be the set of all nested sequences $L_{p_1} \subset L_{p_2} \subset \dots \subset L_{p_k} \subset \mathbb{C}^n$ of linear subspaces of $\mathbb{C}^n$ with dimensions given by the subscripts. Clearly $F(p, n) = G_{p,n}(\mathbb{C})$. $F(p_1, \dots, p_k, n)$ has the structure of a compact complex manifold of dimension $\sum_{j=1}^{k-1} p_j(p_{j+1} - p_j)$. The easiest way to prove this is to represent the flag manifold as a homogeneous space as we did for the Grassmann manifold. Flag manifolds are also algebraic. The proof of this is similar to that we gave for the Grassmann manifold. Flag manifolds are important in the representation theory of semi-simple Lie groups. See, for example, Wells and Wolf [1].

Exercises

1. Let $z_0 \in P^n(\mathbb{C})$. Show that if $n > 1$, $P^n(\mathbb{C}) \setminus \{z_0\}$ is an example of a non-compact complex manifold with no non-constant analytic functions.
2. Let $U = \{(z, L) \in \mathbb{C}^n \times G_{k,n}(\mathbb{C}) : z \in L\}$ and define $\pi : U \rightarrow G_{k,n}(\mathbb{C})$ by $\pi(z, L) = L$. Show that U has the natural structure of a k -dimensional holomorphic vector bundle over $G_{k,n}(\mathbb{C})$. We call U the *universal bundle* or *canonical bundle* of $G_{k,n}(\mathbb{C})$. It can be shown that given any k -dimensional vector bundle E over a differential manifold X there exists a positive integer N , depending only on X , such that for some differentiable map $f : X \rightarrow G_{k,N}(\mathbb{C})$ we have $f^*U \cong E$. Moreover, f^*U depends, up to isomorphism, only

on the homotopy class of the map $f : X \longrightarrow G_{k,N}(\mathbb{C})$. Similar results hold for continuous vector bundles over X . See Atiyah [1], Husmoller [1] and Milnor [1] for further details and proofs.

4 Complex tori

Let $\omega = \{\omega_1, \dots, \omega_{2n}\}$ be a real basis of $\mathbb{C}^n$ and let L_ω denote the lattice subgroup of $\mathbb{C}^n$ defined by

$$L_\omega = \left\{ \sum_{j=1}^{2n} m_j \omega_j : (m_1, \dots, m_{2n}) \in \mathbb{Z}^{2n} \right\}.$$

Clearly $L_\omega \cong \mathbb{Z}^{2n}$. We define the *complex n -torus* T_ω to be the quotient space $T_\omega = \mathbb{C}^n / L_\omega$. The quotient map $\pi : \mathbb{C}^n \longrightarrow T_\omega$ is a local homeomorphism and T_ω is a compact Hausdorff space. The projection π induces a complex structure on T_ω with respect to which π is holomorphic, in fact locally biholomorphic (see also Theorem 4.5.2). Clearly T_ω is diffeomorphic to the standard real $2n$ -dimensional torus $T^{2n} = \mathbb{R}^{2n} / \mathbb{Z}^{2n}$. Since L_ω is a subgroup of $\mathbb{C}^n$, T_ω inherits a group structure from that on $\mathbb{C}^n$. The group operations on T_ω are obviously analytic and so T_ω has the structure of a *complex Lie group* (that is, a Lie group which is a complex manifold and whose group operations are analytic).

Example 1. Every compact connected complex Lie group is a complex torus and is, in particular, Abelian: By standard results in Lie group theory it is enough to prove that a compact connected complex Lie group G is Abelian. To see that this is so consider, for $g \in G$, the map $\phi_g : G \longrightarrow G$ defined by $\phi_g(h) = ghg^{-1}h^{-1}$, $h \in G$. Now $\phi_e(h) = e$ for all $h \in G$ and so there exists a neighbourhood U of e and a coordinate chart (V, ζ) for G containing e such that $\phi_g(G) \subset V$ for all $g \in U$. But then, by the argument of Proposition 4.2.4 applied to $\zeta\phi_g$, we see that ϕ_g must be constant on G for all $g \in U$. Hence, by analytic continuation, ϕ_g is constant on G . Therefore, $\phi_g(h) = e$ for all $g, h \in G$ and so G is Abelian.

Proposition 4.4.1. Let $\{\omega_1, \dots, \omega_{2n}\}, \{\omega'_1, \dots, \omega'_{2n}\}$ be real bases of $\mathbb{C}^n$ and L, L' and T, T' respectively denote the corresponding lattices and tori. Then T is biholomorphic to T' if and only if there exists $A \in GL(n, \mathbb{C})$ such that $A(L) = L'$.

Proof. Suppose $f : T \longrightarrow T'$ is biholomorphic. It is no loss of generality to suppose that f maps the identity of T to the identity of T' (otherwise replace f by $f(e)^{-1}f$). Let $F : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ be any lifting (necessarily biholomorphic) of f to the universal cover $\mathbb{C}^n$ of T, T' :

$$\begin{array}{ccc} \mathbb{C}^n & \xrightarrow{F} & \mathbb{C}^n \\ \pi \downarrow & & \downarrow \pi' \\ T & \xrightarrow{f} & T' \end{array}$$

Clearly $F(z + \omega_j) - F(z) \in L'$, $1 \leq j \leq 2n$, for all $z \in \mathbb{C}^n$. By continuity, these differences are all constant functions. Hence, for $z \in \mathbb{C}^n$, we have

$$\frac{\partial F}{\partial z_i}(z + \omega_j) = \frac{\partial F}{\partial z_i}(z), \quad 1 \leq i \leq n; \quad 1 \leq j \leq 2n.$$

By this periodicity, $\partial F / \partial z_i$ is a bounded holomorphic map and so, by Liouville's theorem, constant. Hence

$$F(z) = Ax + b,$$

for some $A \in GL(n, \mathbb{C})$, $b \in \mathbb{C}^n$. Since $f(e) = e'$, $F(0) \in L'$ and so $b \in L'$. We may suppose $b = 0$ (otherwise replace F by $F - b$). By our construction, $A(L) = L'$. The converse is trivial. $\square$

Remark. Notice that the proof above shows that two complex tori are biholomorphic if and only if they are isomorphic as complex Lie groups.

We shall now examine complex tori of dimension one in somewhat greater detail. Let $\omega = \{\omega_1, \omega_2\}$ be a real basis for $\mathbb{C}$ and set $\tau = \{\omega_2/\omega_1, 1\} = \{\tau, 1\}$. Certainly $\text{Im}(\tau) \neq 0$ and by Proposition 4.4.1, T_ω is biholomorphic to T_τ . Since $\{\tau, 1\}$ and $\{-\tau, 1\}$ generate the same lattice, we may without loss of generality suppose that $\text{Im}(\tau) > 0$. Hence the set of complex one-dimensional tori is parametrized by points in the upper half plane $H = \{\tau : \text{Im}(\tau) > 0\}$. We shall now determine when two points in the upper half plane determine the same torus. Let $\tau, \tau' \in H$ and L, L', T, T' denote the corresponding lattices and tori. T is biholomorphic to T' if and only if there exists $a \in GL(1, \mathbb{C}) \approx \mathbb{C}^*$ such that $a(L) = L'$. That is, if and only if there exist $a_{11}, \dots, a_{22} \in \mathbb{Z}$ such that

$$\begin{aligned} a\tau &= a_{11}\tau' + a_{12}, \\ a &= a_{21}\tau' + a_{22}, \\ a_{11}a_{22} - a_{21}a_{12} &= 1. \end{aligned}$$

Therefore τ and τ' define biholomorphic tori if and only if

$$\tau = \frac{a_{11}\tau' + a_{12}}{a_{21}\tau' + a_{22}}. \quad (*)$$

We see that the relation $(*)$ defines an action of $SL(2, \mathbb{Z})$ on H . The orbit space of this action, $H/SL(2, \mathbb{Z})$, gives an effective parametrization of the set of complex structures on the 2-dimensional real torus. We say that a subset $D \subset H$ is a *fundamental region* for the action of $SL(2, \mathbb{Z})$ on H if no two points in D lie on the same $SL(2, \mathbb{Z})$ orbit and every $SL(2, \mathbb{Z})$ orbit meets D . It is not hard to show that the region D in the figure below is a fundamental region for $SL(2, \mathbb{Z})$ acting on H (for details see Lang [2] or Robert [1]).

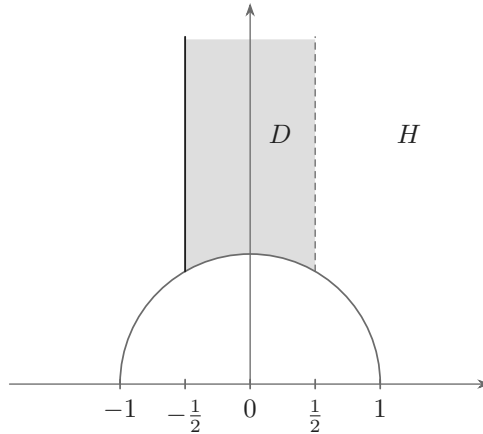


Figure I.7

$$D = \left\{ \tau \in H : \begin{array}{l} \text{Either } -\frac{1}{2} \leq \text{Re}(\tau) < \frac{1}{2} \text{ and } |\tau| > 1 \\ \text{or } |\tau| = 1 \text{ and } \text{Re}(\tau) \leq 0 \end{array} \right\}.$$

It is clear from what we have said above that we can “continuously deform” complex structures on a real 2-dimensional torus to obtain an uncountable family of distinct complex structures. This is very characteristic: There may exist many different complex structures on a given

differential manifold. The classification of all such structures is generally a problem of great difficulty and only partial results are known. For example, if M is a compact Riemann surface of genus $g \geq 2$, then it is known by the work of Riemann, Teichmüller [1], Rauch [1], Ahlfors [2] and Bers [1] that the set of complex structures on M has the natural structure of a complex analytic space of dimension $(3g-3)$. This space of complex structures—The moduli space—will have singularities. Unfortunately, no such global existence theorem holds in general for higher dimensional compact manifolds. One approach to a global theory is due to Griffiths [1]. There is an infinitesimal theory of deformations of complex structure due to Kodaira, Spencer and Kuranishi. This is described in the book by Kodaira and Morrow [1]. Finally we remark the difficult theorem of Douady's [1] to the effect that the set of all complex analytic subspaces of a given complex analytic space has the natural structure of a complex analytic space.

We shall now briefly consider the question of the existence of meromorphic functions on a complex torus. First we look at 1-dimensional tori.

Let $\{\omega_1, \omega_2\}$ be a real basis of $\mathbb{C}$ and let L denote the corresponding lattice. Set $T = \mathbb{C}/L$ and let $\pi : \mathbb{C} \rightarrow T$ denote the quotient map. Given $a \in \mathbb{C}$, we let P_a denote the parallelogram with vertices $a, a + \omega_1, a + \omega_2, a + \omega_1 + \omega_2$. P_a is called a *period parallelogram* (for L) and obviously $\pi(P_a) = T$.

Suppose $m \in \mathcal{M}(T)$. Clearly $\tilde{m} = m\pi \in \mathcal{M}(\mathbb{C})$ and $\tilde{m}$ is L -invariant. That is, given $z \in \mathbb{C}$,

$$\tilde{m}(z + \omega) = \tilde{m}(z), \quad \text{for all } \omega \in L.$$

Equivalently, $\tilde{m}$ is *doubly periodic* with periods ω_1, ω_2 ,

$$m(z + \omega_1) = m(z + \omega_2) = m(z), \quad \text{all } z \in \mathbb{C}.$$

We shall say that an L -invariant meromorphic function on $\mathbb{C}$ or, equivalently, a meromorphic function on T , is (L) -*elliptic*. Clearly the set of poles and zeroes of an elliptic function is finite (mod L).

Theorem 4.4.2. Let $m \in \mathcal{M}^*(T)$ and $\text{div}(m) = \sum_{i=1}^k n_i \cdot z_i$. Then

1. $\sum_{i=1}^k \text{residue}_{z_i}(m) = 0$.
2. $\deg(\text{div}(m)) = 0$.
3. $\sum_{i=1}^k n_i z_i = e$ (relative to the group law on T).

Proof. Choose $a \in \mathbb{C}$ such that the boundary of the period parallelogram P_a is disjoint from the poles and zeroes of m . 1 follows by integrating m (strictly, $\tilde{m} = \pi m$) round ∂P_a and observing that the integrals along opposite sides cancel by the periodicity of m . Similarly 2 follows by integrating the elliptic function m'/m round ∂P_a (see also Lemma 1.4.2). For 3 we integrate zm'/m round ∂P_a . In this case the integrals round opposite sides do not cancel but we do have

$$(2\pi i)^{-1} \left\{ \int_a^{a+\omega_1} - \int_{a+\omega_2}^{a+\omega_1+\omega_2} \frac{zm'}{m} dz \right\} = -(2\pi i)^{-1} \omega_2 \int_a^{a+\omega_1} \frac{m'}{m} dz.$$

Now $(2\pi i)^{-1} \int_a^{a+\omega_1} m'/m dz$ is just the winding number of the closed curve $m([a, a + \omega_1])$ about zero and is therefore an integer, say m_1 . Similarly for the other pair of boundary integrals. Lifting to P_a and choosing $\tilde{z}_i \in P_a$ such that $\pi(\tilde{z}_i) = z_i$, $1 \leq i \leq k$, we see that

$$\sum_{i=1}^k n_i \tilde{z}_i = m_2 \omega_1 - m_1 \omega_2 \in L.$$

□

Remarks. It follows from condition 1 of the Theorem that an elliptic function cannot have a single pole of order 1. The simplest possibilities are a single double pole with zero residue or two single poles with opposite residue. From condition 2 it follows that if $m \in \mathcal{M}^*(T)$, $c \in \mathbb{C}$, then m and $m - c$ have the same number of zeroes. Hence the map $m : T \rightarrow P^1(\mathbb{C})$ (Chapter 1, §4, Example 1) takes all complex values the same number of times. Condition 3 gives a necessary condition on a divisor in order that it be the divisor of an elliptic function. It turns out that conditions 2 and 3 of the theorem are sufficient conditions on a divisor for it to be the divisor of a meromorphic function (Abel's Theorem).

We shall now construct some elliptic functions. We start by defining *Weierstrass' elliptic function* $\wp(z)$:

$$\wp(z) = z^{-2} + \sum' ((z - \omega)^{-2} - \omega^{-2}), \quad z \in \mathbb{C} \setminus L.$$

Here the sum is over all $\omega \in L' = L \setminus \{0\}$. This sum is absolutely convergent at all points of $\mathbb{C} \setminus L$ and converges uniformly on compact subsets of $\mathbb{C} \setminus L$ (See the exercises at the end of this section and note that terms in the sum are $O(\omega^{-3})$). Hence $\wp \in \mathcal{M}(\mathbb{C})$. Clearly $\wp$ is even: $\wp(-z) = \wp(z)$, $z \in \mathbb{C} \setminus L$. To show that $\wp$ is elliptic we consider the derivative $\wp'$. Now

$$\wp'(z) = -2 \sum (z - \omega)^{-3}$$

is obviously elliptic and so, by integrating, we see that

$$\wp(z + \omega) = \wp(z) + C(\omega),$$

where $C(\omega)$ is independent of z . Since $\wp$ is even, $z = -\omega/2$ gives $C(\omega) = 0$ and so $\wp$ is elliptic. Set

$$G_j = \sum' \omega^{-2j}, \quad j \geq 2.$$

The reader may easily verify that the Laurent expansions of $\wp$ and $\wp'$ at zero are given by

$$\begin{aligned} \wp(z) &= z^{-2} \sum_{j \geq 1} (2j+1) G_{2j+2} z^{2j}, \\ \wp'(z) &= -2z^{-3} + \sum_{j \geq 1} (2j+1) 2j G_{2j+2} z^{2j-1} \end{aligned}$$

and converge for $0 < |z| < \min_{L'} |\omega|$.

Let us find the divisor of $\wp'$. Since $\wp'$ is odd and L -invariant $\wp'(z) = 0$ if $-z \equiv z \pmod{L}$. That is, if $2z \in L$. Thus representative zeroes of $\wp'$ in the period parallelogram P_0 are given by $\omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2$. These are the only zeroes in P_0 by Theorem 4.4.2. Now $\wp$ has only two zeroes in P_0 and, by condition 3 of Theorem 4.4.2 they must be $\{a, -a\}$ for some $a \in T$. These zeroes are not known explicitly in terms of the lattice L .

Proposition 4.4.3. The elliptic functions $\wp$ and $\wp'$ are related by the differential equation

$$\wp'^2 = 4\wp^3 - g_2\wp - g_3,$$

where $g_2^3 - 27g_3^2 \neq 0$ and $g_2 = 60G_4$, $g_3 = 140G_6$.

Proof. $\wp'^2 - 4\wp^3$ is certainly elliptic. Using the Laurent series for $\wp$ and $\wp'$ we easily compute that

$$\begin{aligned} \wp'^2 - 4\wp^3 &= -60G_4z^{-2} - 140G_6 + O(z^2) \\ &= -g_2\wp - g_3 + h, \end{aligned}$$

where h is holomorphic, elliptic and $O(z^2)$. Therefore $h \equiv 0$. Now we know the zeroes of $\wp'$ and so those of $4\wp^3 - g_2\wp - g_3$. But the zeroes of $\wp'$ are distinct and so therefore are those of the cubic $4x^3 - g_2x - g_3 = 0$. Therefore the discriminant of the cubic, $-16(g_2^3 - 27g_3^2)$, is non-zero. $\square$

We have already indicated in the previous section that the cubic curve $y^2z - 4x^3 + g_2xz^2 + g_3z^3 = 0$ is non-singular in $P^2(\mathbb{C})$ provided that $g_2^3 - 27g_3^2 \neq 0$. The next theorem gives an explicit embedding of the torus T as an algebraic submanifold of $P^2(\mathbb{C})$.

Theorem 4.4.4. The mapping $\phi : T \longrightarrow P^2(\mathbb{C})$ defined by

$$\phi(z) = (z^3\wp(z), z^3\wp'(z), z^3)$$

is a biholomorphic embedding of T onto the cubic curve $y^2z - 4x^3 + g_2xz^2 + g_3z^3 = 0$.

Proof. By Proposition 4.4.3, ϕ maps T into the given cubic curve. Certainly ϕ is onto since $\wp(z) = A$ has a solution for every $A \in P^1(\mathbb{C})$ (remarks following Theorem 4.4.2). ϕ is holomorphic since $z^3\wp(z)$ and $z^3\wp'(z)$ are holomorphic and the derivative of ϕ is everywhere of maximal rank as is seen by noting that $\wp''(z) \neq 0$ if $z \in \{\omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2\}$ as these points are simple zeroes of $\wp'$. It remains to be proved that ϕ is injective. Suppose $\wp(z) = \wp(y)$. If $z \not\equiv -z, \text{ mod } L$, and $y \not\equiv z, \text{ mod } L$, we see that $y \equiv -z, \text{ mod } L$, since $\wp$ is even and takes each value twice (counting multiplicities). But $\wp'$ is odd and so $\wp'(z) \neq \wp'(y)$ unless $\wp'(z) = 0$ which cannot happen as $z \not\equiv -z, \text{ mod } L$. Now suppose $z \equiv -z, \text{ mod } L$. Since $\wp$ takes each value twice, it is easy to see that $y \equiv -y, \text{ mod } L$. But if $z \equiv -z, \text{ mod } L$, $z \in \{\omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2\}$. Now $\wp(\omega_1/2), \wp(\omega_2/2), \wp(\omega_1 + \omega_2/2)$ are the roots of the cubic $4x^3 - g_2x - g_3 = 0$ and are distinct by our assumption on the discriminant. Hence $z \equiv y, \text{ mod } L$. Our argument proves that ϕ is injective. $\square$

Remarks.

1. Here we have just given a tiny fragment of the theory of elliptic curves. The reader may consult Lang [2], Robert [1] or Siegel [1] for more complete treatments of elliptic curves and functions. Whittaker and Watson [1] is a classical reference.
2. Although we have not proved it here, it is not difficult to show that every meromorphic function on a complex torus is a rational function of $\wp$ and $\wp'$. In particular, any two non-zero meromorphic functions are algebraically dependent (see also Chapter 5, §9).
3. It can be shown that any two meromorphic functions on a compact Riemann surface are algebraically dependent (see Gunning [1, Theorem 26, §10]). Consequently, we may generalise the method of proof of Theorem 4.4.4 to construct a holomorphic map of a compact Riemann surface M onto an algebraic curve in $P^2(\mathbb{C})$ provided that there exist “sufficiently many” meromorphic functions on M . The existence of an abundance of meromorphic functions on M is a consequence of the Riemann–Roch theorem which we prove in Chapter 10 (see also §5, Chapter 7). In general, we cannot embed M onto a non-singular curve in $P^2(\mathbb{C})$ (we always can embed in $P^3(\mathbb{C})$). We shall show in Chapter 7, §5, that every compact Riemann surface can be represented as a branched cover of $P^1(\mathbb{C})$. In case the surface has genus $g > 1$, and admits a 2-fold branched cover over $P^1(\mathbb{C})$, the surface is called *hyperelliptic*. We have more to say about these matters in Chapter 10. See also Griffiths and Harris [1], Hartshorne [2] and Gunning [1].

For the remainder of this section we shall briefly consider complex tori of dimension > 1 .

We no longer have a good moduli space for n -dimensional complex tori if $n > 1$. In fact the natural parameter space for complex structures on a real $2n$ -dimensional torus turns out to be non-Hausdorff—see Kodaira and Spencer [1, pages 408–414] and also Kodaira and Morrow [1, pages 22–23].

Definition 4.4.5. A complex torus which is algebraic is called an *Abelian variety*.

Every 1-dimensional complex torus is an Abelian variety (Theorem 4.4.4). However, it is not the case that every n -dimensional complex torus is algebraic for $n > 1$; in fact “most” are not. An obvious necessary condition for a complex torus to be algebraic is that it admit non-constant meromorphic functions (of course, we might expect to use such functions to construct maps into projective space). A stronger necessary condition is that we can separate points on the torus using meromorphic functions (since we can on projective space).

Example 2. (Siegel [2]). Let T be the complex 2-dimensional torus whose lattice is generated by $\{(1, 0), (0, 1), i(\sqrt{2}, \sqrt{3}), i(\sqrt{5}, \sqrt{7})\} \subset \mathbb{C}^2$. Then T admits no nonconstant meromorphic functions. In particular, T is not algebraic. We shall not prove this result here but instead refer the reader to Siegel [2]. See also Cornalba [1, page 85], de La Harpe [1, page 142], Griffiths and Harris [1] and Swinnerton-Dyer [1].

In fact in every dimension greater than 1 there exist complex tori with no non-constant meromorphic functions. We shall now give necessary and sufficient conditions for a complex torus T with lattice L generated by $\omega_1, \dots, \omega_{2n}$ to be an Abelian variety.

Let $A : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{R}$ be a real skew-symmetric bilinear form (that is, A is real linear in each variable separately and $A(x, y) = -A(y, x)$). We say that A is a *Riemann form* for T (or L) if

1. $A(L, L) \subset \mathbb{Z}$;
2. $A(ix, y)$ is a symmetric positive definite form on $\mathbb{C}^n$.

Example 3. Every lattice $L \subset \mathbb{C}$ admits a Riemann form. Indeed, suppose that L is generated by $\{\omega_1, \omega_2\}$ and that $\text{Im}(\omega_1/\omega_2) > 0$. Denote the area of the period parallelogram defined by ω_1, ω_2 by S . Clearly $S = \text{Im}(\omega_1 \bar{\omega}_2)$. If we define $A : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{R}$ by $A(y, z) = S^{-1} \text{Im}(y \bar{z})$, then A is a Riemann form for L . As an exercise the reader may verify that A is the only Riemann form for L which takes the values ± 1 on any basis of L .

For $n > 1$, lattices do not generally admit a Riemann form or even a non-zero positive semi-definite form taking integer values on the lattice. However, it may be shown that the set of all lattices which admit a non-zero Riemann form is dense in the set of all lattices (obvious topology). If A is a Riemann form for L then we have a positive definite Hermitian form H on $\mathbb{C}^n$ whose imaginary part is integer valued on the lattice. Indeed, we may define

$$H(y, z) = A(iy, z) + iA(y, z), \quad y, z \in \mathbb{C}^n.$$

We have the fundamental result

Theorem. A complex torus is algebraic if and only if it admits a Riemann form. We shall prove that the existence of a Riemann form implies that the torus is algebraic in §6 of Chapter 7 (the converse is not difficult). The theorem will also follow from our results on compact Kähler manifolds presented in Chapter 10. The reader may also find proofs in Cornalba [1], Griffiths and Harris [1], Swinnerton-Dyer [1] and Weil [1]. These references also give further information and references about complex tori. In Chapter 5 we shall have a little to say about the construction of meromorphic functions on complex tori as well as the role played by theta functions and holomorphic line bundles.

Exercises

1. Show that if L is a lattice in $\mathbb{C}$ then $\sum_{L \setminus \{0\}} |\omega|^{-\lambda}$ converges provided that $\lambda > 2$. Deduce that the series for $\wp'(z)$ converges uniformly on compact subsets of $\mathbb{C} \setminus L$.
2. Let L be a lattice in $\mathbb{C}$ and choose $\omega \in L \setminus \{0\}$ as close as possible to the origin. Show that there is a basis for L which contains ω .
3. Show that every even elliptic function may be written uniquely as a product $c \prod_{j=1}^k (\wp(u_j) - \wp(z))^{p_j}$, $c \in \mathbb{C}$ (You will need part 2 of Theorem 4.4.2). By writing an elliptic function as the sum of an even elliptic function plus $\wp' \times$ even elliptic function, show that every elliptic function is a rational combination of $\wp$, $\wp'$ and that the field of elliptic functions is a quadratic extension of $\mathbb{C}(\wp)$ (Use Proposition 4.4.3).
4. Let $F(x, y, z)$ be homogeneous of degree d and $C \subset P^2(\mathbb{C})$ denote the corresponding curve of degree d . Suppose $X \subset P^2(\mathbb{C})$ is an elliptic curve parametrized by $(\wp(u), \wp'(u), 1)$ as in Theorem 4.4.4. Prove that the number of points of intersection of X with C (counting multiplicities) is $3d$ (This is a special case of Bézout's theorem.
Hint: Suppose $F(0, 1, 0) \neq 0$. Consider the elliptic function $F(\wp, \wp', 1)$ and apply Theorem 4.4.2, part 2).
5. Suppose that the elliptic curve $X \subset P^2(\mathbb{C})$ is parametrized by $(\wp, \wp', 1)$. Show that the points $u, v, w \in X$ are collinear if and only if $u + v + w = e$ (relative to the group law on X). Deduce that

$$\wp(u + v) = -\wp(u) - \wp(v) + \frac{1}{4} \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \right)^2$$

provided that $u \not\equiv \pm v \pmod{L}$, where L is the lattice associated to X .

6. Use the methods of questions 4 and 5 to study the intersection of conics with a fixed elliptic curve in $P^2(\mathbb{C})$.
7. Let the lattice $L \subset \mathbb{C}$ have basis $\{1, \tau\}$, $\text{Im}(\tau) > 0$. Show that the set of continuous homomorphisms of the torus $\mathbb{C}/L$ is in bijective correspondence with the set $M(L)$ of complex numbers a such that $aL \subset L$. Clearly $M(L) \supset \mathbb{Z}$. Say that $\mathbb{C}/L$ has *complex multiplication* if $M(L)$ is strictly bigger than $\mathbb{Z}$. Show that if $\mathbb{C}/L$ has complex multiplication then $\tau \in \mathbb{Q}(\sqrt{-p})$, for some positive integer p and that $M(L)$ is then a subring of the ring of integers of the field $\mathbb{Q}(\sqrt{-p})$. Deduce that there are only countably many complex tori that admit complex multiplication.

5 Properly discontinuous actions

Let M be a complex manifold and $\text{Aut}(M)$ denote the group of biholomorphic transformations of M . Denote the identity of $\text{Aut}(M)$ by I . Suppose that Γ is a subgroup of $\text{Aut}(M)$. We say that Γ acts *freely* on M if the only element of Γ that has fixed points is the identity. That is, if $g \in \Gamma$ and $g(x) = x$ for some $x \in M$ then $g = I$. We say that Γ acts *properly discontinuously* on M if for any pair K_1, K_2 of compact subsets of M , the set $\{g \in \Gamma : g(K_1) \cap K_2 \neq \emptyset\}$ is finite.

Lemma 4.5.1. Let Γ act freely, properly discontinuously on M . Then every point $x \in M$ has an open neighbourhood U such that if $g \in \Gamma$ and $g(U) \cap U \neq \emptyset$, then $g = I$.

Proof. Let V be any relatively compact open neighbourhood of x . Since Γ acts properly discontinuously, $P = \{g \in \Gamma : g(V) \cap V \neq \emptyset\}$ is finite. Define

$$U = V \setminus \bigcup_{g \in P \setminus \{I\}} g^{-1}(\overline{g(V) \cap V}).$$

We leave it to the reader to check that U satisfies the conditions of the lemma. $\square$

Theorem 4.5.2. Let M be a complex manifold and Γ be a subgroup of $\text{Aut}(M)$ which acts freely, properly discontinuously on M . Then there exists a natural complex structure on M/Γ such that the orbit map $\pi : M \rightarrow M/\Gamma$ is locally biholomorphic.

Proof. Let $y = \pi(x) \in M/\Gamma$. Choose a chart (U, ϕ) for M containing x such that the conditions of Lemma 4.5.1 hold for the open set U . U is mapped homeomorphically onto $\pi(U)$ by π . Clearly $(\pi(U), \phi^{-1}(\pi|_U)^{-1})$ is a chart on M/Γ containing y . Repeating this construction for every point $y \in M/\Gamma$ we obtain a complex analytic atlas on M/Γ relative to which π is locally biholomorphic. $\square$

Examples.

1. Let M be a Riemann surface (not necessarily compact). Then, by the Uniformization theorem, the universal covering surface $\tilde{M}$ of M is biholomorphic to either the Riemann sphere, upper half plane (or open unit disc) or complex plane. Set $\Gamma = \pi_1(M)$. Then Γ acts freely, properly discontinuously on $\tilde{M}$ and M is biholomorphic to $\tilde{M}/\Gamma$. We remark that with the exception of the Riemann sphere, punctured complex plane, complex plane and torus, $\tilde{M}$ is biholomorphic to the upper half plane. In this case Γ is a discrete subgroup of the group of biholomorphic transformations of the upper half plane, $PSL(2, \mathbb{R})$. ($PSL(2, \mathbb{R}) = SL(2, \mathbb{R})/\{I, -I\}$ and acts on the upper half-plane by mapping $z \mapsto (az + b)/(cz + d)$, $ad - bc = 1$). Viewed in this light, the classification of Riemann surfaces amounts to the classification of all discrete subgroups of $PSL(2, \mathbb{R})$.

2. Let Ω be a bounded domain in $\mathbb{C}^n$ and suppose that $\Gamma \subset \text{Aut}(\Omega)$ acts freely, properly discontinuously on Ω and Ω/Γ is compact. Then Ω/Γ is algebraic. We prove this fundamental result of Kodaira [1] in Chapter 10. We remark that it implies that if the universal cover of a compact complex manifold M is biholomorphic to a bounded domain in $\mathbb{C}^n$ then M is algebraic.

3. If $L \in \mathbb{C}^n$ is a lattice subgroup ($\cong \mathbb{Z}^{2n}$), then L acts freely, properly discontinuously on $\mathbb{C}^n$ and so $\mathbb{C}^n/L$ has the structure of a complex manifold; see §4 of this chapter.

4. Let $a_0, \dots, a_n \in \mathbb{C}^*$ and suppose $|a_j| < 1$, $0 \leq j \leq n$. Let Γ denote the cyclic subgroup of $\text{Aut}(\mathbb{C}^{n+1} \setminus \{0\})$ generated by $g : (z_0, \dots, z_n) \mapsto (a_0 z_0, \dots, a_n z_n)$. Then Γ acts freely, properly discontinuously on $\mathbb{C}^{n+1} \setminus \{0\}$ and so $(\mathbb{C}^{n+1} \setminus \{0\})/\Gamma$ has the structure of a complex manifold which is easily seen to be compact and diffeomorphic to $S^{2n+1} \times S^1$. Any complex manifold diffeomorphic to $S^{2n+1} \times S^1$ is called a *Hopf manifold* and, if $n = 1$, a *Hopf surface*. As we shall see in Chapter 10, Hopf manifolds are never algebraic ($n \neq 0$). Hopf surfaces have been studied in great detail by Kodaira [2]. See also Kodaira and Spencer [1] and Ueno [1, pages 235–238]. Here we shall describe the important (and unpleasant!) phenomenon of “jumping” of complex structure that can occur on complex manifolds of dimension greater than one. Let

D denote the open unit disc in $\mathbb{C}$ and choose $a \in \mathbb{C}^*$, $|a| < 1$. For $t \in D$, let Γ_t be the subgroup of $\text{Aut}(\mathbb{C}^2 \setminus \{0\})$ generated by $g_t : (z_0, z_1) \mapsto (az_0 + z, az_1)$. Then Γ_t acts freely, properly discontinuously on $\mathbb{C}^2 \setminus \{0\}$ and so $M_t = (\mathbb{C}^2 \setminus \{0\})/\Gamma_t$ has the structure of a complex manifold which is easily seen to be a Hopf surface. In this case, we may define $M = (\mathbb{C}^2 \times \{0\}) \times D/\Gamma$, where Γ is generated by $g : ((z_0, z_1), t) \mapsto ((az_0 + tz_1, az_1), t)$. M has the structure of a complex manifold and if we let $\pi : M \rightarrow D$ denote the natural projection we see that π is holomorphic and $\pi^{-1}(t) = M_t$, $t \in D$. Thus we have defined a complex analytic family of Hopf surfaces, parametrized by points in the unit disc. Suppose $f : M_t \rightarrow M_s$ is biholomorphic, $t, s \in D$. Lift f to the universal cover to obtain a biholomorphic map $F : \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{C}^2 \setminus \{0\}$ satisfying $Fg_t = g_sF$. By Hartog's theorem (Theorem 2.3.2), F extends to an analytic map $\bar{F} : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ which is easily seen to be biholomorphic and to preserve the origin. Again we have $\bar{F}g_t = g_s\bar{F}$. Differentiating this relation at zero we see that $D\bar{F}_0 \cdot g_t = g_s \cdot D\bar{F}_0$. Setting $[D\bar{F}_0] = [F_{ij}]$, we easily compute that this relation implies that

$$\begin{aligned}
 tF_{00}z_1 &= sF_{10}z_0 + sF_{11}z_1, \\
 tF_{10}z_1 &= 0.
 \end{aligned}$$

Hence if $st \neq 0$, we must have $F_{10} = 0$ and $tF_{00} = sF_{11}$. If these relations are satisfied, we see immediately that $D\bar{F}_0$ induces a biholomorphic map between M_t and M_s . That is, we have shown that the Hopf surfaces M_t are all biholomorphic provided $t \neq 0$. However, if say $s = 0$, $t \neq 0$, the conditions above imply that $F_{00} = F_{10} = 0$ and so F cannot be biholomorphic. Therefore, M_0 is not biholomorphic to M_t , $t \neq 0$. In other words the complex structure on M_t “jumps” as t passes through zero. Notice that although the analytic map $\pi : M \rightarrow D$ is differentiably trivial it is *not* analytically trivial.

5. Calabi–Eckmann manifolds (Calabi and Eckmann [1]). Fix positive integers $p, q \geq 0$ and consider the projection $\pi : S^{2p+1} \times S^{2q+1} \rightarrow P^p(\mathbb{C}) \times P^q(\mathbb{C})$ obtained as the product of the Hopf fibrations of S^{2p+1} and S^{2q+1} . For each $x \in P^p(\mathbb{C}) \times P^q(\mathbb{C})$, the fibre $\pi^{-1}(x)$ is diffeomorphic to a real two-dimensional torus and so admits a complex structure. Since the base carries a complex structure we might hope to be able to define a complex structure on the product $S^{2p+1} \times S^{2q+1}$ and we shall now show that this is possible.

Fix $\tau \in \mathbb{C}$, $\text{Im}(\tau) > 0$ and let T_τ denote the torus with lattice generated by $\{1, \tau\}$. Given integers r, s , $0 \leq r \leq p$, $0 \leq s \leq q$, define

$$V_{rs} = \{((z_0, \dots, z_p), (w_0, \dots, w_q)) \in S^{2p+1} \times S^{2q+1} : z_r w_s \neq 0\}.$$

V_{rs} is an open subset of $S^{2p+1} \times S^{2q+1}$. We define a map $\phi_{rs} : V_{rs} \rightarrow \mathbb{C}^{p+q} \times T_\tau$ by

$$\begin{aligned}
 \phi_{rs}((z_0, \dots, z_p), (w_0, \dots, w_q)) &= (z_0/z_r, \dots, \widehat{z_r/z_r}, \dots, \\
 &\quad w_0/w_s, \dots, \widehat{w_s/w_s}, \dots, t_{rs}),
 \end{aligned}$$

where

$$t_{rs} = (2\pi i)^{-1}(\log(z_r) + \tau \log(w_s)), \quad \text{mod } 1, \tau.$$

It is a straightforward exercise to verify that ϕ_{rs} is a homeomorphism of V_{rs} onto $\mathbb{C}^{p+q} \times T_\tau$ and that $\{(V_{rs}, \phi_{rs}) : 0 \leq r \leq p, 0 \leq s \leq q\}$ defines a complex analytic structure on $S^{2p+1} \times S^{2q+1}$. Relative to this structure $\pi : S^{2p+1} \times S^{2q+1} \rightarrow P^p(\mathbb{C}) \times P^q(\mathbb{C})$ is holomorphic and the fibres of π are biholomorphic to T_τ . We call $S^{2p+1} \times S^{2q+1}$, with the complex structure defined above, a *Calabi–Eckmann manifold*. Observe that if $P \in S^{2p+1}$, $Q \in S^{2q+1}$ then the complex manifold $X = (S^{2p+1} \setminus \{P\}) \times (S^{2q+1} \setminus \{Q\})$ is homeomorphic to $\mathbb{C}^{p+q+1}$. However, the complex structure induced on $\mathbb{C}^{p+q+1}$ is quite different from the usual structure. We shall show that every analytic function on X is constant. To see this, notice that if $x \in P^p(\mathbb{C}) \times P^q(\mathbb{C})$ then $(\pi|X)^{-1}(x)$ is

biholomorphic to a complex torus provided that $\pi(P, Q) \neq x$. Such tori are dense in X and if $f \in A(X)$, then certainly f is constant on any of these tori (Proposition 4.2.4). Hence, by continuity, f is constant on any fibre $(\pi|X)^{-1}(x)$ and so induces a holomorphic function on $P^p(\mathbb{C}) \times P^q(\mathbb{C})$ which must be constant. Alternatively, if we use Exercise 2, §2, Chapter 2 (locally) we see that f must extend to an analytic function on $S^{2p+1} \times S^{2q+1}$ and so f must be constant. This example shows that a non-compact complex manifold which is homeomorphic to $\mathbb{C}^n$ need not be biholomorphic to any open subset of $\mathbb{C}^n$, $n \geq 3$. This is in sharp contrast to what happens in case $n = 1$.

6. Exotic complex structures (Brieskorn and Van de Ven [1]). Let $a = (a_0, \dots, a_n)$ be an $(n+1)$ -tuple of strictly positive integers. Let $X(a) \subset \mathbb{C}^{n+1}$ denote the zero set of the polynomial $z_0^{a_0} + \dots + z_n^{a_n}$ and set $\Sigma(a) = X(a) \cap S^{2n+1}$. Now $X(a) \setminus \{0\}$ is a complex submanifold of $\mathbb{C}^{n+1} \setminus \{0\}$ and $\Sigma(a)$ is a differential submanifold of S^{2n+1} (for the latter statement see Milnor [2]). If we let Γ denote the cyclic subgroup of $\text{Aut}(\mathbb{C}^{n+1} \setminus \{0\})$ generated by $g : (z_0, \dots, z_n) \mapsto (e^{1/a_0} z_0, \dots, e^{1/a_n} z_n)$, then Γ acts freely, properly discontinuously on $X(a) \setminus \{0\}$ and the resulting complex manifold $M = (X(a) \setminus \{0\}) / \Gamma$ is diffeomorphic to $S^1 \times \Sigma(a)$. If one of the a_j 's equals one it is easy to see that $\Sigma(a)$ is diffeomorphic to S^{2n-1} and so M is a Hopf manifold. Suppose now that no a_j is equal to one. Let $\xi_j = e^{2\pi i/a_j}$, $0 \leq j \leq n$, and set

$$\Delta(t) = \prod_{0 < i_k < a_k} (t - \xi_0^{i_0} \dots \xi_n^{i_n}).$$

It is known (Brieskorn [1]) that if $n \neq 2$ $\Sigma(a)$ is a homotopy $(2n-1)$ -sphere if and only if $\Delta(1) = 1$. Moreover, every homotopy $(2n-1)$ -sphere Σ^{2n-1} which bounds a parallelizable manifold is diffeomorphic to $\Sigma(a)$ for some $(n+1)$ -tuple a ($n \neq 2$). For $n \geq 4$, Σ^{2n-1} is homeomorphic but not necessarily diffeomorphic to S^{2n-1} (Brieskorn [1], Milnor [3]). Now suppose $n \geq 4$ and a is chosen so that $\Sigma(a)$ is a homotopy sphere with non-standard differential structure. In Brieskorn and Van de Ven [1], it is shown that $S^1 \times \Sigma(a)$ is not diffeomorphic to $S^1 \times S^{2n-1}$ and so the resulting complex structure on $S^1 \times \Sigma(a)$ is non-standard or “exotic”. As a specific example, if we take $n = 4$, $a_0 = a_1 = a_2 = 2$, $a_3 = 3$ and $a_4 = 6k - 1$, it can be shown (see Brieskorn [1]) that as k varies from 1 to 28 we obtain 28 distinct complex structures on $S^1 \times S^7$, 27 of which are exotic.

Exercises

1. In the example describing Calabi–Eckmann manifolds show that if $p = 0$, $q > 0$, the resulting manifold is biholomorphic to a Hopf manifold.
2. Let Γ be the subgroup of $GL(3, \mathbb{C})$ consisting of matrices

$$\begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{where } a_j \in \mathbb{Z}[\sqrt{-1}]$$

(that is, the a_j are Gaussian integers). Prove that Γ acts freely, properly discontinuously on $\mathbb{C}^3$. The resulting complex manifold $\mathbb{C}^3/\Gamma$ is called an *Iwasawa manifold*. Note that $\pi_1(\mathbb{C}^3/\Gamma)$ is not Abelian.

6 Analytic hypersurfaces

In this section we make a global study of analytic sets defined locally by the vanishing of a single analytic function.

Definition 4.6.1. Let M be a complex manifold and X be a proper analytic subset of M . We say that X is an *analytic hypersurface* of M if for each $x \in X$ there exists $U \in \mathcal{U}_x$ and $f \in A(U)$ such that $X \cap U = f^{-1}(0)$.

Before stating the next Proposition we recall some notation from §5 of Chapter 3: Suppose X is an analytic subset of M . As in §5 of Chapter 3, we may define the ideal $I_x(X) \triangleleft \mathcal{O}_x$ at each point $x \in M$. If $x \notin X$, $I_x(X) = \mathcal{O}_x$. Otherwise $I_x(X)$ is a non-trivial ideal of $\mathcal{O}_x$.

Proposition 4.6.2. Let X be an analytic subset of the complex manifold M . The following conditions are equivalent.

1. X is an analytic hypersurface.
2. For each $x \in X$, $I_x(X)$ is a principal ideal.

Proof. Suppose X is an analytic hypersurface and $x \in X$. Then there exists an open neighbourhood U of x and $f \in A(U)$ such that $f^{-1}(0) = X \cap U$. By the Nullstellensatz for principal ideals (Corollary 3.5.14), $I_x(X) = \text{Rad}(f_x)$. But $f_x = \prod_{j=1}^k p_j^{r_j}$, where $p_j \in \mathcal{O}_x$ are irreducible and coprime. Hence $I_x(X) = (p_1 \cdots p_k)$ and so is principal. For the converse, note that by Theorem 3.5.9 $Z(I_x(X)) = X_x$. Therefore if $I_x(X)$ is principal, say (f_x) , we see that there exists an open neighbourhood U of x and representative $f \in A(U)$ of f_x such that $X \cap U = f^{-1}(0)$. $\square$

Definition 4.6.3. Let X be an analytic hypersurface in M and $x \in X$. Suppose that $I_x(X) = (f_x)$ and that $df(x) \neq 0$ for some representative f of f_x . Then we say that x is a *regular point* of X . If x is not regular we say it is a *singular point*. We denote the set of regular and singular points of X by $\text{Reg}(X)$ and $\text{Sing}(X)$ respectively.

Remark. A point $x \in X$ is a regular point of X if and only if we can find an open neighbourhood V of x in M such that $V \cap X$ is a complex submanifold of M . Indeed, if the latter condition holds we may find an open neighbourhood W of x contained in V and $g \in A(W)$ such that $g^{-1}(0) = X \cap W$ and $dg(x) \neq 0$. Since $dg(x) \neq 0$ implies that g_x is irreducible, we must have $I_x(X) = (g_x)$. The converse follows from the implicit function theorem.

Theorem 4.6.4. Let X be an analytic hypersurface of the complex manifold M . We have

1. $\text{Reg}(X)$ is an open and dense subset of X and is a codimension one complex submanifold of M .
2. $\text{Sing}(X)$ is an analytic subset of M .

Proof. The remark above already shows that $\text{Reg}(X)$ is an open subset of X and a codimension one complex submanifold of M .

Let $x \in X$. By Theorem 3.5.16 we may find an open neighbourhood V of x in M and $g \in A(V)$ such that $I_y(V) = (g_y)$, for all $y \in V$. Hence $\text{Sing}(X) \cap V = \{y \in V : g(y) = dg(y) = 0\}$. Hence $\text{Sing}(X)$ is an analytic subset of M . Suppose that $\text{Reg}(X) \cap V$ is not dense in $X \cap V$. Then for some $y \in X \cap V$, there exists an open neighbourhood W of y in V such that $g(z) = dg(z) = 0$, $z \in X \cap W$. Since $I_y(X) = (g_y)$, this implies that in a local coordinate system $(z_1, \dots, z_n)$ at y we would have

$$(\partial g / \partial z_j)_y = u_y^j g_y, \quad 1 \leq j \leq n,$$

where $u_y^j \in \mathcal{O}_y$. However such a relation between an analytic function and its derivatives can only hold if the function is identically zero (look at Taylor series of g and dg at y). This

contradiction shows that $\text{Sing}(X) \cap V$ has no interior points in $X \cap V$. Therefore, $\text{Reg}(X)$ is dense in X . Alternatively the reader may use the results at the beginning of §5, Chapter 3 together with Corollary 3.5.15 to prove the density of $\text{Reg}(X)$. $\square$

The next result describes the local structure of an analytic hypersurface.

Proposition 4.6.5. Let X be an analytic hypersurface in the complex manifold M and let $x \in X$. We may find an open neighbourhood U of x in M and analytic hypersurfaces Z_j , $j = 1, \dots, k$, in U such that

1. $X \cap U = \bigcup_{j=1}^k Z_j$.
2. $\text{Reg}(Z_j)$ is connected, $1 \leq j \leq k$.
3. $\text{Reg}(X) \cap U = \bigcup_{j=1}^k \left(\text{Reg}(Z_j) \setminus \bigcup_{l \neq j} Z_l \right)$.

Proof. Choose a chart (U, ϕ) for M at x such that $\phi(x) = 0$. Set $D = \phi(U)$ and $Z = \phi(X \cap U)$. Since ϕ is biholomorphic, Z is an analytic hypersurface in D . Shrinking U if necessary, we may suppose (Corollary 3.5.15 and Theorem 3.5.16) that D is a polydisc and

$$Z = \bigcup_{j=1}^k Z(p^j),$$

where the p^j are coprime, irreducible Weierstrass polynomials on D and $I_y(Z) = (p_y^1 \cdots p_y^k)$ for every $y \in D$. Set $Z_j = Z(p^j)$, $1 \leq j \leq k$. Then $\text{Reg}(Z_j)$ is a connected subset of Z_j , $1 \leq j \leq k$ (Exercise 1, §5, Chapter 3). Since $I_y(Z) = (p_y^1 \cdots p_y^k)$, $y \in D$, we see immediately that

$$\text{Reg}(Z) = \bigcup_{j=1}^k \left(\text{Reg}(Z_j) \setminus \bigcup_{l \neq j} Z_l \right).$$

Let X_j denote the analytic hypersurface $\phi^{-1}(Z_j)$ in D , $1 \leq j \leq k$. Clearly $\{X_j : 1 \leq j \leq k\}$ satisfy the conditions of the Proposition. $\square$

Definition 4.6.6. Let X be an analytic subset of the complex manifold M . We say that X is *reducible* if we can find analytic subsets Y, Z of M , neither of which equals X , such that $X = Y \cup Z$. If X is not reducible, we say X is *irreducible*.

Theorem 4.6.7. Let X be an analytic hypersurface of the complex manifold M and $\{X'_i : i \in I\}$ denote the set of connected components of $\text{Reg}(X)$. Set $X_i = \overline{X'_i}$, $i \in I$. Then

1. X_i is an irreducible analytic hypersurface, $i \in I$.
2. $\{X_i : i \in I\}$ is locally finite. In particular, if M is compact I is finite.
3. $X = \bigcup_{i \in I} X_i$ and this decomposition of X as a (countable) union of irreducible analytic hypersurfaces is unique up to order.
4. If Y is any proper irreducible analytic subset of M and $Y \subset X$ then $Y \subset X_i$ for some $i \in I$.

Proof. Let $x \in X$ and suppose $x \in X_i$. We can find an open neighbourhood U of x in M and hypersurfaces Z_j , $1 \leq j \leq k$, in U satisfying the conditions of Proposition 4.6.5. Define

$$R_j = \text{Reg}(Z_j) \setminus \bigcup_{l \neq j} Z_l.$$

For $1 \leq j \leq k$, $R_j \subset \text{Reg}(X)$ and is connected. Hence for some subset $\Lambda \subset \{1, \dots, k\}$, $R_j \subset X'_i$ if and only if $j \in \Lambda$. But now $X_i \cap U = \bigcup_{j \in \Lambda} R_j = \bigcup_{j \in \Lambda} Z_j$. This proves that X_i is an analytic hypersurface in M . Since only finitely many of the X_i can meet U , the family $\{X_i : i \in I\}$ is locally finite. Next we show that each X_i is irreducible. Suppose that $X_i = Y \cup Z$, where Y and Z are analytic subsets of M . Set $Y' = X'_i \cap Y$, $Z' = X'_i \cap Z$. Since X'_i is a complex submanifold we see that Y', Z' are analytic subsets of X'_i . Now Y', Z' cannot both be proper analytic subsets of X'_i since if they were they would be nowhere dense in X'_i (Corollary 2.2.3) and so X'_i could not be the union of Y' and Z' . Suppose then that $Y' = X'_i$. Taking closures we see that $Y = X_i$ and so X_i is irreducible. Finally suppose that Y is an irreducible analytic subset and $Y \subset X$. Setting $Y_i = X_i \cap Y$, we have $Y = \bigcup_{i \in I} Y_i$. Fix $y \in Y$ and let L be the finite subset of I characterised by $j \in L$ if and only if $y \in X_j$. Since $\{X_i : i \in I\}$ is locally finite, $\Sigma = \bigcup_{i \in I \setminus L} Y_i$ is an analytic subset of M . Clearly $\Sigma \neq Y$, since $y \notin \Sigma$. But now we have expressed Y as a finite union

$$Y = \bigcup_{i \in L} Y_i \cup \Sigma$$

of analytic sets. Since $\Sigma \neq Y$, we must have $Y = Y_i$ for some $i \in L$. That is $Y \subset X_i$. $\square$

As an immediate corollary of Theorem 4.6.7 we have

Theorem 4.6.8. An analytic hypersurface is irreducible if and only if its set of regular points is connected.

Remarks.

1. We call the hypersurfaces X_i constructed in Theorem 4.6.7 the *irreducible components* of X .
2. Theorems 4.6.7 and 4.6.8 hold for arbitrary analytic subsets of a complex manifold. For proofs and further details we refer the reader to Gunning and Rossi [1], R. Narasimhan [3] and Whitney [1].
3. Notice that if X is an irreducible analytic hypersurface it does not follow that X_x is irreducible at every point $x \in X$. For example, the hypersurface $y^2 = x^2(1 - x)$ is irreducible but its germ at the origin is reducible.
4. Theorems 4.6.7 and 4.6.8 do not generalise to real analytic sets. For example, $y^2 = x^3$ is irreducible in $\mathbb{R}^2$ but the set of regular points is not connected. See R. Narasimhan [3, Chapter V] for further examples of the pathology that can occur with real analytic sets.

In §2, 4 of Chapter 1 we defined and discussed divisors on a Riemann surface and their relationship to meromorphic functions. For the remainder of this section we shall show how we may use Theorem 4.6.7 to give a local description of divisors on an arbitrary complex manifold.

Definition 4.6.8. Let M be a complex manifold. A *Weil divisor* on M is a formal sum $\sum_{i \in I} n_i \cdot X_i$, where $\{X_i : i \in I\}$ is a locally finite set of mutually distinct irreducible analytic hypersurfaces in M and $n_i \in \mathbb{Z}$, $i \in I$.

As in Chapter 1, we denote the set of Weil divisors on M by $\mathcal{D}(M)$ and note that $\mathcal{D}(M)$ has the structure of an ordered Abelian group. If $d = \sum_{i \in I} n_i \cdot X_i \in \mathcal{D}(M)$, we let $|d|$ denote the analytic hypersurface $\bigcup_{i \in I} X_i$.

Example 1. Let X be a non-singular analytic hypersurface in $P^n(\mathbb{C})$. Assuming Chow's theorem, we see that X is algebraic and so is the zero set of a homogeneous polynomial P . Since X is non-singular, X is (analytically) irreducible and it is not hard to verify that P can be chosen to be irreducible (as a polynomial). Let $d = \text{degree}(P)$. We say that X is a *hypersurface of degree d* . Now suppose $d > 1$. Let $P^n(\mathbb{C})^*$ denote the set of hyperplanes in $P^n(\mathbb{C})$ (see also §4). Every hyperplane $H \in P^n(\mathbb{C})^*$ will intersect X in a Weil divisor. Indeed, changing coordinates we may assume that H is the hyperplane $z_0 = 0$. $H \cap X$ is then the algebraic set determined by $G(z_1, \dots, z_n) = P(0, z_1, \dots, z_n) = 0$. G may not be irreducible and we let

$$G(z_1, \dots, z_n) = \prod_{i=1}^q G_i(z_1, \dots, z_n)^{n_i}$$

denote the prime factorization of G . If we set $X_i = G_i^{-1}(0) \cap H$, we see that $X \cap H$ determines the divisor

$$d(X \cap H) = \sum_{i=1}^q n_i \cdot X_i.$$

Notice that $\sum_{i=1}^q n_i d_i = d$, where $d_i = \text{degree}(G_i)$, $1 \leq i \leq q$. Let $\Gamma = \{d(X \cap H) : H \in P^n(\mathbb{C})^*\} \subset \mathcal{D}(X)$. Γ is in bijective correspondence with $P^n(\mathbb{C})^*$. Given $d \in \Gamma$, we let $H(d) \in P^n(\mathbb{C})^*$ denote the corresponding hyperplane. The set of divisors Γ on X determines the embedding of X in $P^n(\mathbb{C})$. Indeed, if $x \in X$ let Γ_x denote the set of all divisors $d \in \Gamma$ such that $x \in |d|$. Define $\phi(x) \in P^n(\mathbb{C})$ to be $\bigcap_{d \in \Gamma_x} H(d)$. The reader may easily verify that ϕ embeds X in $P^n(\mathbb{C})$. As we shall see later there is a close relationship between families of divisors on compact complex manifolds and embeddings into projective space.

Our next definition is motivated by the Cousin II problem and the question of constructing a meromorphic function with specified pole and zero set with multiplicities.

Definition 4.6.9. A *Cartier divisor* d on the complex manifold M consists of an open cover $\{U_i : i \in I\}$ of M together with meromorphic functions $m_i \in \mathcal{M}^*(U_i)$ such that for all $i, j \in I$ we have $m_i m_j^{-1} \in A^*(U_{ij})$. We write $d = \{(U_i, m_i) : i \in I\}$.

Remarks.

1. We regard the Cartier divisors $d = \{(U_i, m_i) : i \in I\}$, $d' = \{(V_j, m'_j) : j \in J\}$ as being equal if $m'_j m_i^{-1} \in A^*(V_j \cap U_i)$, $i \in I$, $j \in J$. That is, we think of a Cartier divisor as an equivalence class of local data. Later, in Chapter 6, we shall give a slicker definition of Cartier divisor in terms of sheaves.
2. We let $\tilde{\mathcal{D}}(M)$ denote the set of Cartier divisors on M .
3. Of course, a Cartier divisor is nothing else than the data for the Cousin II problem (Definition 3.4.10). However, we prefer to reserve the term “Cousin II problem” for non-compact complex manifolds, especially domains of holomorphy and Stein manifolds.

Proposition 4.6.10. The set of Cartier divisors $\tilde{\mathcal{D}}(M)$ has the natural structure of an Abelian group.

Proof. Let $d = \{(U_i, m_i) : i \in I\}$, $d' = \{(V_j, m'_j) : j \in J\} \in \tilde{\mathcal{D}}(M)$. We define

$$\begin{aligned} d + d' &= \{(U_i \cap V_j, m_i m'_j) : i \in I, j \in J\}, \\ -d &= \{(U_i, m_i^{-1}) : i \in I\}. \end{aligned}$$

It is straightforward to verify that these definitions do not depend on the particular choice of local data for the divisors. $\square$

Example 2. We have a natural group homomorphism $\text{div} : \mathcal{M}^*(M) \longrightarrow \tilde{\mathcal{D}}(M)$ defined by $\text{div}(m) = \{(M, m)\} \in \tilde{\mathcal{D}}(M)$. $\text{div}(m)$ is called the *divisor* of m .

Theorem 4.6.11. There is a natural group isomorphism between $\mathcal{D}(M)$ and $\tilde{\mathcal{D}}(M)$.

Proof. Let $d = \sum_{i \in I} n_i \cdot X_i \in \mathcal{D}(M)$. Given $x \in M$, choose an open neighbourhood U_x of x and $p^i \in A(U_x)$, $i \in I$, such that

- a) All but finitely many of the p^i are identically equal to 1.
- b) $I_y(X_i) = (p_y^i)$, $y \in U_x$, $i \in I$.

Set

$$m_x = \prod_{i \in I} (p^i)^{n_i} \in \mathcal{M}^*(U_x).$$

Clearly $\phi(d) = \{(U_x, m_x) : x \in M\}$ is a Cartier divisor on M . The map $\phi : \mathcal{D}(M) \longrightarrow \tilde{\mathcal{D}}(M)$ is easily seen to be an injective group homomorphism.

Conversely, suppose $d = \{(U_i, m_i) : i \in I\}$ is a Cartier divisor on M . Define $X = \bigcup_{i \in I} (Z(m_i) \cup P(m_i))$. X is a well defined analytic hypersurface in M . Indeed, $X \cap U_i = Z(m_i) \cup P(m_i)$ since the condition $m_i m_j^{-1} \in A^*(U_{ij})$ implies that m_i and m_j have the same pole and zero sets on U_{ij} . Let $\{X_\alpha : \alpha \in \Lambda\}$ be the decomposition of X into its irreducible components. Refining the cover $\{U_i : i \in I\}$ if necessary, we may find for each $i \in I$, $p_\alpha \in A(U_i)$ such that

- a) All but finitely many of the p_α are identically equal to 1.
- b) $U_i \cap X_\alpha = p_\alpha^{-1}(0)$, $\alpha \in \Lambda$.
- c) $I_y(X_\alpha) = (p_{\alpha, y})$, $y \in U_i$, $\alpha \in \Lambda$.

Given $y \in U_i$, we see that there exist a unit $u_y \in \mathcal{O}_y$ and integers n_α , $\alpha \in \Lambda$, such that

$$m_y = u_y \prod_{\alpha \in \Lambda} p_{\alpha, y}^{n_\alpha}.$$

Hence for some $u \in A^*(U_i)$, we have $m_i = u \prod_{\alpha \in \Lambda} p_\alpha^{n_\alpha}$. (We assume here that U_i is connected, otherwise we could only claim that the n_α were constant on connected components of U_i).

We claim that the integers n_α do not depend on $i \in I$. This follows since n_α is constant on the connected open set $\text{Reg}(X_\alpha) \cap \text{Reg}(X)$. Since $\text{Reg}(X_\alpha) \cap \text{Reg}(X)$ is dense in X_α , n_α is constant on X_α . We now define

$$\gamma(d) = \sum_{\alpha \in \Lambda} n_\alpha \cdot X_\alpha \in \mathcal{D}(M).$$

By our construction it is clear that $\gamma = \phi^{-1}$ and so $\mathcal{D}(M)$ and $\tilde{\mathcal{D}}(M)$ are naturally isomorphic. $\square$

Remarks.

1. In the sequel we regard $\mathcal{D}(M)$ and $\tilde{\mathcal{D}}(M)$ as identified by the isomorphism constructed above and write $\mathcal{D}(M)$ for the set of divisors on M , omitting the prefix “Weil” or “Cartier”.
2. If we attempt to define divisors on more general objects, such as analytic sets with singularities, we find that the sets of Weil and Cartier divisors need not agree and in practice we tend to work with the less geometrical Cartier divisors (see Hartshorne [2, pages 140–142]).

Example 3. Let $m \in \mathcal{M}^*(P^n(\mathbb{C}))$. Then $\text{div}(m) = \sum_{i=1}^q n_i \cdot X_i$, where the X_i are irreducible analytic hypersurfaces in $P^n(\mathbb{C})$. By Chow's theorem and Example 1, each X_i is an irreducible algebraic hypersurface of degree d_i , say. Suppose that X_i is the zero locus of the irreducible homogeneous polynomial P_i of degree d_i , $1 \leq i \leq q$. Define $R = \prod_{i=1}^q P_i^{n_i}$. Regarding m and R as defining meromorphic functions on $\mathbb{C}^{n+1} \setminus \{0\}$, we clearly have $\text{div}(m) = \text{div}(R)$ and so $\text{div}(R^{-1}m) = 0$. But therefore $R^{-1}m$ is an analytic function on $\mathbb{C}^{n+1} \setminus \{0\}$ and so extends by Hartog's theorem to an analytic function on the whole of $\mathbb{C}^{n+1}$. Taking the Taylor expansion of $R^{-1}m$ at 0 and noting that m is homogeneous of degree zero, we see that $R^{-1}m$ is a homogeneous polynomial of degree $-\sum_{i=1}^q n_i d_i$. Hence $\sum_{i=1}^q n_i d_i = 0$ (otherwise $\text{div}(R^{-1}m)$ could not vanish). Consequently, $R^{-1}m = c$, for some $c \in \mathbb{C}^*$ and so

$$m = c \prod_{i=1}^q P_i^{n_i} \quad \text{on } P^n(\mathbb{C}).$$

We have shown that every meromorphic function on $P^n(\mathbb{C})$ is rational. Our arguments further prove that a divisor $d = \sum_{i=1}^q n_i \cdot X_i$ on $P^n(\mathbb{C})$ is the divisor of a meromorphic (rational) function on $P^n(\mathbb{C})$ if and only if $\sum_{i=1}^q n_i d_i = 0$, where d_i is the degree of the hypersurface X_i . Given $d = \sum_{i=1}^q n_i \cdot X_i \in \mathcal{D}(P^n(\mathbb{C}))$, we may define the degree of d , $\deg(d)$, to be the sum $\sum_{i=1}^q n_i d_i$, where d_i is the degree of the hypersurface X_i . Degree defines a homomorphism $\deg : \mathcal{D}(P^n(\mathbb{C})) \rightarrow \mathbb{Z}$ and the kernel of the degree map is precisely the set of divisors of rational functions on $P^n(\mathbb{C})$. (For an alternative proof of the rationality of meromorphic functions on $P^n(\mathbb{C})$, avoiding the use of Chow's theorem, see Jackson [1]).

7 Blowing up

In this section we describe an operation on complex manifolds that is of the greatest importance in the study of singularities of analytic sets and the classification theory of complex manifolds.

Let $\Sigma \subset \mathbb{C}^n \times P^{n-1}(\mathbb{C})$ be the set of points $((z_1, \dots, z_n), (\zeta_1, \dots, \zeta_n))$ satisfying the equations

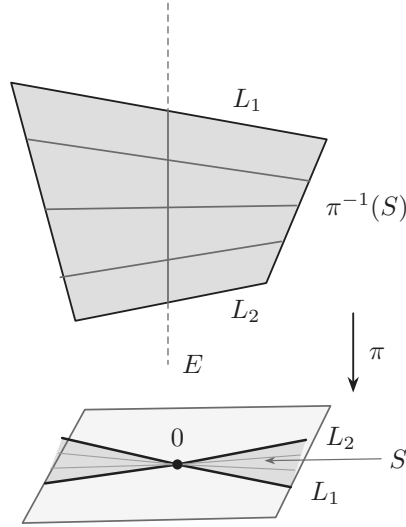
$$z_i \zeta_j = z_j \zeta_i, \quad 1 \leq i, j \leq n.$$

(We suppose in what follows that $n \geq 2$). Notice that $((z_1, \dots, z_n), (tz_1, \dots, tz_n)) \in \Sigma$, for all $(z_1, \dots, z_n) \neq 0$ and $t \in \mathbb{C}^*$. In other words, given $X \in \mathbb{C}^n$, $X \neq 0$, Σ contains the point corresponding to X and the line through X . Since $\{0\} \times P^{n-1}(\mathbb{C}) \subset \Sigma$, we see that Σ contains the point corresponding to zero and all the lines through zero. The reader may easily verify that we have described all the points in Σ . Let $\pi : \Sigma \rightarrow \mathbb{C}^n$ denote the restriction of the projection of $\mathbb{C}^n \times P^{n-1}(\mathbb{C})$ on $\mathbb{C}^n$ to Σ . If we set $E = \pi^{-1}(0)$, then E is biholomorphic to $P^{n-1}(\mathbb{C})$ and π maps $\Sigma \setminus E$ bijectively onto $\mathbb{C}^n \setminus \{0\}$. We call Σ the *blowing up* of $\mathbb{C}^n$ at 0. Amongst many other commonly used terms to describe Σ are: $\mathbb{C}^n$ blown up at zero; the *quadratic transform* of $\mathbb{C}^n$ at zero; the *monoidal transform* of $\mathbb{C}^n$ at zero; the *Hopf σ -process* of $\mathbb{C}^n$ at zero. We call 0 the *centre* of the blowing up; E is called the *exceptional variety*.

It is easy to see that Σ is a complex submanifold of $\mathbb{C}^n \times P^{n-1}(\mathbb{C})$ and we shall now construct an explicit atlas for Σ . For $1 \leq j \leq n$, define $\gamma_j : \mathbb{C}^n \rightarrow \Sigma \subset \mathbb{C}^n \times P^{n-1}(\mathbb{C})$ by

$$\begin{aligned} \gamma_j(X_1, \dots, X_n) = & ((X_1 X_j, \dots, X_{j-1} X_j, X_j, X_{j+1} X_j, \dots, X_n X_j), \\ & (X_1, \dots, X_{j-1}, 1, \dots, X_n)). \end{aligned}$$

The reader may verify that γ_j maps $\mathbb{C}^n$ homeomorphically onto an open subset of Σ . If we set $U_j = \gamma_j(\mathbb{C}^n)$, $\phi_j = \gamma_j^{-1}$, then $\{(U_j, \phi_j) : 1 \leq j \leq n\}$ is a complex analytic atlas on Σ . Relative to this complex structure on Σ , $\pi : \Sigma \rightarrow \mathbb{C}^n$ is holomorphic and π maps $\Sigma \setminus E$ biholomorphically onto $\mathbb{C}^n \setminus \{0\}$.


 Figure I.8. *Blowing up*

In the figure the cone $S \subset \mathbb{C}^n$ is blown up into an open subset of Σ . That is, the cone is “untwisted” into an open set. Notice how lines passing through the origin of $\mathbb{C}^n$ lift to distinct lines in Σ . We can also describe this phenomenon using charts of the atlas we constructed above for Σ . Thus if $n = 2$ and we take any open cone $S \subset \mathbb{C}^2$ which contains the z_1 -axis but omits the z_2 -axis then $\pi^{-1}(S)$ is described by the figure below if we use coordinates given by the chart (U_1, ϕ_1) .

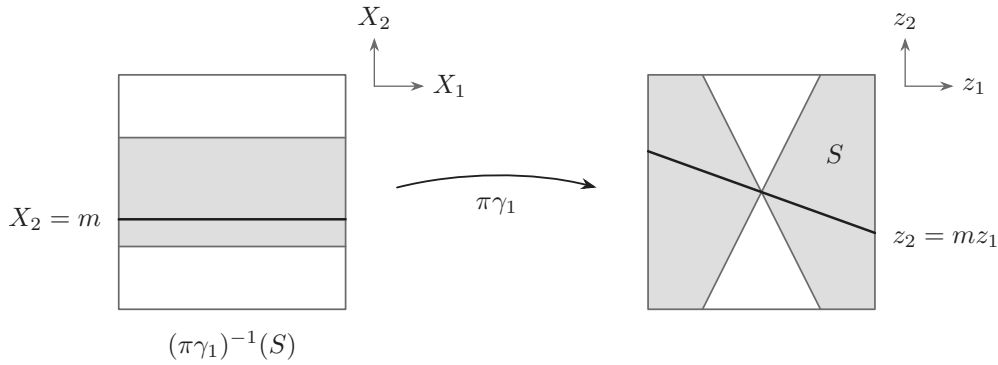


Figure I.9

In the figure, the line $z_2 = mz_1$ in the (z_1, z_2) -plane corresponds to the affine line $X_2 = m$ in the (X_1, X_2) -plane.

The construction we have described above works equally well if instead of $\mathbb{C}^n$ we take an open neighbourhood U of $0 \in \mathbb{C}^n$ and define $\Sigma_U \subset U \times P^{n-1}(\mathbb{C})$ to be the intersection of Σ with $U \times P^{n-1}(\mathbb{C})$. We refer to Σ_U as U blown up at zero. The projection again restricts to a holomorphic map $\pi : \Sigma_U \rightarrow U$ which induces a biholomorphic map between $\Sigma_U \setminus E$ and $U \setminus \{0\}$.

We now wish to generalise the construction above so that we can blow up arbitrary complex manifolds at a point. First, a lemma.

Lemma 4.7.1. Let M and Σ be complex manifolds with closed submanifolds N and X respectively. Suppose that for some open neighbourhood U of N in M there exists a biholomorphism $\phi : U \setminus N \rightarrow \Sigma \setminus X$. Then if we define

$$M^* = (M \setminus N) \cup \Sigma,$$

where we identify $x \in M \setminus N$ with $y \in \Sigma \setminus X$ if $\phi(x) = y$, M^* has the natural structure of a complex manifold such that $M^* \setminus X$ is biholomorphic to $M \setminus N$. Furthermore, if ϕ is the

restriction of a holomorphic map $\pi : U \rightarrow \Sigma$, then π induces a holomorphic map $\pi : M^* \rightarrow M$ such that π maps $M^* \setminus X$ biholomorphically onto $M \setminus N$.

Proof. The proof is quite elementary and we leave it to the reader. $\square$

Suppose M is a complex manifold of dimension n , $n > 1$, and let $p \in M$. Choose a coordinate chart (V, ϕ) for M such that $p \in V$ and $\phi(p) = 0$. Set $U = \phi(V)$. Let Σ denote the blow up of U at 0 (Σ_U in the notation above). Then $\phi^{-1}\pi : \Sigma \setminus E \rightarrow V \setminus \{p\}$ is a biholomorphism and so we may apply Lemma 4.7.1 to construct the complex manifold

$$B_p(M) = (M \setminus \{p\}) \cup \Sigma$$

together with the holomorphic map $\pi : B_p(M) \rightarrow M$. We say that $B_p(M)$ is M blown up at the point p (or any of the other descriptions we gave previously for blowing up $\mathbb{C}^n$). We call p the centre of the blowing up and $E = \pi^{-1}(p)$ the exceptional variety. Clearly E is biholomorphic to $P^{n-1}(\mathbb{C})$. The projection map $\pi : B_p(M) \rightarrow M$ is a proper holomorphic map and restricts to a biholomorphism of $B_p(M) \setminus E$ with $M \setminus \{p\}$.

We may generalise the above construction to include centres which are closed complex submanifolds of M . First we describe the local situation. Regard $\mathbb{C}^m$ as the subspace $\mathbb{C}^m \times \{0\}$ of $\mathbb{C}^{m+n}$. We define $\mathbb{C}^{m+n}$ blown up along $\mathbb{C}^m$ to be the space $\mathbb{C}^m \times \Sigma$, where Σ is $\mathbb{C}^n$ blown up at zero. In other words, we just blow up normal to $\mathbb{C}^m$ in $\mathbb{C}^{m+n}$. More generally, if X is a closed submanifold of M we may blow up M along X to construct a complex manifold $B_X(M)$ together with a holomorphic projection $\pi : B_X(M) \rightarrow M$ which restricts to a biholomorphic map of $B_X(M) \setminus \pi^{-1}(X)$ onto $M \setminus X$. In this case the exceptional variety $\pi^{-1}(X)$ is a bundle over X with fibre biholomorphic to $P^q(\mathbb{C})$, where $q = \dim(M) - \dim(X) - 1$. We give an example of this process below.

We shall now give some examples to show how blowing up may be used to “desingularize” analytic sets. Suppose that X is an analytic subset of the complex manifold M . Let Y be a closed complex submanifold of M which is a subset of $\text{Sing}(X)$. Let us blow up M along Y to give the new complex manifold $B_Y(M)$. The set $\pi^{-1}(X)$ is an analytic subset of $B_Y(M)$. Necessarily $\pi^{-1}(X)$ contains the exceptional variety $E = \pi^{-1}(Y)$ and we define the *strict transform* of X to be the analytic subset $X^* = \pi^{-1}(X) \setminus E$ of $B_Y(M)$. As our examples will show a careful choice of Y will often result in X^* having “simpler” singularities than X .

Examples.

1. Let $Z \subset \mathbb{C}^2$ denote the curve defined by $P(z_1, z_2) = (z_1 - z_2)(z_1 + z_2) = 0$. Z has an isolated singularity at zero. As above we let Σ denote $\mathbb{C}^2$ blown up at zero and $\pi : \Sigma \rightarrow \mathbb{C}^2$ the projection map. Set $\tilde{Z} = \pi^{-1}(Z)$ and let Z^* denote the strict transform of Z . To describe $\tilde{Z}$ we use the atlas $\{(U_i, \phi_i) : i = 1, 2\}$ that we previously constructed on Σ . For $i = 1, 2$, set $\tilde{Z}_i = \phi_i(\tilde{Z} \cap U_i) \subset \mathbb{C}^2$. Then $\tilde{Z}_1$ is the zero set of $P(X_1, X_1X_2)$ and $\tilde{Z}_2$ is the zero set of $P(X_1X_2, X_2)$. We have

$$P(X_1, X_1X_2) = (X_1 - X_1X_2)(X_1 + X_1X_2) = X_1^2(1 - X_2)(1 + X_2).$$

Now $X_1 = 0$ is just the equation of the exceptional variety (intersected with U_1) and so the strict transform Z^* of Z is the pair of distinct lines $X_2 = \pm 1$ (again intersected with U_1). A similar description holds in the chart (U_2, ϕ_2) . Hence Z^* consists of two distinct lines and, in particular, is non-singular.

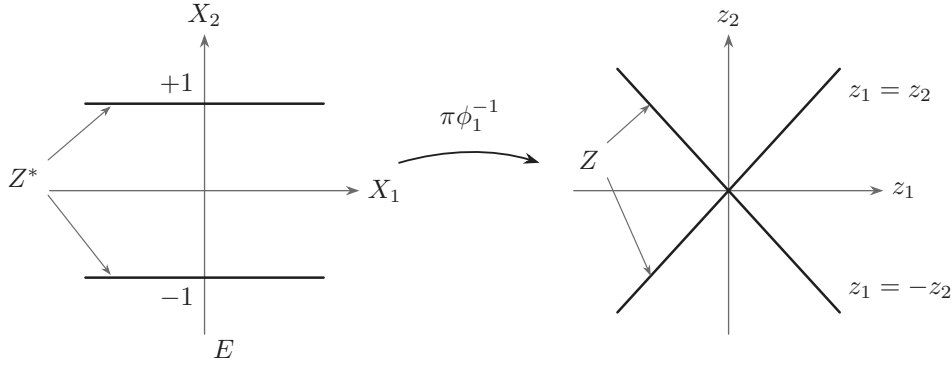


Figure I.10

2. Let $Z \subset \mathbb{C}^2$ be the curve defined by $P(z_1, z_2) = z_1^2 - z_2^3 = 0$. Z has an isolated singularity at zero. Repeating the argument above, we see that $\tilde{Z}_1 \subset \mathbb{C}^2$ is the zero locus of $X_1^2 - X_1^3 X_2^3 = X_1^2(1 - X_1 X_2^3)$. That is, the intersection of Z^* with U_1 is the non-singular curve $1 = X_1 X_2^3$ (notice that this curve does not meet the exceptional variety $X_1 = 0$ —in U_1). Similarly, $\tilde{Z}_2$ is the zero locus of $X_1^2 X_2^2 - X_2^3 = X_2^2(X_1^2 - X_2)$. Now $X_2 = 0$ is the equation of the exceptional variety in U_2 and so the intersection of Z^* with U_2 is the parabola $X_2 = X_1^2$, which is of course non-singular. Hence Z^* is non-singular.

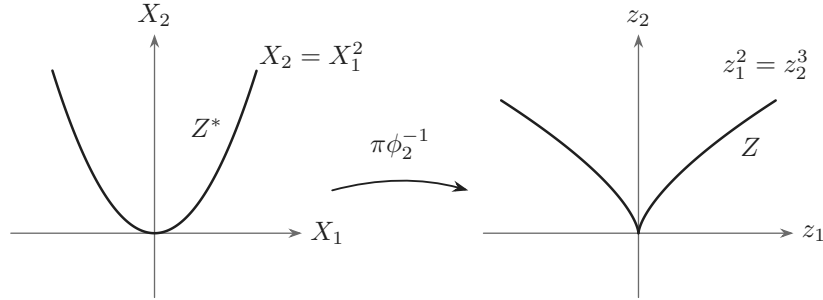


Figure I.11

If instead we had started with the curve $z_1^2 = z_2^5$, we would find that $Z^* \cap U_2$ was the curve $X_1^2 = X_2^5$. Hence a further blowing up, this time of the manifold Σ , would remove the singularity. Without too much difficulty this technique will effectively resolve the singularities of all curves. Details may be found in Walker [1], Mumford [1] or Hartshorne [2].

Fundamental results of Hironaka [1, 2], following on work of Walker and Zariski assert that we can always resolve singularities of complex analytic sets. Specifically, if X is a complex analytic subset of a complex manifold M , there exists a (locally finite) sequence

$$M_\Lambda \xrightarrow{\pi_\Lambda} M_{\Lambda-1} \xrightarrow{\pi_{\Lambda-1}} \cdots \longrightarrow M_1 \xrightarrow{\pi_1} M$$

of blowing ups of M , with non-singular centres, such that the iterated strict transform of X is a non-singular submanifold of M_Λ . The proof of this result is extremely hard. Much of the difficulty lies with choosing the centres so that when we blow up along them the singularities are somehow simplified. A proof of the desingularization theorem, together with many helpful examples is given in Hironaka [2].

The next example shows that one cannot hope to give simple proofs of the desingularization theorem by working with a restricted class of analytic sets such as hypersurfaces with isolated singularities.

Example 3. Let $Z \subset \mathbb{C}^3$ be the hypersurface defined by $z_1^5 = z_2 z_3^6 - z_2^6 z_3$. Z has an isolated singularity at 0. Blow up $\mathbb{C}^3$ at zero. We now describe $\tilde{Z}_i = \tilde{Z} \cap U_i$, $1 \leq i \leq 3$.

$\tilde{Z}_1$ is the zero locus of $X_1^5(1 - X_1^2 X_2 X_3^6 + X_1^2 X_2^6 X_3)$ and so the strict transform Z^* intersects U_1 in the non-singular hypersurface $1 = X_1^2 X_2 X_3^6 - X_1^2 X_2^6 X_3$.

$\tilde{Z}_2$ is the zero locus of $X_2^5(X_1^5 - X_2^2 X_3^6 - X_2^2 X_3)$. Therefore $Z^* \cap U_2$ is the hypersurface $X_1^5 = X_2^2 X_3^6 - X_2^2 X_3$. Now $\text{Sing}(Z^* \cap U_2)$ is the set $X_1 = X_2 = 0$. In particular, $Z^* \cap U_2$ no longer has an isolated singularity.

Just as for $\tilde{Z}_2$, we find that $Z^* \cap U_3$ is the curve $X_1^5 = X_3^2 X_2 - X_3^2 X_2^6$ with singular locus $X_1 = X_3 = 0$.

It follows from the above that $\text{Sing}(Z^*)$ is a subset of the exceptional variety $P^2(\mathbb{C})$ and is in fact the hyperplane $X_1 = 0$ (Here we take homogeneous coordinates (X_1, X_2, X_3) on $P^2(\mathbb{C})$). In other words $\text{Sing}(Z^*)$ is biholomorphic to the Riemann sphere $P^1(\mathbb{C})$.

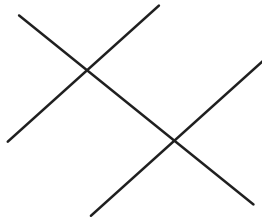
We now blow up Σ along $\text{Sing}(Z^*)$. Let us work in U_2 . $Z^* \cap U_2$ is the hypersurface $X_1^5 = X_2^2 X_3^6 - X_2^2 X_3$ with singular set $X_1 = X_2 = 0$. Blowing up along the X_3 -axis, we make the transformations $X_1 = Y_1 Y_2$, $X_2 = Y_2$, $X_3 = Y_3$ to obtain the hypersurface

$$Y_1^5 Y_2^5 - Y_2^2 Y_3^6 + Y_2^2 Y_3 = Y_2^2 (Y_1^5 Y_2^3 - Y_3^6 + Y_3) = 0.$$

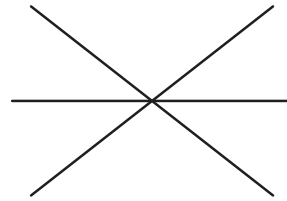
The strict transform is the hypersurface $Y_3 = Y_3^6 - Y_1^5 Y_2^3$ which is of course non-singular. A similar result holds in U_3 and so the strict transform of Z^* obtained by blowing up Σ along $\text{Sing}(Z^*)$ is a non-singular hypersurface.

We now wish to say a few words about the *embedded resolution of singularities*. First we give a definition: Suppose that Y is an analytic hypersurface in the complex manifold N . For each $x \in Y$, we may write $I_x(Y) = (p_1, \dots, p_k)$, where $p_1, \dots, p_k$ are the local defining equations for the irreducible components of Y passing through x . We say that Y is a *hypersurface with (or in) normal crossings* if (1) Each irreducible component of Y is non-singular; (2) For each $x \in Y$, $\{dp_1(x), \dots, dp_k(x)\}$ form a linearly independent set.

Notice that the second condition implies that not more than $\dim(N)$ irreducible components of Y pass through any point of N . Condition (2) may also be given in terms of the tangent spaces to the irreducible components of Y at x . We require $\dim(T_x N \cap \bigcap T_x Y_j) = n - k$, where we assume that there are k irreducible components Y_j of Y passing through x . Yet another equivalent formulation is that given $x \in Y$, we can choose complex analytic coordinates $(z_1, \dots, z_n)$ at x such that Y is locally the zero set of the function $u(z) z_1^{m_1} \cdots z_k^{m_k}$, $u(0) \neq 0$.



Hypersurface in normal crossings



Not in normal crossings

Figure I.12

Suppose $d \in \mathcal{D}(N)$. We say that d is a *divisor in normal crossings* if the hypersurface $|d|$ is in normal crossings. If d is in normal crossings and $x \in |d|$, we may choose complex analytic coordinates at x so that d is locally given as the divisor of the meromorphic function $u(z) z_1^{m_1} \cdots z_k^{m_k}$, $u(0) \neq 0$.

Now suppose X is an analytic subset of the complex manifold M . The embedded resolution of singularities theorem asserts that we can find a proper map $\pi : \tilde{M} \rightarrow M$ of complex manifolds which is a (locally finite) composition of blowing ups with non-singular centres such that $\tilde{X} = \pi^{-1}(X)$ is non-singular and $\pi^{-1}(\text{Sing}(X))$ is a divisor in $\tilde{X}$ in normal crossings. A proof may be found in Hironaka [2].

Examples.

4. The resolution of the curve $(z_1 - z_2)(z_1 + z_2) = 0$ given in example 1 above is an embedded resolution.
5. The resolution of the curve $z_1^2 - z_2^3 = 0$ given in example 2 above is not an embedded resolution. Two further blowing ups are needed to obtain an embedded resolution:

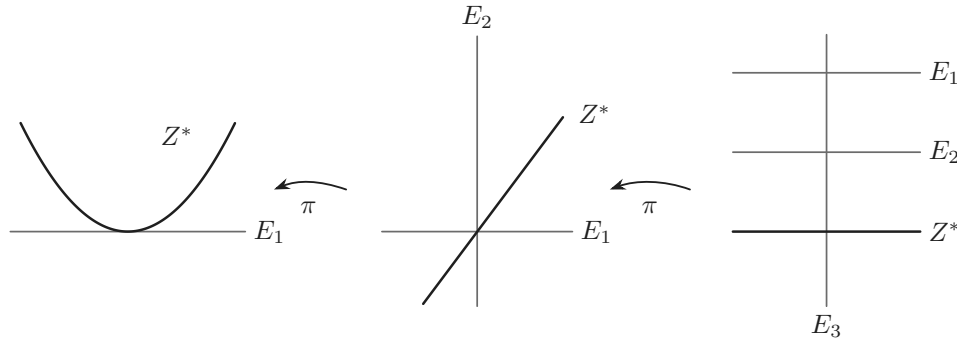


Figure I.13

We conclude this chapter with some remarks on the role of blowing up in the classification theory of compact complex manifolds, especially complex surfaces.

First notice that we can use blowing up to define a relation between complex manifolds of the same dimension. Thus we shall say that M is B-related to N if M may be obtained by blowing up N a finite number of times. If M and N are complex surfaces then the centres of the blowing ups must always be points and the exceptional varieties are always biholomorphic to $P^1(\mathbb{C})$. Now suppose that E is a complex submanifold of the complex surface M which is biholomorphic to $P^1(\mathbb{C})$. It follows from a theorem of Grauert [1], that if the self-intersection number of E is -1 (" $E^2 = -1$ ") then we can *blow down* E to a point. That is, there exists a complex surface N such that for some $p \in N$, $B_p(N) = M$ and the exceptional variety of the blowing up is E . We call such a variety E an *exceptional curve of first kind*. We can now define an equivalence relation between compact complex surfaces. We say that the surfaces M and N are B-equivalent if M can be obtained from N by a finite number of blowing ups and blowing downs. We say that a complex surface is a *relatively minimal model* if and only if it does not contain any exceptional curves of the first kind. Every compact complex surface can be obtained from a relatively minimal model by a finite succession of blowing ups.

In Kodaira [2] there is a description of the relatively minimal models of compact complex surfaces as well as a description of the classification of compact complex surfaces. A useful survey of Kodaira's work, together with further references, may be found in Sundararaman [1]. See also Ueno [1].

It is an important problem to find properties of complex manifolds invariant under blowing up or blowing down (in algebraic geometry the corresponding problem is that of finding birational invariants). We shall describe one invariant here (see also §9, Chapter 5) and refer the reader to the references cited above for the description of other invariants and proofs. Let M be a compact complex manifold with field of meromorphic functions $\mathcal{M}(M)$. We let $t(M)$ denote the transcendence degree of $\mathcal{M}(M)$ over $\mathbb{C}$. A result of Thimm [1] (see also Remmert [1], Siegel [2]) asserts that $t(M) \leq \text{dimension}(M)$. If M is algebraic, $t(M) = \text{dimension}(M)$. We have the important result that $t(M)$ is invariant under blowing ups. Thus surfaces with different transcendence degrees cannot have the same relatively minimal models. If $t(M) = \text{dimension}(M)$ we say that M is a *Moishezon manifold*. If $\text{dimension}(M) = 2$, every Moishezon manifold is algebraic by a theorem of Kodaira [2]. The theory of Moishezon manifolds is expounded in Moishezon [1]. We remark that even though a Moishezon manifold M need not be algebraic

(at least if $\dim(M) > 2$), there exists a finite sequence of blowing ups of M such that the resulting manifold is algebraic (For a proof see Moishezon [1, Chapter 2]). Of course, if M is algebraic, then any finite sequence of blowing ups of M with non-singular centres (necessarily algebraic) will give an algebraic manifold.

Exercises

1. Blow up the surface $zx^2 = y^2$ at zero. Observe that blowing up at the (geometrically) most singular points does not always work. Now find a desingularization of the surface.
2. Verify that the surface $x^2y^2 - z^6 + x^6 + y^6 = 0$ has an isolated singularity at zero. Show that when we blow up at zero, the new surface has a singular set with singularities.

Part II

Calculus, Sheaf Theory and Coherent Sheaves

Several Complex Variables and Complex Manifolds II

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Chapter 5

Calculus on Complex Manifolds

Introduction In sections 1 and 2 we cover basic linear algebra and calculus on differential manifolds. We define the complexification of a real vector space in section 3 and in section 4 we develop the main results of complex linear algebra that we need in the sequel. After giving some general facts about complex and holomorphic vector bundles in section 5 and constructing the tangent and cotangent bundles of a complex manifold in section 6 we reach the heart of the chapter in section 7 with the construction of the $\bar{\partial}$ -operator on an arbitrary complex manifold. In section 8 we prove the Dolbeault–Grothendieck Lemma and, with a little more effort, deduce that the Cousin I and II problems are always solvable on polydiscs. Section 9 is devoted to the discussion of a number of important examples on compact complex manifolds. Thus we construct the Euler sequence for projective space, prove some basic results about linear systems and their relationship with holomorphic line bundles and conclude with a discussion of theta functions and complex tori. Finally, in section 10, we define the concepts of q -pseudoconvexity, q -completeness and weakly positive vector bundles.

1 Review of linear algebra

In this section we shall review some elementary linear algebra that does not depend on the assumption of an inner product structure. We shall generally omit proofs (usually simple) referring the reader to any one of the many texts in linear algebra. All vector spaces in this section will be assumed real and finite dimensional.

Definition 5.1.1. A *graded vector space* $\mathbf{E}$ is a collection $\{E_0, E_1, \dots\}$ of vector spaces indexed by the positive integers. We write $\mathbf{E} = \{E_0, E_1, \dots\}$ and set $\mathbf{E}_n = E_n$. We call E_n the n th. component of $\mathbf{E}$. A *morphism* $\mathbf{T} : \mathbf{E} \longrightarrow \mathbf{F}$, of degree $r \geq 0$, of graded vector spaces is a collection of linear maps $T_i : E_i \rightarrow F_{i+r}$ indexed by the positive integers.

Examples.

1. Associated to the vector space E we may define the graded vector spaces $\otimes E$, ΛE and $\odot E$ whose n th. components are respectively $\otimes^n E$, $\Lambda^n E$ and $\odot^n E$ (n th. tensor, exterior and symmetric powers of E respectively).
2. A linear map $T : E \rightarrow F$ induces degree zero morphisms $\otimes T : \otimes E \longrightarrow \otimes F$, $\Lambda T : \Lambda E \longrightarrow \Lambda F$, $\odot T : \odot E \longrightarrow \odot F$ whose n th. components are $\otimes^n T$, $\Lambda^n T$ and $\odot^n T$ respectively.

Definition 5.1.2. Let $\mathbf{E}$ and $\mathbf{F}$ be graded vector spaces. Then

1. The *direct sum* of $\mathbf{E}$ and $\mathbf{F}$ is the graded vector space $\mathbf{E} \oplus \mathbf{F}$ whose n th. component is $E_n \oplus F_n$.
2. The *tensor product* of $\mathbf{E}$ and $\mathbf{F}$ is the graded vector space $\mathbf{E} \otimes \mathbf{F}$ whose n th. component is $\bigoplus_{r+s=n} E_r \otimes F_s$.

Definition 5.1.3. A *graded vector space algebra* consists of a graded vector space $\mathbf{E}$ together with a morphism of degree zero $\phi : \mathbf{E} \otimes \mathbf{E} \rightarrow \mathbf{E}$, written $\phi(A \otimes B) = AB$, satisfying the following properties

1. The multiplication defined by ϕ is associative.
2. There exists a unit element in $\mathbf{E}$ for the multiplication.

The algebra is said to be *commutative* if $AB = (-1)^{pq}BA$, $A \in E_p$, $B \in E_q$.

Remark. The universal factorization property for the tensor product implies that we may equivalently suppose that ϕ is a bilinear map on $\mathbf{E} \times \mathbf{E}$.

Example 3. Given a vector space E , $\otimes E$, ΛE and $\odot E$ are graded vector space algebras with respective algebra operations of tensor product, exterior or wedge product and symmetric product. ΛE is commutative.

Proposition 5.1.4. Let E and F be vector spaces. Then $\Lambda E \otimes \Lambda F$ has the natural structure of a commutative graded algebra with wedge product defined by

$$(X \otimes Y) \wedge (X' \otimes Y') = (-1)^{pq}(X \wedge X') \otimes (Y \wedge Y'),$$

where $X \in \Lambda E$, $Y \in \Lambda^p F$, $X' \in \Lambda^q E$, $Y' \in \Lambda F$.

Proof. We remark only that in all results of this type it is enough to define the operation or map on a set of generators for the algebra. $\square$

Before stating the next result we remark that we say a morphism of graded vector spaces is an *isomorphism* if it is invertible and of degree zero. An isomorphism of graded vector space algebras will, in addition, preserve the algebra structures.

Theorem 5.1.5. If E and F are vector spaces we have a canonical isomorphism of commutative graded algebras

$$\Lambda(E \oplus F) \approx \Lambda E \otimes \Lambda F.$$

In particular,

$$\Lambda(E \oplus F)_p \approx \bigoplus_{r+s=p} \Lambda^r E \otimes \Lambda^s F, \quad p \geq 0.$$

Proof. For $r, s \geq 0$ we define $\mu_{r,s} : \Lambda^r E \otimes \Lambda^s F \rightarrow \Lambda(E \oplus F)_{r+s}$ by

$$\begin{aligned} \mu_{r,s}(e_1 \wedge \cdots \wedge e_r \otimes f_1 \wedge \cdots \wedge f_s) &= (e_1 + 0) \wedge \cdots \wedge (e_r + 0) \\ &\quad \wedge (f_1 + 0) \wedge \cdots \wedge (f_s + 0), \end{aligned}$$

where $e_i \in E$, $1 \leq i \leq r$, $f_j \in F$, $1 \leq j \leq s$, and the wedge product on the right hand side of the above relation is taken in $\Lambda^{r+s}(E \oplus F)$. We leave it to the reader to verify that the maps $\mu_{r,s}$ define the required morphism. $\square$

Given a vector space E it is possible to identify ΛE and $\odot E$ with graded subspaces (*not* subalgebras) of $\otimes E$ and we now indicate how to do this in the case of exterior powers of E . Fix a positive integer p and let S_p denote the symmetric group on p symbols. We define a representation $T : S_p \rightarrow \text{GL}(\otimes^p E)$ of S_p by

$$T(\sigma)(e_1 \otimes \cdots \otimes e_p) = \text{sign}(\sigma) e_{\sigma(1)} \otimes \cdots \otimes e_{\sigma(p)},$$

where $e_j \in E$, $1 \leq j \leq p$, $\sigma \in S_p$. Let $\text{Alt}_p(E)$ denote the fixed point set of the corresponding action of S_p on $\otimes^p E$. Elements of $\text{Alt}_p(E)$ are called *alternating tensors* of order p . The map $\wedge : \otimes^p E \rightarrow \Lambda^p E$ defined by mapping $e_1 \otimes \cdots \otimes e_p$ to $e_1 \wedge \cdots \wedge e_p$ induces a linear isomorphism of $\text{Alt}_p(E)$ with $\Lambda^p E$. To see this observe that we have a projection map $\text{Alt} : \otimes^p E \rightarrow \text{Alt}_p(E)$ defined by

$$\text{Alt}(e_1 \otimes \cdots \otimes e_p) = \frac{1}{p!} \sum_{\sigma \in S_p} T(\sigma)(e_1 \otimes \cdots \otimes e_p).$$

Clearly $\text{Kernel}(\text{Alt}) = \text{Kernel}(\wedge)$ and so $\text{Alt}_p(E) \approx \Lambda^p E$.

Remark. Under the identification of $\Lambda^p E$ with $\text{Alt}_p(E)$ given by this isomorphism we see that

$$e_1 \wedge \cdots \wedge e_p = \frac{1}{p!} \sum_{\sigma \in S_p} \text{sign}(\sigma) e_{\sigma(1)} \otimes \cdots \otimes e_{\sigma(p)}.$$

Let $E' = L_{\mathbb{R}}(E, \mathbb{R})$ denote the dual space of E and $\langle \cdot, \cdot \rangle : E \times E' \rightarrow \mathbb{R}$ denote the dual pairing between E and E' . For $p \geq 0$, we have dual pairings $\otimes^p E \times \otimes^p E' \rightarrow \mathbb{R}$, $\Lambda^p E \times \Lambda^p E' \rightarrow \mathbb{R}$ respectively defined by

$$\langle e_1 \otimes \cdots \otimes e_p, \phi_1 \otimes \cdots \otimes \phi_p \rangle = \prod_{i=1}^p \langle e_i, \phi_i \rangle$$

$$\langle e_1 \wedge \cdots \wedge e_p, \phi_1 \wedge \cdots \wedge \phi_p \rangle = \det[\langle e_i, \phi_j \rangle],$$

where $e_i \in E$, $\phi_i \in E'$, $1 \leq i \leq p$. Using these pairings we identify the dual spaces of $\otimes^p E$, $\Lambda^p E$ with $\otimes^p E'$, $\Lambda^p E'$ respectively. Notice that the dual pairing of $\otimes^p E$ with $\otimes^p E'$ induces a pairing of $\text{Alt}_p(E)$ with $\text{Alt}_p(E')$ and so a pairing of $\Lambda^p E$ with $\Lambda^p E'$. This pairing differs from the pairing we have defined between $\Lambda^p E$ and $\Lambda^p E'$ by a factor $p!$

We leave the case of the dual pairing for the symmetric powers of E as an exercise.

For $p > 0$, we define the linear map $U : \Lambda^p E \rightarrow E \otimes \Lambda^{p-1} E$ by

$$U(e_1 \wedge \cdots \wedge e_p) = \sum_{j=1}^p (-1)^{j+1} e_j \otimes e_1 \wedge \cdots \wedge \widehat{e_j} \wedge \cdots \wedge e_p,$$

where $e_j \in E$, $1 \leq j \leq p$, and “ $\widehat{}$ ” as usual denotes omission.

Properties of the map U .

1. The operator U is the transpose of $\wedge$, that is,

$$\langle UX, \phi \otimes \psi \rangle = \langle X, \phi \wedge \psi \rangle,$$

for all $X \in \Lambda^p E$, $\phi \in E'$, $\psi \in \Lambda^{p-1} E'$.

2. If $X \in \Lambda^p E$, $\wedge U(X) = pX$.

3. If $Z \in E \otimes \Lambda^p E$, then $U(\wedge Z) = Z - \wedge^{13}(I \otimes U)(Z)$, where $\wedge^{13}$ denotes the operation of wedging the first factor of $E \otimes E \otimes \Lambda^{p-1} E$ into the third factor.

4. If we iterate U , the map $U^p : \Lambda^q E \rightarrow \otimes^p E \otimes \Lambda^{q-p} E$ satisfies the relation $\wedge U^p = q!/(q-p)! I$, where $\wedge : \otimes^p E \otimes \Lambda^{q-p} E \rightarrow \Lambda^q E$ is just the operation of wedge product. In particular, if we identify $\Lambda^p E$ with $\text{Alt}_p(E)$, we see that $\text{Alt } U^p = p! I$, $p \geq 0$.

Properties 1–3 are easily proved by working on generators, property 4 may be proved by induction on p .

We conclude this section by describing the operation of *contraction* or *interior product* of tensors. First observe that since the dual pairing $E' \times E \rightarrow \mathbb{R}$ is bilinear it corresponds to a linear map $E' \otimes E \rightarrow \mathbb{R}$. This map is usually referred to as “trace” as if we use the natural identification of $E' \otimes E$ with $L(E, E)$ it corresponds to the operation of taking the trace of a linear map. Trace is the simplest example of a contraction operation and we shall now consider generalisations. With a view to later applications we work with tensor products of *exterior* powers of vector spaces. Actually this is no loss of generality since $\Lambda^1 E = E$, $\Lambda^1 E' = E'$.

Suppose $p \geq q \geq 0$. We define the linear map

$$C : \Lambda^p E \otimes \Lambda^q E' \longrightarrow \Lambda^{p-q} E$$

by requiring that

$$\langle C(X \otimes \phi), \psi \rangle = \langle \phi \wedge \psi, X \rangle$$

for all $X \in \Lambda^p E$, $\psi \in \Lambda^{p-q} E'$, $\phi \in \Lambda^q E'$. We similarly define

$$C : \Lambda^p E \otimes \Lambda^q E' \rightarrow \Lambda^{q-p} E'$$

in case $q \geq p$.

Given $X \in \Lambda^p E$ and $q \geq p$ we define

$$C_X : \Lambda^q E' \rightarrow \Lambda^{q-p} E'$$

by $C_X(\phi) = C(X \otimes \phi)$, $\phi \in \Lambda^q E'$. We similarly define C_ϕ for $\phi \in \Lambda^p E'$. We call C a *contraction* (operator) and C_X *contraction with X* .

Properties of the contraction operators C , C_X , C_ϕ .

1. For $p \geq 0$, the map $C : \Lambda^p E \otimes \Lambda^p E' \rightarrow \mathbb{R}$ is equal to the dual pairing of $\Lambda^p E$ with $\Lambda^p E'$.
2. C_X is the transpose of the operator “wedge product with X ”. That is,

$$\langle C_X \phi, Y \rangle = \langle \phi, X \wedge Y \rangle, \quad X \in \Lambda^p E, \quad Y \in \Lambda^{s-p} E, \quad \phi \in \Lambda^s E'.$$

3. $C_X C_Y = C_{Y \wedge X} = (-1)^{rs} C_{X \wedge Y}$, $X \in \Lambda^r E$, $Y \in \Lambda^s E$.
4. For $X \in E$, $C_X : \Lambda^p E' \rightarrow \Lambda^{p-1} E'$ is defined on generators by

$$C_X(\phi_1 \wedge \cdots \wedge \phi_p) = \sum_{j=1}^p (-1)^{j+1} \langle X, \phi_j \rangle \phi_1 \wedge \cdots \wedge \widehat{\phi_j} \wedge \cdots \wedge \phi_p.$$

We have similar properties holding for contraction with forms. We omit proofs of the above properties which all follow straightforwardly from the definition and standard properties of wedge products.

We now wish to define contractions between factors of an arbitrary finite tensor product of exterior powers of E and E' . Suppose that $V = \otimes_{i=1}^n V_i$, where each V_i is either $\Lambda^{p_i} E$ or $\Lambda^{p_i} E'$. Assume that the j th. and k th. factors are equal to $\Lambda^{p_j} E$ and $\Lambda^{p_k} E'$ respectively. Set $\tilde{V} = \otimes_{i=1}^n \tilde{V}_i$, where $\tilde{V}_i = V_i$ if $i \neq j, k$ and $\tilde{V}_j = \mathbb{R}$, $\tilde{V}_k = \Lambda^{p_k - p_j} E'$ if $p_j \leq p_k$ and $\tilde{V}_j = \Lambda^{p_j - p_k} E$, $\tilde{V}_k = \mathbb{R}$ if $p_j \geq p_k$ (Notice that $\tilde{V}$ is really a tensor product of $n-1$ factors but inclusion of the

trivial factor for the present simplifies our indexing). The contraction between the j th. and k th. factors of V will be the linear map $C_k^j : V \rightarrow V$ defined on generators by

$$\begin{aligned} C_k^j(\cdots \otimes X_j \otimes \cdots \otimes \phi_k \otimes \cdots) &= \cdots \otimes 1 \otimes \cdots \otimes C(X_j \otimes \phi_k) \otimes \cdots, \quad p_j \leq p_k \\ &= \cdots \otimes C(X_j \otimes \phi_k) \otimes \cdots \otimes 1 \otimes \cdots, \quad p_j \geq p_k. \end{aligned}$$

Notice that our convention is that superscripts refer to factors which are exterior powers of E , subscripts to factors which are exterior powers of E' . The *product* $C_k^j C_m^l$ of the contractions C_k^j and C_m^l is defined if $\{i, j\} \cap \{l, m\} = \emptyset$ and is the simultaneous contraction of the j th. and k th. and l and m th. factors of V . Necessarily, $C_k^j C_m^l = C_m^l C_k^j$. The *composition* $C_k^j \cdot C_m^l$ is defined to be the composite of the contraction C_m^l with the contraction C_k^j , where C_k^j is a contraction of the space $\tilde{V}$ not V . We may also use brackets to describe compositions of contractions. Thus if $Z \in V$, $C_k^j \cdot C_m^l(Z) = C_k^j(C_m^l(Z))$. Notice that $\tilde{V}$ has fewer factors than V and so, in general, $C_k^j \cdot C_m^l \neq C_k^j C_m^l$.

Example 4. Let $p \geq r + s$. Then the following diagram of contractions commutes.

$$\begin{array}{ccc} \Lambda^r E \otimes \Lambda^p E' \otimes \Lambda^s E & \xrightarrow{C_2^3} & \Lambda^r E \otimes \Lambda^{p-s} E' \\ \downarrow C_2^1 & & \downarrow C \\ \Lambda^{p-r} E' \otimes \Lambda^s E & \xrightarrow{(-1)^{rs} C} & \Lambda^{p-s-r} E'. \end{array}$$

Indeed, let $X \in \Lambda^r E$, $Y \in \Lambda^s E$, $Z \in \Lambda^{p-r-s} E$, $\phi \in \Lambda^p E'$. Then

$$\begin{aligned} \langle C(C_2^3(X \otimes \phi \otimes Y)), Z \rangle &= \langle C_{X \wedge Y} \phi, Z \rangle \\ &= (-1)^{rs} \langle C_Y C_X \phi, Z \rangle, \quad \text{Property 2 of contractions.} \\ &= (-1)^{rs} \langle C_2^1(C(X \otimes \phi \otimes Y)), Z \rangle. \end{aligned}$$

Since this is true for all $Z \in \Lambda^{p-r-s} E$, our assertion follows. In our notation above we have shown $C \cdot C_2^3 = (-1)^{rs} C \cdot C_2^1$.

We now list, without proof, some additional properties of contraction operators.

Properties of the contraction operators C , C_X , C_ϕ continued.

5. Let $X \in \Lambda^p E$, $\phi \in \Lambda^q E'$ and suppose $q \geq p$. Then

$$C_X \phi = \frac{1}{(q-p)! p!} C_{p+1}^1 \cdots C_{2p}^p (U^p X \otimes U^p \phi).$$

In particular, if $X \in E$,

$$C_X \phi = C_2^1(X \otimes U\phi).$$

6. Let $X \in \Lambda^p E$, $\phi \in \Lambda^{p+1} E'$ then

$$C_X \phi = (-1)^p C_1^3(\phi \otimes UX).$$

7. Let $X \in E$, $\phi \in \Lambda^p E'$, $\psi \in \Lambda^q E'$. Then

$$C_X(\phi \wedge \psi) = C_X(\phi) \wedge \psi + (-1)^p \phi \wedge C_X \psi.$$

8. Let $p \geq q > 0$ and define

$$C_1 : \Lambda^p E \otimes \Lambda^q E' \longrightarrow \Lambda^{p-1} E \otimes \Lambda^{q-1} E'$$

by

$$C_1(X \otimes \phi) = C_3^1(UX \otimes U\phi), \quad X \in \Lambda^p E, \quad \phi \in \Lambda^q E'.$$

Then $C = (C_1)^p / p! : \Lambda^p E \otimes \Lambda^q E' \rightarrow \Lambda^{p-q} E$.

We end this section with an example giving another characterization of trace.

Example 5. Let $A \in L(E, E)$ and $\dim(E) = n$. Then the map $\tilde{A} : \Lambda^n E \rightarrow \Lambda^n E$, defined on generators by

$$\tilde{A}(Z_1 \wedge \cdots \wedge Z_n) = \sum_{i=1}^n Z_1 \wedge \cdots \wedge A(Z_i) \wedge \cdots \wedge Z_n,$$

is equal to scalar multiplication by $-\text{trace}(A)$. Working on generators in $L(E, E)$, let $A = \phi \otimes X \in E' \otimes E \approx L(E, E)$ and $Z \in \Lambda^n E$. Then $\tilde{A}(Z) = X \wedge C_\phi Z = -(C_\phi X)Z$, Property 7 of contractions. But $C_\phi X$ is just $\text{trace}(A)$.

Exercises.

1. Verify that $\otimes^p E'$ is naturally isomorphic to the space $L^p(E; \mathbb{R})$ of p -linear real valued maps on E . Show also that $\Lambda^p E'$ and $\Lambda^p E'$ are naturally isomorphic to the spaces of p -linear alternating and symmetric real valued maps on E respectively.
2. Let $p \leq q$. Show that $C : \Lambda^p E \otimes \Lambda^q E' \rightarrow \Lambda^{q-p} E'$ is defined in terms of generators by the formula

$$\begin{aligned} C(X_1 \wedge \cdots \wedge X_p \otimes \phi_1 \wedge \cdots \wedge \phi_q) \\ = \sum_{I, J} \text{sign}(I, J) \langle X_1, \phi_{i_1} \rangle \cdots \langle X_p, \phi_{i_p} \rangle \phi_{j_1} \wedge \cdots \wedge \phi_{j_{q-p}}, \end{aligned}$$

where the sum is taken over all p -tuples $I = (i_1, \dots, i_p)$ satisfying $1 \leq i_1, \dots, i_p \leq q$ and $(q-p)$ -tuples $J = (j_1, \dots, j_{q-p})$ satisfying $1 \leq j_1 < \cdots < j_{q-p} \leq q$, subject to the condition that $\{i_1, \dots, i_p, j_1, \dots, j_{q-p}\}$ is a permutation of $\{1, \dots, q\}$. $\text{Sign}(I, J)$ denotes the signature of this permutation.

3. Work out $C_1 : \Lambda^2 E \otimes \Lambda^2 E' \rightarrow E \otimes E'$ in terms of generators and verify that $(C_1)^2 = 2C$.
4. Generalise the example at the end of the section to find other invariants of the map A .
5. Let $0 \rightarrow E \xrightarrow{A} F \xrightarrow{B} G \rightarrow 0$ be a short exact sequence of vector spaces and linear maps. Suppose that the dimensions of E, F, G are p, q, r respectively. Prove that there exists a canonical isomorphism $\Lambda^q F \approx \Lambda^p E \otimes \Lambda^r G$ (Hint: Prove that there exists a natural monomorphism $k : \Lambda^r G' \otimes \Lambda^q F \rightarrow \Lambda^{q-r} F$ defined by $\langle k(\phi \otimes X), \psi \rangle = \langle \Lambda^r B'(\phi) \wedge \psi, X \rangle$, $\phi \in \Lambda^r G', X \in \Lambda^q F, \psi \in \Lambda^{q-r} F'$. Show that $\text{Image}(k) = \text{Image}(\Lambda^p A)$). More generally, show that for $n \geq 1$ we have a natural exact sequence

$$0 \rightarrow \Lambda^n E \rightarrow \Lambda^n F \rightarrow \Lambda^{n-1} E \otimes F \rightarrow 0.$$

6. Let $\dim(E) = n$ and $\phi \in E', \phi \neq 0$. Show that the sequence

$$0 \rightarrow \Lambda^n E \xrightarrow{C_\phi} \Lambda^{n-1} E \rightarrow \cdots \xrightarrow{C_\phi} \Lambda^1 E \xrightarrow{C_\phi} \mathbb{R} \rightarrow 0$$

is exact (Hint: Choose an orthonormal basis for E such that ϕ is an element of the dual orthonormal basis for E').

7. Continuing with the assumptions of question 6, verify that

- a) $L(\Lambda^p E, \mathbb{R}) \approx \Lambda^p E' \approx \Lambda^n E' \otimes \Lambda^{n-p} E, 0 \leq p \leq n$.
- b) The diagram

$$\begin{array}{ccc} L(\Lambda^p E, \mathbb{R}) & \xrightarrow{\approx} & \Lambda^n E' \otimes \Lambda^{n-p} E \\ (C_\phi)' \downarrow & & \downarrow I \otimes C_\phi \\ L(\Lambda^{p+1} E, \mathbb{R}) & \xrightarrow{\approx} & \Lambda^n E' \otimes \Lambda^{n-p-1} E \end{array}$$

commutes, $0 \leq p \leq n$.

(Part b) amounts to saying that the sequence of question 6 is “self-dual”).

2 Calculus on differential manifolds

In this section we review that portion of calculus on manifolds that does not depend on a choice of Riemannian metric. Proofs and further details may be found in Kobayashi and Nomizu [1] and Abraham and Marsden [1]. For the theory of vector bundles we refer in addition to the books by Husemoller [1] and Lang [1] and to the basic theory outlined in §5 of Chapter 1.

Throughout this section M will denote a C^∞ differential manifold of dimension m . For this section only, $C^r(M)$ will denote the space of *real* valued C^r functions on M , $0 \leq r \leq \infty$.

Let E be a smooth (that is, C^∞) vector bundle on M . For $r \geq 0$ we let $C^r(E)$ denote the vector space of C^r sections of E and $C_c^r(E)$ denote the space of C^r sections with compact support.

Notation and examples (see also §5 of Chapter 1).

1. We let $\underline{\mathbb{R}} = M \times \mathbb{R}$ denote the trivial real line bundle over M . Notice that $C^r(\underline{\mathbb{R}}) = C^r(M)$.
2. We let $\mathcal{T}M$ and $\mathcal{T}'M$ respectively denote the tangent and cotangent bundles of M .

As described in §5 of Chapter 1, we may form the dual bundle E' of a vector bundle E and tensor products of tensor, exterior and symmetric powers of E and E' . The contraction operations described in §1 of this chapter all extend to these bundles and their sections. By way of example, there is a natural vector bundle map $\text{trace} : E \otimes E' \rightarrow \underline{\mathbb{R}}$ defined by: $\text{trace}(e_x \otimes \phi_x) = \langle e_x, \phi_x \rangle$, $e_x \in E_x$, $\phi_x \in E'_x$, $x \in M$. If S and ϕ are C^r sections of E and E' respectively, we may define the C^r section $\text{trace}(S \otimes \phi)$ of $\underline{\mathbb{R}}$ by: $\text{trace}(S \otimes \phi)(x) = \text{trace}(S(x) \otimes \phi(x))$. In the sequel we use the same notation for contractions on vector bundles and their sections that we developed in the previous section for vector spaces.

Differential forms. A section of the bundle $\Lambda^p \mathcal{T}'M$ is called a *differential p -form* (on M). For $p \geq 0$ we have the operation $d : C^\infty(\Lambda^p \mathcal{T}'M) \rightarrow C^\infty(\Lambda^{p+1} \mathcal{T}'M)$ of *exterior differentiation* and the corresponding sequence

$$C^\infty(M) \xrightarrow{d} C^\infty(\mathcal{T}'M) \xrightarrow{d} \cdots \xrightarrow{d} C^\infty(\Lambda^p \mathcal{T}'M) \xrightarrow{d} \cdots$$

Properties of exterior differentiation.

1. d is $\mathbb{R}$ -linear.
2. $d^2 = 0$.
3. $d(\phi \wedge \zeta) = d\phi \wedge \zeta + (-1)^p \phi \wedge d\zeta$, $\phi \in C^\infty(\Lambda^p \mathcal{T}'M)$, $\zeta \in C^\infty(\Lambda^q \mathcal{T}'M)$.
4. If $f \in C^\infty(M)$, $x \in M$, then $df(x) = T_x f \in L(T_x M, \mathbb{R}) = \mathcal{T}'_x M$.
5. If $f : M \rightarrow N$ is C^∞ and $\phi \in C^\infty(\Lambda^p \mathcal{T}'N)$, then $d(f^* \phi) = f^* d\phi$ (f^* denotes the operation of pull-back of differential forms induced by f).
6. In local coordinates, if $\phi = \sum_{1 \leq i_1 < \cdots < i_p \leq m} \phi_{i_1 \cdots i_p} dx_{i_1} \wedge \cdots \wedge dx_{i_p}$, where the coefficients $\phi_{i_1 \cdots i_p}$ are C^∞ functions, then

$$d\phi = \sum_{j=1}^m \sum_{1 \leq i_1 < \cdots < i_p \leq m} \frac{\partial \phi_{i_1 \cdots i_p}}{\partial x_j} dx_j \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_p}.$$

We say that a p -form ϕ is *closed* if $d\phi = 0$ and that it is *exact* if, in addition, there exists a $(p-1)$ -form ζ such that $d\zeta = \phi$. A closed form need not be exact though by Poincaré's lemma it is always *locally exact*. That is, if ϕ is an exact p -form on M and $x \in M$ we may find an open neighbourhood U of x (typically contractible) and a $(p-1)$ -form ζ defined on U such that $d\zeta = \phi|_U$.

The bundles $\Lambda^p \mathcal{T}'M$ are zero bundles for $p > m$ and $\Lambda^m \mathcal{T}'M$ is a real line bundle. We say that M is *orientable* if $\Lambda^m \mathcal{T}'M$ is isomorphic to the trivial line bundle $\mathbb{R}$. If M is orientable we have a natural $\mathbb{R}$ -linear map $\int_M : C_c^0(\Lambda^m \mathcal{T}'M) \rightarrow \mathbb{R}$ called *integration*. If M has boundary ∂M (necessarily orientable) then we have *Stokes' theorem*:

$$\int_{\partial M} \phi = \int_M d\phi, \quad \phi \in C_c^1(\Lambda^{m-1} \mathcal{T}'M).$$

For $p \geq 0$, we define the *p*th. de Rham cohomology group, $H_{\text{DR}}^p(M, \mathbb{R})$, of M to be the quotient vector space

$$H_{\text{DR}}^p(M, \mathbb{R}) = (\text{Ker } d : C^\infty(\Lambda^p \mathcal{T}'M) \rightarrow C^\infty(\Lambda^{p+1} \mathcal{T}'M)) / dC^\infty(\Lambda^{p-1} \mathcal{T}'M).$$

Clearly, $H_{\text{DR}}^p(M, \mathbb{R}) = 0$, $p > m$. It is true, though by no means trivial, that if M is compact the de Rham cohomology groups are all finite dimensional vector spaces. It is also true that integration defines a dual pairing between $H_{\text{DR}}^p(M, \mathbb{R})$ and the singular homology group $H_p(M, \mathbb{R})$, $p \geq 0$. As a consequence, $H_{\text{DR}}^p(M, \mathbb{R})$ is isomorphic to $H^p(M, \mathbb{R})$, $p \geq 0$. We give a proof of this isomorphism in Chapter 6, §3.

Vector fields and Lie derivatives. We say that a map $\delta : C^\infty(M) \rightarrow C^\infty(M)$ is a *derivation* if δ is $\mathbb{R}$ -linear and

$$\delta(fg) = (\delta f)g + f(\delta g),$$

for all $f, g \in C^\infty(M)$. We denote the set of derivations of $C^\infty(M)$ by $D(M)$. We have a natural map of $C^\infty(\mathcal{T}M)$ into $D(M)$, denoted $X \mapsto L_X$, defined by

$$L_X f = C_X df, \quad f \in C^\infty(M).$$

$L_X f$ is called the *Lie derivative* of f with respect to X . It is a basic result that this map of $C^\infty(\mathcal{T}M)$ into $D(M)$ is a bijection. This allows us to think of vector fields as (1st. order) linear partial differential operators on $C^\infty(M)$.

Given $X, Y \in C^\infty(\mathcal{T}M)$, the map $L_X L_Y - L_Y L_X$ is a derivation of $C^\infty(M)$ and so there exists a vector field Z such that $L_X L_Y - L_Y L_X = L_Z$. In the sequel we call Z the *Lie bracket* of X and Y and denote it by $[X, Y]$.

Properties of the Lie bracket.

1. $[\ , \]$ is $\mathbb{R}$ -bilinear.
2. $[X, Y] = -[Y, X]$, $X, Y \in C^\infty(\mathcal{T}M)$.
3. $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for all $X, Y, Z \in C^\infty(\mathcal{T}M)$ (Jacobi identity).
4. If $f : M \rightarrow N$ is a C^∞ diffeomorphism then $[f_* X, f_* Y] = f_* [X, Y]$, for all $X, Y \in C^\infty(\mathcal{T}M)$ (f_* is the operation of push-forward of vector fields induced by f).

We may give an invariant definition of exterior differentiation in terms of Lie derivatives and brackets. Suppose that ϕ is a differential p -form and $X_0, \dots, X_p$ are vector fields on M then

$$\begin{aligned} \langle d\phi, X_0 \wedge \dots \wedge X_p \rangle &= \sum_{i=0}^p (-1)^i L_{X_i} \langle \phi, X_0 \wedge \dots \wedge \widehat{X}_i \wedge \dots \wedge X_p \rangle \\ &\quad + \sum_{0 \leq i < j} (-1)^{i+j} \langle \phi, [X_i, X_j], X_0 \wedge \dots \wedge \widehat{X}_i \wedge \dots \wedge \widehat{X}_j \wedge \dots \wedge X_p \rangle. \end{aligned}$$

For each $X \in C^\infty(\mathcal{T}M)$ define $L_X : C^\infty(\mathcal{T}M) \rightarrow C^\infty(\mathcal{T}M)$ by $L_X Y = [X, Y]$. We refer to $L_X Y$ as the *Lie derivative* of Y with respect to X . We now extend the operator L_X to C^∞ sections of arbitrary finite tensor and exterior products of $\mathcal{T}M$ and $\mathcal{T}'M$. We first define L_X on sections of $\mathcal{T}'M$. Suppose $\phi \in C^\infty(\mathcal{T}'M)$ then $L_X \phi$ will be the differential 1-form characterised by

$$L_X \langle \phi, Y \rangle = \langle L_X \phi, Y \rangle + \langle \phi, L_X Y \rangle, \quad \text{for all } Y \in C^\infty(\mathcal{T}M).$$

It is quite straightforward, using the basic properties of Lie derivatives, to check that this relation does indeed define $L_X \phi$ as a section of $\mathcal{T}'M$. We shall extend L_X to arbitrary tensor and exterior powers by requiring that it is a derivation. Thus, for $p \geq 0$, we define $L_X : C^\infty(\Lambda^p \mathcal{T}M) \rightarrow C^\infty(\Lambda^p \mathcal{T}M)$ by

$$L_X(X_1 \wedge \cdots \wedge X_p) = \sum_{i=1}^p X_1 \wedge \cdots \wedge L_X X_i \wedge \cdots \wedge X_p$$

and $L_X : C^\infty(\Lambda^p \mathcal{T}'M) \rightarrow C^\infty(\Lambda^p \mathcal{T}'M)$ by

$$L_X(\phi_1 \wedge \cdots \wedge \phi_p) = \sum_{i=1}^p \phi_1 \wedge \cdots \wedge L_X \phi_i \wedge \cdots \wedge \phi_p,$$

where X_i and ϕ_i are C^∞ vector fields and 1-forms on M respectively. Suppose that V and W are vector bundles which are finite tensor products of tensor and exterior powers of $\mathcal{T}M$ and $\mathcal{T}'M$ and that we have defined L_X on C^∞ sections of V and W . We define $L_X : C^\infty(V \otimes W) \rightarrow C^\infty(V \otimes W)$ by

$$L_X(S \otimes T) = L_X S \otimes T + S \otimes L_X T, \quad S \in C^\infty(V), \quad T \in C^\infty(W).$$

Again it is not hard to verify that L_X is well defined. Since we have already defined L_X on sections of exterior powers of $\mathcal{T}M$ and $\mathcal{T}'M$ the above construction allows us to extend L_X to arbitrary finite tensor products of tensor and exterior powers of $\mathcal{T}M$ and $\mathcal{T}'M$.

A most important property of Lie derivatives is that they commute with contractions. Indeed this property is built into the definition of Lie differentiation on differential 1-forms for we have

$$L_X \mathbf{C}(\phi \otimes Y) = \mathbf{C}(L_X \phi \otimes Y) + \mathbf{C}(\phi \otimes L_X Y),$$

$\phi \in C^\infty(\mathcal{T}'M)$, $X, Y \in C^\infty(\mathcal{T}M)$. We leave it to the reader to verify that Lie derivatives commute with any of the contraction operations we have so far defined on tensor products of $\mathcal{T}M$ and $\mathcal{T}'M$.

Exercises.

1. Let $X \in C^\infty(\mathcal{T}M)$, $\phi \in C^\infty(\Lambda^p \mathcal{T}'M)$, $f \in C^\infty(M)$. Verify the identities
 - i) $L_X \phi = d\mathbf{C}_X \phi + \mathbf{C}_X d\phi$
 - ii) $L_{fX} \phi = f L_X \phi + df \wedge \mathbf{C}_X \phi$.
2. Let $A \in C^\infty(\mathcal{T}'M \otimes \mathcal{T}M)$. Define $\tilde{A} : C^\infty(\mathcal{T}M) \rightarrow C^\infty(\mathcal{T}M)$ by $A(X) = \mathbf{C}_2^1(X \otimes A)$. Show that we can extend $\tilde{A}$, as a derivation, to sections of any finite tensor product of tensor and exterior powers of $\mathcal{T}M$ and $\mathcal{T}'M$ in such a way that $\tilde{A}$ is the zero operator on $C^\infty(M)$ and commutes with contractions. Show further that if Ω is any operator defined on sections of tensor products of $\mathcal{T}M$ and $\mathcal{T}'M$ that is a derivation and commutes with contractions then there exist unique $X \in C^\infty(\mathcal{T}M)$ and $A \in C^\infty(\mathcal{T}'M \otimes \mathcal{T}M)$ such that $\Omega = L_X + \tilde{A}$.
3. Verify the following local forms for the Lie derivative and Lie bracket

$$\begin{aligned} \text{a) } L_X f &= \sum_{i=1}^n X^i \frac{\partial f}{\partial x_i}. \\ \text{b) } [X, Y]^i &= \sum_{j=1}^n \left(X_j \frac{\partial Y_i}{\partial x_j} - Y_j \frac{\partial X_i}{\partial x_j} \right). \end{aligned}$$

3 Complexification

Throughout this section tensor products over $\mathbb{R}$ and $\mathbb{C}$ will be denoted by $\otimes_{\mathbb{R}}$ and $\otimes_{\mathbb{C}}$ respectively. We use similar notation for exterior and symmetric powers. Omission of a subscript will generally indicate a product over $\mathbb{C}$ unless the contrary is clearly indicated.

Definition 5.3.1. Let E be a real vector space. The *complexification* ${}_cE$ of E is the vector space $E \otimes_{\mathbb{R}} \mathbb{C}$.

Properties of complexification.

1. ${}_cE$ has the natural structure of a complex vector space with scalar multiplication defined by $c(e \otimes_{\mathbb{R}} d) = e \otimes_{\mathbb{R}} cd$, $e \in E$, $c, d \in \mathbb{C}$.
2. $\dim_{\mathbb{C}}({}_cE) = \dim_{\mathbb{R}} E$.
3. ${}_cE$ has a natural splitting $E_R \oplus E_I$ into real and imaginary parts defined by $E_R = \{e \otimes_{\mathbb{R}} 1 : e \in E\}$; $E_I = \{e \otimes_{\mathbb{R}} i : e \in E\}$.
4. The operation of complexification commutes with tensor, exterior and symmetric products. For example, if $p \geq 0$, ${}_c(\otimes_{\mathbb{R}}^p E) \approx \otimes^p({}_cE)$ (as complex vector spaces).
5. ${}_c(E') \approx L_{\mathbb{R}}(E, \mathbb{C})$. This isomorphism is defined by mapping $\phi \otimes_{\mathbb{R}} c$ to $c\phi$. In the sequel we set ${}_c(E') = {}_cE'$.
6. The dual pairing $E \times E' \rightarrow \mathbb{R}$ complexifies to the dual pairing ${}_cE \times {}_cE' \rightarrow \mathbb{C}$. On generators, this pairing is defined by $\langle e \otimes_{\mathbb{R}} c, \phi \otimes_{\mathbb{R}} d \rangle = cd \langle \phi, e \rangle$, $c, d \in \mathbb{C}$, $\phi \in E'$, $e \in E$. Using this dual pairing we identify the dual spaces of $\otimes^p {}_cE$, $\Lambda^p {}_cE$ with $\otimes^p {}_cE'$, $\Lambda^p {}_cE'$ respectively. We should stress that we shall *always* use this dual pairing in the sequel.

One of the main reasons that we introduce complexification is so that we can define conjugation.

Definition 5.3.2. The map $S : {}_cE \rightarrow {}_cE$ defined by $S(e \otimes_{\mathbb{R}} c) = e \otimes_{\mathbb{R}} \bar{c}$, $e \in E$, $c \in \mathbb{C}$, is called *conjugation*. We usually write $S(X) = \bar{X}$, $X \in {}_cE$.

Properties of conjugation.

1. $Si = -iS$ (conjugation is a conjugate complex linear map).
2. Let $X \in {}_cE$. Then $X = \bar{X}$ iff $X \in E_R$; $X = -\bar{X}$ iff $X \in E_I$.
3. Conjugation commutes with the operations of tensor, exterior and symmetric products. For example, conjugation on $\Lambda^p {}_cE$ may be defined as $\Lambda^p S$ or, equivalently, as conjugation on ${}_c(\Lambda_{\mathbb{R}}^p E)$. In particular, notice that if $X_1 \wedge \cdots \wedge X_p \in \Lambda^p {}_cE$, then $\bar{X}_1 \wedge \cdots \wedge \bar{X}_p = \overline{X_1 \wedge \cdots \wedge X_p} \in \Lambda^p {}_cE$.
4. Using the natural isomorphism between ${}_cE'$ and $L_{\mathbb{R}}(E, \mathbb{C})$, we may regard conjugation as conjugation of functions. That is, if $f \in {}_cE'$, we define $\bar{f} \in {}_cE'$ by $\bar{f}(e) = \overline{f(e)}$, $e \in E$.

5. Conjugation commutes with the dual pairing ${}_cE \times {}_cE' \rightarrow \mathbb{C}$ and so for all $X \in {}_cE$, $\phi \in {}_cE'$ we have

$$\overline{\langle X, \phi \rangle} = \langle \bar{X}, \bar{\phi} \rangle.$$

6. The conjugate of a map $A \in L_{\mathbb{C}}({}_cE, {}_cF)$ is given by $\bar{A} = S.A.S$.

Finally, we remark that if ϕ is an element of some finite tensor product of tensor and exterior powers of ${}_cE$, ${}_cE'$ then ϕ is said to be *real* if $\phi = \bar{\phi}$. By property 2 above this is clearly equivalent to the existence of a real tensor γ such that $\phi = \gamma \otimes_{\mathbb{R}} 1$.

Exercises.

1. Let $X \in {}_cE$. Show that $X + \bar{X}$ is real, $X - \bar{X}$ is imaginary (that is, a point of E_I).
2. Let $A \in L_{\mathbb{R}}(E, F)$ and ${}_cA = A \otimes_{\mathbb{R}} 1 \in L_{\mathbb{C}}({}_cE, {}_cF)$ denote the complexification of A . Show that $\det_{\mathbb{C}}({}_cA) = \det_{\mathbb{R}}(A)$.
3. Let $A \in L_{\mathbb{C}}({}_cE, {}_cF)$. Show that $\det_{\mathbb{C}}(\bar{A}) = \overline{\det_{\mathbb{C}}(A)}$.

4 Complex linear algebra

This section, which may be regarded as a synthesis of §§1,3, summarizes the main results of complex linear algebra that we need in the sequel. Again we defer any consideration of inner product structures.

Suppose that E is a complex vector space. We let E^* denote the complex dual space $L_{\mathbb{C}}(E, \mathbb{C})$ of E . The theory of contractions that we described in §1 immediately extends to (complex) tensor products of tensor, exterior and symmetric powers of E and E^* . We continue to use the notation developed in §1.

Definition 5.4.1. (See also §5, Chapter 1). Let E be a real vector space. An endomorphism J of E is said to define a *complex structure* on E if $J^2 = -I$.

If E has complex structure J then E may be given the structure of a complex vector space if we define $(a+ib)e = a+eJ(e)$, $a, b \in \mathbb{R}$, $e \in E$. In particular, $\dim_{\mathbb{R}} E$ must be even. Conversely, if E is a complex vector space we may define a complex structure J on E by $J(e) = ie$, $e \in E$.

Definition 5.4.2. If E is a vector space with complex structure J , we define $\bar{E}$, the *conjugate* of E , to be the vector space E with complex structure $-J$.

Remark. Suppose E is a complex vector space and that $X \in E$ has coordinates $(z_1, \dots, z_n)$ relative to some (complex) basis B of E . Then B is also a basis of $\bar{E}$ and the coordinates of $X \in \bar{E}$ are $(\bar{z}_1, \dots, \bar{z}_n)$. In this sense, taking the conjugate of a complex vector space corresponds to taking conjugates of the coordinates. Shortly, we shall use the device of complexification to give a “coordinate-free” version of this important operation. Indeed, much of the formalism we now develop is directed towards obtaining a satisfactory definition of conjugation for complex vector spaces that does not depend on a choice of coordinate system and can be extended to complex manifolds.

Properties of complex structures and conjugate spaces.

1. Let E, F be vector spaces with complex structures J_E, J_F respectively. A map $A \in L_{\mathbb{R}}(E, F)$ is complex linear iff $A.J_E = J_F.A$. The space $L_{\mathbb{C}}(E, F)$ has the natural complex structure J defined by $J(A) = A.J_E = J_F.A$. We may define two *distinct* complex structures J_1, J_2 on $L_{\mathbb{R}}(E, F)$ by $J_1(A) = A.J_E$; $J_2(A) = J_F.A$.

From now on assume that E is a vector space with complex structure J .

2. The complex structure on E^* is defined by $J(\phi) = \phi.J = i\phi$, $\phi \in E^*$.
3. $\bar{E}^* \approx \overline{E^*}$ as complex vector spaces. The isomorphism is defined by mapping $\phi \in \bar{E}^*$ ($= L(\bar{E}, \mathbb{C})$) to $\bar{\phi} \in E^*$, where $\bar{\phi}(e) = \overline{\phi(e)}$, $e \in E$. Notice that the complex structure on $\bar{E}^*$ is defined by $J(\phi) = \phi.(-J) = i\phi$. In particular, $\bar{E}^*$ is the space of conjugate complex linear maps on E .
4. The operation of taking the conjugate space commutes with the operations of taking dual, tensor, exterior and symmetric powers. For example $\overline{\otimes^p E} \approx \otimes^p \bar{E}$ as complex vector spaces.

As we described in §3, we may give ${}_cE$ the structure of a complex vector space. Clearly the complexification of J , which we continue to denote by J , also defines a complex structure on ${}_cE$. As we shall soon see these two complex structures on ${}_cE$ are different. In the sequel, we always give ${}_cE$ the complex vector space structure induced from $\mathbb{C}$ and never that induced from J .

Define $P, \bar{P} : {}_cE \rightarrow {}_cE$ by

$$P = \frac{1}{2}(I - iJ), \quad \bar{P} = \frac{1}{2}(I + iJ).$$

Clearly, $P, \bar{P}$ are complementary projections: $P^2 = P$, $\bar{P}^2 = \bar{P}$ and $P + \bar{P} = I$. We set $\mathcal{E} = P({}_cE)$, $\bar{\mathcal{E}} = \bar{P}({}_cE)$. Then $\mathcal{E}, \bar{\mathcal{E}}$ are complementary complex subspaces of ${}_cE$ and so ${}_cE = \mathcal{E} \oplus \bar{\mathcal{E}}$. Observe that $J|_{\mathcal{E}} = +i$, $J|_{\bar{\mathcal{E}}} = -i$ and $S(\mathcal{E}) = \bar{\mathcal{E}}$ (hence the notation). We have natural isomorphisms $j : E \rightarrow \mathcal{E}$; $\bar{j} : \bar{E} \rightarrow \bar{\mathcal{E}}$ defined by

$$j(e) = \frac{1}{2}(e \otimes 1 - Je \otimes i); \quad \bar{j}(e) = \frac{1}{2}(e \otimes 1 + Je \otimes i).$$

Hence we see that ${}_cE \approx E \oplus \bar{E}$. Notice that if we regard this isomorphism as an identification then $E, \bar{E}$ become complementary complex subspaces of ${}_cE$ in such a way that the complex structure on ${}_cE$ restricts to the complex structure J on E and the conjugate complex structure $-J$ on $\bar{E}$.

Let us now examine how conjugation fits into this framework. We have the commutative diagram

$$\begin{array}{ccc} E & \xrightarrow{j} & \mathcal{E} \oplus \bar{\mathcal{E}} = {}_cE \\ I \downarrow & & \downarrow S \\ \bar{E} & \xrightarrow{\bar{j}} & \mathcal{E} \oplus \bar{\mathcal{E}} = {}_cE. \end{array}$$

We see that the identity map $I : E \rightarrow \bar{E}$, which is associated to taking conjugates of coordinates, corresponds to the invariantly defined operation of conjugation on ${}_cE$.

Properties of complexified linear maps.

1. Suppose E, F are complex vector spaces and $A \in L_{\mathbb{R}}(E, F)$. Let ${}_cA = A \otimes_{\mathbb{R}} 1 \in L_{\mathbb{C}}({}_cE, {}_cF)$ denote the complexification of A . We say that ${}_cA$ respects the splittings $E \oplus \bar{E}, F \oplus \bar{F}$ of ${}_cE, {}_cF$ (or just *splits*) if ${}_cA = A_1 \oplus A_2$, where $A_1 : E \rightarrow F$, $A_2 : \bar{E} \rightarrow \bar{F}$. We have the useful result that ${}_cA$ splits iff $A \in L_{\mathbb{C}}(E, F)$. Moreover, if ${}_cA$ splits, $A_2 = \bar{A}_1$. In the sequel, if $A \in L_{\mathbb{C}}(E, F)$ we write ${}_cA = A \oplus \bar{A}$ and observe that the following diagrams commute

$$\begin{array}{ccc} E & \xrightarrow{j} & \mathcal{E} \oplus \bar{\mathcal{E}} \\ A \downarrow & & \downarrow {}_cA = A \oplus \bar{A} \\ F & \xrightarrow{j} & \mathcal{F} \oplus \bar{\mathcal{F}} \end{array} \quad \begin{array}{ccc} \bar{E} & \xrightarrow{\bar{j}} & \mathcal{E} \oplus \bar{\mathcal{E}} \\ \bar{A} \downarrow & & \downarrow {}_cA = A \oplus \bar{A} \\ \bar{F} & \xrightarrow{\bar{j}} & \mathcal{F} \oplus \bar{\mathcal{F}}. \end{array}$$

The map $\bar{A} : \bar{E} \rightarrow \bar{F}$ is defined to be equal to A on the underlying real vector spaces of $\bar{E}$ and $\bar{F}$.

2. If $A \in L_{\mathbb{R}}(E, F)$, then $\det_{\mathbb{C}}({}_c A) = \det_{\mathbb{R}}(A)$ (Exercise 2, §3). Hence if $A \in L_{\mathbb{C}}(E, F)$,

$$\begin{aligned} \det_{\mathbb{R}}(A) &= \det_{\mathbb{C}}(A \oplus \bar{A}) = \det_{\mathbb{C}} A \det_{\mathbb{C}} \bar{A} \\ &= \det_{\mathbb{C}} A \overline{\det_{\mathbb{C}} A} = |\det_{\mathbb{C}} A|^2 = |\det_{\mathbb{C}} A|^2, \end{aligned}$$

since $A = jAj^{-1}$. In particular, the real determinant of a complex linear map is always positive (this implies, for example, the orientability of complex manifolds).

Next we turn to dual spaces. Recall from §3 that ${}_c E'$ may be identified with $L_{\mathbb{R}}(E, \mathbb{C})$. As described above we have complementary projection maps $P, \bar{P} : {}_c E' \rightarrow {}_c E'$ defined by

$$P(\phi) = \frac{1}{2}(\phi - i\phi.J), \quad \bar{P}(\phi) = \frac{1}{2}(\phi + i\phi.J), \quad \phi \in {}_c E'.$$

Observe that for all $\phi \in {}_c E'$, $P(\phi) \in E^*$, $\bar{P}(\phi) \in \bar{E}^*$. Hence we have the splitting

$${}_c E' = E^* \oplus \bar{E}^*.$$

This splitting amounts to saying that every complex valued $\mathbb{R}$ -linear form on E has a unique decomposition as a sum of a complex linear and conjugate complex linear form.

Let us see how conjugation fits into this picture. The conjugation map $S : E^* \rightarrow \bar{E}^*$, defined by taking conjugates of linear forms, is the restriction to E^* of the conjugate operator $S : {}_c E' \rightarrow {}_c E'$ defined in §3. Hence we have the commutative diagram

$$\begin{array}{ccccc} & E^* & \xrightarrow{j} & E^* \oplus \bar{E}^* = {}_c E' & \\ & \swarrow I & & \downarrow S & \\ \bar{E}^* & & & & \\ & \searrow \approx & & \downarrow S & \\ & \bar{E}^* & \xrightarrow{\bar{j}} & E^* \oplus \bar{E}^* = {}_c E' & \end{array}$$

The maps $j, \bar{j}$ are just inclusion maps. Notice that the conjugation map $S : E^* \rightarrow \bar{E}^*$ factors through $\bar{E}^*$ and compare with the previous diagram that we gave for conjugation on ${}_c E$.

As described in §3, the dual pairing $E \times E' \rightarrow \mathbb{R}$ complexifies to the dual pairing $\langle , \rangle : {}_c E \times {}_c E' \rightarrow \mathbb{C}$. We now investigate how this pairing behaves with respect to the factors $\mathcal{E}, \bar{\mathcal{E}}$ of ${}_c E$ and $E^*, \bar{E}^*$ of ${}_c E'$.

Proposition 5.4.3. The dual pairing $\langle , \rangle : {}_c E \times {}_c E' \rightarrow \mathbb{C}$ induces

1. The zero pairing between $\mathcal{E}$ and $\bar{E}^*$.
1. The zero pairing between $\bar{\mathcal{E}}$ and E^* .
2. A dual pairing of $\mathcal{E}$ and E^* .
2. A dual pairing of $\bar{\mathcal{E}}$ and $\bar{E}^*$.

Moreover, the dual pairings between $\mathcal{E}, E^*$ and $\bar{\mathcal{E}}, \bar{E}^*$ are given explicitly by

$$\langle j(e), \phi \rangle = \phi(e), \quad e \in E, \quad \phi \in E^*$$

$$\langle \bar{j}(e), \psi \rangle = \psi(e), \quad e \in E, \quad \psi \in \bar{E}^*.$$

Proof. It is enough to verify statements 1, 2 as $\bar{1}$, $\bar{2}$ follow by conjugation. Suppose $X \in \mathcal{E}$, $\phi \in \bar{E}^*$. Then there exist $e \in E$, $\zeta \in E'$ such that $X = j(e) = \frac{1}{2}(e \otimes 1 - Je \otimes i)$, $\phi = \bar{j}(e) = \frac{1}{2}(\zeta \otimes 1 + \zeta.J \otimes i)$. Now $\langle \phi, X \rangle = \frac{1}{4}(\langle \zeta, e \rangle - i^2 \langle \zeta, J^2 e \rangle + i \langle \zeta, Je \rangle - i \langle \zeta, Je \rangle) = 0$, proving 1. If instead $\phi \in E^*$, there exists $\zeta \in E'$ such that $\phi = \frac{1}{2}(\zeta \otimes 1 - \zeta.J \otimes i)$ and computing we find that

$$\langle \phi, X \rangle = \frac{1}{2}(\zeta(e) - i\zeta(Je)) = \phi(e).$$

□

We now use Proposition 5.4.3 to examine how complex linear maps and their duals behave under complexification and conjugation. If $A : E \rightarrow F$ is $\mathbb{C}$ -linear, we have induced $\mathbb{C}$ -linear maps $\bar{A} : \bar{E} \rightarrow \bar{F}$, $A^* : F^* \rightarrow E^*$, $\bar{A}^* : \bar{F}^* \rightarrow \bar{E}^*$ defined by

1. $\bar{A}(e) = A(e)$, $e \in \bar{E}$ (where, as real vector spaces, $\bar{E} = E$).
2. $\langle A^*(f^*), e \rangle = f^*(A(e))$, $e \in E$, $f^* \in F^*$.
3. $\langle \bar{A}^*(\bar{f}^*), e \rangle = \bar{f}^*(A(e))$, $e \in E$, $\bar{f}^* \in \bar{F}^*$.

Since A is assumed $\mathbb{C}$ -linear, ${}_c A = \mathcal{A} \oplus \bar{\mathcal{A}}$, ${}_c A' = A^* \oplus \bar{A}^*$ and we have, by Proposition 5.4.3, the following alternative characterizations of $\bar{A}$, A^* , $\bar{A}^*$.

- 1'. $\bar{A} = \bar{j}^{-1} \bar{\mathcal{A}} \bar{j}$.
- 2'. $\langle A^*(f^*), e \rangle = \langle \mathcal{A}(e), f^* \rangle$, $e \in \mathcal{E}$, $f^* \in F^*$.
- 3'. $\langle \bar{A}^*(\bar{f}^*), \bar{e} \rangle = \langle \bar{\mathcal{A}}(\bar{e}), \bar{f}^* \rangle$, $\bar{e} \in \bar{\mathcal{E}}$, $\bar{f}^* \in \bar{F}^*$.

The pairings here are all induced from the dual pairing of E and E' . Observe how the definitions are now much more natural. In particular, 3' is just the conjugate of 2'. This naturality, that allows us to commute conjugation with other operations such as contraction or tensor product, is one of the main advantages of working with the complexifications of E and E' .

Next we look at the exterior algebras of ${}_c E$ and ${}_c E'$. By Theorem 5.1.5 we have the canonical isomorphisms

$$\begin{aligned} \mu : \Lambda^p {}_c E &\longrightarrow \bigoplus_{r+s=p} \Lambda^r \mathcal{E} \otimes \Lambda^s \bar{\mathcal{E}} \\ \mu : \Lambda^p {}_c E' &\longrightarrow \bigoplus_{r+s=p} \Lambda^r E^* \otimes \Lambda^s \bar{E}^*. \end{aligned}$$

We set $\Lambda^{r,s}(E) = \mu^{-1}(\Lambda^r \mathcal{E} \otimes \Lambda^s \bar{\mathcal{E}})$; $\Lambda^{r,s}(E') = \mu^{-1}(\Lambda^r E^* \otimes \Lambda^s \bar{E}^*)$, $r, s \geq 0$. For $p \geq 0$ we therefore have the direct sum decomposition

$$\Lambda^p {}_c E = \bigoplus_{r+s=p} \Lambda^{r,s}(E); \quad \Lambda^p {}_c E' = \bigoplus_{r+s=p} \Lambda^{r,s}(E').$$

An element of $\Lambda^{r,s}(E)$ (resp. $\Lambda^{r,s}(E')$) is called a *complex* (r, s) -*vector* (resp. a *complex* (r, s) -*form*).

Properties of the exterior algebras of ${}_c E$ and ${}_c E'$. For properties 1 and 2 below we suppose $\dim_{\mathbb{C}}(E) = m$.

1. $\Lambda^{r,s}(E) = 0$ if either $r > m$ or $s > m$. Similarly for forms.
2. $\Lambda^{2m} {}_c E = \Lambda^{m,m}(E)$. Similarly for forms.
3. $\overline{\Lambda^{r,s}(E)} = \Lambda^{s,r}(E)$, $r, s \geq 0$. A necessary condition for a complex (r, s) -vector to be real is $r = s$. Similarly for forms.

4. If $X \in \Lambda^{r,s}(E)$, $Y \in \Lambda^{u,v}(E)$ then $X \wedge Y \in \Lambda^{r+u,s+v}(E)$, $r, s, u, v \geq 0$. Similarly for forms.
5. The dual pairing $\Lambda^p_c E \times \Lambda^p_c E' \rightarrow \mathbb{C}$ restricts to a dual pairing $\Lambda^{r,s}(E) \times \Lambda^{r,s}(E') \rightarrow \mathbb{C}$ which is given on generators by

$$\begin{aligned} & \langle X_1 \wedge \cdots \wedge X_r \wedge Y_{\bar{1}} \wedge \cdots \wedge Y_{\bar{s}}, \phi_1 \wedge \cdots \wedge \phi_r \wedge \psi_{\bar{1}} \wedge \cdots \wedge \psi_{\bar{s}} \rangle \\ &= \langle X_1 \wedge \cdots \wedge X_r, \phi_1 \wedge \cdots \wedge \phi_r \rangle \langle Y_{\bar{1}} \wedge \cdots \wedge Y_{\bar{s}}, \psi_{\bar{1}} \wedge \cdots \wedge \psi_{\bar{s}} \rangle, \end{aligned}$$

where $X_j \in \mathcal{E}$, $Y_{\bar{i}} \in \bar{\mathcal{E}}$, $\phi_j \in E^*$, $\psi_{\bar{i}} \in \bar{E}^*$, $1 \leq j \leq r$, $1 \leq i \leq s$. If $r + s = u + v$ but $r \neq u$, then the induced pairing $\Lambda^{r,s}(E) \times \Lambda^{u,v}(E') \rightarrow \mathbb{C}$ is always zero (both properties follow from Proposition 5.4.3).

6. If $A \in L_{\mathbb{C}}(E, F)$, then $\Lambda^p_c A : \Lambda^p_c E \rightarrow \Lambda^p_c F$ induces maps $\Lambda^{r,s}(A) : \Lambda^{r,s}(E) \rightarrow \Lambda^{r,s}(F)$, $r + s = p$. Similarly for forms.
7. If $r \geq u$, $s \geq v$, we have the contraction operation

$$C : \Lambda^{r,s}(E) \otimes \Lambda^{u,v}(E') \longrightarrow \Lambda^{r-u,s-v}(E)$$

obtained as the restriction of the contraction between $\Lambda^{r+s}_c E$ and $\Lambda^{u+v}_c E$. Similar remarks hold for the other contraction operations defined in §1.

We conclude this section by looking at bases for the spaces we have been considering.

Suppose $B = \{e_1, \dots, e_m\}$ is a complex basis for E . Then B is a complex basis for $\bar{E}$ and $B_R = \{e_1, Je_1, \dots, e_m, Je_m\}$ is a real basis for E . The dual real basis B'_R for E' is given by

$$\begin{aligned} B'_R &= \{e'_1, (Je_1)', \dots, e'_m, (Je_m)'\} \\ &= \{e'_1, -Je'_1, \dots, e'_m, -Je'_m\}, \quad \text{since } (Je_j)' = -Je'_j. \end{aligned}$$

We now define bases $\mathcal{B}$, $\bar{\mathcal{B}}$, B^* , $\bar{B}^*$ for $\mathcal{E}$, $\bar{\mathcal{E}}$, E^* , $\bar{E}^*$ respectively. Set

$$\begin{aligned} \mathcal{B} &= \{f_j \in \mathcal{E} : f_j = j(e_j), 1 \leq j \leq m\}. \\ \bar{\mathcal{B}} &= \{f_{\bar{j}} \in \bar{\mathcal{E}} : f_{\bar{j}} = \bar{j}(e_j), 1 \leq j \leq m\} \\ B^* &= \{f_j^* \in E^* : f_j^* = e'_j - i(Je_j)' = e'_j + iJe'_j, 1 \leq j \leq m\} \\ \bar{B}^* &= \{f_{\bar{j}}^* \in \bar{E}^* : f_{\bar{j}}^* = e'_j + i(Je_j)' = e'_j - iJe'_j, 1 \leq j \leq m\}. \end{aligned}$$

Notice that $\bar{f}_{\bar{j}} = f_{\bar{j}}$, $\overline{f_j^*} = f_{\bar{j}}^*$, $1 \leq j \leq m$. Hence our notation for the bases $\mathcal{B}$, $\bar{\mathcal{B}}$ and B^* , $\bar{B}^*$ (such pairs of bases are called *self-conjugate*). We see also that $\mathcal{B}$ and B^* are dual bases (for the pairing $\langle, \rangle$) since

$$\begin{aligned} \langle f_j, f_j^* \rangle &= \frac{1}{2} \langle e_j - iJe_j, e'_j - i(Je_j)' \rangle = \frac{1}{2} + \frac{1}{2} = 1 \\ \langle f_j, f_k^* \rangle &= 0, \quad j \neq k. \end{aligned}$$

Similarly, $\bar{\mathcal{B}}$ and $\bar{B}^*$ are dual bases.

With respect to the self-conjugate basis that we have constructed on $_c E$, every $X \in _c E$ may be written uniquely in the form

$$X = \sum_{j=1}^m z^j f_j + \sum_{j=1}^m \bar{z}^j f_{\bar{j}}$$

where $z^j, \bar{z}^j \in \mathbb{C}$. The coordinates $(z^1, \dots, z^m, \bar{z}^1, \dots, \bar{z}^m)$ are called *self-conjugate* coordinates on $_c E$.

Suppose that F is another complex vector space with complex basis C and associated bases $\mathcal{C}, \bar{\mathcal{C}}, C^*, \bar{C}^*$ as described above for the basis B of E . Let $A \in L_{\mathbb{C}}(E, F)$ have matrix $[a_{ij}]$ with respect to the bases B and C . Then

$$[\bar{A}] = [\bar{a}_{ij}]; \quad [A^*] = [a_{ji}]; \quad [\bar{A}^*] = [\bar{a}_{ji}]; \quad [\mathcal{A}] = [a_{ij}]; \quad [\bar{\mathcal{A}}] = [\bar{a}_{ij}],$$

where the matrices are computed relative to the appropriate bases associated to B and C .

Finally suppose that $X \in \Lambda^{r,s}(E)$, $\phi \in \Lambda^{r,s}(E')$. In coordinates we may write X and ϕ uniquely in the form

$$X = \sum_{I,J} X^{I\bar{J}} f_I f_{\bar{J}}; \quad \phi = \sum_{I,J} \phi_{I\bar{J}} f_I^* f_{\bar{J}}^*,$$

where the summations are over all r -tuples $I = (i_1, \dots, i_r)$ satisfying $1 \leq i_1 < \dots < i_r \leq m$ and s -tuples $J = (j_1, \dots, j_s)$ satisfying $1 \leq j_1 < \dots < j_s \leq m$ and we have used the abbreviated notations $f_I f_{\bar{J}}$ and $f_I^* f_{\bar{J}}^*$ for $f_{i_1} \wedge \dots \wedge f_{i_r} \wedge f_{\bar{j}_1} \wedge \dots \wedge f_{\bar{j}_s}$ and $f_{i_1}^* \wedge \dots \wedge f_{i_r}^* \wedge f_{\bar{j}_1}^* \wedge \dots \wedge f_{\bar{j}_s}^*$ respectively.

Notice that we use subscripts for coordinates of forms and superscripts for coordinates of vectors.

Examples.

1. Suppose X and ϕ are as above. Then

$$\bar{\phi} = \sum_{I,J} \overline{\phi_{I\bar{J}}} \bar{f}_I^* \bar{f}_{\bar{J}}^* = \sum_{I,J} \overline{\phi_{I\bar{J}}} f_I^* f_J^* = (-1)^{rs} \sum_{I,J} \overline{\phi_{I\bar{J}}} f_J^* f_I^*.$$

A similar formula holds for $\bar{X}$.

2. Same assumptions on X and ϕ . We have

$$\langle X, \phi \rangle = \sum_{I,J} X^{I\bar{J}} \phi_{I\bar{J}}.$$

Exercises.

1. Show that ${}_c E^* (= ({}_c E)^*)$ is naturally isomorphic to ${}_c E'$ and deduce that ${}_c E' \approx E^* \oplus \bar{E}^*$.
2. Let E be a real vector space, F a complex vector space. Show that there is a natural operation of conjugation $S : {}_c E \otimes_{\mathbb{C}} F \rightarrow {}_c E \otimes_{\mathbb{C}} \bar{F}$ that is the identity if $E = \mathbb{R}$ and conjugation on ${}_c E$ if $F = \mathbb{C}$.
3. Show that the set of complex structures on $\mathbb{R}^{2m}$ is in bijective correspondence with $\mathrm{GL}(2m, \mathbb{R}) / \mathrm{GL}(m, \mathbb{C})$ (Here we take the standard complex structure on $\mathbb{R}^{2m}$ and regard $\mathrm{GL}(m, \mathbb{C})$ as a subgroup of $\mathrm{GL}(2m, \mathbb{R})$).

5 Generalities on complex vector bundles.

In this section we collect together a number of definitions and elementary facts about complex and holomorphic vector bundles. The reader should be familiar with §5, Chapter 1.

Complexification. Let $E \xrightarrow{p} M$ be a (smooth) real vector bundle over the differential manifold M . The *complexification* ${}_cE$ of E is defined to be the complex vector bundle $E \otimes_{\mathbb{R}} \mathbb{C}$ over M . Notice that if E has transition functions $\phi_{ij} : U_{ij} \rightarrow \mathrm{GL}(m, \mathbb{R})$ then ${}_cE$ has transition functions ${}_c\phi_{ij} : U_{ij} \rightarrow \mathrm{GL}(m, \mathbb{C})$, where ${}_c\phi_{ij}(x) = \phi_{ij}(x) \otimes_{\mathbb{R}} 1$, $x \in U_{ij}$. We remark that all the results on complexification described in §3 extend immediately to vector bundles and their sections. In particular, we have a conjugation operator S defined on ${}_cE$ and $C^\infty({}_cE)$ and this operator commutes with tensor product operations and duals in the manner outlined in §3.

Complex structures. Let $E \xrightarrow{p} M$ be a (smooth) real vector bundle over the differential manifold M . A *complex structure* J on E is a vector bundle morphism $J : E \rightarrow E$ satisfying $J^2 = -I$. Equivalently, a complex structure on E is a C^∞ section J of the vector bundle $L(E, E)$ over M such that $J(x)^2 = -I|_{E_x}$ for all $x \in M$ (see exercise 6, §5, Chapter 1). If E is a complex vector bundle over M then E has a complex structure defined by scalar multiplication by i in the fibres of E . Conversely, it is not hard to show that if E has a complex structure then E has the structure of a complex vector bundle (the proof uses Exercise 3, §4, together with the fact that the quotient map $\mathrm{GL}(2m, \mathbb{R}) \rightarrow \mathrm{GL}(2m, \mathbb{R}) / \mathrm{GL}(m, \mathbb{C})$ admits local sections).

Definition 5.5.1. Let M be a differential manifold. A complex structure J on $\mathcal{T}M$ is called an *almost complex structure* on M . We refer to M as an *almost complex manifold*.

Example 1. Let M be a complex manifold with atlas $\{(U_i, \phi_i) : i \in I\}$. The transition functions for the tangent bundle $\mathcal{T}M$ of M are given by $\phi_{ij} = D(\phi_i \phi_j^{-1})\phi_j$. Since $\phi_i \phi_j^{-1}$ is biholomorphic, $\phi_{ij} : U_{ij} \rightarrow \mathrm{GL}(m, \mathbb{C})$ for all $i, j \in I$. Hence $\mathcal{T}M$ has the structure of a complex vector bundle and so M has the structure of an almost complex manifold.

Holomorphic and anti-holomorphic vector bundles. For the remainder of this section we shall suppose that M is a complex manifold.

Definition 5.5.2. An m -dimensional *holomorphic vector bundle* E over M consists of a complex manifold E and holomorphic map $p : E \rightarrow M$ together with a family $\phi_i : E|_{U_i} \rightarrow U_i \times \mathbb{C}^m$ of biholomorphic trivialisations of E .

Remarks.

1. A holomorphic vector bundle necessarily has the structure of a complex vector bundle.
2. An m -dimensional holomorphic vector bundle may equivalently be described by specifying a family $\phi_{ij} : U_{ij} \rightarrow \mathrm{GL}(m, \mathbb{C})$ of transition functions such that ϕ_{ij} is holomorphic for all i, j . It is clear from the transition function description of holomorphic vector bundles that if E is a holomorphic vector bundle then so is E^* and any finite tensor product of tensor, exterior and symmetric powers of E and E^* .
3. We denote the space of holomorphic sections of E by $\Omega(M, E)$ or just $\Omega(E)$, if M is implicit from the context.

Example 2. Let M be a complex manifold. Then $\mathcal{T}M$ has the structure of a holomorphic vector bundle.

Suppose that E is a holomorphic vector bundle with transition functions $\phi_{ij} : U_{ij} \rightarrow \mathrm{GL}(\mathbb{C}^m) (= \mathrm{GL}(m, \mathbb{C}))$. If E has complex structure J we let $\bar{E}$ denote the complex vector bundle over M with complex structure $-J$. The transition functions for $\bar{E}$ are given by

$$\bar{\phi}_{ij} : U_{ij} \rightarrow \mathrm{GL}(\overline{\mathbb{C}^m}),$$

where $\bar{\phi}_{ij}(x) = \phi_{ij}(x)$, $x \in U_{ij}$, as real linear maps of $\mathbb{C}^m$. Obviously the $\bar{\phi}_{ij}$ are no longer holomorphic maps. Instead they are *anti-holomorphic*. That is, in a local holomorphic coordinate system on M we have $\partial\bar{\phi}_{ij}/\partial z_k = 0$, $1 \leq k \leq \dim(M)$.

In the sequel we shall say that a complex vector bundle E over M is *anti-holomorphic* if the transition functions for E are anti-holomorphic or, equivalently, if $\bar{E}$ is a holomorphic vector bundle.

Remark. The dual of an anti-holomorphic vector bundle is anti-holomorphic as are finite tensor, exterior and symmetric products.

6 Tangent and cotangent bundles of a complex manifold.

In this section we show how the theory of §4 can be applied to the study of the tangent and cotangent bundles of a complex manifold. Our results provide the framework we need for the construction of the global Cauchy–Riemann operators on an arbitrary complex manifold that we carry out in §7.

We start this section by briefly indicating how the theory outlined in §2 for differential manifolds can be “complexified”.

Let M be a differential manifold with tangent bundle $\mathcal{T}M$. We call the complex vector bundles ${}_c\mathcal{T}M$ and ${}_c\mathcal{T}'M$ the *complex tangent* and *complex cotangent bundles* of M respectively. Sections of the bundle $\Lambda^p {}_c\mathcal{T}'M$ are called complex differential p -forms on M , $p \geq 0$. Exterior differentiation complexifies to give an operator on complex differential forms which we shall continue to denote by d . We note that this operator on complex differential forms obeys all the properties described in §3, or rather their complexified analogues, and in addition is a real operator:

$$\overline{d\phi} = d\bar{\phi}, \quad \phi \in C^\infty(\Lambda^p {}_c\mathcal{T}'M), \quad p \geq 0.$$

The Lie bracket complexifies to give a Lie bracket on $C^\infty({}_c\mathcal{T}M)$ defined by

$$[X_1 + iY_1, X_2 + iY_2] = [X_1, X_2] - [Y_1, Y_2] + i([Y_1, X_2] + [X_1, Y_2]),$$

where $X_1, X_2, Y_1, Y_2 \in C^\infty(\mathcal{T}M)$. Similarly the Lie derivative complexifies to give a derivation L_Z of the full tensor algebra of ${}_c\mathcal{T}M$ for all $Z \in C^\infty({}_c\mathcal{T}M)$. We remark that the Lie bracket and derivative that we have constructed are real operators, that is they commute with conjugation. For example, if $X, Y \in C^\infty({}_c\mathcal{T}M)$ we have $\overline{[X, Y]} = [\bar{X}, \bar{Y}]$.

Suppose now that M is an m -dimensional complex manifold with atlas $\{(U_i, \phi_i) : i \in I\}$. We let $\phi_{ij} = D(\phi_i \phi_j^{-1})\phi_j : U_{ij} \rightarrow \text{GL}(m, \mathbb{C})$ denote the transition functions for the tangent bundle $\mathcal{T}M$ of M and J denote the complex structure on $\mathcal{T}M$. Applying the theory of §4, we see that the transition function ${}_c\phi_{ij}$ of ${}_c\mathcal{T}M$ splits as a sum $\theta_{ij} \oplus \bar{\theta}_{ij}$, where $\theta_{ij} : U_{ij} \rightarrow \text{GL}(\mathbb{C}^m)$ and $\bar{\theta}_{ij} : U_{ij} \rightarrow \text{GL}(\overline{\mathbb{C}^m})$ (here we have identified ${}_c\mathbb{C}^m$ with $\mathbb{C}^m \oplus \overline{\mathbb{C}^m}$ using the maps $j, \bar{j}$ described in §4). Clearly θ_{ij} and $\bar{\theta}_{ij}$ are the transition functions for complex vector bundles on M which we shall denote by TM and $\overline{TM}$ respectively. By our construction we see that TM and $\overline{TM}$ are complementary complex subbundles of ${}_c\mathcal{T}M$ and so we have ${}_c\mathcal{T}M = TM \oplus \overline{TM}$. Moreover $J = +i$ on TM , $J = -i$ on $\overline{TM}$ and $S(TM) = \overline{TM}$ (Hence the “bar” notation).

We have the natural inclusion map $j : \mathbb{C}^m \rightarrow {}_c\mathbb{C}^m \approx \mathbb{C}^m \oplus \overline{\mathbb{C}^m}$ and projection $P : {}_c\mathbb{C}^m \rightarrow \mathbb{C}^m$. Since $\theta_{ij} = j\phi_{ij}P$ and j and P are complex linear the θ_{ij} are holomorphic and so TM has the structure of a holomorphic vector bundle. Indeed the map j induces a holomorphic vector bundle isomorphism between $\mathcal{T}M$ and TM . Similarly $\overline{TM}$ has the structure of an anti-holomorphic vector bundle. We call TM the *holomorphic tangent bundle* of M and $\overline{TM}$ the *anti-holomorphic tangent bundle* of M . We reserve these terms for the appropriate subbundles of ${}_c\mathcal{T}M$ and continue to refer to $\mathcal{T}M$ as the real tangent bundle of M even though it is isomorphic to TM .

Taking the standard basis of $\mathbb{C}^m$ we can easily compute the matrices of the transition functions θ_{ij} , $\bar{\theta}_{ij}$. To simplify notation, set $\phi = \phi_i \phi_j^{-1}$, $\theta = \theta_{ij}$ and let $(z_1, \dots, z_m)$ denote the coordinate system on U_{ij} given by the chart (U_j, ϕ_j) . We follow the basis notation given in §4. The q th. column of $[\theta(z)]$ is the vector

$$\begin{aligned} \theta(z)(f_q) &= \theta(z)(\tfrac{1}{2}(e_q \otimes 1 - J e_q \otimes i)) \\ &= \tfrac{1}{2}(c D \phi_z(e_q \otimes 1) - i c D \phi_z(J e_q \otimes 1)) \\ &= \tfrac{1}{2}(\partial \phi / \partial x_q - i \partial \phi / \partial y_q) = \partial \phi / \partial z_q. \end{aligned}$$

Hence $[\theta] = [\partial \phi_p / \partial z_q]$. Conjugating we have $[\bar{\theta}] = [\partial \bar{\phi}_p / \partial \bar{z}_q] = [\overline{\partial \phi_p / \partial z_q}]$. Notice that our expression for $[\theta]$ gives an alternative verification that TM has the structure of a holomorphic vector bundle.

Turning now to dual bundles we let TM^* and $\overline{TM}^*$ denote the complex dual and conjugate complex dual bundles of TM respectively. We have the direct sum decomposition ${}_c\mathcal{T}'M = TM^* \oplus \overline{TM}^*$ and, as above, TM^* is a holomorphic vector bundle, $\overline{TM}^*$ an antiholomorphic vector bundle. We call TM^* and $\overline{TM}^*$ the *holomorphic cotangent* and *antiholomorphic cotangent bundles* of M respectively. The transition functions ψ_{ij} for ${}_c\mathcal{T}'M$ are given by $\psi_{ij} = {}_c\phi'_{ji}$ and so the transition functions for TM^* and $\overline{TM}^*$ are given by θ_{ji}^* and $\bar{\theta}_{ji}^*$ respectively. In local coordinates, the matrices of θ_{ji}^* and $\bar{\theta}_{ji}^*$ are the transpose and conjugate transpose of the matrix $[\theta_{ji}]$ respectively.

Next we consider the exterior algebras of ${}_c\mathcal{T}M$ and ${}_c\mathcal{T}'M$. Working with transition functions we may construct for $r, s \geq 0$, $r + s = p$, subbundles $\Lambda^{r,s}(M)$ and $\Lambda^{r,s}(M)'$ of $\Lambda^p {}_c\mathcal{T}M$ and $\Lambda^p {}_c\mathcal{T}'M$ respectively such that

$$\Lambda^p {}_c\mathcal{T}M = \bigoplus_{r+s=p} \Lambda^{r,s}(M); \quad \Lambda^p {}_c\mathcal{T}'M = \bigoplus_{r+s=p} \Lambda^{r,s}(M)'.$$

We call $\Lambda^{r,s}(M)$ (resp. $\Lambda^{r,s}(M)'$) the bundle of (r, s) -vectors (resp. (r, s) -forms) on M . If $s = 0$, we see that $\Lambda^{r,0}(M) \approx \Lambda^r TM$ and $\Lambda^{r,0}(M)' \approx \Lambda^r TM^*$. In particular, these bundles are holomorphic vector bundles on M , $r \geq 0$.

Notation. For $r, s, p \geq 0$, we let $C^p(M)$ (resp. $C_p(M)$) denote the space of C^∞ sections of $\Lambda^p {}_c\mathcal{T}'M$ (resp. $\Lambda^p {}_c\mathcal{T}M$). (From now on we shall never need to refer to C^p functions on M unless $p = \infty$, when we write $C^\infty(M)$). We let $C^{r,s}(M)$ (resp. $C_{r,s}(M)$) denote the space of C^∞ sections of $\Lambda^{r,s}(M)'$ (resp. $\Lambda^{r,s}(M)$). We let $\Omega^p(M)$ (resp. $\Omega_p(M)$) denote the space of holomorphic sections of $\Lambda^{p,0}(M)'$ (resp. $\Lambda^{p,0}(M)$).

All the theory described in §4 extends immediately to the exterior algebras of ${}_c\mathcal{T}M$ and ${}_c\mathcal{T}'M$ and the corresponding spaces of sections. In particular for $r, s \geq 0$ we have a conjugation operator $S : \Lambda^{r,s}(M)' \rightarrow \Lambda^{s,r}(M)'$ and induced conjugation operator $S : C^{r,s}(M) \rightarrow C^{s,r}(M)$ (similarly for (r, s) -vectors). We usually write $S(\phi) = \bar{\phi}$, for ϕ a complex form or vector. Notice that a necessary condition for an (r, s) -form ϕ to be real— $\phi = \bar{\phi}$ —is that $r = s$.

Suppose that N is a complex manifold and $f : M \rightarrow N$ is a holomorphic map with tangent map $\mathcal{T}f : TM \rightarrow TN$. The complexification ${}_c\mathcal{T}f$ of $\mathcal{T}f$ splits as a sum $\mathcal{T}f \oplus \overline{\mathcal{T}f} : TM \oplus \overline{TM} \rightarrow TN \oplus \overline{TN}$. Similarly the complexification ${}_c\mathcal{T}'f$ of the cotangent map $\mathcal{T}'f : \mathcal{T}'M \rightarrow \mathcal{T}'N$ splits as a sum $\mathcal{T}'f^* \oplus \overline{\mathcal{T}'f}^* : TM^* \oplus \overline{TM}^* \rightarrow TN^* \oplus \overline{TN}^*$. Consequently, for $r, s \geq 0$, the tangent and cotangent maps of f induce vector bundle maps

$$\Lambda^{r,s}(f) : \Lambda^{r,s}(M) \rightarrow \Lambda^{r,s}(N) \quad \text{and} \quad \Lambda^{r,s}(f)' : \Lambda^{r,s}(M)' \rightarrow \Lambda^{r,s}(N)'.$$

We now give a local description of complex (r, s) -vectors and forms in terms of the self-conjugate bases given in §4.

Suppose $\phi \in C^{r,s}(M)$ and (V, ζ) is a chart on M . Then $\zeta_*\phi$ is an (r, s) -form on the open subset $\zeta(V)$ of $\mathbb{C}^m$. Set $U = \zeta(V)$ and $\eta = \zeta_*(\phi)$. As is conventional, we denote the

standard basis of $\mathbb{C}^{m'}$ by $\{dx_1, dy_1, \dots, dx_m, dy_m\}$. For $1 \leq j \leq m$ we set $dz_j = dx_j + i dy_j \in \mathbb{C}^{m*}$, $d\bar{z}_j = dx_j - i dy_j \in \overline{\mathbb{C}^{m*}}$. Then $\{dz_j, d\bar{z}_j : 1 \leq j \leq m\}$ is the self-conjugate basis of $\mathbb{C}^{m*} \oplus \overline{\mathbb{C}^{m*}}$ described in §4. Moreover, if we think of $dz_j, d\bar{z}_j$ as defining sections of $T\mathbb{C}^{m*}, \overline{T\mathbb{C}^{m*}}$ respectively, $\{dz_j : 1 \leq j \leq m\}$ and $\{d\bar{z}_j : 1 \leq j \leq m\}$ give bases for $C^\infty(TU^*)$ and $C^\infty(\overline{TU^*})$ over $C^\infty(U)$ respectively. Hence we may write $\eta \in C^{r,s}(U)$ uniquely in the form

$$\eta = \sum_{I,J} \eta_{I\bar{J}} dz_I d\bar{z}_J,$$

where $\eta_{I\bar{J}} \in C^\infty(U)$ and the summation over the r -tuples I and s -tuples J is as described in §4. Next we turn to the local form for complex (r,s) -vectors. Identifying (complex) vector fields with (complex) derivations, it is clear that if we set $\partial/\partial z_j = \frac{1}{2}(\partial/\partial x_j - i\partial/\partial y_j)$ and $\partial/\partial \bar{z}_j = \frac{1}{2}(\partial/\partial x_j + i\partial/\partial y_j)$ then $\{\partial/\partial z_j : 1 \leq j \leq m\}$ and $\{\partial/\partial \bar{z}_j : 1 \leq j \leq m\}$ form bases over $C^\infty(U)$ for $C^\infty(TU)$ and $C^\infty(\overline{TU})$ respectively. Hence we may write $X \in C_{r,s}(U)$ uniquely in the form

$$X = \sum_{I,J} X^{I\bar{J}} \partial/\partial z_I \partial/\partial \bar{z}_J,$$

where $X^{I\bar{J}} \in C^\infty(U)$ and we again follow the notational conventions of §4.

Remark. Much of what we have done in this section goes over to almost complex manifolds. Thus if M is an almost complex manifold with almost complex structure J we may define $TM = \text{Kernel}(J - i)$ and $\overline{TM} = \text{Kernel}(J + i)$. TM and $\overline{TM}$ are complementary complex subbundles of ${}_{\mathbb{C}}TM$ though now, of course, we can no longer say that TM is a holomorphic vector bundle as we are not assuming that M has a complex structure. In the next section we shall discuss the important question of when an almost complex structure on M is associated to a complex structure on M .

7 Calculus on a complex manifold.

Suppose that M is an almost complex manifold with complex structure J on M . We start this section by investigating the relationships, if any, between J and exterior differentiation and Lie brackets.

Definition 5.7.1. The *torsion* of the almost complex structure J on M is the tensor field $N \in C^\infty(\Lambda^2 \mathcal{T}'M \otimes TM)$ characterised by

$$\langle N, X \wedge Y \rangle = [JX, JY] - [X, Y] - J[X, JY] - J[JX, Y],$$

where $X, Y \in C^\infty(TM)$.

Remarks.

1. As usual the reader should verify, using exercise 6, §5, Chapter 1, that N is a well-defined tensor field on M .
2. In the literature N is usually defined as a section of $\otimes^2 \mathcal{T}'M \otimes TM$ and differs from the torsion field as we have defined it by a factor of 4.
3. In the sequel we usually abbreviate an expression like $\langle N, X \wedge Y \rangle$ to $N(X \wedge Y)$ or just $N(X, Y)$. Note that the pairing is that between exterior and not tensor powers.

The significance of the torsion of an almost complex structure may be gauged from

Proposition 5.7.2. The spaces $C^\infty(TM)$, $C^\infty(\overline{TM})$ are Lie subalgebras of $C^\infty({}_c\mathcal{TM})$ if and only if the torsion tensor field N vanishes.

Proof. Let $U, V \in C^\infty(TM)$. We may write U, V uniquely in the form $U = X - iJX$, $V = Y - iJY$, $X, Y \in C^\infty(TM)$. Computing we see that $[U, V] = A + iB$, where $A = [X, Y] - [JX, JY]$, $B = [JX, Y] + [X, JY]$. Now $A + iB \in C^\infty(TM)$ iff $JB = A$. That is, iff $J[JX, Y] + J[X, JY] = [X, Y] - [JX, JY]$. But this is precisely the condition that the torsion field N vanishes. Conjugating we see that if one of $C^\infty(TM)$, $C^\infty(\overline{TM})$ is a Lie subalgebra of $C^\infty({}_c\mathcal{TM})$ so is the other. $\square$

Proposition 5.7.3. The torsion of the almost complex structure associated to a complex manifold vanishes.

Proof. Choose local analytic coordinates and compute N on the basis vector fields $\partial/\partial x_j$, $\partial/\partial y_k$, $1 \leq j, k \leq m$. The Lie bracket of any pair of these constant fields vanishes and since $J(\partial/\partial x_j) = \partial/\partial y_j$, we see that N must vanish identically. $\square$

Remark. It is true, by a fundamental theorem of Newlander and Nirenberg, that if the torsion of an almost complex structure on M vanishes then the almost complex structure is associated to a complex structure on M . We say that the almost complex structure is *integrable*. This result is not difficult to prove if M is real analytic (a proof may be found in Kobayashi and Nomizu [2, Appendix 8]). For the general case we refer to Hörmander [1]. Although we shall not make any systematic study of almost complex manifolds in these notes we should point out that there are topological obstructions on a differential manifold for it to admit an almost complex structure and on an almost complex manifold for it to admit an integrable complex structure. Specifically, a theorem of Hirzebruch and Hopf [1] gives necessary and sufficient conditions on a compact, oriented 4-manifold for it to admit an almost complex structure. These conditions imply, for example, that S^4 does not admit an almost complex structure and so cannot be given the structure of a complex manifold. Borel and Serre [1] prove that S^n can admit an almost complex structure only if $n = 2, 4, 6$. Of course, if $n = 2$ we obtain the Riemann sphere. This leaves the case $n = 6$. It is well known that S^6 admits an almost complex structure (see Kobayashi and Nomizu [2, page 139]) which is, however, not integrable. As yet it is unknown whether S^6 admits an integrable almost complex structure. Results of Van der Ven [1] show that there are topological obstructions to the existence of integrable almost complex structures on an almost complex manifold. For a useful survey of results on 4-manifolds see Pittie [1].

Theorem 5.7.4. Let M be a complex manifold. Then for $r, s \geq 0$ we have

$$d(C^{r,s}(M)) \subset C^{r+1,s}(M) + C^{r,s+1}(M).$$

Proof. Let $\phi \in C^p(M)$ and $X_0, \dots, X_p \in C^\infty({}_c\mathcal{TM})$. Then, as in §2, we have

$$\begin{aligned} \langle d\phi, X_0 \wedge \cdots \wedge X_p \rangle &= \sum_{i=0}^p (-1)^i L_{X_i} \langle \phi, X_0 \wedge \cdots \wedge \widehat{X}_i \wedge \cdots \wedge X_p \rangle \\ &\quad + \sum_{0 \leq i < j} (-1)^{i+j} \langle \phi, [X_i, X_j] \wedge X_0 \wedge \cdots \wedge \widehat{X}_i \wedge \cdots \wedge \widehat{X}_j \wedge \cdots \wedge X_p \rangle. \end{aligned}$$

Now suppose $p = r + s$, ϕ is an (r, s) -form and each X_j lies in $C^\infty(TM)$ or $C^\infty(\overline{TM})$. By Proposition 5.7.2 we see that if more than $(r + 1)$ of the X_j 's lie in $C^\infty(TM)$ or more than $(s + 1)$ of the X_j 's lie in $C^\infty(\overline{TM})$ then the right hand side of the above expression vanishes (remember that the pairings between TM , $\overline{TM}^*$ and TM , $\overline{TM}^*$ are zero). $\square$

Remark. If M is an almost complex manifold it is easily seen that $d(C^{r,s}(M)) \subset C^{p+2,q-1}(M) + C^{p+1,q}(M) + C^{p,q+1}(M) + C^{p-1,q+2}(M)$. A straightforward calculation shows that the result of Theorem 5.7.4 holds if and only if the torsion of the almost complex structure vanishes. See Kobayashi and Nomizu [2] for further details.

It follows from Theorem 5.7.4 that if M is a complex manifold we can define for $r, s \geq 0$ operators $\partial : C^{r,s}(M) \rightarrow C^{r+1,s}(M)$ and $\bar{\partial} : C^{r,s}(M) \rightarrow C^{r,s+1}(M)$ characterised by the identity $d = \partial + \bar{\partial}$. We remark also that for $p \geq 0$, $\partial, \bar{\partial}$ induce operators $\partial, \bar{\partial} : C^p(M) \rightarrow C^{p+1}(M)$ satisfying $d = \partial + \bar{\partial}$.

Properties of the operators $\partial, \bar{\partial}$.

1. $\partial^2 = 0, \bar{\partial}^2 = 0, \partial\bar{\partial} + \bar{\partial}\partial = 0$.
2. $\partial, \bar{\partial}$ are conjugate operators: $\bar{\partial}\phi = \overline{\partial\bar{\phi}}, \phi \in C^p(M)$.
3. $\partial(\phi \wedge \zeta) = \partial\phi \wedge \zeta + (-1)^p \phi \wedge \partial\zeta, \phi \in C^p(M), \zeta \in C^q(M)$. Similarly for $\bar{\partial}$.
4. If $f : M \rightarrow N$ is holomorphic and $\phi \in C^p(N)$ then $f^*\partial\phi = \partial(f^*\phi)$. Similarly for $\bar{\partial}$.
5. In local coordinates, suppose $\phi = \sum_{I,J} \phi_{I\bar{J}} dz_I d\bar{z}_J$. Then

$$\partial\phi = \sum_{j=1}^m \sum_{I,J} \frac{\partial\phi_{I\bar{J}}}{\partial z_j} dz_j dz_I d\bar{z}_J \quad \text{and} \quad \bar{\partial}\phi = \sum_{j=1}^m \sum_{I,J} \frac{\partial\phi_{I\bar{J}}}{\partial \bar{z}_j} d\bar{z}_j dz_I d\bar{z}_J.$$

6. For $p \geq 0$, we have $\Omega^p(M) = \text{Kernel } \bar{\partial} : C^{p,0}(M) \rightarrow C^{p,1}(M)$. In particular, $A(M) = \text{Kernel } \bar{\partial} : C^\infty(M) \rightarrow C^{0,1}(M)$.

Properties 1–4 follow immediately from the corresponding properties of d together with Theorem 5.7.4. For property 5 we use the local identity $df = \partial f / \partial z_j dz_j + \partial f / \partial \bar{z}_j d\bar{z}_j$, f a C^∞ function. Property 6 is immediate from Property 5.

We see from Properties 5 and 6 that the operator $\bar{\partial}$ is our required generalisation and globalization of the Cauchy–Riemann equations discussed in Chapter 2. We observe that for $p \geq 0$ we have the sequences

$$\Omega^p(M) \hookrightarrow C^{p,0}(M) \xrightarrow{\bar{\partial}} C^{p,1}(M) \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} C^{p,m}(M) \rightarrow 0.$$

Since $\bar{\partial}^2 = 0$, we may for $p, q \geq 0$ define the vector spaces

$$H_{\bar{\partial}}^{p,q}(M) = (\text{Kernel } \bar{\partial} : C^{p,q}(M) \rightarrow C^{p,q+1}(M)) / \bar{\partial}C^{p,q-1}(M).$$

These spaces measure the degree of unsolvability of our generalised Cauchy–Riemann equations. Like the de Rham groups they turn out to be important invariants though now they reflect analytic rather than topological properties of a complex manifold. We return to these matters in the next Chapter.

Example 1. Let $\alpha \in \Omega^p(M), \beta \in \Omega^q(M)$. Then $\alpha \wedge \beta \in \Omega^{p+q}(M)$. Indeed by properties 3 and 6 we have $\bar{\partial}(\alpha \wedge \beta) = \bar{\partial}\alpha \wedge \beta + (-1)^p \alpha \wedge \bar{\partial}\beta = 0$. Hence $\alpha \wedge \beta \in \Omega^{p+q}(M)$.

We conclude this section by extending $\bar{\partial}$ to holomorphic vector bundle valued differential forms. Suppose that E is a holomorphic vector bundle on M . For $r, s \geq 0$, we let $\Lambda^{r,s}(M, E)'$ denote the complex vector bundle $\Lambda^{r,s}(M)' \otimes E$ and $C^{r,s}(M, E)$ denote the space of C^∞ sections of $\Lambda^{r,s}(M)' \otimes E$. We claim that for $r, s \geq 0$, $\bar{\partial}$ extends to a map

$$\bar{\partial}_E : C^{r,s}(M, E) \rightarrow C^{r,s+1}(M, E).$$

To construct $\bar{\partial}_E$ we work locally. Suppose that $\theta_U : E|U \rightarrow U \times \mathbb{C}^p$ is a trivialisation of E over the open subset U of M and that we are given a complex analytic coordinate system on U . Given $s \in C^{r,s}(M, E)$, set $s_U = s|U$. We have

$$s_U^a = \sum_{I,J} s_{U,I\bar{J}}^a dz_I d\bar{z}_J, \quad a = 1, \dots, p,$$

where $s_U = (s_U^1, \dots, s_U^p) : U \rightarrow \Lambda^{r,s}(\mathbb{C}^m)' \otimes \mathbb{C}^p$. We define $\bar{\partial}_E s$ by

$$(\bar{\partial}_E s)_U^a = \sum_{j=1}^m \sum_{I,J} \frac{\partial s_{U,I\bar{J}}^a}{\partial \bar{z}_j} d\bar{z}_j dz_I d\bar{z}_J, \quad a = 1, \dots, p.$$

We must check that our definition of $\bar{\partial}_E s$ does not depend on our choices of trivialisation of E . Suppose that $\theta_V : E|V \rightarrow V \times \mathbb{C}^p$ is another trivialisation of E and let θ_{UV} denote the transition function associated to the trivialisations θ_U and θ_V . On $U \cap V$ we have

$$s_U = \theta_{UV} s_V.$$

Since θ_{UV} is analytic we see that $\partial \theta_{UV} / \partial \bar{z}_j = 0$, $j = 1, \dots, m$, and so $(\bar{\partial}_E s)_U = \theta_{UV} (\bar{\partial}_E s)_V$. Hence $\bar{\partial}_E s$ is a well defined section of $\Lambda^{r,s+1}(M, E)'$.

If E is an anti-holomorphic vector bundle on M we may similarly define an operator $\partial_E : C^{r,s}(M, E) \rightarrow C^{r+1,s}(M, E)$, $r, s \geq 0$. Indeed, using conjugation we may define $\partial_E \phi = \overline{(\bar{\partial}_E \bar{\phi})}$, $\phi \in C^{r,s}(M, E)$, where conjugation is induced from the conjugation map $S : \Lambda^{r,s}(M, E)' \rightarrow \Lambda^{s,r}(M, \bar{E})'$ (see Exercise 2, §4).

In the sequel we usually drop the subscripts from ∂_E and $\bar{\partial}_E$ and just write ∂ and $\bar{\partial}$.

Properties of the operators ∂ , $\bar{\partial}$ on bundle valued forms. We shall only state properties for $\bar{\partial}$; those for ∂ follow by conjugation. In what follows we assume that E is a holomorphic vector bundle on M .

1. $\bar{\partial}^2 = 0$.
2. Kernel $\bar{\partial} : C^{p,0}(M, E) \rightarrow C^{p,1}(M, E)$ is the space $\Omega^p(M, E)$ of holomorphic sections of $\Lambda^p TM^* \otimes E$. In particular, Kernel $\bar{\partial} : C^\infty(E) \rightarrow C^{0,1}(M, E)$ is the space $\Omega(M, E)$ of holomorphic sections of E .
3. $\bar{\partial}$ commutes with contractions on finite tensor products of tensor, exterior and symmetric powers of E and E^* . For example, if $\phi \in C^{r,s}(M, \Lambda^p E \otimes \Lambda^q E^*)$ and $p > q$, $\mathbf{C}(\bar{\partial}\phi) = \bar{\partial}(\mathbf{C}\phi) \in C^{r,s}(M, \Lambda^{p-q} E)$. Here $\mathbf{C}$ is induced from the contraction $\mathbf{C} : \Lambda^p E \otimes \Lambda^q E^* \rightarrow \Lambda^{p-q} E$.
4. $\bar{\partial}$ commutes with contractions between the bundles $\Lambda^{r,s}(M)'$ and exterior powers of the holomorphic tangent bundle of M . That is, if $\phi \in C^{r,s}(M, \Lambda^p TM)$ and $r \geq p$, we have $\bar{\partial}(\mathbf{C}\phi) = \mathbf{C}(\bar{\partial}\phi) \in C^{r-p,s}(M)$.

These properties all follow immediately from our local description of $\bar{\partial}$.

As a consequence of property 1 we have for $p \geq 0$ the sequences

$$\Omega^p(M, E) \hookrightarrow C^{p,0}(M, E) \xrightarrow{\bar{\partial}} C^{p,1}(M, E) \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} C^{p,m}(M, E) \rightarrow 0$$

and the corresponding vector spaces

$$H_{\bar{\partial}}^{p,q}(M, E) = (\text{Kernel } \bar{\partial} : C^{p,q}(M, E) \rightarrow C^{p,q+1}(M, E)) / \bar{\partial} C^{p,q-1}(M, E).$$

Examples.

2. Let E be a holomorphic vector bundle and $J \in C^\infty(L(E, E))$ denote the complex structure on E . Then $\bar{\partial}J = 0$ and so J is a holomorphic section of $L(E, E)$. That $\bar{\partial}J = 0$ is a consequence of the fact that locally J is a constant section. For the same reason $\bar{\partial}I = 0$, where I is the identity section.
3. Let $X \in C^\infty(TM)$, $Y \in C^\infty(\overline{TM})$. Then $[X, Y] = C_X \partial Y - C_Y \bar{\partial}X$. We shall give an invariant proof of this identity using property 4 above. Let $f \in C^\infty(M)$, then

$$\begin{aligned}
 L_{[X, Y]}f &= \langle d\langle df, Y \rangle, X \rangle - \langle d\langle df, X \rangle, Y \rangle \\
 &= \langle \partial\langle \bar{\partial}f, Y \rangle, X \rangle - \langle \bar{\partial}\langle \partial f, X \rangle, Y \rangle \\
 &= (C_X C_Y \partial \bar{\partial}f - C_Y C_X \bar{\partial} \partial f) + (\langle \bar{\partial}f, C_X \partial Y \rangle - \langle \partial f, C_Y \bar{\partial}X \rangle) \\
 &= \langle df, C_X \partial Y - C_Y \bar{\partial}X \rangle, \quad \text{proving our assertion.}
 \end{aligned}$$

Exercises.

1. Starting with the local description of $\bar{\partial}$ (property 5), prove directly that if f is a holomorphic map then $\bar{\partial}(f^*\phi) = f^*(\bar{\partial}\phi)$ (property 4). Deduce that $\bar{\partial}$ may be defined invariantly on complex differential forms on a complex manifold.
2. Let E be a holomorphic vector bundle on M . Suppose $s \in C^\infty(E)$, $f \in C^\infty(M)$. Prove $\bar{\partial}(fs) = \bar{\partial}f \otimes s + f\bar{\partial}s$. More generally, if $\phi \in C^{r,s}(M)$, prove that $\bar{\partial}(\phi \otimes s) = \bar{\partial}\phi \otimes s + (-1)^{r+s}\phi \wedge \bar{\partial}s$.

8 The Dolbeault–Grothendieck Lemma.

This section is devoted to the proof of an important result that plays the same rôle in the theory of complex manifolds as the Poincaré lemma does in the cohomology of differential manifolds.

Theorem 5.8.1. (Dolbeault–Grothendieck lemma). Let D be an open polydisc in $\mathbb{C}^n$ and suppose that $f \in C^{p,q+1}(D)$ satisfies $\bar{\partial}f = 0$ ($p, q \geq 0$). Then if W is any relatively compact open subset of D there exists $u \in C^{p,q}(W)$ such that $\bar{\partial}u = f$ on W .

Proof. The theorem is proved inductively. The k th. step of the induction is to prove the theorem true if f is independent of $d\bar{z}_{k+1}, \dots, d\bar{z}_n$. The theorem is trivially true when $k = 0$ and the theorem is obtained for $k = n$.

Let us assume that the theorem has been proved for $k - 1$ and that f does not involve $d\bar{z}_{k+1}, \dots, d\bar{z}_n$. We may write f uniquely in the form

$$f = d\bar{z}_k \wedge g + h,$$

where $g \in C^{p,q}(D)$, $h \in C^{p,q+1}(D)$ and g and h are independent of $d\bar{z}_k, \dots, d\bar{z}_n$. Set $g = \sum_{I, J} g_{I\bar{J}} dz_I d\bar{z}_J$. Since $\bar{\partial}f = 0$, we have

$$\partial g_{I\bar{J}} / \partial \bar{z}_j = 0, \quad j > k. \quad (\text{A})$$

We now find a solution $G_{I\bar{J}}$ of the equation

$$\partial G_{I\bar{J}} / \partial \bar{z}_k = g_{I\bar{J}}.$$

To do this suppose $D = \prod_{j=1}^n D_j$, $D_j \subset \mathbb{C}$, and choose $\phi \in C_c^\infty(D_k)$ such that $\phi(z_k) = 1$ on a neighbourhood W' of $\bar{W}$. Define

$$\begin{aligned} G_{I\bar{J}}(z) &= (2\pi i)^{-1} \int_{\mathbb{C}} (t - z_k)^{-1} \phi(t) g_{I\bar{J}}(z_1, \dots, z_{k-1}, t, z_{k+1}, \dots, z_n) dt d\bar{t} \\ &= -(2\pi i)^{-1} \int_{\mathbb{C}} t^{-1} \phi(z_k - t) g_{I\bar{J}}(z_1, \dots, z_{k-1}, t, z_{k+1}, \dots, z_n) dt d\bar{t}. \end{aligned}$$

The second expression for $G_{I\bar{J}}$ implies that $G_{I\bar{J}} \in C^\infty(D)$. Theorem A1.6 implies that $\partial G_{I\bar{J}}/\partial \bar{z}_k = g_{I\bar{J}}$ on W' and, by (A), that $\partial G_{I\bar{J}}/\partial \bar{z}_j = 0$, $j > k$. Set $G = \sum_{I,J} G_{I\bar{J}} dz_I d\bar{z}_J$. Then on W' we have $\bar{\partial}G = d\bar{z}_k \wedge g + h_1$, where h_1 is independent of $d\bar{z}_k, \dots, d\bar{z}_n$. Hence, on W' , $h - h_1 = f - \bar{\partial}G$ is independent of $d\bar{z}_k, \dots, d\bar{z}_n$. Since $\bar{\partial}(h - h_1) = 0$, we may apply the inductive hypothesis to find $v \in C^{p,q}(W)$ such that $\bar{\partial}v = f - \bar{\partial}G$ on W . Setting $u = v + g$, we have $\bar{\partial}u = f$ on W . $\square$

Remark. It is easy to see, using bump functions, that we may construct $u \in C^{p,q}(D)$ satisfying $\bar{\partial}u = f$ on W .

Theorem 5.8.1 is sufficient for the development of cohomology theory in Chapter 6. However, we shall now prove a stronger version of the Dolbeault–Grothendieck Lemma and obtain a particular case of a result that holds on arbitrary Stein manifolds and to which we shall return in Chapter 11.

Theorem 5.8.2. Let D be an open, not necessarily relatively compact, polydisc in $\mathbb{C}^n$ and suppose $f \in C^{p,q+1}(D)$ satisfies $\bar{\partial}f = 0$. Then there exists $u \in C^{p,q}(D)$ such that $\bar{\partial}u = f$. Here we assume $p, q \geq 0$.

Proof. We divide the proof into two cases: $q = 0$, $q \geq 1$.

Case 1. $q = 0$. Choose a sequence D_j , $j \geq 1$, of relatively compact open polydiscs in $\mathbb{C}^n$ which have the same centres as D and which satisfy:

$$\text{A. } \bar{D}_j \subset D_{j+1}, \quad j \geq 1 \quad \text{and} \quad \text{B. } \bigcup_{j \geq 1} D_j = D.$$

If $u = \sum u_I dz_I \in C^{p,0}(D)$ and $K \subset D$, we define $|u|_K = \max_I \|u_I\|_K$.

We shall construct inductively a sequence $u_j \in C^{p,0}(D)$ satisfying

1. $\bar{\partial}u_j = f$ on some open neighbourhood of $\bar{D}_j$, $j \geq 1$.
2. $|u_{j+1} - u_j|_{D_j} < 2^{-j}$, $j \geq 1$.

The existence of u_1 follows from Theorem 5.8.1. Assume that we have constructed $u_1, \dots, u_k$ satisfying the conditions above. By Theorem 5.8.1, there exists $u'_{k+1} \in C^{p,0}(D)$ such that $\bar{\partial}u'_{k+1} = f$ on an open neighbourhood of $\bar{D}_{k+1}$. The difference $u'_{k+1} - u_k$ is holomorphic on an open neighbourhood of $\bar{D}_k$ since $\bar{\partial}(u'_{k+1} - u_k) = f - f = 0$ on an open neighbourhood of $\bar{D}_k$. Hence, if we take Taylor's expansion of the coefficients of $u'_{k+1} - u_k$ at the centre of D , we may find $P \in \Omega^p(D)$, with polynomial coefficients, such that

$$|u'_{k+1} - u_k - P|_{\bar{D}_k} < 2^{-k}.$$

Now we define $u_{k+1} = u'_{k+1} - P$ and see that u_{k+1} satisfies the required conditions and so the inductive step is completed.

The sequence (u_k) has coefficients which converge uniformly on each $\bar{D}_k$ and so (u_k) converges to a continuous $(p,0)$ -form, u . Now $u = u_1 + \sum_{j=1}^\infty (u_{j+1} - u_j)$ and only finitely many of the differences $u_{j+1} - u_j$ are not holomorphic on any given D_k . Hence, by Corollary 2.1.8, u must be C^∞ on each D_k and so $u \in C^{p,0}(D)$. Finally, on each D_k , $u = u_k + a_k$, $a_k \in A(D_k)$, and so $\bar{\partial}u = \bar{\partial}u_k = f$ on each D_k . Hence $\bar{\partial}u = f$ on D .

Case 2. $q \geq 1$. We choose a sequence D_j of polydiscs in $\mathbb{C}^n$ satisfying the conditions of Case 1. We shall construct inductively a sequence $u_j \in C^{p,q}(D)$ such that

1. $f = \bar{\partial}u_j$ on some open neighbourhood of $\bar{D}_j$, $j \geq 1$.
2. $u_{j+1}|_{D_j} = u_j$, $j \geq 1$.

The existence of u_1 follows from Theorem 5.8.1. Suppose we have constructed $u_1, \dots, u_k$ satisfying the conditions above. By Theorem 5.8.1, there exists $u'_{k+1} \in C^{p,q}(D)$ such that $\bar{\partial}u'_{k+1} = f$ on some open neighbourhood of $\bar{D}_{k+1}$. Now $\bar{\partial}(u'_{k+1} - u_k) = 0$ on some open neighbourhood of $\bar{D}_k$ and so, since $q \geq 1$, another application of Theorem 5.8.1 implies that there exists $\phi \in C^{p,q-1}(D)$ such that $u'_{k+1} - u_k = \bar{\partial}\phi$ on some open neighbourhood W of D_k . Define $u_{k+1} = u'_{k+1} - \bar{\partial}\phi$. Then, since $\bar{\partial}^2 = 0$, we see that $\bar{\partial}u_{k+1} = \bar{\partial}u'_{k+1} = f$ on an open neighbourhood of $\bar{D}_{k+1}$ and $u_{k+1}|_{\bar{D}_k} = u'_{k+1} - \bar{\partial}\phi = u_k$. This completes the inductive step and we now define $u \in C^{p,q}(D)$ by $u|_{D_j} = u_j$, $j \geq 1$. Clearly $\bar{\partial}u = f$. $\square$

Remarks. Both Theorems 5.8.1 and 5.8.2 hold for $\mathbb{C}^m$ -valued forms. The proofs are identical to those given above.

Corollary 5.8.3. Every open polydisc in $\mathbb{C}^n$ is a Cousin I, II, A and B domain. In particular, $\mathbb{C}^n$ is a Cousin I, II, A and B domain.

Proof. Propositions 2.7.1 and 2.7.3. $\square$

Corollary 5.8.4. Every holomorphic line bundle on an open polydisc in $\mathbb{C}^n$ is holomorphically trivial.

Proof. Let $\phi_{ij} : U_{ij} \rightarrow \mathbb{C}^\bullet$ be the transition functions for the holomorphic line bundle L on the polydisc D . Then $\phi_{ij} \cdot \phi_{jk} = \phi_{ik}$ for all i, j, k and so $\{\phi_{ij}\}$ is the data for a Cousin B problem on D . By Corollary 5.8.3, there exists $a_i \in A^*(U_i)$ such that $\phi_{ij} = a_j/a_i$. But, by §5 of Chapter 1, this implies that L is holomorphically trivial. $\square$

Exercises.

1. Suppose $p, q \geq 0$ and $m > 1$. Show that if $f \in C^{p,q+1}_c(\mathbb{C}^m)$ and $\bar{\partial}f = 0$ then there exists $u \in C^{p,q}_c(\mathbb{C}^m)$ such that $\bar{\partial}u = f$.
2. Show that the open Euclidean disc $E(z; r)$ in $\mathbb{C}^n$ is a Cousin I, II, A and B domain (Use Exercise 1, §1, Chapter 2).

9 Holomorphic vector bundles on compact complex manifolds.

In this section we present a number of important examples of holomorphic vector bundles on compact complex manifolds. We shall pay particular attention to the spaces of holomorphic sections of such bundles which, by the theory of §7, may be represented as the kernel of the $\bar{\partial}$ -operator.

We start by proving an elementary special case of a rather general finiteness theorem that we return to in Chapter 7.

Theorem 5.9.1. Let E be a holomorphic vector bundle on the compact complex manifold M . Then $\dim_{\mathbb{C}} \Omega(E) < \infty$.

Proof. Suppose that we are given a finite open cover $\{U_i : i = 1, \dots, r\}$ of M such that over each U_i , E has a holomorphic trivialisation $\theta_i : E|_{U_i} \rightarrow U_i \times \mathbb{C}^p$, $p = \dim(E)$. Suppose also that $\{V_i\}$ is an open refinement of the cover $\{U_i\}$ such that for all i , V_i is a relatively compact subset of U_i . Given $s \in \Omega(E)$, we let $s_i : V_i \rightarrow \mathbb{C}^p$ denote the local representative of s on V_i .

If $\|\cdot\|$ denotes the standard Euclidean norm on $\mathbb{C}^p$, we may define a norm $|\cdot|$ on $\Omega(E)$ by

$$|s| = \sum_{i=1}^r \|s_i\|_{V_i}, \quad s \in \Omega(E).$$

Observe that $\|s_i\|_{V_i} < \infty$ since $\bar{V}_i$ is compact. We shall prove that the closed unit ball B in the normed vector space $(\Omega(E), |\cdot|)$ is compact. This implies that $(\Omega(E), |\cdot|)$ is locally compact and hence finite dimensional by F. Riesz' theorem (for an elementary proof of Riesz' theorem see Field [1, page 54]). Suppose then that (s^j) is a sequence in B . Given i , $1 \leq i \leq r$, we have corresponding sequences $(s_i^j) \subset A(V_i, \mathbb{C}^p)$ of local representatives. By our definition of $|\cdot|$, it is clear that for all i we have

$$\|s_i^j(z)\| \leq 1, \quad z \in V_i, \quad j \geq 1.$$

In particular the sequence (s_i^j) is bounded on $\bar{V}_i$ and so by Montel's theorem (Corollary 2.1.9) we may find a subsequence $(t(1)^j)$ of (s^j) which converges uniformly on V_1 . Proceeding inductively, suppose that we have constructed a subsequence $(t(k)^j)$ of (s^j) which converges uniformly on $V_1 \cup \dots \cup V_k$. Applying Montel's theorem we may find a subsequence $(t(k+1)^j)$ of $(t(k)^j)$ which converges uniformly on V_{k+1} and hence uniformly on $V_1 \cup \dots \cup V_{k+1}$. Hence we may find a subsequence (t^j) of (s^j) which converges uniformly on $V_1 \cup \dots \cup V_r = M$. Hence B is sequentially compact and therefore compact. $\square$

Holomorphic vector bundles on projective space. To each point $\ell \in P^n(\mathbb{C})$ is naturally associated a complex line in $\mathbb{C}^{n+1}$. This suggests that we should be able to construct a complex line bundle over $P^n(\mathbb{C})$ whose fibre at the point $\ell \in P^n(\mathbb{C})$ is the line $\ell \subset \mathbb{C}^{n+1}$. We start this subsection by constructing this “tautological” line bundle (see also Exercise 2, §3, Chapter 4). Let $L \subset P^n(\mathbb{C}) \times \mathbb{C}^{n+1}$ denote the set $\{(\ell, z) : z \in \ell\}$. The projection on $P^n(\mathbb{C})$ induces a projection $\pi : L \rightarrow P^n(\mathbb{C})$. Recalling §7 of Chapter 4, we see that L is $\mathbb{C}^{n+1}$ blown up at the origin and so, in particular, L has the structure of a complex manifold of dimension $n+1$ and π is holomorphic. We claim that L has the natural structure of a holomorphic line bundle over $P^n(\mathbb{C})$. For this we have only to observe that the maps $\theta_i : L|_{U_i} \rightarrow U_i \times \mathbb{C}$ defined by

$$\theta_i((z_0, \dots, z_n), (a_0, \dots, a_n)) = ((z_0, \dots, z_n), a_i)$$

define holomorphic trivialisations for L . The corresponding transition functions $\theta_{ij} : U_{ij} \rightarrow \text{GL}(1, \mathbb{C}) = \mathbb{C}^\bullet$ are given by

$$\theta_{ij}(z_0, \dots, z_n) = z_i/z_j.$$

We call the holomorphic line bundle L the *tautological* or *universal* line bundle on $P^n(\mathbb{C})$.

We let H denote the dual bundle L^* of L . For reasons that will soon become clear we call H the *hyperplane section bundle* of $P^n(\mathbb{C})$. For $p \in \mathbb{Z}$ we define

$$H^p = \begin{cases} \otimes^p H, & p \geq 0, \\ \otimes^{-p} L, & p \leq 0. \end{cases}$$

Note that the transition functions θ_{ij}^p for H^p are given by $\theta_{ij}^p = (z_j/z_i)^p$.

The next proposition gives an indication of the important rôle that the hyperplane section bundle plays in projective algebraic geometry and also indicates an important bridge that exists between complex analysis and algebra.

Proposition 5.9.2. For $p \geq 0$, $\Omega(H^p)$ is canonically isomorphic to the space $P^{(p)}(\mathbb{C}^{n+1})$ of homogeneous polynomials of degree p on $\mathbb{C}^{n+1}$. For $p < 0$, $\Omega(H^p)$ consists of the zero section.

Proof. Suppose $p \geq 0$. If $s \in \Omega(H^p)$, we let $s_i : U_i \rightarrow \mathbb{C}$ denote the local representatives of s relative to the standard trivialisation of H^p . For $0 \leq i, j \leq n$ we have $\theta_{ij}^p s_j = s_i$ and so

$$s_j(z_0, \dots, z_n) z_j^p = s_i(z_0, \dots, z_n) z_i^p.$$

Hence we may define the holomorphic map $S : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{C}$ by $S(z_0, \dots, z_n) = s_i(z_0, \dots, z_n) z_i^p$, $z_i \neq 0$. Since the s_i are homogeneous of degree zero, S is homogeneous of degree p . By Hartog's theorem, S extends to $\mathbb{C}^{n+1}$ as an analytic function which we continue to denote by S . Since S is homogeneous of degree p , we see by taking Taylor's series at the origin of $\mathbb{C}^{n+1}$, that S must be a homogeneous polynomial of degree p . Hence we have defined a map of $\Omega(H^p)$ into $P^{(p)}(\mathbb{C}^{n+1})$, $p \geq 0$. If $S \in P^{(p)}(\mathbb{C}^{n+1})$, define $s_i = S/z_i^p$, $i = 0, \dots, n$. The s_i are the local representatives of a holomorphic section of H^p . Since the maps between $\Omega(H^p)$ and $P^{(p)}(\mathbb{C}^{n+1})$ are clearly inverses of one another we have shown that $\Omega(H^p)$ is canonically isomorphic to $P^{(p)}(\mathbb{C}^{n+1})$.

We leave the case $p < 0$ as an exercise for the reader which makes use of the isomorphism $H^p \otimes H^{-p} \approx \underline{\mathbb{C}}$. $\square$

As a special case of the proposition we see that if $p = 1$, sections of H correspond to linear functionals on $\mathbb{C}^{n+1}$. In particular the zero sets of such sections are hyperplanes in $P^n(\mathbb{C})$.

Next we show that there exist natural exact sequences

$$0 \rightarrow \underline{\mathbb{C}} \rightarrow (n+1)H \rightarrow TP^n \rightarrow 0$$

$$0 \rightarrow TP^{n*} \rightarrow (n+1)L \rightarrow \underline{\mathbb{C}} \rightarrow 0.$$

Here $(n+1)H$ denotes the $(n+1)$ -fold direct sum of H ; similarly for $(n+1)L$. Since the second sequence is the dual of the first it suffices to construct the first sequence. First, however, we need to prove some results about holomorphic vector fields on projective space.

Let $q : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow P^n(\mathbb{C})$ denote the quotient map. Suppose X is a vector field on $\mathbb{C}^{n+1} \setminus \{0\}$ which is homogeneous of degree 1: $X(\lambda v) = \lambda X(v)$, $\lambda \in \mathbb{C}^\bullet$, $v \in \mathbb{C}^{n+1} \setminus \{0\}$. Then q_*X is well-defined as a vector field on $P^n(\mathbb{C})$. Indeed, since $q = q \cdot \lambda$, $Dq_v = Dq_{\lambda v} \cdot \lambda$, $\lambda \in \mathbb{C}^\bullet$. Hence $(q_*X)(q(v)) = Dq_v(X(v)) = Dq_{\lambda v}X(\lambda v)$, $v \in \mathbb{C}^{n+1} \setminus \{0\}$, $\lambda \in \mathbb{C}^\bullet$, proving that q_*X is well-defined on $P^n(\mathbb{C})$. We define the *Euler vector field* E on $\mathbb{C}^{n+1} \setminus \{0\}$ by

$$E(z_0, \dots, z_n) = \sum_{i=0}^n z_i \partial / \partial z_i.$$

Certainly E is homogeneous of degree 1 and so q_*E is a well-defined holomorphic vector field on $P^n(\mathbb{C})$. Since $E(z) \in T_z L_z$, where L_z denotes the line through z and 0, we see that $q_*E = 0$. Let $\{s_0, \dots, s_n\}$ be a basis of $\Omega(H)$. Since $\{s_0(z), \dots, s_n(z)\}$ spans H_z for all $z \in P^n(\mathbb{C})$, to define a vector bundle morphism $(n+1)H \rightarrow TP^n$ it is enough to specify the map on holomorphic sections of $(n+1)H$. So suppose $(\phi_0, \dots, \phi_n) \in \Omega((n+1)H)$. We define $\xi(\phi_0, \dots, \phi_n) \in \Omega(TP^n)$ by

$$\xi(\phi_0, \dots, \phi_n)(z) = \sum_{i=0}^n q_*(\phi_i(z) \partial / \partial z_i),$$

where we have used the identification between sections of H and linear functionals on $\mathbb{C}^{n+1}$. Now the kernel of ξ is the image of the map

$$T : \underline{\mathbb{C}} \rightarrow (n+1)H$$

defined on sections by $T(1)(z) = (z_0, \dots, z_n)$, where $z_0, \dots, z_n$ denote the coordinate functions. In view of the fact that $T(1) = q_*E$, we see that the “Euler sequence”

$$0 \rightarrow \underline{\mathbb{C}} \xrightarrow{T} (n+1)H \xrightarrow{\xi} TP^n \rightarrow 0$$

is exact.

Taking the highest exterior power of the Euler sequence and using the result of Exercise 5, §1, we see that

$$\Lambda^n TP^n \approx \Lambda^{n+1}((n+1)H) \approx H^{n+1}.$$

Taking duals, we deduce that

$$\Lambda^n TP^{n*} \approx H^{-n-1}.$$

In the sequel we call the n th. exterior power of the cotangent bundle of an n -dimensional complex manifold M the *canonical bundle* of M and denote it by $K(M)$. Thus we have shown that $K(P^n(\mathbb{C})) = H^{-n-1}$. As we shall see later the canonical bundle plays a central rôle in the theory of compact complex manifolds. If X is a closed complex submanifold of $P^n(\mathbb{C})$, we may pull back (restrict) the hyperplane section bundle of $P^n(\mathbb{C})$ to X . We denote the resulting bundle on X by H_X . We remark that the zero sets of holomorphic sections of H_X are intersections of X with hyperplanes in $P^n(\mathbb{C})$. We refer to H_X as the *hyperplane section bundle* of X .

Divisors, holomorphic line bundles and linear systems. Recall from §6 of Chapter 4 that the group $\mathcal{D}(M)$ of divisors on a complex manifold M is the set of (locally finite) formal sums $\sum_{\alpha \in \Lambda} n_\alpha V_\alpha$, where the n_α are integers and the V_α are irreducible analytic hypersurfaces of M . Here we suppose that M is compact and so we may assume that Λ is finite. By Theorem 4.6.11 every divisor on M may be specified by a Cartier divisor $\{(U_i, d_i) : i \in I\} = \{d_i \in \mathcal{M}^*(U_i) : d_i d_j^{-1} \in A^*(U_{ij})\}$ and two Cartier divisors $\{(U_i, d_i) : i \in I\}$ and $\{(V_j, e_j) : j \in J\}$ determine the same element of $\mathcal{D}(M)$ if and only if $d_i e_j^{-1} \in A^*(U_i \cap V_j)$ for all $i \in I, j \in J$.

Before stating the next proposition we recall from Chapter 1 that the set $\text{HLB}(M)$ of isomorphism classes of holomorphic line bundles on a complex manifold M has the natural structure of an Abelian group with composition defined by tensor product and inverse by dual. As in Chapter 1, we shall use the abbreviated notations $E.F$ and E^{-1} for $E \otimes F$ and E^* respectively. As usual $\underline{\mathbb{C}}$ will denote the trivial holomorphic line bundle.

Proposition 5.9.3. There is a canonical group homomorphism

$$[\] : \mathcal{D}(M) \rightarrow \text{HLB}(M).$$

Proof. Let $d = \{(U_i, d_i) : i \in I\} \in \mathcal{D}(M)$. We let $[d]$ denote the holomorphic line bundle on M with transition functions $\theta_{ij} : U_{ij} \rightarrow \text{GL}(1, \mathbb{C}) \approx \mathbb{C}^\bullet$ defined by $\theta_{ij} = d_i/d_j$. We must show that $[d]$ depends only on d and not on our particular representation of d as a Cartier divisor. Suppose then that $\{(U_i, d'_i) : i \in I\}$ also defines the divisor d . The corresponding transition functions are given by $\theta'_{ij} = d'_i/d'_j$. But now $d_i/d'_i \in A^*(U_i)$ and so, setting $a_i = d_i/d'_i$, we have

$$a_i \theta_{ij} = \theta'_{ij} a_j, \quad i, j \in I.$$

Hence $\theta_{ij}, \theta'_{ij}$ define isomorphic holomorphic line bundles (see Chapter 1, §5). The fact that $[\]$ is a group homomorphism is immediate from our definition of $[\]$ using Cartier divisors. $\square$

Remark. As a consequence of Proposition 5.9.3 we see that if $d, d' \in \mathcal{D}(M)$, then $[d + d'] = [d][d']$ and $[d]^* = [-d]$.

Proposition 5.9.4. The sequence

$$\mathcal{M}^*(M) \xrightarrow{\text{div}} \mathcal{D}(M) \xrightarrow{[\]} \text{HLB}(M)$$

is exact.

Proof. Let $d = \{(U_i, d_i) : i \in I\} \in \mathcal{D}(M)$ and suppose that $[d] = \mathbb{C}$. Then there exist $a_i \in A^*(U_i)$ such that $a_i \theta_{ij} = a_j$, where $\theta_{ij} = d_i/d_j$ are the transition functions for $[d]$. Hence $a_i d_i/d_j = a_j$ and so $a_i d_i = a_j d_j$ on U_{ij} . Therefore we may define $m \in \mathcal{M}^*(M)$ by $m|_{U_i} = a_i d_i$. Clearly $\text{div}(m) = d$ since $\text{div}(a_i d_i) = \text{div}(d_i)$, $i \in I$. Obviously $[\text{div}(m)] = \mathbb{C}$ for all $m \in \mathcal{M}^*(M)$ and so we have shown that $\text{div}(\mathcal{M}^*(M)) = \text{Kernel}[\]$. $\square$

Definition 5.9.5. Let $d, d' \in \mathcal{D}(M)$. We say that d and d' are *linearly equivalent* if $d - d'$ is the divisor of a meromorphic function. We denote the group of linear equivalence classes of divisors on M by $L(M)$. Thus $L(M) = \mathcal{D}(M)/\text{div}(\mathcal{M}^*(M))$. Given $d \in \mathcal{D}(M)$, we let $L(d)$ denote the set of all divisors on M linearly equivalent to d .

Next we wish to define *meromorphic sections* of a holomorphic line bundle. Suppose that $E \in \text{HLB}(M)$ has transition functions $\theta_{ij} : U_{ij} \rightarrow \mathbb{C}^\bullet$. We say that a family $m_i \in \mathcal{M}(U_i)$, $i \in I$, defines a meromorphic section of E if $\theta_{ij} m_j = m_i \in \mathcal{M}(U_{ij})$, $i, j \in I$. We let $\mathcal{M}(E)$ denote the space of meromorphic sections of E and $\mathcal{M}^*(E)$ denote the space of non-zero meromorphic sections of E .

Remark. The group law in $\text{HLB}(M)$ induces corresponding maps on spaces of sections. For example, if $E \in \text{HLB}(M)$ and $s \in \mathcal{M}^*(E)$ has local representatives $s_i \in \mathcal{M}^*(U_i)$, we may define $s^{-1} \in \mathcal{M}^*(E^{-1})$ to be the section with local representatives $(s^{-1})_i = s_i^{-1}$. This construction defines an inversion map $\mathcal{M}^*(E) \rightarrow \mathcal{M}^*(E^{-1})$. Similarly if $E, F \in \text{HLB}(M)$ and $s, t \in \mathcal{M}(E) \times \mathcal{M}(F)$ we may define the composition $s.t \in \mathcal{M}(E.F)$ by $s.t = s \otimes t$.

Proposition 5.9.6. Let $E \in \text{HLB}(M)$ and suppose that $\mathcal{M}^*(E) \neq \emptyset$. We have a natural group homomorphism $\text{div} : \mathcal{M}^*(E) \rightarrow \mathcal{D}(M)$ satisfying

$$[\text{div}(s)] = E, \quad s \in \mathcal{M}^*(E).$$

Proof. Let E have transition functions $\theta_{ij} : U_{ij} \rightarrow \mathbb{C}^\bullet$ and $s \in \mathcal{M}^*(E)$ have corresponding local representatives $s_i \in \mathcal{M}^*(U_i)$. Since $\theta_{ij} s_j = s_i$, we have $s_i s_j^{-1} \in A^*(U_{ij})$ and so we may define $\text{div}(s)$ to be the Cartier divisor $\{(U_i, s_i) : i \in I\}$. Clearly $\text{div}(s)$ depends only on s and not on our particular choice of transition functions for E . It is immediate from our local description of $\text{div}(s)$ that $[\text{div}(s)] = E$. Finally, $\text{div} : \mathcal{M}^*(E) \rightarrow \mathcal{D}(M)$ is obviously a group homomorphism. $\square$

Remark. We call $\text{div}(s)$ the *divisor* of the section s .

Example 1. Let $m \in \mathcal{M}^*(P^n(\mathbb{C}))$. Then $\deg(\text{div}(m)) = 0$ (Example 3, §6, Chapter 4). Suppose that E is a holomorphic line bundle on $P^n(\mathbb{C})$ and $s, t \in \mathcal{M}^*(E)$. Since $st^{-1} \in \mathcal{M}^*(P^n(\mathbb{C}))$, we see at once that $\deg(\text{div}(s)) = \deg(\text{div}(t))$. Hence we may define the degree of E , $\deg(E)$, to be the degree of any non-trivial meromorphic section of E . Clearly, $\deg : \text{HLB}(P^n(\mathbb{C})) \rightarrow \mathbb{Z}$ is a homomorphism. See also Proposition 1.5.7. We shall give another interpretation of the degree map in §3, Chapter 6.

Proposition 5.9.7. Let $d \in \mathcal{D}(M)$. Then

1. There exists $s(d) \in \mathcal{M}^*([d])$ such that $\text{div}(s(d)) = d$. The section $s(d)$ is unique up to multiplication by elements of $\mathbb{C}^\bullet$.
2. $\text{div}(\mathcal{M}^*([d])) = L(d)$. Moreover the map $\text{div} : \mathcal{M}^*([d]) \rightarrow L(d)$ induces a bijection of $L(d)$ with $\mathcal{M}^*([d])/\mathbb{C}^\bullet$.

Proof. The proof of 1 is the same as the proof of Proposition 1.5.4. Let us prove part 2. Suppose that $s \in \mathcal{M}^*([d])$. Then $[\text{div}(s)] \cong [d]$ and so $s(d)^{-1}s \in \mathcal{M}^*(M)$. Therefore, $\text{div}(s) \in L(d)$. Conversely, if $d' \in L(d)$, there exists $m \in \mathcal{M}^*(M)$ such that $d - d' = \text{div}(m)$. Hence $[d] \cong [d']$ and $s(d')$ determines a section of $[d]$ with divisor $d' = d - \text{div}(m) \in L(d)$. The remaining assertion of part 2 is immediate from part 1. $\square$

Proposition 5.9.8. A holomorphic line bundle E lies in the image of $[\] : \mathcal{D}(M) \rightarrow \text{HLB}(M)$ if and only if E has a non-trivial meromorphic section.

Proof. Immediate from proposition 5.9.6 and 5.9.7. $\square$

Remark. Proposition 5.9.8 shows that the study of divisors on M is closely related to the problem of finding which holomorphic line bundles on a complex manifold admit non-trivial meromorphic sections. Two fundamental results that we prove later show that if M is projective or Stein then every holomorphic vector bundle on M admits a non-trivial meromorphic section. It must be stressed that an arbitrary complex manifold of dimension greater than 1 need not have any holomorphic line bundles which admit meromorphic sections (equivalently, the manifold need not have any divisors).

Definition 5.9.9. We say that a divisor $d = \sum_{\alpha \in \Lambda} n_\alpha V_\alpha$ is *effective* or *positive* if $n_\alpha \geq 0$, $\alpha \in \Lambda$. We write $d \geq 0$.

Suppose that $d \in \mathcal{D}(M)$. We let $L(d)$ denote the vector subspace of $\mathcal{M}(M)$ defined by

$$m \in L(d) \quad \text{iff} \quad d + \text{div}(m) \geq 0 \quad \text{or} \quad m = 0.$$

Proposition 5.9.10. The vector space $L(d)$ is isomorphic to $\Omega([d])$. In particular $\dim_{\mathbb{C}} L(d) < \infty$.

Proof. By Proposition 5.9.7, there exists $s \in \mathcal{M}^*([d])$ such that $\text{div}(s) = d$. Let $\gamma : \mathcal{M}(M) \rightarrow \mathcal{M}([d])$ denote the map defined by $\gamma(m) = s.m$, $m \in \mathcal{M}(M)$. Now $\text{div}(\gamma(m)) = \text{div}(s) + \text{div}(m)$ and so if $m \in L(d)$ we see that $\text{div}(\gamma(m)) \geq 0$. That is, $\gamma(m) \in \Omega([d])$. The map γ clearly restricts to a linear isomorphism between $L(d)$ and $\Omega([d])$ with inverse defined by $\gamma^{-1}(m) = s^{-1}.m$, $m \in \Omega([d])$. $\square$

Remarks.

1. It follows from Proposition 5.9.7, part 1, that the isomorphism between $L(d)$ and $\Omega([d])$ is uniquely determined by d up to scalar multiplication by elements of $\mathbb{C}^\bullet$.
2. It follows from Proposition 5.9.10 that every non-zero meromorphic function on M can be expressed as a quotient of holomorphic sections of some holomorphic line bundle on M . Indeed, if $m \in \mathcal{M}^*(M)$ let $P_m = \min(0, \text{div}(m))$ denote the polar divisor of m and choose $s \in \Omega([-P_m])$ such that $\text{div}(s) = -P_m$. By Proposition 5.9.10 the map $\gamma^{-1} : \Omega([-P_m]) \rightarrow L(-P_m)$ defined by $\gamma^{-1}(t) = t/s$ is an isomorphism. But $m \in L(-P_m)$ and so m can be written as a quotient of holomorphic sections of the line bundle $[-P_m]$.

Example 2. Let $m \in \mathcal{M}^*(P^n(\mathbb{C}))$. Assuming Chow's theorem, we may write m as a quotient P/Q , where P and Q are homogeneous polynomials of the same degree. If the common degree of P and Q is d , we see from Proposition 5.9.2, that m is the quotient of the holomorphic sections of H^d determined by P and Q .

Given $d \in \mathcal{D}(M)$, we let $E(d)$ denote the set of all effective divisors linearly equivalent to d . The map $\text{div} : \Omega([d]) \rightarrow E(d)$ induces an isomorphism between $E(d)$ and $P(\Omega([d]))$. Hence for every divisor d on M we have natural isomorphisms

$$E(d) \approx P(L(d)) \approx P(\Omega([d])).$$

A family Σ of effective divisors on M is called a *linear system of divisors* on M if there exists a holomorphic line bundle E on M and a (projective) linear subspace V of $P(\Omega(E))$ such that $\Sigma = \text{div}(V)$. We say that Σ is a *complete linear system* if $\Sigma = \text{div}(P(\Omega(E)))$ for some $E \in \text{HLB}(M)$.

If Σ is the linear system of divisors on M corresponding to the subspace V of $P(\Omega(E))$, we define the dimension of Σ , $\dim(\Sigma)$, to be $\dim_{\mathbb{C}}(V)$. We see that if $\dim_{\mathbb{C}}(V) = n$, then Σ is parametrized by $P^n(\mathbb{C})$ and we may write $\Sigma = \{d_t : t \in P^n(\mathbb{C})\}$.

Let $d \in \mathcal{D}(M)$. If V is a linear subspace of $P(\Omega([d]))$, then $\text{div}(V)$ is a linear system on M . In case $V = P(\Omega([d]))$, we see that $\dim(\Sigma) = \dim_{\mathbb{C}} \Omega([d]) - 1 = \dim_{\mathbb{C}} L(d) - 1$ and the linear system equals $E(d)$. By the remarks above it is clear that linear systems on M always correspond to a subspace of $E(d)$ for some divisor d on M and that a linear system is complete if and only if it equals $E(d)$ for some divisor d on M . Suppose $\Sigma = \{d_t : t \in P^n(\mathbb{C})\}$ is a linear system on M . We define the *base locus* B_{Σ} of Σ to be the analytic set

$$B_{\Sigma} = \bigcap_{t \in P^n(\mathbb{C})} |d_t|.$$

Let $t_0, \dots, t_n$ be linearly independent in $P^n(\mathbb{C})$. Then

$$B_{\Sigma} = \bigcap_{j=0}^n |d_{t_j}|.$$

Since Σ is a linear system, there exists $L \in \text{HLB}(M)$ and a linear subspace V of $P(\Omega(L))$ such that $\Sigma = \text{div}(V)$. Let $\{s^0, \dots, s^n\}$ be a basis for V then we clearly have

$$B_{\Sigma} = \bigcap_{j=0}^n (s^j)^{-1}(0).$$

Suppose $B_{\Sigma} = \emptyset$. Then for each $x \in M$, the $(n+1)$ -tuple $(s^0(x), \dots, s^n(x))$ defines a unique point in $P^n(\mathbb{C})$. Indeed, if $s_i^0, \dots, s_i^n$ and $s_j^0, \dots, s_j^n$ are local representations of $s^0, \dots, s^n$ with respect to trivialisations over U_i and U_j respectively, we see that $s_i^p(x) = \theta_{ij}(x) s_j^p(x)$, $0 \leq p \leq n$, $x \in U_{ij}$, where $\theta_{ij} \in A^*(U_{ij})$. Since $\theta_{ij}(x) \neq 0$, $(s_i^0(x), \dots, s_i^n(x)) = (s_j^0(x), \dots, s_j^n(x)) \in P^n(\mathbb{C})$. Hence provided $B_{\Sigma} = \emptyset$, we have a holomorphic map $P : M \rightarrow P^n(\mathbb{C})$ defined by $P(x) = (s^0(x), \dots, s^n(x))$, $x \in M$.

It is of great interest to know whether a given holomorphic line bundle L on M has “enough” holomorphic sections to determine an *embedding* of M in projective space. Notice that if X is a complex submanifold of $P^n(\mathbb{C})$, H_X denotes the hyperplane section bundle of $P^n(\mathbb{C})$ restricted to X and Σ denotes the complete linear system corresponding to $\Omega(H_X)$, then $B_{\Sigma} = \emptyset$ and the corresponding map $P : X \rightarrow P^n(\mathbb{C})$ is an embedding onto the submanifold X .

In conclusion, we see that the theory of divisors and meromorphic functions on a compact complex manifold is intimately related with the theory of holomorphic line bundles and their holomorphic and meromorphic sections. We mention the following basic problems:

1. The existence of non-trivial meromorphic sections of a given holomorphic line bundle.
2. Relations between the dimension of the space of holomorphic sections of a holomorphic line bundle L on M and other invariants of L and M .
3. Conditions for the existence of sufficiently many holomorphic sections of a holomorphic line bundle L on M for it to determine an embedding of M in projective space.

Geometric genus. Let M be a compact complex manifold of dimension m and let $K(M)$ denote the canonical bundle $\Lambda^m TM^*$ of M . We define the *geometric genus* $p_g(M)$ of M to be $\dim_{\mathbb{C}}(\Omega(K(M)))$.

The geometric genus is obviously a biholomorphic invariant.

Proposition 5.9.11. The geometric genus is invariant under blowing ups with non-singular centres.

Proof. Let $\pi : \widetilde{M} \rightarrow M$ denote the blow-up of M with centre p and exceptional variety E . We have an induced map $\pi^* : \Omega(K(M)) \rightarrow \Omega(K(\widetilde{M}))$. Since π restricts to a biholomorphic map between $\widetilde{M} \setminus E$ and $M \setminus \{p\}$ we see by uniqueness of analytic continuation that π^* is injective. On the other hand if $\phi \in \Omega(K(\widetilde{M}))$, we may define $\tilde{\phi} = (\pi|_{\widetilde{M} \setminus E})_* \phi \in K(M \setminus \{p\})$. By Hartog's theorem, $\tilde{\phi}$ extends to a holomorphic section of $K(M)$ which we denote by $\pi_* \phi$. Clearly $\pi^* \pi_* \phi = \phi$, $\phi \in K(\Omega(\widetilde{M}))$ and so π^* is a linear isomorphism. Hence $p_g(M) = p_g(\widetilde{M})$. The proof for general non-singular centres is similar, using the second Riemann removable singularities theorem (Exercise 5, §2, Chapter 4) and we leave details to the reader. $\square$

Remark. The proof given for proposition 5.9.11 also shows that the numbers

$$\dim_{\mathbb{C}} \Omega(\Lambda^p TM^*), \quad p > 0,$$

are also invariant under blowings up with non-singular centres.

The geometric genus is in fact a *bimeromorphic* invariant. Whilst we shall not define bimeromorphic maps here (see Ueno [1] for details and references) we point out that bimeromorphic maps are the complex analytic analogue of the birational maps of algebraic geometry. Moreover, if two complex analytic *surfaces* are bimeromorphic then one is obtained from the other by a finite sequence of blow ups and blow downs (see Kodaira [1]). This result is, however, false in higher dimensions. We refer to Hartshorne [1, pages 412–414] for a discussion of the higher dimensional case and references. Other bimeromorphic invariants related to the geometric genus are the *plurigenera* $p_r(M)$ defined by $p_r(M) = \dim_{\mathbb{C}} \Omega(\otimes^r K(M))$, $r > 0$. The proof that the plurigenera are invariant under blowing ups is similar to that of Proposition 5.9.11.

Holomorphic line bundles on complex tori and theta functions. We conclude this section by indicating the rôle of holomorphic line bundles and their sections in the study of meromorphic functions on complex tori. For further details, proofs and references the reader may consult Cornalba [1], Griffiths and Harris [1], Swinnerton-Dyer [1] and Weil [1].

Let $T = \mathbb{C}^n / \Lambda$ be an n -dimensional complex torus with period lattice Λ and $\pi : \mathbb{C}^n \rightarrow T$ denote the quotient map. If L is a holomorphic line bundle on T then $\pi^* L$ is a holomorphic line bundle on $\mathbb{C}^n$ (As usual, $\pi^* L$ denotes the pull-back of the bundle L by π —see the exercises at the end of §5, Chapter 1). By Corollary 5.8.4, every holomorphic line bundle on $\mathbb{C}^n$ is holomorphically trivial and so $\pi^* L \cong \underline{\mathbb{C}} = \mathbb{C}^n \times \mathbb{C}$. Fixing an isomorphism of $\pi^* L$ with $\underline{\mathbb{C}}$, we may regard L as the quotient of $\mathbb{C}^n \times \mathbb{C}$ under the identifications

$$(z, v) \sim (z + \lambda, f_{\lambda}(z)v), \quad z \in \mathbb{C}^n, \quad v \in \mathbb{C}, \quad \lambda \in \Lambda,$$

where the functions $f_\lambda \in A^*(\mathbb{C}^n)$ satisfy the relations

$$f_\lambda(z + \mu)f_\mu(z) = f_{\lambda+\mu}(z), \quad z \in \mathbb{C}^n, \quad \lambda, \mu \in \Lambda. \quad (\text{A})$$

A (holomorphic) section of L corresponds to a $\mathbb{C}$ -valued (holomorphic) function θ on $\mathbb{C}^n$ which satisfies the functional equation

$$\theta(z + \lambda) = f_\lambda(z)\theta(z), \quad z \in \mathbb{C}^n, \quad \lambda \in \Lambda.$$

Such functions are called *theta functions* (relative to the family f_λ). The quotient of any two non-zero theta functions defines a Λ -periodic meromorphic function on $\mathbb{C}^n$ and hence a meromorphic function on T . Conversely, by Remark 2 following Proposition 5.9.10, every meromorphic function on T may be represented as the quotient of two theta functions (associated to the same family f_λ). The study of meromorphic functions on T may therefore be reduced to the study of theta functions on $\mathbb{C}^n$. In fact for $n > 1$, this approach to the theory of meromorphic functions on complex tori is much more effective than any direct attempt to construct meromorphic functions as we did in case $n = 1$ with the Weierstrass $\wp$ -function and its derivative.

Suppose that α, β are isomorphism of π^*L with $\underline{\mathbb{C}}$. Then there exists $\phi \in A^*(\mathbb{C}^n)$ such that $\beta = \phi \cdot \alpha$. If α, β correspond to the families $\{f_\lambda : \lambda \in \Lambda\}, \{g_\lambda : \lambda \in \Lambda\}$ respectively, then it is easily verified that f_λ and g_λ are related by

$$g_\lambda(z) = \phi(z + \lambda)\phi(z)^{-1}f_\lambda(z), \quad z \in \mathbb{C}^n, \quad \lambda \in \Lambda.$$

By suitable choice of ϕ we might hope to put the functions f_λ into a “standard” form. This amounts to obtaining a classification of $\text{HLB}(T)$.

Example 3. The trivial line bundle on T is defined by the family $f_\lambda \equiv 1, \lambda \in \Lambda$. Given $\phi \in A^*(\mathbb{C}^n)$, we may write $\phi = \exp(f)$, for some $f \in A(\mathbb{C}^n)$. The family $g_\lambda(z) = \exp(f(z + \lambda) - f(z))$ also defines the trivial line bundle on T . Since $\phi(z + \lambda) = g_\lambda(z)\phi(z)$, we see that ϕ is a theta function for this family. Any theta function of this type is called a *trivial theta function* by virtue of the fact that it corresponds to a constant (non-zero) section of the trivial holomorphic line bundle on T . Of special interest to us will be the case when $f(z) = az^2 + bz + c, a, b, c \in \mathbb{C}$. We have $g_\lambda(z) = \exp(2az + z\lambda^2 + b\lambda)$ and the corresponding trivial theta function is $\exp(az^2 + bz + c)$.

If L is a holomorphic line bundle on T it is natural to try to define L by functions f_λ where $f_\lambda(z) = \exp F(z, \lambda)$ and $F(z, \lambda)$ is affine linear in z . Suppose L is defined by a family of functions of this type. The relations (A) impose conditions on the $F(z, \lambda)$ and, multiplying by suitable non-zero analytic functions, it is not hard to show that the f_λ can be put in the form

$$f_\lambda(z) = m(\lambda) \exp(\pi H(z, \lambda) + \frac{1}{2}\pi H(\lambda, \lambda)),$$

where H is an Hermitian form whose imaginary part E is integer valued on $\Lambda \times \Lambda$ and $m : \Lambda \rightarrow S^1 \subset \mathbb{C}$ satisfies $m(\lambda)m(\mu) = m(\lambda + \mu) \exp(\pi i E(\lambda, \mu))$ for all $\lambda, \mu \in \Lambda$.

We denote the line bundle on T corresponding to H and m by $L(H, m)$. By a theorem of Apell and Humbert every holomorphic line bundle on T is isomorphic to a line bundle of the form $L(H, m)$. Moreover, $L(H_1, m_1) \cong L(H_2, m_2)$ iff $H_1 = H_2$ and $m_1 = m_2$. The proof of this result may be found in the references. Here we only remark that $L(H_1, m_1)$ and $L(H_2, m_2)$ are isomorphic as *complex* line bundles iff $H_1 = H_2$.

A holomorphic section of $L(H, m)$ corresponds to a theta function θ which satisfies the functional equation

$$\theta(z + \lambda) = m(\lambda) \exp(\pi H(z, \lambda) + \frac{1}{2}\pi H(\lambda, \lambda))\theta(z).$$

Example 4. (The Weierstrass σ -function). We follow the notation and assumptions of §4 of Chapter 4. Thus we assume $n = 1$ and let $\wp(z)$ denote the Weierstrass elliptic function associated to the lattice L generated by $\{\omega_1, \omega_2\}$. Integrating $\wp$ we obtain the Weierstrass zeta function

$$\zeta(z) = z^{-1} - \sum' [(z + \omega)^{-1} - \omega^{-1} + 2\omega^2]$$

(the prime denotes that the sum is over non-zero elements of L). $\zeta(z + \omega) - \zeta(z)$ is constant, not necessarily zero, for all $\omega \in L$ and we define

$$\eta_1 = \zeta(z + \omega_1) - \zeta(z); \quad \eta_2 = \zeta(z + \omega_2) - \zeta(z). \quad (\text{B})$$

As in the proof of part 3 of Theorem 4.4.2, the integral of ζ round a period parallelogram for L equals $2\pi i = \omega_1 \eta_2 - \omega_2 \eta_1$ (Legendre's relation).

Exponentiating the integral of ζ we obtain a (single valued) analytic function called the Weierstrass σ -function. We have

$$\begin{aligned} \sigma(z) &= \exp \left(\log z + \int_0^z (\zeta(t) - t^{-1}) dt \right) \\ &= 2 \prod' \left(\frac{z + \omega}{\omega} \exp \left(-\frac{z}{\omega} + \frac{z^2}{2\omega^2} \right) \right) \end{aligned}$$

Exponentiating the integrals of the relations (B) above, substituting $z = -\frac{1}{2}\omega_1, -\frac{1}{2}\omega_2$ and using the fact that σ is an odd function which does not vanish at $\frac{1}{2}\omega_1, \frac{1}{2}\omega_2$, we find that

$$\sigma(z + \omega) = (-1)^n \sigma(z) \exp(\eta(z + \tfrac{1}{2}\omega)),$$

where $\omega = n_1\omega_1 + n_2\omega_2$, $\eta = n_1\eta_1 + n_2\eta_2$, $n = n_1n_2 + n_1 + n_2$ and $n_1, n_2 \in \mathbb{Z}$. Hence σ is a theta function on $\mathbb{C}$ corresponding to the family $\{g_\omega \in A^*(\mathbb{C}) : \omega \in L\}$ defined by

$$g_\omega(z) = (-1)^n \exp(\eta(z + \tfrac{1}{2}\omega)), \quad \omega \in L.$$

However, the functions g_ω are not in the standard form that we gave above. Recall from §4 of Chapter 4 that the lattice L has Riemann form defined by $A(y, z) = S^{-1} \text{Im}(y\bar{z})$, where $S = \text{Im}(\omega_1\bar{\omega}_2)$. Associated to A we have the hermitian form H defined by $H(y, z) = y\bar{z}/S$. The imaginary part of H equals A and is integer valued on $L \times L$. We now define

$$f_\omega(z) = (-1)^n \exp(\pi H(z, \omega) + \tfrac{1}{2}\pi H(\omega, \omega)), \quad \omega \in L.$$

Set $a = (2\omega_1)^{-1}(S^{-1}\pi\bar{\omega}_1 - \eta_1)$ and let $\theta_0(z)$ denote the trivial theta function $\exp(az^2)$. The reader may verify, using Legendre's relation, that if we define $\tilde{\sigma}(z) = \theta_0(z)\sigma(z)$, then $\tilde{\sigma}(z + \omega) = f_\omega(z)\tilde{\sigma}(z)$. Hence we have put the Weierstrass σ -function in standard form. Set $T = \mathbb{C}/L$. Because the holomorphic line bundle on T associated to the family f_ω actually generates $\text{HLB}(T)$ we are able to give a particularly simple description of meromorphic functions on T . Thus if $d = \sum_{k=1}^n n_k z_k \in \mathcal{D}(T)$, $\deg(d) = 0$ and $\sum_{k=1}^n n_k z_k = e$, we may define

$$m_d = \prod_{k=1}^n \sigma(z - a_k)^{n_k}, \quad \text{where } \pi(a_k) = z_k, \quad k = 1, \dots, n, \quad \text{and } \sum_{k=1}^n n_k a_k = 0.$$

It is easily verified that m_d is L -elliptic and defines a meromorphic function on T with divisor d .

The theory of theta functions of more than 1 complex variable is highly developed and we conclude by mentioning just two important results:

1. If there exists $L(H, m) \in \text{HLB}(T)$ such that H is positive definite (that is, H is associated to a Riemann form), then the complete linear system defined by $L(3H, m) = L(H, m)^3$ gives a projective embedding of T .
2. The dimension of $\Omega(L(H, m))$ can be computed and is equal to $\sqrt{\det E}$, where E is the imaginary part of H and the determinant is computed relative to any basis of the period lattice of T .

Exercises.

1. Let s be a meromorphic section of the holomorphic line bundle E . Define the zero and pole sets $Z(s)$, $P(s)$ of s and show that s is holomorphic if and only if $P(s) = \emptyset$.
2. Let $p \geq 1$ and Σ denote the complete linear system on $P^n(\mathbb{C})$ corresponding to $\Omega(H^p)$. Show that

- a) $\dim(\Sigma) = \binom{n+p}{n} - 1$.
- b) The base locus of Σ is empty.
- c) Σ determines an embedding of $P^n(\mathbb{C})$ in $P^N(\mathbb{C})$, $N = \dim(\Sigma)$ (The “ p -tuple embedding”).

(In case $n = 2$, $p = 2$, we obtain an embedding of $P^2(\mathbb{C})$ in $P^5(\mathbb{C})$. The corresponding surface in $P^5(\mathbb{C})$ is called the *Veronese surface*).

3. Let E be a holomorphic line bundle on the complex manifold M . Show that, as vector spaces, $\mathcal{M}(E) \cong \mathcal{M}(M)$, provided that E admits a non-trivial meromorphic section.
4. Suppose V is a smooth analytic hypersurface in the compact complex manifold M (that is, V is a closed submanifold of M of codimension 1). Let $N_V = (TM|V)/TV$ denote the normal bundle of V . Show that

- a) $N_V^* \approx \{v \in TM^*|V : v \text{ is zero on } TV^*\}$.
- b) $N_V^* \cong [-V]|V$.
- c) $K(V) \cong (K(M) \otimes [V])|V$.

(Hints: For b), show that the local defining equations for V determine a non-zero holomorphic section of $N_V^* \otimes [V]|V$; for c), use the exact sequence $0 \rightarrow N_V^* \rightarrow TM^*|V \rightarrow TV^* \rightarrow 0$ and exercise 5, §1).

5. Suppose that $f_\lambda \in A^*(\mathbb{C}^n)$, $\lambda \in \Lambda$, define the holomorphic line bundle L on $\mathbb{C}^n/\Lambda$. Show that f_λ^{-1} , $\bar{f}_\lambda$, $\bar{f}_\lambda^{-1}$ respectively define the complex line bundles L^* , $\bar{L}$, $\bar{L}^*$ on $\mathbb{C}^n/\Lambda$.
6. Let Λ be a lattice in $\mathbb{C}^n$ and H be an hermitian form whose imaginary part is integer valued on $\Lambda \times \Lambda$. Show that $L(H, 1)$ is a holomorphic line bundle on $\mathbb{C}^n/\Lambda$. Now set $L = L(H, 1)$. Show that a function $\theta : \mathbb{C}^n \rightarrow \mathbb{C}$ determines a section of $L^* \otimes \bar{L}^*$ if and only if for all $z \in \mathbb{C}^n$ we have

$$\theta(z + \lambda) = \exp(-\pi(2 \operatorname{Re}(H(z, \lambda)) + H(\lambda, \lambda)))\theta(z), \quad \lambda \in \Lambda.$$

Deduce that $\phi(z) = \exp(-\pi H(z, z))$ determines a nowhere vanishing smooth section of $L^* \otimes \bar{L}^*$ (We may think of ϕ as determining a canonical hermitian form on $L = L(H, 1)$).

10 Pseudoconvexity and Stein manifolds.

In this section we shall show how some of the pseudoconvexity definitions we discussed in Chapter 2 may be generalised to arbitrary non-compact complex manifolds.

We start with a few remarks about Hermitian forms on an m -dimensional complex vector space E . Recall that $H : E \times E \rightarrow \mathbb{C}$ is said to be an *Hermitian form* if

1. $H(x, y) = \overline{H(y, x)}$, $x, y \in E$.
2. $H(ax_1 + bx_2, y) = aH(x_1, y) + bH(x_2, y)$, $a, b \in \mathbb{C}$, $x_1, x_2, y \in E$.

We say that H is *positive definite* if, in addition,

3. $H(x, x) > 0, x \neq 0$.

Conditions 1 and 2 imply that H is conjugate complex linear in the second variable. Consequently, an Hermitian form may be regarded as a complex bilinear map $H : E \times \bar{E} \rightarrow \mathbb{C}$ satisfying the conjugate symmetry condition 1. Since the space of complex bilinear maps of $E \times \bar{E}$ to $\mathbb{C}$ is naturally isomorphic to $E^* \otimes \bar{E}^*$, we may also regard H as lying in $E^* \otimes \bar{E}^*$. Thus, relative to a basis of E , we may write H in coordinates as

$$H = \sum_{i,j=1}^m h_{i\bar{j}} dz_i \otimes d\bar{z}_j.$$

The conjugate symmetry condition amounts to requiring that the matrix $[h_{i\bar{j}}]$ be Hermitian. That is, $\overline{h_{i\bar{j}}} = h_{j\bar{i}}, 1 \leq i, j \leq m$.

Next observe that $E^* \otimes \bar{E}^* \approx \Lambda^{1,1}(E')$. If we regard H as lying in $\Lambda^{1,1}(E')$, the conjugate symmetry condition amounts to requiring that $\bar{H} = -H$ (conjugate the form $\sum h_{i\bar{j}} dz_i \wedge d\bar{z}_j$).

Finally note that since $E^* \otimes \bar{E}^* \approx L(E, \bar{E}^*)$, we may regard H as an element of $L(E, \bar{E}^*)$. In this case conjugate symmetry amounts to $H = \bar{H}^*$.

Recall that if we choose a complex basis for E and let H have matrix $[h_{i\bar{j}}]$ relative to this basis, then the integers

$$\begin{aligned} n(H) &= \text{number of negative eigenvalues of } [h_{i\bar{j}}], \\ z(H) &= \text{number of zero eigenvalues of } [h_{i\bar{j}}], \\ p(H) &= \text{number of positive eigenvalues of } [h_{i\bar{j}}], \end{aligned}$$

are invariants of H which do not depend on the choice of basis for E .

Definition 5.10.1. Let M be a complex manifold. An Hermitian form on M is a section H of $TM^* \otimes \bar{T}M^*$ such that $H(x)$ is an Hermitian form on $T_x M$ for all $x \in M$.

Remark. We may equivalently define an Hermitian form on M to be a $(1, 1)$ -form H satisfying $\bar{H} = -H$.

Definition 5.10.2. Let M be a complex manifold and $\phi \in C_{\mathbb{R}}^2(M)$. The *Levi form* of ϕ is the $(1, 1)$ -form defined by

$$L(\phi) = \partial \bar{\partial} \phi.$$

Since $\overline{\partial \bar{\partial} \phi} = \bar{\partial} \partial \phi = -\partial \bar{\partial} \phi$, we see that $L(\phi)$ is an Hermitian form on M . In local coordinates,

$$L(\phi) = \sum_{i,j=1}^m \frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j} dz_i \otimes d\bar{z}_j,$$

and so $L(\phi)$ is an invariant version of the Levi form we discussed in Chapter 2. Suppose now that M is a relatively compact domain in the m -dimensional complex manifold $\widetilde{M}$ and that M has C^2 boundary ∂M . We say that $\phi \in C_{\mathbb{R}}^2(\widetilde{M})$ is a *defining function* for M if

1. $M = \{x \in M : \phi(x) < 0\}$.
2. $\partial M = \phi^{-1}(0)$.
3. $d\phi \neq 0$ on ∂M .

Given a defining function ϕ for M we may define the *holomorphic tangent space* $T_x\partial M$ to ∂M at x by $T_x\partial M = \{v \in T_xM : d\phi(x)(v) = 0\}$. Set

$$T\partial M = \bigcup_{x \in \partial M} T_x\partial M.$$

It is straightforward to verify that $T\partial M$ is an $(m-1)$ -dimensional complex vector bundle which is defined independently of choices of defining function for M (Use Lemmas 2.5.1, 2.5.7).

Given $x \in \partial M$, we set $L(\phi)(x) = L(\phi)|_{T_x\partial M}$ and define

$$n(x) = n(L(\phi)(x))$$

$$z(x) = z(L(\phi)(x))$$

$$p(x) = p(L(\phi)(x)).$$

Clearly $n(x) + z(x) + p(x) = m - 1$ and it is a straightforward exercise to verify that $n(x)$, $z(x)$ and $p(x)$ depend only on $x \in \partial M$ and not the choice of defining function ϕ (see §5, Chapter 2).

Definition 5.10.3. Let M be a relatively compact domain of the complex manifold $\widetilde{M}$ and assume that M has C^2 boundary. Suppose that M has defining function ϕ . Then we say that M is *q-pseudoconvex* (respectively, *strictly q-pseudoconvex*) if for all $x \in \partial M$ we have

$$n(x) \leq q \quad (\text{respectively, } n(x) + z(x) \leq q).$$

Remark. As in Proposition 2.5.12, we may show that q -pseudoconvexity is a local property of the boundary.

Example 1. Let Ω be an L.p. (respectively, s.L.p.) domain in $\mathbb{C}^n$. Then Ω is 0-pseudoconvex (respectively, strictly 0-pseudoconvex).

Remark. In the sequel we shall always refer to 0-pseudoconvex (respectively, strictly 0-pseudoconvex) domains as L.p. (respectively, s.L.p.) domains.

Proposition 5.10.4. Let M be an s.L.p. domain in $\widetilde{M}$. Then there exists a C^2 defining function ϕ for M such that $L(\phi)|_{\partial M}$ is positive definite.

Proof. Same as the proofs of Propositions 2.5.5 and Lemma 2.5.11. □

In Chapter 7 we shall prove the basic result of Grauert to the effect that an s.L.p. domain is holomorphically convex. However, an s.L.p. domain need not be Stein.

Examples.

2. The unit Euclidean disc $E(1) \subset \mathbb{C}^n$ is s.L.p. (take $\phi(z) = \sum |z_i|^2 - 1$). Suppose $n > 1$ and let $M, \widetilde{M}$ denote the result of blowing up $E(1), \mathbb{C}^n$ at zero. Then M will be an s.L.p. domain in $\widetilde{M}$ as we may choose a defining function for M which is equal to ϕ on a neighbourhood of ∂M in $\widetilde{M}$ which does not contain the exceptional variety of the blowing up. However, M cannot be Stein as the exceptional variety of the blowing up is a compact complex submanifold of M biholomorphic to $P^{n-1}(\mathbb{C})$. In particular, we cannot separate points on $P^{n-1}(\mathbb{C})$ by holomorphic functions on M (Proposition 4.2.4).
3. Let L denote the universal line bundle on $P^n(\mathbb{C})$. Then L is biholomorphic to $\mathbb{C}^{n+1}$ blown up at zero. Hence, by example 2, there is an s.L.p. neighbourhood of the zero section of L . More generally, if X is any compact complex submanifold of $P^n(\mathbb{C})$ then $L_X = L|_X$ has an s.L.p. neighbourhood of the zero section.

Motivated by example 2, we now give an important definition due to Grauert [1].

Definition 5.10.5. Let E be a holomorphic vector bundle on the compact complex manifold M . We say that E is *weakly negative* if there is an s.L.p. neighbourhood of the zero section of E . We say that E is *weakly positive* if E^* is weakly negative.

Remark. One of the main theorems we prove in Chapter 7 is that if a compact complex manifold M admits a weakly positive vector bundle then M is algebraic.

Our definition of the q -pseudoconvexity of a complex manifold M depended on representing M as a domain in some larger complex manifold. Our next aim is to present an intrinsic definition of pseudoconvexity.

Definition 5.10.6. Suppose that M is an m -dimensional complex manifold and let $\phi \in C_{\mathbb{R}}^{\infty}(M)$. We say that ϕ is *strictly q -plurisubharmonic* (abbreviated, strictly q -psh) if $L(\phi)(x)$ has at least $m - q$ positive eigenvalues at every point x of M .

Recall from §5, Chapter 2, that $\phi \in C_{\mathbb{R}}^{\infty}(M)$ is said to be an *exhaustion function* for M if for all $c \in \mathbb{R}$,

$$M_c = \{x \in M : \phi(x) < c\}$$

is a relatively compact subset of M .

Definition 5.10.7. A complex manifold M is said to be *q -complete* if there exists a strictly q -psh exhaustion function ϕ on M .

Remarks.

1. Clearly q -complete implies $(q + 1)$ -complete.
2. We often refer to 0-complete manifolds as being *holomorphically complete*.

Theorem 5.10.8. Every Stein manifold is 0-complete.

Proof. Exactly the same as the proof of Theorem 2.5.20. □

Remark. We shall prove in Chapter 11 that every 0-complete manifold is Stein. Next we wish to say a few words about the relationship between q -pseudoconvexity and q -completeness. Suppose that M is a domain in $\widetilde{M}$ and that M has C^2 boundary and $\widetilde{M}$ is Stein. Then it can be shown that if M is q -pseudoconvex then M is q -complete (see Eastwood and Vigna Suria [1]). We only remark here that if $q = 0$ then the proof that q -pseudoconvexity implies q -completeness is similar to the proof of Proposition 2.5.15 and makes use of the elementary fact that we can find a Stein neighbourhood of $\bar{M}$ in $\widetilde{M}$ which embeds in the unit Euclidean disc in some $\mathbb{C}^N$ (cf. Lemma 7.2.13). Conversely, it can be shown that q -completeness implies q -pseudoconvexity.

In Chapter 11 we shall show that q -completeness implies existence theorems for the $\bar{\partial}$ -operator. To be precise, we shall show that if M is q -complete, E is a holomorphic vector bundle on E and $\phi \in C^{r,s}(M, E)$ is $\bar{\partial}$ -closed then there exists $\psi \in C^{r,s-1}(M, E)$ such that $\bar{\partial}\psi = \phi$ provided that $s \geq q + 1$, $r \geq 0$. In particular, if M is Stein we can always solve the generalised Cauchy–Riemann equations on M .

Although it is not our intention to say very much here about q -pseudoconvexity or q -completeness in case $q \neq 0$, we remark that q -pseudoconvexity may be regarded as a measure of how far away a domain is from being s.L.p. Moreover, the concept may be related to extension problems in complex analysis. See, for example, Eastwood and Vigna Suria [1] and Andreotti and Grauert [1]. We give one example to show how we may construct q -complete spaces, $q \neq 0$.

Example 4. (see also Griffiths [1, Theorem H], Serre [1], Simha [1], Vesentini [1]).

Let M be a p -complete manifold and $f_1, \dots, f_{q+1} \in A(M)$. Set $Z = Z(f_1, \dots, f_{q+1})$. Then we claim that $Y = M \setminus Z$ is $(p+q)$ -complete. In particular if M is Stein and $q = 0$, Y is 0-complete and therefore Stein by the result cited above. Of course, in this case it is easy to verify directly that Y is holomorphically convex and therefore Stein as $f_1^{-1} \in A(Y)$.

Suppose ϕ is a strictly p -psh exhaustion function on M . Choose a C^∞ function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

1. $g(t), g'(t), g''(t) > 0, t \in \mathbb{R}$.
2. $g(t) \rightarrow \infty$ as $t \rightarrow \infty$.
3. $g\phi > \sum_{i=1}^{q+1} |f_i|^2$ on Y .

Set $\theta = g\phi - \log \sum_{i=1}^{q+1} |f_i|^2$. Since $L(\log \sum_{i=1}^{q+1} |f_i|^2)$ is positive semi-definite with at most q positive eigenvalues we see easily that θ is strictly $(p+q)$ -psh on Y . It is sufficient to show that θ is an exhaustion function on Y . That is, $Y_c = \{x \in Y : \theta(x) < c\}$ is a relatively compact subset of Y for all $c \in \mathbb{R}$. Suppose $\theta(x) < c$. Certainly,

$$\sum_{i=1}^{q+1} |f_i|^2 \geq e^{g(\phi)-c} > e^{-c} > 0.$$

Consequently, $Y_c \subset M \setminus U$, where U is an open neighbourhood of Z . On the other hand

$$e^{g(\phi)}/g(\phi) < e^c$$

and so $g(\phi) < e^c$ on Y_c . Hence $Y_c \subset M_C, C = g^{-1}(e^c)$. It follows that Y_c is a relatively compact subset of Y .

Finally, we conclude this section with a few brief remarks about the Bergman kernel function of a complex manifold. Given an m -dimensional complex manifold M , let

$$L^2(M) = \{f \in \Omega^m(M) : \int_M f \wedge \bar{f} < \infty\}.$$

Then $L^2(M)$ has the structure of a Hilbert space with inner product defined by $(f, g) = \int_M f \wedge \bar{g}$. As in §6, Chapter 2 we can construct a Bergman kernel function $K(z, \bar{\zeta})$ for $L^2(M)$ and then $K(z, \bar{z})$ defines a smooth section of $C^{m,m}(M)$ (cf. Proposition 2.6.6). If instead, we start with an Hermitian metric on M and corresponding measure $d\lambda$ on M , we may define $L^2(M) = \{f \in A(M) : \int |f|^2 d\lambda < \infty\}$. In this case we may show that there exists a Bergman kernel function $K(z, \bar{\zeta})$ for $L^2(M)$ and that for important examples $\log K(z, \bar{z})$ will be a strictly q -psh exhaustion function on M (cf. Proposition 2.6.8).

Chapter 6

Sheaf Theory

Introduction

In Section 1 of this chapter we present the basic definitions and constructions of sheaf theory with many motivating examples. In section 2 we give an application of sheaf theory to prove the existence and uniqueness of the envelope of holomorphy of a Riemann domain. In section 3 we define the sheaf cohomology groups of a sheaf of groups over a paracompact space using fine resolutions. Amongst the most important results we prove are Leray's theorem and the existence of a canonical, natural isomorphism between Čech cohomology and sheaf cohomology. We conclude with a number of important examples and computations involving the 1st. Chern class.

1 Sheaves and presheaves

Our aim in this section is to develop the theory of sheaves and show how it provides a unifying topological framework for the study of a diverse range of structures on topological spaces. Our presentation will be geared towards applications in complex analysis and the reader may consult Godement [1] or Tennison [1] for more extensive and general expositions of the theory of sheaves.

Let X be a topological space with topology of open sets $\mathcal{U}$.

Definition 6.1.1. A *presheaf of groups* on X is a collection of groups $G(U)$, one for each $U \in \mathcal{U}$, together with group homomorphisms $r_{VU} : G(U) \rightarrow G(V)$, defined for $V, U \in \mathcal{U}$ and $V \subset U$, such that

1. If $U = \emptyset$, $G(U)$ is the zero group.
2. For all $U \in \mathcal{U}$, r_{UU} is the identity.
3. For $U, V, W \in \mathcal{U}$ and $W \subset V \subset U$, we have $r_{WV}r_{VU} = r_{WU}$.

We usually denote the presheaf by $\{G(U), r_{VU}\}$ or just G .

Remarks.

1. Replacing the word “group” everywhere by “set”, “ring”, “field”, “algebra”, etc. we may define presheaves of sets, rings, fields, algebras, etc. We shall assume these definitions in the sequel.
2. The homomorphisms r_{VU} occurring in the definition are usually called *restriction homomorphisms*. In all our examples they will be restriction maps and we therefore generally omit any explicit specification.

As we shall soon see, many basic structures in analysis can be formulated in terms of presheaves.

Examples.

1. For each $U \in \mathcal{U}$, let $C(U)$ denote the ring of continuous $\mathbb{C}$ -valued functions on U . Given $U, V \in \mathcal{U}$, with $V \subset U$, define $r_{VU} : C(U) \rightarrow C(V)$ to be restriction of continuous functions on U to V . Then $C_X = \{C(U), r_{VU}\}$ is a presheaf of rings on X : The *presheaf of continuous $\mathbb{C}$ -valued functions* on X .
2. Let Γ be a ring with discrete topology. For each $U \in \mathcal{U}$, let $C(U, \Gamma)$ denote the ring of continuous Γ -valued functions on U . Defining r_{VU} as restriction, $V \subset U$, the set $\Gamma_X = \{C(U, \Gamma), r_{VU}\}$ is a presheaf of rings on X : The *presheaf of locally constant Γ -valued functions* on X .
3. Suppose X has the structure of a differential manifold. Let $C^k(U)$ denote the ring of C^k $\mathbb{C}$ -valued functions on U , $U \in \mathcal{U}$, $1 \leq k < \infty$. Then $C_X^k = \{C^k(U), r_{VU}\}$ is a presheaf of rings on X : The *presheaf of C^k $\mathbb{C}$ -valued functions* on X . We shall let $D_X = \{C^\infty(U), r_{VU}\}$ denote the *presheaf of C^∞ $\mathbb{C}$ -valued functions* on X .
4. Suppose X has the structure of a complex manifold. We let $O_X = \{A(U), r_{VU}\}$ denote the presheaf of analytic $\mathbb{C}$ -valued functions on X . In the sequel we usually refer to O_X as the *Oka presheaf of X* . For each U let $S(U)$ denote the multiplicatively closed subset of $A(U)$ consisting of all analytic functions on U which do not vanish identically on any component of U . Set $M(U) = A(U)_{S(U)}$. That is, $M(U)$ is the quotient ring of $A(U)$ with respect to the multiplicative system $S(U)$ (See Zariski and Samuel [1, page 46] and observe that if U is connected $M(U)$ is just the quotient field of $A(U)$). Then, defining r_{VU} as restriction, $M_X = \{M(U), r_{VU}\}$ is a presheaf of rings on X : The *presheaf of meromorphic functions* on X .
5. Let Z be an analytic subset of the complex manifold X . For each $U \in \mathcal{U}$, let $I_Z(U) = \{f \in A(U) : f|_Z \equiv 0\}$. Then $I_Z = \{I_Z(U), r_{VU}\}$ is a *presheaf of ideals* of O_X . That is, for each $U \in \mathcal{U}$, $I_Z(U)$ is an ideal of $A(U)$. We call I_Z the *ideal presheaf* of Z .
6. Let E denote a holomorphic vector bundle over X . For each $U \in \mathcal{U}$, we let $\underline{E}(U)$ denote the space of holomorphic sections of E over U . Each $\underline{E}(U)$ is an $A(U)$ -module in the obvious way and the presheaf $\underline{E} = \{\underline{E}(U), r_{VU}\}$ is thus an example of a *presheaf of O_X -modules*. We call $\underline{E}$ the *presheaf of holomorphic sections* of E . In case E is a smooth, not necessarily holomorphic, vector bundle over X , we let $\underline{E}_\infty = \{\underline{E}_\infty(U), r_{VU}\}$ denote the presheaf of C^∞ sections of E . Thus, $\underline{E}_\infty$ is an example of a presheaf of D_X -modules and, of course, the construction works for any differential manifold X .

Whilst a presheaf contains essentially all the information about a particular structure on a topological space, it is a large, seemingly cumbersome, object. We now describe the process of “sheafification” whereby out of every presheaf we can construct a topological space in such a way that, for all important examples, no information is lost.

Let $R = \{R(U), r_{VU}\}$ be a presheaf of rings on the topological space X . Fix $x \in X$. We define an equivalence relation $\sim_x$ on the rings $R(U)$ for which $x \in U$. Suppose that $U, V \in \mathcal{U}_x$ and $f \in R(U)$, $g \in R(V)$. We say f is equivalent to g at x , $f \sim_x g$, if and only if there exists $W \in \mathcal{U}_x$, $W \subset U \cap V$, such that

$$r_{WU}(f) = r_{WV}(g).$$

Using conditions 2 and 3 of Definition 6.1.1, the reader may easily verify that $\sim_x$ is an equivalence relation. We denote the set of $\sim_x$ equivalence classes by R_x . The set R_x inherits the structure of a ring from the rings $R(U)$ and we let $r_{U,x} : R(U) \rightarrow R_x$ denote the corresponding

“equivalence class” ring homomorphisms, defined for $U \in \mathcal{U}_x$. In the sequel we often write f_x for $r_{U,x}(f)$, $f \in R(U)$, and call f_x the *germ* of f at x .

Examples.

7. If we let $\mathcal{O}$ denote the Oka presheaf of the complex manifold X , then $\mathcal{O}_x$ is just the ring of germs of analytic functions at x (see also Chapter 3, §1; Chapter 4, §1).
8. If we let $\mathcal{M}$ denote the presheaf of meromorphic functions on the complex manifold X , then $\mathcal{M}_x$ is the field of germs of meromorphic functions at x (see Chapter 3, §4; Chapter 4, §1).

Set

$$\mathcal{R} = \bigcup_{x \in X} \mathcal{R}_x$$

and let $\pi : \mathcal{R} \rightarrow X$ denote the projection defined by mapping points in $\mathcal{R}_x$ to x . We now topologise $\mathcal{R}$. Given $f \in R(U)$, we have a section $\tilde{f}$ (relative to π) of $\mathcal{R}$ over U defined by $\tilde{f}(x) = f_x$, $x \in U$. For a base of open sets for the topology of $\mathcal{R}$, we take the collection of sets $\tilde{f}(U) \subset \mathcal{R}$, over all $U \in \mathcal{U}$ and $f \in R(U)$. The reader may easily verify that this defines the base for a topology on $\mathcal{R}$. Clearly the local sections $\tilde{f} : U \rightarrow \mathcal{R}$, $f \in R(U)$, are continuous in this topology.

Lemma 6.1.2. With the above notation we have

1. $\pi : \mathcal{R} \rightarrow X$ is a local homeomorphism.
2. The induced topology on $\mathcal{R}_x \subset \mathcal{R}$ is discrete for all $x \in X$.

Proof. Property 2 is immediate from 1. For 1 we note that $\pi\tilde{f}$ is the identity map on U for all $f \in R(U)$ and so $\tilde{f}$ maps U homeomorphically onto $\tilde{f}(U)$ with inverse $\pi|_{\tilde{f}(U)}$. $\square$

We call the topological space $\mathcal{R}$, together with the projection map $\pi : \mathcal{R} \rightarrow X$, the *sheafification* of the presheaf R or the sheaf associated to the presheaf R . We denote the sheaf by the triple $(\mathcal{R}, \pi, X)$ or, more usually, by the symbol $\mathcal{R}$. The ring $\mathcal{R}_x = \pi^{-1}(x)$ is called the *stalk* of the sheaf at x .

Let us summarise our construction. Given a presheaf $R = \{R(U), r_{VU}\}$ of rings on X we have a ring $\mathcal{R}_x$ naturally defined at each point $x \in X$. The disjoint union $\mathcal{R}$ of the rings $\mathcal{R}_x$ has the natural structure of a topological space in such a way that the local sections $\tilde{f}$ associated to $f \in R(U)$ are continuous and the projection $\pi : \mathcal{R} \rightarrow X$ is a local homeomorphism. Each stalk $\mathcal{R}_x$ has the structure of a ring with discrete topology. The triple $(\mathcal{R}, \pi, X)$ is called a *sheaf of rings* on X . Clearly our construction works equally well for presheaves of sets, groups, algebras, etc. to yield sheaves of sets, groups, algebras, etc. Shortly we shall give a general definition of a sheaf which does not depend, a priori, on the existence of a presheaf.

Examples.

9. Let $\mathcal{C}_X$ denote the sheaf of rings associated to the presheaf C_X of continuous $\mathbb{C}$ -valued functions on X . We call $\mathcal{C}_X$ the *sheaf of germs of continuous $\mathbb{C}$ -valued functions* on X . Often we drop the subscript X and just write $\mathcal{C}$ (this remark applies also to subsequent examples). The stalk $\mathcal{C}_x$ is the ring of germs of continuous $\mathbb{C}$ -valued functions at x , $x \in X$. Suppose that $A : U \rightarrow \mathbb{C}$ is a continuous section of $\mathcal{C}$ over the open subset U of X . We claim that there exists a unique $a \in C(U)$ such that $\tilde{a} = A$. In other words, continuous sections of the sheaf correspond to continuous $\mathbb{C}$ -valued functions. First notice that A determines a function $a : U \rightarrow \mathbb{C}$ defined by $a(x) = A(x)(x)$ (evaluation of the germ $A(x)$

at x). We must prove that a is continuous. Let $x \in U$ and observe that by definition of the topology on $\mathcal{C}$ we may find an open neighbourhood V of x , $V \subset U$, and $s \in C(V)$ such that $\tilde{s} = A|_V$. But now $s = a|_V$ and so a is continuous at x . Since x was an arbitrary point in U it follows that a is continuous on U . Finally a is the unique $\mathbb{C}$ -valued function on U satisfying $\tilde{a} = A$ since the germ of a at x determines the value of a at x , $x \in U$. Clearly what we have said above for the presheaf C_X and corresponding sheaf $\mathcal{C}_X$ works equally well for the presheaves C_X^k , $\mathcal{D}_X$ and $\mathcal{O}_X$ and so we obtain the sheaves of rings

$\mathcal{C}_X^k$: Sheaf of germs of C^k $\mathbb{C}$ -valued functions on the differential manifold X .

$\mathcal{D}_X$: Sheaf of germs of C^∞ $\mathbb{C}$ -valued functions on the differential manifold X .

$\mathcal{O}_X$: Sheaf of germs of analytic functions on the complex manifold X .

The sheaf $\mathcal{O}_X$ is often referred to as the *Oka sheaf* of X .

We remark the important fact that a continuous local section of $\mathcal{C}_X^k$ (resp. $\mathcal{D}_X$, $\mathcal{O}_X$) over an open subset U corresponds to a unique C^k (resp. C^∞ , analytic) function on U . The proof is the same as for $\mathcal{C}_X$. Finally, we observe that $\mathcal{O}_X$ is an open subset of $\mathcal{D}_X$ and that each stalk $\mathcal{O}_x$ is a subring of $\mathcal{D}_x$. We say that $\mathcal{O}_X$ is a *subsheaf* (of rings) of $\mathcal{D}_X$. Generally, if $(\mathcal{R}, \pi, X)$ and $(\mathcal{S}, \eta, X)$ are sheaves of rings on X , we say that $\mathcal{R}$ is a *subsheaf* (of rings) of $\mathcal{S}$ if $\mathcal{R}$ is an open subset of $\mathcal{S}$, $\eta|_{\mathcal{R}} = \pi$ and for all $x \in X$, $\mathcal{R}_x$ is a subring of $\mathcal{S}_x$. Thus all the sheaves constructed above are subsheaves of $\mathcal{C}_X$. The reader may care to formulate the analogous concept of *subpresheaf*.

10. Let Γ denote a ring with discrete topology and Γ_X be the presheaf of locally constant Γ -valued function on X . Then the corresponding sheaf, which we shall also denote by Γ_X , is homeomorphic to $X \times \Gamma$. We say that Γ_X is a *constant sheaf* (that is, topologically a product). Sections of Γ_X are Γ -valued functions on X which are constant on connected components of X .
11. Let Z be an analytic subset of the complex manifold X . We let $\mathcal{I}_Z$ denote the sheaf associated to the ideal presheaf I_Z of Z . Dropping the subscript Z , we see that for each $x \in X$, $\mathcal{I}_x$ is an ideal in $\mathcal{O}_x$. For this reason we refer to $\mathcal{I}$ as a *sheaf of ideals* (of $\mathcal{O}$). Observe that for $x \notin Z$, $\mathcal{I}_x = \mathcal{O}_x$ whilst if $x \in Z$, $\mathcal{I}_x \subsetneq \mathcal{O}_x$.
12. If E is a holomorphic vector bundle over the complex manifold X , we let $\underline{\mathcal{E}}$ denote the sheaf of germs of holomorphic sections of E associated to the presheaf of holomorphic sections of E . We see here that for each $x \in X$, $\underline{\mathcal{E}}_x$ is an $\mathcal{O}_x$ -module and we refer to $\underline{\mathcal{E}}$ as a *sheaf of $\mathcal{O}$ -modules*. For our particular example, we see that $\underline{\mathcal{E}}_x$ is a free $\mathcal{O}_x$ -module of rank equal to the fibre dimension of E .
13. Let $\mathcal{M}$ denote the *sheaf of germs of meromorphic functions* on X associated to the presheaf $M = \{M(U), r_{VU}\}$ of meromorphic functions on X . Since $\mathcal{M}_x$ is a field for all $x \in X$, $\mathcal{M}$ is an example of a sheaf of fields. A continuous section of $\mathcal{M}$ over X is called a meromorphic function on X – see Definition 4.1.4. However, a continuous section of $\mathcal{M}$ over an open subset U of X need not correspond to an element of $M(U)$. This is a reflection of the fact that elements of $M(U)$ are all quotients of analytic functions defined on U whilst meromorphic functions on U need not be representable globally as a quotient of analytic functions. The simplest example is found by taking $X = U = P^1(\mathbb{C})$ and m any non-constant meromorphic function on $P^1(\mathbb{C})$.
14. Our final example concerns the topology of the sheaves $\mathcal{D}_X$ and $\mathcal{O}_X$. We shall prove that the topology on $\mathcal{O}_X$ is Hausdorff whilst that on $\mathcal{D}_X$ is not. First we prove that $\mathcal{O}_X$ is Hausdorff. Let $A, B \in \mathcal{O}_X$, $A \neq B$. Set $x = \pi(A)$, $y = \pi(B)$. If $x \neq y$, choose disjoint open neighbourhoods U, V of x, y and $a \in A(U)$, $b \in A(V)$ such that $\tilde{a}(x) = A$,

$\tilde{b}(y) = B$. Clearly $\tilde{a}(U)$, $\tilde{b}(V)$ are disjoint open neighbourhoods of A , B . If $x = y$, choose a connected open neighbourhood U of x and $a, b \in A(U)$ such that $\tilde{a}(x) = A$, $\tilde{b}(x) = B$. If $\tilde{a}(U) \cap \tilde{b}(U) \neq \emptyset$, uniqueness of analytic continuation implies that $a = b$ on U contradicting our assumption that $A \neq B$. Hence $\mathcal{O}_X$ is Hausdorff. To show that $\mathcal{D}_X$ is not Hausdorff it is enough to observe that for $n \geq 1$, we cannot separate the zero germ from the germ at zero, in $\mathbb{R}^n$, of the function γ defined by

$$\gamma(x_1, \dots, x_n) = \begin{cases} 0, & x_n \leq 0, \\ \exp(-1/x_n^2), & x_n > 0. \end{cases}$$

Since $\mathcal{O}_X$ is Hausdorff it follows that $\mathcal{O}_X$ has the structure of a complex manifold spread over X (the complex structure on $\mathcal{O}_X$ is induced from that on X via the local homeomorphism π). We shall exploit this fact in our construction of the envelope of holomorphy in §2.

Next we shall give the general definition of a sheaf and show how a presheaf is naturally associated to every sheaf.

Suppose that we are given a topological space $\mathcal{F}$ and local homeomorphism $\pi : \mathcal{F} \rightarrow X$. We shall say that $(\mathcal{F}, \pi, X)$ is a *sheaf of rings* on X if

1. The stalks $\mathcal{F}_x (= \pi^{-1}(x))$ have the structure of a ring for each $x \in X$.
2. The ring operations are continuous in the topology on $\mathcal{F}$.

Condition 2 needs further elaboration: Suppose that U is any open subset of X and s, t are continuous sections of $\mathcal{F}$ over U . Then we require that $s \pm t$, st are continuous sections of $\mathcal{F}$ over U , where we define addition, subtraction and multiplication of sections using the ring structure in the stalks. Equivalently, we may take the product $\mathcal{F} \times \mathcal{F}$ over $X \times X$ and restrict to the diagonal $\Delta \subset X \times X$. Addition, subtraction and multiplication then define maps of $\mathcal{F} \times \mathcal{F}|_{\Delta}$ to $\mathcal{F}$ which should be continuous. Again it is straightforward to define sheaves of sets, groups, algebras, etc. and we omit formal definitions. In case $\mathcal{R}$ is a sheaf of rings over X , we say that a sheaf $\mathcal{S}$ on X is a *sheaf of $\mathcal{R}$ -modules* if each stalk $\mathcal{S}_x$ has the structure of an $\mathcal{R}_x$ -module and the module operations are continuous in the sense described above.

Suppose that $(\mathcal{F}, \pi, X)$ is a sheaf of rings. For each $U \in \mathcal{U}$, we let $\mathcal{F}(U)$ denote the space of continuous sections of $\mathcal{F}$ over U . Then $\mathcal{F}(U)$ is a ring and, defining restriction homomorphisms in the obvious way, we see that $\mathcal{F}' = \{\mathcal{F}(U), r_{VU}\}$ is a presheaf of rings on X .

Proposition 6.1.3. Let $(\mathcal{F}, \pi, X)$ denote a sheaf of rings on X . Then the sheafification of the presheaf $\mathcal{F}' = \{\mathcal{F}(U), r_{VU}\}$ is equal to $\mathcal{F}$.

Proof. We leave this as an elementary exercise for the reader. □

Example 15. The presheaf $\mathcal{M}'$ associated to the sheaf $\mathcal{M}$ of germs of meromorphic functions on X is not generally equal to the presheaf M of meromorphic functions on X . However, it is clear that M and $\mathcal{M}'$ have the common sheafification $\mathcal{M}$.

Suppose that R is a presheaf of rings on X with associated sheaf $\mathcal{R}$. We have already seen that for all our examples, except that of meromorphic functions, $\mathcal{R}' = R$. Necessary and sufficient conditions for $\mathcal{R}'$ to equal R are given by the following elementary lemma the proof of which we omit.

Proposition 6.1.4. Let R be a presheaf of rings on X with associated sheaf $\mathcal{R}$. Then $\mathcal{R}' = R$ if and only if given any family $\{U_i : i \in I\} \subset \mathcal{U}$ and corresponding $s_i \in R(U_i)$ such that $s_i = s_j$ on U_{ij} for all $i, j \in I$, there exists a unique $s \in R(\bigcup U_i)$ such that $s|_{U_i} = s_i$, $i \in I$.

Remarks on terminology and notation. In the literature a sheaf is often defined to be a presheaf which is equal to the presheafification of its associated sheaf. What we have called a sheaf is then referred to as the *espace étalé* of the sheaf (or presheaf). In the sequel we usually use the same notation for the sheaf and its associated presheaf. By virtue of Propositions 6.1.3, 6.1.4 this will not lead to confusion as, with the exception of the sheaf of germs of meromorphic functions, all our basic examples satisfy the conditions of Proposition 6.1.4. Aside from sheaves of sections of vector bundles and constant sheaves, we generally use script letters to denote sheaves and follow the notation developed earlier for our examples on sheaves.

For the remainder of this section we shall be considering morphisms of sheaves and various constructions involving sheaves. For the sake of brevity we restrict attention to sheaves of rings and modules noting that all our definitions generalise straightforwardly to sheaves of groups, fields, algebras, etc.

Morphisms of sheaves. Let $(\mathcal{F}, \pi, X)$ and $(\mathcal{F}', \pi', X)$ be sheaves of rings on X . A *sheaf morphism* from $\mathcal{F}$ to $\mathcal{F}'$ is a continuous map $A : \mathcal{F} \rightarrow \mathcal{F}'$ covering the identity on X such that if $A_x : \mathcal{F}_x \rightarrow \mathcal{F}'_x$ denotes the map induced by A on the stalks at x then A_x is a ring homomorphism for all $x \in X$. We say that A is a *sheaf isomorphism* if A is a homeomorphism and A and A^{-1} are sheaf morphisms.

Remarks.

1. We shall often refer to a sheaf morphism between sheaves of rings (or groups, algebras, etc.) as a *sheaf homomorphism* or just *homomorphism*. In case $\mathcal{F}, \mathcal{F}'$ are $\mathcal{S}$ -modules, where $\mathcal{S}$ is a sheaf of rings on X , we refer to a sheaf morphism between $\mathcal{F}$ and $\mathcal{F}'$ as an *$\mathcal{S}$ -module homomorphism*.
2. Notice that a sheaf morphism is a local homeomorphism and therefore an open mapping.

We may similarly define morphisms between presheaves. Indeed, suppose $R = \{R(U), r_{VU}\}$ and $S = \{S(U), s_{VU}\}$ are presheaves of rings on X . Then a morphism $a : R \rightarrow S$ consists of a family $\{a_U : R(U) \rightarrow S(U) : U \in \mathcal{U}\}$ of ring homomorphisms which are compatible with the restriction homomorphisms. That is, for $U, V \in \mathcal{U}$, $U \supset V$, we have the commutative diagram

$$\begin{array}{ccc} R(U) & \xrightarrow{a_U} & S(U) \\ r_{VU} \downarrow & & \downarrow s_{VU} \\ R(V) & \xrightarrow{a_V} & S(V). \end{array}$$

The reader may easily verify that a morphism $a : R \rightarrow S$ of presheaves induces a unique sheaf morphism $A : \mathcal{R} \rightarrow \mathcal{S}$ between the sheaves associated to R and S . Conversely, any sheaf morphism $A : \mathcal{R} \rightarrow \mathcal{S}$ is associated to a unique presheaf morphism $a : \mathcal{R}' \rightarrow \mathcal{S}'$ between the presheaves associated to $\mathcal{R}$ and $\mathcal{S}$. We frequently use these observations in our construction of sheaf morphisms and indeed in the following examples all the sheaf morphisms are constructed first at the presheaf level.

Examples.

16. Let E and F be holomorphic vector bundles over the complex manifold X and $A : E \rightarrow F$ be a holomorphic vector bundle map. Then $\underline{E}$ and $\underline{F}$ are $\mathcal{O}_X$ -modules and A induces in the obvious way an $\mathcal{O}_X$ -module homomorphism $\underline{A} : \underline{E} \rightarrow \underline{F}$.
17. Let X be a differential manifold and for $p \geq 0$ let $\underline{C}^p$ denote the sheaf of germs of C^∞ sections of the bundle $\bigwedge^p \mathbb{C}T'X$ of complex p -forms. Note that $\underline{C}^0 = \mathcal{D}_X$ and each $\underline{C}^p$ has the structure of a $\mathcal{D}_X$ -module. The constant sheaf $\mathbb{C}$ is a subsheaf of $\mathcal{D}_X$ and so the sheaves $\underline{C}^p$ have the structure of $\mathbb{C}$ -modules. For $p \geq 0$, exterior differentiation induces a morphism $d : \underline{C}^p \rightarrow \underline{C}^{p+1}$ of $\mathbb{C}$ -modules. Observe that d is certainly not a morphism of $\mathcal{D}_X$ -modules.
18. Let X be a complex manifold and for $p, q \geq 0$ let $\underline{C}^{p,q}$ denote the sheaf of germs of C^∞ sections of the bundle $\bigwedge^{p,q}(X)'$ of complex (p, q) -forms. As in example 17, the operators ∂ and $\bar{\partial}$ induce $\mathbb{C}$ -module homomorphisms

$$\begin{aligned}\partial : \underline{C}^{p,q} &\longrightarrow \underline{C}^{p+1,q} \\ \bar{\partial} : \underline{C}^{p,q} &\longrightarrow \underline{C}^{p,q+1}.\end{aligned}$$

Since $\bar{\partial}\mathcal{O}_X = 0$, we see that $\bar{\partial} : \underline{C}^{p,q} \rightarrow \underline{C}^{p,q+1}$ is actually an $\mathcal{O}_X$ -module homomorphism. Similarly ∂ is an $\overline{\mathcal{O}}_X$ -module homomorphism where $\overline{\mathcal{O}}_X$ denotes the sheaf of germs of anti-holomorphic functions on X .

19. Let X be a complex manifold and for $p \geq 0$ let Ω^p denote the sheaf of germs of holomorphic sections of $\bigwedge^{p,0}(M)'$. Since $\bar{\partial}(\Omega^p) = 0$, we see that the operator ∂ induces a $\mathbb{C}$ -module homomorphism

$$\partial : \Omega^p \longrightarrow \Omega^{p+1}, \quad p \geq 0.$$

Definition 6.1.5. A sequence $\mathcal{F} \xrightarrow{A} \mathcal{G} \xrightarrow{B} \mathcal{H}$ of sheaves is said to be *exact* at $\mathcal{G}$ if the sequence $\mathcal{F}_x \xrightarrow{A_x} \mathcal{G}_x \xrightarrow{B_x} \mathcal{H}_x$ of rings is exact for all $x \in X$.

Remark. We may define a sequence $R \xrightarrow{a} S \xrightarrow{b} T$ of presheaves to be exact at S if the sequence $R(U) \xrightarrow{a_U} S(U) \xrightarrow{b_U} T(U)$ of rings is exact for all $U \in \mathcal{U}$. It is most important to note that presheaf exactness is not equivalent to sheaf exactness and a sheaf exact sequence will *not* generally give rise to a presheaf exact sequence (the converse is always true). In fact sheaf exactness is very much a local matter whilst presheaf exactness is essentially global. In §3 we show how sheaf cohomology enables us to measure the deviation from presheaf exactness of a given sheaf exact sequence. We should also mention the important class of coherent $\mathcal{O}_X$ -modules that we introduce in Chapter 7 for which it is true that sheaf exactness implies “local” presheaf exactness.

A sequence $0 \rightarrow \mathcal{F} \xrightarrow{A} \mathcal{G} \xrightarrow{B} \mathcal{H} \rightarrow 0$ of sheaves is said to be *short exact* if each of the sequences $0 \rightarrow \mathcal{F}_x \xrightarrow{A_x} \mathcal{G}_x \xrightarrow{B_x} \mathcal{H}_x \rightarrow 0$ is a short exact sequence, $x \in X$. We may similarly define exactness for general sequences and we follow the usual notational conventions.

Examples.

20. Let $0 \rightarrow E \xrightarrow{A} F \xrightarrow{B} G \rightarrow 0$ be a short exact sequence of holomorphic vector bundles over the complex manifold X . Then the corresponding sequence $0 \rightarrow \underline{E} \xrightarrow{\underline{A}} \underline{F} \xrightarrow{\underline{B}} \underline{G} \rightarrow 0$ of sheaves is a short exact sequence of $\mathcal{O}_X$ -modules.

21. Let X be an n -dimensional differential manifold. The *de Rham complex* is the sequence of $\mathbb{C}$ -modules given by exterior differentiation:

$$0 \rightarrow \mathbb{C} \xrightarrow{i} \mathcal{D}_X \xrightarrow{d} \underline{C}^1 \xrightarrow{d} \dots \xrightarrow{d} \underline{C}^{n-1} \xrightarrow{d} \underline{C}^n \rightarrow 0.$$

(i denotes inclusion). The Poincaré lemma implies that the de Rham complex is sheaf exact. However, the de Rham complex is generally not presheaf exact as we shall now show. Suppose that X is compact, oriented and without boundary. We prove that the de Rham complex is not presheaf exact at $\underline{C}^n$. For this it is enough to find an n -form on X (that is, continuous section of $\underline{C}^n$ over X) which is not the exterior derivative of an $(n-1)$ -form on X . Choose any n -form ϕ on X such that $\int_X \phi \neq 0$ (Such forms always exist with support in a coordinate chart). Now $d\phi = 0$ and so if the de Rham complex is presheaf exact at $\underline{C}^n$ there exists an $(n-1)$ -form ψ on X such that $d\psi = \phi$. But this cannot be since by Stokes' theorem $\int_X \phi = \int_X d\psi = \int_{\partial X} \psi = 0$. We shall see in §3 that the obstruction to ϕ being the boundary of an $(n-1)$ -form is topological and lies in $H^n(X, \mathbb{C})$ – the n th. cohomology group of X . More precisely, ϕ determines an element $[\phi] \in H^n(X, \mathbb{C})$ and ϕ is a boundary if and only if $[\phi] = 0$.

22. Let X be an n -dimensional complex manifold. For $p, q \geq 0$ we have the *Dolbeault complexes*

$$\begin{aligned} 0 \rightarrow \Omega^p \xrightarrow{i} \underline{C}^{p,0} \xrightarrow{\bar{\partial}} \underline{C}^{p,1} \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} \underline{C}^{p,n-1} \xrightarrow{\bar{\partial}} \underline{C}^{p,n} \rightarrow 0, \\ 0 \rightarrow \bar{\Omega}^q \xrightarrow{i} \underline{C}^{0,q} \xrightarrow{\bar{\partial}} \underline{C}^{1,q} \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} \underline{C}^{n-1,q} \xrightarrow{\bar{\partial}} \underline{C}^{n,q} \rightarrow 0. \end{aligned}$$

Here i denotes inclusion and $\bar{\Omega}^q$ denotes the sheaf of germs of anti-holomorphic sections of the anti-holomorphic bundle $\wedge^{0,q}(X)'$. In case $p = 0$, we obtain the important complex

$$0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{D}_X \xrightarrow{\bar{\partial}} \underline{C}^{0,1} \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} \underline{C}^{0,n-1} \xrightarrow{\bar{\partial}} \underline{C}^{0,n} \rightarrow 0$$

which relates the Oka sheaf to the sheaves $\underline{C}^{0,q}$. It is an immediate consequence of the Dolbeault–Grothendieck lemma (Theorem 5.8.1) that the Dolbeault complexes are all exact.

23. It follows from example 19 that if X is an n -dimensional complex manifold we have a complex

$$0 \rightarrow \mathbb{C} \rightarrow \mathcal{O}_X \xrightarrow{\partial} \Omega^1 \xrightarrow{\partial} \dots \xrightarrow{\partial} \Omega^{n-1} \xrightarrow{\partial} \Omega^n \rightarrow 0$$

of $\mathbb{C}$ -modules. We shall now prove that this sequence is exact. Exactness at $\mathcal{O}_X$ is clear since $d|_{\mathcal{O}_X} = \partial$. Now suppose $p > 0$ and let $\phi \in \Omega^p$ be ∂ -closed. Since $\bar{\partial}\phi = 0$ and $d = \partial + \bar{\partial}$, $d\phi = 0$ and we see by the Poincaré lemma that there exists $\psi \in \underline{C}^p$ such that $d\psi = \phi$. Since $d\psi = \partial\psi + \bar{\partial}\psi \in \underline{C}^{p,0}$ we see that $\psi \in \underline{C}^{p-1,0}$ and $\bar{\partial}\psi = 0$. That is $\psi \in \Omega^{p-1}$ and $\phi = \partial\psi$. Hence the sequence is exact.

For the next few paragraphs we consider some general constructions involving sheaves, presheaves and morphisms.

Definition 6.1.6. Let $A : \mathcal{F} \rightarrow \mathcal{G}$ be a sheaf homomorphism and $\{a_U\} = \{a_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)\}$ denote the corresponding morphism of presheaves. We define the *presheaf kernel* of A , *presheaf cokernel* of A and *presheaf image* of A to be the presheaves given by $U \mapsto \text{Ker}(a_U)$, $U \mapsto \text{Coker}(a_U)$ and $U \mapsto \text{Im}(a_U)$ respectively. We denote the associated sheaves by $\text{Ker}(A)$, $\text{Coker}(A)$ and $\text{Im}(A)$ respectively and refer to them as the (sheaf) *kernel*, *cokernel* and *image* of A respectively.

Remarks.

1. It is easy to see that $\text{Ker}(A)$ is always equal to the presheaf kernel of A but that, in general, $\text{Im}(A)$ and $\text{Coker}(A)$ are not equal to the image and cokernel presheaves of A .
2. For all $x \in X$, we have $\text{Ker}(A)_x \approx \text{Ker}(A_x)$, $\text{Coker}(A)_x \approx \text{Coker}(A_x)$ and $\text{Im}(A)_x \approx \text{Im}(A_x)$.
3. We say that A is *injective* if $\text{Ker}(A) = 0$; *surjective* if $\text{Coker}(A) = 0$. By remark 2 this is equivalent to injectivity or surjectivity at the stalk level.
4. A sheaf sequence $\mathcal{F} \xrightarrow{A} \mathcal{G} \xrightarrow{B} \mathcal{H}$ is exact at $\mathcal{G}$ if and only if $\text{Im}(A) = \text{Ker}(B)$ (Here we are regarding $\text{Im}(A)$, $\text{Ker}(B)$ as subsheaves of $\mathcal{G}$).
5. Associated to a sheaf homomorphism $A : \mathcal{F} \rightarrow \mathcal{G}$ we have the exact sheaf sequence

$$0 \rightarrow \text{Ker}(A) \xrightarrow{i} \mathcal{F} \xrightarrow{A} \mathcal{G} \xrightarrow{q} \text{Coker}(A) \rightarrow 0.$$

(Here i and q are induced from the inclusion and quotient maps respectively).

Suppose $i : \mathcal{F} \rightarrow \mathcal{G}$ is the inclusion map of $\mathcal{F}$ as a subsheaf of $\mathcal{G}$. We define the *quotient sheaf* $\mathcal{G}/\mathcal{F}$ to be the sheaf associated to the presheaf $U \mapsto \mathcal{G}(U)/\mathcal{F}(U)$. We have the corresponding short exact sequence

$$0 \rightarrow \mathcal{F} \xrightarrow{i} \mathcal{G} \xrightarrow{q} \mathcal{G}/\mathcal{F} \rightarrow 0.$$

We remark that $\mathcal{G}/\mathcal{F}$ is isomorphic to $\text{Coker}(i)$ and that for all $x \in X$, $(\mathcal{G}/\mathcal{F})_x \approx \mathcal{G}_x/\mathcal{F}_x$.

Examples.

24. Let $\mathcal{O}$, $\mathcal{M}$ respectively denote the Oka-sheaf and sheaf of germs of meromorphic functions on the complex manifold X . We have the short exact sequence of sheaves of Abelian groups

$$0 \rightarrow \mathcal{O} \rightarrow \mathcal{M} \xrightarrow{q} \mathcal{M}/\mathcal{O} \rightarrow 0.$$

We see that local sections of $\mathcal{M}/\mathcal{O}$ are just the principal parts of meromorphic functions. That is, if $m, m' \in \mathcal{M}(U)$ then m, m' determine the same section of $\mathcal{M}/\mathcal{O}$ if and only if $m - m' \in \mathcal{O}(U)$ ($= A(U)$). We can now give a sheaf theoretic formulation of the Cousin I problem (see Definition 3.4.9): The *data* for the Cousin I problem on X is a continuous section P of $\mathcal{M}/\mathcal{O}$ over X . The *Cousin I problem* (for P) is then to find a continuous section m of $\mathcal{M}$ over X such that $q(m) = P$.

25. Let $\mathcal{O}^*$, $\mathcal{M}^*$ denote the multiplicative sheaves of groups of units of $\mathcal{O}$, $\mathcal{M}$ on the complex manifold X . Thus, $\mathcal{O}_x^*$, $\mathcal{M}_x^*$ will be the groups of invertible germs in $\mathcal{O}_x$, $\mathcal{M}_x$ respectively (see §4, Chapter 3). We let $\mathcal{D}$ denote the quotient sheaf $\mathcal{M}^*/\mathcal{O}^*$. Then $\mathcal{D}$ is a sheaf of Abelian groups called the *sheaf of germs of divisors* on X . We have the short exact sequence

$$0 \rightarrow \mathcal{O}^* \rightarrow \mathcal{M}^* \rightarrow \mathcal{D} \rightarrow 0.$$

Observe that a continuous section of $\mathcal{D}$ over X is a *Cartier divisor* (Definition 4.6.9). We may now give a sheaf theoretic formulation of the Cousin II problem (Definition 3.4.10): The *data* for the Cousin II problem is a continuous section d of $\mathcal{D}$ over X . The *Cousin II problem* (for d) is to find a continuous section m of $\mathcal{M}^*$ such that $q(m) = d$. If we take X to be a Riemann surface we see that $\mathcal{D}_x \cong \mathbb{Z}$ for all $x \in X$. However, $\mathcal{D}$ is far from being the constant sheaf $\mathbb{Z}$ as continuous sections of $\mathcal{D}$ are always non-zero on a *discrete* subset of X . In fact the “zero set” of a continuous section of $\mathcal{D}$ is always an *open* subset of X !

26. Let Z be an analytic subset of the complex manifold X . We start by defining the Oka or structure sheaf of Z . Let U be an open subset of X and $f, g \in \mathcal{O}_X(U)$. We say f and g are Z -equivalent if $f = g$ on $Z \cap U$. That is, if $f - g \in \mathcal{I}_Z(U)$ (see Example 5). The set of Z -equivalence classes associated to U is isomorphic to $\mathcal{O}_X(U)/\mathcal{I}_Z(U)$ and $U \mapsto \mathcal{O}_X(U)/\mathcal{I}_Z(U)$ defines a presheaf of rings on X . We denote the associated sheaf by $\mathcal{O}_Z$ and observe that $\mathcal{O}_{Z,x} = 0$, $x \notin Z$. We call $\mathcal{O}_Z$ the *Oka* or *structure sheaf* of Z . Now $\mathcal{O}_Z$ is just the quotient sheaf $\mathcal{O}_X/\mathcal{I}_Z$ and we have the short exact sequence

$$0 \rightarrow \mathcal{I}_Z \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_Z \rightarrow 0.$$

Any element of $\mathcal{O}_X(U)/\mathcal{I}_Z(U)$ determines a well defined continuous function on $Z \cap U$. Consequently, a continuous section of $\mathcal{O}_Z$ over an open set U of X determines a well-defined continuous function on $Z \cap U$. We call such functions *analytic functions* on $Z \cap U$. It is easy to see that a function $f : Z \cap U \rightarrow \mathbb{C}$ will be analytic if and only if for every $x \in Z \cap U$ there exists an open neighbourhood V of x in X and $g \in A(V)$ such that $g|_{Z \cap U \cap V} = f|_{Z \cap V}$.

Definition 6.1.7. Let $\mathcal{R}$ be a sheaf of rings on X and $\mathcal{F}, \mathcal{G}$ be sheaves of $\mathcal{R}$ -modules.

1. The *direct sum* $\mathcal{F} \oplus \mathcal{G}$ of $\mathcal{F}$ and $\mathcal{G}$ is the sheaf of $\mathcal{R}$ -modules associated to the presheaf $U \mapsto \mathcal{F}(U) \oplus \mathcal{G}(U)$.
2. The *tensor product* $\mathcal{F} \otimes_{\mathcal{R}} \mathcal{G}$ of $\mathcal{F}$ and $\mathcal{G}$ over $\mathcal{R}$ is the sheaf of $\mathcal{R}$ -modules associated to the presheaf $U \mapsto \mathcal{F}(U) \otimes_{\mathcal{R}(U)} \mathcal{G}(U)$.

Remarks.

1. It is easily seen that $(\mathcal{F} \oplus \mathcal{G})_x \approx \mathcal{F}_x \oplus \mathcal{G}_x$ and $(\mathcal{F} \otimes_{\mathcal{R}} \mathcal{G})_x \approx \mathcal{F}_x \otimes_{\mathcal{R}_x} \mathcal{G}_x$, $x \in X$ and we assume these properties in the sequel.
2. We denote the p -fold direct sum of the sheaf of $\mathcal{R}$ -modules $\mathcal{F}$ by $\mathcal{F}^p$.

Examples.

27. Let E, F be holomorphic vector bundles over the complex manifold X . Then $\underline{E} \oplus \underline{F} = \underline{E \oplus F}$ and $\underline{E} \otimes \underline{F} = \underline{E \otimes F}$ (equality here means, of course, up to natural isomorphism). Similar relations hold for smooth or continuous vector bundles.
28. The sheaf of germs of $\mathbb{C}^p$ -valued holomorphic functions on a complex manifold is isomorphic to $\mathcal{O}^p$ (we say the sheaf is free of rank p – see Definition 6.1.8 below)
29. Let E be a holomorphic vector bundle of dimension p over the complex manifold X . Then $\underline{E}$ is *locally isomorphic* to $\mathcal{O}^p$. That is, we may find an open neighbourhood U of each $x \in X$ such that $\underline{E}|_U \cong \mathcal{O}_U^p$. Here the restriction of sheaves to an open set has the obvious interpretation. In particular, $\mathcal{O}_X|_U = \mathcal{O}_U$.

Definition 6.1.8. Suppose $\mathcal{F}$ is a sheaf of $\mathcal{R}$ -modules on X . We say that $\mathcal{F}$ is a *locally free* sheaf of $\mathcal{R}$ -modules of rank p if $\mathcal{F}$ is locally isomorphic to $\mathcal{R}^p$. We say that $\mathcal{F}$ is *free* of rank p if $\mathcal{F} \cong \mathcal{R}^p$.

The next proposition is valid, with the same proof, for smooth or continuous vector bundles.

Proposition 6.1.9. Let X be a complex manifold. There is a bijective correspondence between isomorphism classes of locally free sheaves of $\mathcal{O}$ -modules of finite rank over X and holomorphic vector bundles over X .

Proof. We have already indicated in Example 29 that the sheaf of holomorphic sections of a holomorphic vector bundle is a locally free sheaf of $\mathcal{O}$ -modules of finite rank. Suppose now that $\mathcal{F}$ is a locally free sheaf of $\mathcal{O}$ -modules on X of rank p . Thus we have an open cover $\{U_i\}$ of X and corresponding $\mathcal{O}_{U_i}$ -isomorphisms $\phi_i : \mathcal{F}|_{U_i} \rightarrow \mathcal{O}_{U_i}^p$.

Define $\phi_{ij} : \mathcal{O}_{U_{ij}}^p \rightarrow \mathcal{O}_{U_{ij}}^p$ by $\phi_{ij} = \phi_i \phi_j^{-1}$. Now ϕ_{ij} is an isomorphism of the $\mathcal{O}_{U_{ij}}$ -module $\mathcal{O}_{U_{ij}}^p$ and so is given by a $p \times p$ -matrix with holomorphic entries defined on U_{ij} . That is, ϕ_{ij} determines a holomorphic map $\phi_{ij} : U_{ij} \rightarrow GL(p, \mathbb{C})$. Clearly the $\{\phi_{ij}\}$ are the transition functions for a holomorphic vector bundle E on X . We leave it to the reader to check that $\underline{E} = \mathcal{F}$. $\square$

Example 30. In this example we follow the notation and assumptions of Examples 18 and 22. Thus for $p, q \geq 0$ we have an $\mathcal{O}$ -homomorphism $\bar{\partial} : \underline{C}^{p,q} \rightarrow \underline{C}^{p,q+1}$ of sheaves over the complex manifold X . If E is a holomorphic vector bundle over X , $\underline{E}$ is a sheaf of $\mathcal{O}$ -modules and so, since $\bar{\partial}$ is an $\mathcal{O}$ -homomorphism, we may form

$$\bar{\partial} (= \bar{\partial} \otimes I) : \underline{C}^{p,q} \otimes_{\mathcal{O}} \underline{E} \rightarrow \underline{C}^{p,q+1} \otimes_{\mathcal{O}} \underline{E}.$$

But $\underline{C}^{p,q} \otimes_{\mathcal{O}} \underline{E} = \underline{C}^{p,q} \otimes_{\mathcal{O}} \underline{E}_{\infty} = \underline{\Lambda}^{p,q}(M, E)'_{\infty}$. Hence, as in §7 of Chapter 5, we have extended the $\bar{\partial}$ -operator to E -valued forms. In the sequel we set $\underline{\Lambda}^{p,q}(M, E)'_{\infty} = \underline{C}^{p,q}(E)$ and $\underline{\Lambda}^{p,0}(M, E)' = \Omega^p(\underline{E})$. Using the Dolbeault–Grothendieck lemma as in Example 22, we find that the Dolbeault complexes

$$0 \rightarrow \Omega^p(\underline{E}) \xrightarrow{i} \underline{C}^{p,0}(E) \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} \underline{C}^{p,n-1}(E) \xrightarrow{\bar{\partial}} \underline{C}^{p,n}(E) \rightarrow 0$$

are exact for $p \geq 0$.

Next we see how sheaves transform under maps of the underlying topological spaces.

Suppose $f : X \rightarrow Y$ is a continuous map of topological spaces and $(\mathcal{F}, \pi, Y)$ is a sheaf of rings on Y . We shall define a sheaf $f^{-1}\mathcal{F}$ of rings on X called the *inverse image sheaf* of $\mathcal{F}$. To this end, the stalk of $f^{-1}\mathcal{F}$ at $x \in X$ will be equal, as a ring, to $\mathcal{F}_{f(x)}$. We let $f^{-1}\mathcal{F}$ be the disjoint union of the rings $\mathcal{F}_{f(x)}$ over $x \in X$. It remains to topologise $f^{-1}\mathcal{F}$. For a basis of open sets for the topology on $f^{-1}\mathcal{F}$ we take the set of all images $s(U)$, where U is an open subset of X and s is a section of $f^{-1}\mathcal{F}$ over U such that $\pi s : U \rightarrow Y$ is continuous.

Examples.

31. Let Z be a subset of X with induced topology and $i : Z \rightarrow X$ denote the inclusion map. If $\mathcal{F}$ is a sheaf on X we call $i^{-1}\mathcal{F}$ the *restriction* of $\mathcal{F}$ to Z and denote it by $\mathcal{F}|_Z$ or $\mathcal{F}_Z$.
32. Let $\mathcal{F}$ be a sheaf of $\mathcal{R}$ -modules on Y and $f : X \rightarrow Y$ be continuous. Then $f^{-1}\mathcal{F}$ is a sheaf of $f^{-1}\mathcal{R}$ -modules on X . Suppose that X, Y are complex manifolds, f is holomorphic and E is a holomorphic vector bundle over Y . It is natural to ask for the relation between $f^{-1}\underline{E}$ and the sheaf of holomorphic sections of the pull-back bundle f^*E . Now $f^{-1}\underline{E}$ is an $f^{-1}\mathcal{O}_Y$ -module rather than an $\mathcal{O}_X$ -module. However, $\mathcal{O}_X$ is naturally an $f^{-1}\mathcal{O}_Y$ -module – indeed we have a sheaf homomorphism of $f^{-1}\mathcal{O}_Y$ into $\mathcal{O}_X$ defined in the obvious way by composition of elements of $f^{-1}\mathcal{O}_Y$ with f . Hence we may form the sheaf

$$f^*\underline{E} = f^{-1}\underline{E} \otimes_{f^{-1}(\mathcal{O}_Y)} \mathcal{O}_X.$$

Now f^*E is a sheaf of $\mathcal{O}_X$ -modules and the reader may verify that $f^*E = f^*E$. Generally, for any sheaf $\mathcal{F}$ of $\mathcal{O}_Y$ -modules, we define $f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}(\mathcal{O}_Y)} \mathcal{O}_X$ and $f^*\mathcal{F}$ will then be a sheaf of $\mathcal{O}_X$ -modules on X .

Next we turn to the push-forward of sheaves. Suppose $f : X \rightarrow Y$ is a continuous map and $(\mathcal{F}, \pi, X)$ is a sheaf of rings on X . We define the *direct image sheaf* $f_*\mathcal{F}$ to be the sheaf of rings on Y associated to the presheaf $U \mapsto \mathcal{F}(f^{-1}(U))$.

Example 33. If $f : X \rightarrow Y$ is a holomorphic map of complex manifolds and $\mathcal{F}$ is a sheaf of $\mathcal{O}_X$ -modules on X then $f_*\mathcal{F}$ is a sheaf of $f_*\mathcal{O}_X$ -modules on Y . The reader may verify that we have a canonical sheaf morphism of $\mathcal{O}_Y$ into $f_*\mathcal{O}_X$ and so $f_*\mathcal{F}$ has the natural structure of an $\mathcal{O}_Y$ -module. If E is a holomorphic vector bundle over X , f_*E will not generally be the sheaf of sections of a holomorphic vector bundle over Y . A simple example is found by taking the sheaf of sections $\underline{\mathbb{C}} (= \mathcal{O})$ of the trivial bundle $\underline{\mathbb{C}}$ over $\mathbb{C}$ and the map $f : \mathbb{C} \rightarrow \mathbb{C}$ defined by $f(z) = z^2$. It is easily verified that $f_*\underline{\mathbb{C}}$ is not locally free (see also the discussion below and Chapter 7).

Direct image sheaves play a most important role in complex analysis but are considerably more difficult to describe and analyse than inverse image sheaves. The full analysis of direct image sheaves requires the machinery of spectral sequences (see Godement [1] and also Griffiths and Harris [1, Chapter 3]). Here we shall only describe the stalks of direct image sheaves and then only in the case when the underlying spaces and map satisfy additional conditions.

Proposition 6.1.10. Let $(\mathcal{F}, \pi, X)$ be a sheaf of rings on X and $f : X \rightarrow Y$ be continuous. Suppose that X, Y are locally compact and that f is proper (that is, inverse images of compact sets are compact). Then $f_*\mathcal{F}_y$ is naturally isomorphic to $\mathcal{F}(f^{-1}(y))$, all $y \in Y$. ($\mathcal{F}(f^{-1}(y))$ denotes the spaces of continuous sections of the sheaf $\mathcal{F}$, restricted to $f^{-1}(y)$, over $f^{-1}(y)$).

Our proof follows that given in Godement [1]. First we need some preliminary lemmas which are of interest in their own right.

Lemma 6.1.11. Let $\{M_i : i \in I\}$ be a locally finite cover by closed sets of the topological space X . Suppose that $(\mathcal{F}, \pi, X)$ is a sheaf of rings on X and that for each $i \in I$ we are given a continuous section s_i of $\mathcal{F}_{M_i}$. Then, if the $\{s_i\}$ satisfy the compatibility condition $s_i = s_j$ on $M_i \cap M_j$, there exists $s \in \mathcal{F}(X)$ such that $s|_{M_i} = s_i$.

Proof. First we remark that the lemma is trivial if the $\{M_i\}$ form an *open* cover of X . Clearly our conditions on the $\{s_i\}$ imply that there exists a unique section s of $\mathcal{F}$ over X which restricts to s_i on M_i . We must show that s is continuous. Now, given $x \in X$, we may find an open neighbourhood U of x which meets only finitely many M_i , say $M_1, \dots, M_p$. As the M_i are closed, we may assume that U is chosen so that $x \in M_1 \cap \dots \cap M_p$. Shrinking U further if necessary we may also suppose that there exists $t \in \mathcal{F}(U)$ such that $t(x) = s(x) = s_1(x) = \dots = s_p(x)$. By definition of the sheaf topology on $\mathcal{F}$, there exist open neighbourhoods U_j of x such that $t = s_j$ on U_j , $1 \leq j \leq p$. We may suppose $U = U_j$, $1 \leq j \leq p$. Hence s and t are equal in $U \cap (M_1 \cup \dots \cup M_p) = U$ and so s is continuous at x . $\square$

Lemma 6.1.12. Let S be a closed subset of the paracompact space X and suppose that $\mathcal{F}$ is a sheaf on X and $s \in \mathcal{F}(S)$ ($= \mathcal{F}_S(S)$). Then there exists an open neighbourhood V of S in X and $\tilde{s} \in \mathcal{F}(V)$ such that $\tilde{s}|_S = s$.

Proof. By definition of the topology on $\mathcal{F}$, we may find an open neighbourhood U in X of every point $x \in S$ and $t \in \mathcal{F}(U)$ such that $t|_{U \cap S} = s$. Hence, by the paracompactness of X , we may find a locally finite open cover $\{U_i : i \in I\}$ of S and sections $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i \cap S} = s$ for all $i \in I$. Take a refinement $\{V_i\}$ of U_i such that $\bar{V}_i \subset U_i$ for all i and let W be the subset of X consisting of all points x in $\bigcup_i \bar{V}_i$ such that if $x \in \bar{V}_i \cap \bar{V}_j$ then $s_i(x) = s_j(x)$. By

Lemma 6.1.11, applied to $\mathcal{F}|_W$, the $\{s_i\}$ define a continuous section $\tilde{s}$ of $\mathcal{F}$ over W . We claim that W contains an open neighbourhood of S . Let $x \in S$. There exists an open neighbourhood V of x which meets only finitely many of the $\bar{V}_i$'s, say $\bar{V}_1, \dots, \bar{V}_p$. Shrinking V we may suppose that $x \in \bar{V}_1, \dots, \bar{V}_p$. Now $s_1(x) = \dots = s_p(x)$ and so, shrinking V further if necessary, we may suppose that $s_1, \dots, s_p$ are equal on V . Now observe that $V \subset W$. $\square$

Lemma 6.1.13. Let $f : X \rightarrow Y$ be a proper continuous map between locally compact spaces. Fix $y \in Y$ and let V be any open neighbourhood of $f^{-1}(y)$ in X . Then there exists an open neighbourhood U of y in Y such that $f^{-1}(U) \subset V$. In other words, a fundamental system of open neighbourhoods for $f^{-1}(y)$ is given by $\{f^{-1}(U) : U \text{ an open neighbourhood of } y\}$.

Proof. The intersection $\bigcap f^{-1}(\bar{U})$ over all relatively compact neighbourhoods U of y in Y is clearly equal to $f^{-1}(y)$. Since each $f^{-1}(\bar{U})$ is compact, it follows that for some relatively compact neighbourhood U of y we must have $f^{-1}(\bar{U}) \cap (X \setminus V) = \emptyset$. (For an alternative proof of this lemma, see the exercises at the end of the section). $\square$

Proof of Proposition 6.1.10. We first remark that we have a natural homomorphism $\theta : f_*\mathcal{F}_y \rightarrow \mathcal{F}(f^{-1}(y))$ defined in the obvious way. We must show that $\mathcal{F}$ is bijective. Let $\gamma \in \mathcal{F}(f^{-1}(y))$. By Lemma 6.1.12, there exists an open neighbourhood V of $f^{-1}(y)$ and $\tilde{\gamma} \in \mathcal{F}(V)$ such that $\tilde{\gamma}|_{f^{-1}(y)} = \gamma$. By Lemma 6.1.13, we may assume that $V = f^{-1}(U)$, for some open neighbourhood U of y . But now $\tilde{\gamma}$ lies in the presheaf generating $f_*\mathcal{F}$ and so determines an element of $f_*\mathcal{F}_y$. Clearly this construction gives the required inverse to θ . $\square$

We now briefly look at the use of sheaf formalism in defining structures on topological spaces. Suppose X is a topological space and $\mathcal{F}$ is a subsheaf of $\mathbb{C}$ -algebras of $\mathcal{C}_X$ (regarded as a sheaf of $\mathbb{C}$ -algebras). We may think of $\mathcal{F}$ as defining a structure on X : Structure of all functions of type $\mathcal{F}$ (that is, functions of type $\mathcal{F}$ would be given by continuous sections of $\mathcal{F}$). All the structures so far considered – analytic, smooth, etc. – are locally defined and we shall now exploit this fact to give sheaf theoretic definitions of various structures on topological spaces.

Example 34. Let X be a topological space and $\mathcal{F}$ be a subsheaf of $\mathbb{C}$ -algebras of $\mathcal{C}_X$. Suppose that $\mathcal{F}$ is *locally isomorphic* to the sheaf of germs of C^∞ functions on $\mathbb{R}^n$. Then X may be given the unique structure of a differential manifold with $\mathcal{D}_X = \mathcal{F}$. Before proving our assertion we need to explain what is meant by (local) isomorphism of sheaves defined over different topological spaces. For our example, we require that we can find an open neighbourhood U of each point $x \in X$ and a homeomorphism $\phi : U \rightarrow \phi(U) \subset \mathbb{R}^n$ such that the induced map $\phi^* : \mathcal{C}_{\phi(U)} \rightarrow \mathcal{C}_U$ restricts to an isomorphism of $\mathcal{D}_{\phi(U)}$ with $\mathcal{F}_U$. We see that if $\mathcal{F}$ is locally isomorphic to $\mathcal{D}$, we may find an open cover $\{U_i\}$ of X and corresponding homeomorphism $\phi_i : U_i \rightarrow \phi(U_i) \subset \mathbb{R}^n$ such that ϕ_i induces an isomorphism of $\mathcal{D}_{\phi(U_i)}$ with $\mathcal{F}_{U_i}$ for all i . It is now a straightforward matter to verify that $\{(U_i, \phi_i)\}$ defines a differential atlas on X and that the associated structure sheaf $\mathcal{D}_X = \mathcal{F}$. The same argument works if $\mathcal{F}$ is locally isomorphic to the Oka-sheaf of $\mathbb{C}^n$ and in this case we find that X has the structure of an n -dimensional complex manifold with Oka sheaf equal to $\mathcal{F}$.

Motivated by the example above we may now give an intrinsic definition of a (reduced) analytic space which generalises our earlier definition of analytic set. First we define the *local models* for analytic spaces: The local models for analytic spaces will be the set of all pairs $(Z, \mathcal{O}_Z)$, where Z is an analytic subset of an open subset of some $\mathbb{C}^n$ and $\mathcal{O}_Z$ is the structure sheaf of Z (restricted to Z). A *(reduced) analytic space* will then be a pair $(X, \mathcal{O}_X)$, where X is a topological space and $\mathcal{O}_X$ is a subsheaf of $\mathbb{C}$ -algebras of $\mathcal{C}_X$ which is locally isomorphic to a local model. That is, we may find an open neighbourhood V of any point $x \in X$, a local model $(Z, \mathcal{O}_Z)$ and a homeomorphism of V onto Z such that ϕ induces an isomorphism of $\mathcal{O}_V$ with $\mathcal{O}_Z$ (“isomorphism” in the sense described in Example 34).

Remark. Unfortunately fibre products of (reduced) analytic spaces cannot generally be constructed within the category of (reduced) analytic spaces. The problem lies with the fact that the structure sheaves we obtain when attempting these constructions may have nilpotents. Thus, the category of analytic spaces has to be enlarged so that it is closed under fibre products. For this, instead of considering subsheaves of $\mathcal{C}_X$, we consider sheaves of local $\mathbb{C}$ -algebras on X : A sheaf $\mathcal{F}$ on X is said to be a sheaf of local $\mathbb{C}$ -algebras on X if $\mathcal{F}$ is a sheaf of $\mathbb{C}$ -algebras and each $\mathcal{F}_x$ has a unique maximal ideal m_x with $\mathcal{F}_x/m_x \cong \mathbb{C}$ for all $x \in X$. A simple example of a sheaf of local $\mathbb{C}$ -algebras which is not a subsheaf of $\mathcal{C}_X$ is given by taking X to be the origin of $\mathbb{C}$ and $\mathcal{F} = \mathcal{O}_0/(z^2)$ ($\mathcal{O}_0$ denotes the Oka sheaf of $\mathbb{C}$ restricted to 0). Let us now describe the local models for (unreduced) analytic spaces. A local model will be a pair $(X, \mathcal{O}_X)$, where X is an analytic subset of an open subset U of $\mathbb{C}^n$ and $\mathcal{O}_X = \mathcal{O}_U/(f_1, \dots, f_k)|_X$, where $f_1, \dots, f_k \in \mathcal{I}_X(U)$ and $X = Z(f_1, \dots, f_k)$. An *analytic space* is then defined to be a pair $(X, \mathcal{O}_X)$, where $\mathcal{O}_X$ is a sheaf of local $\mathbb{C}$ -algebras on X which is locally isomorphic to a local model. For further details and examples the reader may refer to the introductory article by Malgrange [1].

Exercises.

1. For each open subset U of the topological space X let $B(U)$ denote the ring of continuous bounded $\mathbb{C}$ -valued functions on U . Show that $B_X = \{B(U), r_{VU}\}$ is a presheaf of rings on X with sheafification $\mathcal{C}_X$. Hence deduce that the presheaf associated to the sheafification of B_X is not in general equal to B_X .
2. Let $\mathcal{F}, \mathcal{G}$ be sheaves of rings on X and $\text{Hom}(\mathcal{F}, \mathcal{G})$ be the sheaf associated to the presheaf $U \mapsto \text{Hom}(\mathcal{F}_U, \mathcal{G}_U)$, where $\text{Hom}(\mathcal{F}_U, \mathcal{G}_U)$ is the ring of sheaf homomorphisms from $\mathcal{F}_U$ to $\mathcal{G}_U$. Show that for all $x \in X$, we have a natural map $\text{Hom}(\mathcal{F}, \mathcal{G})_x \rightarrow \text{Hom}(\mathcal{F}_x, \mathcal{G}_x)$ which is, in general, neither injective nor surjective.
3. Let X be a complex manifold with Oka sheaf $\mathcal{O}$ and $\mathcal{F}$ be a locally free sheaf of $\mathcal{O}$ -modules of finite rank on X . Define the *dual* $\mathcal{F}^*$ of $\mathcal{F}$ to be the sheaf $\text{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{O})$. Prove that
 - A) $\mathcal{F}^*$ is the sheaf of germs of sections of the dual of the holomorphic bundle associated to $\mathcal{F}$ (up to isomorphism).
 - B) $\mathcal{F}^{**} \approx \mathcal{F}$.
 - C) $\text{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) \approx \mathcal{F}^* \otimes \mathcal{G}$, for any sheaf of $\mathcal{O}$ -modules $\mathcal{G}$.
 - D) $\text{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{G})_x \approx \text{Hom}(\mathcal{F}_x, \mathcal{G}_x)$, $x \in X$.

Show that similar results hold for locally free sheaves of $\mathcal{C}_X$ - or $\mathcal{D}_X$ -modules.

4. Let Z be a closed subset of the topological space X and $i : Z \rightarrow X$ denote the inclusion map. If $\mathcal{F}$ is a sheaf on Z , set $\tilde{\mathcal{F}} = i_*\mathcal{F}$. We call $\tilde{\mathcal{F}}$ the *trivial extension* of $\mathcal{F}$ to X or the sheaf on X obtained by extending $\mathcal{F}$ by zero outside Z . Show that
 - A) $\tilde{\mathcal{F}}_x = 0$, $x \notin Z$, $= \mathcal{F}_x$, $x \in Z$.
 - B) $i^{-1}\tilde{\mathcal{F}} \approx \mathcal{F}$.
5. Let $(\mathcal{F}, \pi, X)$ be a sheaf of rings and $s \in \mathcal{F}(X)$. Define the *support of* s , $\text{supp}(s)$, to be $\{x \in X : s(x) \neq 0\}$. Show that $\text{supp}(s)$ is a closed subset of X .
6. Let $(\mathcal{F}, \pi, X)$ be a sheaf of rings. Define the *support of* $\mathcal{F}$, $\text{supp}(\mathcal{F})$, to be $\{x \in X, \mathcal{F}_x \neq 0\}$. Show that $\text{supp}(\mathcal{F})$ need be neither open nor closed.

7. Let $\mathcal{F}$ be a locally free sheaf of $\mathcal{O}_Y$ -modules on the complex manifold Y and suppose that $f : X \rightarrow Y$ is a holomorphic map of complex manifolds. Prove that

$$f_*(\mathcal{G} \otimes_{\mathcal{O}} f^*\mathcal{F}) \sim f_*\mathcal{G} \otimes_{\mathcal{O}} \mathcal{F}, \quad \text{for any sheaf } \mathcal{G} \text{ of } \mathcal{O}_X\text{-modules on } X.$$

8. Let X, Y be complex manifolds and $\mathcal{F}, \mathcal{G}$ be sheaves of $\mathcal{O}_X$ -, $\mathcal{O}_Y$ -modules, respectively. Show that for any holomorphic map $f : X \rightarrow Y$, $\text{Hom}(f^*\mathcal{G}, \mathcal{F}) \approx \text{Hom}(\mathcal{G}, f_*\mathcal{F})$ (We say that f^* and f_* are *adjoint functors* between the categories of $\mathcal{O}_X$ - and $\mathcal{O}_Y$ -modules). Deduce that we have canonical homomorphisms $\mathcal{G} \rightarrow f_*f^*\mathcal{G}$ and $f^*f_*\mathcal{F} \rightarrow \mathcal{F}$.

9. Let X be a complex manifold and set $K = \text{Kernel } \partial\bar{\partial} : \underline{C}^{0,0} \rightarrow \underline{C}^{1,1}$. Show that the complex

$$0 \rightarrow K \rightarrow \underline{C}^{0,0} \xrightarrow{\partial\bar{\partial}} \underline{C}^{1,1} \xrightarrow{d} \underline{C}^{1,2} \oplus \underline{C}^{2,1} \xrightarrow{d} \dots$$

is sheaf exact (Hint: For exactness at $\underline{C}^{1,1}$ the argument used in Example 23 may be helpful).

10. Let $f : X \rightarrow Y$ be a continuous map of topological spaces. Prove

- If $a : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves over Y then we have a canonical sheaf map $f^{-1}(a) : f^{-1}\mathcal{F} \rightarrow f^{-1}\mathcal{G}$ of sheaves over X .
- f^{-1} is *exact*. That is, given a short exact sequence $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H} \rightarrow 0$ of sheaves over Y , the sequence

$$0 \rightarrow f^{-1}\mathcal{F} \xrightarrow{f^{-1}(a)} f^{-1}\mathcal{G} \xrightarrow{f^{-1}(b)} f^{-1}\mathcal{H} \rightarrow 0 \text{ is exact.}$$

11. Same assumptions as Q10. Show that if $a : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves over X then we

- have a canonical sheaf map $f_*(a) : f_*\mathcal{F} \rightarrow f_*\mathcal{G}$ of sheaves over Y .
- f_* is *left exact*. That is, given an exact sequence $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H}$ of sheaves over X , the sequence

$$0 \rightarrow f_*\mathcal{F} \xrightarrow{f_*(a)} f_*\mathcal{G} \xrightarrow{f_*(b)} f_*\mathcal{H} \text{ is exact.}$$

- In general f_* is not exact.

(Hint for c): Take $X = \mathbb{C}^2 \setminus \{0\}$, $Y = \mathbb{C}$ and define $f(x, y) = x$. Let $\mathcal{I}$ denote the ideal sheaf of $f^{-1}(0)$. Show that the f_* image of the sequence $\mathcal{O}_X \rightarrow \mathcal{O}_X/\mathcal{I} \rightarrow 0$ is not exact).

12. Suppose that $f : X \rightarrow Y$ is a holomorphic map of complex manifolds. Prove that

- If $a : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves of $\mathcal{O}_Y$ -modules over Y then we have a canonical sheaf morphism $f^*a : f^*\mathcal{F} \rightarrow f^*\mathcal{G}$ of sheaves of $\mathcal{O}_X$ -modules.
- f^* is right exact.
- f^* is generally not left exact.

13. Let X, Y be metrizable topological spaces and $f : X \rightarrow Y$ be a proper continuous map. Show that f is closed (that is, f -images of closed sets are closed). Deduce Lemma 6.1.13 in case X, Y are metrizable (as will always be the case if X, Y are manifolds).

14. (Koszul complex). Let E be a holomorphic vector bundle of rank q on the complex manifold M and suppose $s \in \Omega(E)$. Show that we have complexes of $\mathcal{O}$ -modules

$$0 \rightarrow \mathcal{O} \xrightarrow{\beta_0} \bigwedge^1 E \xrightarrow{\beta_1} \cdots \xrightarrow{\beta_{q-1}} \bigwedge^q E \rightarrow 0$$

$$0 \leftarrow \mathcal{O} \xleftarrow{\alpha_0} \bigwedge^1 E^* \xleftarrow{\alpha_1} \cdots \xleftarrow{\alpha_{q-1}} \bigwedge^q E^* \leftarrow 0$$

where $\beta_j(t) = s \wedge t$; $\alpha_j(t) = \iota_s t$, $0 \leq j \leq q-1$, and that these complexes are exact if (and only if) s is nowhere zero (see exercise 6, §1, Chapter 5).

2 Envelope of holomorphy

In this section we wish to consider the following problem: Given a domain Ω in $\mathbb{C}^n$ can we find a “maximal” n -dimensional complex manifold $\hat{\Omega}$ containing Ω such that every analytic function on Ω has a unique extension to $\hat{\Omega}$? Example 6 of §4, Chapter 2 shows that we cannot require $\hat{\Omega}$ to be a domain in $\mathbb{C}^n$. It turns out that to solve our problem it is best to work within the category of domains spread over $\mathbb{C}^n$. It then becomes possible to give a particularly elegant solution using the formalism of sheaf theory. First we recall some definitions.

Definition 6.2.1. A manifold *spread over* $\mathbb{C}^n$ is a pair (Ω, π) , where Ω is a connected, separable, Hausdorff space and $\pi : \Omega \rightarrow \mathbb{C}^n$ is a local homeomorphism (not necessarily surjective). We call π the *spreading* of Ω in $\mathbb{C}^n$.

Given a spread manifold (Ω, π) , the map π induces a complex structure on Ω . In the sequel we always assume that Ω comes with this complex structure. We also often refer to (Ω, π) as “the manifold spread over $\mathbb{C}^n$ ”, omitting reference to π . We remark that given a complex manifold Ω there may exist many different spreadings of Ω in $\mathbb{C}^n$ all of which induce the given complex structure on Ω .

Suppose (Ω, π) , (Ω', π') are manifolds spread over $\mathbb{C}^n$ and $\theta : \Omega \rightarrow \Omega'$ is continuous. We say that θ is a *morphism* of spread manifolds if $\pi'\theta = \pi$. That is, if $\theta(\Omega_x) = \Omega'_x$ for all $x \in \pi(\Omega)$. Here we have set $\Omega_x = \pi^{-1}(x)$, $\Omega'_x = \pi'^{-1}(x)$. The reader may easily verify that a morphism of spread manifolds is *analytic* with respect to the induced complex structures and is also an *open mapping*.

Definition 6.2.2. Let Ω, Ω' be manifolds spread over $\mathbb{C}^n$ and $\theta : \Omega \rightarrow \Omega'$ be a morphism. We say that the pair (Ω', θ) is an *analytic extension* of Ω if every $f \in A(\Omega)$ extends uniquely across θ to an element $\tilde{f} \in A(\Omega')$. That is, there exists $\tilde{f} \in A(\Omega')$ such that $f = \tilde{f}\theta$.

Remark. We sometimes say that Ω' is an analytic extension of Ω if there exists a morphism θ such that (Ω', θ) is an analytic extension of Ω in the sense of Definition 6.2.2.

Definition 6.2.3. A manifold Ω spread over $\mathbb{C}^n$ is called a *domain of holomorphy* if for every analytic extension (Ω', θ) of Ω , θ is an isomorphism.

Remark. The reader may easily verify that a domain Ω in $\mathbb{C}^n$ is a domain of holomorphy according to Definition 6.2.3 if and only if it is according to Definition 2.4.1.

We may now state the main result of this section.

Theorem 6.2.4. Let (Ω, π) be a manifold spread over $\mathbb{C}^n$. Then there exists an analytic extension $(\widehat{\Omega}, \theta)$ of Ω such that

1. $\widehat{\Omega}$ is a maximal analytic extension of Ω in the sense that if $(\widetilde{\Omega}, \gamma)$ is any other analytic extension of Ω , there exists a morphism $\eta : \widetilde{\Omega} \rightarrow \widehat{\Omega}$ making $(\widehat{\Omega}, \eta)$ an analytic extension of Ω .
2. $\widehat{\Omega}$ is independent of the spreading π of Ω in $\mathbb{C}^n$. That is, if $\pi_1, \pi_2 : \Omega \rightarrow \mathbb{C}^n$ are two spreadings of Ω in $\mathbb{C}^n$, compatible with the given complex structure on Ω , then the corresponding maximal analytic extensions $\widehat{\Omega}_1, \widehat{\Omega}_2$ given by 1 are isomorphic.

Definition 6.2.5. We call for any analytic extension $(\widehat{\Omega}, \theta)$ of Ω satisfying the conditions of Theorem 6.2.4 the *envelope of holomorphy* of Ω .

Proof of Theorem 6.2.4. Our proof follows that in Malgrange [2].

Step 1. Existence of $\widehat{\Omega}$.

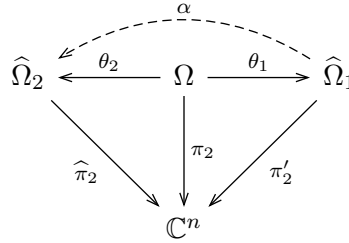
Let $\{f_i\}$ denote the set of elements in $A(\Omega)$ indexed by the set I . We let $\mathbb{C}^I$ denote the vector space of all functions from I to $\mathbb{C}$; addition and scalar multiplication defined coordinatewise. Let U be an open subset of $\mathbb{C}^n$. We say that a map $h = (h_i) : U \rightarrow \mathbb{C}^I$ is analytic if each component function $h_i : U \rightarrow \mathbb{C}$ is analytic. As in §1 we may construct the sheaf $(\mathcal{O}^I, p, \mathbb{C}^n)$ of germs of analytic functions from $\mathbb{C}^n$ to $\mathbb{C}^I$. Exactly as in Example 14 of §1, $(\mathcal{O}^I, p)$ is a spread manifold over $\mathbb{C}^n$ (of course, $\mathcal{O}^I$ will not be connected). We shall show that the envelope of holomorphy of Ω may be represented as a connected component of $\mathcal{O}^I$. First we define a morphism $\theta : \Omega \rightarrow \mathcal{O}^I$. Given $x \in \Omega$, choose an open neighbourhood U of x in Ω such that π restricts to a homeomorphism of U on $\pi(U) \subset \mathbb{C}^n$. Set $\phi = (\pi|U)^{-1} : \pi(U) \rightarrow U$. For $i \in I$, set $\tilde{f}_i = f_i \phi \in A(\pi(U))$. Define $\theta(x) = (\tilde{f}_{i,x}) \in \mathcal{O}^I$, where $\tilde{f}_{i,x}$ denotes the germ of $\tilde{f}_i$ at x . It is clear from the definition of the topology on $\mathcal{O}^I$ that θ is continuous and so defines a morphism of spread manifolds. We let $\widehat{\Omega}$ denote the connected component of $\mathcal{O}^I$ containing $\theta(\Omega)$ and set $\widehat{p} = p|_{\widehat{\Omega}}$. Thus $(\widehat{\Omega}, \widehat{p})$ is a manifold spread over $\mathbb{C}^n$.

Step 2. $(\widehat{\Omega}, \theta)$ is an analytic extension of Ω . We show that each $f_i \in A(\Omega)$ extends uniquely to an element $\widehat{f}_i \in A(\widehat{\Omega})$. First observe that for each $i \in I$ we have a projection map $\pi_i : \mathbb{C}^I \rightarrow \mathbb{C}$ which induces a morphism $\eta_i : \mathcal{O}^I \rightarrow \mathcal{O}$ of spread manifolds. Thus η_i maps the germ of a map of $\mathbb{C}^n$ into $\mathbb{C}^I$ to its i th. component. Let $z \in \widehat{\Omega}$ and choose an open neighbourhood U of z in $\widehat{\Omega}$ such that $\widehat{p}|U$ is a homeomorphism. Set $\gamma = (\widehat{p}|U)^{-1}$. Certainly $\pi_i \gamma$ is a section of $\mathcal{O}$ over $\widehat{p}(U)$ and we let γ_i denote the corresponding analytic function on $\widehat{p}(U)$. We define $\widehat{f}_i|U = \gamma_i \widehat{p}$. This construction clearly defines an element $\widehat{f}_i \in A(\widehat{\Omega})$ for each $i \in I$. Next we must prove that $\widehat{f}_i \theta = f_i$ and that $\widehat{f}_i$ is the unique element of $A(\widehat{\Omega})$ satisfying this relation. Uniqueness is obvious by uniqueness of analytic continuation and the openness of $\theta(\Omega)$ in $\widehat{\Omega}$. For the extension property suppose $x \in \Omega$ and choose an open neighbourhood U of x such that $\pi|U$ is a homeomorphism. Now $\theta(x) = (\tilde{f}_{i,x})$ and $\pi_i((\tilde{f}_{i,x})) = \tilde{f}_{i,x}$. Hence

$$\widehat{f}_i(\theta(x)) = ((f_i \pi^{-1})|_{\pi(U)}) \widehat{p} \theta(x) = (f_i \pi^{-1}|_{\pi(U)}) \pi(x) = f_i(x).$$

Step 3. $\widehat{\Omega}$ is a maximal analytic extension of Ω . Suppose that $(\widetilde{\Omega}, \eta)$ is any analytic extension of Ω . We may repeat the construction of Step 1 to obtain an analytic extension $\widehat{\widetilde{\Omega}}$ of $\widetilde{\Omega}$, with $\widehat{\widetilde{\Omega}}$ a connected component of $\mathcal{O}^I$. Observe that the construction used implies that $\widehat{\widetilde{\Omega}}$ necessarily contains the image of Ω by θ in $\mathcal{O}^I$. Since $\widehat{\Omega}$ is defined as the connected component $\theta(\Omega)$ in $\mathcal{O}^I$ it follows immediately that $\widehat{\widetilde{\Omega}} = \widehat{\Omega}$. Hence $\widehat{\Omega}$ is a maximal analytic extension of Ω .

Step 4. $\widehat{\Omega}$ is independent, up to isomorphism, of π . We first note that if $\widetilde{\Omega}$ is an analytic extension of Ω then every analytic map $F : \Omega \rightarrow \mathbb{C}^m$, $m \geq 1$, extends uniquely to $\widetilde{\Omega}$. This follows by applying the definition componentwise to F . Let us suppose that we are given two spreadings π_1, π_2 of Ω with corresponding maximal analytic extensions $(\widehat{\Omega}_1, \theta_1), (\widehat{\Omega}_2, \theta_2)$. By Step 3 and the above remark, π_2 extends to $\pi'_2 : \widehat{\Omega}_1 \rightarrow \mathbb{C}^n$. We claim that π'_2 spreads $\widehat{\Omega}_1$ in $\mathbb{C}^n$. For this we first note that the tangent bundle $T\widehat{\Omega}_1$ is analytically trivial since $\widehat{\Omega}_1$ is spread in $\mathbb{C}^n$ by π_1 . But if $T\widehat{\Omega}_1 \cong \widehat{\Omega}_1 \times \mathbb{C}^n$, it follows that we can take the derivative of π'_2 to obtain a map $D\pi'_2 : \widehat{\Omega}_1 \rightarrow L(\mathbb{C}^n, \mathbb{C}^n)$. Since π_2 is a spreading it follows that $q(x) = D\pi_2(\theta_1(x))$ is an isomorphism for all $x \in \Omega$. Hence we may define an analytic map $p : \Omega \rightarrow L(\mathbb{C}^n, \mathbb{C}^n)$ by $p(x) = q(x)^{-1}$, $x \in \Omega$. Applying our remark again we see that p extends to $\widehat{p} : \widehat{\Omega}_1 \rightarrow L(\mathbb{C}^n, \mathbb{C}^n)$. Form the composite maps $p \cdot D\pi'_2, D\pi'_2 \cdot p$ (composition in $L(\mathbb{C}^n, \mathbb{C}^n)$). We see that both compositions are analytic and equal to the identity on $\theta_1(\Omega)$. Hence, by uniqueness of analytic continuation, they are equal to the identity on the whole of $\widehat{\Omega}_1$. It follows that $D\pi'_2$ is invertible on $\widehat{\Omega}_1$ and so, by the inverse function theorem, π'_2 defines a spreading of $\widehat{\Omega}_1$ in $\mathbb{C}^n$, compatible with the given complex structure on $\widehat{\Omega}_1$.



Now $(\widehat{\Omega}_1, \theta_1)$ is an analytic extension of (Ω, π_2) , where we choose the spreading of $\widehat{\Omega}_1$ given by π'_2 . By the maximality property of $(\widehat{\Omega}_2, \theta_2)$ it follows that there exists a morphism $\alpha : \widehat{\Omega}_1 \rightarrow \widehat{\Omega}_2$ satisfying $\alpha\theta_1 = \theta_2$. The map α is unique by uniqueness of analytic continuation. We may now repeat the above constructions to obtain a morphism $\beta : (\widehat{\Omega}_2, \pi'_1) \rightarrow (\widehat{\Omega}_1, \pi_1)$. It remains to prove that α and β are inverses of one another. This follows, again by uniqueness of analytic continuation, once we have noticed that

$$\beta\alpha(\theta_1(x)) = \theta_1(x); \quad \alpha\beta(\theta_2(x)) = \theta_2(x), \quad x \in \Omega.$$

□

Corollary 6.2.5. Let $\widetilde{\Omega}$ be an analytic extension of Ω . If $\widetilde{\Omega}$ is a domain of holomorphy, $\widetilde{\Omega}$ is the envelope of holomorphy of Ω .

Proof. Let $\widehat{\widetilde{\Omega}}$ denote the envelope of holomorphy of $\widetilde{\Omega}$ given by Theorem 6.2.4. If $\widetilde{\Omega}$ is a domain of holomorphy then $\widehat{\widetilde{\Omega}}$ is isomorphic to $\widetilde{\Omega}$. But $\widehat{\widetilde{\Omega}}$ is a maximal analytic extension of $\widetilde{\Omega}$ and hence of Ω . It follows that $\widetilde{\Omega}$ is the envelope of holomorphy of Ω . □

Remarks.

1. The map $\theta : \Omega \rightarrow \widehat{\Omega}$ constructed in Step 1 of the proof of Theorem 6.2.4 will not be injective unless $A(\Omega)$ separates points in Ω . This requirement is fulfilled of course if Ω is a domain in $\mathbb{C}^n$.
2. It should be noticed that the proof of Theorem 6.2.4 works for any subset of $A(\Omega)$. In particular, it gives us the maximum domain of continuation of any analytic function defined on Ω , representing it as a manifold spread over $\mathbb{C}^n$.

3. It should be appreciated that Theorem 6.2.4 is very much an existence and uniqueness theorem that gives no information about how to construct and represent envelopes of holomorphy in practice. Of course, our existence proof is very formal and elementary and not too much should be expected. In this regard it should be compared with the classical continuation proof such as is given in Hörmander [1, Theorem 5.4.5].
4. The envelope of holomorphy of a manifold spread over $\mathbb{C}^n$ may also be represented as the spectrum of the algebra $A(\Omega)$. That is, the envelope of holomorphy is isomorphic to the space of non-zero continuous homomorphisms of $A(\Omega)$ into $\mathbb{C}$. For this approach to the construction of the envelope of holomorphy we refer to Gunning and Rossi [1].
5. Finally we remark the fundamental result that a manifold spread over $\mathbb{C}^n$ is a domain of holomorphy if and only if it is holomorphically convex and if and only if it is isomorphic to its envelope of holomorphy. In particular, every domain of holomorphy is a Stein manifold. Proofs of these results, which use pseudo-convexity methods, may be found in Gunning and Rossi [1] and Hörmander [1].

Exercises.

1. Let (Ω, π) be a manifold spread over $\mathbb{C}^n$. Show that if we let $\tilde{\Omega}$ denote the quotient space of Ω defined by the relation “ x equivalent to y iff x and y cannot be separated by elements of $A(\Omega)$ ”, then
 - A) $\tilde{\Omega}$ is naturally a manifold spread over $\mathbb{C}^n$.
 - B) $A(\tilde{\Omega})$ separates points in $\tilde{\Omega}$.
 - C) The envelopes of holomorphy of Ω and $\tilde{\Omega}$ are isomorphic.
2. Prove that a holomorphically convex manifold spread over $\mathbb{C}^n$ is a domain of holomorphy (Hint: Follow the proof given in Chapter 2).
3. Let (Ω, π) be a manifold spread over $\mathbb{C}^n$ and suppose $\Omega \neq \mathbb{C}^n$. Given a compact subset K of Ω , define $d(K) = \inf\{|z - \pi(\zeta)| : z \in \partial(\pi\Omega), \zeta \in K\}$ (see also §4, Chapter 2). Prove that if (Ω, π) is a domain of holomorphy then $d(K) = d(\widehat{K})$ for all compact subsets of Ω . Show also that if (Ω, π) is finitely sheeted and $d(K) = d(\widehat{K})$ for all compact subsets K of Ω , then (Ω, π) is holomorphically convex and so a domain of holomorphy (the non-finitely sheeted case is more difficult and is treated in Gunning and Rossi [1]).

3 Sheaf cohomology

In §1 we showed how sheaf formalism provided a unifying topological framework for the description of a wide range of structures on topological spaces. In this section we introduce a powerful computational machine for the analysis of sheaves: *Sheaf Cohomology*. In essence our methods allow us to apply the highly systematised and powerful methods of homological algebra to problems in global complex analysis and algebraic geometry. The ideas we describe were introduced into complex analysis by H. Cartan (see H. Cartan [1, 2]) and into algebraic geometry by J.P. Serre (see J.P. Serre [2]). This use of sheaves and sheaf cohomology has undoubtedly revolutionised and clarified both fields.

Our approach to sheaf cohomology will be to first develop a rather abstract, non-computable, theory which is valid for sheaves defined over paracompact spaces. We then relate this theory to the computable Čech theory by means of Leray’s theorem. Throughout this

section we shall assume that all topological spaces are paracompact and, as always, Hausdorff. “Sheaf” will always refer to a sheaf of abelian groups unless the contrary is indicated.

Suppose that $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H} \rightarrow 0$ is a short exact sequence of sheaves over X . Given an open set $U \subset X$, it is easily seen that the sequence

$$0 \rightarrow \mathcal{F}(U) \xrightarrow{a_U} \mathcal{G}(U) \xrightarrow{b_U} \mathcal{H}(U)$$

is exact but that, in general, $b_U : \mathcal{G}(U) \rightarrow \mathcal{H}(U)$ will not be surjective (The section functor is left- but not right-exact). Just by reference to the Cousin I and II problems (Examples 24, 25, §1) we see the importance of finding a measure of how far the map $b_U : \mathcal{G}(U) \rightarrow \mathcal{H}(U)$ fails to be surjective. A satisfactory solution to this problem is the primary aim of sheaf cohomology theory. Our first task will be to describe a class of sheaves for which it is true that short exact sequences of sheaves transform into short exact sequences of groups under the section functor. That is, we shall be describing a class of sheaves for which sheaf and presheaf exactness are equivalent.

Definition 6.3.1. A sheaf $(\mathcal{F}, \pi, X)$ is said to be *soft* if for all closed subsets K of X and sections $s \in \mathcal{F}(K)$, s extends to a continuous section of $\mathcal{F}$ over X . That is, the natural map $\mathcal{F}(X) \rightarrow \mathcal{F}(K)$ is surjective for all closed subsets K of X .

Proposition 6.3.2. Let $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H} \rightarrow 0$ be a short exact sequence of sheaves over X . Provided $\mathcal{F}$ is soft, the corresponding sequence of sections

$$0 \rightarrow \mathcal{F}(X) \xrightarrow{a} \mathcal{G}(X) \xrightarrow{b} \mathcal{H}(X) \rightarrow 0$$

is exact.

Proof. First we remark that we shall now generally use the same notation for the map induced on sections and the corresponding sheaf morphism. If it is necessary to distinguish them, we use a “star” superscript for the map induced on sections. We must show that for every $s \in \mathcal{H}(X)$, there exists $t \in \mathcal{G}(X)$ such that $b(t) = s$. Since $b : \mathcal{G} \rightarrow \mathcal{H}$ is surjective, we may find an open neighbourhood U_x of every $x \in X$ and $t_x \in \mathcal{G}(U_x)$ such that $b(t_x) = s|_{U_x}$. Hence, by the paracompactness of X , we may find a locally finite open cover $\{U_i : i \in I\}$ of X and family $\{t_i \in \mathcal{G}(U_i)\}$ such that $b(t_i) = s|_{U_i}$, $i \in I$. Choose an open refinement $\{V_i\}$ of $\{U_i\}$ such that $\overline{V_i} \subset U_i$, $i \in I$. Consider the set Λ of pairs (g, J) where $J \subset I$ and, setting $V_J = \bigcup_{j \in J} \overline{V_j}$, we have $g \in \mathcal{G}(V_J)$ and $b(g) = s|_{V_J}$. Now $\Lambda \neq \emptyset$ and is partially ordered by inclusion. The requirements of Zorn’s lemma are clearly satisfied and so Λ contains a maximal element, say (t, K) . It is sufficient to show $K = I$. Suppose $i \in I \setminus K$. Then $b(t - t_i) = 0$ on $V(K) \cap \overline{V_i}$ and so there exists $r_i \in \mathcal{F}(V(K) \cap \overline{V_i})$ such that $t - t_i = a(r_i)$. Since $\mathcal{F}$ is soft, r_i extends to U_i . But now $(t, K) \cup (a(r_i) + t_i, i)$ extends (t, K) contradicting the maximality of (t, K) . Therefore $K = I$. $\square$

Corollary 6.3.3. Let $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H} \rightarrow 0$ be a short exact sequence of sheaves over X . If $\mathcal{F}$ and $\mathcal{G}$ are soft so is $\mathcal{H}$.

Proof. Let $K \subset X$ be closed. We must show that every $s \in \mathcal{H}(K)$ extends to a section of $\mathcal{H}$ over X . First observe that $\mathcal{F}_K$ is soft. Hence $0 \rightarrow \mathcal{F}_K \xrightarrow{a} \mathcal{G}_K \xrightarrow{b} \mathcal{H}_K \rightarrow 0$ is a short exact sequence of sheaves for which $\mathcal{F}_K$ is soft and, applying Proposition 6.3.2, there exists a section t of $\mathcal{G}$ over K such that $b(t) = s$. Since $\mathcal{G}$ is soft, t extends to a section of $\mathcal{G}$ over X . $\square$

Corollary 6.3.4. Let $0 \rightarrow \mathcal{F}_0 \xrightarrow{a_0} \mathcal{F}_1 \xrightarrow{a_1} \mathcal{F}_2 \rightarrow \cdots$ be a long exact sequence of soft sheaves over X . Then the corresponding complex

$$0 \rightarrow \mathcal{F}_0(X) \xrightarrow{a_0} \mathcal{F}_1(X) \xrightarrow{a_1} \mathcal{F}_2(X) \xrightarrow{a_2} \cdots$$

is exact.

Proof. For $i \geq 0$, let $\mathcal{K}_i$ denote the sheaf $\text{Ker}(a_i)$. The exactness of the given long exact sequence is then equivalent to the exactness of the short exact sequences

$$0 \rightarrow \mathcal{K}_i \rightarrow \mathcal{F}_i \xrightarrow{a_{i+1}} \mathcal{K}_{i+1} \rightarrow 0, \quad i \geq 0.$$

For $i = 0$, $\mathcal{K}_0 = \mathcal{F}_0$ and so is soft. Hence, by Corollary 6.3.3, $\mathcal{K}_1$ is soft. By induction, every $\mathcal{K}_i$ is soft. Hence by Proposition 6.3.2, the sequences

$$0 \rightarrow \mathcal{K}_i(X) \rightarrow \mathcal{F}_i(X) \xrightarrow{a_{i+1}} \mathcal{K}_{i+1}(X) \rightarrow 0$$

are all exact. But this is equivalent to the required result. $\square$

Definition 6.3.5. A sheaf $(\mathcal{F}, \pi, X)$ is *fine* if it admits a partition of unity of the identity morphism of $\mathcal{F}$ subordinate to any locally finite open cover of X . That is, given a locally finite open cover $\{U_i\}$ of X , there exist sheaf morphisms $\eta_i : \mathcal{F} \rightarrow \mathcal{F}$ satisfying

1. $\eta_i = 0$ outside of some closed subset of X contained in U_i .
2. $\sum_{i \in I} \eta_i = I$, the identity morphism of $\mathcal{F}$.

Proposition 6.3.6. Every fine sheaf is soft.

Proof. Let $\mathcal{F}$ be a fine sheaf over X , $K \subset X$ be closed and $\tilde{s} \in \mathcal{F}(K)$. By Lemma 6.1.12, we may find an open neighbourhood U of K in X and $\tilde{s} \in \mathcal{F}(U)$ extending s . Take the open cover $\{U, X \setminus K\}$ of X and a partition of unity $\{\theta, \eta\}$ of the identity morphism of $\mathcal{F}$ subordinate to this cover. Observe that $\theta|_K = I$. Setting $\tilde{s} = 0$ outside U , we see that $\theta\tilde{s}$ is the required section of $\mathcal{F}$. $\square$

Examples.

1. Let E be a smooth vector bundle over the differential manifold X . Then $\underline{E}_\infty$ is fine. Indeed, suppose that $\{U_i\}$ is any locally finite open cover of X and let $\{\eta_i\}$ be a C^∞ partition of unity subordinate to $\{U_i\}$. The η_i induce sheaf morphisms $\eta_i : \underline{E}_\infty \rightarrow \underline{E}_\infty$ and clearly $\{\eta_i\}$ is a partition of unity of the identity morphism of $\underline{E}_\infty$ which is subordinate to $\{U_i\}$. Hence $\underline{E}_\infty$ is fine. Consequently all the sheaves $\underline{C}^p$, $\underline{C}^{p,q}$ that we have defined on differential and complex manifolds are fine and, therefore, soft. On the other hand, if $\underline{E}$ is the sheaf of holomorphic sections of a holomorphic vector bundle E then $\underline{E}$ is never soft.
2. Let $\mathcal{F}$ be a sheaf on X . Let $\mathcal{F}^*$ denote the sheaf of germs of not necessarily continuous sections of $\mathcal{F}$. Thus, $\mathcal{F}^*(U)$ will be the set of *all* sections of $\mathcal{F}$ over U . The sheaf $\mathcal{F}^*$ is obviously fine. In fact it is also *flabby*: Every section of $\mathcal{F}^*$ over an open subset of X extends to X . Flabby sheaves are used in the development of sheaf cohomology in algebraic geometry where the spaces are not even Hausdorff, let alone paracompact (see also the exercises at the end of this section).

We need one more definition before we can define the sheaf cohomology groups of a sheaf on X .

Definition 6.3.7. Let $\mathcal{F}$ be a sheaf on X . A *resolution* of $\mathcal{F}$ is a long exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{F}_0 \xrightarrow{d_0} \mathcal{F}_1 \xrightarrow{d_1} \mathcal{F}_2 \xrightarrow{d_2} \dots$$

A resolution of $\mathcal{F}$ will be called *soft* (resp. *fine*) if each of the sheaves $\mathcal{F}_i$ is soft (resp. fine).

Proposition 6.3.8. Every sheaf $\mathcal{F}$ on X has a fine resolution.

Proof. Let $\mathcal{F}^*$ denote the sheaf of germs of not necessarily continuous sections of $\mathcal{F}$ (example 2 above) and let $\epsilon : \mathcal{F} \rightarrow \mathcal{F}^*$ denote the corresponding inclusion map. Set $\mathcal{F}_0 = \mathcal{F}^*$ and let $\overline{\mathcal{F}}_0 = \mathcal{F}^*/\mathcal{F}$. We have the short exact sequence $0 \rightarrow \mathcal{F} \rightarrow \mathcal{F}_0 \xrightarrow{q_0} \overline{\mathcal{F}}_0 \rightarrow 0$. Proceeding inductively, let $\mathcal{F}_{j+1} = (\overline{\mathcal{F}}_j)^*$ and $\overline{\mathcal{F}}_{j+1} = \mathcal{F}_{j+1}/\mathcal{F}_{j+1}^*$. For $j \geq 0$ we have the short exact sequences

$$0 \rightarrow \overline{\mathcal{F}}_j \xrightarrow{\epsilon_{j+1}} \mathcal{F}_{j+1} \xrightarrow{q_{j+1}} \overline{\mathcal{F}}_{j+1} \rightarrow 0.$$

Here, of course, $\overline{\mathcal{F}}_0 = \mathcal{F}$. Hence we have the corresponding long exact sequence

$$0 \rightarrow \mathcal{F} \xrightarrow{\epsilon} \mathcal{F}_0 \xrightarrow{d_0} \mathcal{F}_1 \xrightarrow{d_1} \dots$$

where $d_j = \epsilon_{j+1}q_j$ and $\epsilon = \epsilon_0$. Since the sheaves $\mathcal{F}_j$ are all fine, we have constructed a fine resolution of $\mathcal{F}$. $\square$

Remark. We call the resolution of $\mathcal{F}$ constructed in the proof of Proposition 6.3.8 the *canonical resolution* of $\mathcal{F}$.

Let $\mathcal{F}$ be a sheaf on X and $0 \rightarrow \mathcal{F} \xrightarrow{\epsilon} \mathcal{F}_0 \xrightarrow{d_0} \mathcal{F}_1 \xrightarrow{d_1} \dots$ be the canonical resolution of $\mathcal{F}$. Associated to the canonical resolution of $\mathcal{F}$ we have the complex

$$0 \rightarrow \mathcal{F}(X) \xrightarrow{\epsilon^*} \mathcal{F}_0 \xrightarrow{d_0^*} \mathcal{F}_1 \xrightarrow{d_1^*} \dots$$

We set

$$H^0(X, \mathcal{F}) = \text{Ker}(d_0^*); \quad H^p(X, \mathcal{F}) = \text{Ker}(d_p^*)/\text{Im}(d_{p-1}^*), \quad p > 0.$$

Each $H^p(X, \mathcal{F})$ is an abelian group.

Definition 6.3.9. The group $H^p(X, \mathcal{F})$ constructed above is called the *p th. sheaf cohomology group of X with coefficients in the sheaf $\mathcal{F}$* .

Theorem 6.3.10. The sheaf cohomology groups of X satisfy the following basic properties:

1. Given a sheaf $\mathcal{F}$ on X then
 - A) $H^0(X, \mathcal{F}) \approx \mathcal{F}(X)$.
 - B) $H^p(X, \mathcal{F}) = 0$ for $p > 0$ if $\mathcal{F}$ is fine.
2. If $a : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves over X then for $p \geq 0$ there are induced homomorphisms $a^p : H^p(X, \mathcal{F}) \rightarrow H^p(X, \mathcal{G})$ satisfying
 - A) $a^0 : H^0(X, \mathcal{F}) \rightarrow H^0(X, \mathcal{G})$ is precisely the map on sections induced by a .
 - B) If $a : \mathcal{F} \rightarrow \mathcal{F}$ is the identity, then a^p is the identity, $p \geq 0$.
 - C) If $a : \mathcal{F} \rightarrow \mathcal{G}$, $b : \mathcal{G} \rightarrow \mathcal{H}$ are morphisms of sheaves over X then for $p \geq 0$ we have $(ba)^p = b^p a^p : H^p(X, \mathcal{F}) \rightarrow H^p(X, \mathcal{H})$.
3. If $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H} \rightarrow 0$ is a short exact sequence of sheaves on X then for $p \geq 0$ there is a connecting homomorphism $\delta : H^p(X, \mathcal{H}) \rightarrow H^{p+1}(X, \mathcal{F})$ satisfying

A) The cohomology sequence

$$0 \rightarrow H^0(X, \mathcal{F}) \xrightarrow{a^0} H^0(X, \mathcal{G}) \xrightarrow{b^0} H^0(X, \mathcal{H}) \xrightarrow{\delta} H^1(X, \mathcal{F}) \xrightarrow{a^1} \dots$$

is exact.

B) Given a commutative diagram of short exact sequences on X

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F} & \xrightarrow{a} & \mathcal{G} & \xrightarrow{b} & \mathcal{H} & \longrightarrow & 0 \\ & & \eta \downarrow & & \downarrow \phi & & \downarrow \omega & & \\ 0 & \longrightarrow & \mathcal{A} & \xrightarrow{c} & \mathcal{B} & \xrightarrow{d} & \mathcal{C} & \longrightarrow & 0 \end{array}$$

the corresponding cohomology diagram

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & H^0(X, \mathcal{F}) & \xrightarrow{a^0} & H^0(X, \mathcal{G}) & \xrightarrow{b^0} & H^0(X, \mathcal{H}) & \xrightarrow{\delta} & H^1(X, \mathcal{F}) & \xrightarrow{a^1} & \dots \\ & & \eta^0 \downarrow & & \downarrow \phi^0 & & \downarrow \omega^0 & & \downarrow \eta^1 & & \\ 0 & \longrightarrow & H^0(X, \mathcal{A}) & \xrightarrow{c^0} & H^0(X, \mathcal{B}) & \xrightarrow{d^0} & H^0(X, \mathcal{C}) & \xrightarrow{\delta} & H^1(X, \mathcal{A}) & \xrightarrow{c^1} & \dots \end{array}$$

commutes.

Proof. 1A is obvious and 1B follows from Corollary 6.3.4. In our proof of 2 we follow the notational conventions of the proof of Proposition 6.3.8. Observe that a morphism $a : \mathcal{F} \rightarrow \mathcal{G}$ induces a morphism $a^0 : \mathcal{F}_0 \rightarrow \mathcal{G}_0$ and so a morphism $\bar{a}^0 : \bar{\mathcal{F}}_0 \rightarrow \bar{\mathcal{G}}_0$. Clearly the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F}_0 & \longrightarrow & \bar{\mathcal{F}}_0 \longrightarrow 0 \\ & & a \downarrow & & \downarrow a_0 & & \downarrow \bar{a}_0 \\ 0 & \longrightarrow & \mathcal{G} & \longrightarrow & \mathcal{G}_0 & \longrightarrow & \bar{\mathcal{G}}_0 \longrightarrow 0 \end{array}$$

commutes. Proceeding inductively, we see that for $j \geq 0$ we have morphisms $a^j : \mathcal{F}_j \rightarrow \mathcal{G}_j$, $\bar{a}_j : \bar{\mathcal{F}}_j \rightarrow \bar{\mathcal{G}}_j$ such that the diagrams

$$\begin{array}{ccccccc} 0 & \longrightarrow & \bar{\mathcal{F}}_j & \longrightarrow & \mathcal{F}_j & \longrightarrow & \bar{\mathcal{F}}_{j+1} \longrightarrow 0 \\ & & \bar{a}^j \downarrow & & \downarrow a^j & & \downarrow \bar{a}^{j+1} \\ 0 & \longrightarrow & \bar{\mathcal{G}}_j & \longrightarrow & \mathcal{G}_j & \longrightarrow & \bar{\mathcal{G}}_{j+1} \longrightarrow 0 \end{array}$$

commute. Hence we have the commutative ladder of long exact sequences:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F} & \xrightarrow{\epsilon} & \mathcal{F}_0 & \xrightarrow{d_0} & \mathcal{F}_1 \xrightarrow{d_1} \dots \\ & & a \downarrow & & \downarrow a^0 & & \downarrow a^1 \\ 0 & \longrightarrow & \mathcal{G} & \xrightarrow{\epsilon} & \mathcal{G}_0 & \xrightarrow{d_0} & \mathcal{G}_1 \xrightarrow{d_1} \dots \end{array}$$

and hence the corresponding commutative diagram of sections:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}(X) & \xrightarrow{\epsilon} & \mathcal{F}_0(X) & \xrightarrow{d_0^*} & \mathcal{F}_1(X) \xrightarrow{d_1^*} \dots \\ & & a \downarrow & & \downarrow a^0 & & \downarrow a^1 \\ 0 & \longrightarrow & \mathcal{G}(X) & \xrightarrow{\epsilon} & \mathcal{G}_0(X) & \xrightarrow{d_0^*} & \mathcal{G}_1(X) \xrightarrow{d_1^*} \dots \end{array}$$

Suppose $s \in \text{Ker}(d_j^*)$. The commutativity of the diagram implies that $d_j^*(a^j s) = a^j d_j^* s = 0$. Hence $a^j(\text{Ker}(d_j^*)) \subset \text{Ker}(d_j^*)$. Similarly, $a^j(\text{Im}(d_{j-1}^*)) \subset \text{Im}(d_{j-1}^*)$. Therefore, a^j induces a homomorphism $a^j : H^j(X, \mathcal{F}) \rightarrow H^j(X, \mathcal{G})$. Properties 2A, B, C all follow straightforwardly from the definition of the induced maps on cohomology and the naturality of our constructions and we leave their verification to the reader.

It remains to prove 3. Take canonical resolutions of the sheaves $\mathcal{F}, \mathcal{G}, \mathcal{H}$. As in the proof of 2 we have an induced sequence between these resolutions and a corresponding commutative diagram of sections, a typical portion of which is displayed below.

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & \mathcal{F}_{j-1}(X) & \xrightarrow{a^{j-1}} & \mathcal{G}_{j-1}(X) & \xrightarrow{b^{j-1}} & \mathcal{H}_{j-1}(X) & \longrightarrow & 0 \\
 & & \downarrow d_{j-1} & & \downarrow d_{j-1} & & \downarrow d_{j-1} & & \\
 0 & \longrightarrow & \mathcal{F}_j(X) & \xrightarrow{a^j} & \mathcal{G}_j(X) & \xrightarrow{b^j} & \mathcal{H}_j(X) & \longrightarrow & 0 \\
 & & \downarrow d_j & & \downarrow d_j & & \downarrow d_j & & \\
 0 & \longrightarrow & \mathcal{F}_{j+1}(X) & \xrightarrow{a^{j+1}} & \mathcal{G}_{j+1}(X) & \xrightarrow{b^{j+1}} & \mathcal{H}_{j+1}(X) & \longrightarrow & 0 \\
 & & \downarrow d_{j+1} & & \downarrow d_{j+1} & & \downarrow d_{j+1} & & \\
 0 & \longrightarrow & \mathcal{F}_{j+2}(X) & \xrightarrow{a^{j+2}} & \mathcal{G}_{j+2}(X) & \xrightarrow{b^{j+2}} & \mathcal{H}_{j+2}(X) & \longrightarrow & 0
 \end{array}$$

Observe that the rows in the diagram are all exact since the sheaves $\mathcal{F}_j, \mathcal{G}_j$ and $\mathcal{H}_j$ are all soft. We now show how to define the connecting homomorphism $\delta : H^j(X, \mathcal{H}) \rightarrow H^{j+1}(X, \mathcal{F})$. Let $\alpha \in H^j(X, \mathcal{H})$ and $H \in \text{Ker}(d_j) \subset \mathcal{H}_j(X)$ be a representative for α . Now $H = b^j(G)$ for some $G \in \mathcal{G}_j(X)$. Since $d_j b^j = b^{j+1} d_j$, we see that $b^{j+1} d_j(G) = 0$ and so there exists $F \in \mathcal{F}_{j+1}(X)$ such that $a^{j+1}(F) = d_j(G)$. But $a^{j+2} d_{j+1}(F) = d_{j+1} a^{j+1}(F) = d_{j+1} d_j(G) = 0$. Since a^{j+2} is injective it follows that $d_{j+1}(F) = 0$ and so F defines an element of $H^{j+1}(X, \mathcal{F})$. We claim that the class of F in $H^{j+1}(X, \mathcal{F})$ depends only on α and not on our choices of H, G and F . Granted this, our construction defines the connecting homomorphism $\delta : H^j(X, \mathcal{H}) \rightarrow H^{j+1}(X, \mathcal{F})$. Suppose that H', G' and F' are the result of another sequence of choices. Since $H - H'$ defines the zero element of $H^j(X, \mathcal{H})$, there exists $\tilde{H} \in \mathcal{H}_{j-1}(X)$ such that $H - H' = d_{j-1}(\tilde{H})$. Certainly $\tilde{H} = b^{j-1}(\tilde{G})$ for some $\tilde{G} \in \mathcal{G}_{j-1}(X)$ and so $b^j(\tilde{G} - G' - d_{j-1}(\tilde{G})) = 0$. Hence $G - G' - d_{j-1}(\tilde{G}) = a^j \tilde{F}$, for some $\tilde{F} \in \mathcal{F}_j(X)$. Now $a^{j+1} d_j(F) = d_j a^j(F) = d_j(G) - d_j(G') = a^{j+1}(F) - a^{j+1}(F') = a^{j+1}(F - F')$ and so, since a^{j+1} is injective, we have $F - F' = d_j(F)$. Therefore, F and F' define the same class in $H^{j+1}(X, \mathcal{F})$.

The proof of 3A involves a straightforward diagram chase and we leave details to the reader. To prove 3B we take canonical resolutions of the sequences and form the corresponding 3-dimensional commutative diagram of sequences of sections. Given $\alpha \in H^j(X, \mathcal{H})$, we construct, as above, elements H, G, F such that the cohomology classes of H and F define α and $\delta\alpha$ respectively. It now suffices to observe that the class of $\omega^j H$ is $\eta^j \alpha$ and that $\omega^j H, \phi^j G, \eta^{j+1} F$ is a sequence defining $\delta(\omega^j \alpha)$. Hence $\delta\omega^j = \omega^{j+1}\delta$. The commutativity of the squares not involving δ is, of course, immediate from 2C. $\square$

Theorem 6.3.11. Let $0 \rightarrow \mathcal{F} \xrightarrow{\epsilon} \mathcal{F}_0 \xrightarrow{d_0} \mathcal{F}_1 \xrightarrow{d_1} \mathcal{F}_2 \xrightarrow{d_2} \cdots$ be a resolution of the sheaf $\mathcal{F}$. Suppose that $H^j(X, \mathcal{F}_k) = 0, j > 0, k > 0$. Then

$$H^0(X, \mathcal{F}) \cong \text{Ker}(d_0^*); \quad H^p(X, \mathcal{F}) \cong \text{Ker}(d_p^*) / \text{Im}(d_{p-1}^*), \quad p > 0.$$

Proof. Let $\mathcal{K}_j = \text{Ker}(d_j)$, $j \geq 0$. The exactness of the resolution of $\mathcal{F}$ is equivalent to the exactness of the sequences

$$0 \rightarrow \mathcal{K}_j \rightarrow \mathcal{F}_j \xrightarrow{d_j} \mathcal{K}_{j+1} \rightarrow 0, \quad j \geq 0.$$

Take the long exact cohomology sequences of these short exact sequences:

$$\rightarrow H^p(X, \mathcal{F}_j) \rightarrow H^p(X, \mathcal{K}_{j+1}) \xrightarrow{\delta} H^{p+1}(X, \mathcal{K}_j) \rightarrow H^{p+1}(X, \mathcal{F}_j) \rightarrow .$$

Since $H^p(X, \mathcal{F}_j) = 0$, $p > 0$, we see that

$$H^p(X, \mathcal{K}_{j+1}) \cong H^{p+1}(X, \mathcal{K}_j), \quad p > 0, \quad j \geq 0. \quad (\text{A})$$

We also have the initial portion of the long exact cohomology sequences:

$$\rightarrow H^0(X, \mathcal{F}_j) \xrightarrow{d_j^*} H^0(X, \mathcal{K}_{j+1}) \xrightarrow{\delta} H^1(X, \mathcal{K}_j) \rightarrow 0$$

and so $H^1(X, \mathcal{K}_j) \cong H^0(X, \mathcal{K}_{j+1})/d_j^*H^0(X, \mathcal{F}_j)$. Now $H^0(X, \mathcal{K}_{j+1}) = \text{Ker}(d_{j+1}^*)$ and so

$$H^1(X, \mathcal{K}_j) \cong \text{Ker}(d_{j+1}^*)/\text{Im}(d_j^*), \quad j \geq 0. \quad (\text{B})$$

By repeated application of A and one application of B we see that for $p > 0$

$$H^p(X, \mathcal{F}) = H^p(X, \mathcal{K}_0) \cong H^{p-1}(X, \mathcal{K}_1) \cong \cdots \cong H^1(X, \mathcal{K}_{p-1}) \cong \text{Ker}(d_p^*)/\text{Im}(d_{p-1}^*).$$

Finally the result for $p = 0$ follows from the exactness of the sequence

$$0 \rightarrow H^0(X, \mathcal{F}) \xrightarrow{\epsilon^*} H^0(X, \mathcal{F}_0) \rightarrow H^0(X, \mathcal{K}_1).$$

□

As an immediate consequence of Theorem 6.3.11 together with 1B of Theorem 6.3.10 we have

Corollary 6.3.12. We can compute the cohomology of X with coefficients in $\mathcal{F}$ using any fine (or soft) resolution of $\mathcal{F}$. In particular, Properties 1–3 of Theorem 6.3.10 determine the cohomology groups $H^j(X, \mathcal{F})$ up to isomorphism.

Remark. Suppose that $\tilde{H}^j(X, \mathcal{F})$, $j \geq 0$, are the groups of another sheaf cohomology theory for X . That is, we suppose that the groups $\tilde{H}^j(X, \mathcal{F})$ satisfy all the properties listed in Theorem 6.3.10. By Corollary 6.3.12, we have $\tilde{H}^j(X, \mathcal{F}) \cong H^j(X, \mathcal{F})$ for all sheaves $\mathcal{F}$ on X . It can be shown that this isomorphism is natural – in particular, commutes with connecting homomorphisms. For the general proof the reader may consult Godement [1]. We shall prove the existence of this natural isomorphism for the case of Čech cohomology later in the section.

Examples.

- Let X be a topological manifold. Taking the canonical resolution of the constant sheaf $\mathbb{Z}$ we may define the cohomology groups $H^p(X, \mathbb{Z})$, $p \geq 0$. We may also define the singular cohomology groups $H_{\text{sing}}^p(X, \mathbb{Z})$ of X (see Spanier [1], Greenberg [1]). We claim that $H^p(X, \mathbb{Z}) \cong H_{\text{sing}}^p(X, \mathbb{Z})$, $p \geq 0$. First, let $S_p(U, \mathbb{Z})$ denote the abelian groups of integral singular p -chains on the open subset U of X . Then $S^p(U, \mathbb{Z}) = \text{Hom}_{\mathbb{Z}}(S_p(U, \mathbb{Z}), \mathbb{Z})$ is the group of singular integral p -cochains on U . We let $D = D_p : S^p(U, \mathbb{Z}) \rightarrow S^{p+1}(U, \mathbb{Z})$ denote the coboundary homomorphism. The assignment $U \mapsto S^p(U, \mathbb{Z})$ defines a presheaf $S^p(\mathbb{Z})$ on X and we let $\underline{S}^p(\mathbb{Z})$ denote the corresponding sheaf, $p \geq 0$. Since D commutes with restrictions, we arrive at the sheaf complex

$$0 \rightarrow \mathbb{Z} \rightarrow \underline{S}^0(\mathbb{Z}) \xrightarrow{D_0} \underline{S}^1(\mathbb{Z}) \xrightarrow{D_1} \cdots . \quad (\text{A})$$

By definition,

$$H_{\text{sing}}^0(X, \mathbb{Z}) = \text{Ker}(D_0^*)$$

$$H_{\text{sing}}^0(X, \mathbb{Z}) = \text{Ker}(D_p^*) / \text{Im}(D_{p-1}^*), \quad p \geq 1.$$

We claim that (A) is a fine resolution of $\mathbb{Z}$. Exactness follows since every point of X has a neighbourhood base of contractible open neighbourhoods which have vanishing singular cohomology by standard theory. To prove fineness of the sheaves $\underline{S}^j(\mathbb{Z})$ first observe that $\underline{S}^0(\mathbb{Z}) = \mathbb{Z}^*$ – sheaf of germs of discontinuous sections of $\mathbb{Z}$. Hence $\underline{S}^0(\mathbb{Z})$ is fine. Suppose that $p > 0$ and $s \in \underline{S}^p(\mathbb{Z})(K)$, $K \subset X$ closed. By Lemma 6.1.12, s is the restriction of a continuous section $\tilde{s}$ of $\underline{S}^p(\mathbb{Z})$ over some open neighbourhood U of K . Let 1_K be the (continuous!) section of $\underline{S}^0(\mathbb{Z})$ over X defined by $1_K|K \equiv 1$, $1_K|X \setminus K \equiv 0$. Since $\underline{S}^p(\mathbb{Z})$ is a sheaf of $\underline{S}^0(\mathbb{Z})$ -modules, $1_K \tilde{s}$ is a continuous section of $\underline{S}^p(\mathbb{Z})$ over X extending s . We may now apply Corollary 6.3.12 to obtain the required isomorphisms between $H^p(X, \mathbb{Z})$ and $H_{\text{sing}}^p(X, \mathbb{Z})$. We conclude this example by making two additional remarks. First, replacing $\mathbb{Z}$ by any abelian group G (or commutative ring with 1), we can repeat the proof to obtain isomorphisms between $H^p(X, G)$ and $H_{\text{sing}}^p(X, G)$. Secondly, if X is a differential manifold we may define the sheaves $\underline{S}_{\infty}^p(\mathbb{Z})$ of smooth (that is C^{∞}) p -cochains on X . It is a basic and well known fact in differential topology that the complex $0 \rightarrow \mathbb{Z} \rightarrow \underline{S}_{\infty}^0(\mathbb{Z}) \xrightarrow{D_0} \underline{S}_{\infty}^1(\mathbb{Z}) \rightarrow \dots$ continues to define the singular cohomology groups of X . In particular, the complex is a fine resolution of $\mathbb{Z}$.

4. Let X be an n -dimensional differential manifold and $0 \rightarrow \mathbb{C} \rightarrow \underline{C}^0 \xrightarrow{d} \dots \xrightarrow{d} \underline{C}^n \rightarrow 0$ be the de Rham complex of X . Since the sheaves $\underline{C}^p$ are all fine, the de Rham complex is a fine resolution of the constant sheaf $\mathbb{C}$. Let

$$H_{\text{DR}}^0(X, \mathbb{C}) = \text{Ker } d : C^0(X) \rightarrow C^1(X)$$

$$H_{\text{DR}}^p(X, \mathbb{C}) = \frac{\text{Ker } d : C^p(X) \rightarrow C^{p+1}(X)}{\text{Im } d : C^{p-1}(X) \rightarrow C^p(X)}, \quad p \geq 1.$$

denote the *de Rham cohomology* groups of X . Corollary 6.3.12 implies that

$$H_{\text{DR}}^p(X, \mathbb{C}) \cong H^p(X, \mathbb{C}), \quad p \geq 0.$$

In particular, $H^p(X, \mathbb{C}) = 0$, $p > n$.

We showed in Example 3 that $H^p(X, \mathbb{C}) \cong H_{\text{sing}}^p(X, \mathbb{C})$ and we shall now describe an explicit isomorphism between $H_{\text{DR}}^p(X, \mathbb{C})$ and $H_{\text{sing}}^p(X, \mathbb{C})$. For $p \geq 0$ we define a sheaf morphism $I^p : \underline{C}^p \rightarrow \underline{S}^p(\mathbb{C})$ by integration of chains:

$$I^p(f)(c) = \int_c f, \quad f \in C^p(U), \quad c \in S_p(U, \mathbb{C}).$$

By Stokes' theorem, we see that I^p commutes with the differentials d , D and so we obtain a commutative ladder of exact sequences:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{C} & \longrightarrow & \underline{C}^0 & \xrightarrow{d} & \underline{C}^1 & \xrightarrow{d} & \dots \\ & & \downarrow I & & \downarrow I^0 & & \downarrow I^1 & & \\ 0 & \longrightarrow & \mathbb{C} & \longrightarrow & \underline{S}^0(\mathbb{C}) & \xrightarrow{D} & \underline{S}^1(\mathbb{C}) & \xrightarrow{D} & \dots \end{array}$$

The morphisms I^p induce the de Rham isomorphisms between $H_{\text{DR}}^p(X, \mathbb{C})$ and $H_{\text{sing}}^p(X, \mathbb{C})$.

Finally we should point out that the de Rham isomorphisms actually give an *algebra* isomorphism between the de Rham and singular cohomology groups (the algebra structure on the de Rham and singular cohomology groups is given by wedge and cup product respectively). The reader may find a proof of this stronger statement in Warner [1].

5. Let X be an n -dimensional complex manifold. For $p \geq 0$, we have the Dolbeault complex

$$0 \rightarrow \Omega^p \rightarrow \underline{C}^{p,0} \xrightarrow{\bar{\partial}} \underline{C}^{p,1} \xrightarrow{\bar{\partial}} \cdots \rightarrow \underline{C}^{p,n} \rightarrow 0.$$

Since the sheaves $\underline{C}^{p,q}$ are fine, $p, q \geq 0$, we therefore obtain the *Dolbeault isomorphisms*

$$H^q(X, \Omega^p) \cong \frac{\text{Ker } \bar{\partial} : C^{p,q}(X) \rightarrow C^{p,q+1}(X)}{\text{Im } \bar{\partial} : C^{p,q-1}(X) \rightarrow C^{p,q}(X)}, \quad q \geq 0.$$

In particular,

$$H^q(X, \mathcal{O}) = \frac{\text{Ker } \bar{\partial} : C^{0,q}(X) \rightarrow C^{0,q+1}(X)}{\text{Im } \bar{\partial} : C^{0,q-1}(X) \rightarrow C^{0,q}(X)}, \quad q \geq 0.$$

If E is a holomorphic vector bundle on X , we have the Dolbeault complex

$$0 \rightarrow \Omega^p(E) \rightarrow \underline{C}^{p,0}(E) \xrightarrow{\bar{\partial}} \underline{C}^{p,1}(E) \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \underline{C}^{p,n}(E) \rightarrow 0$$

and corresponding Dolbeault isomorphisms

$$H^q(X, \Omega^p(E)) \cong \frac{\text{Ker } \bar{\partial} : C^{p,q}(M, E) \rightarrow C^{p,q+1}(M, E)}{\text{Im } \bar{\partial} : C^{p,q-1}(M, E) \rightarrow C^{p,q}(M, E)}, \quad p, q \geq 0.$$

6. We have the short exact sequence $0 \rightarrow \mathcal{O} \rightarrow \mathcal{M} \rightarrow \mathcal{M}/\mathcal{O} \rightarrow 0$ of sheaves over any complex manifold X . Take the initial portion of the long exact cohomology sequence¹:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(X, \mathcal{O}) & \longrightarrow & H^0(X, \mathcal{M}) & \longrightarrow & H^0(X, \mathcal{M}/\mathcal{O}) \xrightarrow{\delta} H^1(X, \mathcal{O}) \\ & & \parallel & & \parallel & & \\ & & A(X) & & M(X) & & \end{array}$$

Given the data $\alpha \in H^0(X, \mathcal{M}/\mathcal{O})$ for the Cousin I problem on X , we see that we can solve the Cousin I problem for α if and only if $\delta\alpha = 0$ in $H^1(X, \mathcal{O})$. In particular, X will be a Cousin I domain if and only if $\text{Im } \delta = 0$ in $H^1(X, \mathcal{O})$. Later we shall see that $H^1(X, \mathcal{O}) = 0$ whenever X is a Stein manifold and so Stein manifolds are Cousin I domains.

7. Let Z be an analytic subset of the complex manifold X . We have the short exact sequence

$$0 \rightarrow \mathcal{I}_Z \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_Z \rightarrow 0$$

of sheaves over X and corresponding initial portion of the long exact cohomology sequence:

$$\begin{array}{ccccccc} \longrightarrow & H^0(X, \mathcal{O}_X) & \longrightarrow & H^0(X, \mathcal{O}_Z) & \xrightarrow{\delta} & H^1(X, \mathcal{I}_Z) & \longrightarrow \cdots \\ & \parallel & & \parallel & & & \\ & A(X) & & A(Z) & & & \end{array}$$

Suppose $f \in A(Z)$. Then f is the restriction of an analytic function on X if and only if $\delta f = 0$. In particular, if $H^1(X, \mathcal{I}_Z) = 0$ every analytic function on Z extends to an analytic function on X . We shall see later that this cohomology group vanishes whenever X is Stein.

¹Diagram correction: $M(X)$ is aligned with $H^0(X, \mathcal{M})$. The original diagram on Part II, p. 106 places it one column too far to the right.

Examples 6 and 7 above should convince the reader of the importance of computing sheaf cohomology groups in complex analysis.

For the remainder of this section we develop Čech cohomology theory for sheaves over a paracompact space. As we shall see Čech theory is computable – at least for all the examples that interest us – and is also isomorphic to the sheaf cohomology theory we have already constructed.

Let $\mathcal{U} = \{U_i : i \in I\}$ be an open cover of X and p be a non-negative integer. Given $s = (s_0, \dots, s_p) \in I^{p+1}$, set $U_s = U_{s_0} \cap \dots \cap U_{s_p}$. A p -cochain of $\mathcal{U}$ with values in $\mathcal{F}$ is a map c which assigns to each $s \in I^{p+1}$ a section $c_s \in \mathcal{F}(U_s)$ and for which c_s is an alternating function of s . That is, $c_{s_0 \dots s_i \dots s_j \dots s_p} = -c_{s_0 \dots s_j \dots s_i \dots s_p}$, $0 \leq i < j \leq p$. We let $C^p(\mathcal{U}, \mathcal{F})$ denote the abelian group of all p -cochains of $\mathcal{U}$ with values in $\mathcal{F}$.

For $p \geq 0$, we define a coboundary operator $D : C^p(\mathcal{U}, \mathcal{F}) \rightarrow C^{p+1}(\mathcal{U}, \mathcal{F})$ by

$$(Dc)_s = \sum_{j=0}^{p+1} (-1)^j c_{s_0 \dots \widehat{s}_j \dots s_{p+1}}, \quad s \in I^{p+2}.$$

A simple computation shows that $D^2 = 0$. We let

$$Z^p(\mathcal{U}, \mathcal{F}) = \{c \in C^p(\mathcal{U}, \mathcal{F}) : Dc = 0\}; \quad B^p(\mathcal{U}, \mathcal{F}) = \{Dc : c \in C^{p-1}(\mathcal{U}, \mathcal{F})\}$$

denote the groups of p -cocycles and p -coboundaries respectively. Here we take $C^{-1}(\mathcal{U}, \mathcal{F}) = 0$ and $B^0(\mathcal{U}, \mathcal{F}) = 0$. Since $D^2 = 0$, $B^p(\mathcal{U}, \mathcal{F})$ is a subgroup of $Z^p(\mathcal{U}, \mathcal{F})$ and we let $H^p(\mathcal{U}, \mathcal{F})$ denote the quotient group $Z^p(\mathcal{U}, \mathcal{F})/B^p(\mathcal{U}, \mathcal{F})$. We call $H^p(\mathcal{U}, \mathcal{F})$ the p th. cohomology group of $\mathcal{U}$ with values in $\mathcal{F}$.

Example 8. The data for the Cousin A (resp. Cousin B) problem on a complex manifold defines a class in $H^1(\mathcal{U}, \mathcal{O})$ (resp. $H^1(\mathcal{U}, \mathcal{O}^*)$).

Lemma 6.3.13. For any sheaf $\mathcal{F}$ on X we have

$$H^0(\mathcal{U}, \mathcal{F}) \approx \mathcal{F}(X).$$

Proof. Certainly $Z^0(\mathcal{U}, \mathcal{F}) = H^0(\mathcal{U}, \mathcal{F})$. But if $c \in Z^0(\mathcal{U}, \mathcal{F})$, $c_i - c_j = 0$ on U_{ij} for all $i, j \in I$. Hence we may define $f \in \mathcal{F}(X)$ by $f|_{U_i} = c_i$. $\square$

Lemma 6.3.14. Let $a : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves over X . For $p \geq 0$ we have induced maps $a_{\mathcal{U}}^p : H^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(\mathcal{U}, \mathcal{G})$ satisfying

- A. $a_{\mathcal{U}}^0 : H^0(\mathcal{U}, \mathcal{F}) \rightarrow H^0(\mathcal{U}, \mathcal{G})$ is precisely the map on sections induced by a .
- B. If a is the identity map of $\mathcal{F}$, then $a_{\mathcal{U}}^p$ is the identity, $p \geq 0$.
- C. If $a : \mathcal{F} \rightarrow \mathcal{G}$, $b : \mathcal{G} \rightarrow \mathcal{H}$ are morphisms of sheaves over X then $(ba)_{\mathcal{U}}^0 = b_{\mathcal{U}}^p a_{\mathcal{U}}^p$, $p \geq 0$.

Proof. Elementary and left to the reader. $\square$

Lemma 6.3.15. Suppose that $\mathcal{F}$ is fine. Then $H^p(\mathcal{U}; \mathcal{F}) = 0$, $p > 0$.

Proof. Let $\mathcal{V} = \{V_j, j \in J\}$ be a locally finite open refinement of $\mathcal{U}$ chosen such that there is a map $\phi : J \rightarrow I$ with $\overline{V_j} \subset U_{\phi(j)}$, $j \in J$. Choose a partition of unity $\{\eta_j\}$ for the sheaf $\mathcal{F}$ which is subordinate to $\mathcal{V}$. Let $c \in Z^p(\mathcal{U}, \mathcal{F})$. For $j \in J$, define $b^j \in C^{p-1}(\mathcal{U}, \mathcal{F})$ by

$$b_s^j = \begin{cases} 0 & \text{if } U_s \cap V_j = \emptyset, \\ (-1)^p \eta_j c_{s_0 \dots s_{p-1} \phi(j)} & \text{if } U_s \cap V_j \neq \emptyset. \end{cases}$$

(Here $\eta_j c_{s_0 \dots \phi(j)}$ is defined to be zero outside V_j). A simple computation shows that for all $s \in I^{p+1}$ we have

$$(Db^j)_s = \eta_j c_s.$$

Set $b = \sum_{j \in J} b^j$. Since $\{\eta_j\}$ is a partition of unity, we see that $DB = c$. Hence $H^p(\mathcal{U}, \mathcal{F}) = 0$. $\square$

Remark. To prove this result we needed to know that $\mathcal{F}$ was fine and not just soft. It is for this reason that we put “fine” rather than “soft” in 1B of Theorem 6.3.10. See also Godement [1, Theorem 5.2.3, Chapter 2].

Theorem 6.3.16. Let $\mathcal{F}$ be a sheaf on X and $\mathcal{U}$ be an open cover of X . For $p \geq 0$ we have a canonical homomorphism $\rho(\mathcal{U}) : H^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$ satisfying

A. If $a : \mathcal{F} \rightarrow \mathcal{G}$ is a homomorphism then

$$\begin{array}{ccc} H^p(\mathcal{U}, \mathcal{F}) & \xrightarrow{\rho(\mathcal{U})} & H^p(X, \mathcal{F}) \\ \downarrow a_{\mathcal{U}}^p & & \downarrow a^p \\ H^p(\mathcal{U}, \mathcal{G}) & \xrightarrow{\rho(\mathcal{U})} & H^p(X, \mathcal{G}) \end{array}$$

commutes.

B. If $H^p(U_s, \mathcal{F}) = 0$ for all $s \in I^{p+1}$ and $p > 0$, then $\rho(\mathcal{U})$ is an isomorphism (*Leray's Theorem*).

Proof. Undoubtedly the most elegant proof of this result uses spectral sequences – see, for example, Godement [1]. Our proof is an elementary diagram chase:

Set $\mathcal{F}_{-1} = \mathcal{F}$ and let $0 \rightarrow \mathcal{F}_{-1} \xrightarrow{d_{-1}} \mathcal{F}_0 \xrightarrow{d_0} \mathcal{F}_1 \xrightarrow{d_1} \dots$ denote the canonical resolution of $\mathcal{F}$. For $s \in I^{p+1}$ we have the sequence of sections

$$0 \rightarrow \mathcal{F}_{-1}(U_s) \xrightarrow{d_{-1}} \mathcal{F}_0(U_s) \xrightarrow{d_0} \dots$$

and so, taking the direct sum over I^{p+1} , we have for $p \geq 0$ sequences

$$0 \rightarrow C^p(\mathcal{U}, \mathcal{F}_{-1}) \xrightarrow{d_{-1}} C^p(\mathcal{U}, \mathcal{F}_0) \xrightarrow{d_0} \dots \quad (\text{A})$$

Associated to each of the sheaves $\mathcal{F}_j$, $j \geq -1$, we have sequences

$$0 \rightarrow \mathcal{F}_j(X) \xrightarrow{D_{-1}} C^0(\mathcal{U}, \mathcal{F}_j) \xrightarrow{D_0} C^1(\mathcal{U}, \mathcal{F}_j) \rightarrow \dots \quad (\text{B})$$

Here D_{-1} denotes the inclusion map and D_j is the appropriate coboundary operator for $j \geq 0$. Since $\mathcal{F}_j$ is fine, $j \geq 0$, the sequences B are exact for $j \geq 0$. Combining the sequences A and

B we arrive at a commutative diagram of sequences the initial portion of which is displayed below.

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \mathcal{F}_{-1}(X) & \xrightarrow{d_{-1}} & \mathcal{F}_0(X) & \xrightarrow{d_0} & \mathcal{F}_1(X) \xrightarrow{d_1} \dots \\
 & & \downarrow D_{-1} & & \downarrow D_{-1} & & \downarrow D_{-1} \\
 0 & \longrightarrow & C^0(\mathcal{U}, \mathcal{F}_{-1}) & \xrightarrow{d_{-1}} & C^0(\mathcal{U}, \mathcal{F}_0) & \xrightarrow{d_0} & C^0(\mathcal{U}, \mathcal{F}_1) \xrightarrow{d_1} \dots \\
 & & \downarrow D_0 & & \downarrow D_0 & & \downarrow D_0 \\
 0 & \longrightarrow & C^1(\mathcal{U}, \mathcal{F}_{-1}) & \xrightarrow{d_{-1}} & C^1(\mathcal{U}, \mathcal{F}_0) & \xrightarrow{d_0} & C^1(\mathcal{U}, \mathcal{F}_1) \xrightarrow{d_1} \dots \\
 & & \downarrow D_1 & & \downarrow D_1 & & \downarrow D_1 \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

The columns of this diagram are all exact, save the first. We now show how to construct the required homomorphism $\rho(\mathcal{U}) : H^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$. Let $c_0 \in Z^p(\mathcal{U}, \mathcal{F})$ represent the class $\alpha \in H^p(\mathcal{U}, \mathcal{F})$. We construct inductively a sequence $c_j \in C^{p-j}(\mathcal{U}, \mathcal{F}_{j-1})$ which satisfies

$$d_{p-j}c_j = D_{p-j+1}c_{j+1}, \quad 0 \leq j \leq p.$$

Suppose that we have constructed c_j , $j \leq r < p+1$. Then $D_{p-r}d_{r-1}c_r = d_{r-1}D_{p-r}c_r = d_{r-1}d_{r-2}c_{r-1} = 0$. Hence by exactness of the $(r+1)$ th. column, there exists $c_{r+1} \in C^{p-r-1}(\mathcal{U}, \mathcal{F}_r)$ such that $D_{p-r-1}c_{r+1} = d_{r-1}c_r$. This completes the inductive step. Now $c_{p+1} \in \mathcal{F}_p(X)$. Observe that $D_{-1}d_p c_{p+1} = d_p D_{-1}c_{p+1} = d_p d_{p-1}c_p = 0$ and so, since D_{-1} is injective, $d_p c_{p+1} = 0$. Therefore $c_{p+1} \in \text{Ker}(d_p)$ defines a class in $H^p(X, \mathcal{F})$. It is straightforward to verify that this class depends only on α and not on our choices of $c_0, \dots, c_p, c_{p+1}$. Our construction therefore defines the required homomorphism $\rho(\mathcal{U})$. Property A of the Proposition follows immediately since the sequence $c_0, \dots, c_{p+1}$ is mapped by a into a sequence $c_0, \dots, c_{p+1}$ defining $\rho(\mathcal{U})(a(\alpha))$.

If the conditions of B hold then rows, as well as columns, of our diagram are exact (excluding initial row and column) and so, by symmetry, we may construct an inverse for $\rho(\mathcal{U})$. $\square$

Remarks.

1. We call a cover satisfying the conditions of 1B a *Leray cover* (for $\mathcal{F}$). Notice that the existence of a Leray cover for $\mathcal{F}$ may imply that higher dimensional sheaf cohomology groups vanish. Thus, if $U_s = \emptyset$, $s \in I^{p+1}$, $p \geq p_0$, we see that $H^p(X, \mathcal{F}) = 0$, $p \geq p_0$.
2. It is clearly important to find Leray covers for a given sheaf $\mathcal{F}$. For example one can show, using a Riemannian metric, that every differential manifold has a cover by convex open sets (Helgason [1, page 54]). Since intersections of convex sets are convex and convex sets are contractible, it follows that differential manifolds admit Leray covers for the constant sheaves $\mathbb{Z}$, $\mathbb{R}$, $\mathbb{C}$, etc. At a much deeper level, we shall show later that Stein open covers of a complex manifold are Leray covers for an important class of $\mathcal{O}$ -modules.
3. The proof of Theorem 6.3.16 clearly works for any fine resolution of $\mathcal{F}$ – not just the canonical resolution. We frequently use this observation in the sequel.

Example 9. Let $\mathcal{U} = \{U_i\}$ be an open cover of the differential manifold X . We give an explicit computation for the map $\rho(\mathcal{U}) : H^p(\mathcal{U}, \mathbb{C}) \rightarrow H_{\text{DR}}^p(X, \mathbb{C})$ in case $p = 2$. First choose a C^∞ partition of unity $\{\theta_i\}$ of X subordinate to the cover $\mathcal{U}$. Let $\{c_{ijk}\} \in Z^2(\mathcal{U}, \mathbb{C})$ represent the class $\alpha \in H^2(\mathcal{U}, \mathbb{C})$. Define $\{\phi_{jk}\} \in C^1(\mathcal{U}, \underline{C}^0)$ by

$$\phi_{jk} = \sum_i \theta_i c_{ijk}.$$

Certainly we have $D_0(\{\phi_{jk}\}) = d_{-1}\{c_{ijk}\} (= \{c_{ijk}\})$. Again define $\{\phi_k\} \in C^0(\mathcal{U}, \underline{C}^1)$ by

$$\phi_k = \sum_j \theta_j d\phi_{jk}.$$

We have $D_1(\{\phi_j\}) = d\{\phi_{jk}\}$. Finally observe that $d\phi_k = d\phi_l$ on U_{kl} and so $\gamma = \{d\phi_k\}$ is a well defined closed 2-form on X which represents the class $\rho(\mathcal{U})(\alpha)$ in $H_{\text{DR}}^2(X, \mathbb{C})$. The construction for $p \neq 2$ is similar.

Proposition 6.3.17. Let $\mathcal{U}, \mathcal{V}$ be open covers of X and $\mathcal{V}$ be a refinement of $\mathcal{U}$. Given a sheaf $\mathcal{F}$ on X there exists a canonical homomorphism

$$\rho(\mathcal{U}, \mathcal{V}) : H^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(\mathcal{V}, \mathcal{F})$$

satisfying

1. The diagram

$$\begin{array}{ccc} H^p(\mathcal{U}, \mathcal{F}) & \xrightarrow{\rho(\mathcal{U}, \mathcal{V})} & H^p(\mathcal{V}, \mathcal{F}) \\ & \searrow \rho(\mathcal{U}) \quad \swarrow \rho(\mathcal{V}) & \\ & H^p(X, \mathcal{F}) & \end{array}$$

commutes, $p \geq 0$.

2. If $a : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism, we have

$$\rho(\mathcal{U}, \mathcal{V}) a_{\mathcal{U}}^p = a_{\mathcal{V}}^p \rho(\mathcal{U}, \mathcal{V}), \quad p \geq 0.$$

3. If $\mathcal{W}$ is a refinement of $\mathcal{V}$, we have $\rho(\mathcal{U}, \mathcal{W}) = \rho(\mathcal{V}, \mathcal{W}) \rho(\mathcal{U}, \mathcal{V})$.

Proof. Let $\mathcal{V} = \{V_j : j \in J\}$, $\mathcal{U} = \{U_i : i \in I\}$ and fix a *refinement mapping* $\phi : J \rightarrow I$ satisfying $V_j \subset U_{\phi(j)}$, $j \in J$. Given $c \in C^p(\mathcal{U}, \mathcal{F})$, define $\tilde{\phi}c \in C^p(\mathcal{V}, \mathcal{F})$ by

$$(\tilde{\phi}c)_s = c_{\phi(s_0)\dots\phi(s_p)}|V_s, \quad s = (s_0, \dots, s_p) \in I^{p+1}.$$

Since $\tilde{\phi}$ obviously commutes with the coboundary operators D , $\tilde{\phi}$ induces a map $\phi_* : H^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(\mathcal{V}, \mathcal{F})$. We claim that ϕ_* is independent of the choice of ϕ . Suppose that ϕ' is another refinement mapping. We construct a “homotopy” operator $H : C^{p+1}(\mathcal{U}, \mathcal{F}) \rightarrow C^p(\mathcal{V}, \mathcal{F})$ between $\tilde{\phi}$ and $\tilde{\phi}'$. To do this we first give J a total ordering. Then if $s_0 < \dots < s_p$ and $c \in C^{p+1}(\mathcal{U}, \mathcal{F})$ we define

$$(Hc)_s = \sum_{j=0}^p (-1)^j c_{\phi(s_0)\dots\phi(s_j)\phi'(s_j)\dots\phi'(s_p)}.$$

Extend H to all cochains by requiring that H is alternating in s . Computing we find that

$$HD + DH = \tilde{\phi}' - \tilde{\phi}.$$

Hence, if $c \in Z^p(\mathcal{U}, \mathcal{F})$, we have $(\tilde{\phi}' - \tilde{\phi})(c) \in B^p(\mathcal{V}, \mathcal{F})$. Therefore $\phi_* = \phi'_*$ and we may set $\rho(\mathcal{U}, \mathcal{V}) = \phi_*$ for any choice of refinement mapping ϕ . The proofs of the remaining statements of the proposition are elementary and we leave details to the reader. $\square$

An immediate corollary of Proposition 6.3.17 is that $\{H^p(\mathcal{U}, \mathcal{F}), \rho(\mathcal{U}, \mathcal{V})\}$ constitutes a direct system. Hence we may define

$$\check{H}^p(X, \mathcal{F}) = \text{dirlim}_{\mathcal{U}} H^p(\mathcal{U}, \mathcal{F}).$$

We call $\check{H}^p(X, \mathcal{F})$ the *p*th. Čech cohomology group of X with coefficients in $\mathcal{F}$.

By part 1 of Proposition 6.3.17, we see that for $p \geq 0$ we have canonical homomorphisms

$$\begin{aligned} \chi(\mathcal{U}) : H^p(\mathcal{U}, \mathcal{F}) &\rightarrow \check{H}^p(X, \mathcal{F}) \\ \chi : \check{H}^p(X, \mathcal{F}) &\rightarrow H^p(X, \mathcal{F}). \end{aligned}$$

These homomorphisms satisfy the usual naturality properties. For example, if $a : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves we have induced homomorphisms $\check{a}^p : \check{H}^p(X, \mathcal{F}) \rightarrow \check{H}^p(X, \mathcal{G})$ satisfying the conditions of part 2 of Theorem 6.3.10 and

$$\chi \check{a}^p = \chi a^p, \quad p \geq 0.$$

Theorem 6.3.18. For paracompact spaces, Čech and sheaf cohomology groups are isomorphic.

To prove this result, it is sufficient to show that the Čech groups satisfy all the properties of Theorem 6.3.10. In view of what we have proved already, it remains to construct a long exact cohomology sequence for $\check{H}$. Actually, we shall prove a little more. We shall show that Čech and sheaf cohomology theory are canonically isomorphic with canonical isomorphisms being given by the maps χ . In particular, we shall show that the maps χ commute with connecting homomorphisms.

Theorem 6.3.19. Let $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H} \rightarrow 0$ be a short exact sequence of sheaves on X . For $p \geq 0$ we have a connecting homomorphism $\delta : \check{H}^p(X, \mathcal{H}) \rightarrow \check{H}^{p+1}(X, \mathcal{F})$ satisfying

A. The Čech cohomology sequence

$$0 \rightarrow \check{H}^0(X, \mathcal{F}) \xrightarrow{\check{a}^0} \check{H}^0(X, \mathcal{G}) \xrightarrow{\check{b}^0} \check{H}^0(X, \mathcal{H}) \xrightarrow{\delta} \check{H}^1(X, \mathcal{F}) \xrightarrow{\check{a}^1} \dots$$

is exact.

B. The diagram

$$\begin{array}{ccccccc} 0 \rightarrow & \check{H}^0(X, \mathcal{F}) & \xrightarrow{\check{a}^0} & \check{H}^0(X, \mathcal{G}) & \xrightarrow{\check{b}^0} & \check{H}^0(X, \mathcal{H}) & \xrightarrow{\delta} \check{H}^1(X, \mathcal{F}) \xrightarrow{\check{a}^1} \dots \\ & \downarrow \chi & & \downarrow \chi & & \downarrow \chi & \downarrow \chi \\ 0 \rightarrow & H^0(X, \mathcal{F}) & \xrightarrow{a^0} & H^0(X, \mathcal{G}) & \xrightarrow{b^0} & H^0(X, \mathcal{H}) & \xrightarrow{\sigma} H^1(X, \mathcal{F}) \xrightarrow{a^1} \dots \end{array}$$

commutes.

Proof. Let $\mathcal{U}$ be an open cover of X and $p \geq 0$. Set $\tilde{C}^p(\mathcal{U}, \mathcal{H}) = bC^p(\mathcal{U}, \mathcal{G})$. Then the sequence

$$0 \rightarrow C^p(\mathcal{U}, \mathcal{F}) \rightarrow C^p(\mathcal{U}, \mathcal{G}) \rightarrow \tilde{C}^p(\mathcal{U}, \mathcal{H}) \rightarrow 0$$

is exact. Since $D\tilde{C}^{p-1}(\mathcal{U}, \mathcal{H}) \subset \tilde{C}^p(\mathcal{U}, \mathcal{H})$ we may define the “cohomology” group $\tilde{H}^p(\mathcal{U}, \mathcal{H})$ to be $\tilde{Z}^p(\mathcal{U}, \mathcal{H})/\tilde{B}^p(\mathcal{U}, \mathcal{H})$. Exactly as in the proof of 3 in Theorem 6.3.10 we may define a connecting homomorphism $\tilde{\delta} : \tilde{H}^p(\mathcal{U}, \mathcal{H}) \rightarrow H^{p+1}(\mathcal{U}, \mathcal{F})$ and obtain a long exact sequence for the cohomology groups $H^*(\mathcal{U}, \mathcal{F})$, $H^*(\mathcal{U}, \mathcal{G})$, $\tilde{H}^*(\mathcal{U}, \mathcal{H})$. Everything commutes with direct limits and so we arrive at the long exact cohomology sequence

$$\dots \rightarrow \tilde{H}^p(X, \mathcal{H}) \rightarrow \check{H}^{p+1}(X, \mathcal{F}) \xrightarrow{\check{a}^{p+1}} \check{H}^{p+1}(X, \mathcal{G}) \xrightarrow{\check{b}^{p+1}} \check{H}^{p+1}(X, \mathcal{H}) \rightarrow \dots$$

We claim that $\tilde{H}^p(X, \mathcal{H}) \approx \check{H}^p(X, \mathcal{H})$. For this it suffices to show that if $h \in C^p(\mathcal{U}, \mathcal{H})$, we can find a refinement $\mathcal{V} = \{V_j : j \in J\}$ of $\mathcal{U} = \{U_i : i \in I\}$ and refinement map $\phi : J \rightarrow I$ such that $\tilde{\phi}h \in \tilde{C}^p(\mathcal{V}, \mathcal{H})$. Since X is paracompact, we may assume that $\mathcal{U}$ is a locally finite cover of X . Let $\mathcal{W} = \{W_i\}$ be a refinement of $\mathcal{U}$ such that $\overline{W}_i \subset U_i$, $i \in I$. Choose an open neighbourhood V_x of each point $x \in X$ such that

- a) If $x \in W_i$, $V_x \subset W_i$ and if $V_x \cap W_j \neq \emptyset$, then $V_x \subset U_j$.
- b) If $x \in U_s$, $s \in I^{p+1}$, then $V_x \subset U_s$.

Observe that a) and b) continue to hold for open neighbourhoods of x contained in V_x . Since $b : \mathcal{G} \rightarrow \mathcal{H}$ is surjective we may, shrinking V_x if necessary, find $g_s^x \in \mathcal{G}(V_x)$ such that $bg_s^x = h_s|_{V_x}$. Choose a refining map $\phi : X \rightarrow I$ for the covers $\mathcal{V} = \{V_x : x \in X\}$ and $\mathcal{U}$ satisfying $x \in W_{\phi(x)}$ for all $x \in X$. Let $t = (t_0, \dots, t_p) \in X^{p+1}$ and suppose $V_{t_i} \neq \emptyset$. By choice of ϕ , $V_{t_0} \cap W_{\phi(t_j)} \neq \emptyset$, $0 \leq j \leq p$, and so by a) $V_{t_0} \subset U_{\phi(t_j)}$, $0 \leq j \leq p$. Hence $V_{t_0} \subset U_{\phi(t)}$, where we have used the abbreviated notation $\phi(t)$ for $\phi(t_0) \cdots \phi(t_p)$. For $t \in X^{p+1}$, we define $g_t \in \mathcal{G}(V_t)$ by

$$g_t = g_{t_0}^{t_0}.$$

Clearly $bg_t = h_{\phi(t)}|_{V_t}$. Hence $\tilde{\phi}h \in \tilde{C}^p(\mathcal{V}, \mathcal{H})$.

All that remains to be proved is statement B of the theorem. For this we take canonical resolutions of $\mathcal{F}$, $\mathcal{G}$ and $\mathcal{H}$, fix an open cover $\mathcal{U}$ of X and take appropriate complexes of cochains over each term in the canonical resolutions as in the proof of Theorem 6.3.13. We obtain a 3-dimensional commutative diagram of sequences. It is then a straightforward diagram chase to verify that the definition of $\tilde{\delta} : \tilde{H}^p(\mathcal{U}, \mathcal{H}) \rightarrow H^{p+1}(\mathcal{U}, \mathcal{F})$ is compatible with that of $\delta : H^p(X, \mathcal{H}) \rightarrow H^{p+1}(X, \mathcal{F})$. Essentially we have to check that the maps $\rho(\mathcal{U})$ map the defining sequences for $\tilde{\delta}$ down to defining sequences for δ . We omit the lengthy details. Finally take direct limits. $\square$

Remarks.

1. An alternative construction of the long exact sequence of Čech cohomology, based on sheaves of cochains (cf. Example 3) may be found in Godement [1].
2. The paracompactness assumption implicit in Theorem 6.3.19 is essential: Čech cohomology need not be exact for non-paracompact spaces. However, Leray's theorem continues to hold and this fact was exploited by Serre in his foundational paper on coherent sheaves in algebraic geometry (Serre [2]).

We end this chapter with some important examples and computations.

Examples.

10. Let X be a differential manifold with structure sheaf $\mathcal{D}$ and let $\mathcal{D}^*$ denote the (multiplicative) sheaf of groups of units of $\mathcal{D}$. We claim that the group $\text{CLB}(X)$ of isomorphism classes of complex line bundles on X is canonically isomorphic to $H^1(X, \mathcal{D}^*)$. Let $\xi \in H^1(X, \mathcal{D}^*)$. Now $H^1(X, \mathcal{D}^*) \approx \check{H}^1(X, \mathcal{D}^*)$ and so we may find an open cover $\mathcal{U} = \{U_i\}$ of X and $\{\phi_{ij}\} \in Z^2(\mathcal{U}, \mathcal{D}^*)$ such that $\chi(\mathcal{U})$ maps the cohomology class of $\{\phi_{ij}\}$ to ξ . The cocycle conditions on $\{\phi_{ij}\}$ imply that $\phi_{ij}\phi_{jk} = \phi_{ik}$, $i, j, k \in I$. Since $\phi_{ij} : U_{ij} \rightarrow \mathbb{C}^* \approx \text{GL}(1, \mathbb{C})$, we see that the ϕ_{ij} are the transition functions for a complex line bundle $L(\xi)$ on X . We leave it to the reader to check that $L(\xi)$ depends only on ξ and not on our particular choice of cover or cocycle and that the map $\xi \mapsto L(\xi)$ is an isomorphism of $H^1(X, \mathcal{D}^*)$ with $\text{CLB}(X)$.

We may use this isomorphism between $\text{CLB}(X)$ and $H^1(X, \mathcal{D}^*)$ to define an important topological invariant of complex line bundles. First observe that we have an exact sheaf sequence

$$0 \rightarrow \mathbb{Z} \xrightarrow{i} \mathcal{D} \xrightarrow{\exp} \mathcal{D}^* \rightarrow 0.$$

Here $i : \mathbb{Z} \rightarrow \mathcal{D}$ denotes the inclusion and $\exp : \mathcal{D} \rightarrow \mathcal{D}^*$ is defined by $\exp_U(f)(x) = \exp(2\pi i f(x))$, $x \in U$, $f \in \mathcal{D}(U)$. The surjectivity of $\exp$ follows by noting that $\exp_U$ has inverse $(2\pi i)^{-1} \log$ on simply connected open subsets U . Consider the following portion of the long exact cohomology sequence

$$\cdots \rightarrow H^1(X, \mathcal{D}) \rightarrow H^1(X, \mathcal{D}^*) \xrightarrow{\delta} H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathcal{D}) \rightarrow \cdots.$$

Since $\mathcal{D}$ is a fine sheaf, $H^1(X, \mathcal{D}) = H^2(X, \mathcal{D}) = 0$ and so $\delta : H^1(X, \mathcal{D}^*) \approx H^2(X, \mathbb{Z})$. Suppose $L \in H^1(X, \mathcal{D}^*)$ is a complex line bundle. Define $c_1(L) = -\delta(L) \in H^2(X, \mathbb{Z})$. We call $c_1(L)$ the *first Chern class* of L . We see that $L = \underline{\mathbb{C}}$ – the trivial line bundle – if and only if $c_1(L) = 0$. Fix an open cover $\mathcal{U} = \{U_i : i \in I\}$ of X such that all intersections U_s , $s \in I^{p+1}$, $p \geq 0$, are contractible. For example, we can choose a cover of X by contractible open sets (see Remark 2 following Theorem 6.3.16). Applying Leray's theorem, we can see that $H^2(X, \mathbb{Z}) \approx H^2(\mathcal{U}, \mathbb{Z})$ and $H^1(X, \mathcal{D}^*) \approx H^1(\mathcal{U}, \mathcal{D}^*)$. Given $L \in \text{CLB}(X)$ we may therefore find transition functions $\{\phi_{ij}\}$ for L defined relative to the cover $\mathcal{U}$. Every intersection U_{ij} is simply connected and so we may choose a branch of $\log \phi_{ij}$ on each U_{ij} . The cocycle conditions on $\{\phi_{ij}\}$ imply that for all $i, j, k \in I$,

$$c_{ijk} = (2\pi i)^{-1}(\log \phi_{ij} + \log \phi_{jk} + \log \phi_{ki}) \in \mathbb{Z},$$

and so $\{c_{ijk}\} \in Z^2(\mathcal{U}, \mathbb{Z})$. By the construction of the connecting homomorphism, it follows that $\{c_{ijk}\}$ is a representative for $c_1(L) \in H^2(X, \mathbb{Z})$.

11. Let X be a complex manifold. As in Example 10 we have a short exact sequence

$$0 \rightarrow \mathbb{Z} \xrightarrow{i} \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \rightarrow 0$$

of sheaves on X and we see that $H^1(X, \mathcal{O}^*)$ is isomorphic to the group $\text{HLB}(X)$ of holomorphic line bundles on X . We have the commutative diagram

$$\begin{array}{ccccccc} \cdots & \longrightarrow & H^1(X, \mathcal{O}) & \longrightarrow & H^1(X, \mathcal{O}^*) & \xrightarrow{\delta} & H^2(X, \mathbb{Z}) \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \parallel \\ \cdots & \longrightarrow & H^1(X, \mathcal{D}) & \longrightarrow & H^1(X, \mathcal{D}^*) & \xrightarrow{\delta} & H^2(X, \mathbb{Z}) \longrightarrow \cdots \end{array}$$

Here the vertical maps are induced by the natural inclusions of $\mathcal{O}$ in $\mathcal{D}$, $\mathcal{O}^*$ in $\mathcal{D}^*$ and the identity map of $\mathbb{Z}$. Hence, as in Example 10, the map $\delta : H^1(X, \mathcal{O}^*) \rightarrow H^2(X, \mathbb{Z})$ is minus the first Chern class map. However, the map will generally not be an isomorphism as neither of $H^1(X, \mathcal{O})$ and $H^2(X, \mathcal{O})$ need vanish. This is just a reflection of the fact that a holomorphic line bundle may be trivial in $\text{CLB}(X)$ but not trivial in $\text{HLB}(X)$. The existence of a Leray cover for $\mathcal{O}^*$ is now a non-trivial matter as it depends on being able to find covers which are Leray for $\mathcal{O}$ as well as $\mathbb{Z}$.

12. We continue with the notation and assumptions of Example 11. Let $j : \mathbb{Z} \rightarrow \mathbb{C}$ denote the inclusion map of constant sheaves. We have the commutative diagram

$$\begin{array}{ccc} \text{HLB}(X) \approx H^1(X, \mathcal{O}^*) & \xrightarrow{\delta} & H^2(X, \mathbb{Z}) \\ & \searrow \theta & \downarrow j \\ & & H^2(X, \mathbb{C}) \\ & & \cong \\ & & H_{\text{DR}}^2(X, \mathbb{C}) \end{array}$$

Here $\theta : H^1(X, \mathcal{O}^*) \rightarrow H_{\text{DR}}^2(X, \mathbb{C})$ is defined to be the composition of δ with j and the canonical isomorphism between $H^2(X, \mathbb{C})$ and $H_{\text{DR}}^2(X, \mathbb{C})$. We let $c_1(L)_{\mathbb{C}} = -\theta(L) \in H_{\text{DR}}^2(X, \mathbb{C})$, $L \in \text{HLB}(X)$. We are going to give an explicit representation of $c_1(L)_{\mathbb{C}}$ as a closed 2-form on X . Suppose that $\{\phi_{ij}\}$ are the transition functions of $L \in \text{HLB}(X)$ relative to some open cover $\mathcal{U} = \{U_i\}$ of X . As in Example 10 we may and shall assume that all intersections U_s are contractible. As we showed in Example 10, $c_1(L) \in H^2(X, \mathbb{Z})$ is represented by the cocycle

$$\{c_{ijk}\} = \{(2\pi i)^{-1}(\log \phi_{ij} + \log \phi_{jk} + \log \phi_{ki})\} \in Z^2(\mathcal{U}, \mathbb{Z}).$$

Let us assume for the moment that there exist $\{a_i\} \in C^0(\mathcal{U}, \mathcal{D}^*)$ such that

$$|\phi_{ij}|^2 = a_j / a_i.$$

(This amounts to claiming that $L^* \otimes \bar{L}^* = \mathbb{C}$ in $\text{CLB}(X)$). As in the proof of Theorem 6.3.16 (see also Example 9), we first construct $\{f_i\} \in C^0(\mathcal{U}, \mathbb{C}^1)$ such that

$$\frac{1}{2\pi i} d \log \phi_{ij} = f_j - f_i.$$

For this we may take $f_i = \frac{1}{2\pi i} \partial \log a_i$, since

$$\log \phi_{ij} + \overline{\log \phi_{ij}} = \log a_j - \log a_i$$

and so

$$d \log \phi_{ij} = \partial \log \phi_{ij} = \partial \log a_j - \partial \log a_i.$$

Our required 2-form representing $c_1(L)_{\mathbb{C}}$ is then given by

$$\begin{aligned} \psi &= -\{df_i\} \\ &= -\left\{ \frac{1}{2\pi i} \bar{\partial} \partial \log a_i \right\} \\ &= \left\{ -\frac{i}{2\pi} \partial \bar{\partial} \log a_i \right\}. \end{aligned}$$

Notice that $c_1(L)_{\mathbb{C}}$ is represented by a $(1,1)$ -form. In Example 15 we shall show that every “integral” $(1,1)$ -form is the Chern class of some holomorphic line bundle. The reader should also observe that we do not really need to assume that the domains U_{ij} are simply connected as the indeterminacy in $\log \phi_{ij}$ drops out when we take exterior derivations.

Finally, let us justify our assumption that $L^* \otimes \bar{L}^*$ is isomorphic to the trivial line bundle. First observe that $L_z^* \otimes \bar{L}_z^*$ is the space of Hermitian forms on L_z , $z \in X$. Now any convex

combination of positive definite Hermitian quadratic forms is positive definite. Therefore, taking trivialisations of L over $\mathcal{U}$, we may choose a smooth section of $L^* \otimes \bar{L}^*$ over each U_i which defines a positive definite Hermitian form on the fibres. Glueing together using a smooth partition of unity we obtain a global section of $L^* \otimes \bar{L}^*$ which restricts to a positive definite Hermitian form on the fibres. Hence we have obtained a non-vanishing smooth section of $L^* \otimes \bar{L}^*$.

13. We compute the 1st. Chern class of the hyperplane section bundle H of $P^1(\mathbb{C})$. Let $\mathcal{U} = \{U_0, U_1\}$ be the open cover of $P^1(\mathbb{C})$ corresponding to the canonical atlas. The transition functions for the hyperplane section bundle are given by

$$\phi_{01}(z_0, z_1) = z_1/z_0$$

and setting

$$\begin{aligned} a_0(z_0, z_1) &= |z_0|^2 / (|z_0|^2 + |z_1|^2) \\ a_1(z_0, z_1) &= |z_1|^2 / (|z_0|^2 + |z_1|^2) \end{aligned}$$

we see that $|\phi_{01}|^2 = a_1/a_0$. Hence our representative μ for $c_1(H)_{\mathbb{C}}$ is given by $\mu = \{\mu_i\} = \{-\frac{i}{2\pi} \partial \bar{\partial} \log a_i\}$. Setting $z_0/z_1 = t$, we see that

$$\begin{aligned} \mu_1(t) &= -\frac{i}{2\pi} \partial \bar{\partial} \log(1/(1 + |t|^2)) \\ &= \frac{i}{2\pi} (1 + |t|^2)^{-2} dt d\bar{t}. \end{aligned}$$

Now for compact Riemann surfaces, integration defines a canonical isomorphism $H^2(X, \mathbb{C}) \approx \mathbb{C}$ and so we may regard $c_1(H)_{\mathbb{C}}$ as lying in $\mathbb{C}$. Thus

$$\begin{aligned} c_1(H)_{\mathbb{C}} &= \int_{P^1(\mathbb{C})} \mu = \int_{\mathbb{C}} \mu_1 \\ &= \frac{i}{2\pi} \int_{\mathbb{C}} (1 + |t|^2)^{-2} dt d\bar{t} \\ &= \frac{1}{\pi} \int_0^\infty \int_0^{2\pi} (1 + r^2)^{-2} r dr d\theta \\ &= +1. \end{aligned}$$

Since $H^2(X, \mathbb{Z}) \approx \mathbb{Z}$ for compact Riemann surfaces, we have proved $c_1(H) = +1$. Our choice of sign for c_1 was made precisely so that the hyperplane section bundle of $P^1(\mathbb{C})$ had positive Chern class...

14. Let L be a holomorphic line bundle on the compact Riemann surface M and suppose that $M^*(L) \neq \emptyset$. Recall (proposition 1.5.7) that if $s \in M^*(L)$ then $\deg(\text{div}(s))$ is independent of s and that the degree of L , $\deg(L)$, is defined to be $\deg(\text{div}(s))$. Using the canonical isomorphism of $H^2(M, \mathbb{Z})$ with $\mathbb{Z}$ we may regard $c_1(L)$ as lying in $\mathbb{Z}$. We claim that $c_1(L) = \deg(L)$. Fix $s \in M^*(L)$ and let $\text{div}(s) = \sum_{i=1}^n n_i \cdot z_i$. Choose a finite cover $\{U_i : i = 1, \dots, m\}$ of M satisfying

- a) $L|_{U_i}$ is holomorphically trivial.
- b) $z_i \in U_i$, $1 \leq i \leq n$.
- c) There exists an open neighbourhood V_i of each z_i such that V_i is biholomorphic to the unit disc in $\mathbb{C}$ and $V_i \cap \bigcup_{j \neq i} U_j = \emptyset$.

Denote the corresponding set of transition functions for L by $\{\phi_{jk} : U_{jk} \rightarrow \mathbb{C}^*\}$. Let s_j denote the local representative of s on U_j . Then s_j, s_k will be holomorphic on U_{jk} , $j \neq k$, and

$$|\phi_{jk}|^2 |s_k|^2 = |s_j|^2 \text{ on } U_{jk}.$$

For $i > n$, set $g_i = |s_i|^2$. For $i \leq n$, we may, using bump functions, modify $|s_i|^2$ to obtain a C^∞ positive function g_i on U_i satisfying

$$g_i|_{U_i \setminus V_i} = |s_i|^2.$$

In particular, $g_i|_{\partial V_i} = |s_i|^2$ and, by condition c) above, we will have

$$|\phi_{jk}|^2 g_k = g_j \text{ on } U_{jk} \text{ for all } j, k.$$

Hence, by Example 12, we have

$$\begin{aligned} c_1(L)_{\mathbb{C}} &= c_1(L) = -\frac{i}{2\pi} \int_M \partial \bar{\partial} \log g_j^{-1} \\ &= \frac{1}{2\pi i} \int_M \bar{\partial} \partial \log g_j, \\ c_1(L) &= \frac{1}{2\pi i} \sum_{i=1}^n \int_{V_i} \bar{\partial} \partial \log g_i. \end{aligned}$$

Now $d\partial \log g_i = \bar{\partial} \partial \log g_i$ and so, by Stokes' theorem,

$$\begin{aligned} c_1(L) &= \frac{1}{2\pi i} \sum_{i=1}^n \int_{\partial V_i} \partial \log g_i \\ &= \frac{1}{2\pi i} \sum_{i=1}^n \int_{\partial V_i} d \log s_i \\ &= \sum_{i=1}^n n_i, \quad \text{by the residue theorem.} \\ &= \deg(L). \end{aligned}$$

15. We showed in Example 12 that the Chern class of a holomorphic line bundle on X could be represented, via the de Rham isomorphism, by a closed $(1, 1)$ -form on X . We shall now show that every integral closed $(1, 1)$ -form is the Chern class of some holomorphic line bundle on X (modulo torsion). Let $j : \mathbb{Z} \rightarrow \mathbb{C}$ denote the inclusion map of constant sheaves and $H_{\text{DR}}^{1,1}(X, \mathbb{C})$ denote the subgroup of $H_{\text{DR}}^2(X, \mathbb{C})$ admitting representatives by closed $(1, 1)$ -forms. Set $H^{1,1}(X, \mathbb{Z}) = H_{\text{DR}}^{1,1}(X, \mathbb{C}) \cap H^2(X, \mathbb{Z})$. We claim that $c_1(H^1(X, \mathcal{O}^*))_{\mathbb{C}} = H^{1,1}(X, \mathbb{Z})$. Let $i : \mathbb{Z} \rightarrow \mathcal{O}$, $k : \mathbb{C} \rightarrow \mathcal{O}$ denote the inclusion maps of the constant sheaves $\mathbb{Z}, \mathbb{C}$ in $\mathcal{O}$. We have the commutative diagram²

$$\begin{array}{ccccc} H^1(X, \mathcal{O}^*) & \xrightarrow{c_1} & H^2(X, \mathbb{Z}) & \xrightarrow{i} & H^2(X, \mathcal{O}) \\ & & \searrow j & & \nearrow k \\ & & & H^2(X, \mathbb{C}) & \end{array}$$

²Diagram correction: the arrow labelled k points from $H^2(X, \mathbb{C})$ to $H^2(X, \mathcal{O})$, in accordance with the inclusion $k : \mathbb{C} \rightarrow \mathcal{O}$ just defined. Its direction is reversed in the original diagram on Part II, p. 122.

Since we already know that $c_1(H^1(X, \mathcal{O}^*))_{\mathbb{C}} \subset H^{1,1}(X, \mathbb{Z})$, it is enough to prove that $kH_{\text{DR}}^{1,1}(X, \mathbb{C}) = 0$. Taking the de Rham and Dolbeault resolutions of $\mathbb{C}$ and $\mathcal{O}$ we have the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{C} & \longrightarrow & \underline{C}^0 & \xrightarrow{d} & \underline{C}^1 & \xrightarrow{d} & \underline{C}^2 & \xrightarrow{d} & \dots \\ & & \downarrow k & & \downarrow I & & \downarrow Q_1 & & \downarrow Q_2 & & \\ 0 & \longrightarrow & \mathcal{O} & \longrightarrow & \underline{C}^{0,0} & \xrightarrow{\bar{\partial}} & \underline{C}^{0,1} & \xrightarrow{\bar{\partial}} & \underline{C}^{0,2} & \xrightarrow{\bar{\partial}} & \dots \end{array}$$

where the morphism $Q_p : \underline{C}^p \rightarrow \underline{C}^{0,p}$ is just projection of p -forms onto the $(0, p)$ -component. The map $k^p : H^p(X, \mathbb{C}) \rightarrow H^p(X, \mathcal{O})$ is therefore represented by

$$Q_p : \begin{array}{ccc} \text{Ker}(d) & \longrightarrow & \text{Ker}(\bar{\partial}) \\ \downarrow & & \downarrow \\ C^p(M) & & C^{0,p}(M) \end{array}$$

Since Q_2 maps $(1, 1)$ -forms to zero it follows that the image of $H_{\text{DR}}^{1,1}(X, \mathbb{C})$ in $H^2(X, \mathcal{O})$ is zero.

16. Associated to the divisor sequence $0 \rightarrow \mathcal{O}^* \rightarrow \mathcal{M}^* \rightarrow \mathcal{D} \rightarrow 0$ on a complex manifold X we have the long exact sequence

$$\begin{array}{ccccccc} 0 & \rightarrow & H^0(X, \mathcal{O}^*) & \rightarrow & H^0(X, \mathcal{M}^*) & \rightarrow & H^0(X, \mathcal{D}) \xrightarrow{\delta} H^1(X, \mathcal{O}^*) \rightarrow \dots \\ & & \parallel & & \parallel & & \parallel & & \parallel \\ & & A^*(X) & & M^*(X) & & \mathcal{D}(X) & & \text{HLB}(X) \end{array}$$

Recall from §9, Chapter 5, that we have a homomorphism

$$[\] : \mathcal{D}(X) \rightarrow \text{HLB}(X).$$

By the definition of δ in Čech theory it is clear that

$$\delta(d) = -[d], \quad d \in \mathcal{D}(X).$$

So we see again that a divisor d is the divisor of a meromorphic function if and only if $[d]$ is a holomorphically trivial line bundle. Recalling from Example 12 that the Chern class map $c_1 : \text{HLB}(X) \rightarrow H^2(X, \mathbb{Z})$ is defined to be *minus* the connecting homomorphism $\delta : \text{HLB}(X) \rightarrow H^2(X, \mathbb{Z})$, we see that $c_1([d]) = \delta\delta(d)$. In summary, we see that for a divisor d to be the divisor of a meromorphic function there are precisely two obstructions. Firstly a topological obstruction: $c_1([d])$ must be zero; secondly an analytic obstruction: $[d]$ must be holomorphically trivial.

For the remainder of this example, we shall assume that X is projective algebraic. We shall show later that every holomorphic line bundle on X admits a non-trivial meromorphic section. This clearly implies that the image of $H^1(X, \mathcal{O}^*)$ in $H^1(X, \mathcal{M}^*)$ is zero and so we obtain the exact sequence

$$0 \rightarrow A^*(X) \rightarrow M^*(X) \rightarrow \mathcal{D}(X) \xrightarrow{\delta} \text{HLB}(X) \rightarrow 0.$$

Hence, $\text{HLB}(X) \cong \mathcal{D}(X)/L(X)$, where $L(X)$ is the group of linear equivalence classes of divisors on X . From Example 14, we see that given $\eta \in H^{1,1}(X, \mathbb{Z})$, there exists

$d \in \mathcal{D}(X)$ such that $c_1([d])_{\mathbb{C}} = \eta$. It may be shown (see Griffiths and Harris [1]) that d defines a class in $H_{2n-2}(X, \mathbb{Z})$ which is the Poincaré dual of η with respect to the pairing $H_{2n-2}(X, \mathbb{Z}) \times H^2(X, \mathbb{Z}) \rightarrow H^{2n}(X, \mathbb{Z}) \approx \mathbb{Z}$. So we have a special case of the *Hodge conjecture* due to Lefschetz: Every class in $H^{1,1}(X, \mathbb{Z})$ has a Poincaré dual represented by an integral combination of analytic hypersurfaces in X (modulo torsion).

Finally let $\text{Pic}(X)$ denote the subgroup of $\text{HLB}(X)$ consisting of holomorphic line bundles which are trivial as complex line bundles. Thus $\text{Pic}(X) = \text{Ker}(c_1)$. Since X is compact it is easily seen from the cohomology sequence of $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \rightarrow 0$ that $\text{Pic}(X) \cong H^1(X, \mathcal{O})/H^1(X, \mathbb{Z})$. Furthermore it can be shown that $\text{Pic}(X)$ has the natural structure of a compact connected complex Lie group. That is, a complex torus. $\text{Pic}(X)$ is called the *Picard variety* of X . The Picard variety is an important birational invariant of X . It is an Abelian variety (for this it is sufficient that X be Moishezon). Proofs of these statements, together with further references, may be found in Ueno [1] and Griffiths and Harris [1].

Exercises.

1. Verify that
 - a) $H^p(D, \mathcal{O}) = 0$, $p \geq 1$, for every domain D in $\mathbb{C}$.
 - b) $H^p(D, \Omega^q) = 0$, $p \geq 1$, $q \geq 0$, for every open polydisc D in $\mathbb{C}^n$.
2. Let G be a group, not necessarily abelian. Show how to define $\check{H}^1(X, G)$ and prove that $\check{H}^1(X, \text{GL}(n, \mathbb{C}))$ is isomorphic to the set of isomorphism classes of n -dimensional complex vector bundles on X .
3. Prove that the map $\rho(\mathcal{U}) : H^1(\mathcal{U}, \mathcal{F}) \rightarrow H^1(X, \mathcal{F})$ is injective for all open covers $\mathcal{U}$ of X .
4. Verify that the de Rham cohomology groups of a differential manifold are topological invariants (as opposed to differential topological invariants) without using the isomorphism between singular and de Rham groups.
5. Let X be a Riemann surface. Verify that the sheaves $\mathcal{D}$ and $\mathcal{M}/\mathcal{D}$ are soft. Is this result true for general complex manifolds?
6. Show that $\text{Pic}(X) = \{\mathbb{C}\}$ in case X is any domain in $\mathbb{C}$ or an open polydisc in $\mathbb{C}^n$.
7. Prove that $c_1(TP^1(\mathbb{C})) = +2$.
8. $H^2(P^n(\mathbb{C}), \mathbb{Z}) \cong \mathbb{Z}$. Let H denote the hyperplane section bundle of $P^n(\mathbb{C})$. Prove that $c_1(H)$ generates $H^2(P^n(\mathbb{C}), \mathbb{Z})$.
9. Let $f : X \rightarrow Y$ be holomorphic and $L \in \text{HLB}(Y)$. Prove that $f^*(c_1(L)_{\mathbb{C}}) = c_1(f^*L)_{\mathbb{C}}$.
10. Let $T = \mathbb{C}^n/\Lambda$ be an n -dimensional complex torus with period lattice Λ and suppose that $L(H, m)$ is a holomorphic line bundle on T (for notation, see §9, Chapter 5 and recall that H is an Hermitian form on $\mathbb{C}^n$ whose imaginary part is integral on $\Lambda \times \Lambda$). Prove
 - a) The function $a(z, t) = \exp(-\pi H(z, z))|t|^2$ on $\mathbb{C}^n \times \mathbb{C}$ is invariant under the action

$$(z, t) \rightarrow (z + \lambda, \exp(-\pi(2 \text{Re}(H(z, \lambda) + H(\lambda, \lambda)))|t|^2), \quad \lambda \in \Lambda.$$

Deduce that a induces a smooth map $\tilde{a} : L(H, m) \rightarrow \mathbb{R}$ which is quadratic on fibres (see also Exercise 6, §9, Chapter 5).

- b) The first Chern class of $L(H, m)$ is represented by the form

$$-\frac{i}{2\pi}\partial\bar{\partial}\log\exp(-\pi H(z, z)) = \frac{i}{2}H.$$

- c) Every 1-dimensional complex torus admits a holomorphic line bundle of Chern class +1 (Integrate $\frac{i}{2}H$ over a period parallelogram).
11. Recall that a sheaf $\mathcal{F}$ of groups over a topological space X (not necessarily paracompact) is *flabby* if for all open subsets U of X the sequence $\mathcal{F}(X) \rightarrow \mathcal{F}(U) \rightarrow 0$ is exact. Prove
- a) If $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ is an exact sequence of sheaves over X and $\mathcal{F}$ is flabby then the sequence is presheaf exact. If, in addition $\mathcal{G}$ is flabby, deduce that $\mathcal{H}$ is flabby.
- b) If X is paracompact, then $\mathcal{F}$ flabby implies $\mathcal{F}$ soft.
12. Let X be a closed subset of the topological space Z and $\mathcal{F}$ be a sheaf of groups on X . As in exercise 4, §1, let $\tilde{\mathcal{F}}$ denote the trivial extension of $\mathcal{F}$ to Z . Prove that for $p \geq 0$, $H^p(Z, \tilde{\mathcal{F}}) \approx H^p(X, \mathcal{F})$. (Look at the trivial extension of the canonical resolution of $\mathcal{F}$).
- 13*. Let $f : X \rightarrow Y$ be a holomorphic map between complex manifolds and $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules on X . For $p \geq 0$, let $R^p f_* \mathcal{F}$ be the sheaf of $\mathcal{O}_Y$ -modules on Y associated to the presheaf $U \mapsto H^p(f^{-1}(U), \mathcal{F})$. Show that for any exact sequence $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ of $\mathcal{O}_X$ -modules on X , the induced sequence

$$0 \rightarrow f_* \mathcal{F} \rightarrow f_* \mathcal{G} \rightarrow f_* \mathcal{H} \rightarrow R^1 f_* \mathcal{F} \rightarrow R^1 f_* \mathcal{G} \rightarrow \dots$$

is exact.

14. Let $f : X \rightarrow Y$ be a homeomorphism between topological spaces X and Y , $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X and Y respectively and $a : \mathcal{G} \rightarrow \mathcal{F}$ be a morphism of sheaves covering f^{-1} . Show that for $p \geq 0$, we have naturally induced homomorphisms

$$f(a)^* : H^p(Y, \mathcal{G}) \rightarrow H^p(X, \mathcal{F})$$

(Use Čech theory).

15. Let $f : X \rightarrow Y$ be a continuous map between topological spaces X, Y and $\mathcal{F}$ be a sheaf on Y . Show that there exist homomorphisms $f^* : H^p(Y, \mathcal{F}) \rightarrow H^p(X, f^{-1}\mathcal{F})$ satisfying
- a) If $f = \text{Id}$, then $f^* = \text{Id}$.
- b) If $g : Z \rightarrow X$ is continuous then $(fg)^* = g^* f^*$.
- c) The construction of f^* is compatible with the construction of Q14, in case f is a homeomorphism and a is the inverse of the natural map $\tilde{f} : f^{-1}\mathcal{F} \rightarrow \mathcal{F}$.

Chapter 7

Coherent Sheaves

Introduction In section 1 we define coherence for sheaves of $\mathcal{O}$ -modules and prove Oka's theorem. In section 2 we prove Cartan's theorems A and B for Stein manifolds assuming exactness of the $\bar{\partial}$ -sequence. The remainder of the Chapter is devoted to applications of Theorems A and B. We prove Cartan and Serre's finiteness theorem in section 3 and Grauert's finiteness theorem for coherent sheaves on s.l.p. domains in section 4. Using the finiteness theorem of Cartan-Serre, we prove Serre's theorems A and B for coherent sheaves on complex projective space in section 5. We give a number of applications including Grothendieck's theorem on the splitting of holomorphic vector bundles on the Riemann sphere. Finally in section 6 we prove Kodaira's embedding theorem, following Grauert, and conclude by showing that complex tori that admit a Riemann form are algebraic.

1 Coherent sheaves

Throughout this section we shall be studying sheaves of $\mathcal{O}$ -modules on a complex manifold M (Here, as in the sequel, we usually drop the subscript “ M ” from the Oka sheaf $\mathcal{O}_M$). In future we call a sheaf of $\mathcal{O}$ -modules an *analytic sheaf*. We have already seen some examples where algebraic conditions holding at a point continue to hold in a neighbourhood of the point. Thus the condition that germs of analytic functions be relatively prime is an open condition (Proposition 3.4.2). Another important example came from the theory of analytic hypersurfaces where we showed that if Z was an analytic hypersurface in M with $I_z(Z) = (g_z)$, then $I_y(Z) = (g_y)$ for y in some neighbourhood of z (Theorem 3.5.16). The main aim of this section will be to describe the class of coherent analytic sheaves for which this type of behaviour is characteristic. For example, we shall show that if $\mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H}$ is a sequence of coherent sheaves with $ba = 0$ and if the sequence is exact (at stalk level) at $x \in M$, then it is exact in a neighbourhood of x . Of course, if we wish to use topological methods to get global results then our local results need to be framed in terms of open sets – not points – and this is why coherence is such an important concept in the cohomology theory of sheaves. Our proofs will be close to those in the original works on coherence by H. Cartan [1, 2] and J.P. Serre [2].

Definition 7.1.1. An analytic sheaf $\mathcal{F}$ on M is said to be of *finite type* if we can find an open neighbourhood U of every point x in M and a finite number of sections $s_1, \dots, s_k \in \mathcal{F}(U)$ such that $\{s_{1,y}, \dots, s_{k,y}\}$ generates $\mathcal{F}_y$ as an $\mathcal{O}_y$ -module for every $y \in U$.

Example 1. The sheaf of sections of a holomorphic vector bundle on M is of finite type.

Proposition 7.1.2. Let $\mathcal{F}$ be an analytic sheaf of finite type and $f_1, \dots, f_p$ be continuous sections of $\mathcal{F}$ defined over some open neighbourhood of $x \in M$ such that $f_{1,x}, \dots, f_{p,x}$ generate $\mathcal{F}_x$. Then $f_{1,y}, \dots, f_{p,y}$ generate $\mathcal{F}_y$ for y in some neighbourhood of x .

Proof. Since $\mathcal{F}$ is of finite type we can find an open neighbourhood U of x and sections $s_1, \dots, s_k \in \mathcal{F}(U)$ such that $s_{1,y}, \dots, s_{k,y}$ generate $\mathcal{F}_y$ for $y \in U$. By assumption, there exist $a_{ij,x} \in \mathcal{O}_x$ such that

$$s_{i,x} = \sum_{j=1}^p a_{ij,x} f_{j,x}, \quad i = 1, \dots, k.$$

Choosing representatives for the germs $a_{ij,x}$, we see that on some neighbourhood V of x we have

$$s_i = \sum_{j=1}^p a_{ij} f_j, \quad i = 1, \dots, k.$$

Hence for all $z \in V$ we have

$$s_{i,z} = \sum_{j=1}^p a_{ij,z} f_{j,z}.$$

□

For $q \geq 0$, recall that $\mathcal{O}^q$ denotes the p -fold direct sum of $\mathcal{O}$ and that $\mathcal{O}^q$ is equal to the sheaf of germs of $\mathbb{C}^q$ -valued holomorphic functions on M . We have a canonical $\mathcal{O}$ -module basis of $\mathcal{O}^q$ given by the constant functions

$$E_1(z) = (1, 0, \dots, 0), \dots, E_q(z) = (0, 0, \dots, 1).$$

Suppose that $\mathcal{F}$ is an analytic sheaf on M and that $f_1, \dots, f_p$ are continuous sections of $\mathcal{F}$ over some open subset U of M . Let $f = (f_1, \dots, f_p) : \mathcal{O}_U^p \rightarrow \mathcal{F}_U$ denote the sheaf homomorphism defined by

$$f(g_1, \dots, g_p) = \sum_{j=1}^p g_j f_{j,z}, \quad (g_1, \dots, g_p) \in \mathcal{O}_z^p, \quad z \in U.$$

The kernel of f is a subsheaf $R(f_1, \dots, f_p)$ of $\mathcal{O}_U^p$ called the *sheaf of relations between $f_1, \dots, f_p$* .

Theorem 7.1.3. Let U be an open subset of M and $f_1, \dots, f_p \in \mathcal{O}^q(U)$. Then the sheaf of relations $R(f_1, \dots, f_p)$ is a subsheaf of $\mathcal{O}_U^p$ of finite type.

This theorem, due to Oka, is the fundamental result upon which the theory of coherent analytic sheaves rests.

In view of the local nature of Theorem 7.1.3 it suffices to prove

Theorem 7.1.4. Let Ω be an open subset of $\mathbb{C}^n$ and $f_1, \dots, f_p \in A(\Omega)^q$. We can find an open neighbourhood U of any point $z \in \Omega$ and finitely many functions $C_1, \dots, C_r \in A(U)^p$ such that $C_{1,x}, \dots, C_{r,x}$ generate $R(f_1, \dots, f_p)_x$ as an $\mathcal{O}_x$ -module for all $x \in U$.

Proof. First of all notice that

$$R(f_1, \dots, f_p)_z = \{(a_1, \dots, a_p) \in \mathcal{O}_z^p : \sum_{j=1}^p a_j f_{j,z} = 0\}, \quad z \in \Omega.$$

Since $\mathcal{O}_z$ is Noetherian, $R(f_1, \dots, f_p)_z$ is certainly finitely generated. We have to find generators which also generate the stalks in a neighbourhood of z .

We may clearly assume that $0 \in \Omega$ and that $z = 0$. Our proof goes by induction on q and n . First, suppose that the theorem has been proved for all q in the $(n-1)$ -dimensional case (note that the case $n = 0$ is trivial). We shall prove that the theorem is true for n and $q = 1$.

Without loss of generality we may assume that $f_1, \dots, f_p$ are normalised in direction z_n and so, by the Weierstrass Preparation theorem, we may suppose that $f_1, \dots, f_p$ are Weierstrass polynomials in z_n with coefficients in $A(\Omega')$, where Ω' is an open neighbourhood of $0 \in \mathbb{C}^{n-1}$. In what follows we may suppose $\Omega = \Omega' \times \mathbb{C}$. Let d denote the maximum of the degrees of the polynomials $f_1, \dots, f_p$. We may write

$$f_i = \sum_{j=0}^d f_{ij} z_n^j, \quad i = 1, \dots, p,$$

where $f_{ij} \in A(\Omega')$. We let ${}^d A'[z_n]$ denote the group of polynomials in z_n of degree $\leq d$ which have coefficients in $A(\Omega')$. For $\zeta = (\zeta', \zeta_n) \in \mathbb{C}^n$, we let $\mathcal{O}'_{\zeta'}$ denote the ring of germs of analytic functions on $\mathbb{C}^{n-1}$ at ζ' and ${}^d A'[z_n]_{\zeta}$ denote the germs at ζ of functions in ${}^d A'[z_n]$. Finally, let ${}^d R(f_1, \dots, f_p)$ denote the subset of $R(f_1, \dots, f_p)$ defined by ${}^d R(f_1, \dots, f_p)_{\zeta} = R(f_1, \dots, f_p)_{\zeta} \cap {}^d A'[z_n]_{\zeta}$, $\zeta \in \Omega$. Note that ${}^d R(f_1, \dots, f_p)$ has the structure of an $\mathcal{O}'$ -module.

We shall prove the inductive step by first showing that for all $\zeta \in \Omega$, ${}^d R(f_1, \dots, f_p)_{\zeta}$ generates $R(f_1, \dots, f_p)_{\zeta}$ as an $\mathcal{O}_{\zeta}$ -module. Then, using the inductive hypothesis, we show that we can find a finite set of generators for ${}^d R(f_1, \dots, f_p)$ (as an $\mathcal{O}'$ -module) over some neighbourhood of zero.

Step 1: The $\mathcal{O}$ -module $R(f_1, \dots, f_p)_{\zeta}$ is generated by the elements of ${}^d R(f_1, \dots, f_p)_{\zeta}$, $\zeta \in \Omega$.

Suppose f_p has degree d . Then, by the Weierstrass Preparation theorem, we have for $\zeta \in \Omega$

$$f_{p,\zeta} = f' f'',$$

where $f', f'' \in \mathcal{O}_{\zeta}$, f' is the germ of a Weierstrass polynomial in $z_n - \zeta_n$ and $f''(\zeta) \neq 0$. By Lemma 3.4.1, f'' is a polynomial in z_n with leading coefficient 1. Let d', d'' denote the degrees of f', f'' with respect to z_n . Given $(a_1, \dots, a_p) \in R(f_1, \dots, f_p)$, we can by the Weierstrass division theorem write

$$a_i = f_{p,\zeta} b_i + c_i, \quad i = 1, \dots, p-1,$$

where $b_i \in \mathcal{O}_{\zeta}$ and $c_i \in \mathcal{O}'_{\zeta'}[z_n]$ is of degree $< d'$. Set

$$c_p = a_p + \sum_{i=1}^{p-1} f_{i,\zeta} b_i$$

and observe that we have the identity

$$\begin{aligned} (a_1, \dots, a_p) &= (f_p, 0, \dots, 0, -f_1)_{\zeta} b_1 + \dots + (0, \dots, 0, f_p, -f_{p-1})_{\zeta} b_p \\ &\quad + (c_1, \dots, c_p). \end{aligned} \tag{*}$$

Now all the terms in this identity, except possibly $(c_1, \dots, c_p)$, lie in $R(f_1, \dots, f_p)_{\zeta}$. Hence $(c_1, \dots, c_p) \in R(f_1, \dots, f_p)_{\zeta}$. Therefore

$$\sum_{i=1}^{p-1} c_i f_{i,\zeta} + (c_p f'') f' = 0.$$

But the sum is a polynomial in z_n of degree $< d + d'$. Consequently, by Lemma 3.4.1, $c_p f''$ is a polynomial in z_n of degree $< d$. But since

$$(c_1, \dots, c_p) = (1/f'')(f'' c_1, \dots, f'' c_p)$$

and $f'' c_1, \dots, f'' c_p$ have degree $< d$, it follows that $(c_1, \dots, c_p) \in {}^d R(f_1, \dots, f_p)_{\zeta}$. This, together with (*), proves step 1.

Step 2: We may find an open neighbourhood U of $0 \in \mathbb{C}^n$ and sections $C_1, \dots, C_r \in {}^dR(f_1, \dots, f_p)(U)$ such that for all $\zeta \in U$, $C_{1,\zeta}, \dots, C_{r,\zeta}$ generate ${}^dR(f_1, \dots, f_p)_\zeta$ as an $\mathcal{O}'_\zeta$ -module.

Suppose $a = (a_1, \dots, a_p) \in {}^dA'[z_n]_\zeta$. Then

$$a_i = \sum_{j=0}^d c_{ij}(z_n^j)_\zeta, \quad c_{ij} \in \mathcal{O}'_{\zeta'}.$$

Now $a \in {}^dR(f_1, \dots, f_p)_\zeta$ if and only if $\sum_{i=0}^p a_i f_i = 0$. That is, if and only if

$$\sum_{k=0}^d \sum_{j=0}^d \sum_{i=1}^p c_{ij} f_{ik}(z_n^{k+j}) = 0.$$

Equating coefficients of powers of z_n to zero we see that $a \in {}^dR(f_1, \dots, f_p)_\zeta$ if and only if

$$\sum_{j+k=r} \sum_{i=1}^p c_{ij} f_{ik} = 0, \quad r = 0, \dots, 2d.$$

That is, ${}^dR(f_1, \dots, f_p)$ is isomorphic to the kernel of the homomorphism $F = (F_1, \dots, F_{p(d+1)}) : \mathcal{O}'^{p(d+1)} \rightarrow \mathcal{O}'^{2d+1}$ defined by

$$(F([c_{ij}]))_r = \sum_{i,j} c_{ij} f_{i,r-j}, \quad r = 0, \dots, 2d.$$

In other words, ${}^dR(f_1, \dots, f_p)$ is isomorphic to $R(F_1, \dots, F_{p(d+1)})$ and so, by the inductive hypothesis, we may find an open neighbourhood U' of $0 \in \mathbb{C}^{n-1}$ and $C_1, \dots, C_r \in A(U')^{p(d+1)}$ such that $C_{1,\zeta'}, \dots, C_{r,\zeta'}$ generate $R(F_1, \dots, F_{p(d+1)})_{\zeta'}$ for all $\zeta' \in U'$. Taking $U = U' \times \mathbb{C} \subset \mathbb{C}^n$, we see that $C_1, \dots, C_r$ give the required generators of ${}^dR(f_1, \dots, f_p)$ as an $\mathcal{O}'$ -module.

To complete our induction we now show that if the result is true for n and $q = 1$ it is true for n and $q > 1$.

Setting $f_j = (f_{j1}, \dots, f_{jq})$, we define $\tilde{f}_j = (f_{j1}, \dots, f_{jq-1})$, $j = 1, \dots, p$. For $\zeta \in \Omega$, we have

$$R(f_1, \dots, f_p)_\zeta \subset R(\tilde{f}_1, \dots, \tilde{f}_p)_\zeta.$$

Now by the inductive hypothesis for n and $q - 1$, there exists a neighbourhood $V \subset \Omega$ of 0 and $g_1, \dots, g_r \in A(V)^p$ such that $g_{1,\zeta}, \dots, g_{r,\zeta}$ generate the $\mathcal{O}_\zeta$ -module $R(\tilde{f}_1, \dots, \tilde{f}_p)_\zeta$ for all $\zeta \in V$. For $\zeta \in V$, we have $R(f_1, \dots, f_p)_\zeta \subset \{\sum_{j=1}^r c_j g_{j,\zeta} : c_j \in \mathcal{O}_\zeta\}$. Setting $g_j = (g_{j1}, \dots, g_{jp})$, $1 \leq j \leq r$, and taking components as we see that $\sum_{j=1}^r c_j g_{j,\zeta} \in R(f_1, \dots, f_p)_\zeta$ if and only if

$$\sum_{j=1}^r \sum_{k=1}^p c_j (g_{jk} f_{ki})_\zeta = 0, \quad i = 1, \dots, q.$$

But the first $q-1$ of these equations automatically hold and so only the q th. equation remains to be satisfied. By the inductive hypothesis for n and $q = 1$, there exists a neighbourhood $U \subset V$ of 0 and $h_1, \dots, h_s \in A(U)^r$ such that the solutions $(c_1, \dots, c_r) \in \mathcal{O}'_\zeta$ of $\sum_{j=1}^r \sum_{k=1}^p c_j (g_{jk} f_{kq})_\zeta = 0$ are generated by $h_{1,\zeta}, \dots, h_{s,\zeta}$, $\zeta \in U$. Defining $C_1, \dots, C_s \in A(U)^p$ by $C_i = \sum_{j=1}^r h_{ij} g_j$, $1 \leq i \leq s$, we see that $C_1, \dots, C_s$ are the required set of generators for $R(f_1, \dots, f_p)_U$. $\square$

Definition 7.1.5. An analytic sheaf $\mathcal{F}$ on the complex manifold M is said to be *coherent* if

- a) $\mathcal{F}$ is of finite type.
- b) Given any open subset U of M and sections $f_1, \dots, f_p \in \mathcal{F}(U)$, then the sheaf of relations $R(f_1, \dots, f_p)$ is of finite type.

Theorem 7.1.6. Every analytic subsheaf of $\mathcal{O}^q$ which is of finite type is coherent.

Proof. Theorem 7.1.3. □

Corollary 7.1.7. Let $\mathcal{F}$ be a coherent sheaf on M , U be an open subset of M and $f_1, \dots, f_p \in \mathcal{F}(U)$. Then $R(f_1, \dots, f_p)$ is coherent.

Proof. $R(f_1, \dots, f_p)$ is a subsheaf of $\mathcal{O}_U^p$ of finite type. □

Corollary 7.1.8. Let $\mathcal{F}$ be a coherent sheaf on M and $x \in M$. Then we may find a free resolution of $\mathcal{F}$, of length $m = \dim(M)$, over some open neighbourhood U of x :

$$0 \rightarrow \mathcal{O}_U^{p_m} \xrightarrow{s_m} \mathcal{O}_U^{p_{m-1}} \rightarrow \dots \xrightarrow{s_1} \mathcal{O}_U^{p_0} \xrightarrow{s_0} \mathcal{F}_U \rightarrow 0.$$

Proof. Since $\mathcal{F}$ is of finite type, we may find an open neighbourhood U_0 of x and $s_1^0, \dots, s_{p_0}^0 \in \mathcal{F}(U_0)$ such that for all $\zeta \in U_0$, $s_{1,\zeta}^0, \dots, s_{p_0,\zeta}^0$ generate $\mathcal{F}_\zeta$ as an $\mathcal{O}_\zeta$ -module. That is, setting $s_0 = (s_1^0, \dots, s_{p_0}^0)$ we have the exact sequence

$$\mathcal{O}_{U_0}^{p_0} \xrightarrow{s_0} \mathcal{F}_{U_0} \rightarrow 0.$$

Now $\text{Ker}(s_0) = R(s_1^0, \dots, s_{p_0}^0)$ and so, since $R(s_1^0, \dots, s_{p_0}^0)$ is of finite type, we may find an open neighbourhood $U_1 \subset U_0$ of x and $s_1^1, \dots, s_{p_1}^1 \in \mathcal{O}_{U_1}^{p_0}$ such that $s_{1,\zeta}^1, \dots, s_{p_1,\zeta}^1$ generate $R(s_1^0, \dots, s_{p_0}^0)_\zeta$, $\zeta \in U_1$. Setting $s_1 = (s_1^1, \dots, s_{p_1}^1)$, we obtain the exact sequence

$$\mathcal{O}_{U_1}^{p_1} \xrightarrow{s_1} \mathcal{O}_{U_1}^{p_0} \xrightarrow{s_0} \mathcal{F}_{U_1} \rightarrow 0.$$

Proceeding inductively, we obtain after m steps the exact sequence

$$\mathcal{O}_{U_{m-1}}^{p_{m-1}} \xrightarrow{s_{m-1}} \dots \xrightarrow{s_1} \mathcal{O}_{U_{m-1}}^{p_0} \xrightarrow{s_0} \mathcal{F}_{U_{m-1}} \rightarrow 0.$$

By the Hilbert Syzygy theorem (Theorem 3.6.1), $\text{Ker}(s_{m-1})_y$ is a free $\mathcal{O}_y$ -module, $y \in U_{m-1}$. Suppose $\text{Ker}(s_{m-1})_x \approx \mathcal{O}_x^{p_m}$. A set of generators for $\mathcal{O}_x^{p_m}$ is given by the constant functions E_j , $1 \leq j \leq p_m$ and, since $\text{Ker}(s_{m-1})$ is of finite type, it follows from Proposition 7.1.2 that the E_j generate $\text{Ker}(s_{m-1})_y$ for y in some open neighbourhood U_m of x . Taking $U = U_m$, we therefore obtain the exact sequence

$$\mathcal{O}_U^{p_m} \xrightarrow{s_m} \mathcal{O}_U^{p_{m-1}} \rightarrow \dots \xrightarrow{s_0} \mathcal{F}_U \rightarrow 0.$$

By the Hilbert Syzygy theorem, $\text{Ker}(s_m) = 0$. □

Remarks. One consequence of Corollary 7.1.8 is that every coherent sheaf is locally isomorphic to the cokernel of a sheaf homomorphism $a : \mathcal{O}^p \rightarrow \mathcal{O}^q$. Later on in this chapter, we shall examine to what extent coherent sheaves admit global resolutions by free, and more especially locally free, sheaves of $\mathcal{O}$ -modules.

Example 2. The sheaf of holomorphic sections of a holomorphic vector bundle E is coherent. Indeed, $\underline{E}$ is locally isomorphic to $\mathcal{O}^q$, $q = \dim(E)$.

Theorem 7.1.9. We have the following basic properties of coherent sheaves:

1. Every analytic subsheaf of a coherent sheaf which is of finite type is coherent.
2. Suppose $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H} \rightarrow 0$ is a short exact sequence of analytic sheaves. If any two of the sheaves $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are coherent, so is the third.
3. The direct sum of a finite family of coherent sheaves is coherent.
4. Let $a : \mathcal{F} \rightarrow \mathcal{G}$ be a homomorphism of coherent sheaves. Then $\text{Ker}(a)$, $\text{Im}(a)$ and $\text{Coker}(a)$ are coherent.
5. Let $\mathcal{F}, \mathcal{G}$ be coherent subsheaves of a coherent sheaf $\mathcal{H}$. Then the sheaves $\mathcal{F} + \mathcal{G}$, $\mathcal{F} \cap \mathcal{G}$ are coherent.
6. Let $\mathcal{F}, \mathcal{G}$ be coherent sheaves. Then $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G}$ is coherent.

Proof. 1. Clearly every analytic subsheaf of a sheaf satisfying (b) of Definition 7.1.5 also satisfies condition (b).

2.a) Suppose $\mathcal{G}$ and $\mathcal{H}$ are coherent. Since $\mathcal{G}$ is of finite type, may find an open neighbourhood U of any point $x \in M$ and surjective homomorphism $c : \mathcal{O}_U^p \rightarrow \mathcal{G}_U$. Since $\mathcal{H}$ satisfies condition (b), $\text{Ker}(bc)$ is of finite type and so, by 1., $c(\text{Ker}(bc))$ is a coherent subsheaf of $\mathcal{G}_U$. But a maps $\mathcal{F}_U$ isomorphically onto $c(\text{Ker}(bc))$ and so $\mathcal{F}$ is coherent.

b) Suppose $\mathcal{F}$ and $\mathcal{G}$ are coherent. Since $\mathcal{G}$ is of finite type and b is surjective $\mathcal{H}$ is of finite type. We must show that $\mathcal{H}$ satisfies condition (b). Let $x \in M$, U be an open neighbourhood of x and $s_1, \dots, s_p \in \mathcal{H}(U)$. Shrinking U if necessary, we may choose $t_1, \dots, t_p \in \mathcal{G}(U)$ such that $s_j = b(t_j)$, $j = 1, \dots, p$. Shrinking U further if necessary, we may find $u_1, \dots, u_q \in \mathcal{F}(U)$ such that $u_{1,y}, \dots, u_{q,y}$ generate $\mathcal{F}_y$ as an $\mathcal{O}_y$ -module for all $y \in U$. Now given $y \in U$ and $(f_1, \dots, f_p) \in \mathcal{O}_y^p$, $(f_1, \dots, f_p) \in R(s_1, \dots, s_p)_y$ if and only if $\sum_{i=1}^p f_i t_i \in \text{Im}(a)$. That is, if and only if there exist $g_1, \dots, g_q \in \mathcal{O}_y$ such that $\sum_{i=1}^p f_i t_i = \sum_{j=1}^q g_j a(u_j)$. But $R(t_1, \dots, t_p, a(u_1), \dots, a(u_q))$ is of finite type since $\mathcal{G}$ is coherent and since $R(s_1, \dots, s_p)$ is the image of $R(t_1, \dots, a(u_q))$ under the canonical projection of $\mathcal{O}^{p+q}$ on $\mathcal{O}^p$ it follows that $R(s_1, \dots, s_p)$ is of finite type.

c) We leave the case $\mathcal{F}$ and $\mathcal{H}$ coherent implies $\mathcal{G}$ coherent as an exercise (details may be found in Serre [2] or Gunning and Rossi [1]).

3. The finite direct sum of coherent sheaves is coherent. This follows by an easy induction from 2.c) or directly and we omit details.

4. Suppose $a : \mathcal{F} \rightarrow \mathcal{G}$ is a homomorphism of coherent sheaves. Now $\text{Im}(a)$ is of finite type since $\mathcal{F}$ is of finite type and so, by 1, $\text{Im}(a)$ is coherent. Applying 2.a) and b) to the exact sequences

$$0 \rightarrow \text{Ker}(a) \rightarrow \mathcal{F} \rightarrow \text{Im}(a) \rightarrow 0$$

$$0 \rightarrow \text{Im}(a) \rightarrow \mathcal{G} \rightarrow \text{Coker}(a) \rightarrow 0$$

we see that $\text{Ker}(a)$ and $\text{Coker}(a)$ are coherent.

5. The sheaf $\mathcal{F} + \mathcal{G}$ is of finite type and so coherent by 1. The sheaf $\mathcal{F} \cap \mathcal{G}$ is the kernel of the quotient map $\mathcal{F} \rightarrow \mathcal{H}/\mathcal{G}$ and is therefore coherent by 2.a).

6. Let $x \in M$. Since $\mathcal{F}$ is coherent, there exists an open neighbourhood U of x and exact sheaf sequence

$$\mathcal{O}_U^p \rightarrow \mathcal{O}_U^q \rightarrow \mathcal{F}_U \rightarrow 0.$$

Tensoring with $\mathcal{G}_U$ we obtain the exact sequence

$$\mathcal{G}_U \otimes_{\mathcal{O}} \mathcal{O}_U^p \rightarrow \mathcal{G}_U \otimes_{\mathcal{O}} \mathcal{O}_U^q \rightarrow (\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G})_U \rightarrow 0.$$

Now $\mathcal{G}_U \otimes_{\mathcal{O}} \mathcal{O}_U^p \approx \mathcal{G}_U^p$; $\mathcal{G}_U \otimes_{\mathcal{O}} \mathcal{O}_U^q \approx \mathcal{G}_U^q$ and so we arrive at the exact sequence

$$\mathcal{G}_U^p \rightarrow \mathcal{G}_U^q \rightarrow (\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G})_U \rightarrow 0.$$

But by 3, $\mathcal{G}_U^p, \mathcal{G}_U^q$ are coherent and so, by 4, $(\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G})_U$ is coherent. Hence $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G}$ is coherent. $\square$

Remark. Theorem 7.1.9 suggests that set of coherent sheaves on M is the smallest class of analytic sheaves on M which contains the locally free sheaves (holomorphic vector bundles) and is closed under the operations of quotient, kernel and image. Unfortunately this is not generally true, even when M is compact (the ideal sheaf of a point need not have a resolution by locally free sheaves in case every holomorphic vector bundle on M is flat). However, if M is projective algebraic, then every coherent sheaf on M has a global resolution by locally free sheaves and so the coherent sheaves on M are the smallest class containing the locally free sheaves and which are closed under the operations of quotient, kernel and image. We shall return to this question later in the chapter.

Examples. 3. Let X be a complex submanifold of the complex manifold M . Then the sheaves $\mathcal{I}_X, \mathcal{O}_X$ are coherent sheaves on M . First we prove $\mathcal{I}_X$ is coherent. The question being local we may suppose that X is the subspace $\mathbb{C}^k$ of $\mathbb{C}^m$ defined by $z_{k+1} = \dots = z_m = 0$. If $z \notin \mathbb{C}^k$, $\mathcal{I}_{X,z} = \mathcal{O}_z$. For $z \in \mathbb{C}^k$, $\mathcal{I}_{X,z} = (z_{k+1}, \dots, z_m)$ and clearly $z_{k+1}, \dots, z_m$ give a finite set of generators for $\mathcal{I}_X$ in a neighbourhood of z . Hence $\mathcal{I}_X$ is of finite type and so coherent by Theorem 7.1.6. Since $\mathcal{O}_X = \mathcal{O}_M / \mathcal{I}_X$, we see from Theorem 7.1.9, 2, that $\mathcal{O}_X$ is coherent.

4. Let X be an analytic hypersurface in the complex manifold M . Then $\mathcal{I}_X, \mathcal{O}_X$ are coherent. Theorem 3.5.16 implies that $\mathcal{I}_X$ is coherent. $\mathcal{O}_X$ is coherent as in Example 3.

5. Let X be an analytic subset of M . Then $\mathcal{I}_X, \mathcal{O}_X$ are coherent. The proof of this result is outside the scope of these notes depending, as it does, on the local parametrization theorem for analytic sets. The proof may be found in H. Cartan [1], Gunning [1, page 43], Gunning and Rossi [1], R. Narasimhan [1], Whitney [1].

6. Let $f : M \rightarrow N$ be a holomorphic map of complex manifolds and $\mathcal{F}$ be a coherent sheaf on N . Then $f^*\mathcal{F}$ is a coherent sheaf on M . This result follows by representing $\mathcal{F}$ locally as the cokernel of a map $\mathcal{O}^p \rightarrow \mathcal{O}^q$ and then using the right exactness of f^* – see Exercise 12, §1, Chapter 6.

7. Suppose $\mathcal{F}$ is a coherent sheaf on M and $f : M \rightarrow N$ is holomorphic. In general $f_*\mathcal{F}$ will not be a coherent sheaf on N . However, if f is proper, a deep and difficult theorem of Grauert asserts that $f_*\mathcal{F}$ is coherent. Proofs of this important result may be found in Forster and Knorr [1], Kiehl and Verdier [1]. See also R. Narasimhan [2]. In case f is a *finite* map, the reader may consult Grauert and Remmert [1], Gunning [1], R. Narasimhan [1].

8. Let $\mathcal{F}$ be a coherent sheaf on M . Then $\text{supp}(\mathcal{F})$ is an analytic subset of M (for the definition of $\text{supp}(\mathcal{F})$, see Exercise 6, §1, Chapter 6). To see this note that we can find an open neighbourhood U of any $x \in M$ and exact sequence

$$\mathcal{O}_U^p \xrightarrow{s_0} \mathcal{O}_U^q \xrightarrow{s_1} \mathcal{F}_U \rightarrow 0.$$

Now $\text{supp}(\mathcal{F}) \cap U = \{x \in U : s_{1,x} \neq 0\} = \{x \in U : s_{0,x} \text{ is not of maximal rank}\}$. Now s_0 may be represented as a $q \times p$ matrix with holomorphic entries defined on U . The condition that $s_{0,x}$ is not of maximal rank is given by a finite set of algebraic equations in the components of $s_{0,x}$. Hence the set of points where $s_{0,x}$ fails to be of maximal rank is an analytic subset of U . A similar argument will show that the set of points in M where $\mathcal{F}$ is not locally free is an analytic subset of M .

9. Let X be a complex submanifold of M and let $\mathcal{O}_X$ denote $\mathcal{O}_M|_X$ as well as the sheaf $\mathcal{O}_X$ on M . Suppose $\mathcal{F}$ is a sheaf of $\mathcal{O}_X$ -modules on X and $\tilde{\mathcal{F}}$ denote the trivial extension of

$\mathcal{F}$ to M . Clearly $\tilde{\mathcal{F}}$ has the structure of an $\mathcal{O}_X$ -module and, since $\mathcal{O}_X$ is an $\mathcal{O}_M$ -module, $\tilde{\mathcal{F}}$ has the structure of an $\mathcal{O}_M$ -module. We claim that $\tilde{\mathcal{F}}$ is coherent as a sheaf of $\mathcal{O}_M$ -modules if and only if $\mathcal{F}$ is coherent as a sheaf of $\mathcal{O}_X$ -modules. Well suppose $\mathcal{F}$ is coherent as a sheaf of $\mathcal{O}_X$ -modules. Obviously $\tilde{\mathcal{F}}|_{M \setminus X}$ is coherent. Let $x \in X$. We may find an open neighbourhood U of x in X and exact sequence

$$\mathcal{O}_U^q \rightarrow \mathcal{O}_U^p \rightarrow \mathcal{F}_U \rightarrow 0.$$

Taking any open neighbourhood V of x in M such that $V \cap X = U$ and taking trivial extensions we obtain the exact sequence

$$(\mathcal{O}_X|V)^q \rightarrow (\mathcal{O}_X|V)^p \rightarrow \tilde{\mathcal{F}}_V \rightarrow 0.$$

But $\mathcal{O}_X^p, \mathcal{O}_X^q$ are coherent sheaves on M and so by Theorem 7.1.9, 4, $\tilde{\mathcal{F}}_V$ is coherent. Hence $\tilde{\mathcal{F}}$ is coherent. The converse follows easily from Example 6.

10. Let $\mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H}$ be a sequence of coherent sheaves on M and suppose $ba = 0$ and the sequence is exact at the point $x \in M$ (that is at the stalk level). Then the sequence is exact on some open neighbourhood of x . To see this we note that $\ker(b)/\text{Im}(a)$ is coherent with zero stalk at x . Now apply the result of Example 8.

Exercises.

1. Let $\mathcal{F}, \mathcal{G}$ be analytic sheaves on X with $\mathcal{F}$ coherent. Show that for all $x \in X$, $\text{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{G})_x \approx \text{Hom}_{\mathcal{O}_x}(\mathcal{F}_x, \mathcal{G}_x)$. Deduce that if $\mathcal{F}$ and $\mathcal{G}$ are coherent then so is $\text{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$.
2. Remmert's proper mapping theorem states that the image of an analytic subset by a proper analytic map is an analytic set. Deduce this result from Grauert's direct image theorem and Example 8.
3. Let $(X, \mathcal{O}_X)$ be an analytic space (§1, Chapter 6). Using the result described in Example 5, show that $\mathcal{O}_X^p$ is a coherent sheaf of $\mathcal{O}_X$ -modules, $p \geq 1$. Describe the appropriate extension of Example 9 to this more general framework.
4. Show that $f_*\mathcal{F}$ need not be of finite type if f is not proper, $\mathcal{F}$ coherent. (Hint: Take $X = \mathbb{C}$, Y a point and $\mathcal{F} = \mathcal{O}_X$).
5. Let $\mathcal{F}$ be a coherent sheaf of $\mathcal{O}_X$ -modules on the complex manifold X . Suppose that $\mathcal{F}_x$ is a free $\mathcal{O}_x$ -module for every $x \in X$. Prove that $\mathcal{F}$ is locally free.

2 Coherent sheaves on a Stein manifold

Suppose that E is a holomorphic vector bundle on the m -dimensional complex manifold M . We have the associated $\bar{\partial}$ -complex

$$0 \rightarrow \Omega(E) \rightarrow C^{0,0}(M, E) \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} C^{0,m}(M, E) \rightarrow 0.$$

For this section and the remainder of the chapter we shall assume that the $\bar{\partial}$ -complex of any holomorphic vector bundle on a Stein manifold is *exact*. In Chapter 11 we shall give a proof of this fundamental result that depends on the theory of elliptic operators. For the present we remark that we have proved the exactness of the $\bar{\partial}$ -sequence in case M is a polydisc or $\mathbb{C}^n$ and E is trivial (Theorem 5.8.2) and indicated an elementary proof in case M is the Euclidean disc and E is trivial (Exercise, 2, §8, Chapter 5).

The Dolbeault isomorphisms (Example 5, §3, Chapter 6) imply that our assumption is equivalent to the vanishing of $H^p(M, \underline{E})$, $p \geq 1$, for every holomorphic vector bundle E on a Stein manifold M . Since the sheaf of sections of a holomorphic vector bundle is coherent we see that our assumption amounts to a special case of the second of the remarkable and famous Theorems A and B of H. Cartan:

Theorem A. Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M . Given $x \in M$, we may find $s_1, \dots, s_p \in H^0(M, \mathcal{F}) = \mathcal{F}(M)$ such that $s_{1,x}, \dots, s_{p,x}$ generate $\mathcal{F}_x$ as an $\mathcal{O}_x$ -module.

Theorem B. Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M . Then $H^q(M, \mathcal{F}) = 0$, $q \geq 1$.

The main aim of this section is to prove Theorems A and B, granted our assumption that Theorem B is true for locally free sheaves of $\mathcal{O}$ -modules.

Theorems A and B have many profound applications.

Examples. 1. Let X be an analytic subset of the Stein manifold M . Then X may be represented as the common zero locus of a set of analytic functions on M . For this it is enough to show that for each $x \in M \setminus X$, there exists $f \in A(M)$ such that $f \in \mathcal{I}_X(M)$ and $f(x) \neq 0$. Set $\mathcal{I} = \mathcal{I}_X$ and let $\mathcal{I}_x$ denote the ideal sheaf of the analytic set $X \cup \{x\}$. Then $\mathcal{I}_x$ is a subsheaf of $\mathcal{O}$ and $\mathcal{I}/\mathcal{I}_x = \mathbb{C}(x)$, where $\mathbb{C}(x)$ denotes the “skyscraper” sheaf whose stalk is zero except at x where it equals $\mathbb{C}$. Now, by Cartan’s coherence theorem, $\mathcal{I}_x$ is coherent and so, by Theorem B, $H^1(M, \mathcal{I}_x) = 0$. Therefore, taking the cohomology sequence of the short exact sequence $0 \rightarrow \mathcal{I}_x \rightarrow \mathcal{I} \rightarrow \mathbb{C}(x) \rightarrow 0$ we obtain the exact sequence

$$\mathcal{I}(M) \rightarrow \mathbb{C} \rightarrow 0.$$

Hence, for any $a \in \mathbb{C}$, there exists $f \in \mathcal{I}(M) \subset A(M)$ such that $f(x) = a$. Choosing $a \neq 0$, our proof is complete. In fact a much sharper result is true. It can be shown that if M is of dimension m , there exist $f_1, \dots, f_m \in A(M)$ such that $X = Z(f_1, \dots, f_m)$. The reader may find a proof in Forster and Ramspott [1] (see also Grauert [2]).

2. Let M be a complex manifold and $\mathcal{U} = \{U_i : i \in I\}$ be an open cover of M by Stein manifolds. Then, exactly as in the proof of Example 15, §4, Chapter 2, $U_{i_0 \dots i_k} = U_{i_0} \cap \dots \cap U_{i_k}$ is Stein for all $i_0, \dots, i_k \in I$. It follows from Theorem B that if $\mathcal{F}$ is any coherent sheaf on M , then any Stein open cover of M is a Leray cover for $\mathcal{F}$. Before we start on the main work of this section, we show how Theorem B for free sheaves enables us to give a complete solution to the Cousin problems on a Stein manifold.

Theorem 7.2.1. A Stein manifold is a Cousin I and Cousin A domain.

Proof. Our standing assumption implies that $H^1(M, \mathcal{O}) (= H^1(M, \mathbb{C})) = 0$, M Stein. Hence, as in Example 6, §3, Chapter 6, M is a Cousin I and Cousin A domain. $\square$

Theorem 7.2.2. Let d be a divisor on the Stein manifold M . Then d is a divisor of a meromorphic function on M if and only if $c_1([d]) = 0$. In particular, M is a Cousin I and Cousin B domain if and only if $H^2(M, \mathbb{Z}) = 0$.

Proof. Our standing assumption implies that $H^1(M, \mathcal{O}) = H^2(M, \mathcal{O}) = 0$. Hence the Chern class map $c_1 : H^1(M, \mathcal{O}^*) \rightarrow H^2(M, \mathbb{Z})$ is an isomorphism (see Example 11, §3, Chapter 6). The result now follows from Example 16, §3, Chapter 6. $\square$

As an immediate consequence of Theorem 7.2.2 we have

Theorem 7.2.3. Let X be an analytic hypersurface in the Stein manifold M . Provided that $H^2(M, \mathbb{Z}) = 0$, there exists $f \in A(M)$ such that

1. $X = Z(f)$.
2. $\mathcal{I}_{X,x} = (f_x)$, for all $x \in M$.

Remark. If X is an analytic hypersurface in the m -dimensional Stein manifold M and $H^2(M, \mathbb{Z}) \neq 0$, then it can be shown that X is representable as the common zero locus of not more than $1 + [\frac{m}{2}]$ analytic functions on M . See Forster and Ramspott [1, Satz 3].

We shall give further applications of Theorems A and B in the remainder of the chapter and also in Chapter 12.

Proposition 7.2.4. Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M . Suppose that $\mathcal{F}$ admits the (projective) resolution

$$0 \rightarrow \underline{E}_m \xrightarrow{a_m} \cdots \xrightarrow{a_1} \underline{E}_0 \xrightarrow{a_0} \mathcal{F} \rightarrow 0,$$

where $\underline{E}_m, \dots, \underline{E}_0$ are holomorphic vector bundles on M . Then

1. $H^q(M, \mathcal{F}) = 0$, $q \geq 1$.
2. The map $a_0^0 : \Omega(\underline{E}_0) \rightarrow \mathcal{F}(M)$ is onto.

Proof. Set $\mathcal{K}_j = \text{Im}(a_j)$, $0 \leq j \leq m$. The exactness of the resolution of $\mathcal{F}$ is equivalent to the exactness of the sequences

$$0 \rightarrow \mathcal{K}_{j+1} \rightarrow \underline{E}_j \rightarrow \mathcal{K}_j \rightarrow 0, \quad j = 0, \dots, m-1.$$

Our standing assumption implies that $H^q(M, \underline{E}_j) = 0$, $q \geq 1$, $0 \leq j \leq m$. Consider the cohomology sequence of the short exact sequence $0 \rightarrow \mathcal{K}_m \rightarrow \underline{E}_{m-1} \rightarrow \mathcal{K}_{m-1} \rightarrow 0$. Since $\mathcal{K}_m \cong \underline{E}_m$, we see easily that $H^q(M, \mathcal{K}_{m-1}) = 0$, $q \geq 1$. Proceeding inductively, we deduce that $H^q(M, \mathcal{K}_j) = 0$, $q \geq 1$, $j = 0, \dots, m-1$. But $\mathcal{K}_0 = \mathcal{F}$ and so we have proved 1. Now take the cohomology sequence of $0 \rightarrow \mathcal{K}_1 \rightarrow \underline{E}_0 \xrightarrow{a_0} \mathcal{F} \rightarrow 0$ to obtain the exact sequence

$$0 \rightarrow \mathcal{K}_1(M) \rightarrow \Omega(\underline{E}_0) \xrightarrow{a_0^0} \mathcal{F}(M) \rightarrow 0.$$

This proves 2. □

Examples. 3. Let X be an analytic hypersurface in the Stein manifold M and suppose that $H^2(M, \mathbb{Z}) = 0$. We claim that $H^q(M, \mathcal{O}_X) = 0$, $q \geq 1$. Indeed by Theorem 7.2.3, there exists $f \in A(M)$ such that $\mathcal{I}_{X,x} = (f_x)$ for all $x \in M$. Therefore $\mathcal{O}_X$ has the resolution

$$0 \rightarrow \mathcal{O}_M \xrightarrow{\times f} \mathcal{O}_M \rightarrow \mathcal{O}_M/\mathcal{I}_X = \mathcal{O}_X \rightarrow 0.$$

Now apply Proposition 7.2.4.

4. Let $\mathcal{I}_0$ denote the ideal sheaf of the point $(0,0) \in \mathbb{C}^2$ and set $\mathcal{O} = \mathcal{O}_{\mathbb{C}^2}$, $\mathcal{O}_0 = \mathcal{O}/\mathcal{I}_0$. Then $\mathcal{O}_0$ has the resolution

$$0 \rightarrow \mathcal{O} \xrightarrow{a} \mathcal{O}^2 \xrightarrow{b} \mathcal{O} \rightarrow \mathcal{O}_0 \rightarrow 0,$$

where $a(f) = (z_2 f, z_1 f)$, $b(f_1, f_2) = z_1 f_1 - z_2 f_2$. Hence, by Proposition 7.2.4, $H^q(\mathbb{C}^2, \mathcal{O}_0) = 0$, $q \geq 1$.

Remarks. 1. Both the examples above are special instances of a general technique for constructing resolutions of ideal or structure sheaves based on the *Koszul complex*. For details we refer the reader to Griffiths and Harris [1] and Hartshorne [1]. See also the exercises at the end of §5.

2. Unfortunately it is not true that every coherent sheaf on a Stein manifold admits a resolution by free or even locally free sheaves (see exercise 9). However, we shall prove that a coherent sheaf on a Stein manifold M admits a free resolution over any relatively compact subset of M and this will be a main step in the proof of Theorems A and B.

The next few paragraphs are devoted to a study of the space of sections of a holomorphic vector bundle defined over a domain of holomorphy.

Proposition 7.2.5. Let E be a holomorphic vector bundle over the domain of holomorphy Ω in $\mathbb{C}^n$. Then

1. Given $e \in E_x$, $x \in \Omega$, there exists $s \in \Omega(E)$ such that $s(x) = e$.
2. If ω is any relatively compact open subset of Ω , there exists an exact sequence

$$\mathcal{O}_\omega^r \rightarrow \underline{E}_\omega \rightarrow 0.$$

Proof. Our proof of 1 goes by induction on n . Suppose $n = 1$. Let $\mathcal{I}$ denote the ideal sheaf of $\{x\}$. As in example 3 we have the free resolution $0 \rightarrow \mathcal{O} \xrightarrow{a} \mathcal{O} \rightarrow 0$ of $\mathcal{I}$, where a is defined as multiplication by $z - x$. Tensoring with $\underline{E}$, we obtain the exact sequence

$$0 \rightarrow \underline{E} \rightarrow \underline{E}^x \rightarrow 0,$$

where $\underline{E}^x = \mathcal{I} \otimes_{\mathcal{O}} \underline{E}$ and is the subsheaf of sections of $\underline{E}$ which vanish at x . Taking the cohomology sequence and applying our standing assumption we see that $H^p(\Omega, \underline{E}^x) = 0$, $p \geq 1$. Since $\underline{E}^x$ is a subsheaf of $\underline{E}$, we have the short exact sequence

$$0 \rightarrow \underline{E}^x \rightarrow \underline{E} \xrightarrow{P} E(x) \rightarrow 0$$

where $E(x)$ is the skyscraper sheaf with stalk E_x at x and zero stalk everywhere else and P evaluates sections at x . Taking the cohomology sequence and using the vanishing of $H^1(\Omega, \underline{E}^x)$, we arrive at the exact sequence

$$0 \rightarrow \Omega(\underline{E}^x) \rightarrow \Omega(\underline{E}) \xrightarrow{P} E_x \rightarrow 0$$

and so P is onto, proving 1. Now suppose the result proven for $n - 1$. Without loss of generality suppose $x = 0 \in \Omega$. Let H denote the intersection of the hyperplane $z_1 = 0$ with Ω . Since H is obviously holomorphically convex, H is a domain of holomorphy (in $\mathbb{C}^{n-1}$). We let $\mathcal{O}'$ denote the Oka sheaf of H and remark that $\mathcal{O}' \approx {}_{n-1}\mathcal{O}|_H$. We have the short exact sequence $0 \rightarrow \mathcal{O} \xrightarrow{a} \mathcal{O} \xrightarrow{r} \mathcal{O}' \rightarrow 0$ of sheaves over Ω where a corresponds to multiplication by z_1 and r is restriction of germs to H . Tensoring this sequence with $\underline{E}$ we obtain the short exact sequence

$$0 \rightarrow \underline{E} \rightarrow \underline{E} \xrightarrow{r} \underline{E}_H \rightarrow 0$$

where $\underline{E}_H$ is the sheaf of sections of the holomorphic bundle E restricted to H . Taking the cohomology sequence we arrive at the short exact sequence

$$0 \rightarrow \Omega(E) \rightarrow \Omega(E) \xrightarrow{r} \Omega(E|_H) \rightarrow 0.$$

By our inductive assumption, there exists $\tilde{s} \in \Omega(E|_H)$ such that $\tilde{s}(x) = e$. Since r is onto, it follows that there exists $s \in \Omega(E)$ such that $s(x) = e$, completing the inductive step and proving 1.

For the proof of 2 we note that for each $x \in \omega$, there exist, by 1, $s_1, \dots, s_q \in \Omega(E)$ such that $\{s_1(x), \dots, s_q(x)\}$ is a basis for E_x $q = \dim(E)$. By the compactness of $\bar{\omega}$, we may therefore find sections $s_1^1, \dots, s_q^1, s_1^2, \dots, s_q^p \in \Omega(E)$ such that for any $x \in \bar{\omega}$, $\{s_1^1(x), \dots, s_q^p(x)\}$ spans E_x . Taking $r = pq$, we see that the map $c = (s_1^1, \dots, s_q^p) : \mathcal{O}^r \rightarrow \underline{E}$ is onto. $\square$

The next theorem provides a key step towards the proof of Theorem B.

Theorem 7.2.6. Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M . Then there exists a free resolution of $\mathcal{F}$ over any relatively compact open subset ω of M :

$$0 \rightarrow \mathcal{O}_\omega^{p_m} \rightarrow \cdots \rightarrow \mathcal{O}_\omega^{p_0} \rightarrow \mathcal{F} \rightarrow 0.$$

Our proof of Theorem 7.2.6 will depend on several lemmas. Essentially, we shall first prove the theorem in case M is a domain of holomorphy (hard) and then, using a lemma of Rossi, deduce the general case (easy).

Definition 7.2.7. An open subset P of the Stein manifold M is called an *analytic polyhedron* if there exist $f_1, \dots, f_k \in A(M)$ such that P is a union of connected components of the set $\{z \in M : |f_j(z)| < 1, j = 1, \dots, k\}$.

Remark. Just as in §4, Chapter 2, it is easily verified that an analytic polyhedron is holomorphically convex and therefore a Stein manifold.

Lemma 7.2.8. Let U be an open neighbourhood of the compact subset K of the Stein manifold M . Suppose that K is $A(M)$ -convex (that is, $\widehat{K} = K$). Then we may find an analytic polyhedron $P \subset U$ which is a neighbourhood of K .

Proof. Without loss of generality we may suppose that U is relatively compact. For each $x \in \partial U$, there exists $F \in A(M)$ such that $|F(x)| > 1$, $\|F\|_K < 1$. By the compactness of ∂U , we may therefore find $F_1, \dots, F_q \in A(M)$ such that $\|F_1\|_K, \dots, \|F_q\|_K < 1$ and $\max |F_j(x)| > 1$ for every $x \in \partial U$. Since $Q = \{z \in M : |F_j(z)| < 1, j = 1, \dots, q\}$ is disjoint from ∂U , we may take $P = Q \cap U$. $\square$

Suppose K is a compact $A(M)$ -convex subset of the Stein manifold M . We say that K possesses property (R) if given any open neighbourhood U of K and coherent sheaf $\mathcal{F}$ on U , we can find an open neighbourhood V of K contained in U and exact sequence

$$\mathcal{O}_V^p \rightarrow \mathcal{F}_V \rightarrow 0$$

(V and p will depend on $\mathcal{F}$).

Lemma 7.2.9. Suppose that the compact $A(M)$ -convex subset K possesses property (R). Then given an open neighbourhood U of K and coherent sheaf $\mathcal{F}$ on U , we can find a Stein open neighbourhood $\omega \subset U$ of K and free resolution of $\mathcal{F}$ over ω :

$$0 \rightarrow \mathcal{O}_\omega^{p_m} \rightarrow \cdots \rightarrow \mathcal{O}_\omega^{p_0} \xrightarrow{a_0} \mathcal{F}_\omega \rightarrow 0.$$

Proof. Since K possesses property (R), we may find an open neighbourhood $U_0 \subset U$ of K and exact sequence $\mathcal{O}_{U_0}^{p_0} \xrightarrow{a_0} \mathcal{F}_{U_0} \rightarrow 0$. Now $\text{Ker}(a_0)$ is a coherent sheaf on U_0 and so, since K possesses property (R), we have an exact sequence $\mathcal{O}_{U_1}^{p_1} \xrightarrow{a_1} \text{Ker}(a_0)_{U_1} \rightarrow 0$ over some open neighbourhood $U_1 \subset U_0$ of K . That is, we have the exact sequence $\mathcal{O}_{U_1}^{p_1} \xrightarrow{a_1} \mathcal{O}_{U_1}^{p_0} \xrightarrow{a_0} \mathcal{F}_{U_1} \rightarrow 0$. We now proceed inductively, just as in the proof of Corollary 7.1.8, to obtain a free resolution

$$0 \rightarrow \mathcal{O}_{U_m}^{p_m} \rightarrow \cdots \rightarrow \mathcal{O}_{U_m}^{p_0} \rightarrow \mathcal{F}_{U_m} \rightarrow 0 \quad (*)$$

of $\mathcal{F}$ over some open neighbourhood $U_m \subset U$ of K . Finally, by Lemma 7.2.8 we may find a Stein neighbourhood $\omega \subset U_m$ of K and restricting $(*)$ to ω we obtain the required result. $\square$

Corollary 7.2.10. Suppose that the compact $A(M)$ -convex compact subset K possesses property (R) and that $\mathcal{F}$ is a coherent sheaf defined on some open neighbourhood of K . Then

1. For every open neighbourhood V of K , we may find a Stein open neighbourhood $\omega \subset V$ of K such that $H^q(\omega, \mathcal{F}_\omega) = 0$, $q \geq 1$.
2. If $f, f_1, \dots, f_k$ are sections of $\mathcal{F}$ over some open neighbourhood V of K and if $f_1, \dots, f_k$ generate $\mathcal{F}_V$, we may find a (Stein) open neighbourhood $\omega \subset V$ of K and $a_1, \dots, a_k \in A(\omega)$ such that

$$f = \sum_{j=1}^k a_j f_j \quad \text{on } \omega.$$

Proof. Both 1 and 2 are immediate from Lemma 7.2.9 and Proposition 7.2.4. $\square$

We now come to our main lemma.

Lemma 7.2.11 (H. Cartan). Let K be a compact $A(\Omega)$ -convex subset of the domain of holomorphy Ω . Suppose that $f \in A(\Omega)$ and that $K_a = \{z \in K : \operatorname{Re}(f(z)) = a\}$ possesses property (R) for all $a \in \mathbb{R}$. Then K possesses property (R).

Proof. Our proof is a combination of that due to H. Cartan together with a twist due to Hörmander [1] which makes use of Theorem B for locally free sheaves.

Let $\mathcal{F}$ be a coherent sheaf defined on some neighbourhood of K . For $-\infty \leq a \leq b \leq +\infty$ we set $K_{a,b} = \{z \in K : a \leq \operatorname{Re}(f(z)) \leq b\}$. Certainly $K_{a,b}$ is $A(\Omega)$ -convex since the condition $a \leq \operatorname{Re}(f(z)) \leq b$ may be equivalently written as $|\exp(f(z))| \leq \exp(b)$; $|\exp(-f(z))| \leq \exp(-a)$.

For sufficiently large negative numbers a , $K_{-\infty,a} = \emptyset$ and so $K_{-\infty,a}$ possesses property (R). Let S denote the supremum of numbers a such that $K_{-\infty,a}$ possesses property (R). It is enough to prove $S = +\infty$, since $K_{-\infty,+\infty} = K$.

Suppose $S < +\infty$. Since $K_{S,S} = K_S$ possesses property (R), Proposition 7.2.4 and Corollary 7.2.10 imply that there exists a Stein open neighbourhood U_1 of K_S and $f_1 = (f_1^1, \dots, f_1^p) \in \mathcal{F}(U_1)^p$, such that $f_1^1, \dots, f_1^p$ generate $\mathcal{F}$ as an $\mathcal{O}$ -module over U_1 . Choose $a < S < b$ so that $K_{a,b} \subset U_1$. By definition of S and Proposition 7.2.4, Corollary 7.2.10, there exists a Stein open neighbourhood U_2 of $K_{-\infty,a}$ and $f_2 = (f_2^1, \dots, f_2^q) \in \mathcal{F}(U_2)^q$ such that $f_2^1, \dots, f_2^q$ generate $\mathcal{F}$ as an $\mathcal{O}$ -module over U_2 . By Corollary 7.2.10, 2, we may find an open Stein neighbourhood U_3 of $K_{a,a}$ and $q \times p$ matrix θ_1 , with coefficients holomorphic on U_3 , such that

$$\theta_1 f_1 = f_2 \quad \text{on } U_3.$$

Similarly, we may find a $p \times q$ matrix θ_2 , with coefficients holomorphic on some Stein open neighbourhood U_4 of $K_{a,a}$, such that

$$\theta_2 f_2 = f_1 \quad \text{on } U_4.$$

Shrinking $U_1, \dots, U_4$ we may suppose that $U_1 \cap U_2 = U_3 = U_4$ (though, of course, the neighbourhoods need no longer be Stein). Since $K_{-\infty,b}$ is $A(\Omega)$ -convex and contained in $U_1 \cup U_2$, there exists a Stein open neighbourhood U of $K_{-\infty,b}$ which is contained in $U_1 \cup U_2$ (Lemma 7.2.8). Replacing the neighbourhoods U_j by their intersections with U , we may assume that $U_1 \cup U_2 = U$ and $U_1 \cap U_2 = U_3 = U_4$.

For $j = 1, 2$, we define homomorphisms $F_j : \mathcal{O}_{U_j}^{p+q} \rightarrow \mathcal{F}_{U_j}$ by

$$\begin{aligned} F_1(u_1^1, \dots, u_{p+q}^1) &= \sum_{i=1}^p u_i^1 f_1^i, & u^1 &= (u_1^1, \dots, u_{p+q}^1) \in A(U_1)^{p+q}, \\ F_2(u_1^2, \dots, u_{p+q}^2) &= \sum_{i=1}^q u_{i+p}^2 f_2^i; & u^2 &= (u_1^2, \dots, u_{p+q}^2) \in A(U_2)^{p+q}. \end{aligned}$$

Our construction guarantees that the sequences

$$\mathcal{O}_{U_j}^{p+q} \xrightarrow{F_j} \mathcal{F}_{U_j} \rightarrow 0$$

are exact, $j = 1, 2$. In the remainder of the proof we show how to identify the sequences over U_{12} to obtain a locally free resolution of $\mathcal{F}$ over U .

Let I_p, I_q denote the identity $p \times p, q \times q$ matrices respectively. We see that

$$\begin{pmatrix} I_p & 0 \\ \theta_1 & I_q \end{pmatrix} \begin{pmatrix} f_1 \\ 0 \end{pmatrix} = \begin{pmatrix} I_p & \theta_2 \\ 0 & I_q \end{pmatrix} \begin{pmatrix} 0 \\ f_2 \end{pmatrix} \quad \text{on } U_{12}.$$

Hence

$$\begin{pmatrix} f_1 \\ 0 \end{pmatrix} = \begin{pmatrix} I_p & 0 \\ -\theta_1 & I_q \end{pmatrix} \begin{pmatrix} I_p & \theta_2 \\ 0 & I_q \end{pmatrix} \begin{pmatrix} 0 \\ f_2 \end{pmatrix}.$$

Let $\theta : U_{12} \rightarrow \text{GL}(p+q, \mathbb{C})$ denote the holomorphic matrix function defined by

$$\theta = \begin{pmatrix} I_p & 0 \\ -\theta_1 & I_q \end{pmatrix} \begin{pmatrix} I_p & \theta_2 \\ 0 & I_q \end{pmatrix}.$$

Observe that, over U_{12} , we have $F'_1 = \theta F'_2$, where the prime denotes transpose. Therefore, $F_1 = F_2 \theta'$. Hence, if $u^j \in A(U_j)^{p+q}$, $j = 1, 2$, $F_1 u^1 = F_2 u^2$ if and only if $F_2(\theta' u^1) = F_2 u^2$. In particular, we have equality if $u^2 = \theta' u^1$. Setting $\phi_{12} = (\theta')^{-1}$, $\phi_{21} = \phi_{12}^{-1}$ we see that $\{\phi_{12}, \phi_{21}\}$ are the transition functions for a holomorphic vector bundle E over $U = U_1 \cup U_2$ and that the condition $\phi_{21} u^1 = u^2$ amounts to saying that u^1, u^2 are local representatives of a holomorphic section u of E (over U). The morphisms F_1, F_2 determine a homomorphism $\underline{E} \rightarrow \mathcal{F}_U$ which is surjective since both F_1 and F_2 are surjective. We now apply Proposition 7.2.5, 2, to deduce that $K_{-\infty, b}$ possesses property (R). Contradiction. Therefore $S = +\infty$ and K possesses property (R). $\square$

Lemma 7.2.12. Every compact $A(\Omega)$ -convex subset K of a domain of holomorphy Ω in $\mathbb{C}^n$ possesses property (R).

Proof. The set $\{z \in K : \text{Re}(z_j) = a_j, \text{Im}(z_j) = b_j, j = 1, \dots, n\}$ possesses property (R) for arbitrary $a_j, b_j \in \mathbb{R}$ since it either is empty or consists of a single point. By Lemma 7.2.11, the set obtained by dropping one of the conditions $\text{Re}(z_j) = a_j, \text{Im}(z_j) = b_j$ still has property (R). Iteration of this argument $2n$ times gives the result. $\square$

The first part of the next lemma will enable us easily to extend Lemma 7.2.12 to an arbitrary Stein manifold. We shall use the second part to prove a key approximation theorem needed in the proof of Theorem B.

Lemma 7.2.13. Let K be a compact $A(M)$ -convex subset of the Stein manifold M . Then, given an open neighbourhood U of K , we may find a Stein open neighbourhood $V \subset U$ of K and holomorphic map $F : M \rightarrow \mathbb{C}^N$ such that

1. F maps V biholomorphically onto a closed submanifold P of the unit Euclidean disc $E = \{(z_1, \dots, z_N) : \sum_{i=1}^N |z_i|^2\}$ of $\mathbb{C}^N$.
2. For $p \geq 1$, $H^p(E, \mathcal{I}_P) = 0$ ($\mathcal{I}_P$ denotes the ideal sheaf of P).

Proof (Rossi [1]). Without loss of generality we may suppose that U is relatively compact. For each $x \in \bar{U}$, we may find $f_1, \dots, f_m \in A(M)$ which define local coordinates at x . Hence, by the compactness of $\bar{U}$, we may find a finite set $h_1, \dots, h_p \in A(M)$ which give local coordinates at every point $x \in \bar{U}$. In particular, we may find an open neighbourhood W of the diagonal

in $\bar{U} \times \bar{U}$ such that if $x, y \in W$, $x \neq y$, there exists an h_j with $h_j(x) \neq h_j(y)$. For each $x, y \in \bar{U} \times \bar{U} \setminus W$, we can find $f \in A(M)$ with $f(x) \neq f(y)$. Therefore since $\bar{U} \times \bar{U} \setminus W$ is compact we may find a finite set $h_{p+1}, \dots, h_r \in A(M)$ which separates points in $\bar{U} \times \bar{U} \setminus W$. We see at once that the set $h_1, \dots, h_r$ separates points of $\bar{U}$ and gives local coordinates at every point of $\bar{U}$. In particular, the map $h = (h_1, \dots, h_r) : M \rightarrow \mathbb{C}^r$ restricts to an injective holomorphic immersion on some neighbourhood of $\bar{U}$. Multiplying h by a sufficiently small scalar we may in addition suppose that

$$\|h\|_K^2 = \sum_{j=1}^r \|h_j\|_K^2 < \frac{1}{2}.$$

Just as in the proof of Lemma 7.2.8, we may find $f_1, \dots, f_k \in A(M)$ such that $\|f_j\|_K < 1$, $1 \leq j \leq k$ and $\max_j |f_j(z)| > 1$, all $z \in \partial U$. For positive integers q , define $g_q : M \rightarrow \mathbb{R}$ by

$$g_q(z) = \sum_{j=1}^k |f_j^q|^2.$$

Fix a value of q so large that $\|g_q\|_K < \frac{1}{2}$ and $|g_q(z)| > 1$, $z \in \partial U$. For $z \in M$, define

$$G(z) = g_q(z) + \sum_{j=1}^r |h_j(z)|^2$$

and let $V = \{z \in U : G(z) < 1\}$. Our construction guarantees that V is an open neighbourhood of K with $\bar{V} \subset U$. The map $F : M \rightarrow \mathbb{C}^{r+k}$ defined by $F = (h_1, \dots, h_r, f_1^q, \dots, f_k^q)$, maps V biholomorphically onto a closed submanifold P of E . In particular P is Stein, since E is Stein, and so V is Stein. To complete the proof, notice that for ϵ small and positive, $V_\epsilon = \{z \in U : G(z) < 1 + \epsilon\}$ will be mapped biholomorphically by F onto a closed submanifold P_ϵ of the Euclidean disc E_ϵ of radius $1 + \epsilon$. Now $\mathcal{I}_{P_\epsilon}|E = \mathcal{I}_P$ and so, since $\bar{E}$ is an $A(E_\epsilon)$ -convex compact subset of E_ϵ , Lemma 7.2.12 and Corollary 7.2.10 imply that $H^p(E, \mathcal{I}_P) = 0$, $p \geq 1$. $\square$

Proof of Theorem 7.2.6. Let ω be a relatively compact Stein open subset of M . Then $K = \hat{\omega}$ is a compact $A(M)$ -convex subset of M . By Lemma 7.2.13, we may find a Stein open neighbourhood V of K and holomorphic map F of M into some $\mathbb{C}^N$ such that F maps V biholomorphically onto a closed submanifold P of the unit Euclidean disc E in $\mathbb{C}^N$. Since F maps V biholomorphically onto P , $F_*\mathcal{F}_V$ is a coherent sheaf of $F_*\mathcal{O}_V = \mathcal{O}_P$ -modules on P . Let $\tilde{\mathcal{F}}$ denote the trivial extension of $F_*\mathcal{F}_V$ to E . $\tilde{\mathcal{F}}$ is a coherent sheaf of $\mathcal{O}_E$ -modules. Choose $\epsilon > 0$ so that $(F|V)^{-1}E_\epsilon \supset K$, where E_ϵ denotes the open Euclidean disc of radius $1 - \epsilon$ in $\mathbb{C}^N$. By Lemmas 7.2.9, 7.2.11, we may find a free resolution $0 \rightarrow \mathcal{O}_{E_\epsilon}^{p_m} \rightarrow \dots \rightarrow \mathcal{O}_{E_\epsilon}^{p_0} \rightarrow \tilde{\mathcal{F}}_{E_\epsilon} \rightarrow 0$ of $\tilde{\mathcal{F}}$ over E_ϵ . Restrict the resolution to $P_\epsilon = E_\epsilon \cap P$ and apply Exercise 9, §1 to obtain a free $\mathcal{O}_P$ -resolution of $F_*\mathcal{F}$ over P_ϵ . Pull back by F to obtain a free $\mathcal{O}_M$ -resolution of $\mathcal{F}$ over $(F|V)^{-1}(E_\epsilon) \supset K$. We have shown that $\mathcal{F}$ has a free resolution over some open neighbourhood of K and, a fortiori, over ω . $\square$

As an immediate corollary of Theorem 7.2.6 we have

Lemma 7.2.14. Every compact $A(M)$ -convex subset K of a Stein manifold M possesses property (R).

Proposition 7.2.15. Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M . Then $H^q(M, \mathcal{F}) = 0$, $q \geq 2$.

Proof. Using Lemma 7.2.8 we may choose an open cover $\mathcal{U} = \{U_i : i = 1, 2, \dots\}$ of M by relatively compact Stein open subsets of M satisfying

$$1. \quad \bar{U}_n \subset U_{n+1}, \quad n \geq 1; \quad 2. \quad \bigcup_{n=1}^{\infty} U_n = M.$$

By Example 2 and Theorem 7.2.6, $\mathcal{U}$ is a Leray cover of M for $\mathcal{F}$. Hence $H^q(M, \mathcal{F}) \approx H^q(\mathcal{U}, \mathcal{F})$, $q \geq 0$. Suppose $A \in Z^q(\mathcal{U}, \mathcal{F})$. For $n \geq 1$, let $\mathcal{U}_n = \{U_j : j = 1, \dots, n\}$ and A_n denote the restriction of A to $\mathcal{U}_n$. Since $\mathcal{U}_n$ is a Leray cover of U_n for $\mathcal{F}$ and $H^q(U_n, \mathcal{F}) = 0$, $q \geq 1$, we have $H^q(\mathcal{U}_n, \mathcal{F}) = 0$, $q \geq 1$. Now $A_n \in Z^q(\mathcal{U}_n, \mathcal{F})$ and so there exists $b_n \in C^{q-1}(\mathcal{U}_n, \mathcal{F})$ such that $D(b_n) = A_n$. Extend b_n to $C^{q-1}(\mathcal{U}, \mathcal{F})$ by setting $b_n|_{U_j} = 0$, $j > n$. We construct inductively a sequence $B_n \in C^{q-1}(\mathcal{U}, \mathcal{F})$ satisfying

$$1. \quad B_n|_{U_j} = B_j|_{U_j}, \quad j \leq n; \quad 2. \quad A_n = D(B_n)|_{\mathcal{U}_n}.$$

Define $B_1 = b_1$ and suppose $B_1, \dots, B_n$ have been constructed. We have $D((b_{n+1} - B_n)|_{\mathcal{U}_n}) = 0$. So, provided that $q - 1 \geq 1$, there exists $C_n \in C^{q-2}(\mathcal{U}_n, \mathcal{F})$ such that $D(C_n) = (b_{n+1} - B_n)|_{\mathcal{U}_n}$. Extend C_n to $C^{q-2}(\mathcal{U}, \mathcal{F})$ by taking $C_n|_{U_j} = 0$, $j > n$. We define $B_{n+1} = b_{n+1} - D(C_n)$. Finally define $B \in C^{q-1}(\mathcal{U}, \mathcal{F})$ by $B|_{U_j} = B_j$. Clearly $D(B) = A$. $\square$

The proof of Proposition 7.2.15 clearly breaks down if $q = 1$ and we have to use an approximation theorem for this case. First we need to topologise the space of sections of a coherent sheaf.

Let K be a compact $A(M)$ -convex subset of the Stein manifold M and $\mathcal{F}$ be a coherent sheaf on M . By Proposition 7.2.4 and Theorem 7.2.6 we may find sections $s_1, \dots, s_k \in \mathcal{F}(K)$ which generate $\mathcal{F}$ as an $\mathcal{O}$ -module over K (see also Lemma 6.1.12). Let $s \in \mathcal{F}(K)$. Define

$$|s|_K = \inf \left\{ \max_j \|c_j\|_K : s = \sum_{j=1}^k c_j s_j, \quad c_j \in \mathcal{O}(K) \right\}.$$

The seminorm $|\cdot|_K$ may depend on the choice of generators $s_1, \dots, s_k$ but another choice of generators gives an equivalent seminorm since the two sets of generators are related by a matrix with holomorphic entries analytic on K , Corollary 7.2.10.

Lemma 7.2.16. Let $s \in \mathcal{F}(K)$ and suppose that $|s|_K = 0$. Then $s_z = 0$, for all $z \in \mathring{K}$.

Proof. Choose generators $s_1, \dots, s_k \in \mathcal{F}(K)$ for $\mathcal{F}$ over K and suppose $s = \sum_{j=1}^k c_j s_j$ on K . If $|s|_K = 0$, we may find for every $\epsilon > 0$, $c_j^\epsilon \in \mathcal{O}(K)$ such that $s = \sum_{j=1}^k c_j^\epsilon s_j$ and $\|c_j^\epsilon\|_K < \epsilon$, $1 \leq j \leq k$. We certainly have

$$(c_1 - c_1^\epsilon, \dots, c_k - c_k^\epsilon)_z \in \mathcal{R}(s_1, \dots, s_k)_z, \quad z \in K. \quad (*)$$

Fix $z \in \mathring{K}$ and let $P_1, \dots, P_q \in \mathcal{R}(s_1, \dots, s_k)_z$ be a set of generators over $\mathcal{O}_z$ for $\mathcal{R}(s_1, \dots, s_k)_z$. By Theorem 3.6.2 and the Oka Theorem (Theorem 7.1.6), there exists an open neighbourhood $D \subset \mathring{K}$ of z such that the sequence

$$A_B(D)^q \xrightarrow{P} \mathcal{R}_B(s_1, \dots, s_k)(D) \rightarrow 0$$

is *split* exact (“B” denotes bounded sections). In particular, letting $\epsilon \rightarrow 0$ in (*), we see that $(c_1, \dots, c_k)|_D \in \text{Im}(P) = \mathcal{R}(s_1, \dots, s_k)(D)$. But therefore $s_y = 0$, $y \in D$. Hence the result. Of course, our argument is just the closure of modules theorem, Exercise 3, §6, Chapter 3. $\square$

Let $(K_n)_{n \geq 1}$ be a normal exhaustion of M by compact $A(M)$ -convex subsets K_n (see §4, Chapter 2 and note that we require $K_n \subset \mathring{K}_{n+1}$, $n \geq 1$). For each n choose sections in $\mathcal{F}(K_n)$ which generate $\mathcal{F}$ over K_n and let $|\cdot|_n = |\cdot|_{K_n}$ denote the corresponding seminorm. We take the topology on $H^0(M, \mathcal{F})$ defined by the seminorms $|\cdot|_n$, $n \geq 1$. It is quite straightforward to verify that any two normal exhaustions of M will give rise to the same topology on $H^0(M, \mathcal{F})$.

Lemma 7.2.17. Suppose that we are given sections $f_n \in \mathcal{F}(K_n)$, $n \geq 1$, such that for $p \geq 1$, $|f_n - f_m|_p \rightarrow 0$, $n, m \rightarrow \infty$. Then there exists a unique section $F \in \mathcal{F}(M)$ such that $|F - f_n|_p \rightarrow 0$, $n \rightarrow \infty$, $p \geq 1$.

Proof. Fix $p \geq 1$ and let $s_1, \dots, s_k \in \mathcal{F}(K_{p+1})$ be sections defining $|_{p+1}$. Choose integers $n_1 \leq n_2 \leq \dots$ such that

$$|f_{n_{i+1}} - f_{n_i}|_{p+1} \leq 2^{-i}, \quad i \geq 1.$$

Set $u_0 = f_{n_1}$, $u_i = f_{n_{i+1}} - f_{n_i}$, $i \geq 1$. Choose $c_{ij} \in \mathcal{O}(K_{p+1})$ such that $u_i = \sum_{j=1}^k c_{ij} s_j$ and $\max_j \|c_{ij}\|_{K_{p+1}} \leq |u_i|_{p+1} + 2^{-i} \leq 2^{-i+1}$. Since $\sum_{i=1}^{\infty} \|c_{ij}\|_{K_{p+1}} < \infty$, the series $\sum_{i=0}^{\infty} c_{ij}$ converges uniformly on K_{p+1} to a function C_j which is analytic on $\mathring{K}_{p+1}$, $1 \leq j \leq k$. Since $K_p \subset \mathring{K}_{p+1}$, $C_j \in \mathcal{O}(K_p)$. Define $F_p = \sum_{j=1}^k C_j s_j$ and observe that

$$|F_p - f_{n_{q+1}}|_p = \left| F_p - \sum_{i=0}^q u_i \right|_p \rightarrow 0 \quad \text{as } q \rightarrow \infty.$$

Now by the triangle-inequality it is clear that $|F_p - f_n|_p \rightarrow 0$, $n \rightarrow \infty$. Moreover by Lemma 7.2.16, this condition determines F_p uniquely on $\mathring{K}_p$. In particular, $F_{p+r}|_{\mathring{K}_p} = F_p|_{\mathring{K}_p}$, $r \geq 0$. Define $F \in \mathcal{F}(M)$ by $F|_{\mathring{K}_p} = F_p$. $\square$

Theorem 7.2.18. Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M and give $H^0(M, \mathcal{F})$ the topology described above. Then $H^0(M, \mathcal{F})$ is a Fréchet space.

Proof. Lemmas 7.2.16, 7.2.17. $\square$

Next we prove our main approximation theorem.

Theorem 7.2.19. Let K be an $A(M)$ -convex subset of the Stein manifold M and suppose that $f \in \mathcal{O}(K)$. Then there exists a sequence $f_n \in A(M)$ such that $\|f_n - f\|_K \rightarrow 0$, $n \rightarrow \infty$.

Proof. We may suppose that $f \in A(U)$, where U is an open neighbourhood of K . By Lemma 7.2.13, we may find a Stein open neighbourhood $V \subset U$ of K and analytic map $F : M \rightarrow \mathbb{C}^N$ such that $P = F(V)$ is a closed submanifold of the unit Euclidean disc E in $\mathbb{C}^N$ and $H^p(E, \mathcal{I}_P) = 0$, $p \geq 1$. Let $g = f(F|V)^{-1} \in A(P)$. Taking the cohomology sequence of $0 \rightarrow \mathcal{I}_P \rightarrow \mathcal{O}_E \rightarrow \mathcal{O}_P \rightarrow 0$ we see that the restriction map $A(E) \rightarrow A(P)$ is onto. Therefore there exists $G \in A(E)$ such that $G|_P = g$. Now polynomials are dense in $A(E)$ (Exercise 1, §2, Chapter 2). Therefore, for $n \geq 1$ we may find a polynomial p_n on $\mathbb{C}^N$ such that $|p_n - G|_{F(K)} \leq 1/n$. Set $f_n = p_n F \in A(M)$. Then $\|f_n - f\|_K \rightarrow 0$, $n \rightarrow \infty$. $\square$

Theorem 7.2.20. Let K be a compact $A(M)$ -convex subset of the Stein manifold M and $\mathcal{F}$ be a coherent sheaf on M . Suppose $f \in \mathcal{F}(K)$. Then there exists a sequence $f_n \in \mathcal{F}(M)$ such that $|f - f_n|_K \rightarrow 0$, $n \rightarrow \infty$.

Proof. Choose a normal exhaustion (K_p) of M with $K_1 = K$. Fix $n \geq 1$. For $p > 0$, we shall construct $g_p \in \mathcal{F}(K_p)$ such that

$$g_1 = f \quad \text{and} \quad |g_p - g_{p+1}|_r \leq 2^{-p}/n, \quad r \leq p.$$

By Lemma 7.2.17 this will imply that there exists $f_n \in \mathcal{F}(M)$ such that $|f_n - g_j|_p \rightarrow 0$, $j \rightarrow \infty$, $p \geq 1$. Taking $p = 1$ and noting that $f_n = \sum_{j=1}^{\infty} (g_{j+1} - g_j) + f$ on K , we therefore have $|f_n - f|_K \leq 1/n$. To construct the sections g_p we proceed inductively and suppose $g_1, \dots, g_p$ constructed. Let $s_1, \dots, s_k \in \mathcal{F}(K_{p+1})$ generate $\mathcal{F}$ over K_{p+1} . Then $g_p = \sum_{j=1}^k c_j s_j$, $c_j \in \mathcal{O}(K_p)$. By Theorem 7.2.19, we may find $d_j \in A(M)$ such that $\|c_j - d_j\|_{K_p} \leq 2^{-p}/n$. Clearly $g_{p+1} = \sum_{j=1}^k d_j s_j$ satisfies our requirements. $\square$

Theorem 7.2.21 (Theorem B of Cartan). Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M . Then $H^q(M, \mathcal{F}) = 0$, $q \geq 1$.

Proof. We have already proved that $H^q(M, \mathcal{F}) = 0$, $q \geq 2$ (Proposition 7.2.15). There remains the case $q = 1$.

Choose a normal exhaustion $(K_n)_{n \geq 1}$ of M . For each n we may, by Lemma 7.2.8, find a relatively compact Stein open neighbourhood $U_n \subset \mathring{K}_{n+1}$ of K_n . Certainly $U_n \subset U_{n+1}$, $n \geq 1$. As in the proof of Proposition 7.2.15, $\mathcal{U} = \{U_n\}$ is a Leray cover of M for $\mathcal{F}$. Set $\mathcal{U}_n = \{U_i : i = 1, \dots, n\}$, $n \geq 1$. Given $A \in Z^1(\mathcal{U}, \mathcal{F})$, let A_n denote the restriction of A to $\mathcal{U}_n$. Since $H^1(\mathcal{U}_n, \mathcal{F}) = 0$, there exists $b_n \in C^0(\mathcal{U}_n, \mathcal{F})$ such that $D(b_n) = A_n$, $n \geq 1$. We construct inductively a sequence $B_n \in C^0(\mathcal{U}_n, \mathcal{F})$ such that

1. $D(B_n) = A_n$.
2. $|B_n - B_{n+1}|_r \leq 2^{-n}$, $1 \leq r \leq n$.

(For 2 note that 1 implies $B_n - B_{n+1} \in \mathcal{F}(U_n)$). Take $B_1 = b_1$ and suppose $B_1, \dots, B_n$ have been constructed. Condition 1 implies $b_{n+1} - B_n \in \mathcal{F}(U_n)$ and so by Theorem 7.2.20 there exists $\eta \in \mathcal{F}(M)$ such that $|B_n - b_{n+1} - \eta|_n \leq 2^{-n}$. Now take $B_{n+1} = b_{n+1} + \eta$. Next we shall construct $F_p \in \mathcal{F}(\mathring{K}_p)$ such that $B_p + F_p = B_{p+1} + F_{p+1}$ on $\mathring{K}_p$, $p \geq 1$. Given $p \geq 1$, let $s_1, \dots, s_k \in \mathcal{F}(K_p)$ define the seminorm $|\cdot|_p$. We may choose $c_{ij} \in \mathcal{O}(K_p)$ such that for $i \geq p$, $(B_{i+1} - B_i)|_{K_p} = \sum_{j=1}^k c_{ij}s_j$ and $\max_j \|c_{ij}\|_{K_p} \leq 2^{-i+1}$. Just as in the proof of Lemma 7.2.17 these conditions imply that for $1 \leq j \leq k$, $\sum_{i=p}^\infty c_{ij}$ converges uniformly on K_p to a function C_j which is analytic on $\mathring{K}_p$. Set $F_p = \sum_{j=1}^k C_j s_j \in \mathcal{F}(\mathring{K}_p)$. For every $A(M)$ -convex compact subset K of $\mathring{K}_p$ we have

$$\lim_{n \rightarrow \infty} |(F_p + B_p) - B_{n+1}|_K = \lim_{n \rightarrow \infty} \left| F_p - \sum_{i=p}^n (B_{i+1} - B_i) \right|_K = 0.$$

It follows from Lemma 7.2.16 that $B_p + F_p = B_{p+1} + F_{p+1}$ on $\mathring{K}_p$. Hence we may define $B \in C^0(\mathcal{U}, \mathcal{F})$ by $B|_{\mathring{K}_p} = B_p + F_p$, $p \geq 1$. Now as $D(B_{n+1})|_{\mathcal{U}_n} = A_n$ and F_{n+1} restricts to a section of $\mathcal{F}$ over U_n – that is a cocycle – we see that $D(B)|_{\mathcal{U}_n} = A_n$, $n \geq 1$. Hence $D(B) = A$. $\square$

Theorem 7.2.22 (Theorem A of Cartan). Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M . For each $x \in M$, there exists $s_1, \dots, s_p \in \mathcal{F}(M)$ such that $s_{1,x}, \dots, s_{p,x}$ generate $\mathcal{F}_x$ as an $\mathcal{O}_x$ -module.

Proof. Let $\mathcal{I}$ denote the ideal sheaf of $\{x\}$. Then $\mathcal{IF}$ is a coherent subsheaf of $\mathcal{F}$, Theorem 7.1.9, 1. Therefore by Theorem B, $H^1(M, \mathcal{IF}) = 0$ and so taking the cohomology sequence of $0 \rightarrow \mathcal{IF} \rightarrow \mathcal{F} \rightarrow \mathcal{F}/\mathcal{IF} \rightarrow 0$ we obtain the exact sequence

$$\mathcal{F}(M) \rightarrow (\mathcal{F}/\mathcal{IF})(M) \rightarrow 0. \quad (*)$$

For $y \neq x$, $(\mathcal{IF})_y = \mathcal{F}_y$ and so $(\mathcal{F}/\mathcal{IF})_y = 0$. Hence $(\mathcal{F}/\mathcal{IF})(M) = \mathcal{F}_x/\mathfrak{m}_x \mathcal{F}_x$, where $\mathfrak{m}_x$ denotes the maximal ideal of $\mathcal{O}_x$ at x . Let N denote the $\mathcal{O}_x$ -submodule of $\mathcal{F}_x$ generated by $\{s_x : s \in \mathcal{F}(M)\}$. Since $(*)$ is exact we have $N + \mathfrak{m}_x \mathcal{F}_x = \mathcal{F}_x$. It now follows by Nakayama's Lemma that $N = \mathcal{F}_x$. $\square$

We may now give a global version of Corollary 7.2.10.

Theorem 7.2.23. Let $\mathcal{F}$ be a coherent sheaf on the Stein manifold M . Suppose that $s_1, \dots, s_p \in \mathcal{F}(M)$ generate $\mathcal{F}_x$ for every $x \in M$. Then given $S \in \mathcal{F}(M)$, there exist $f_j \in A(M)$ such that

$$S = \sum_{j=1}^p f_j s_j.$$

Proof. The sheaf map $s = (s_1, \dots, s_p) : \mathcal{O}_M^p \rightarrow \mathcal{F} \rightarrow 0$ is onto. Since $\text{Ker}(s)$ is coherent, Corollary 7.1.7, $H^1(M, \text{Ker}(s)) = 0$ by Theorem B. Hence taking the cohomology sequence of $0 \rightarrow \text{Ker}(s) \rightarrow \mathcal{O}_M^p \xrightarrow{s} \mathcal{F} \rightarrow 0$ we obtain the exact sequence $A(M)^p \xrightarrow{s} \mathcal{F}(M) \rightarrow 0$. $\square$

Remarks. Our proof of Theorem B is close to that of Hörmander [1] in that it makes use of Theorem B for locally free sheaves of $\mathcal{O}$ -modules. The main difference is that we make use of a device of Rossi [1] to give an elementary proof of the approximation theorem, Theorem 7.2.19. Cartan's original proof starts by establishing Theorems A and B for cubes in $\mathbb{C}^n$ and then extends the result to Stein manifolds. For expositions of this approach to Theorems A and B see H. Cartan [1], Grauert and Remmert [1] and Gunning and Rossi [1]. Rossi [1] gives a proof of Theorems A and B by starting from the relatively elementary result of Oka to the effect that $H^q(D, \mathcal{O}_D) = 0$, $q \geq 1$, for all polynomially convex domains D in $\mathbb{C}^n$. Using this result he constructs arbitrarily fine Leray covers for coherent sheaves and then proves Grauert's finiteness theorem for strictly pseudoconvex domains (see §4). Rossi then shows that the cohomology of a coherent sheaf on a strictly pseudoconvex domain is supported on the compact analytic subsets of the domain (see also Rossi [2], R. Narasimhan [3]). Since Stein manifolds have no non-trivial compact analytic subsets this is sufficient to deduce the vanishing theorem for relatively compact strictly pseudoconvex domains in a Stein manifold. The rest of his proof is similar to what we have presented here.

One point about our proof should be noticed: We really only assume Theorem B for locally free sheaves defined over *contractible* domains in $\mathbb{C}^n$. In fact by a theorem of Grauert, such locally free sheaves are free. Of course if we knew this, we could apply an appropriate version of the (elementary) Dolbeault-Grothendieck lemma to prove Theorems A and B. However, a direct proof that locally free sheaves over contractible domains in $\mathbb{C}^n$ are free is not easy. It is a main step in the original proof of H. Cartan (and then only for a restricted class of contractible domains in $\mathbb{C}^n$). A proof of the triviality of locally free sheaves over contractible domains in $\mathbb{C}^n$ which uses Theorem B for locally free sheaves may be found in Adams and Griffiths [1].

One merit of the partial differential equation techniques used in establishing the exactness of the $\bar{\partial}$ -sequence on a Stein manifold is that they give good estimates on the growth of solutions to $\bar{\partial}f = g$. The resulting "cohomology theory with bounds" has important applications to the theory of partial differential equations and is described in Hörmander [1].

Exercises.

1. Show that Theorem 7.2.20 is false if K is not $A(M)$ -convex.
2. Prove that if M is a complex manifold such that $H^1(M, \mathcal{I}) = 0$ for all coherent sheaves of ideals $\mathcal{I}$ of $\mathcal{O}_M$ then M is Stein (Hint: For the holomorphic convexity of M show that given any discrete subset $\{x_i : i \geq 1\}$ of M there exists $f \in A(M)$ such that $f(x_i) = i$, $i \geq 1$. See also Seminar number 20 by J. Serre in H. Cartan [2]).
3. Show that if M is an m -dimensional Stein manifold then $H^r(M, \mathbb{C}) = 0$, $r > m$ (Use Example 23, §1, Chapter 6).

4. Let M be a Stein manifold and suppose $H^2(M, \mathbb{Z}) = 0$. Show that any non-zero meromorphic function on M may be written in the form f/g , $f, g \in A(M)$ and $(f_x, g_x) = 1$, $x \in M$.
5. Let M be a Stein manifold and suppose m is a meromorphic function on M . Show that there exist $f, g \in A(M)$ such that $m = f/g$ (Hint: for $z \in M$, let $\widehat{\mathcal{F}}_z$ denote the ideal of $\mathcal{O}_z$ consisting of all germs g_z such that $g_z m_z \in \mathcal{F}_z$. Prove that the sheaf $\widehat{\mathcal{F}}$ is coherent and find a non-trivial section of $\widehat{\mathcal{F}}$). In Chapter 12 we give examples to show that we cannot generally require $(f_z, g_z) = 1$, $z \in M$.
6. Let M be a Stein manifold with $H^2(M, \mathbb{Z}) = 0$ and suppose that H is an analytic hypersurface in M . Prove that $M \setminus H$ is Stein.
7. Suppose that $\{x_i\}$ is a discrete subset of the Stein manifold M and that for each i we are given a Laurent series

$$L_i = \sum_{m=-N(i)}^{P(i)} a_m(z - x_i)^m, \quad 0 \leq P(i), N(i) < \infty.$$

Show that there exists a meromorphic function m on M such that for all i , $(m - L_i)$ is holomorphic on some neighbourhood of x_i and has a zero of order $P(i) + 1$ at z_i (see Exercise, §3, Chapter 1).

8. Let H be an analytic hypersurface in the Stein manifold M . Show that $\mathcal{O}_H$ has a resolution by locally free sheaves of the form

$$0 \rightarrow [H]^{-1} \rightarrow \mathcal{O}_M \rightarrow \mathcal{O}_H \rightarrow 0.$$

9. Show that a coherent sheaf on $\mathbb{C}^n$, $n > 1$, need not have a resolution by (locally) free sheaves of $\mathcal{O}_{\mathbb{C}^n}$ -modules (Hint: Given $z \in \mathbb{C}^n$ and $p \geq 1$, consider resolutions of the coherent sheaf $\mathcal{O}_{\mathbb{C}^n}/\mathcal{I}_{\{z\}}^p$).

3 The finiteness theorem of Cartan and Serre

The aim of this section is to prove that $\dim H^q(M, \mathcal{F}) < \infty$, $q \geq 0$, for all coherent sheaves $\mathcal{F}$ on a compact complex manifold M . We start by reviewing some definitions and results about Fréchet spaces. A general reference for Fréchet spaces is Rudin [1]. See also Appendix B in Gunning and Rossi [1].

Lemma 7.3.1. Finite direct sums and countable products of Fréchet spaces are Fréchet. If G is a closed subspace of the Fréchet space E then G and E/G are Fréchet.

Proof. We shall prove that E/G is Fréchet if G is a closed subspace of E and leave the remaining assertions as elementary exercises for the reader. Suppose that the topology on E is defined by the seminorms $|\cdot|_p$, $p \geq 1$. Let $q : E \rightarrow E/G$ denote the quotient map. For $p \geq 1$, define

$$|q(u)|'_p = \inf_{g \in G} |u - g|_p, \quad u \in E.$$

Then $|\cdot|'_p$ are seminorms on E/G defining the quotient topology on E/G . Since G is a closed subspace of E , E/G is Hausdorff and it is straightforward to verify that the seminorms $|\cdot|'_p$ define the structure of a Fréchet space on E/G . $\square$

Theorem 7.3.2 (Open mapping theorem). Let $A : E \rightarrow F$ be a continuous surjective linear map between Fréchet spaces. Then A is open. In particular, if A is a continuous linear bijection, A is a homeomorphism.

Proof. Rudin [1, Corollary 2.12], Gunning and Rossi [1, Appendix B]. $\square$

Theorem 7.3.3. A locally compact Fréchet space is finite dimensional.

Proof. Rudin [1, Theorem 1.22], Gunning and Rossi [1, Appendix B]. $\square$

Definition 7.3.4. Let $A : E \rightarrow F$ be a continuous linear map between Fréchet spaces. We say A is *compact* if there exists an open neighbourhood V of 0 in E such that $A(V)$ is relatively compact.

Theorem 7.3.5 (L. Schwartz). Let $A, B : E \rightarrow F$ be continuous linear maps between Fréchet spaces and suppose that A is surjective and B is compact. Then $\text{Im}(A + B)$ is a closed subspace of F and $F/\text{Im}(A + B)$ is finite dimensional.

Proof. Gunning and Rossi [1, Appendix B]. $\square$

Suppose that U is an open subset of the complex manifold M . We may give $\mathcal{O}(U)$ the structure of a Fréchet space by taking as seminorms

$$|f|_p = \|f\|_{K_p}, \quad f \in \mathcal{O}(U),$$

where $(K_p)_{p \geq 1}$ is any increasing family of compact subsets of U satisfying $\bigcup K_p = U$ and $K_p \subset \overset{\circ}{K}_{p+1}$, $p \geq 1$. Indeed, the topology defined by the seminorms is just the topology of uniform convergence on compact subsets of U . In particular, it is independent of the choice of sequence $(K_p)_{p \geq 1}$ and corresponding seminorms. It follows immediately from Lemma 7.3.1, that $\mathcal{O}(U)^p$ has the structure of a Fréchet space, $p \geq 1$.

Now suppose $\mathcal{K}$ is a coherent subsheaf of $\mathcal{O}_U^p$. Just as in the proof of Lemma 7.2.16, the closure of modules theorem implies that $\mathcal{K}(U)$ is a closed subspace of $\mathcal{O}(U)^p$. Hence, by Lemma 7.3.1, $\mathcal{K}(U)$ is Fréchet.

We shall say that an open relatively compact subset U of M is *C-admissible* if U is a Stein open subset of M and there exists another Stein open subset V of M such that $\bar{U} \subset V$.

Suppose that U is *C-admissible* and $\mathcal{F}$ is a coherent sheaf on M . By Theorem 7.2.6 and Proposition 7.2.4 there exist $s_1, \dots, s_k \in \mathcal{F}(U)$ which generate $\mathcal{F}(U)$ as an $\mathcal{O}(U)$ -module. Taking the cohomology sequence of the exact sequence $0 \rightarrow \mathcal{R}(s_1, \dots, s_k) \rightarrow \mathcal{O}_U^k \rightarrow \mathcal{F}_U \rightarrow 0$ and applying Theorem B, we arrive at the exact sequence

$$0 \rightarrow \mathcal{R}(s_1, \dots, s_k)(U) \rightarrow \mathcal{O}(U)^k \rightarrow \mathcal{F}(U) \rightarrow 0.$$

As noted above, $\mathcal{R}(s_1, \dots, s_k)$ is a closed subspace of $\mathcal{O}(U)^k$ and so, by Lemma 7.3.1, $\mathcal{F}(U)$ has the structure of a Fréchet space.

We have already defined a Fréchet topology on $\mathcal{F}(U)$, Theorem 7.2.18. Noting the definition of the seminorms on the quotient $\mathcal{F}(U) = \mathcal{O}^k(U)/\mathcal{R}(s_1, \dots, s_k)(U)$, it is clear that the two topologies coincide. In particular, the topology we have defined on $\mathcal{F}(U)$ is independent of the choice of generators $s_1, \dots, s_k$ (see the discussion in §2).

From now on, assume that $\mathcal{F}(U)$ is topologised as a Fréchet space for all *C-admissible* subsets U of M .

Lemma 7.3.6. Let U, V be *C-admissible* subsets of M with $V \subset U$. Then the restriction map $r_{VU} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is continuous.

Proof. Certainly, the restriction maps are continuous if $\mathcal{F} = \mathcal{O}^p$. Hence they are continuous if $\mathcal{F}$ is a coherent subsheaf of $\mathcal{O}_U^p$ and so, taking quotients, the result follows in general. $\square$

Suppose now that U is an arbitrary open subset of M . Since M has a basis of open sets consisting of C -admissible sets (for example, biholomorphic images of polydiscs), we may write

$$U = \bigcup_{j=1}^{\infty} U_j,$$

where the U_j are C -admissible open subsets of M . By Lemma 7.3.1, $\prod_{j=1}^{\infty} \mathcal{F}(U_j)$ has the structure of a Fréchet space. Let $Z \subset \prod_{j=1}^{\infty} \mathcal{F}(U_j)$ be the subset defined by $Z = \{(f_j) : f_j = f_k \text{ on } U_{jk}, j, k \geq 1\}$ and $K : \prod_{j=1}^{\infty} \mathcal{F}(U_j) \rightarrow \prod_{1 \leq j < k < \infty} \mathcal{F}(U_{jk})$ be the linear map defined by $K((f_j)) = ((f_j - f_k)|_{U_{jk}})$. By Lemma 7.3.6, K is continuous and therefore $Z = K^{-1}(0)$ is a closed subspace of $\prod_{j=1}^{\infty} \mathcal{F}(U_j)$. Hence by Lemma 7.3.1, Z is Fréchet. Now define $\chi : \mathcal{F}(U) \rightarrow \prod_{j=1}^{\infty} \mathcal{F}(U_j)$ by $\chi(f) = (f|_{U_j})$. We see that χ maps $\mathcal{F}(U)$ bijectively onto Z and so $\mathcal{F}(U)$ inherits the Fréchet space structure of Z . Moreover, the Fréchet space structure we have defined on $\mathcal{F}(U)$ is independent of the decomposition of U as a countable union of C -admissible sets. For suppose $U = \bigcup_{i=1}^{\infty} U_i^1 = \bigcup_{j=1}^{\infty} U_j^2$ are two such decompositions of U . Then $U = \bigcup_{i,j} U_i^1 \cap U_j^2$ is also a decomposition of U as C -admissible sets. Denote the corresponding closed subspaces of $\prod \mathcal{F}(U_i^1)$, $\prod \mathcal{F}(U_j^2)$, $\prod \mathcal{F}(U_i^1 \cap U_j^2)$ by Z_1 , Z_2 , Z_{12} respectively. We have the commutative diagram

$$\begin{array}{ccccc} & & \prod \mathcal{F}(U_i^1) & & \\ & \nearrow \chi_1 & & \searrow r_1 & \\ \mathcal{F}(U) & & & & \prod \mathcal{F}(U_i^1 \cap U_j^2) \\ & \searrow \chi_2 & & \nearrow r_2 & \\ & & \prod \mathcal{F}(U_j^2) & & \end{array}$$

where r_1 and r_2 are induced by restriction and are therefore continuous by Lemma 7.3.6. Now r_1 and r_2 restrict to continuous bijections of Z_1 and Z_2 on Z_{12} . Therefore by the open mapping theorem (Theorem 7.3.2), Z_1 is homeomorphic to Z_{12} which in turn is homeomorphic to Z_2 .

From now on assume that $\mathcal{F}(U)$ is topologised as a Fréchet space according to the recipe above for all open subsets U of M .

Suppose that $\mathcal{G}$ is another coherent sheaf on M and $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is a homomorphism. We claim that for all open subsets U of M the induced map $\phi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is continuous. By our definition of the topologies on $\mathcal{F}(U)$, $\mathcal{G}(U)$ it is clearly enough to verify that ϕ_U is continuous in case U is C -admissible. If U is C -admissible, there exist exact sequences $\mathcal{O}(U)^p \xrightarrow{s} \mathcal{F}(U) \rightarrow 0$, $\mathcal{O}(U)^q \xrightarrow{t} \mathcal{G}(U) \rightarrow 0$. Take the standard bases $\{E_1, \dots, E_p\}$, $\{E_1, \dots, E_q\}$ of $\mathcal{O}(U)^p$, $\mathcal{O}(U)^q$ (see §1). For $1 \leq j \leq p$, there exist $r_{ij} \in \mathcal{O}(U)$ such that

$$\phi_U(s(E_j)) = \sum_{i=1}^q r_{ij} t(E_i).$$

Defining $\psi_U : \mathcal{O}(U)^p \rightarrow \mathcal{O}(U)^q$ by the matrix $[r_{ij}]$ we obtain the commutative diagram

$$\begin{array}{ccccc} \mathcal{O}(U)^p & \xrightarrow{s} & \mathcal{F}(U) & \longrightarrow & 0 \\ \psi_U \downarrow & & \downarrow \phi_U & & \\ \mathcal{O}(U)^q & \xrightarrow{t} & \mathcal{G}(U) & \longrightarrow & 0. \end{array}$$

Now ψ_U is obviously continuous and so, since the quotient maps s and t are continuous, it follows from the Open Mapping theorem that ϕ_U is continuous.

Definition 7.3.7. A sheaf $\mathcal{F}$ of $\mathcal{O}$ -modules on a complex manifold M is said to be a *Fréchet sheaf* if

- a) For each open subset U of M , $\mathcal{F}(U)$ is a Fréchet space.
- b) The restriction maps $r_{VU} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$, $V \subset U$, are all continuous.

Theorem 7.3.8. Let M be a complex manifold. Then there is a unique way of giving every coherent sheaf defined over an open subset of M the structure of a Fréchet sheaf satisfying:

- a) If U is an open subset of M and $\mathcal{F}$ is a coherent subsheaf of $\mathcal{O}_U^p$, then $\mathcal{F}(U) \subset \mathcal{O}(U)^p$ has the topology of uniform convergence on compact subsets.
- b) If $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is a homomorphism of coherent sheaves, defined over some open subset V of M , then the induced maps $\phi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ are continuous for every open subset $U \subset V$.

Proof. The existence part of the theorem follows from the discussion preceding Definition 7.3.7. For the uniqueness it is easily seen that it is enough to verify that conditions a) and b) determine the Fréchet space structure on $\mathcal{F}(U)$ for U C -admissible. If U is C -admissible we have an exact sequence

$$0 \rightarrow \mathcal{R}(U) \xrightarrow{s_U} \mathcal{O}^p(U) \xrightarrow{t_U} \mathcal{F}(U) \rightarrow 0.$$

Condition b) implies that s_U , t_U are continuous. Condition a) determines the topology on $\mathcal{R}(U)$, $\mathcal{O}^p(U)$ and hence, by the open mapping theorem, the topology on $\mathcal{F}(U)$. $\square$

Remark. Our discussion of the Fréchet sheaf structure on coherent sheaves is based on Gunning and Rossi [1]. For much more extensive treatments of the topologisation of spaces of sections of coherent sheaves over complex manifolds and analytic sets see Gunning and Rossi [1] and Grauert and Remmert [1]. In the latter text a very nice characterisation of the Fréchet sheaf structure on coherent sheaves is given based on the Fréchet space structure on the stalks (see Exercise 4, §6, Chapter 3).

Lemma 7.3.9. Let U and V be open subsets of M and suppose $\bar{V} \subset U$, $\bar{V}$ compact. Then $r_{VU} : \mathcal{O}^p(U) \rightarrow \mathcal{O}^p(V)$ is compact.

Proof. Montel's theorem. $\square$

Lemma 7.3.10. Let $\mathcal{F}$ be a coherent sheaf on M and suppose that U, V are open subsets of M with $\bar{V} \subset U$, $\bar{V}$ compact. Then $r_{VU} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is compact.

Proof. First suppose U is C -admissible. We have a commutative diagram of continuous maps

$$\begin{array}{ccccc} \mathcal{O}^p(U) & \longrightarrow & \mathcal{F}(U) & \longrightarrow & 0 \\ r_{VU} \downarrow & & \downarrow r_{VU} & & \\ \mathcal{O}^p(V) & \longrightarrow & \mathcal{F}(V) & \longrightarrow & 0. \end{array}$$

Since $r_{VU} : \mathcal{O}^p(U) \rightarrow \mathcal{O}^p(V)$ is compact, Lemma 7.3.9, it follows that $r_{VU} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is compact.

For the general case, choose covers $\{U_j : j = 1, \dots, n\}$, $\{V_j : j = 1, \dots, n\}$ of V by C -admissible sets such that $\bar{V}_j \subset U_j$ is compact and $U_j \subset U$, $j = 1, \dots, n$. Then $r_{V_j U_j} : \mathcal{F}(U_j) \rightarrow \mathcal{F}(V_j)$ is compact and so therefore is

$$\prod_{j=1}^n r_{V_j U_j} : \prod_{j=1}^n \mathcal{F}(U_j) \rightarrow \prod_{j=1}^n \mathcal{F}(V_j).$$

Set $\tilde{U} = \bigcup_{j=1}^n U_j$. The map $r_{V\tilde{U}} : \mathcal{F}(\tilde{U}) \rightarrow \mathcal{F}(V)$ factors through $\prod r_{V_j U_j}$ and so is compact. Hence $r_{VU} = r_{V\tilde{U}} r_{\tilde{U}U}$ is compact. $\square$

Theorem 7.3.11. Let $\mathcal{F}$ be a coherent sheaf on the compact complex manifold M . Then $\dim_{\mathbb{C}} H^0(M, \mathcal{F}) < \infty$.

Proof. Take $U = V = M$ in Lemma 7.3.10 and apply Theorem 7.3.3. $\square$

Theorem 7.3.12 (Cartan-Serre). Let $\mathcal{F}$ be a coherent sheaf on the compact complex manifold M . Then $\dim_{\mathbb{C}} H^q(M, \mathcal{F}) < \infty$, $q \geq 0$.

Proof. Choose Leray covers $\mathcal{U} = \{U_1, \dots, U_n\}$, $\mathcal{U}' = \{U'_1, \dots, U'_n\}$ of M for $\mathcal{F}$ such that $\bar{U}'_j \subset U_j$, $j = 1, \dots, n$. For $p \geq 0$, let

$$C^p = \bigoplus \mathcal{F}(U_s),$$

where the (finite) direct sum is taken over all distinct $(p+1)$ -tuples $s = (s_0, \dots, s_p)$ of integers satisfying $1 \leq s_0, \dots, s_p \leq n$. By Lemma 7.3.1, C^p has the structure of a Fréchet space. Now $C^p(\mathcal{U}, \mathcal{F})$ clearly defines a closed subspace of C^p (remember that $C^p(\mathcal{U}, \mathcal{F})$ is the space of *alternating* cochains). Hence, $C^p(\mathcal{U}, \mathcal{F})$ has the structure of a Fréchet space. Similarly, $C^p(\mathcal{U}', \mathcal{F})$ has the structure of a Fréchet space. Let $R : C^p(\mathcal{U}, \mathcal{F}) \rightarrow C^p(\mathcal{U}', \mathcal{F})$ denote the restriction homomorphism. By our assumptions on the covers $\mathcal{U}$, $\mathcal{U}'$ and Lemma 7.3.10, R is compact. Since the covers $\mathcal{U}$, $\mathcal{U}'$ are Leray for $\mathcal{F}$, the natural map

$$Z^p(\mathcal{U}, \mathcal{F}) \rightarrow Z^p(\mathcal{U}', \mathcal{F}) / B^p(\mathcal{U}', \mathcal{F})$$

is surjective and so

$$C^{p-1}(\mathcal{U}', \mathcal{F}) \oplus Z^p(\mathcal{U}, \mathcal{F}) \xrightarrow{D \oplus R} Z^p(\mathcal{U}', \mathcal{F})$$

is surjective. But now take $A = D \oplus R$, $B = -0 \oplus R$ in Schwartz' finiteness theorem (Theorem 7.3.5) and we see that

$$H^p(M, \mathcal{F}) = H^p(\mathcal{U}', \mathcal{F}) = C^{p-1}(\mathcal{U}', \mathcal{F}) \oplus Z^p(\mathcal{U}, \mathcal{F}) / \text{Im}(D)$$

is finite dimensional. $\square$

Remarks.

1. We shall give an important application of the finiteness theorem in §5.
2. For generalisations of the finiteness theorem to coherent sheaves over compact analytic spaces see Gunning and Rossi [1] and Grauert and Remmert [1] (the original theorem of Cartan and Serre was proved for compact analytic spaces).

Exercises.

1. Show that the finiteness theorem of Cartan-Serre is an immediate consequence of Grauert's direct image theorem (look at constant maps).
- 2*. (Gunning [2], Grauert and Remmert [1]). Let (U, ϕ) be a chart on the compact complex manifold M such that ϕ maps U biholomorphically onto a polydisc in $\mathbb{C}^m$ ($m = \dim(M)$). Set $\mathcal{O}_h(U) = L^2(U)$ and note that $\mathcal{O}_h(U)$ has the structure of a Hilbert space (§6, Chapter 2; $\mathcal{O}_h(U)$ will depend on the chart map ϕ). Given a coherent sheaf $\mathcal{F}$ on M , show how to define the space $C_h^p(\mathcal{U}, \mathcal{F})$ of square integrable cochains on $\mathcal{U}$, where $\mathcal{U}$ will be a cover of M by open polydiscs, and prove that $C_h^p(\mathcal{U}, \mathcal{F})$ has the structure of a Hilbert space. Let

$H_h^p(\mathcal{U}, \mathcal{F})$ denote the corresponding cohomology group defined using square integrable cochains and prove that $\check{H}_h^p(M, \mathcal{F}) \approx \check{H}^p(M, \mathcal{F})$, $p \geq 0$. Finally deduce the finiteness theorem of Cartan-Serre by using the elementary finiteness theorem of Schwartz for Hilbert spaces (we shall prove this finiteness theorem in the appendix to Chapter 10).

4 The finiteness theorem of Grauert

Suppose that M is a strictly Levi pseudoconvex (S.L.p) domain in $\widehat{M}$ and that $\mathcal{F}$ is a coherent sheaf on $\widehat{M}$. In this section we shall prove the theorem of Grauert that the cohomology groups $H^p(M, \mathcal{F})$ are finite dimensional $\mathbb{C}$ -vector spaces, $p \geq 1$. As in the proof of the finiteness theorem of Cartan-Serre, we shall make use of Schwartz' finiteness theorem. We use Grauert's finiteness theorem in §6 to give a proof of Kodaira's embedding theorem and again in Chapter 12 to construct real analytic embeddings.

Let $\| \cdot \|$ denote the standard Euclidean norm on $\mathbb{C}^n$ and $E(r)$ denote the open Euclidean disc centre 0, radius r in $\mathbb{C}^n$. Given an open subset U of $\mathbb{C}^n$ we let $C_B^2(U)$ denote the space of C^2 $\mathbb{R}$ -valued functions on U which, together with derivatives up to order 2, are bounded on U . Define a norm on $C_B^2(U)$ by

$$|\phi| = \sup_{x \in U} (\|\phi(x)\| + \|D^2\phi_x\|), \quad \phi \in C_B^2(U).$$

($\|D^2\phi_x\|$ denotes the polynomial norm of the bilinear map $D^2\phi_x$. That is, $\|D^2\phi_x\| = \sup_{\|v\|=1} \|D^2\phi_x(v^2)\|$ – see Dieudonné [1] or Field [1]). Recall from §10 of Chapter 5 that the Levi form $L(\phi)$ of a C^2 $\mathbb{R}$ -valued map ϕ is the Hermitian quadratic form $\partial\bar{\partial}\phi$ given in local coordinates by the matrix $[\frac{\partial^2\phi}{\partial z_i \partial \bar{z}_j}]$.

Lemma 7.4.1. Let ϕ be a C^2 $\mathbb{R}$ -valued function defined on some neighbourhood of 0 in $\mathbb{C}^n$. Suppose that for all $x \in \phi^{-1}(0)$, we have $d\phi(x) \neq 0$ and $L(\phi)(x)$ positive definite. Then there exists $r > 0$ and an open neighbourhood N of ϕ in $C_B^2(E(r))$ such that

1. For all $\psi \in N$ and $x \in \psi^{-1}(0)$, $d\psi(x) \neq 0$ and $L(\psi)(x)$ is positive definite.
2. For all $\psi \in N$, $E(s) \cap \{z : \psi(z) < 0\}$ is Stein, $0 < s \leq r$.

Proof. Choose $R > 0$ so that ϕ is defined on an open neighbourhood of $\overline{E(R)}$. Certainly $\phi|_{E(R)} \in C_B^2(E(R))$ and moreover there exist $C(\phi), M(\phi) > 0$ such that for all $z \in \phi^{-1}(0) \cap E(R)$ we have

$$\begin{aligned} \|d\phi(z)\| &> C(\phi) \\ L(\phi)(z)(v) &> M(\phi)\|v\|^2, \quad v \in \mathbb{C}^n. \end{aligned} \tag{*}$$

From now on assume that ϕ is defined on $E(R)$ and that (*) holds. By scalar valued Taylor's theorem, we have for $y \in \phi^{-1}(0)$ and $z \in E(R)$

$$\begin{aligned} \phi(z) &= 2 \operatorname{Re} \left(\sum_i \frac{\partial \phi}{\partial z_i}(y)(z_i - y_i) + \sum_{i,j} \frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j}(y)(z_i - y_i)(z_j - y_j) \right) \\ &\quad + L(\phi)(y)(z - y) + R_\phi(z, y), \end{aligned}$$

where $R_\phi(z, y) = \frac{1}{2}(D^2\phi_{y+\theta(z-y)} - D^2\phi_y)((z-y)^2)$ and $0 < \theta < 1$. Since ϕ is C^2 , there exists $r > 0$ such that $\|D^2\phi_y - D^2\phi_0\| \leq M(\phi)/2$ for all $y \in E(r)$. Consequently for $y \in \phi^{-1}(0) \cap E(r)$, $z \in E(r)$ we have

$$|R_\phi(z, y)| \leq \frac{M(\phi)}{2} \|z - y\|^2.$$

Hence

$$L(\phi)(y)(z - y) + R_\phi(z, y) > \frac{M(\phi)}{2} \|z - y\|^2, \quad y \in \phi^{-1}(0) \cap E(r), \quad z \in E(r). \quad (\#)$$

Estimates (*), (#) are open conditions in $C_B^2(E(r))$ and so there exists an open neighbourhood N of ϕ in $C_B^2(E(r))$, such that (*), (#) hold for all $\psi \in N$. We claim that our choice of r , N implies the remaining statement of the Lemma. Let $\psi \in N$. We must prove that $D_s(\psi) = E(s) \cap \{z : \psi(z) < 0\}$ is Stein, $0 < s \leq r$. For this it is enough to prove that $D_s(\psi)$ is holomorphically convex. Suppose that $\{y_n : n \geq 1\}$ is a discrete subset of $D_s(\psi)$ converging to the point $y \in \partial D_s(\psi)$. If $y \in \bar{E}(s)$, there certainly exists $f \in A(E(s))$ which is unbounded on $\{y_n\}$. So suppose $\psi(y) = 0$ and let

$$F_y(z) = \sum_i \frac{\partial \psi}{\partial z_i}(y)(z_i - y_i) + \sum_{i,j} \frac{\partial^2 \psi}{\partial z_i \partial z_j}(y)(z_i - y_i)(z_j - y_j).$$

The quadratic polynomial $F_y(z)$ is non-vanishing in $D_s(\psi)$. Indeed, if $F_y(z) = 0$ and $\psi(z) < 0$ we would have from the Taylor expansion of ψ at y that $L(\psi)(y)(z - y) + R_\psi(z, y) < 0$, violating (#). Observing that $F_y(y) = 0$, we see that $F_y^{-1} \in A(D_s(\psi))$ and is unbounded on any sequence of points of $D_s(\psi)$ converging to y . Hence $D_s(\psi)$ is holomorphically convex, $0 < s \leq r$. $\square$

Theorem 7.4.2 (Grauert [3]). Let M be an S.L.p. domain in $\widehat{M}$. Then for any coherent sheaf $\mathcal{F}$ on $\widehat{M}$ we have $\dim_{\mathbb{C}} H^p(M, \mathcal{F}) < \infty$, $p \geq 1$.

Proof. Let $\phi \in C_{\mathbb{R}}^2(M)$ define M . That is, we suppose $M = \{z \in \widehat{M} : \phi(z) < 0\}$, $d\phi \neq 0$ on ∂M and $L(\phi)$ is positive definite on ∂M (see §10, Chapter 5). Since ∂M is compact, we may find a finite open cover $\{U_i : i = 1, \dots, n\}$ of ∂M , biholomorphic maps $\gamma_i : U_i \rightarrow E(r_i)$ and open neighbourhoods N_i of $\phi\gamma_i^{-1} \in C_B^2(E(r_i))$ such that the conclusions of Lemma 7.4.1 hold for $\phi\gamma_i^{-1}$, r_i and N_i , $i = 1, \dots, n$. In particular, the open sets U_i (respectively $U_i \cap M$) will be Stein open subsets of $\widehat{M}$ (respectively M). Choose $0 < s_i < r_i$ so that $\{V_i = \gamma_i^{-1}(E(s_i)) : i = 1, \dots, n\}$ is an open cover of ∂M . By the openness assertions of Lemma 7.4.1, we may clearly inductively choose positive functions $\eta_i \in C_c^\infty(U_i)$, $1 \leq i \leq n$, such that

1. $(\phi - \sum_{i=1}^r \eta_i)\gamma_j^{-1} \in N_j$, $r, j = 1, \dots, n$.
2. η_i is strictly positive on $V_i \cap \partial M$.

Set $\phi_0 = \phi$, $\phi_j = \phi - \sum_{i=1}^j \eta_i$, $1 \leq j \leq n$. Observe that, $d\phi_j \neq 0$ on $\phi_j^{-1}(0)$ and $L(\phi_j)$ is positive definite on $\phi_j^{-1}(0)$, $0 \leq j \leq n$. Set $M_0 = M$, $M_j = \{z : \phi_j(z) < 0\}$. We see that

$$M = M_0 \subset M_1 \subset \dots \subset M_n \subset \bigcup_{i=1}^n U_i.$$

Moreover, M is relatively compact in M_n since ϕ_n is strictly negative on ∂M .

Step 1. The natural restriction map $H^p(M_{j+1}, \mathcal{F}) \rightarrow H^p(M_j, \mathcal{F})$ is surjective, $p \geq 1$.

Fix j , $0 \leq j < n$. For $1 \leq i \leq n$, define

$$\begin{aligned} W_i &= U_i \cap M_j, \\ W'_i &= U_i \cap M_j, \quad i \neq j+1 \\ &= U_{j+1} \cap M_{j+1}, \quad i = j+1. \end{aligned}$$

Observe that $W_i = W'_i$ unless $i = j+1$ and that, by Lemma 7.4.1, W_i, W'_i are Stein open subsets of $\widehat{M}$. Adjoin Stein open subsets $W_{n+1} = W'_{n+1}, \dots, W_p = W'_p$ of M so that $\mathcal{W} = \{W_1, \dots, W_p\}$, $\mathcal{W}' = \{W'_1, \dots, W'_p\}$ are open covers of M_j, M_{j+1} respectively.

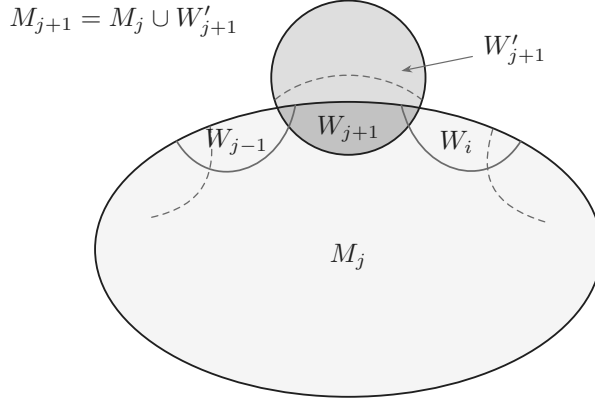


Figure II.1

The covers $\mathcal{W}$, $\mathcal{W}'$ are Leray covers of M_j , M_{j+1} for $\mathcal{F}$. Observe that any p -fold intersection of distinct elements of $\mathcal{W}$ is equal to a p -fold intersection of elements of $\mathcal{W}'$ provided only that $p > 1$. Hence

$$Z^p(\mathcal{W}, \mathcal{F}) \approx Z^p(\mathcal{W}', \mathcal{F}), \quad p \geq 1.$$

Therefore the natural restriction map $H^p(\mathcal{W}', \mathcal{F}) \rightarrow H^p(\mathcal{W}, \mathcal{F})$ is surjective. Step 1 now follows by Leray's theorem.

Step 2. For $1 \leq j \leq n$, set $W_j = V_j \cap M$, $W'_j = U_j \cap M_n$ and observe that W_j is a relatively compact subset of W'_j . Choose Stein open subsets $W_{n+1}, W'_{n+1}, \dots, W_p, W'_p$ of M so that W_j is a relatively compact subset of W'_j , $n+1 \leq j \leq p$, and $\mathcal{W} = \{W_1, \dots, W_p\}$, $\mathcal{W}' = \{W'_1, \dots, W'_p\}$ are open covers of M , M_n respectively. Just as in the proof of the finiteness theorem of Cartan-Serre the restriction map

$$R : Z^p(\mathcal{W}', \mathcal{F}) \rightarrow Z^p(\mathcal{W}, \mathcal{F})$$

is a compact operator between Fréchet spaces. From Step 1, the restriction map $H^p(\mathcal{W}', \mathcal{F}) \rightarrow H^p(\mathcal{W}, \mathcal{F})$ is surjective, $p \geq 1$, and so, for $p \geq 1$, the map

$$D \oplus R : C^{p-1}(\mathcal{W}, \mathcal{F}) \oplus Z^p(\mathcal{W}', \mathcal{F}) \rightarrow Z^p(\mathcal{W}, \mathcal{F})$$

is a surjective map between Fréchet spaces. Taking $A = D \oplus R$, $B = -0 \oplus R$ in Schwartz' finiteness theorem we deduce the finiteness of $H^p(\mathcal{W}, \mathcal{F})$, $p \geq 1$. $\square$

Theorem 7.4.3 (Grauert [3]). An S.L.p. domain is holomorphically convex.

Proof. Let M be an S.L.p. domain in $\widehat{M}$. It is enough to show that given $p \in \partial M$, there exists $f \in A(M)$ which is unbounded on any sequence of points of M converging to p . The method described in Exercise 2, §2 will not work here as the ideal sheaf of an infinite discrete subset of M does not extend to a coherent sheaf on $\widehat{M}$. Our proof follows that in R. Narasimhan [3]. As in the proof of Theorem 7.4.2 we suppose M is defined by the C^2 function ϕ . By Lemma 7.4.1, we may choose an open neighbourhood U of p in M , biholomorphic map $\gamma : U \rightarrow E(r)$ and neighbourhood N of $\phi\gamma^{-1}$ in $C_B^2(E(r))$ such that for all $\psi\gamma^{-1} \in N$, $D_r(\psi) = \{z : \psi(z) < 0\}$ is Stein. Choose $\eta \in C_c^\infty(U)$ such that η is positive, $\eta(p) > 0$ and $(\phi - \eta)\gamma^{-1} \in N$. Let $\widetilde{M} = \{z \in \widehat{M} : \phi(z) - \eta(z) < 0\}$. Certainly $\widetilde{M} \supset M$. As in the proof of Lemma 7.4.1, there exists $F \in A(U)$ such that $F(p) = 0$ and F is non-zero in $D_r(\phi)$. Take the open cover $\mathcal{U} = \{U \cap \widetilde{M}, \widetilde{M} \setminus F^{-1}(0)\}$ of $\widetilde{M}$. Since the natural map $H^1(\mathcal{U}, \mathcal{O}_{\widetilde{M}}) \rightarrow H^1(\widetilde{M}, \mathcal{O}_{\widetilde{M}})$ is injective (Exercise 4, §3, Chapter 6), and $\widetilde{M}$ is S.L.p., we have $\dim_{\mathbb{C}} H^1(\mathcal{U}, \mathcal{O}_{\widetilde{M}}) < \infty$. Let L denote the infinite dimensional linear subspace of $Z^1(\mathcal{U}, \mathcal{O}_{\widetilde{M}})$ defined by

$$L = \left\{ \sum_{j=1}^{\infty} c_j F^{-j} : c_j \in \mathbb{C} \text{ and all but finitely many } c_j \text{'s are zero} \right\}.$$

Since $\dim_{\mathbb{C}} H^1(\mathcal{U}, \mathcal{O}_{\widetilde{M}}) < \infty$, there exist elements of L which are boundaries. Therefore we may find a combination

$$G = \sum_{j=1}^n g_j F^{-j} \in L,$$

with not all the g_j 's vanishing, such that $G = D(H)$, for some $H \in C^0(\mathcal{U}, \mathcal{O}_{\widetilde{M}})$. Hence there exist $H_0 \in \mathcal{O}(U \cap \widetilde{M})$, $H_1 \in \mathcal{O}(\widetilde{M} \setminus F^{-1}(0))$ with $G = H_1 - H_0$ on $(U \cap \widetilde{M}) \setminus F^{-1}(0)$. That is, we have

$$H_1 = H_0 + \sum_{j=1}^n g_j F^{-1} \quad \text{on } (U \cap \widetilde{M}) \setminus F^{-1}(0).$$

Clearly $f = H_1|_M \in A(M)$ is unbounded on every sequence of points of M converging to p . $\square$

We may now give a solution to Levi's problem.

Theorem 7.4.4. An S.L.p. domain of a Stein manifold is Stein.

Proof. Immediate from Theorem 7.4.3. $\square$

Remarks.

1. Suppose that Ω is a Levi pseudoconvex domain in $\mathbb{C}^n$ which is not S.L.p. Then it can easily be proved that Ω is the union of an increasing family of S.L.p. domains (see Oka [1] or Gunning and Rossi [1, Lemma 2, Section D, Chapter 10]). Hence, by Theorem 7.4.4, Ω is the limit of an increasing family of domains of holomorphy and so, by a theorem of Behnke and Stein [1], a domain of holomorphy (see also Gunning and Rossi [1, section D, Chapter 10]). This result was first obtained by Oka in case $n = 2$ and then for general n independently by Oka [1], Bremermann [1] and Norguet [1]. Their result generalises to the case when Ω is a Levi pseudoconvex domain of a Riemann domain spread over $\mathbb{C}^n$ (see Gunning and Rossi [1, section D, Chapter 10]). Moreover, it is not necessary to assume that the boundary of Ω is smooth. All that is required is that the function $-\log(d(z, \partial\Omega))$ is plurisubharmonic in Ω (see the above references and also Hörmander [1]). It is not generally true that a Levi pseudoconvex domain of an arbitrary complex manifold is holomorphically convex. For an example of a non-holomorphically convex Levi pseudoconvex domain see Grauert [4] and also the survey article by Siu [1], especially §7.
2. R. Narasimhan [4] has shown that if M is a complex manifold then $\dim H^p(M, \mathcal{F}) < \infty$, $p \geq 1$, for all coherent sheaves $\mathcal{F}$ on M if and only if M is holomorphically convex. The proof that finiteness of cohomology implies holomorphic convexity is similar to that indicated in Exercise 2, §2 and makes use of the observation that the space of bounded infinite sequences of complex numbers is of infinite codimension in the space of all infinite complex sequences. The converse is much deeper and uses Grauert's direct image theorem together with results of H. Cartan [3] and Remmert [1] to the effect that if M is holomorphically convex then M possesses a maximal non-trivial compact analytic subset, A . Narasimhan proves that the inclusion $A \rightarrow M$ induces isomorphisms $H^p(A, \mathcal{F}_A) \rightarrow H^p(M, \mathcal{F})$, $p \geq 1$. The finiteness theorem of Cartan-Serre for compact analytic spaces then gives the result. In particular, M will be Stein if and only if M has no non-trivial compact analytic subsets. See Rossi [2] for a discussion of the case of S.L.p. domains.
3. Suppose M is a strictly pseudoconvex manifold. That is, there exists a C^∞ function $\phi : M \rightarrow \mathbb{R}$ such that $L(\phi)$ is everywhere positive definite and $M_a = \{z \in M : \phi(z) < a\}$ is relatively compact for all $a \in \mathbb{R}$. By Sard's theorem, M_a is S.L.p. for a dense set

$\Sigma \subset \mathbb{R}$ of values of a . Now it can be shown (see Gunning and Rossi [1, Section C, Chapter 10]) that plurisubharmonic functions satisfy a maximal principle and so M cannot have any non-trivial compact analytic subsets. It follows from Remark 2 that M_a is Stein, $a \in \Sigma$. Thus we have expressed M as a union of an increasing family of Stein manifolds parametrized by points in a dense subset of $\mathbb{R}$. It is shown in Docquier and Grauert [1] that this is enough to prove M Stein. See also Siu [1] and note that it is not generally true that an increasing union of Stein open sets, parametrized by the positive integers, need be Stein. Of course the union will be Stein if all the sets are domains in $\mathbb{C}^n$ or a Riemann domain (see Remark 1). If they are all subdomains of a Stein manifold it is not yet known whether their union must be Stein (see also Markoe [1]). We shall prove in Chapter 11 that every strictly pseudoconvex manifold is Stein. Our proof will depend on the existence theory for the $\bar{\partial}$ -operator and follows the approach of Kohn [1], Andreotti-Vesentini [1], Hörmander [1] and Vesentini [1].

Exercise. Let M be an S.L.p. domain in the complex manifold $\widehat{M}$ and suppose that M has C^2 defining function ϕ . Show

- a) There exists $C > 0$ such that $M_c = \{x \in M : \phi(x) < -c\}$ is S.L.p. for $0 \leq c < C$.
- b)* If M has no non-trivial compact analytic subvarieties then M_c is Stein, $0 < c < C$ (See Rossi [1] and note that we do not need to assume a maximal principle for strictly psh functions).

5 Coherent sheaves on projective space

In this section we shall prove theorems A and B of Serre for coherent sheaves on projective space. These fundamental theorems play a similar rôle in the theory of projective varieties to that of Cartan's theorems A and B in Stein manifold theory.

Throughout this section $\mathcal{U} = \{U_i\}$ will denote the standard open cover of $\mathbb{P}^n(\mathbb{C})$ by the open sets $U_i = \{(z_0, \dots, z_n) : z_i \neq 0\}$, $0 \leq i \leq n$. Suppose that $\mathcal{F}$ is a coherent sheaf on $\mathbb{P}^n(\mathbb{C})$. Since each U_i is biholomorphic to $\mathbb{C}^n$ and is therefore Stein, $\mathcal{U}$ is a Leray cover of $\mathbb{P}^n(\mathbb{C})$ for $\mathcal{F}$. It follows immediately from Leray's theorem that

$$H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{F}) = 0, \quad p > n.$$

Let H denote the hyperplane section bundle of $\mathbb{P}^n(\mathbb{C})$. Relative to the cover $\mathcal{U}$, H has transition functions $\phi_{ij} = z_j/z_i$. For $m \in \mathbb{Z}$, we let H^m denote the holomorphic line bundle with transition functions $(z_j/z_i)^m$ on U_{ij} . We may regard H^m as the line bundle associated to the divisor $z_0^m = 0$. That is, if we let $\mathbb{P}^{n-1}(\mathbb{C}) \subset \mathbb{P}^n(\mathbb{C})$ denote the hyperplane $z_0 = 0$, we have $H^m = [m \cdot \mathbb{P}^{n-1}(\mathbb{C})]$. With this convention, H^m has the "canonical" section s^m given locally by $s_i^m = (z_0/z_i)^m$ and $\text{div}(s^m) = m \cdot \mathbb{P}^{n-1}(\mathbb{C})$.

As is conventional, we let $\mathcal{O}(m)$ denote the sheaf of germs of holomorphic sections of H^m , $m \in \mathbb{Z}$.

Lemma 7.5.1. Let Σ be a hyperplane in $\mathbb{P}^n(\mathbb{C})$. For $m \in \mathbb{Z}$, we have a non-zero $\mathcal{O}$ -morphism

$$\phi_\Sigma : \mathcal{O}(m) \rightarrow \mathcal{O}(m+1)$$

satisfying $\phi_{\Sigma,z} = 0$ if and only if $z \in \Sigma$. In particular, ϕ_Σ restricts to an isomorphism of $\mathcal{O}(m)$ with $\mathcal{O}(m+1)$ over $\mathbb{P}^n(\mathbb{C}) \setminus \Sigma$.

Proof. Let Σ have equation $s(z_0, \dots, z_n) = 0$. Define $\phi_\Sigma(f) = s_z f$, $f \in \mathcal{O}(m)_z$. □

Remark. The morphism ϕ_Σ is given locally by $\phi_\Sigma(f_i) = f_i s / z_i$, $f_i \in \mathcal{O}(m)(U_i)$. Let $P^{(m)}(\mathbb{C}^{n+1})$ denote the space of homogeneous polynomials of degree m on $\mathbb{C}^{n+1}$. Then (Proposition 5.9.2),

$$\begin{aligned} H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{O}(m)) &\approx P^{(m)}(\mathbb{C}^{n+1}), \quad m \geq 0 \\ &= 0, \quad m < 0. \end{aligned}$$

Theorem 7.5.2.

1. For $m \geq 0$, we have

$$\begin{aligned} H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{O}(m)) &= 0, \quad p \neq 0 \\ &\approx P^{(m)}(\mathbb{C}^{n+1}), \quad p = 0. \end{aligned}$$

2. For $m < 0$, we have

$$\begin{aligned} H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{O}(m)) &= 0, \quad p \neq n \\ &\approx P^{(-m-n-1)}(\mathbb{C}^{n+1}), \quad p = n. \end{aligned}$$

Proof. Since we have already covered the cases $p = 0$, $p > n$, we shall assume from now on that $1 \leq p \leq n$. Let us start by considering the case $m = 0$. Suppose $c \in Z^p(\mathcal{U}, \mathcal{O})$. Then $c = \{c(s) = c(s_0, \dots, s_p)\}$, where $c(s) : U \rightarrow \mathbb{C}$ is holomorphic and $s = (s_0, \dots, s_p)$ is a $(p+1)$ -tuple of (distinct) integers lying between 0 and n . We may regard each $c(s)$ as a holomorphic function on $U_s \subset \mathbb{C}^{n+1}$ which is homogeneous of degree zero. By Theorem 2.1.10, we may take Laurent expansions of the $c(s)$. Thus

$$\begin{aligned} c(s)(z_0, \dots, z_n) &= \sum_{r_0 + \dots + r_n = 0} c(s)_{r_0 \dots r_n} z_0^{r_0} \dots z_n^{r_n} \\ &= \sum_{|r|=0} c(s)_r z^r, \quad \text{using multi-index notation.} \end{aligned}$$

Observe that the coefficient $c(s)_{r_0 \dots r_n}$ will be zero if any $r_j < 0$ with $j \notin \{s_0, \dots, s_p\}$. Define

$$a(s_0, \dots, s_{p-1}) = \frac{1}{n-p+1} \sum' c(j, s_0, \dots, s_{p-1})^+,$$

where $\sum'$ denotes the sum over all $j \notin \{s_0, \dots, s_{p-1}\}$ and $c(j, s_0, \dots, s_{p-1})^+$ is the holomorphic function on $U_{s_0} \cap \dots \cap U_{s_{p-1}}$ with Laurent series

$$\sum_{\substack{|r|=0 \\ r_j \geq 0}} c(j, s_0, \dots, s_{p-1})_r z^r.$$

That is, we omit terms from the Laurent expansion of $c(j, s_0, \dots, s_{p-1})$ having $r_j < 0$. Our construction defines an element $a \in C^{p-1}(\mathcal{U}, \mathcal{O})$. Now

$$\begin{aligned} (Da)_{s_0 \dots s_p} &= \frac{1}{n-p+1} \left(\sum' (c(j, s_1, s_2, \dots, s_p) - \dots \pm c(j, s_0, \dots, s_{p-1}))^+ \right) \\ &= \frac{1}{n-p+1} \left(\sum' c(s_0, \dots, s_p) \right)^+, \quad \text{since } Dc = 0 \\ &= c(s_0, \dots, s_p). \end{aligned}$$

We have shown that $Z^p(\mathcal{U}, \mathcal{O}) = DC^{p-1}(\mathcal{U}, \mathcal{O})$ and so $H^p(\mathcal{U}, \mathcal{O}) = 0$, $p > 0$.

Exactly the same proof shows that if $m > 0$ then $H^p(\mathcal{U}, \mathcal{O}(m)) = 0$, $p > 0$. Indeed, the only difference is that a cochain will now be a collection of holomorphic functions which are

homogeneous of degree m . The same proof also works for $m < 0$ provided that $p < n$. We conclude by considering the case $p = n$, $m < 0$. Suppose $c \in Z^n(\mathcal{U}, \mathcal{O}(m))$. Then

$$c : U_{01\dots n} \rightarrow \mathbb{C}$$

is holomorphic and homogeneous of degree m . Thus

$$c(z_0, \dots, z_n) = \sum_{|r|=m} c_r z^r.$$

Write $c = C_0 + C_1$, where C_0 is the sum over all terms $c_{r_0\dots r_n} z_0^{r_0} \dots z_n^{r_n}$ where at least one index r_j is positive and C_1 is the sum over terms such that every index r_j is negative. Notice that $C_1 = 0$ if $-m \leq n$. As above it is easily seen that $C_0 = Da$ for some $(n-1)$ -cochain a . A simple Laurent series argument shows that a non-zero C_1 can never be a coboundary. So suppose $m \leq -n-1$ and set

$$E = \left\{ \sum_{|r|=m} c_r z^r : c_r \in \mathbb{C}, \text{ every index } r_j < 0 \right\}.$$

By what we have shown above $E \approx H^n(\mathcal{U}, \mathcal{O}(m))$. Given $c \in E$, we may write

$$c = (z_0 \dots z_n)^{-1} \sum_{\substack{r_0, \dots, r_n \leq 0 \\ r_0 + \dots + r_n = m + n + 1}} c_r z_0^{r_0+1} \dots z_n^{r_n+1}.$$

It follows that $H^n(\mathcal{U}, \mathcal{O}(m)) \approx P^{(-m-n-1)}(\mathbb{C}^{n+1})$ where the isomorphism is given explicitly by mapping $P \in P^{(-m-n-1)}(\mathbb{C})$ to

$$c(z_0, \dots, z_n) = (z_0 \dots z_n)^{-1} P(z_0^{-1}, \dots, z_n^{-1}).$$

□

Remark. For an alternative proof of Theorem 7.5.2, using non-trivial facts from the Hodge theory of Kähler manifolds, see Seminar 18 by Serre in H. Cartan [2].

Given a coherent sheaf $\mathcal{F}$ on $\mathbb{P}^n(\mathbb{C})$ we let

$$\mathcal{F}(m) = \mathcal{F} \otimes_{\mathcal{O}} \mathcal{O}(m), \quad m \in \mathbb{Z}.$$

We call $\mathcal{F}(m)$ the sheaf $\mathcal{F}$ “twisted by $\mathcal{O}(m)$ ”. One feature of twisting is that we expect $\dim_{\mathbb{C}} H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ to be an increasing function of m . To explain why this should be so, let us consider the case when $\mathcal{F}$ is the sheaf of sections of a holomorphic vector bundle E . We claim that $H^0(\mathbb{P}^n(\mathbb{C}), E(m)) \cong \{s \in \mathcal{M}(E) : \text{div}(s) + m \cdot \mathbb{P}^{n-1}(\mathbb{C}) \geq 0\}$. Certainly it follows from this isomorphism that $\dim_{\mathbb{C}} H^0(\mathbb{P}^n(\mathbb{C}), E(m))$ is an increasing function of m . Suppose that E has transition functions ϕ_{ab} relative to a cover $\mathcal{W}$ of $\mathbb{P}^n(\mathbb{C})$ where we suppose that $\mathcal{W}$ is a refinement of $\mathcal{U}$. Let $s \in \mathcal{M}^*(E)$ and $\text{div}(s) + m \cdot \mathbb{P}^{n-1}(\mathbb{C}) \geq 0$. If $s_a \in \mathcal{M}^*(W_a)$ is the local representative of s on $W_a \subset U_{i(a)}$, we have

$$(z_0/z_{i(a)})^m s_a \in A(W_a).$$

Clearly if $W_b \subset U_{i(b)}$, we have

$$\phi_{ab}(z_{i(b)}/z_{i(a)})^m (z_0/z_{i(b)})^m s_b = (z_0/z_{i(a)})^m s_a \quad \text{on } W_{ab}.$$

Therefore $\{(z_0/z_{i(a)})^m s_a\}$ defines a holomorphic section of $E(m)$. Reversing the argument shows that every holomorphic section of $E(m)$ gives rise to a meromorphic section of E such that $\text{div}(s) + m \cdot \mathbb{P}^{n-1}(\mathbb{C}) \geq 0$. Of course, our argument depended on being able to show that E admits at least one non-trivial meromorphic section.

Remark. We should point out that a holomorphic vector bundle E on $\mathbb{P}^n(\mathbb{C})$ always restricts to a holomorphically trivial bundle over U_i , $0 \leq i \leq n$, Serre [2]. This fact also follows from a general and difficult theorem of Grauert to the effect that a holomorphic vector bundle on a Stein manifold is holomorphically trivial if and only if it is topologically trivial. See also Adams and Griffiths [1] for a proof that holomorphic vector bundles over polydiscs in $\mathbb{C}^n$ are holomorphically trivial as well as references and discussion concerning Grauert's theorems.

Theorem 7.5.3 (Theorems A and B of Serre). Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n(\mathbb{C})$. Then there exists $m_0 = m_0(\mathcal{F}) \in \mathbb{Z}$ such that

A. For each $z \in \mathbb{P}^n(\mathbb{C})$, $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ generates $\mathcal{F}(m)_z$ as a $\mathcal{G}_z$ -module, $m \geq m_0$.

B. For $m \geq m_0$, $H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)) = 0$, $p \geq 1$.

Proof (Seminar 19 by Serre, H. Cartan [2]). We start by looking at some special cases. Suppose $\mathcal{F} \cong \mathcal{O}(q)$. Take $m_0 = -q$. Since $\mathcal{O}(q)(m) = \mathcal{O}(q+m)$, we see immediately from Theorem 7.5.2 that $H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)) = 0$, $p \geq 1$, $m \geq m_0$. Clearly for all $z \in \mathbb{P}^n(\mathbb{C})$, $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ generates $\mathcal{F}(m)_z$, $m \geq m_0$. Since cohomology commutes with direct sums, A and B hold whenever $\mathcal{F} \cong \bigoplus_{i=1}^k \mathcal{O}(a_i)$ and we may take $m_0 = -\min\{a_i\}$.

We prove the theorem in general by an induction on n . Let A_n, B_n denote statements A and B for dimension n . We shall show that A_{n-1} and B_{n-1} imply A_n and A_n implies B_n . The Theorem is, of course, trivial for $n = 0$.

Step 1. A_{n-1} and B_{n-1} imply A_n . Let $z \in \mathbb{P}^n(\mathbb{C})$ and choose any hyperplane $\Sigma \subset \mathbb{P}^n(\mathbb{C})$ not containing z . For $m \in \mathbb{Z}$, let

$$\tilde{\phi}_\Sigma = \text{id} \otimes \phi_\Sigma : \mathcal{F}(m) \longrightarrow \mathcal{F}(m+1),$$

where ϕ_Σ is the map given by Lemma 7.5.1. Observe that $\tilde{\phi}_\Sigma$ restricts to an isomorphism of $\mathcal{F}(m)$ with $\mathcal{F}(m+1)$ over $\mathbb{P}^n(\mathbb{C}) \setminus \Sigma$. Suppose $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m_0))$ generates $\mathcal{F}(m_0)_z$. Then $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ generates $\mathcal{F}(m)_z$ for $m \geq m_0$. Indeed, $\tilde{\phi}_\Sigma^{m-m_0} : \mathcal{F}(m_0) \rightarrow \mathcal{F}(m)$ restricts to an isomorphism over $\mathbb{P}^n(\mathbb{C}) \setminus \Sigma$ and so maps any set of generators for $\mathcal{F}(m_0)_z$ to a set of generators for $\mathcal{F}(m)_z$. Let $A_n(z)$ (respectively $A'_n(z)$) be the statement “There exists $m_0 = m_0(\mathcal{F}, z)$ such that $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m_0))$ (respectively $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$) generates $\mathcal{F}(m_0)_z$ (respectively $\mathcal{F}(m)_z$, $m \geq m_0$)”. By the coherence of $\mathcal{F}$, $A_n(z)$ implies $A_n(y)$ for y in some open neighbourhood of z . Since $A_n(y)$ implies $A'_n(y)$, it follows from the compactness of $\mathbb{P}^n(\mathbb{C})$ that $A_n(z)$ for all $z \in \mathbb{P}^n(\mathbb{C})$ implies A_n . Therefore it is enough to prove that A_{n-1} and B_{n-1} imply $A_n(z)$.

Without loss of generality suppose $z \in \mathbb{P}^{n-1}(\mathbb{C}) \subset \mathbb{P}^n(\mathbb{C})$. Set $\phi = \tilde{\phi}_{\mathbb{P}^{n-1}(\mathbb{C})} : \mathcal{F}(-1) \rightarrow \mathcal{F}$ and let

$$\begin{aligned} \mathcal{K} &= \text{Ker } \phi : \mathcal{F}(-1) \rightarrow \mathcal{F} \\ \mathcal{G} &= \text{Coker } \phi : \mathcal{F}(-1) \rightarrow \mathcal{F}. \end{aligned}$$

We have the exact sequence

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{F}(-1) \xrightarrow{\phi} \mathcal{F} \rightarrow \mathcal{G} \rightarrow 0.$$

Tensoring with $\mathcal{O}(m)$ we obtain the exact sequence

$$0 \rightarrow \mathcal{K}(m) \rightarrow \mathcal{K}(m-1) \xrightarrow{\phi_m} \mathcal{F}(m) \rightarrow \mathcal{G}(m) \rightarrow 0.$$

Since $\phi_m|_{U_0}$ is an isomorphism we see that $\mathcal{K}_{U_0} = \mathcal{G}_{U_0} = 0$. Let i denote the inclusion map of $\mathbb{P}^{n-1}(\mathbb{C})$ in $\mathbb{P}^n(\mathbb{C})$. For any sheaf $\mathcal{H}$ on $\mathbb{P}^n(\mathbb{C})$ we let $\mathcal{H}^* = i^{-1}\mathcal{H}$ denote the restriction of $\mathcal{H}$ to $\mathbb{P}^{n-1}(\mathbb{C})$. Observe that $\mathcal{I}_{\mathbb{P}^{n-1}(\mathbb{C})} \subset \mathcal{O}_{\mathbb{P}^n(\mathbb{C})}$ acts trivially on $\mathcal{K}$ and $\mathcal{G}$. Indeed if $s_z \in \mathcal{I}_{\mathbb{P}^{n-1}(\mathbb{C})}$, $f_z \in \mathcal{K}$, then $s_z f_z = 0$ since $\mathcal{K} = \text{Ker}(\phi)$. Similarly for $\mathcal{G}$ since $\mathcal{G} = \text{Coker}(\phi)$. Hence $\mathcal{K}^*, \mathcal{G}^*$ have the natural structure of $\mathcal{O}_{\mathbb{P}^{n-1}(\mathbb{C})}^* = \mathcal{O}_{\mathbb{P}^n(\mathbb{C})}^* / \mathcal{I}_{\mathbb{P}^{n-1}(\mathbb{C})}^*$ -modules and the reader may easily

verify that $\mathcal{K}^*, \mathcal{G}^*$ are coherent $\mathcal{O}_{\mathbb{P}^{n-1}(\mathbb{C})}$ -modules. Moreover, it is clear that $\mathcal{K}(m)^* \approx \mathcal{K}^*(m)$, $\mathcal{G}(m)^* \approx \mathcal{G}^*(m)$, $m \in \mathbb{Z}$.

By Exercise 12, §3, Chapter 6,

$$H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{K}(m)) = H^p(\mathbb{P}^{n-1}(\mathbb{C}), \mathcal{K}(m)^*), \quad p \geq 0,$$

similarly for $\mathcal{G}(m)$. Applying our inductive assumption to $\mathcal{K}^*, \mathcal{G}^*$, there exists $m_1 \in \mathbb{Z}$ such that for $p \geq 1$

$$\begin{aligned} H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{K}(m)) &= H^p(\mathbb{P}^{n-1}(\mathbb{C}), \mathcal{K}^*(m)) = 0, \quad m \geq m_1 \\ H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{G}(m)) &= H^p(\mathbb{P}^{n-1}(\mathbb{C}), \mathcal{G}^*(m)) = 0, \quad m \geq m_1. \end{aligned}$$

We have the short exact sequences

$$\begin{aligned} 0 \rightarrow \mathcal{K}(m) \rightarrow \mathcal{F}(m-1) \rightarrow \text{Im}(\phi_{m-1}) \rightarrow 0 \\ 0 \rightarrow \text{Im}(\phi_{m-1}) \rightarrow \mathcal{F}(m) \rightarrow \mathcal{G}(m) \rightarrow 0. \end{aligned}$$

Taking the cohomology sequences of these short exact sequences we obtain the exact sequences

$$\begin{aligned} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m-1)) \rightarrow H^1(\mathbb{P}^n(\mathbb{C}), \text{Im}(\phi_{m-1})) \rightarrow H^2(\mathbb{P}^n(\mathbb{C}), \mathcal{K}(m)) \\ H^1(\mathbb{P}^n(\mathbb{C}), \text{Im}(\phi_{m-1})) \rightarrow H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)) \rightarrow H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{G}(m)). \end{aligned}$$

Since $H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{G}(m)), H^2(\mathbb{P}^n(\mathbb{C}), \mathcal{K}(m)) = 0$, $m \geq m_1$, we see that

$$\begin{aligned} \dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m-1)) &\geq \dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \text{Im}(\phi_{m-1})) \\ &\geq \dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)), \end{aligned}$$

for $m \geq m_1$. Hence $\dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ is a decreasing function of m , $m \geq m_1$. But now by the finiteness theorem of Cartan–Serre, $\dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)) < \infty$, and so we deduce that $\dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ is independent of m for $m > m_2$, say. In particular, for $m \geq m_2$ we have

$$\begin{aligned} \dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m-1)) &= \dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \text{Im}(\phi_{m-1})) \\ &= \dim_{\mathbb{C}} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)). \end{aligned}$$

Since the surjective homomorphism $H^1(\mathbb{P}^n(\mathbb{C}), \text{Im}(\phi_{m-1})) \rightarrow H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ is between spaces of the same finite dimension, it must be injective, $m \geq m_2$. Therefore, from the cohomology sequence we see that $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)) \rightarrow H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{G}(m))$ is surjective, $m \geq m_2$. But $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{G}(m)) = H^0(\mathbb{P}^{n-1}(\mathbb{C}), \mathcal{G}^*(m))$ and so by A_{n-1} we see that $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{G}(m))$ generates $\mathcal{G}^*(m)_z = \mathcal{G}(m)_z$ for $m \geq m_0$, where we may suppose $m_0 \geq m_2$. In particular, $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ generates $\mathcal{G}(m)_z$ for $m \geq m_0$. We claim that for $m \geq m_0$, $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ generates $\mathcal{F}(m)_z$. Suppose $z \in U_i$, $i \neq 0$. Let $\psi : \mathcal{F} \rightarrow \mathcal{F}(1)$ be the homomorphism defined by $\psi(f) = z_i f$. As described above, ψ^{-m} restricts to an isomorphism of $\mathcal{F}(m)_{U_i}$ with $\mathcal{F}_{U_i}$. Under this identification of $\mathcal{F}(m)_{U_i}$ with $\mathcal{F}_{U_i}$, the map ϕ corresponds to multiplication by $t_0 = z_0/z_i$ and $\mathcal{G}(m)_z$ is therefore identified with $\mathcal{F}_z/t_0\mathcal{F}_z$. Let $M_z \subset \mathcal{F}_z$ be the submodule generated by the elements of $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$. Since $\mathcal{G}(m)_z$ is generated by the elements of $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$, $m \geq m_0$, the image of M_z in $\mathcal{F}_z/t_0\mathcal{F}_z$ is the whole of $\mathcal{F}_z/t_0\mathcal{F}_z$. That is, $\mathcal{F}_z = t_0\mathcal{F}_z + M_z$. Since $t_0 \in \mathfrak{m}_z \subset \mathcal{O}_z$, it follows from Nakayama's lemma that $\mathcal{F}_z = M_z$.

Step 2. A_n implies B_n .

For $p > n$, we have already shown that $H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)) = 0$ for all m . We prove B_n by decreasing induction on p . So suppose that for every coherent sheaf $\mathcal{F}$ on $\mathbb{P}^n(\mathbb{C})$ there exists $m_0 = m_0(\mathcal{F}, p)$ such that $H^{p+1}(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m)) = 0$, $m \geq m_0$. By A_n there exists $m \in \mathbb{Z}$ such that $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$ generates $\mathcal{F}(m)_z$ for all $z \in \mathbb{P}^n(\mathbb{C})$. Choose a basis $s_1, \dots, s_k$ of $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m))$. We have the exact sequence

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{O}^k \xrightarrow{s} \mathcal{F}(m) \rightarrow 0,$$

where $s = (s_1, \dots, s_k)$ and $\mathcal{K} = \text{Ker}(s)$. Tensoring by $\mathcal{O}(q)$ yields the exact sequence

$$0 \rightarrow \mathcal{K}(q) \rightarrow \mathcal{O}^k(q) \rightarrow \mathcal{F}(m+q) \rightarrow 0$$

and corresponding portion of the cohomology sequence

$$H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{O}^k(q)) \rightarrow H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m+q)) \rightarrow H^{p+1}(\mathbb{P}^n(\mathbb{C}), \mathcal{K}(q)).$$

By the special case of B described at the beginning of the proof, $H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{O}^k(q)) = 0$ for $p > 0$, $q \geq 0$. By our inductive assumption, $H^{p+1}(\mathbb{P}^n(\mathbb{C}), \mathcal{K}(q)) = 0$ for sufficiently large q . Hence $H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{F}(m+q)) = 0$ for sufficiently large q . This completes the inductive step and the proof of the theorem. $\square$

Remark. The reader should note the crucial rôle played by the finiteness theorem of Cartan–Serre in the proof of Theorem 7.5.3. In §6, we give another proof of Theorem 7.5.3 depending on Grauert’s finiteness theorem.

Now for some applications of Serre’s theorem.

Theorem 7.5.4. Every holomorphic vector bundle E on $\mathbb{P}^n(\mathbb{C})$ has a non-trivial meromorphic section.

Proof. By statement A of Theorem 7.5.3, there exists m_0 such that $H^0(\mathbb{P}^n(\mathbb{C}), \underline{E}(m_0))$ generates $\underline{E}(m_0)_z$ for all $z \in \mathbb{P}^n(\mathbb{C})$. In particular, $\dim_{\mathbb{C}} H^0(\mathbb{P}^n(\mathbb{C}), \underline{E}(m_0)) \geq 1$. But as we showed earlier,

$$H^0(\mathbb{P}^n(\mathbb{C}), \underline{E}(m_0)) \cong \{s \in \mathcal{M}(E) : \text{div}(s) + m_0 \cdot \mathbb{P}^{n-1}(\mathbb{C}) \geq 0\}.$$

$\square$

Remark. Of course, we can prove much more than that E has a non-trivial meromorphic section. Using statement B, we can show that for any $v \in E_z$, there exists $s \in \mathcal{M}^*(E)$ such that $s(z) = v$. See the exercises at the end of the section.

Theorem 7.5.5. (Chow’s theorem). Every analytic subset of $\mathbb{P}^n(\mathbb{C})$ is algebraic.

Proof. Let X be an analytic subset of $\mathbb{P}^n(\mathbb{C})$ with ideal sheaf $\mathcal{I}$. Then $\mathcal{I}$ is coherent subsheaf of $\mathcal{O}$ (see §1 for discussion and references). We prove: Given $z \in \mathbb{P}^n(\mathbb{C}) \setminus X$, there exists a homogeneous polynomial $p = p(z_0, \dots, z_n)$ which vanishes on X and does not vanish at x . Granted this, we consider the set of all homogeneous polynomials which vanish on X . The common zero locus of these polynomials is X and by the Hilbert basis theorem we may choose a finite subset of these polynomials with common zero locus X .

Suppose then that $\mathcal{I}_z$ denotes the ideal sheaf of $X \cup \{z\}$. For $m \in \mathbb{Z}$, we have the exact sequence

$$0 \rightarrow \mathcal{I}_z(m) \rightarrow \mathcal{I} \rightarrow \mathbb{C}(z)(m) \rightarrow 0,$$

where $\mathbb{C}(z)$ is the skyscraper sheaf supported at z . Note that $\mathbb{C}(z)(m) = \mathbb{C}(z)$. By B of Theorem 7.5.3, there exists m_0 such that $H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{I}_z(m)) = 0$, $m \geq m_0$. Taking the cohomology sequence of our short exact sequence we obtain for $m \geq m_0$ the exact sequence

$$H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{I}(m)) \rightarrow H^0(\mathbb{P}^n(\mathbb{C}), \mathbb{C}(z)(m)) \rightarrow 0.$$

Now $H^0(\mathbb{P}^n(\mathbb{C}), \mathbb{C}(z)(m)) \cong \mathbb{C}$ and so there exists $p \in H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{I}(m))$ such that $p(z) \neq 0$. But since $\mathcal{I} \subset \mathcal{O}$, it follows that $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{I}(m))$ is a subspace of $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{O}(m)) \approx P^{(m)}(\mathbb{C}^{n+1})$. Indeed, it is just the (non-empty!) subspace of homogeneous polynomials of degree m which vanish on X . $\square$

Remark. Chow's theorem is a special instance of a general relationship between global analytic and algebraic structures on projective space. This relationship is explained fully in Serre's G.A.G.A. paper, Serre [3]. In particular, Serre shows that there is an equivalence between coherent algebraic sheaves and coherent (analytic) sheaves on projective space. Chow's theorem is one corollary of this correspondence. Another is that every holomorphic vector bundle on projective space is "algebraic".

Theorem 7.5.6. Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n(\mathbb{C})$. Then $\mathcal{F}$ has a resolution

$$0 \rightarrow \underline{E}_n \rightarrow \underline{E}_{n-1} \rightarrow \cdots \rightarrow \underline{E}_0 \rightarrow \mathcal{F} \rightarrow 0$$

by locally free sheaves. We may require that for $0 \leq j < n$ the sheaf $\underline{E}_j$ is isomorphic to a direct sum $\mathcal{O}(a_j)^{k_j}$.

Proof. Just as in the proof of Step 2 of Theorem 7.5.3, statement A of Theorem 7.5.3 implies that there exists an exact sequence

$$0 \rightarrow \mathcal{K}_1 \rightarrow \mathcal{O}^{k_0} \rightarrow \mathcal{F}(m_0) \rightarrow 0.$$

Setting $\mathcal{F} = \mathcal{K}_0$ and iterating this construction we obtain for $0 \leq j \leq n$ exact sequences

$$0 \rightarrow \mathcal{K}_{j+1} \rightarrow \mathcal{O}^{k_j} \rightarrow \mathcal{K}_j(m_j) \rightarrow 0.$$

Tensoring each sequence by an appropriate power of $\mathcal{O}(1)$ we obtain exact sequences

$$\begin{aligned} 0 \rightarrow \mathcal{K}_{j+1}(-m_0 - \cdots - m_j) \rightarrow \mathcal{O}^{k_j}(-m_0 - \cdots - m_j) \\ \rightarrow \mathcal{K}_j(-m_0 - \cdots - m_{j-1}) \rightarrow 0 \end{aligned}$$

and hence a long exact sequence

$$0 \rightarrow \mathcal{K}_n(p_{n-1}) \rightarrow \mathcal{O}^{k_{n-1}}(p_{n-1}) \rightarrow \cdots \rightarrow \mathcal{O}^{k_0}(p_0) \rightarrow \mathcal{F} \rightarrow 0.$$

The coherence of $\mathcal{K}_n(p_{n-1})$ together with the Hilbert Syzygy theorem imply that $\mathcal{K}_n(p_{n-1})$ is locally free (see the proof of Corollary 7.1.8). $\square$

Remarks.

1. In the exercises at the end of the section we show how, in certain cases, we can construct explicitly a resolution of a coherent sheaf on $\mathbb{P}^n(\mathbb{C})$ by locally free sheaves.
2. To appreciate the significance of Theorem 7.5.6 we first need to discuss *Serre's duality theorem*. In Chapter 10 we shall prove Serre's duality theorem for complex manifolds: Let M be a compact complex manifold of dimension m and E be a holomorphic vector bundle on M . Then $H^p(M, \Omega^q(\underline{E}))$ is isomorphic (not canonically) to $H^{m-p}(M, \Omega^{m-q}(\underline{E}^*))$, $p, q \geq 0$. In particular, if $q = 0$, $H^p(M, \underline{E}) \cong H^{m-p}(M, \underline{K} \otimes \underline{E}^*)$, where $\underline{K}$ denotes the canonical bundle of M . Serre duality plays an important rôle in complex analysis. In particular, it gives an easy proof of the Riemann–Roch theorem for compact Riemann surfaces (see Chapter 10). Suppose that $M = \mathbb{P}^n(\mathbb{C})$. Then the canonical bundle of $\mathbb{P}^n(\mathbb{C})$ is canonically isomorphic to H^{-n-1} (§9, Chapter 5). Theorem 7.5.2 implies immediately that

$$H^p(\mathbb{P}^n(\mathbb{C}), \mathcal{O}(m)) \approx H^{n-p}(\mathbb{P}^n(\mathbb{C}), \mathcal{O}(-m) \otimes \underline{K}), \quad p \geq 0, \quad m \in \mathbb{Z}.$$

A simple computation shows that we have a (canonical) Serre duality for any sheaf on $\mathbb{P}^n(\mathbb{C})$ which is a direct sum of sheaves $\mathcal{O}(m)$. Theorem 7.5.6, together with some basic facts from homological algebra, now allows us to verify Serre duality for any locally free

sheaf on $\mathbb{P}^n(\mathbb{C})$. Even more, we may prove a duality theorem for arbitrary coherent sheaves on $\mathbb{P}^n(\mathbb{C})$. However, the formulation of this duality theorem will now involve “Ext” groups. These matters are explained further in Griffiths and Harris [1] and Hartshorne [1, 2]. We should also mention that Ramis and Ruget [1] and Ramis, Ruget and Verdier [1] have developed a general duality theorem applicable to analytic spaces and proper analytic maps.

For the remainder of this section we make a preliminary study of holomorphic vector bundles on projective space.

Proposition 7.5.7. The group of holomorphic line bundles on $\mathbb{P}^n(\mathbb{C})$ is isomorphic to the infinite cyclic group generated by the hyperplane section bundle H .

Proof. First note that the group generated by H in $\text{HLB}(\mathbb{P}^n(\mathbb{C}))$ is infinite cyclic. Indeed, H^p is isomorphic to $\mathbb{C}$ if and only if $H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{O}(p)) \cong \mathbb{C}$ and by Theorem 7.5.2 this happens if and only if $p = 0$. It follows from the cohomology sequence of $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \rightarrow \mathcal{O}^* \rightarrow 0$ and the vanishing of $H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{O})$, $H^2(\mathbb{P}^n(\mathbb{C}), \mathcal{O})$ that

$$H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{O}^*) \cong H^2(\mathbb{P}^n(\mathbb{C}), \mathbb{Z}),$$

where the isomorphism is given by c_1 , 1st. Chern class map. By topology, $H^2(\mathbb{P}^n(\mathbb{C}), \mathbb{Z}) \cong \mathbb{Z}$, $n \geq 1$. Let $\mathbb{P}^1(\mathbb{C}) \subset \mathbb{P}^n(\mathbb{C})$. We have the commutative diagram

$$\begin{array}{ccc} H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{O}^*) & \xrightarrow{c_1} & H^2(\mathbb{P}^n(\mathbb{C}), \mathbb{Z}) \cong \mathbb{Z} \\ \downarrow r & & \downarrow \\ H^1(\mathbb{P}^1(\mathbb{C}), \mathcal{O}^*) & \xrightarrow{c_1} & H^2(\mathbb{P}^1(\mathbb{C}), \mathbb{Z}) \cong \mathbb{Z} \end{array}$$

where the vertical maps are induced by inclusion. Now we know from Example 13, §3, Chapter 6 that H generates $H^1(\mathbb{P}^1(\mathbb{C}), \mathcal{O}^*)$. But r maps the hyperplane section bundle of $\mathbb{P}^n(\mathbb{C})$ to the hyperplane section bundle of $\mathbb{P}^1(\mathbb{C})$ and so $c_1 r(H)$ generates $H^2(\mathbb{P}^1(\mathbb{C}), \mathbb{Z})$. By the commutativity of the diagram it follows that H generates $H^1(\mathbb{P}^n(\mathbb{C}), \mathcal{O}^*)$. $\square$

We now work towards a classification of holomorphic vector bundles on $\mathbb{P}^1(\mathbb{C})$. Suppose $L \in \text{HLB}(\mathbb{P}^1(\mathbb{C}))$ and $s \in \mathcal{M}^*(L)$. Now $\deg(\text{div}(s))$ is independent of s and depends only on L —§5, Chapter 1. We set $\deg(L) = \text{div}(s)$, where s is any non-trivial meromorphic section of L . By Proposition 7.5.7, $L \cong H^d$ for some $d \in \mathbb{Z}$. Since H^d admits a meromorphic section whose divisor had degree d , we see that $\deg(L) = d$. Indeed, $\deg(L) = c_1(L) = c_1(H^d) = d$, where we make the usual identification of $H^2(\mathbb{P}^1(\mathbb{C}), \mathbb{Z})$ with $\mathbb{Z}$. Suppose that E is a holomorphic vector bundle on $\mathbb{P}^1(\mathbb{C})$ of rank k and let $s \in \mathcal{M}^*(E)$ (such sections exist by Theorem 7.5.4). Since the zeroes and poles of s are isolated points, $\text{div}(s)$ is a well defined element of $\mathcal{D}(\mathbb{P}^1(\mathbb{C}))$ (this would not be the case if E were a holomorphic vector bundle over a complex manifold of dimension greater than 1). For each $z \in \mathbb{P}^1(\mathbb{C})$, there exists $f_z \in \mathcal{M}_z$, $h_z \in \underline{E}_z$ such that $s_z = f_z h_z$ and $h_z(z) \neq 0$. Define a subset $\mathcal{L}^s \subset \underline{E}$ by $\mathcal{L}_z^s = h_z \mathcal{O}_z$, $z \in \mathbb{P}^1(\mathbb{C})$. It is easily verified that $\mathcal{L}^s$ is a locally free subsheaf of $\underline{E}$ of rank 1. Hence $\mathcal{L}^s$ is the sheaf of holomorphic sections of a holomorphic line subbundle L^s of E . By the construction $s_z \in \mathcal{M}^*(L^s)_z$ for all $z \in \mathbb{P}^1(\mathbb{C})$ and so we may regard s as defining a meromorphic section of L^s . Clearly the bundle L^s is uniquely determined by these conditions. Set $d(s) = \deg(L^s)$. We claim that $\max\{d(s) : s \in \mathcal{M}^*(E)\} < \infty$. Since $\mathcal{L}^s \cong \mathcal{O}(d(s))$, we have $\dim_{\mathbb{C}} H^0(\mathbb{P}^1(\mathbb{C}), \mathcal{L}^s) = d(s) + 1$ (that is, the dimension of $P^{(d(s))}(\mathbb{C}^2)$). Clearly, $\dim_{\mathbb{C}} H^0(\mathbb{P}^1(\mathbb{C}), \underline{E}) \geq \dim_{\mathbb{C}} H^0(\mathbb{P}^1(\mathbb{C}), \mathcal{L}^s)$ for all $s \in \mathcal{M}^*(E)$. So if $d(s)$ were unbounded, this would contradict the finiteness of $\dim_{\mathbb{C}} H^0(\mathbb{P}^1(\mathbb{C}), \underline{E})$. Set $a_1 = \max_s d(s)$. Choose a holomorphic line subbundle L of E of degree a_1 . Thus $\underline{L} \cong \mathcal{O}(a_1)$ and we have the exact sequence

$$0 \rightarrow \mathcal{O}(a_1) \rightarrow \underline{E} \rightarrow \underline{F} \rightarrow 0,$$

where $F = E/L$ is a vector bundle of rank $k - 1$ on $\mathbb{P}^1(\mathbb{C})$.

Theorem 7.5.8. (Grothendieck). Let E be a holomorphic vector bundle on $\mathbb{P}^1(\mathbb{C})$ of rank k . Then there exists a unique sequence $a_1 \geq \dots \geq a_k$ of integers such that

$$\underline{E} \cong O(a_1) \oplus \dots \oplus O(a_k).$$

Proof. We prove by induction on k . Suppose true $k - 1$. Then as we showed above there exists a holomorphic line subbundle $L \subset E$ of maximal degree, say a_1 , and exact sequence

$$0 \rightarrow O(a_1) \rightarrow \underline{E} \rightarrow \underline{F} \rightarrow 0, \quad (*)$$

where F is a holomorphic vector bundle of rank $k - 1$. By the inductive assumption

$$\underline{F} \cong O(a_2) \oplus \dots \oplus O(a_k), \quad a_2 \geq \dots \geq a_k.$$

We claim $a_1 \geq a_2$. If not, $a_2 \geq a_1 + 1$ and tensoring $(*)$ by $O(-a_1 - 1)$ we obtain the exact sequence

$$0 \rightarrow O(-1) \rightarrow \underline{E}(-a_1 - 1) \rightarrow \underline{F}(a_1 - 1) \rightarrow 0.$$

Taking the cohomology sequence, together with the vanishing of $H^1(\mathbb{P}^1(\mathbb{C}), O(-1))$, we deduce that

$$H^0(\mathbb{P}^1(\mathbb{C}), \underline{E}(-a_1 - 1)) \cong H^0(\mathbb{P}^1(\mathbb{C}), \underline{F}(a_1 - 1)).$$

Now $\underline{F}(-a_1 - 1) \cong \bigoplus_{j=2}^k O(a_j - a_1 - 1)$. Therefore since $a_2 - a_1 - 1 \geq 0$, $H^0(\mathbb{P}^1(\mathbb{C}), \underline{F}(-a_1 - 1)) \neq 0$ and so $\underline{E}(-a_1 - 1)$ admits a non-trivial holomorphic section. Hence $\underline{E}(-a_1 - 1)$ contains a holomorphic line bundle isomorphic to H^p , $p \geq 0$. Consequently, E contains a holomorphic line bundle isomorphic to H^{p+a_1+1} which is of degree $p + a_1 + 1$, contradicting the definition of a_1 .

Next we claim that the sequence $(*)$ splits. This is a consequence of a general splitting lemma.

Lemma 7.5.9. Let $0 \rightarrow \mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H} \rightarrow 0$ be an exact sequence of coherent sheaves on the complex manifold M . Suppose that $\mathcal{H}$ is locally free and that $H^1(M, \text{Hom}(\mathcal{H}, \mathcal{F})) = 0$. Then the sequence splits.

Proof. We refer to the exercises at the end of §1 for the definition of $\text{Hom}(\mathcal{F}, \mathcal{G})$. See also the exercises at the end of §1, Chapter 6. Since $\mathcal{H}$ is locally free the sequence

$$0 \rightarrow \text{Hom}(\mathcal{H}, \mathcal{F}) \rightarrow \text{Hom}(\mathcal{H}, \mathcal{G}) \rightarrow \text{Hom}(\mathcal{H}, \mathcal{H}) \rightarrow 0$$

is exact. Taking the cohomology sequence we deduce that

$$H^0(M, \text{Hom}(\mathcal{H}, \mathcal{G})) \rightarrow H^0(M, \text{Hom}(\mathcal{H}, \mathcal{H})) \rightarrow 0$$

is exact. Therefore there exists $c : \mathcal{H} \rightarrow \mathcal{G}$ which is mapped to the identity homomorphism of $\mathcal{H}$. That is, $bc = \text{Id}$. $\square$

Returning to the proof of our theorem we see that

$$\text{Hom}(\underline{F}, O(a_1)) = \underline{F}^* \otimes O(a_1) \cong \bigoplus_{j=2}^k O(a_1 - a_j).$$

Now $a_1 - a_j \geq 0$ and so $H^1(\mathbb{P}^1(\mathbb{C}), \text{Hom}(\underline{F}, O(a_1))) = 0$. Hence we may apply the splitting lemma to deduce that $(*)$ splits. But then,

$$\underline{E} \cong O(a_1) \oplus \bigoplus_{j=2}^k O(a_j).$$

Finally the uniqueness of the sequence $a_1 \geq \dots \geq a_k$ follows by observing that the number of a_i 's equal to m is precisely $\dim_{\mathbb{C}} H^0(\mathbb{P}^1(\mathbb{C}), \underline{E}(-m))$. $\square$

Remark. Much is known about the classification of holomorphic vector bundles on compact Riemann surfaces. See Gunning [3], M. Narasimhan [1] and Tjurin [1].

Theorem 7.5.8 does *not* generalise to vector bundles over $\mathbb{P}^n(\mathbb{C})$, $n > 1$.

Example 1. $T\mathbb{P}^2(\mathbb{C})$ is not a sum of holomorphic line bundles. Suppose the contrary. Then $T\mathbb{P}^2(\mathbb{C}) \cong O(a) \oplus O(b)$, where we may suppose $a \geq b$. Taking the Euler sequence for $T\mathbb{P}^2(\mathbb{C})$, we therefore have the exact sequence

$$0 \rightarrow \mathcal{O} \rightarrow O(1)^3 \rightarrow O(a) \oplus O(b) \rightarrow 0.$$

Suppose $a \geq 2$. Tensoring the Euler sequence by $O(-2)$ and taking the cohomology sequence of the resulting short exact sequence we find $H^0(\mathbb{P}^2(\mathbb{C}), O(-1)^3) \cong H^0(\mathbb{P}^2(\mathbb{C}), O(a-2) \oplus O(b-2))$, which cannot be since the first cohomology group is zero, the second non-zero. Therefore, $a, b \leq 1$. Now take the cohomology sequence of the Euler sequence and count dimensions of the zero dimensional groups to derive a contradiction. (The same analysis will show that $T\mathbb{P}^n(\mathbb{C})$ is never a direct sum of line bundles for $n \geq 2$).

Suppose that $\ell \subset \mathbb{P}^n(\mathbb{C})$ is a line (that is, $\ell \cong \mathbb{P}^1(\mathbb{C})$). Grothendieck's theorem, together with a similar analysis to that presented in the example above, shows that

$$T\mathbb{P}^n(\mathbb{C})|_{\ell} \cong O(2) \oplus O(1)^{n-1}.$$

We say that a holomorphic vector bundle E on $\mathbb{P}^n(\mathbb{C})$ of rank is *uniform* of splitting type $(a_1, \dots, a_k)$ if for every complex line $\ell \subset \mathbb{P}^n(\mathbb{C})$,

$$E|_{\ell} = \bigoplus_{j=1}^k O(a_j).$$

It can be shown that a uniform bundle of rank $k < n$ is a direct sum of line bundles and that if $k = n$ then it is either a direct sum of line bundles or of the form $T\mathbb{P}^n(\mathbb{C})(m)$, $T\mathbb{P}^n(\mathbb{C})^*(m)$ (see Okonek, Schneider and Spindler [1, pages 70,71]).

However, not every holomorphic vector bundle on $\mathbb{P}^n(\mathbb{C})$ is uniform, $n > 1$, and the problem of classification is still open. Substantial progress has recently been made in the classification of a class of rank 2 vector bundles over $\mathbb{P}^3(\mathbb{C})$ —the so called *instanton bundles*—which give rise to self dual solutions of the Yang–Mills field equations. An important feature here is the appearance of moduli in the classification. For references, together with an up-to-date survey of the theory of holomorphic vector bundles on projective space, we refer to the book by Okonek, Schneider and Spindler [1].

Exercises.

1. Let E be a holomorphic vector bundle on $\mathbb{P}^n(\mathbb{C})$ and $v \in E_z$, $z \in \mathbb{P}^n(\mathbb{C})$. Show that there exists $s \in \mathcal{M}^*(E)$ such that $s(z) = v$.
2. Let $M \subset \mathbb{P}^n(\mathbb{C})$ be algebraic. Show that Serre's Theorems A and B hold for all coherent sheaves on M (twist with the hyperplane section bundle restricted to M).
3. Verify that
 1. $T\mathbb{P}^n(\mathbb{C})$ is not a sum of line bundles, $n > 2$.
 2. $T\mathbb{P}^n(\mathbb{C})$ is uniform of splitting type $(2, 1, \dots, 1)$, $n \geq 1$.

(Hint: It may be useful to recall that $\dim_{\mathbb{C}} P^{(m)}(\mathbb{C}^{n+1}) = \binom{m+n}{n}$).

4. Show that Grothendieck's theorem (Theorem 7.5.8) is equivalent to the following statement about invertible holomorphic matrices: Regarding $\mathbb{P}^1(\mathbb{C})$ as the 1-point compactification of $\mathbb{C}$, let $t > 1$ and set $U_1 = \{z \in \mathbb{P}^1(\mathbb{C}) : |z| < t\}$, $U_2 = \{z \in \mathbb{P}^1(\mathbb{C}) : |z| > 1\}$. Suppose $M \in \mathrm{GL}(n, A(U_{12}))$. Then there exist $P \in \mathrm{GL}(n, A(U_1))$, $Q \in \mathrm{GL}(n, A(U_2))$ such that PMQ is a diagonal matrix with diagonal terms z^{a_j} , $a_j \in \mathbb{Z}$.
5. (Koszul complex) Let A be a commutative ring with 1 and for $m \geq 1$, let $E_1, \dots, E_m$ denote the standard A -module basis of A^m (that is $E_1 = (1, 0, \dots, 0)$, $\dots$, $E_m = (0, 0, \dots, 1)$). The A -module $\Lambda^p A^m$ has A -module basis $\{E_J = E_{j_1} \wedge \dots \wedge E_{j_p} : 1 \leq j_1 < \dots < j_p \leq m\}$. Suppose that $f_1, \dots, f_m \in A$, set $f = (f_1, \dots, f_m) \in A^m$ and let I denote the ideal in A generated by $f_1, \dots, f_m$. Show

a) We have the *Koszul complex*

$$0 \rightarrow \Lambda^m A^m \xrightarrow{\partial} \Lambda^{m-1} A^m \xrightarrow{\partial} \dots \xrightarrow{\partial} \Lambda^1 A^m \xrightarrow{\partial} A \rightarrow A/I \rightarrow 0$$

of A -module homomorphisms, where $\partial a = C_f a$. That is, if $a = a_J E_J \in \Lambda^p A^m$ then

$$\partial a = \sum_J a_J \sum_{i=1}^p (-1)^{i+1} f_{j_i} E_{j_1} \wedge \dots \wedge \widehat{E}_{j_i} \wedge \dots \wedge E_{j_p}.$$

Show also that $\partial(\Lambda^1 A^m) = I$.

- b) Let $I_k = (f_1, \dots, f_k)$ denote the ideal in A generated by $f_1, \dots, f_k$. We say that $(f_1, \dots, f_m)$ is a *regular sequence* if f_k is not a zero divisor in A/I_{k-1} , $k = 1, \dots, m$. Show that if $(f_1, \dots, f_m)$ is a regular sequence, then the Koszul complex is exact (Hint: prove by induction on m . See also Griffiths and Harris [1, pages 688–690]).

Now suppose that $p_j \in \mathbb{C}[z_0, \dots, z_n]$ is homogeneous of degree d_j , $1 \leq j \leq m$, and let $p = (p_1, \dots, p_m)$ denote the corresponding section of $E = \mathcal{O}(d_1) \oplus \dots \oplus \mathcal{O}(d_m)$. Set $X = Z(p) \subset \mathbb{P}^n(\mathbb{C})$. Show

- c) If $(p_1, \dots, p_m)$ is a regular sequence in $\mathbb{C}[z_0, \dots, z_n]$ then the Koszul complex

$$0 \rightarrow \Lambda^m \underline{E}^* \xrightarrow{\partial} \Lambda^{m-1} \underline{E}^* \xrightarrow{\partial} \dots \xrightarrow{\partial} \Lambda^1 \underline{E}^* \rightarrow \mathcal{O}_{\mathbb{P}^n(\mathbb{C})} \rightarrow \mathcal{O}_X \rightarrow 0$$

is a locally free resolution of $\mathcal{O}_X$.

(See also Exercise 14, §1, Chapter 6. The assumption on $(p_1, \dots, p_m)$ amounts to saying that X is a *complete intersection*).

6. Let E be a holomorphic vector bundle on the compact Riemann surface M . Given $d \in \mathcal{D}(M)$, we set $E(d) = E \otimes [d]$. Suppose $d, d' \in \mathcal{D}(M)$ and that $d \leq d'$. Set $d' - d = \sum_{i=1}^k p_i \cdot z_i$, where each $p_i > 0$. Show

- a) We have a natural exact sequence

$$0 \rightarrow \underline{E}(d) \rightarrow \underline{E}(d') \rightarrow \mathcal{F} \rightarrow 0$$

where $\mathcal{F}$ is the “Manhattan” sheaf whose stalk is zero except at points $x_i \in |d' - d|$ where we have $\mathcal{F}_{x_i} = \mathbb{C}^{p_i q}$, $q = \dim(E)$.

- b) If we define $\chi(E) = \dim_{\mathbb{C}} H^0(M, E) - \dim_{\mathbb{C}} H^1(M, E)$ then

$$\chi(E(d)) = q \deg(d) + \chi(E).$$

(Hint: The alternating sum of dimensions in an exact sequence $0 \rightarrow L_0 \rightarrow L_1 \rightarrow \dots \rightarrow L_n \rightarrow 0$ of vector spaces is zero).

- c) $\dim_{\mathbb{C}} \Omega(E(d)) \geq q \deg(d) + \chi(E)$.
 - d) E has non-trivial meromorphic sections.
 - e) In case E is a holomorphic line bundle on M , there exists $d \in \mathcal{D}(M)$ such that $E \cong [d]$ and $c_1(E) = \deg(d)$.
 - f) $H^1(M, \mathcal{M}^*) = 0$.
7. Let M be a compact Riemann surface. Given $d \in \mathcal{D}(M)$, set $l(d) = \dim_{\mathbb{C}} H^0(M, [d])$; $i(d) = \dim_{\mathbb{C}} H^1(M, [d])$. Show that

$$l(d) - i(d) = \deg(d) + 1 - g,$$

where $g = \dim_{\mathbb{C}} H^1(M, \mathcal{O})$.

(Riemann–Roch formula).

Deduce

- a) $l(d) \geq \deg(d) + 1 - g$ (Riemann’s inequality).
- b) If $\deg(d) \geq g + 1$, there exists $m \in \mathcal{M}^*(M)$ such that $\operatorname{div}(m) + d \geq 0$.
- c) M may be represented as a branched cover of $\mathbb{P}^1(\mathbb{C})$ with at most $g + 1$ sheets. In particular, if $g = 0$, $M = \mathbb{P}^1(\mathbb{C})$.

(We remark that g is actually the genus of M . In particular, g is a topological rather than analytic invariant of M . To see that g is the genus of M (that is $\frac{1}{2} \dim_{\mathbb{C}} H^1(M, \mathbb{C})$), we note that Serre-duality implies that $H^1(M, \mathcal{O}) = H^0(M, \Omega^1)$. Now take the cohomology sequence of the short exact sequence $0 \rightarrow \mathbb{C} \rightarrow \mathcal{O} \xrightarrow{\partial=d} \Omega^1 \rightarrow 0$ —see Example 23, §1, Chapter 6).

6 The Kodaira embedding theorem

In this section we prove a version of the Kodaira embedding theorem due to Grauert. The main step in the proof is a cohomology vanishing theorem for coherent sheaves on a compact complex manifold admitting a weakly positive vector bundle (for the definition of weak positivity, see §10, Chapter 5).

First some notation. Suppose that E is a holomorphic vector bundle on the complex manifold M . We let $E^{(s)}$ denote the s -fold *symmetric* tensor product of E (this is just the usual tensor product if E is a line bundle). If $\mathcal{F}$ is a coherent sheaf on M , we let $\mathcal{F}^{(s)}(E)$ denote the E -twisted sheaf $\mathcal{F} \otimes_{\mathcal{O}} \underline{E}^{(s)}$.

Theorem 7.6.1. (Grauert [1]). Let E be a weakly positive vector bundle on the compact complex manifold M . Then for any coherent sheaf $\mathcal{F}$ on M , there exists $m_0 = m_0(\mathcal{F})$ such that

$$H^p(M, \mathcal{F}^{(m)}(E)) = 0, \quad p \geq 1, \quad m \geq m_0.$$

Proof. Let $\pi : E^* \rightarrow M$ denote the dual of E . Then E^* is weakly negative and so there exists an s.l.p. neighbourhood D of the zero section of E^* . Set $\tilde{\mathcal{F}} = \pi^* \mathcal{F}|_D$. Then $\tilde{\mathcal{F}}$ is a coherent sheaf on D (Example 6, §1). We shall show that for $N \geq 0$, there exists a canonical linear injection

$$\phi_N : \sum_{s=0}^N H^p(M, \mathcal{F}^{(s)}(E)) \longrightarrow H^p(D, \tilde{\mathcal{F}}).$$

Since Grauert’s finiteness theorem implies that $H^p(D, \tilde{\mathcal{F}})$ is finite dimensional, $p \geq 1$, the existence of such an injection certainly implies that $H^p(M, \mathcal{F}^{(s)}(E))$ is zero dimensional for all sufficiently large s , $p \geq 1$.

Let ω denote the zero section of E^* and $i : M \rightarrow E^*$ the canonical inclusion of M on ω . Set $\hat{\mathcal{O}} = i^{-1}\mathcal{O}_{E^*}$ and note that $\hat{\mathcal{O}}$ has the natural structure of a sheaf of $\mathcal{O}_M$ -modules (not of finite type). For $s \geq 0$, we have a natural $\mathcal{O}_M$ -morphism

$$\chi_s : \underline{E}^{(s)} \longrightarrow \hat{\mathcal{O}}$$

defined by $\chi_s(f_z)(z)(v_z) = f_z(z)(v_z^s)$, $z \in M$, $f_z \in \underline{E}_z^{(s)}$, $v_z \in E_z^*$. That is, $f_z(z) \in E_z^{(s)} \approx (E_z^*)^{*(s)}$ and so defines a homogeneous polynomial map of degree s from E_z^* to $\mathbb{C}$. Evaluate $f_z(z)$ at points of the fibre E_z^* . The map χ_s is obviously injective. Moreover, if we set

$$\chi^N = \bigoplus_{s=0}^N \chi_s : \bigoplus_{s=0}^N \underline{E}^{(s)} \longrightarrow \hat{\mathcal{O}}$$

it is clear that χ^N is also injective since the image of each χ_s consists of analytic germs which are homogeneous of degree s in the fibre coordinates. Furthermore, we can define an $\mathcal{O}_M$ -morphism $j^N : \hat{\mathcal{O}} \rightarrow \bigoplus_{s=0}^N \underline{E}^{(s)}$ which is a right inverse for χ^N (that is, $j^N \chi^N = \text{Id}$). To do this suppose $g_z \in \hat{\mathcal{O}}_z$. Then, by Taylor's theorem, we may write $g_z = \sum_{j=0}^{\infty} f_j P_j$, where $f_j \in \mathcal{O}_{M,z}$, $P_j \in E_z^{**^{(s)}} \approx E_z^{(s)}$. Now define $j^N(g_z) = (f_0 P_0, \dots, f_N P_N)$ ($j^N g_z$ is the N -jet of g_z along the fibres of E^*). Since i is a biholomorphism, we have induced maps

$$\begin{aligned} I_N : \mathcal{F} \otimes \bigoplus_{s=0}^N \underline{E}^{(s)} &\longrightarrow \tilde{\mathcal{F}}_{\omega} \quad \left(= (\pi^{-1}\mathcal{F})_{\omega} \otimes_{\pi^{-1}(\mathcal{O}_M)} \mathcal{O}_{\omega} \right) \\ J_N : \tilde{\mathcal{F}}_{\omega} &\longrightarrow \mathcal{F} \otimes \bigoplus_{s=0}^N \underline{E}^{(s)} \end{aligned}$$

satisfying $J_N I_N = \text{Id}$, and induced maps on cohomology

$$H^p \left(M, \mathcal{F} \otimes \bigoplus_{s=0}^N \underline{E}^{(s)} \right) \xrightleftharpoons[I_N^*]{I_N^*} H^p(\omega, \tilde{\mathcal{F}}_{\omega})$$

(Exercise 14, §3, Chapter 6). In particular, I_N^* is injective, $p, N \geq 0$. Let $r = i^{-1}(\pi|_D) : D \rightarrow \omega$ and $k : \omega \rightarrow D$ denote the inclusion. Now r^*k is the identity map on ω and so we have the commutative diagram

$$\begin{array}{ccc} H^p(\omega, \tilde{\mathcal{F}}_{\omega}) & \xrightarrow{\text{Id}} & H^p(\omega, \tilde{\mathcal{F}}_{\omega}) \\ & \searrow r^* & \nearrow k^* \\ & H^p(D, \tilde{\mathcal{F}}) & \end{array}$$

(see Exercise 15, §3, Chapter 6 and note that $k^{-1}\tilde{\mathcal{F}} = \tilde{\mathcal{F}}_{\omega}$, $r^{-1}\tilde{\mathcal{F}}_{\omega} = \tilde{\mathcal{F}}$). But therefore r^* is injective and so the map

$$\phi_N = r^* I_N^* : H^p \left(M, \mathcal{F} \otimes \bigoplus_{s=0}^N \underline{E}^{(s)} \right) \longrightarrow H^p(D, \tilde{\mathcal{F}})$$

is injective, $p \geq 0$. □

Suppose that F is a holomorphic line bundle on the compact complex manifold M and that for each $z \in M$, there exists $s \in H^0(M, \underline{F})$ such that $s(z) \neq 0$. Choose a $\mathbb{C}$ -basis $s_0, \dots, s_k$ for $H^0(M, \underline{F})$. Then, as we showed in §9, Chapter 5, $s = (s_0, \dots, s_k)$ defines a holomorphic map of M in $\mathbb{P}^k(\mathbb{C})$. We shall say that F is *ample* (respectively *very ample*) if $H^0(M, \underline{F})$ determines a holomorphic map (respectively embedding) of M in some projective space. We remark that if F is ample (respectively, very ample) then so is F^k , $k \geq 1$.

Theorem 7.6.2. (Kodaira embedding theorem). Let M be a compact complex manifold and suppose that E is a weakly positive vector bundle on M . Then M admits an embedding in projective space.

Proof. (Grauert [1]). We shall prove the theorem in case E is a weakly positive line bundle. The proof in case E is a bundle of rank greater than 1 is similar except that M gets embedded in a Grassmann manifold (which, of course, can be embedded in projective space).

First we show that there exists k_0 such that $H^0(M, \underline{E}^k)$ determines a holomorphic immersion of M in projective space, $k \geq k_0$.

Fix $z \in M$ and let $\mathcal{I}_z^2 = \mathcal{I}_z \mathcal{I}_z$, where $\mathcal{I}_z$ is the ideal sheaf of z . The sheaf $\mathcal{I}_z^2$ is coherent and we have the short exact sequence

$$0 \rightarrow \mathcal{I}_z^2 \rightarrow \mathcal{O}_M \xrightarrow{T} J(z) \rightarrow 0, \quad (*)$$

where $J(z)$ is the skyscraper sheaf supported at z with stalk $\mathcal{O}_z/\mathfrak{m}_z^2$. Now $T(f_z) = (f_z(z), df_z(z))$. That is, $\mathcal{O}_z/\mathfrak{m}_z^2$ measures the first two terms in the Taylor expansion of f_z . By Theorem 7.6.1, there exists $k(z)$ such that $H^1(M, \mathcal{I}_z^2 \otimes \underline{E}^k) = 0$, $k \geq k(z)$. Tensoring (*) with $\underline{E}^k$ and taking the cohomology sequence of the short exact sequence we deduce that

$$H^0(M, \underline{E}^k) \xrightarrow{T^*} H^0(M, J(z) \otimes \underline{E}^k)$$

is surjective, $k \geq k(z)$. Since $H^0(M, J(z) \otimes \underline{E}^k) \cong J(z)$, we deduce that $H^0(M, \underline{E}^k)$ determines a holomorphic embedding of some open neighbourhood U_z of z in projective space, $k \geq k(z)$ (U_z may be chosen independently of $k \geq k(z)$). Indeed, a U_z that works for $k(z)$ will work for $k > k(z)$). Doing this for every $z \in M$ and using the compactness of M we obtain a finite cover $U_1, \dots, U_n$ of M , corresponding integers $k_1, \dots, k_n$, such that $H^0(M, \underline{E}^{k_j})$ determines a holomorphic embedding of U_j in projective space. Taking $k_0 = \max_j k_j$, we see that $H^0(M, \underline{E}^k)$ determines a holomorphic immersion of M in projective space, $k \geq k_0$. Set $U = \bigcup_{j=1}^n (U_j \times U_j) \subset M \times M$. For $(x, y) \in (M \times M) \setminus U$, let $\mathcal{I}_{x,y}$ denote the sheaf of germs of holomorphic functions on M vanishing on $\{x, y\}$. Just as we did above, we find that there exists $m(x, y)$ such that for $m \geq m(x, y)$ the natural map

$$H^0(M, \underline{E}^m) \longrightarrow H^0(M, (\mathbb{C}(x) \oplus \mathbb{C}(y)) \otimes \underline{E}^m)$$

is onto. Hence there exists a neighbourhood W of (x, y) in $M \times M$ such that if $(x', y') \in W$, there exists $s \in H^0(M, \underline{E}^m)$ with $s(x') \neq s(y')$, $m \geq m(x, y)$. Since $(M \times M) \setminus U$ is compact, we may find an open cover $W_1, \dots, W_p$ of $(M \times M) \setminus U$ and corresponding integers $m_1, \dots, m_p$ such that given $(x, y) \in W_j$, there exists $s \in H^0(M, \underline{E}^m)$ with $s(x) \neq s(y)$, $m \geq m_j$. Now let $m_0 = \max\{k_0, m_1, \dots, m_p\}$. Our construction guarantees that $\underline{E}^m$ is very ample for $m \geq m_0$. $\square$

Remarks.

1. Kodaira's original proof of Theorem 7.6.2 is rather different from that of Grauert. Theorem 7.6.1 is replaced by a cohomology vanishing theorem for cohomology with coefficients in a holomorphic vector bundle whose curvature satisfies certain positivity conditions. In the construction of the embedding, Kodaira uses blowing up arguments, in combination with his vanishing theorem, rather than the twisting arguments we used. The reader may consult Kodaira and Morrow [1] and Wells [1] for presentations of Kodaira's original proof. In Chapter 10, we shall prove the cohomology vanishing theorem ("Kodaira's vanishing theorem") referred to above. One important feature of Grauert's proof of the Kodaira embedding theorem is that it generalises to compact analytic spaces admitting a weakly positive bundle. For details we refer to Grauert [1].

2. Since $\mathbb{P}^n(\mathbb{C})$ admits a weakly positive line bundle (H^1), Theorem 7.6.2 implies part B of Theorem 7.5.3 of Serre. But part B easily implies part A by the usual cohomology sequence arguments and so we see that Theorem 7.6.1 may be regarded as a generalisation of Theorems A and B of Serre.
3. Chow's theorem implies that the image of M in $\mathbb{P}^N(\mathbb{C})$ given by Kodaira's embedding theorem is an algebraic set. We may actually embed M in $\mathbb{P}^{2m+1}(\mathbb{C})$, $m = \dim_{\mathbb{C}} M$. We indicate briefly why this is so (for details see Griffiths and Harris [1] and also Hartshorne [1] for the case of curves). Let $\Sigma \subset \mathbb{P}^N(\mathbb{C})$ denote the union of all projective lines which are either chords or tangents to M (that is, its image in $\mathbb{P}^N(\mathbb{C})$). It may be shown that Σ is an algebraic subset of $\mathbb{P}^N(\mathbb{C})$ of dimension $\leq 2m + 1$. In particular, $\mathbb{P}^N(\mathbb{C}) \setminus \Sigma$ will not be empty provided $N > 2m + 1$. Choose any $\mathbb{P}^{N-1}(\mathbb{C}) \subset \mathbb{P}^N(\mathbb{C})$ and $P \in \mathbb{P}^N(\mathbb{C}) \setminus (\Sigma \cup \mathbb{P}^{N-1}(\mathbb{C}))$. Project M into $\mathbb{P}^{N-1}(\mathbb{C})$ from P . Clearly the projection is an injective immersion— $P \notin \Sigma$ —and so defines an embedding of M in $\mathbb{P}^{N-1}(\mathbb{C})$.

We end with an example.

Example. Let $\Lambda \subset \mathbb{C}^n$ be a lattice and suppose that Λ admits a Riemann form. That is, we shall assume that there exists a positive definite Hermitian form H on $\mathbb{C}^n$ whose imaginary part is integer valued on Λ (see Chapter 4, §4; Chapter 5, §9). We claim that $T = \mathbb{C}^n/\Lambda$ is an abelian variety. We shall prove this by showing that the holomorphic line bundle $L(H, 1)$ on T is weakly positive and applying Kodaira's embedding theorem. By Exercise 6, §9, Chapter 5, $\theta(z) = \exp \pi H(z, z)$, $z \in \mathbb{C}^n$, determines a smooth nowhere vanishing section of $L(H, 1) \otimes \overline{L(H, 1)}$. Define $\eta : \mathbb{C}^n \times \mathbb{C} \rightarrow \mathbb{R}$ by $\eta(z, t) = |t|^2 \theta(z)$ and observe that η induces a smooth map $\tilde{\eta} : L(H, 1)^* \cong L(-H, 1) \rightarrow \mathbb{R}$ ($\tilde{\eta}$ is the square of the radius function on $L(H, 1)^*$ associated to the hermitian metric on $L(H, 1)^*$ determined by θ). A straightforward computation shows that $L(\eta)(z, t)$ is positive definite provided that $t \neq 0$. Hence $L(\tilde{\eta})$ is certainly positive definite off the zero section of $L(H, 1)^*$. Setting $D = \{v \in L(H, 1)^* : \tilde{\eta}(v) < 1\}$, we see that D is an s.l.p. neighbourhood of the zero section of $L(H, 1)^*$ and so $L(H, 1)^*$ is weakly negative. Hence $L(H, 1)$ is weakly positive.

Exercises.

1. Prove Theorem 7.6.2 in case E is of rank greater than 1.
2. This exercise is a continuation of exercises 6, 7 of §5. Let M be a compact Riemann surface. Show that if $z \in M$ then $\mathcal{I}_z^p = [-p, z]$, $p \geq 0$, where $\mathcal{I}_z$ denotes the ideal sheaf of $\{z\}$. Suppose that E is a holomorphic line bundle on M . Prove
 - a) If $c_1(E) \geq 2g - 1$, then $H^1(M, \underline{E}) = 0$.
 - b) If $c_1(E) \geq 2g$, then E is ample (see the proof of Theorem 7.6.2).
 - c) If $c_1(E) \geq 2g + 1$, then E is very ample.

Deduce that every compact Riemann surface is algebraic.

3. Show that every compact Riemann surface M admits an open covering by two Stein open sets (Hint: Embed M in $\mathbb{P}^N(\mathbb{C})$. Consider the intersection of M with two hyperplanes H_1, H_2 chosen so that $M \cap H_1 \cap H_2 = \emptyset$. Show that $\{M \setminus H_i : i = 1, 2\}$ is a Stein open cover of M). Deduce that $H^p(M, \mathcal{F}) = 0$, $p \geq 2$, for every coherent sheaf $\mathcal{F}$ on M (see also Grauert and Remmert [1, page 210]).

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